Theoretical Chemistry and Computational Modelling

Arnout Jozef Ceulemans

Group Theory Applied to Chemistry

Second Edition

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Theoretical Chemistry and Computational Modelling

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To my grandson Louis

The world is so full of a number of things,
I'm sure we should all be as happy as kings.

—Robert Louis Stevenson

Preface

Symmetry is a general principle, which plays an important role in various areas of knowledge and perception, ranging from arts and aesthetics to natural sciences and mathematics. The first edition of this book was entirely devoted to the use of point groups in the study of chemical and physical phenomena. The second edition contains an additional chapter on line groups and plane groups, which apply to extended structures with translational symmetry in one and two dimensions. This choice was motivated by the growing interest in nanostructures, based on polymers and molecular sheets. The chapter explains the direct relationship between line and plane symmetries and the corresponding band structures. The results are applied to polymers such as polyacetylene and carbon nanotubes. As for the existing chapters on point group applications, the advice was followed not to overload these with new material, but rather devote more time to explain the more difficult parts. Such topics included the bra–ket conjugation relation, the connection between point groups and isometries, the symmetric groups, the practical use of subduction tables, the eigenvalues of Cayley graphs, and the symmetry of Slater determinants. To facilitate reading, summaries of essential background knowledge were inserted in separate text boxes. Extra figures and study problems were also provided to illustrate group theory at work in molecular science.

Leuven, Belgium — Arnout Jozef Ceulemans

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Chapter 1
Operations



Abstract In this chapter, we examine the precise meaning of the statement that a symmetry operation *acts* on a point in space, on a function, and on an operator. The difference between active and passive views of symmetry is explained and a few practical conventions are introduced.

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1.1 Operations and Points

In the usual crystallographic sense, symmetry operations are defined as rotations and reflections that turn a body into a congruent position. This can be realized in two ways. The active view of a rotation is the following. An observer takes a snapshot of a crystal, then the crystal is rotated while the camera is left immobile. A second snapshot is taken. If the two snapshots are identical, then we have performed a symmetry operation. In the passive view, the camera takes a snapshot of the crystal, then the camera is displaced while the crystal is left immobile. From a new perspective, a second snapshot is taken. If this is the same as the first one, we have found a symmetry-related position. Both points of view are equivalent as far as the *relative* positions of the observer and the crystal are concerned. However, viewed in the frame of absolute space there is an important difference: if the rotation of the crystal in the active view is taken to be counterclockwise, the rotation of the observer in the passive alternative will be clockwise. Hence, the transformation from active

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to passive involves a change of the sign of the rotation angle. In order to avoid annoying sign problems, only one choice of definition should be adhered to. In the present monograph, we shall consistently adopt the *active view*, in line with the usual convention in chemistry textbooks. In this script, the part of the observer is played by the set of coordinate axes which defines absolute space in a Cartesian way. They will stay where they are. On the other hand, the structures, which are operated on, are moving on the scene. To be precise, a symmetry operation $\hat{R}$ will move a point P_1 with coordinates[1] $(\mathrm{x}_1, \mathrm{y}_1, \mathrm{z}_1)$ to a new position P_2 with coordinates $(\mathrm{x}_2, \mathrm{y}_2, \mathrm{z}_2)$.

$$\hat{R}\,P_1 = P_2 \tag{1.1}$$

A pure rotation, $\hat{C}_n$ $(n > 1)$, around a given axis through an angle $2\pi/n$ radians displaces all the points, except the ones that are lying on the rotation axis itself. A reflection plane, $\hat{\sigma}_h$, moves all points except the ones lying in the reflection plane itself. A rotation–reflection, $\hat{S}_n$ $(n > 2)$, is a combination in either order of a $\hat{C}_n$ rotation and a reflection through a plane perpendicular to the rotation axis. As a result, only the point of intersection of the plane with the axis perpendicular to it is kept. A special case arises for $n = 2$. The $\hat{S}_2$ operator corresponds to the inversion and will be denoted as $\hat{\imath}$. It maps every point onto its antipode. A plane of symmetry can also be expressed as the result of a rotation through an angle π around an axis perpendicular to the plane, followed by inversion through the intersection point of the axis and the plane. A convenient way to present these operations is shown in Fig. 1.1. Operator products are 'right-justified', so that $\hat{\imath}\,\hat{C}_2^z$ means that $\hat{C}_2^z$ is applied first, and *then* the inversion acts on the intermediate result.

$$\hat{\sigma}_h P_1 = \hat{\imath}\;\hat{C}_2^z P_1 = \hat{\imath}\,P_2 = P_3 \tag{1.2}$$

From the mathematical point of view, the rotation of a point corresponds to a transformation of its coordinates. Consider a right-handed Cartesian coordinate frame, and a point P_1 lying in the xy-plane. The point is being subjected to a rotation about the upright z-axis by an angle α. By convention, a positive value of α will correspond to a counterclockwise direction of rotation. An observer on the pole of the rotation axis and looking down onto the plane will view this rotation as going in the opposite sense to that of the rotation of the hands on his watch. A synonym for counterclockwise here is right-handed. If the reader orients his/her thumb in the direction of the rotational pole, the palm of his/her right hand will indicate the counterclockwise direction. The transformation can be obtained as follows. Let r be the length of the radius-vector, $\mathbf{r}$, from the origin to the point P_1, and let ϕ_1 be the angular coordinate of the point measured in the horizontal plane starting from the x-direction, as shown in Fig. 1.2. The coordinates of P_1 are then given by

[1] The use of upright (roman) symbols for the coordinates is deliberate. Italics will be reserved for variables, but here $\mathrm{x}_1, \mathrm{y}_1, \ldots$ refer to fixed values of the coordinates. The importance of this difference will become clear later (see Eq. (1.15)).

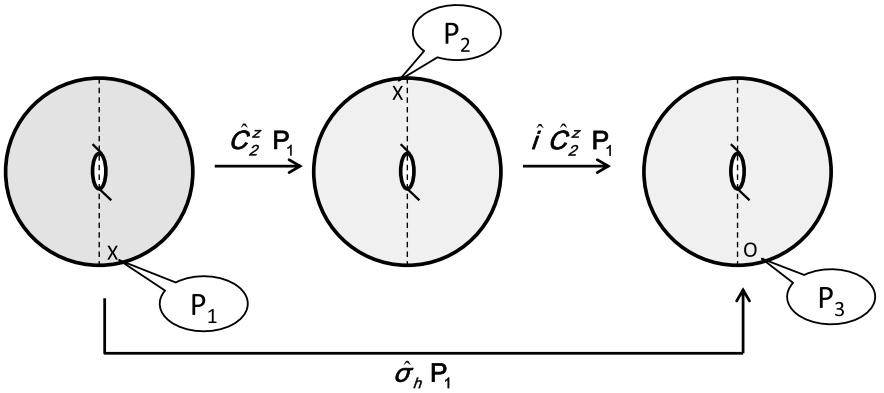


Fig. 1.1 Stereographic view of the reflection plane. The point P_1, indicated by X, is above the plane of the gray disk. The reflection operation in the horizontal plane, $\hat{\sigma}_h$, is the result of the $\hat{C}_2^z$ rotation around the center by an angle of π, followed by inversion through the center of the diagram, to reach the position P_3 below the plane, indicated by the small circle

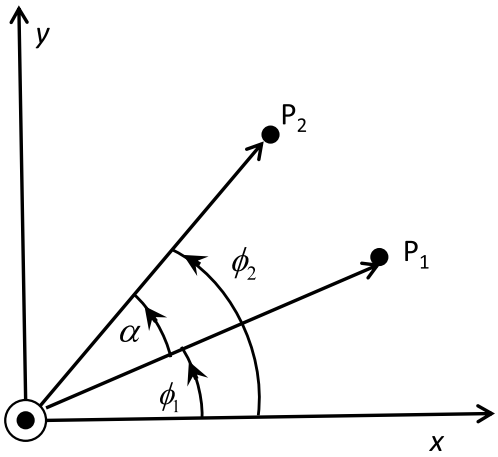


Fig. 1.2 Counterclockwise rotation of the point P_1 by an angle α in the xy-plane; the encircled dot is the pole of the z-axis

$$\begin{aligned} \mathrm{x}_1 &= \mathrm{r}\cos\phi_1 \\ \mathrm{y}_1 &= \mathrm{r}\sin\phi_1 \\ \mathrm{z}_1 &= 0 \end{aligned} \tag{1.3}$$

Rotating the point will not change its distance from the origin, but the angular coordinate will increase by α. The angular coordinate of P_2 will thus be given by $\phi_2 = \phi_1 + \alpha$. The coordinates of the image point in terms of the coordinates of the original point are thus given by

$$\begin{aligned} \mathrm{x}_2 &= \mathrm{r}\cos\phi_2 = \mathrm{r}\cos(\phi_1 + \alpha) \\ &= \mathrm{r}\cos\phi_1\cos\alpha - \mathrm{r}\sin\phi_1\sin\alpha \\ &= \mathrm{x}_1\cos\alpha - \mathrm{y}_1\sin\alpha \\ \mathrm{y}_2 &= \mathrm{r}\sin\phi_2 = \mathrm{r}\sin(\phi_1 + \alpha) \\ &= \mathrm{r}\cos\phi_1\sin\alpha + \mathrm{r}\sin\phi_1\cos\alpha \\ &= \mathrm{x}_1\sin\alpha + \mathrm{y}_1\cos\alpha \\ \mathrm{z}_2 &= 0 \end{aligned} \tag{1.4}$$

In this way, the coordinates of P_2 are obtained as a function of the coordinates of P_1 and the rotation angle. This derivation depends simply on the trigonometric relationships for sums and differences of angles. We may also express this result in the form of a matrix transformation. For this we put the coordinates in a column vector and operate on it (on the left) by means of a transformation matrix $\mathbb{D}(R)$.

$$\begin{pmatrix} \mathrm{x}_2 \\ \mathrm{y}_2 \end{pmatrix} = \mathbb{D}(R) \begin{pmatrix} \mathrm{x}_1 \\ \mathrm{y}_1 \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \mathrm{x}_1 \\ \mathrm{y}_1 \end{pmatrix} \tag{1.5}$$

Having obtained the algebraic expressions, it is always prudent to consider whether the results make sense. Hence, while the point P_1 is rotated as shown in the picture, its x-coordinate will decrease while its y-coordinate will increase. This is reflected by the entries in the first row of the matrix which show how x_1 will change: the $\cos\alpha$ factor is smaller than 1, and thus will reduce the x-value as the acute angle increases, and this will be reinforced by the second term, $-\mathrm{y}_1\sin\alpha$, which will be negative for a point with y_1 and $\sin\alpha$ both positive. In what follows we also need the inverse operation, $\hat{R}^{-1}$, which will undo the operation itself. In the case of a rotation, this is simply the rotation around the same axis by the same angle but in the opposite direction, that is, by an angle $-\alpha$. The combination of clockwise and counterclockwise rotations by the same angle will leave all points unchanged. The resulting nil operation is called the unit operation, $\hat{E}$.

$$\hat{R}\hat{R}^{-1} = \hat{R}^{-1}\hat{R} = \hat{E} \tag{1.6}$$

1.2 Operations and Functions

Chemistry, of course, goes beyond the structural characteristics of molecules, and considers functional properties, associated with the structures. This is certainly the case for the quantum-mechanical description of the molecular world. The primary functions which come to mind are the orbitals which describe the distribution of the electrons in atoms and molecules. A function $f(x, y, z)$ associates a certain property (usually a scalar number) with a particular coordinate position. A displacement of a point will thus induce a change of the function. This can again be defined in several

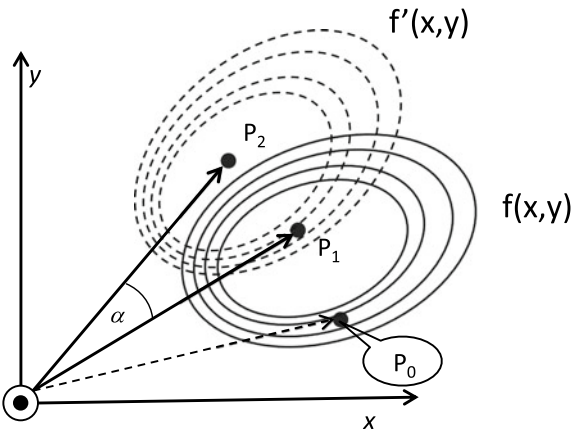


Fig. 1.3 The rotation of the function $f(x, y)$, counterclockwise by an angle α, generates a new function, $f'(x, y)$. The value of the new function at P_2 is equal to the value of the old function at P_1. Similarly, to find the value of the new function at P_1, we have to retrieve the value of the old function at a point P_0, which is the point that will be reached by the clockwise rotation of P_1

ways. Let us agree on the following: when we displace a point, the property associated with that point will likewise be displaced with it. In this way, we create a new property distribution in space, and hence a new function. This new function will be denoted by $\hat{R}f$ (or sometimes as f'), i.e., it is viewed as the result of the action of the operation on the original function. In line with our agreement, a property associated with the displaced point will have the same value as that property had when associated with the original point, hence:

$$\hat{R}f(P_2) = f(P_1) \tag{1.7}$$

or, in general:

$$\hat{R}f(\hat{R}P_1) = f(P_1) \tag{1.8}$$

Note that in this expression the same symbol $\hat{R}$ is used in two different meanings, either as transforming a coordinate or a function, as is evident from the entity that follows the operator. This rule is sufficient to *plot* the transformed function, as shown in Fig. 1.3. In order to determine the mathematical form of the new transformed function, we must be able to compare the value of the new function with the original function *at the same point*, i.e., we must be able to see how the property changes at a given point. Thus we would like to know what would be the value of $\hat{R}f$ in the original point P_1. Equation (1.8) cannot be used to determine this since the transformed function is as yet unknown and we thus do not know the rules for working out the brackets in the left-hand side of the equation. However this relationship must be true for every point; thus we may substitute $\hat{R}^{-1}P_1$ for P_1 on both the left- and the right-hand sides of Eq. (1.8). The equation thus becomes

$$\hat{R}f(\hat{R}(\hat{R}^{-1}P_1)) = \hat{R}f(\hat{E}P_1) = \hat{R}f(P_1) = f(\hat{R}^{-1}P_1) \tag{1.9}$$

This result reads as follows: the transformed function attributes to the original point P_1 the property that the original function attributed to the point $\hat{R}^{-1}P_1$. In Fig. 1.3,

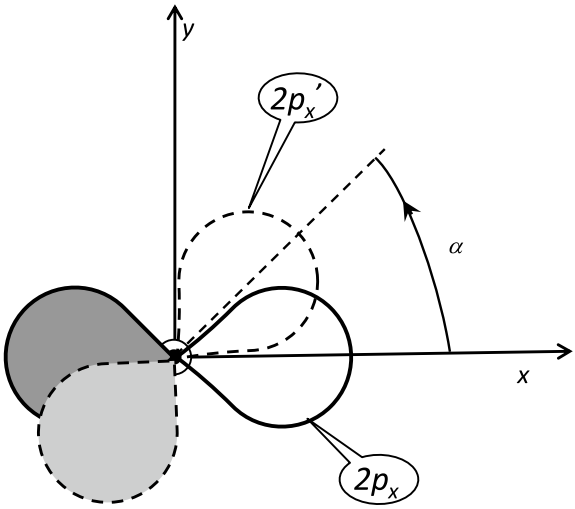


Fig. 1.4 The dashed orbital is obtained by rotating the $2p_x$ orbital counterclockwise through an angle α

this point from which the function value was retrieved is indicated as P_0. Thus the function and the coordinates transform in opposite ways.[2] This connection transfers the operation from the function to the coordinates, and, since the original function is a known function, we can also use the toolbox of corresponding rules to work out the bracket on the right-hand side of Eq. (1.9).

As an example consider the familiar $2p$ orbitals in the xy-plane: $2p_x$, $2p_y$. These orbitals are usually represented by the iconic dumbbell structure.[3] We can easily find out what happens to these upon rotation, simply by inspection of Fig. 1.4, in which we performed the rotation of the $2p_x$ orbital by an angle α around the z-axis. Clearly, when the orbital rotates the overlap with the $2p_x$ function decreases and the $2p_y$ orbital gradually appears. Now let us apply the formula to determine $\hat{R}\,2p_x$. The functional form of the $2p_x$ orbital for a hydrogen atom, in polar coordinates, reads $R_{2p}(r)\Theta_{2p|m_l|}(\theta)\Phi_x(\phi)$, where $R_{2p}(r)$ is the radial part, $\Theta_{2p|m_l|}(\theta)$ with $m_l = \pm 1$ is the part which depends on the azimuthal angle, and $\Phi_x(\phi)$ indicates how the function depends on the angle ϕ in the xy-plane, measured from the positive x-direction. One has

$$\begin{aligned} \Phi_x(\phi) &= \frac{1}{\sqrt{\pi}} \cos\phi \\ \Phi_y(\phi) &= \frac{1}{\sqrt{\pi}} \sin\phi \end{aligned} \tag{1.10}$$

[2] A more general expression for the transportation of a quantum state may also involve an additional phase factor, which depends on the path. See, e.g., [1].

[3] The electron distribution corresponding to the square of these orbitals is described by a lemniscate of Bernoulli. The angular parts of the orbitals themselves are describable by osculating spheres.

Both r and θ are invariant under a rotation around the z-direction, $\theta_1 = \theta_0$, and $r_1 = r_0$; hence, only the ϕ part will matter when we rotate in the plane. The transformed functions are easily determined starting from the general equation and using the matrix expression for the coordinate rotation, where we replace α by $-\alpha$, since we need the inverse operation here:

$$\begin{aligned} \hat{R}\,(\cos\phi_1) &= \cos\phi_0 = \cos(\phi_1 - \alpha) \\ &= \cos\phi_1\cos\alpha + \sin\phi_1\sin\alpha \\ \hat{R}\,(\sin\phi_1) &= \sin\phi_0 = \sin(\phi_1 - \alpha) \\ &= \sin\phi_1\cos\alpha - \cos\phi_1\sin\alpha \end{aligned} \tag{1.11}$$

Multiplying with the radial and azimuthal part, we obtain the desired functional transformation of the in-plane $2p$-orbitals.

$$\begin{aligned} \hat{R}\;2p_x &= 2p_x\cos\alpha + 2p_y\sin\alpha \\ \hat{R}\;2p_y &= -2p_x\sin\alpha + 2p_y\cos\alpha \end{aligned} \tag{1.12}$$

Again we should get accustomed to read these expressions almost visually. For instance, when the angle is 90°, one has $2p'_x = 2p_y$, and $2p'_y = -2p_x$. This simply means: take the $2p_x$ orbital, rotate it over 90° counterclockwise around the z-direction, and it will become $2p_y$. If the same is done with $2p_y$, it will go over into $-2p_x$ since the plus and minus lobes of the dumbbell become congruent with the oppositely signed lobes of the $2p_x$-orbital.

Equation (1.12) further reveals an important point. To express the transformation of a function, one almost automatically encounters the concept of a *function space*. To describe the transformation of the cosine function, one really also needs the sine. The two form a two-dimensional space, which we shall call a vector space. This will be explained in greater depth in Chap. 2. For now we may cast the transformation of the basis components of this space in matrix form. This time we arrange the basis orbitals in a row-vector notation, so that the transformation matrix is written to the right of the basis. Thus:

$$\hat{R}\left(2p_x\;\;2p_y\right) = \left(2p_x\;\;2p_y\right)\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \tag{1.13}$$

The matrix which is used here is precisely the same matrix which we used for the coordinate transformation. How is this possible, since functions and points transform in opposite ways? The reason is of course that we also switched from a column vector for points to a row vector for functions. Indeed, transposition, T, of the entire matrix multiplication simultaneously inverts the transformation matrix and interchanges columns and rows.

$$\left[\mathbb{D}^{-1}\begin{pmatrix} x \\ y \end{pmatrix}\right]^{\mathrm{T}} = \left(x\;\;y\right)\left(\mathbb{D}^{-1}\right)^{\mathrm{T}} = \left(x\;\;y\right)\mathbb{D} \tag{1.14}$$

where we made use of the property that transposition of the rotation matrix changes α into $-\alpha$ and thus is the same as taking the inverse of $\mathbb{D}$. The final point about functions is somewhat tricky, so attention is required. Just like the value of a field, or the amplitude of an orbital, the values of the coordinates themselves are properties associated with points. As an example, the function that yields the x-coordinate of a point P_1 will be denoted as $x(P_1)$. The value of this function is x_1, where we are using different styles to distinguish the function x, which is a variable, and the coordinate x_1, which is a *number*. We can thus write:

$$x(P_1) = \mathrm{x}_1 \tag{1.15}$$

A typical quantum-chemical example of the use of these coordinate functions is the dipole operator, e.g., the x-component of the electric dipole is simply given by $\mu_x = -\mathrm{e}x$, where $-\mathrm{e}$ is the electron charge. We may thus write in analogy with Eq. (1.13):

$$\hat{R}\begin{pmatrix} x & y \end{pmatrix} = \begin{pmatrix} x & y \end{pmatrix}\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \tag{1.16}$$

In summary, we have learned that when a symmetry operator acts on all the points of a space, it induces a change of the functions defined in that space. The transformed functions are the result of a direct action of the symmetry operator in a corresponding function space. Furthermore, there exists a dual relation between the transformations of coordinate points and of functions. They are mutual inverses. Finally, the active picture also applies to the functions: the symmetry operation sets the function itself into motion, as if we were (physically) grasping the orbitals and twisting them.

1.3 Operations and Operators

Besides functions, we must also consider the action of operations on operators. In quantum chemistry, operators, such as the Hamiltonian, $\mathcal{H}$, are usually spatial functions and, as such, are transformed in the same way as ordinary functions, e.g., $\mathcal{H}'(P_1) = \mathcal{H}(\hat{R}^{-1}P_1)$. So why devote a special section to this? Well, operators are different from functions in the sense that they also operate on a subsequent argument, which is itself usually a function. Hence, when symmetry is applied to an operator, it will also affect whatever follows the operator. Symmetry operations act on the entire expression at once. This can be stated for a general operator $\mathcal{O}$ as follows:

$$\hat{R}\mathcal{O}f = \mathcal{O}'\hat{R}f \tag{1.17}$$

From this, we can identify the transformed operator $\mathcal{O}'$, by smuggling $\hat{R}^{-1}\hat{R}$ $(= \hat{E})$ into the left-hand side of the equation:

$$\hat{R}\mathcal{O}\hat{R}^{-1}\hat{R}f = \mathcal{O}'\hat{R}f \tag{1.18}$$

This equality is true for any function f, and thus implies[4] that the operators preceding $\hat{R}f$ on both sides must be the same:

$$\mathcal{O}' = \hat{R}\mathcal{O}\hat{R}^{-1} \tag{1.19}$$

This equation provides the algebraic formalism for the transformation of an operator. In the case when $\mathcal{O}' = \mathcal{O}$, we say that the operator is *invariant* under the symmetry operation. Equation (1.19) then reduces to

$$\hat{R}\mathcal{O} - \mathcal{O}\hat{R} = \left[\hat{R}, \mathcal{O}\right] = 0 \tag{1.20}$$

This bracket is known as the *commutator* of $\hat{R}$ and $\mathcal{O}$. If the commutator vanishes, we say that $\hat{R}$ and $\mathcal{O}$ commute. This is typically the case for the Hamiltonian. As an application, we shall study the functional transformations of the differential operators $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$ under a rotation around the positive z-axis. To find the transformed operators, we have to work out expressions such as $\frac{\partial}{\partial x'}$ where $x' = x(\hat{R}^{-1}P_1)$. Hence, we are confronted with a functional form, *viz.*, the derivative operator, of a transformed argument, x', but this is precisely where classical analysis comes to our rescue: it provides us with the chain rule, needed to work out the coordinate change. We have

$$\frac{\partial}{\partial x'} = \frac{\partial x}{\partial x'}\frac{\partial}{\partial x} + \frac{\partial y}{\partial x'}\frac{\partial}{\partial y} \tag{1.21}$$

In order to evaluate this equation, we have to determine the partial derivatives in the transformed coordinates. Using the result in Eq. (1.16) one obtains

$$\begin{aligned} x' &= x\cos\alpha + y\sin\alpha \\ y' &= -x\sin\alpha + y\cos\alpha \end{aligned} \tag{1.22}$$

Invert these equations to express x and y as a function of x' and y':

$$\begin{aligned} x &= x'\cos\alpha - y'\sin\alpha \\ y &= x'\sin\alpha + y'\cos\alpha \end{aligned} \tag{1.23}$$

The partial derivatives, needed in the chain rule, can now be obtained by direct derivation:

[4] The fact that two operators transform a given function in the same way does not automatically imply that those operators are the same. Operators will be the same if this relationship extends over the entire Hilbert space.

$$\begin{aligned}\frac{\partial x}{\partial x'} &= \cos\alpha \\ \frac{\partial y}{\partial x'} &= \sin\alpha \\ \frac{\partial x}{\partial y'} &= -\sin\alpha \\ \frac{\partial y}{\partial y'} &= \cos\alpha\end{aligned} \tag{1.24}$$

Hence, the transformation of the derivatives is entirely similar to the transformation of the x, y functions themselves:

$$\hat{R}\begin{pmatrix}\frac{\partial}{\partial x} & \frac{\partial}{\partial y}\end{pmatrix} = \begin{pmatrix}\frac{\partial}{\partial x} & \frac{\partial}{\partial y}\end{pmatrix}\begin{pmatrix}\cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha\end{pmatrix} \tag{1.25}$$

In an operator formalism, we should denote this as

$$\begin{aligned}\hat{R}\frac{\partial}{\partial x}\hat{R}^{-1} &= \cos\alpha\frac{\partial}{\partial x} + \sin\alpha\frac{\partial}{\partial y} \\ \hat{R}\frac{\partial}{\partial y}\hat{R}^{-1} &= -\sin\alpha\frac{\partial}{\partial x} + \cos\alpha\frac{\partial}{\partial y}\end{aligned} \tag{1.26}$$

As a further example, consider a symmetry transformation of a symmetry operator itself. Take, for instance, a rotation, $\hat{C}_2^x$, corresponding to a rotation about the x-axis of 180°, and rotate this 90° counterclockwise about the z-direction by $\hat{C}_4^z$. Applying the general expression for an operator transformation yields

$$\left(\hat{C}_2^x\right)' = \hat{C}_4^z\hat{C}_2^x\left(\hat{C}_4^z\right)^{-1} \tag{1.27}$$

The sequence of operators in this transformation is depicted in Fig. 1.5. The result is an equivalent twofold rotation around the y-direction, $\hat{C}_2^y$. The rotation around x is thus mapped onto a rotation around y by a fourfold rotation axis along the z-direction. In Chap. 3, we shall see that this relation installs an equivalence between both twofold rotations, whenever such a $\hat{C}_4^z$ is present.

1.4 An Aide Mémoire

- Use a right-handed coordinate system.
- Always leave the Cartesian directions unchanged.
- Symmetry operations are defined in an active sense.

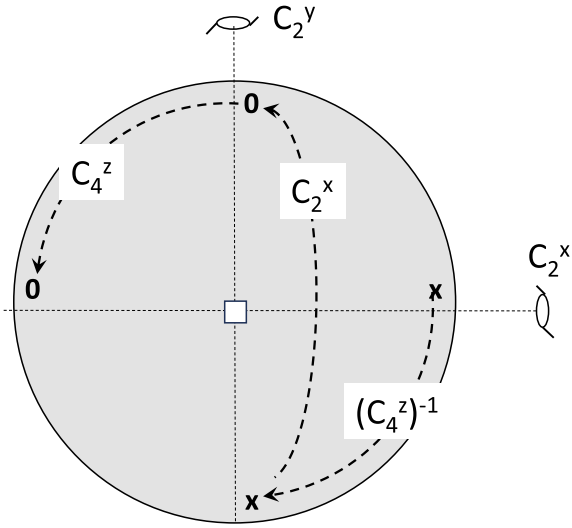


Fig. 1.5 The arrows represent the sequence of operations in Eq. (1.27), transforming a twofold axis around the x-direction to the y direction

- Rotations through positive angles appear counterclockwise, when viewed from the rotational pole; 'counterclockwise is positive'.
- The transformation of the coordinates of a point is written in a column vector notation.
- The transformation of a function space is written in a row-vector notation.
- There is a dual relationship between the transformations of functions and of coordinates: $\hat{R} f(\mathbf{r}) = f(\hat{R}^{-1}\mathbf{r})$.
- The transformation of an operator $\mathcal{O}$ is given by $\hat{R}\mathcal{O}\hat{R}^{-1}$.

1.5 Problems

1.1 Use the stereographic representation of Fig. 1.1 to show: $[\hat{\imath}, \hat{C}_2^z] = 0$.

1.2 The square of the radial distance of the point P_1 in the xy-plane may be obtained by multiplying the coordinate column by its transposed row.

$$\mathrm{r}^2 = \mathrm{x}_1^2 + \mathrm{y}_1^2 = \begin{pmatrix} \mathrm{x}_1 & \mathrm{y}_1 \end{pmatrix} \begin{pmatrix} \mathrm{x}_1 \\ y_1 \end{pmatrix} \tag{1.28}$$

Show that this scalar product is invariant under a rotation about the z-axis.

1.3 Prove: $\hat{R}\hat{S} f(P_1) = f(\hat{S}^{-1}\hat{R}^{-1}P_1)$.

1.4 The translation operator $\mathcal{T}_a$ displaces a point with x-coordinate x_1 to a new position $\mathrm{x}_1 + \mathrm{a}$. Apply this operator to the wavefunction e^{ikx}.

1.5 Construct a differential operator such that its action on the coordinate functions x and y matches the matrix transformation in Eq. (1.16). What is the angular derivative of this operator when the rotation angle tends to zero? Can you relate this limit to the angular momentum operator $\mathcal{L}_z$?

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Chapter 2
Function Spaces and Matrices



Abstract This chapter refreshes such necessary algebraic knowledge as will be needed in this book. It introduces function spaces and their adjoints, the meaning of a linear operator, and the properties of unitary matrices. The homomorphism between operations and matrix multiplications is established, and the Dirac notation for function spaces is defined. For those who might think that the linearity of operators is obvious, the final section introduces time reversal, which is anti-linear.

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2.1 Function Spaces

In the first chapter, we saw that if we wanted to rotate the $2p_x$ function we automatically also needed its companion $2p_y$ function. If this is extended to out-of-plane rotations, the $2p_z$ function will also be needed. The set of the three p-orbitals forms a prime example of what is called a *linear vector space*. In general, this is a space which consists of components that can be combined linearly using real or complex numbers as coefficients. An n-dimensional linear vector space consists of a set of n vectors that are linearly independent. The components or basis vectors will be denoted as f_l, with l ranging from 1 to n. At this point, we shall introduce the Dirac notation [1] and rewrite these functions as $|f_l\rangle$, which characterizes them as so-called *ket* functions. Whenever we have such a set of vectors we can set up a complementary

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set of so-called *bra* functions, denoted as $\langle f_k|$. The scalar product of a bra and a ket yields a number. It is denoted as the bracket: $\langle f_k|f_l\rangle$. In other words, when a bra collides with a ket on its right, it yields a scalar number. A bra vector is completely defined when its scalar product with every ket vector of the vector space is given. For linearly independent functions, we have

$$\forall k \neq l : \ \langle f_k|f_l\rangle = 0 \tag{2.1}$$

The basis is orthonormal if all vectors are in addition normalized to $+1$.

$$\forall k : \langle f_k|f_k\rangle = 1 \tag{2.2}$$

This result can be summarized with the help of the Kronecker delta, δ_{ij}, which is zero unless the subscript indices are identical, in which case it is unity. Hence for an orthonormal basis set:

$$\langle f_k|f_l\rangle = \delta_{kl} \tag{2.3}$$

Since the bracket is a scalar product, the bra part, associated with a linear combination of ket functions, will be the anti-linear combination of the corresponding bra functions. This means that the expansion coefficients are complex conjugate:

$$\langle c_1 f_1 + c_2 f_2| = \bar{c}_1\langle f_1| + \bar{c}_2\langle f_2| \tag{2.4}$$

Accordingly, in the wave formalism of quantum mechanics, the bra function of f_k is simply the complex-conjugate function, $\bar{f_k}$, and the bracket or scalar product is defined as the integral of the product of the functions over space:

$$\langle f_k|f_l\rangle = \iiint \bar{f_k} f_l \mathrm{d}V \tag{2.5}$$

One thus also has

$$\langle \overline{f_k|f_l}\rangle = \langle f_l|f_k\rangle \tag{2.6}$$

2.2 Linear Operators and Transformation Matrices

A linear operator is an operator which commutes with multiplicative scalars and is distributive with respect to summation: this means that when it acts on a sum of functions, it will operate on each term of the sum.

$$\begin{aligned} \hat{R}c|f_k\rangle &= c\hat{R}|f_k\rangle \\ \hat{R}\left(|f_k\rangle + |f_l\rangle\right) &= \hat{R}|f_k\rangle + \hat{R}|f_l\rangle \end{aligned} \tag{2.7}$$

If the transformations of functions under an operator can be expressed as a mapping of these functions onto a linear combination of the basis vectors in the function space, then the operator is said to leave the function space *invariant*. The corresponding coefficients can then be collected in a transformation matrix. For this purpose, we arrange the components in a row vector, $(|f_1\rangle, |f_2\rangle, \cdots |f_n\rangle)$, as agreed upon in Chap. 1. This row precedes the transformation matrix. The usual symbols are $\hat{R}$ for the operator and $\mathbb{D}(R)$ for the corresponding matrix.

$$\hat{R}\ (|f_1\rangle|f_2\rangle \cdots |f_n\rangle) = (|f_1\rangle|f_2\rangle \cdots |f_n\rangle) \begin{pmatrix} \\ \mathbb{D}(R) \\ \\ \end{pmatrix}$$

i.e.:

$$\hat{R}|f_i\rangle = \sum_{j=1}^{n} D_{ji}(R)|f_j\rangle \tag{2.8}$$

When multiplying this equation left and right with a given bra function in an orthonormal basis, one obtains

$$\langle f_k|\hat{R}|f_i\rangle = \sum_{j=1}^{n} D_{ji}(R)\langle f_k|f_j\rangle = \sum_{j=1}^{n} D_{ji}(R)\delta_{kj} = D_{ki}(R) \tag{2.9}$$

where the summation index j has been restricted to k by the Kronecker delta. *Hence, the elements of the transformation matrix are recognized as matrix elements of the symmetry operators.* For convenience, we sometimes abbreviate the row vector of the function space as $|\mathbf{f}\rangle$, so that the transformation is written as

$$\hat{R}|\mathbf{f}\rangle = |\mathbf{f}\rangle\mathbb{D}(R) \tag{2.10}$$

A product of two operators is executed consecutively, thus the one closest to the ket acts first. In detail:

$$\begin{aligned} \hat{R}\hat{S}\,|f_i\rangle &= \hat{R}\sum_j D_{ji}(S)|f_j\rangle \\ &= \sum_{k,j} D_{kj}(R)D_{ji}(S)|f_k\rangle \\ &= \sum_k [\mathbb{D}(R) \times \mathbb{D}(S)]_{ki}\,|f_k\rangle \end{aligned} \tag{2.11}$$

Here, the symbol $\times$ refers to the product of two matrices. This is an important result. It shows that the consecutive action of two operators can be expressed by the product of the corresponding matrices. The matrices are said to *represent* the action of the corresponding operators. The relationship between both is a mapping. In this

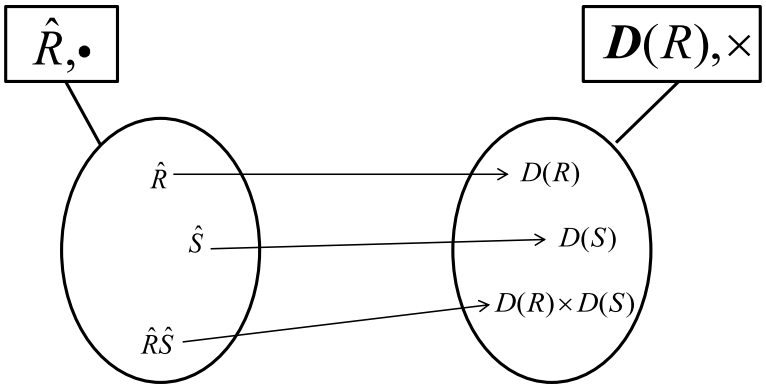


Fig. 2.1 Matrix representation of a group: the operators (left) are mapped onto the transformations (right) of a function space. The consecutive action of two operators on the left (symbolized by $\bullet$) is replaced by the multiplication of two matrices on the right (symbolized by $\times$)

mapping, the operators are replaced by their respective matrices and the product of the operators is mapped onto the product of the corresponding matrices. In this mapping, the order of the elements is kept.

$$\mathbb{D}(RS) = \mathbb{D}(R) \times \mathbb{D}(S) \tag{2.12}$$

In mathematical terms, such a mapping is called a *homomorphism* (see Fig. 2.1). The conservation of the order is an important characteristic. In Eq. (2.12), both the operators and the matrices that represent them are *right-justified*, that is, the operator (matrix) on the *right* is applied first, and then the operator (matrix) immediately to the left of it. In the active picture with the functions in a row matrix, the action of the first operator brings the original function row again in front of the second operator, so that this one can act straightaway on the function row. Equation (2.11) can be expressed as follows:

$$\hat{R}\hat{S}|\mathbf{f}\rangle = \hat{R}|\mathbf{f}\rangle\mathbb{D}(S) = |\mathbf{f}\rangle\mathbb{D}(R) \times \mathbb{D}(S) \tag{2.13}$$

Further consequences of the homomorphism are that the unit element is represented by the unit matrix, $\mathbb{I}$, and that an inverse element is represented by the corresponding inverse matrix:

$$\begin{aligned} \mathbb{D}(E) &= \mathbb{I} \\ \mathbb{D}(R^{-1}) &= [\mathbb{D}(R)]^{-1} \end{aligned} \tag{2.14}$$

2.3 The Adjoint Relation Between Bra's and Kets

The transformation of bra functions runs entirely parallel with the transformation of ket functions, except that the complex conjugate of the transformation matrix has to be taken. The transformed $\langle \hat{R} f_i |$ bra is obtained starting from the ket $|\hat{R} f_i\rangle$ in Eq. (2.8) and applying the anti-linear relationship of Eq. (2.4):

$$\hat{R}\langle f_i| = \langle \hat{R} f_i| = \sum_{j=1}^{n} \overline{D}_{ji}(R)\langle f_j| \tag{2.15}$$

The bra functions also appear to be ordered in a row vector, which we denote as $\langle \mathbf{f}|$.

$$\hat{R}\langle \mathbf{f}| = \langle \mathbf{f}|\overline{\mathbb{D}}(R) \tag{2.16}$$

The relation between transformed bra and ket functions is called the adjoint relationship. Following Eq. (2.6), one has

$$\begin{aligned} \langle f_k|\hat{R} f_l\rangle &= D_{kl}(R) \\ \langle \hat{R} f_k| f_l\rangle &= \overline{\langle f_l|R f_k\rangle} = \overline{D}_{lk}(R) \end{aligned} \tag{2.17}$$

The matrix element $\langle \hat{R} f_k| f_l\rangle$ is thus obtained by taking the complex conjugate and switching row and column indices. The latter operation is called matrix transposition, indicated by a capital superscript T. The resulting matrix is called the *adjoint matrix*, indicated by a dagger:

$$\mathbb{D}^{\dagger} = \overline{\mathbb{D}}^{T} \tag{2.18}$$

Matrices in quantum mechanics are characterized by a special relationship to their adjoint. They are either Hermitian, which means that they are equal to their adjoint, or unitary, which means that their inverse is equal to their adjoint. This has everything to do with the fact that the physics imposes special requirements on eigenvalues (see inset).

Matrices in Quantum Mechanics
Hermitian matrices: $\mathbb{A}^{\dagger} = \mathbb{A} \rightarrow$ eigenvalues are real
Unitary matrices: $\mathbb{A}^{\dagger} = \mathbb{A}^{-1} \rightarrow$ eigenvalues are unimodular

The matrices representing symmetry operations are all unitary. This is derived as follows: Consider the matrix element $\langle f_k| f_l\rangle = \delta_{kl}$. This is a scalar, and as such it is invariant under symmetry operations. We thus have:

$$\begin{aligned}\hat{R}\langle f_k|f_l\rangle &= \langle \hat{R}f_k|\hat{R}f_l\rangle \\ &= \sum_{ij} \overline{D}_{ik}(R)D_{jl}(R)\langle f_i|f_j\rangle \\ &= \sum_{i} \overline{D}_{ik}(R)D_{il}(R) \\ &= [\mathbb{D}^\dagger(R)\times\mathbb{D}(R)]_{kl} = \delta_{kl} \end{aligned} \tag{2.19}$$

The matrix product of $\mathbb{D}^\dagger$ and $\mathbb{D}$ is thus equal to the unit matrix:

$$\mathbb{D}^\dagger(R)\times\mathbb{D}(R) = \mathbb{I} \tag{2.20}$$

The order of this product can be swapped by making use of the inverse matrix[1]:

$$\begin{aligned}\mathbb{D}^\dagger\times\mathbb{D} &= \mathbb{I} \\ \mathbb{D}\times\mathbb{D}^\dagger\times\mathbb{D} &= \mathbb{D} \\ \mathbb{D}\times\mathbb{D}^\dagger\times\mathbb{D}\times\mathbb{D}^{-1} &= \mathbb{D}\times\mathbb{D}^{-1} \\ \mathbb{D}\times\mathbb{D}^\dagger &= \mathbb{I} \end{aligned} \tag{2.21}$$

Note that this derivation is based on the orthonormality of the vector basis, and the fact that symmetry operators have an inverse. As a result, it is shown that left as well as right multiplication of a symmetry matrix with its adjoint yields the unit matrix. Only an inverse matrix has this property, hence the adjoint matrix and the inverse matrix coincide, and symmetry matrices will be unitary. Also the symmetry operators themselves are said to be unitary. This may be expressed as

$$\langle \hat{R}f_k| = \langle f_k|\hat{R}^{-1} \tag{2.22}$$

2.4 Unitary Matrices

In this section, we further explore the properties of unitary matrices. Let $\mathbb{U}$ be any unitary matrix, not necessarily a representation matrix for a point group element. According to the definition in the previous section, it is a matrix that obeys $\mathbb{U}^\dagger = \mathbb{U}^{-1}$. This definition implies that rows are orthonormal to each other, and likewise for columns. Here the scalar product of row k with row l is obtained by summing pairwise products of the corresponding elements, $U_{ki}\overline{U}_{li}$, one of which is taken to be complex conjugate, and similarly for the scalar product of two columns.

[1] Adapted from [2], problem 8, p. 59.

$$\mathbb{U} \times \mathbb{U}^\dagger = \mathbb{I} \rightarrow \sum_i U_{ki} U_{il}^\dagger = \sum_j U_{ki} \overline{U}_{li} = \delta_{kl}$$
$$\mathbb{U}^\dagger \times \mathbb{U} = \mathbb{I} \rightarrow \sum_i U_{ki}^\dagger U_{il} = \sum_j \overline{U}_{ik} U_{il} = \delta_{kl} \tag{2.23}$$

Now consider a function space $|\mathbf{f}\rangle$, and a linear transformation matrix, $\mathbb{U}$, which recombines the basis functions to yield a transformed basis set, say $|\mathbf{f}'\rangle$. Such a linear transformation of an orthonormal vector space preserves orthonormality *if and only if the transformation matrix* $\mathbb{U}$ *is unitary*. As we have shown in Eq. (2.19) of the previous section unitarity of a matrix indeed follows directly from the orthonormality of the vector space which it transforms. The reverse also applies. If the transformation operator $\hat{U}$ is unitary, orthonormality is preserved:

$$\langle f_k' | f_l' \rangle = \langle \hat{U} f_k | \hat{U} f_l \rangle = \langle f_k | \hat{U}^{-1} \hat{U} f_l \rangle = \langle f_k | f_l \rangle \tag{2.24}$$

A unitary matrix has several interesting properties, which can easily be checked from the general definition:

- The inverse of a unitary matrix is obtained by combining complex conjugation and transposition.
- The inverse and the transpose of a unitary matrix are unitary.
- The product of unitary matrices is a unitary matrix.
- The determinant of a unitary matrix has an absolute value of unity.

To prove the final property, we note that the determinant of a product of matrices is equal to the product of the determinants of the individual matrices, and we also note that the determinant does not change upon transposition of a matrix. By definition, $\mathbb{I} = \mathbb{U} \times \mathbb{U}^{-1}$, and it then follows:

$$\begin{aligned} \det(\mathbb{U} \times \mathbb{U}^{-1}) &= \det(\mathbb{U}) \det(\mathbb{U}^{-1}) \\ &= \det(\mathbb{U}) \det(\overline{\mathbb{U}}^{\mathrm{T}}) \\ &= \det(\mathbb{U}) \det(\overline{\mathbb{U}}) \\ &= \det(\mathbb{U}) \overline{\det(\mathbb{U})} \\ &= |\det(\mathbb{U})|^2 = \det(\mathbb{I}) = 1 \end{aligned} \tag{2.25}$$

Last but not least, the eigenvalues of unitary matrices are unimodular. To prove this property, consider an eigenvector $\mathbf{a}$ with associated eigenvalue λ. The eigenvector points to a direction in the vector space $|\mathbf{f}\rangle$:

$$\mathbf{a} = \sum_i a_i |f_i\rangle \tag{2.26}$$

The a_i coefficients are the associated coordinates, and as such they are arranged in a column vector. The eigenvalue equation then reads

$$\begin{pmatrix} \mathbb{U} \end{pmatrix} \begin{pmatrix} \mathbf{a} \end{pmatrix} = \lambda \begin{pmatrix} \mathbf{a} \end{pmatrix} \tag{2.27}$$

Now take the complex-conjugate transpose left and right:

$$\begin{aligned} (\mathbb{U}\mathbf{a})^\dagger &= \bar{\mathbf{a}}^T \mathbb{U}^{-1} \\ (\lambda\mathbf{a})^\dagger &= \overline{\lambda}\, \bar{\mathbf{a}}^T \end{aligned} \tag{2.28}$$

Note that the adjoint of the **a** column vector is a row vector:

$$\bar{\mathbf{a}}^T = (\bar{a}_1 \quad \ldots \quad \bar{a}_n) \tag{2.29}$$

Combining Eqs. (2.27) and (2.28) one obtains

$$\bar{\mathbf{a}}^T \mathbb{U}^{-1} \mathbb{U}\mathbf{a} = \lambda\overline{\lambda}\, \bar{\mathbf{a}}^T \mathbf{a} \tag{2.30}$$

from which follows: $|\lambda| = 1$.

Spatial symmetry operations are linear transformations of a coordinate function space. When choosing the space in orthonormal form, symmetry operations will conserve orthonormality and hence all transformations will be carried out by unitary matrices. This will be the case for all spatial representation matrices in this book. When all elements of a unitary matrix are real it is called an *orthogonal* matrix. Being unitary matrices, real orthogonal matrices have the same properties except that complex conjugation leaves them unchanged. The determinant of an orthogonal matrix will thus be equal to ± 1. The rotation matrices in Chap. 1 are all orthogonal and have determinant $+1$. Orthogonal matrices conserve the dot product of vectors in real space. We will encounter them in the subsequent chapter when discussing the isometry transformations of Euclidean space.

2.5 Time Reversal as an Anti-linear Operator

The fact that an operator cannot change a scalar constant in front of the function on which it operates seems to be evident. However, in quantum mechanics, there is one important operator that does affect a scalar constant, and turns it into its complex conjugate. This is the operator of *time reversal*, i.e., the operator which inverts time, $t \to -t$, and sends the system back to its own past. If we are looking at a stationary state, with no explicit time dependence, time inversion really means reversal of the direction of motion, where all angular momenta will be changing sign, including the 'spinning' of the electrons. We shall denote this operator as $\hat{\vartheta}$. It has the following properties:

$$\begin{aligned}\hat{\vartheta}\,(|f_k\rangle + |f_l\rangle) &= \hat{\vartheta}|f_k\rangle + \hat{\vartheta}|f_l\rangle \\ \hat{\vartheta}\; c|f_k\rangle &= \overline{c}\;\hat{\vartheta}|f_k\rangle\end{aligned} \tag{2.31}$$

These properties are characteristic of an *anti-linear* operator. As a rationale for the complex conjugation upon commutation with a multiplicative constant, we consider a simple case study of a stationary quantum state. The time-dependent Schrödinger equation, describing the time evolution of a wavefunction, Ψ, defined by a Hamiltonian $\mathcal{H}$, is given by

$$-\frac{\hbar}{i}\frac{\partial \Psi}{\partial t} = \mathcal{H}\Psi \tag{2.32}$$

For a stationary state, the Hamiltonian is independent of time, and the wavefunction is characterized by an eigenenergy, E; hence the right-hand side of the equation is given by $\mathcal{H}\Psi = E\Psi$. The solution for the stationary state then becomes

$$\Psi(t) = \Psi(t_0)\exp\left(-\frac{iE(t-t_0)}{\hbar}\right) \tag{2.33}$$

Hence, the phase of a stationary state is 'pulsating' at a frequency given by $E/\hbar$. Now we demonstrate the anti-linear character, using Wigner's argument that a perfect looping in time would bring a system back to its original state [3].

Such a process can be achieved by running backwards in time over a certain interval, and then returning to the original starting time. Let $T_{\Delta t}$ represent a displacement in time towards the future over an interval Δt, and $T_{-\Delta t}$ a displacement over the same interval, but towards the past. The consecutive action of $T_{\Delta t}$ and $T_{-\Delta t}$ certainly describes a perfect loop in time, thus we can write

$$\hat{T}_{\Delta t}\hat{T}_{-\Delta t} = \hat{E} \tag{2.34}$$

The reversal of the translation in time is the result of a reversal of the time variable. We thus can apply the operator transformation under $\hat{\vartheta}$, in line with the previous results in Sect. 1.3.

$$\hat{T}_{-\Delta t} = \hat{\vartheta}\hat{T}_{\Delta t}\hat{\vartheta}^{-1} \tag{2.35}$$

The complete loop can thus be written as follows:

$$T_{\Delta t}\hat{\vartheta}T_{\Delta t}\hat{\vartheta}^{-1} = \hat{E} \tag{2.36}$$

This equation decomposes the closed path in time in four consecutive steps. Reading Eq. (2.36) from right to left one sets off at time t_0 and reverses time ($\hat{\vartheta}^{-1}$). Now time runs for a certain interval Δt, along the reversed time axis. A positive interval actually means that we are returning in time, since the time axis has been oriented towards the past. This operation is presented by the displacement $\hat{T}_{\Delta t}$. Then one applies the time reversal again, and now runs forward over the same interval to close

the loop. The forward translation corresponds to the same $\hat{T}_{\Delta t}$ operator, since again, the interval is positive. Now multiply both sides of the equation, on the right, by $\hat{\vartheta}\hat{T}_{-\Delta t}$.

$$\hat{T}_{\Delta t}\hat{\vartheta} = \hat{\vartheta}\hat{T}_{-\Delta t} \tag{2.37}$$

The actions of the translations on the wavefunction are given by

$$\begin{aligned} T_{\Delta t}\Psi(t_0) &= \Psi(t_0)\exp\left(-\frac{iE\ \Delta t}{\hbar}\right) \\ T_{-\Delta t}\Psi(t_0) &= \Psi(t_0)\exp\left(\frac{iE\ \Delta t}{\hbar}\right) \end{aligned} \tag{2.38}$$

Applying now both sides of Eq. (2.37) to the initial state yields

$$\begin{aligned} T_{\Delta t}\vartheta\Psi(t_0) &= \vartheta T_{-\Delta t}\Psi(t_0) \\ &= \vartheta\exp\left(\frac{iE\ \Delta t}{\hbar}\right)\Psi(t_0) \end{aligned} \tag{2.39}$$

Since the Hamiltonian that we have used is invariant under time reversal, the function $\vartheta\Psi(t_0)$ on the left-hand side of Eq. (2.39) will be characterized by the same energy, E, and thus translate in time with the same phase factor as $\Psi(t_0)$ itself. Then the equation becomes

$$\exp\left(-\frac{iE\ \Delta t}{\hbar}\right)\vartheta\Psi(t_0) = \vartheta\exp\left(\frac{iE\ \Delta t}{\hbar}\right)\Psi(t_0) \tag{2.40}$$

which shows that time reversal will invert scalar constants to their complex conjugate, and hence it will be an anti-linear operator.

Note that in the present derivation we avoided providing an explicit form for the inverse of the time reversal operator. As a matter of fact, while space inversion is its own inverse, applying time reversal twice may give rise to an additional phase factor, which is $+1$ for systems with an even number of electrons, but -1 for systems with an odd number of electrons. We shall demonstrate this point later in Sect. 7.6. Hence, $\vartheta^{-1} = \pm\vartheta$, or

$$\hat{\vartheta}^2 = \pm 1. \tag{2.41}$$

2.6 Problems

2.1 Prove Eq. (2.14).

2.2 A complex number can be characterized by an absolute value and a phase. A 2×2 complex matrix thus contains eight parameters, say

$$\mathbb{C} = \begin{pmatrix} |a|e^{i\alpha} & |b|e^{i\beta} \\ |c|e^{i\gamma} & |d|e^{i\delta} \end{pmatrix}$$

Impose now the requirements that this matrix be unitary. This will introduce relationships between the parameters. Try to solve these by adopting a reduced set of parameters.

2.3 The cyclic waves $e^{ik\phi}$ and $e^{-ik\phi}$ are defined in a circular interval $\phi \in [0, 2\pi[$, with $e^{2\pi ik} = 1$. Normalize these waves over the interval. Are they mutually orthogonal?

2.4 A matrix $\mathbb{H}$ which is equal to its complex-conjugate transpose, $\mathbb{H} = \overline{\mathbb{H}}^{\mathrm{T}}$, is called *Hermitian*. It follows that the diagonal elements of such a matrix are real, while corresponding off-diagonal elements form complex-conjugate pairs:

$$\mathbb{H} \text{ Hermitian } \rightarrow H_{ii} \in \mathcal{R} \; ; \; H_{ij} = \overline{H}_{ji}$$

Prove that the eigenvalues of a Hermitian matrix are real. If the matrix is *skew*-Hermitian, $\mathbb{H} = -\overline{\mathbb{H}}^{\mathrm{T}}$, the eigenvalues are all imaginary.

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Chapter 3
Groups

Abstract The concept of a group is introduced using the example of the symmetry group of the ammonia molecule. In spite of its tiny size this molecule has a structural symmetry which is the same as the symmetry of a macroscopic trigonal pyramid. From the mathematical point of view, a group is an elementary structure, which proves to be a powerful tool for describing molecular properties. Three ways of dividing (and conquering) groups are shown: subgroups, cosets, and classes. Spatial isometries are introduced and an overview of molecular symmetry groups is given. The relationship between rotational groups and chirality is explained, and symmetry lowerings due to applied magnetic and electric fields are determined.

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3.1 The Symmetry of Ammonia

The umbrella shape of the ammonia molecule has a trigonal symmetry axis, with, in addition, three vertical reflection planes through the hydrogen atoms. Together these symmetry elements form a point group, which, in the Schoenflies notation, is denoted as C_{3v}. It is good practice to start the treatment by making a simple sketch of the molecule, and putting it in a right-handed Cartesian frame, as shown in Fig. 3.1. By convention, the z-axis is defined as the principal threefold axis. One of the hydrogens is put in the xz-plane as shown in the figure. We attach labels A, B, C to distinguish the equivalent hydrogen atoms. In the active view, which we keep throughout, the atoms will be displaced while the symmetry elements remain tied to the immobile Cartesian frame. We shall thus not label the reflection planes by A, B, C but we shall instead denote them as $\hat{\sigma}_1, \hat{\sigma}_2, \hat{\sigma}_3$. The $\hat{\sigma}_1$ reflection plane coincides with the xz-coordinate plane. The action of the symmetry elements will be to permute the atoms. The threefold axis, rotating counterclockwise about z, moves the atom A to the position of B, which itself is displaced to the position originally occupied by C. Finally, C travels to the place previously occupied by atom A. The $\hat{\sigma}_1$ plane will leave A unchanged, and will interchange B and C. Now consider the combination $\hat{\sigma}_1\hat{C}_3$ of these two elements. We place the structure to the right of the right-justified

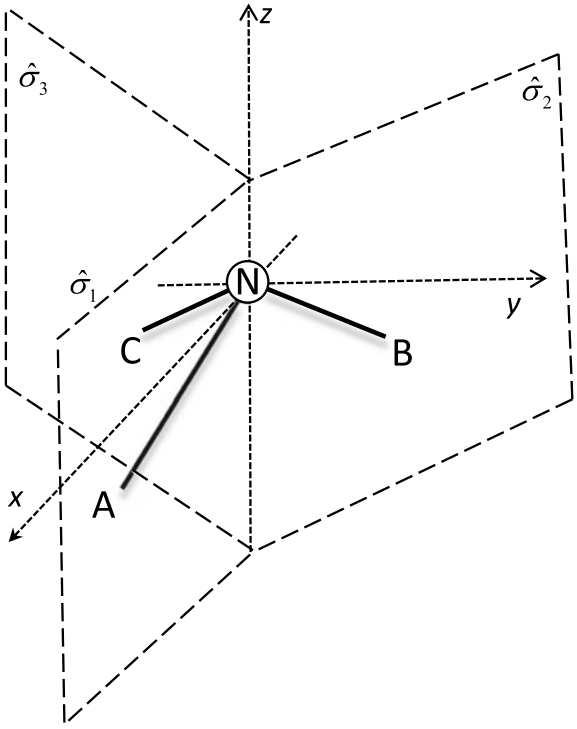


Fig. 3.1 Group theory of the ammonia molecule, with three sets of labels: x, y, z label the Cartesian axes, $\hat{\sigma}_1, \hat{\sigma}_2, \hat{\sigma}_3$ label symmetry planes, and A, B, C label the hydrogen atoms

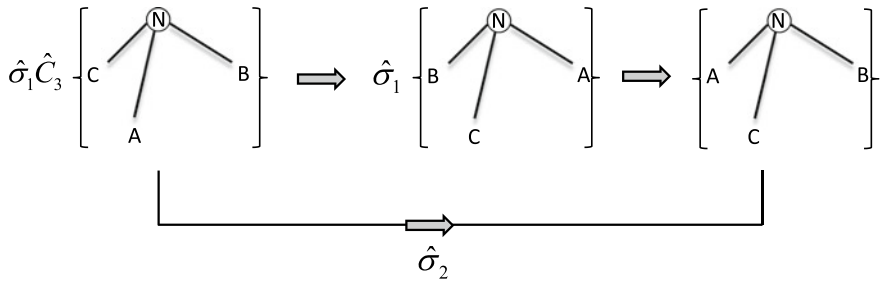


Fig. 3.2 Applying $\hat{\sigma}_1\hat{C}_3$ to the starting structure is equivalent to applying $\hat{\sigma}_2$

Table 3.1 Multiplication table for the point group C_{3v}

C_{3v}	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_1$	$\hat{\sigma}_2$	$\hat{\sigma}_3$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_1$	$\hat{\sigma}_2$	$\hat{\sigma}_3$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{\sigma}_3$	$\hat{\sigma}_1$	$\hat{\sigma}_2$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\hat{\sigma}_2$	$\hat{\sigma}_3$	$\hat{\sigma}_1$
$\hat{\sigma}_1$	$\hat{\sigma}_1$	$\hat{\sigma}_2$	$\hat{\sigma}_3$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
$\hat{\sigma}_2$	$\hat{\sigma}_2$	$\hat{\sigma}_3$	$\hat{\sigma}_1$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$
$\hat{\sigma}_3$	$\hat{\sigma}_3$	$\hat{\sigma}_1$	$\hat{\sigma}_2$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$

operators, and then simply work out the action from right to left; hence, first the $\hat{C}_3$-axis, and then the plane. This is shown in a pictorial way in Fig. 3.2. First the axis will permute the atoms, such that C takes the place of A. Consequently, the $\hat{\sigma}_1$ plane will now conserve C, and interchange A and B. The combined action is itself again one of the symmetry elements, *viz.*, $\hat{\sigma}_2$. The reverse product order yields a different result. In summary:

$$\begin{aligned} \hat{\sigma}_1\hat{C}_3 &= \hat{\sigma}_2 \\ \hat{C}_3\hat{\sigma}_1 &= \hat{\sigma}_3 \end{aligned} \tag{3.1}$$

In this way, we can easily work out the full set of binary products, keeping in mind that applying the threefold rotation three times, or the symmetry planes twice, leaves every atom in place and thus corresponds to the unit element. The results are gathered in the 6×6 multiplication Table 3.1. This table should be read from left to right, i.e., the product $\hat{R}_i\hat{R}_j$ is found in the ith row and jth column. We may symbolically denote the matrix elements in the table as

$$M_{ij} = \hat{R}_i\hat{R}_j \tag{3.2}$$

As has already been shown, these operations can also be performed directly in function space. Choosing the xy-plane $2p$-orbitals on nitrogen, $\{p_x, p_y\}$, as a suitable

Table 3.2 Representation matrices for the (p_x, p_y) basis in C_{3v}

$\mathbb{D}(E)$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\mathbb{D}(\sigma_1)$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
$\mathbb{D}(C_3)$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\mathbb{D}(\sigma_2)$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & +\frac{1}{2} \end{pmatrix}$
$\mathbb{D}(C_3^2)$	$\begin{pmatrix} -\frac{1}{2} & +\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\mathbb{D}(\sigma_3)$	$\begin{pmatrix} -\frac{1}{2} & +\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & +\frac{1}{2} \end{pmatrix}$

basis set, we may represent all the symmetry elements by transformation matrices. The resulting matrices are summarized in Table 3.2. Note that all six matrices are different. The mapping between the symmetry elements and the matrices is therefore one to one, and the representation is said to be *faithful*. For the $\hat{C}_3$-axis, the matrix corresponds to the one in Eq. 1.13, with rotation angle $\alpha = 2\pi/3$, and for the $\hat{C}_3^2$-axis, one has $\alpha = 4\pi/3$, which is equivalent to the inverse angle $\alpha = -2\pi/3$. The $\hat{\sigma}_1$ element leaves p_x unchanged, and inverts p_y. The other reflection planes are similar to $\hat{\sigma}_1$, which means that they can be obtained by a symmetry transformation of this operator, using the results in Sect. 1.3; hence:

$$\begin{aligned} \hat{\sigma}_2 &= \hat{C}_3\, \hat{\sigma}_1 \hat{C}_3^{-1} \\ \hat{\sigma}_3 &= \hat{C}_3\, \hat{\sigma}_2 \hat{C}_3^{-1} \end{aligned} \tag{3.3}$$

The set of the six matrices in Table 3.2 offers an alternative algebraic way of constructing the multiplication table by direct matrix multiplication. The product $\hat{C}_3\hat{\sigma}_1$ is then replaced by the matrix multiplication $\mathbb{D}(C_3) \times \mathbb{D}(\sigma_1)$:

$$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & +\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & +\frac{1}{2} \end{pmatrix} \tag{3.4}$$

which yields the representation matrix for $\hat{\sigma}_3$, in line with Eq. (3.1).

3.2 The Group Structure

The set of symmetry operations of ammonia is said to form a *group*, G. This is a fundamental mathematical structure, consisting of a set of elements and a multiplication rule, with the following characteristics:

- Existence of a unit element, $\hat{E}$, which leaves all elements unchanged:

$$\hat{E}\hat{R} = \hat{R}\hat{E} = \hat{R} \tag{3.5}$$

  In the list of elements, the unit element is placed in front. As a result, the first row and first column of the multiplication table will simply repeat the ordered list of symmetry elements on which the table was based.
- Existence of an inverse element, $\hat{R}^{-1}$, for every element $\hat{R}$:

$$\hat{R}\hat{R}^{-1} = \hat{R}^{-1}\hat{R} = \hat{E} \tag{3.6}$$

  Hence every action can also be undone. If this were not the case we would have produced a monster that can only continue to grow. In the multiplication table, a unit element at position ij indicates that $\hat{R}_i$ and $\hat{R}_j$ are mutual inverses. Clearly, for the unit element, as well as for the reflection planes, $\hat{R}$ and $\hat{R}^{-1}$ coincide, while, for the $\hat{C}_3$-axis, the inverse is equal to $\hat{C}_3^2$, which completes a full turn.
- Closure:

$$\forall \hat{R} \in G \;\&\; \hat{S} \in G \;\Rightarrow\; \hat{R}\hat{S} \in G \;\&\; \hat{S}\hat{R} \in G \tag{3.7}$$

  In the multiplication table, this is apparent by the fact that no entries are left open.
- Associativity:

$$\hat{R}\left(\hat{S}\hat{T}\right) = \left(\hat{R}\hat{S}\right)\hat{T} \tag{3.8}$$

It is difficult to imagine that such a simple set of rules can give rise to such a powerful structure, on which entire properties of molecules will depend. Note that, compared with a number system, a group is an even more primitive concept since it contains only one operation, multiplication, by means of which two elements of the group may be combined. In contrast, the set of numbers allows for addition as well as multiplication, which results in two kinds of unit elements, zero for addition, and unity for multiplication. A closer look at the multiplication table of the group shows that it has very remarkable properties. Each row and each column represents a permutation of the ordered set of elements, but in such a way that every element occurs only once in each row and column. This is a direct consequence of the group properties. As in many group-theoretical proofs, the simplest way to show this is by a *reductio ad absurdum*. Suppose that a given element, $\hat{T}$, occurred at entries ij and ik, with $\hat{R}_k \neq \hat{R}_j$. Then one would have, by applying the rules:

$$\begin{aligned}
\hat{R}_i\hat{R}_j &= \hat{R}_i\hat{R}_k \\
\hat{R}_i^{-1}\left(\hat{R}_i\hat{R}_j\right) &= \hat{R}_i^{-1}\left(\hat{R}_i\hat{R}_k\right) \\
\left(\hat{R}_i^{-1}\hat{R}_i\right)\hat{R}_j &= \left(\hat{R}_i^{-1}\hat{R}_i\right)\hat{R}_k \\
\hat{E}\hat{R}_j &= \hat{E}\hat{R}_k \\
\hat{R}_j &= \hat{R}_k
\end{aligned} \tag{3.9}$$

which would contradict the original supposition. Along the same lines, it is easy to prove that the inverse of a product is equal to the product of the inverses in the opposite order:

$$\left(\hat{R}_i \hat{R}_j\right)^{-1} = \hat{R}_j^{-1} \hat{R}_i^{-1} \tag{3.10}$$

As a matter of principle, the group multiplication table contains everything there is to know about the group. It is, though, not necessary to store the whole multiplication table. A more compact way uses *generators*. The generators are defined as a minimal set of elements capable of generating the whole group. For the present example, two generators are needed, e.g., $\hat{C}_3$ and $\hat{\sigma}_1$. It is sufficient to make all binary combinations of these two operators in order to generate all remaining elements:

$$\begin{aligned} \hat{C}_3 \hat{C}_3 &= \hat{C}_3^2 \\ \hat{\sigma}_1 \hat{\sigma}_1 &= \hat{E} \\ \hat{C}_3 \hat{\sigma}_1 &= \hat{\sigma}_3 \\ \hat{\sigma}_1 \hat{C}_3 &= \hat{\sigma}_2 \end{aligned} \tag{3.11}$$

Alternatively any pair of reflection planes would suffice as generators, say $\hat{\sigma}_1$ and $\hat{\sigma}_2$, but in this case the remaining symmetry plane can be obtained only by a further multiplication.

$$\begin{aligned} \hat{\sigma}_1 \hat{\sigma}_1 &= \hat{E} \\ \hat{\sigma}_2 \hat{\sigma}_2 &= \hat{E} \\ \hat{\sigma}_1 \hat{\sigma}_2 &= \hat{C}_3 \\ \hat{\sigma}_2 \hat{\sigma}_1 &= \hat{C}_3^2 \\ \hat{\sigma}_1 \hat{\sigma}_2 \hat{\sigma}_1 &= \hat{\sigma}_3 \end{aligned} \tag{3.12}$$

A *presentation* of a group is a set of generators, together with a minimal set of relations, that are sufficient to work out any product of two elements. As an example, let us denote the $\hat{C}_3$, $\hat{\sigma}_1$ generators as c, s. Just three relations among these generators are sufficient to derive the whole multiplication table: $c^3 = s^2 = e$, $sc = c^2 s$. The generation of the six elements of the group follows from Eq. (3.11):

$$\begin{aligned} \hat{E} &= e \\ \hat{C}_3 &= c \\ \hat{C}_3^2 &= c^2 \\ \hat{\sigma}_1 &= s \\ \hat{\sigma}_2 &= sc \\ \hat{\sigma}_3 &= cs \end{aligned} \tag{3.13}$$

Any product of these elements can be shown to be closed, using only the presentation. As an example, the rule $\hat{\sigma}_2\hat{\sigma}_3 = \hat{C}_3$ is expressed in the presentation as follows:

$$sccs = s(c^2 s) = s(sc) = s^2 c = c \tag{3.14}$$

In this way, the whole multiplication table can be derived.

The structure of the group can also be encoded in a graph known as the Cayley graph. A graph is an abstract mathematical object, consisting of a set of points, or nodes, and a set of lines connecting pairs of these points. In a *directed graph* these pairs are ordered, which means that directional arrows are added to the connecting lines. In the Cayley graph, every element of the group corresponds to a node. The lines correspond to the action of the group generators. The generator $\hat{g}$ connects a given node $\langle \hat{R}_i \rangle$ by a directed line to the resulting node $\langle \hat{g}\hat{R}_i \rangle$. The action of the group on its own Cayley graph will not only map nodes onto nodes, but will also preserve the directed connections. As a result, the symmetry group will map the graph onto itself. Such a mapping is called an *automorphism*. The group G is thus isomorphic to the automorphism group of its Cayley graph. The Cayley graph corresponding to the group C_{3v}, generated by $\hat{C}_3$ and $\hat{\sigma}_1$, is shown in Fig. 3.3. It resembles a trigonal prism, but with opposite directions in the upper and the lower triangle. The $\hat{\sigma}_1$ generator

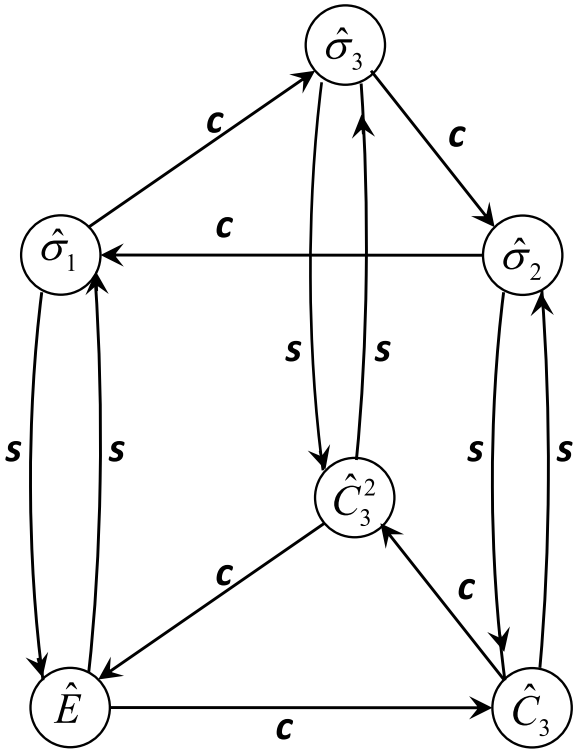


Fig. 3.3 Cayley graph of the C_{3v} point group. The generators are $c = \hat{C}_3$, $s = \hat{\sigma}_1$

corresponds to the upright edges of the prism. Since this generator is its own inverse, these edges can be traversed in both directions, so they are really undirected.

3.3 Some Special Groups

Abelian groups[1] are groups with a commutative multiplication rule, i.e.,

$$\forall \hat{R} \in G \;\&\; \hat{S} \in G \;\Rightarrow\; \hat{R}\hat{S} = \hat{S}\hat{R} \tag{3.15}$$

Hence, in an abelian group, the multiplication table is symmetric about the diagonal. Clearly, our group C_{3v} is not abelian.

Cyclic groups are groups with only one generator. They are usually denoted as C_n. The threefold axis gives rise to the cyclic group C_3. Its elements consist of products of the generator. By analogy with number theory, such multiple products are called powers; hence, $C_3 = \{\hat{C}_3, \hat{C}_3^2, \hat{C}_3^3\}$ where the third power is, of course, the unit element. Similarly, the reflection planes yield a cyclic group of order 2. The standard notation for this group is not C_2 but C_s. Cyclic groups are of course abelian, because the products of elements give rise to a sum of powers, and summation is commutative:

$$\hat{C}^i \hat{C}^j = \hat{C}^{i+j} = \hat{C}^{j+i} = \hat{C}^j \hat{C}^i \tag{3.16}$$

By contrast, not all abelian groups are cyclic. A simple example is the group[2] D_2 of order 4, which is presented in Table 3.3. It needs two perpendicular twofold axes as generators, and thus cannot be cyclic. Nonetheless, it is abelian since its generators commute.

The *symmetric* group, S_n, is the group of all permutations of the elements of a set of cardinality n. The order of S_n is equal to $n!$. As it happens, our C_{3v} group is isomorphic to S_3. The permutations are defined on the ordered set of the three hydrogen atomic labels ⟨ABC⟩. Interchange of A and B means that, in this row, the element A is replaced by B and *vice versa*. Another way to express this is that 'A becomes B, and B becomes A', hence (A → B → A). This interchange is a *transposition* or 2-cycle which will be denoted as (AB). The operation for the entire set is then written as a sequence of two disjunct cycles (C)(AB), where the 1-cycle indicates that the element C remains unchanged. The 3-cycle (ABC) corresponds to a cyclic permutation of all three elements: (A → B → C → A). The successive application of both operations, acting on the letter string, can be worked out as follows:

[1] Named after the Norwegian mathematician Niels Henrik Abel (1802–1829).

[2] This group is isomorphic to Felix Klein's four-group (*Vierergruppe*).

Table 3.3 Multiplication table for the point group D_2

D_2	$\hat{E}$	$\hat{C}_2^x$	$\hat{C}_2^y$	$\hat{C}_2^z$
$\hat{E}$	$\hat{E}$	$\hat{C}_2^x$	$\hat{C}_2^y$	$\hat{C}_2^z$
$\hat{C}_2^x$	$\hat{C}_2^x$	$\hat{E}$	$\hat{C}_2^z$	$\hat{C}_2^y$
$\hat{C}_2^y$	$\hat{C}_2^y$	$\hat{C}_2^z$	$\hat{E}$	$\hat{C}_2^x$
$\hat{C}_2^z$	$\hat{C}_2^z$	$\hat{C}_2^y$	$\hat{C}_2^x$	$\hat{E}$

$$\begin{array}{rc} & \langle \mathrm{A\ B\ C} \rangle \\ (\mathrm{C})(\mathrm{AB}) & \Downarrow \\ & \langle \mathrm{B\ A\ C} \rangle \\ (\mathrm{ABC}) & \Downarrow \\ & \langle \mathrm{C\ B\ A} \rangle \end{array} \tag{3.17}$$

The result is to permute A and C, and leave B invariant. This result defines the product of the two operations as

$$(ABC) \cdot (C)(AB) = (B)(AC) \tag{3.18}$$

The multiplication table for the whole group is given in Table 3.4. The group multiplication tables of S_3 and C_{3v} clearly have the same structure: they are *isomorphic*, but it is important to keep in mind that the two kinds of groups have a very different meaning. The C_{3v} operations refer to spatial symmetry operations of the ammonia molecule, while the permutational group is a set-theoretic concept and acts on elements in an ordered set. A one-to-one mapping between the permutations and the spatial symmetry operations is not as obvious as it seems. For instance, it is tempting to identify the (A)(BC) permutation operation with the action of the $\hat{\sigma}_1$ reflection plane, since it indeed leaves A invariant and swaps B and C. However, in the spatial symmetry group atoms are displaced, and thus atom A may be displaced by other symmetry operations out of the $\hat{\sigma}_1$-plane, while the $(A)(BC)$ operation remains tied to the atom A, irrespective of the other permutations with which it is combined. Likewise the (ABC) operation always involves the cycle A → B → C → A, irrespective of preceding transpositions, as is exemplified in Eq. (3.17). This indicates that the application of the permutation operators on the molecular frame, as defined by the usual cycle index notation, is in line with the passive view of symmetry operations, with symmetry elements attached to fixed atoms. In Problem 3.2 at the end of this chapter, it is discussed how to redefine permutation operations in accordance with the active definition of symmetry elements, as used throughout this book. In Sect. 5.4, we will examine in more detail the relationship between nuclear permutations and spatial symmetry operations, when discussing the molecular symmetry group.

In S_3, the number of transpositions, i.e., pairwise interchanges of atoms, is zero for the unit element, one for the reflection planes, and two for the threefold axes. *Odd permutations* are defined by an odd number of transpositions. The product of two even permutations is an even permutation, and for this reason the even permutations

Table 3.4 Multiplication table for the symmetric group S_3. The unit element can also be expressed as three 1-cycles: (A)(B)(C)

S_3	$\hat{E}$	(ABC)	(ACB)	(A)(BC)	(B)(AC)	(C)(AB)
$\hat{E}$	$\hat{E}$	(ABC)	(ACB)	(A)(BC)	(B)(AC)	(C)(AB)
(ABC)	(ABC)	(ACB)	$\hat{E}$	(C)(AB)	(A)(BC)	(B)(AC)
(ACB)	(ACB)	$\hat{E}$	(ABC)	(B)(AC)	(C)(AB)	(A)(BC)
(A)(BC)	(A)(BC)	(B)(AC)	(C)(AB)	$\hat{E}$	(ABC)	(ACB)
(B)(AC)	(B)(AC)	(C)(AB)	(A)(BC)	(ACB)	$\hat{E}$	(ABC)
(C)(AB)	(C)(AB)	(A)(BC)	(B)(AC)	(ABC)	(ACB)	$\hat{E}$

alone will also form a group, known as the *alternating group*, A_n. In the present example, the alternating group A_3 is isomorphic to the cyclic group C_3. By contrast, the product of two odd permutations is not odd, but even. So odd permutations cannot form a separate group. In Appendix A, one finds the cycle structure and character tables for the symmetric groups with 3, 4, and 5 elements.

The group multiplication table contains all there is to know about a group. It hides a wealth of internal structures which are directly relevant to the physical phenomena to which the group applies. In order to elucidate these structures, three ways of delineating subsets of the group are useful: subgroups, cosets, and classes.

3.4 Subgroups

A subgroup H of G, denoted $H \subset G$, is a subset of elements of G which itself has the group property. Trivial subgroups are the group containing the identity alone, denoted as $C_1 = \{\hat{E}\}$, and the group G itself. Besides these, in the case of the group of ammonia, C_{3v}, there are four non-trivial subgroups: $C_3 = \{\hat{E}, \hat{C}_3, \hat{C}_3^2\}$, and $C_s = \{\hat{E}, \hat{\sigma}_i\}$, with $i = 1, 2$ or 3. The three C_s groups are equivalent. We can construct a simplified genealogical tree, which shows the subgroup structure (Fig. 3.4). In chemistry and physics subgroup structures are highly relevant, since the distortions of a symmetric system can be described as a descent down the genealogical tree. We shall describe this in Sect. 4.6 as the subduction process. For the moment we retain Cayley's theorem:

Theorem 3.1 *Every group of order n is isomorphic with a subgroup of the symmetric group S_n.*

This theorem is immediately clear from the multiplication table. A given row of the table shows how the corresponding element maps the entire set of elements onto itself. This mapping is a permutation of the n elements, and every element gives rise to a different permutation, since no two rows can be the same. Thus, G must

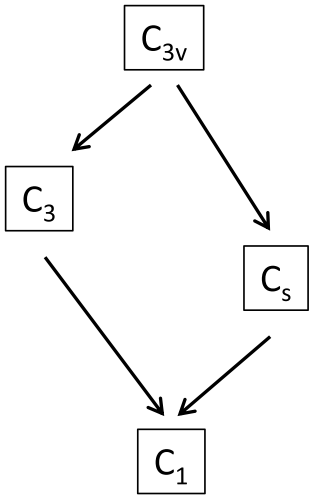


Fig. 3.4 Genealogical tree, representing progressive symmetry breaking of the C_{3v} point group. The C_s box stands for the three equivalent reflections groups

be a subgroup of S_n. The importance of this theorem is especially evident from the mathematical point of view. It tells us that the symmetric groups exhaust all the possible structures of finite groups.

3.5 Cosets

A genuine *partitioning* of a group is achieved when the set of elements is divided into separate subsets, which do not exhibit any overlap, and, together, constitute the whole group. Subgroups clearly do not form a partitioning, since, for instance, they all share the same unit element. On the other hand, cosets do form a partitioning. In molecules, the natural realizations of the cosets are the sets of equivalent sites. These are atoms or groups of atoms which are permuted by the action of the molecular symmetry group. In the example of the ammonia molecule, each of the three hydrogen atoms occupies an equivalent site, with C_s symmetry. The nitrogen atom, however, occupies a unique site, which has the full C_{3v} symmetry. Now consider the site of one particular hydrogen atom, say A. The C_s subgroup which leaves this site invariant consists of only two symmetry elements: $\hat{E}$ and $\hat{\sigma}_1$. This subgroup is called the *stabilizer* of the site. When we multiply each element of this subgroup (on the left) with an element outside it, say $\hat{C}_3$, we obtain two new elements, $\hat{C}_3$ and $\hat{\sigma}_3$, which both share the property that they map A onto B. They form a (left) coset of the original C_s subgroup, and the element that we used to form this coset is the coset representative. There is still another coset, which may be generated by one of the remaining elements, say $\hat{C}_3^2$. In this way, one finds the coset, $\{\hat{C}_3^2, \hat{\sigma}_2\}$, of elements which have the property that they both map A onto C. The sum of all the cosets forms the total set, hence:

$$C_{3v} = \{\hat{E}, \hat{\sigma}_1\} + \{\hat{C}_3, \hat{\sigma}_3\} + \{\hat{C}_3^2, \hat{\sigma}_2\} \tag{3.19}$$

We can rewrite this in general as

$$G = \sum_n \hat{R}_n \, H \tag{3.20}$$

where $\hat{R}_n$ denotes a coset representative, and the product $\hat{R}_n H$ denotes the nth coset, obtained by multiplying every element of the subgroup, on the left, by the generator. The choice of coset representatives is not unique, since every element of a given coset may act as representative. In the case of the present group, we can choose all representatives as powers of the threefold axis, which allows introduction of a cyclic summation:

$$C_{3v} = \sum_{n=1}^{3} (\hat{C}_3)^n \, C_s \tag{3.21}$$

This expression forms a nice illustration of how representatives make different maps of the subgroup. These cosets form a cyclic *orbit* inside the group. Note that the way cosets are defined here is based on left multiplication by the representatives, giving rise to what is also denoted as *left cosets*. An analogous partitioning of the group can also be based on right cosets. The partitioning of a group in cosets is at the basis of the famous Lagrange theorem:

Theorem 3.2 *The order of a subgroup of a finite group is a divisor of the order of the group.*

Consider a group G and subgroup H, with respective orders $|G|$ and $|H|$. The theorem states that $|G|/|H|$ is an integer. The proof is based on two elements. One first has to prove that all elements in a given coset are different, and further that different cosets do not overlap.

Consider a coset $\hat{R}_i H$, with elements $\hat{R}_i \hat{h}_x$. For $\hat{h}_x \neq \hat{h}_y$, $\hat{R}_i \hat{h}_x$ must be different from $\hat{R}_i \hat{h}_y$, simply because two elements in the same row in the multiplication table can never be equal, as was proven in Eq. (3.9). Hence, the size of the coset will be equal to $|H|$. Then we consider an element $\hat{R}_j \notin \hat{R}_i H$. This new element will in turn be the representative of a new coset, $\hat{R}_j H$, and we must prove that this new coset does not overlap with the previous one. This can easily be demonstrated by *reductio ad absurdum*. Suppose that there is an element $\hat{R}_j \hat{h}_x$ in the second coset that also belongs to the first coset, as $\hat{R}_i \hat{h}_y$. We then have:

$$\begin{aligned} \hat{R}_j \hat{h}_x &= \hat{R}_i \hat{h}_y \\ \hat{R}_i^{-1} \hat{R}_j &= \hat{h}_y \hat{h}_x^{-1} \end{aligned} \tag{3.22}$$

Since the subgroup H has the group property, the inverse element $\hat{h}_x^{-1}$ is also an element of H, and so is its product with $\hat{h}_y$. Hence, the product $\hat{R}_i^{-1} \hat{R}_j$ will be an

element of H, say $\hat{h}_z = \hat{h}_y \hat{h}_x^{-1}$. The first coset will of course contain the element $\hat{R}_i \hat{h}_z$, which reduces to

$$\hat{R}_i \hat{h}_z = \hat{R}_i \hat{h}_y \hat{h}_x^{-1} = \hat{R}_i \hat{R}_i^{-1} \hat{R}_j = \hat{R}_j \tag{3.23}$$

Hence, $\hat{R}_j \in \hat{R}_i H$, contrary to the assumption. The expansion of the group in cosets thus leads to a complete partitioning in subsets of equal sizes. It starts by the subset formed by the subgroup H. If H is smaller than G, take an element outside H and form with this element a coset, which will have the same dimension as H. If there are still elements outside use one of these to form a new coset, again containing $|H|$ elements, etc. Each time a coset is formed, a block of size $|H|$ is occupied, till the full territory of the group is occupied by subsets of the same size. Their order must thus be a divisor of the group order. We shall make use of this concept in the induction of representations in Sect. 4.6. In the present example, the point group of ammonia is of order 6, with divisors 1, 2, 3, and 6, and for each of these there are indeed subgroups. This is rather exceptional, though. It is *not* the case that for every divisor there should be a subgroup. As a further corollary, groups with an order which is a prime number have no non-trivial subgroups.

3.6 Classes

Probably the most natural way to partition a group is by putting all elements 'of the same kind' into separate classes. Hence, in the group C_{3v}, we could put the three planes in one class; the unit element is of course a separate class, but what about the $\hat{C}_3$- and $\hat{C}_3^2$-axes? Are they of the same kind or not? Clearly, we need a rigorous definition of what it means for two symmetry elements to be equivalent. We can use as a criterion the symmetry transformations of an operator, as explained in Sect. 1.3. Hence, an element $\hat{B}$ will belong to the same class as an element $\hat{A}$, $\hat{B} \sim \hat{A}$, if there exists (denoted as $\exists$) a symmetry operation $\hat{U}$ which belongs to the group and transforms $\hat{A}$ into $\hat{B}$:

$$\hat{B} \sim \hat{A} \ : \exists \hat{U} \in G \rightarrow \ \hat{B} = \hat{U} \hat{A} \hat{U}^{-1} \tag{3.24}$$

In C_{3v}, the symmetry planes will map $\hat{C}_3$ onto $\hat{C}_3^2$. Hence, we can safely say that the two threefold elements belong to the same class, because of the existence of symmetry planes, which can reverse the direction of rotation. Synonyms for 'to belong to the same class' are 'to be (class-)conjugate' or 'to be similarity transforms'. If the element $\hat{U}$ transforms $\hat{A}$ to $\hat{B}$, then the inverse element, $\hat{U}^{-1}$, which because of the group properties also belong to G, will do the reverse and will transform $\hat{B}$ into $\hat{A}$. Hence, conjugation is reflexive. It is, furthermore, transitive:

$$\left(\hat{A} \sim \hat{B}\right) \ \& \ \left(\hat{B} \sim \hat{C}\right) \rightarrow \left(\hat{A} \sim \hat{C}\right) \tag{3.25}$$

The set of all elements that are conjugate with a given element is a *conjugacy class* or simply a class. A class is fully denoted by specifying any one of its elements, in the same way as a coset is defined by any one of its representatives. The total group is of course the sum of all its classes. In abelian groups, the similarity transformation will always return the element on which it was acting; hence, in this case, all classes will be singletons (sets of order 1). The unit element is unique, so it is always in its own class.

Theorem 3.3 *The number of elements in a class is a divisor of the group order.*

The proof makes use of the coset concept. We start by considering all elements that stabilize a given element, say $\hat{A}_0$, of the group. One can prove that these elements constitute a subgroup $H \subset G$. Hence, we write

$$\hat{h}_x \in H \;\Leftrightarrow \hat{h}_x \hat{A}_0 \hat{h}_x^{-1} = \hat{A}_0 \tag{3.26}$$

The proof consists of a check of the four-group criteria. The closure, for instance, is proven as follows: suppose that both $\hat{h}_x$ and $\hat{h}_y$ stabilize $\hat{A}_0$; then their product will also be a stabilizer:

$$\hat{h}_x \hat{h}_y \hat{A}_0 \left(\hat{h}_x \hat{h}_y\right)^{-1} = \hat{h}_x \hat{h}_y \hat{A}_0 \hat{h}_y^{-1} \hat{h}_x^{-1} = \hat{h}_x \hat{A}_0 \hat{h}_x^{-1} = \hat{A}_0 \tag{3.27}$$

where we use the result of Eq. (3.10) that the inverse of a product is equal to the product of the inverses, in the reverse order. Next, we expand G in cosets of this newly found subgroup H. All elements of a coset $\hat{R}_i H$, with $\hat{R}_i \notin H$, will transform $\hat{A}_0$ into the same new element $\hat{A}_i$:

$$\hat{R}_i \hat{h}_x \hat{A}_0 \left(\hat{R}_i \hat{h}_x\right)^{-1} = \hat{R}_i \hat{h}_x \hat{A}_0 \hat{h}_x^{-1} \hat{R}_i^{-1} = \hat{R}_i \hat{A}_0 \hat{R}_i^{-1} = \hat{A}_i \tag{3.28}$$

The result must be different from $\hat{A}_0$, because, otherwise, $\hat{R}_i$ would be an element of H. In order to prove the theorem, the following remaining questions have to be decided. Do different cosets give rise to different similarity transforms? Does one obtain all elements of a class by finding all transforms of a given starting element? The answers to both questions are affirmative. For the first question, if $\hat{R}_i$ and $\hat{R}_j$ represent different cosets, one should conclude: $\hat{A}_i \neq \hat{A}_j$. Suppose that the opposite is true:

$$\begin{aligned} \hat{R}_i \hat{A}_0 \hat{R}_i^{-1} &= \hat{R}_j \hat{A}_0 \hat{R}_j^{-1} \\ \hat{R}_i^{-1} \hat{R}_j \hat{A}_0 \hat{R}_j^{-1} \hat{R}_i &= \hat{A}_0 \\ \hat{R}_i^{-1} \hat{R}_j \hat{A}_0 \left(\hat{R}_i^{-1} \hat{R}_j\right)^{-1} &= \hat{A}_0 \end{aligned} \tag{3.29}$$

This implies that $\hat{R}_i^{-1} \hat{R}_j$ stabilizes $\hat{A}_0$ and thus must belong to H, where it corresponds to, say, $\hat{h}_z$. But one then has, again:

$$\hat{R}_i \hat{h}_z = \hat{R}_i \hat{R}_i^{-1} \hat{R}_j = \hat{R}_j \tag{3.30}$$

and thus $\hat{R}_j$ is a representative of the same coset as $\hat{R}_i$, which contradicts the starting assumption. Hence, there will be at least as many equivalent elements in the class as there are cosets of the stabilizing subgroup. Have we then generated the entire class? Yes, because by going through all the cosets, we run through the entire group. In this way, we have found all elements that are conjugate to a given one, but, because of transitivity, this also means that there cannot be other conjugate elements. The one-to-one mapping between conjugate elements and cosets implies that the number in a class is equal to the number of cosets of the stabilizing subgroup, and hence—by virtue of Lagrange's theorem—must be a divisor of the group order.

The proof illustrates the connection between cosets and conjugacy classes. A special example of this arises in the case of *normal*, or *invariant*, subgroups. A subgroup H is normal[3] if its left and right cosets coincide, i.e., if $\hat{R}_i H = H \hat{R}_i$. This implies that all the elements of the group will map the subgroup onto itself or, for H a normal subgroup:

$$\forall \hat{U} \in G \;\&\; \hat{h}_x \in H \;:\; \hat{U} \hat{h}_x \hat{U}^{-1} \in H \tag{3.31}$$

Normal subgroups are thus made up of entire conjugacy classes of G. As an example, the group C_3 is a normal subgroup of C_{3v} since it contains the whole conjugacy class of the trigonal elements, as well as the trivial class of the unit element itself. A molecular site which is stabilized by a normal subgroup must be unique since it can be mapped only onto itself. Such is the case for the nitrogen atom in ammonia. By contrast, the C_s subgroups, stabilizing the hydrogen sites in ammonia, are not normal, since they are based on only one reflection plane, while there are three symmetry planes in the corresponding conjugacy class. Accordingly, the hydrogen sites form a set of three. Cosets of normal subgroups form themselves a proper group, which is termed as the quotient or factor group. This property is addressed in Problem 3.3, and will surface again in Chap. 8 as the isogonal point group of extended structures.

3.7 Overview of the Point Groups

Symmetry operations involve translations, rotations, reflections, and their combinations. All these operations are congruence operations or *isometries.* They are transformations which keep fixed distances between all points, i.e., they conserve the metric of space, as opposed to such operations as dilatations. Indeed, since the elements of symmetry groups permute equivalent atoms, they must keep interatomic distances unchanged. Translational symmetry groups apply to extended structures, and will be treated in Chap. 8, which is the final chapter. At present, we look at

[3] Note that the fact that an operation commutes with a normal subgroup as a whole does not imply that it commutes with individual elements of the subgroup.

molecular symmetry groups, with no translations involved. In that case, isometries keep *all* distances unchanged, but moreover leave at least one spatial point immobile. We could think of this point as the center of gravity of the molecule. The isometries of the point groups are described in the box.

Isometries of the point groups
Consider a vector $\mathbf{r}$ in 3D space. The squared length of this vector is given by

$$r^2 = \mathbf{r} \odot \mathbf{r} = \mathbf{r}^T \mathbf{r} = \begin{pmatrix} x & y & z \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$
$$= x^2 + y^2 + z^2$$

The origin of space has coordinates $(0, 0, 0)$. An isometry, $\hat{O}$, is an operation which conserves the distance between the origin and the endpoint of any vector $\mathbf{r}$. This distance is the length of the vector, denoted as r:

$$\hat{O}\, r^2 = \mathbf{r}^T \mathbb{O}^\dagger \mathbb{O}\, \mathbf{r} \equiv \mathbf{r}^T \mathbf{r}$$

Here $\mathbb{O}$ is the transformation matrix, operating on the column vector; $\mathbb{O}^\dagger$ is its adjoint. Since space transformations are real, the adjoint is equal to the transposed: $\mathbb{O}^\dagger = \mathbb{O}^T$. The above equivalence thus requires $\mathbb{O}^T \mathbb{O} = \mathbb{I}$. This implies that the $\mathbb{O}$ transformation matrices are orthogonal matrices. In addition, angles between vectors, as expressed by their scalar product, are conserved as well:

$$\hat{O}\, \mathbf{r} \odot \mathbf{s} = \mathbf{r}^T \mathbb{O}^\dagger \mathbb{O}\, \mathbf{s} \equiv \mathbf{r} \odot \mathbf{s}$$

Conservation of vector length and angle between vectors implies that isometries leave all distances between space points unchanged, and are represented by orthogonal matrices. Moreover, the coordinate origin with coordinates $(0, 0, 0)$ is the stable point, which is not affected by the orthogonal transformation.

As we concluded that all symmetry operations must be represented by orthogonal matrices, our overview of the point group thus will be rooted in the covering group of all orthogonal 3×3 matrices. This group coincides with the highest point group symmetry, which is the spherical symmetry group. The gradual descent in symmetry from the sphere provides a practical tool to determine and classify molecular symmetry groups. Molecules with a symmetry that is closest to the sphere are *isotropic*, in the sense that there is no unique direction to orient them. The corresponding symmetry groups are the cubic and icosahedral groups. Breaking spherical symmetry gives rise to symmetry groups based on the cylinder. Cylindrical symmetry splits 3D space into an axial 1D component and an equatorial 2D space, which remains isotropic. Molecules with cylindrical shapes have a unique anisotropy axis, along which they may be oriented in space. Conventionally, this direction is denoted as the z-direction.

Along this direction they have a principal n-fold rotation or rotation–reflection axis, which is responsible for the remaining in-plane isotropy. Finally, further removal of all $\hat{C}_n$ or $\hat{S}_n$ with $n > 2$ leads to molecules which are completely anisotropic, and which have at most orthorhombic symmetry, D_{2h}. In this section, we will provide a concise overview of the point groups, following a path of descent in symmetry. We thereby refer to the list of point groups, presented in the character tables in Appendix A.

3.7.1 Spherical Symmetry and the Platonic Solids

From a geometrical point of view, the symmetry of the sphere is described as the set of all possible rotations around any axis through the center, all possible reflection planes cutting the sphere in halves, and all combinations of rotations and reflections, including the center of inversion. From an algebraic point of view, the symmetry group of the sphere is identified as the group of all 3×3 orthogonal matrices. This symmetry group is the orthogonal group, O(3). One might wonder if the geometric and algebraic definitions coincide. Is every element of O(3) also a rotation–reflection, or vice versa has every rotation–reflection one unique orthogonal matrix representation? The answer to the first question is affirmative: every orthogonal matrix describes a rotation–reflection isometry. However the opposite is not true: different rotations can give rise to the same orthogonal matrix: e.g., the rotations $\hat{C}_4^3$ and $\hat{C}_4^{-1}$ are different from a geometrical perspective, as they follow paths in opposite directions, but the matrix describing the result is the same. A detailed analysis of this relationship is provided in Sect. 7.1. As the determinants of orthogonal matrices are ± 1, and the product of two matrices with determinant $+1$ will also have determinant $+1$, all matrices with determinant $+1$ will form a separate subgroup: the special orthogonal group SO(3). This group covers all rotations around any axis through the center of the sphere. In turn, SO(3) has a single coset, which consists of all orthogonal matrices with determinant -1. Of course this coset does not have the group structure as it does not contain a unit element. Its elements can be obtained by multiplying all matrices of SO(3) by minus the unit matrix. On the operational level, this corresponds to combining a rotation axis with the inversion center. The resulting elements are the inversion center, reflection through any plane that runs through the center of the sphere, and any rotation axis combined with a perpendicular reflection plane.

Only isolated atoms exhibit spherical symmetry. Molecular shapes that approximate perfect spherical symmetry are based on the *regular polyhedra*, the building blocks of which are *regular polygons*. These polygons are 2D objects, obtained by distributing n points over a circle in such a way that all points are equivalent, i.e., that the distances between any two neighboring points are the same, implying that the angles subtended by adjacent edges are also the same. In this way, the circle may circumscribe an equilateral triangle, a square, a regular pentagon, etc. In fact, for any integer n, with $3 \leq n \leq \infty$, there is a regular n-gon, though not all of them can be drawn by use of a ruler and compass. In 3D things are quite different. Defining regular

Table 3.5 The Platonic solids and their point groups. The numbers count the triangles, squares, and pentagons that intersect in a vertex of the solid

	Triangle	Square	Pentagon
3	Tetrahedron (T_d)	Cube (O_h)	Dodecahedron (I_h)
4	Octahedron (O_h)		
5	Icosahedron (I_h)		

polyhedra by distributing n points over a sphere in such a way that all vertices, edges, and faces of the resulting structures are the same cannot be realized in infinitely many ways. Quite on the contrary: only five solutions are possible.[4] These are known as the Platonic solids, and they are listed in Table 3.5. They played an important role in Pythagorean tradition, as well as in Eastern philosophy and religion. The fact that in 3D only five solutions exist was considered to reveal a fundamental truth about nature, to the extent that Plato, in his *Timaeus*, based his natural philosophy on these solids. The ancient doctrine of the four elements was placed in correspondence with four of the solids. The tetrahedron (or 4-plane), with symmetry T_d, has the most acute angles and was associated with fire. The cube, or hexahedron (6-plane), with symmetry O_h is clearly the most stable structure and refers to the earth. The icosahedron (20-plane), with symmetry I_h, contains 20 faces and is therefore closest to a globular surface, which symbolizes the most fluid element, water. Both cube and icosahedron have a dual partner, which is obtained by replacing vertices by faces and *vice versa*. The dual of the cube is the octahedron (8-plane), which figured for air. In Table 3.5, the octahedron is placed between the tetrahedron and the icosahedron, and this seemed appropriate for air because it is intermediate between fire and water in its mobility, sharpness, and ability to penetrate. The dual of the icosahedron is the regular dodecahedron (12-plane). There being only four elements, the discovery of the fifth solid caused some embarrassment, which found an elegant solution by its being assigned to the substance of the heavens. The dodecahedron and its intriguing symmetry were considered worthy of the stars, with their perfect cosmic order. Besides the three point groups, T_d, O_h and I_h, represented by the Platonic solids, there are four more groups that belong to the isotropic family: T_h, T, O and I. They are subgroups of the octahedral and icosahedral groups, as shown in Fig. 3.5.

The Tetrahedron

T_d symmetry (Fig. 3.6a) plays a crucial role in chemistry. It is the symmetry of the valence structure of aliphatic carbon. The attribution of this symmetry to carbon by van't Hoff [5] as early as 1874, i.e., well before the modern concept of molecular structure, and some two and a half millennia after Pythagoras remains a tribute to the Greek vision that the fundamental structure of matter consists of ideal symmetric

[4] The sum of the angles, subtended at a vertex of a Platonic solid, must be smaller than a full angle of 2π. Hence, no more than 5 triangles, 3 squares, or 3 pentagons can meet in a vertex; regular hexagons are already excluded, since the junction of 3 such hexagons already gives rise to an angle of 2π at the shared vertex.

[5] van't Hoff published his findings in 1874 in Utrecht. In the same year, Le Bel came to the same conclusion, based on the investigation of optical rotatory power. An English translation of the original papers of both chemists can be found in [1].

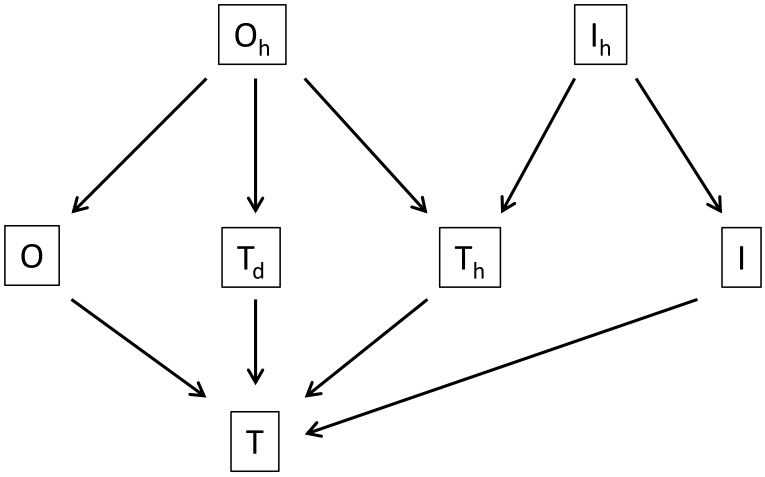


Fig. 3.5 Genealogical tree for the cubic and icosahedral point groups

shapes. The tetrahedron is the most fundamental of the solids, since it is the *simplex* of 3D space. A simplex is a figure consisting of vertices which are all equivalent and are equidistant from each other.[6] In n-dimensional space, the simplex contains $n + 1$ vertices, e.g., in 2D Euclidean space exactly three points can be distributed in such a way that they are equidistant, *viz.*, by occupying the vertices of an equilateral triangle. In 3D space, only four points can be distributed in such a way that they have this property, the solution being the tetrahedron. The perfect permutational symmetry of the vertices of the n-simplex implies that the corresponding symmetry group is isomorphic to the symmetric group S_n. We have already seen that the triangular symmetry of the hydrogens in ammonia could be described by S_3; in the same way, the tetrahedral symmetry group is isomorphic to S_4. The rotational subgroup of T_d is the group T, which, in turn, is isomorphic to the alternating group A_4. An extraordinary member of the tetrahedral family is the group T_h, which has the same order as T_d and also contains T as its rotational subgroup. Molecular examples of this group are quite rare, and will mostly be encountered as symmetry lowering of cubic or icosahedral molecules (see later, Fig. 3.8).

The Cube and Octahedron

The group O_h contains 48 elements and is the symmetry group of the octahedron and the cube (see Fig. 3.6b). The system of Cartesian axes itself has octahedral symmetry, and, as such, this symmetry group is the natural representative of 3D space. It is

[6] In graph theory, the graph of a simplex with n vertices is the complete n-graph, K_n. In such a graph, each of the n vertices is connected to all the other $(n - 1)$ vertices. There is only one simplex for each dimension.

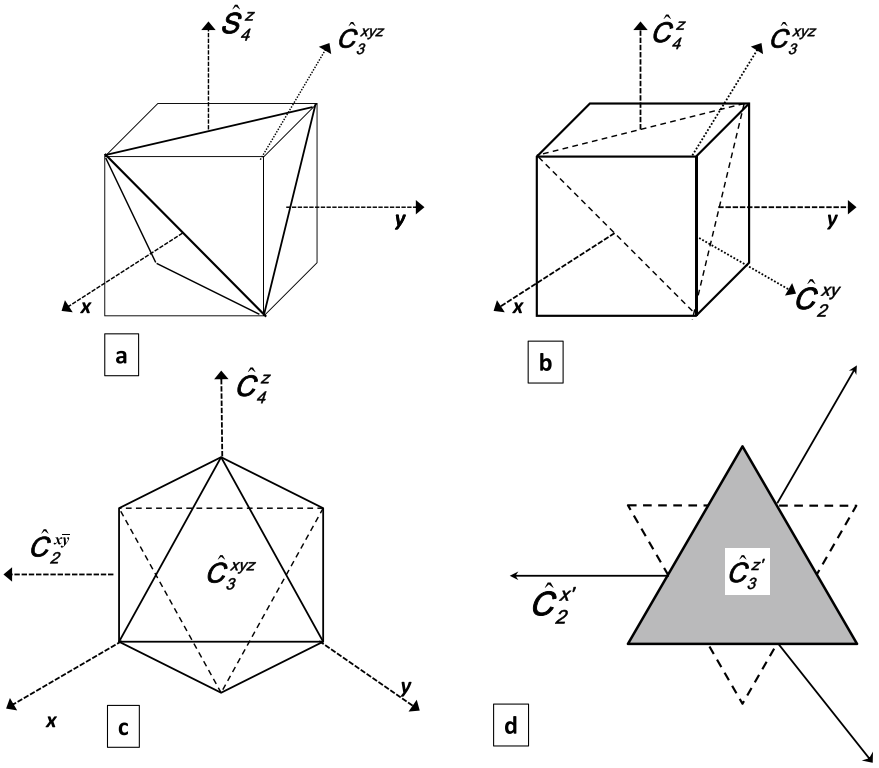


Fig. 3.6 The tetrahedral and octahedral symmetry groups: **a** tetrahedron, inscribed in a cube by connecting four alternating corners, **b** orientation of the cube based on fourfold axes along the Cartesian directions; $\hat{C}_2^{xy}$ lies along the acute bisector of the x- and y-directions; the dashed triangle is one face of the inscribed tetrahedron, **c** construction drawing of the octahedron in the same fourfold setting; the $\hat{C}_2$-axis shown is the bisector of the positive x- and negative y-direction, (**d**) same drawing of the octahedron with a trigonal coordinate orientation: the z'-direction is along the $\hat{C}_3$-axis, and the x'-direction is along the bisector of the positive x- and negative y-direction

ubiquitous in ionic crystals, where it corresponds to coordination numbers 6 (as in rock salt, NaCl) or 8 (as in caesium chloride, CsCl). It is also the dominant symmetry group of coordination compounds. The rotational subgroup is the group O with 24 elements. The octahedron provides us with an insight into the basic architecture of polyhedra. There are three structural elements: vertices, edges, and faces. Through each of these runs a rotational symmetry axis: a $\hat{C}_4$-axis through the vertices, a $\hat{C}_2$-axis through the edge, and a $\hat{C}_3$-axis through the face center. In a triangular face, the product of the three rotations equals the unit element. For the triangle that is turned towards the viewer in Fig. 3.6c, this is

$$\hat{C}_2^{xz}\hat{C}_3^{xyz}\hat{C}_4^z = \hat{E} \tag{3.32}$$

The symmetry elements are labeled by the indices x, y, and z, which refer to their orientation in the Cartesian coordinate system, e.g., $\hat{C}_3^{xyz}$ indicates the $\hat{C}_3$-axis which is the diagonal of the positive Cartesian directions. This notation emphasizes that symmetry elements are tied to the coordinate system, and stay fixed in space.

The Icosahedron and Dodecahedron

Icosahedral symmetry is less obvious and thus more intriguing than cubic symmetry. While tetrahedral and octahedral molecules were already known before the turn of the nineteenth/twentieth century, the first structural study of an icosahedral molecule was the closo-dodecaborane, $B_{12}H_{12}^{2-}$, in 1960 [2]. More examples would follow: in 1984, dodecahedrane, $C_{20}H_{20}$ [3], and then the supermolecule Buckminsterfullerene, C_{60}, which has the shape of a truncated icosahedron [4]. Icosahedral structures can be drawn inside a regular hexagon, as shown in Fig. 3.7a. The coordinate axes in this figure are chosen in such a way that each of the Cartesian directions coincides with a twofold axis. These axes together form a D_{2h} subgroup. This orientation thus corresponds to a D_{2h} setting. Figure 3.7b shows the orientation of the edges through the Cartesian directions. When going from one axis to a neighboring axis, one always rotates the edges by 90°. In fact, there are two ways to realize this D_{2h} setting, either with the top edge in the xz-plane, as in Fig. 3.7b, or in the yz-plane [5]. The reason is that the O_h symmetry of the Cartesian frame is not a subgroup of I_h, since $|O_h|$ is not a divisor of $|I_h|$. Their highest common subgroup is the 24-element group T_h, as indicated in Fig. 3.5. The structure of the six edges in Fig. 3.7b does indeed have T_h symmetry. Note that T_h is a normal subgroup of O_h, but not of I_h. The dual of the icosahedron is the dodecahedron, as shown in Fig. 3.8a. In a dodecahedron, one can select a set of eight vertices which form the corners of a cube. The dashed square in the figure is the top face of such an inscribed cube. Euclid already demonstrated that a cube could be drawn inside a dodecahedron in such a way that its 12 edges each lie in 1 of the 12 faces of the dodecahedron. In turn, in the center of each of the faces of this cube lies a dodecahedral edge, which breaks the fourfold symmetry axes of the cube. The stabilizer of the cube inside I_h is the subgroup T_h. The icosahedral group can be partitioned into five cosets of T_h, which can be generated in a cyclic orbit by a pentagonal generator:

$$I_h = \sum_{n=1}^{5} (\hat{C}_5)^n \, T_h \tag{3.33}$$

Each of the cosets refers to an inscribed cube, exactly as the three C_s cosets in C_{3v} ammonia refer to the three equivalent hydrogen sites. Hence, in a dodecahedron there will be five inscribed cubes. Since the order of the I_h group is 120, which equals 5!, it is tempting to think of this group as isomorphic to S_5, permuting the five inscribed cubes. This, however, is not the case, since I_h contains 10-fold rotation–reflection axes, which are not present in S_5. On the other hand, the rotational subgroup I is indeed isomorphic to the alternating group A_5. The icosahedral rotations act transitively

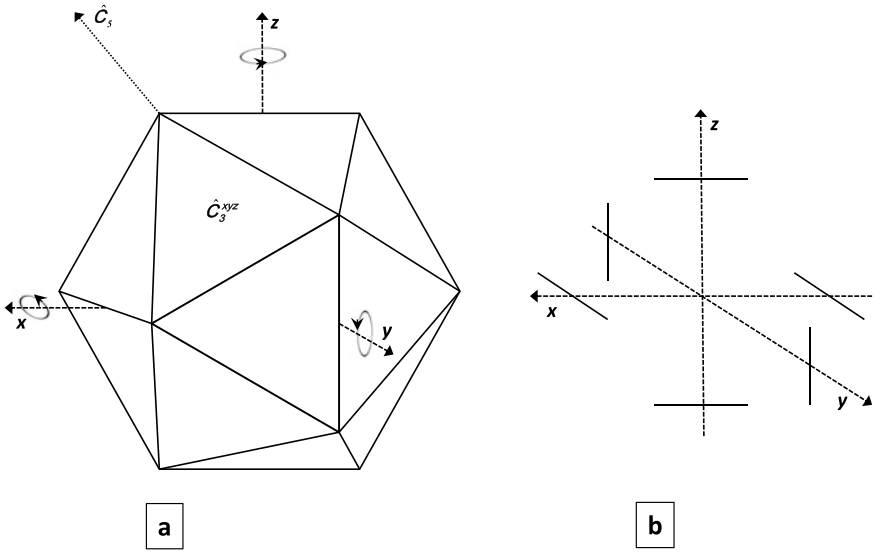


Fig. 3.7 **a** Icosahedron in a D_{2h} setting, **b** T_h structure based on the icosahedral edges through the Cartesian coordinate axes

on the set of five cubes. The icosahedron also contains sixfold rotation–reflection axes, $\hat{S}_6$. These symmetry elements act transitively on the set of the six pentagonal directions. C_{60} (Buckminsterfullerene) has perfect I_h symmetry and corresponds to a truncated icosahedron. Bonds that are adjacent to two hexagons have pronounced double-bonding character. Metal fragments can coordinate [6] to these bonds, as shown in Fig. 3.8b. A hexa-adduct is formed with near T_h symmetry.

3.7.2 Cylindrical Symmetries

Cylinders, Prisms, Antiprisms

The full symmetry group of an ideal cylinder is denoted by $D_{\infty h}$. The D stands for *dihedral*[7] and h for *horizontal*. Like the spherical group, the cylindrical symmetry group is a continuous symmetry group, which has an infinite number of elements. It contains any rotation or rotation–reflection about the z-axis, any $\hat{C}_2$-axis in the equator and any vertical symmetry plane, $\hat{\sigma}_v$. In addition, there are four singleton classes, *viz.*, the identity, the $\hat{C}_2^z$ rotation, the unique horizontal symmetry plane, and spatial

[7] Dihedral means literally 'having two planes'. The dihedral angle is an important molecular descriptor. The dihedral angle of the central B−C bond in an A−B−C−D chain is the angle between the ABC and BCD faces. In the present context, the term dihedral originates from crystallography, such as when two plane faces meet in an apex of a crystal.

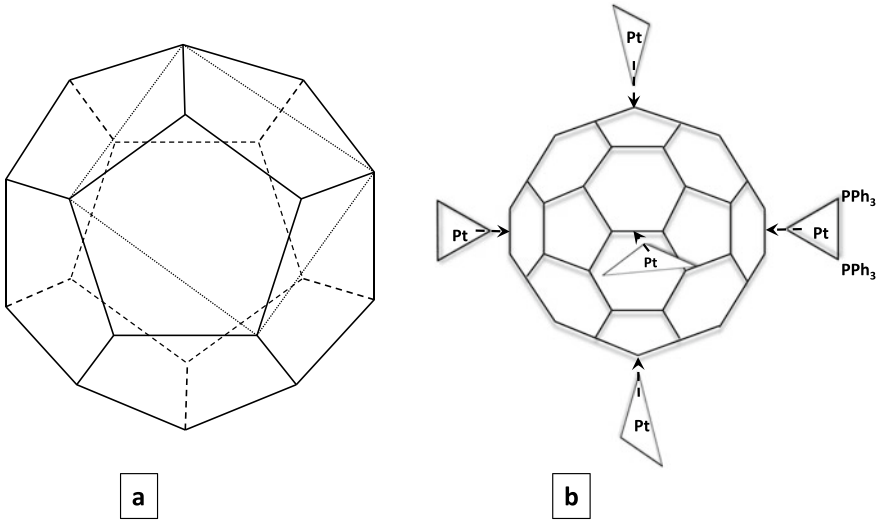


Fig. 3.8 **a** Dodecahedron, the square is the top face of an inscribed cube, **b** supramolecular complex of C_{60} Buckminsterfullerene with six η^2-platina-bis(triphenyl)phosphine adducts on double bonds adjacent to two hexagons; the adducts adopt a D_{2h} setting and reduce the icosahedral symmetry of the buckyball to T_h

inversion. Cylindrical symmetry is met only in linear molecules such as homonuclear diatomics. Obviously, in non-linear molecules, the rotational symmetry of the cylinder is replaced by a finite cyclic symmetry. Two shapes are realizations of maximal finite subgroups of the cylinder: prisms and antiprisms, with respective symmetries D_{nh} and D_{nd}. In both structures, the principal axis is a $\hat{C}_n$-axis, perpendicular to which there are n twofold axes. The horizontal symmetry plane is conserved only in prisms, and not in antiprisms. In Fig. 3.9a, the staggered configuration in ferrocene exemplifies a pentagonal antiprism, while the eclipsed configuration in ruthenocene is a pentagonal prism. The presence of inversion symmetry depends on the parity of n. It is present only in D_{2nh} prisms and $D_{(2n+1)d}$ antiprisms. The cylindrical symmetries reach their lower limit when the principal rotation axis is twofold. In the D_{2h} case, the equatorial directions are no longer equivalent and we have a rectangular parallelepiped which is an *orthorhombic* structure with three different and mutually perpendicular directions. By contrast, in the twofold antiprism, with symmetry D_{2d}, we have a scalenohedron which has two directions perpendicular to the z-axis which are equivalent. In fact, the highest symmetry element in this case is an $\hat{S}_4$ rotation–reflection axis, which is responsible for the equivalence of the in-plane $\hat{C}_2$-axes. The triplet excited state of ethylene adopts a staggered conformation, which has D_{2d} symmetry (Fig. 3.9c). Benzene has prismatic D_{6h} symmetry. In this case, there are two classes of perpendicular twofold axes, which are distinguished in the tables by the labels $\hat{C}_2'$ and $\hat{C}_2''$, and, likewise, two classes of vertical symmetry planes, $\hat{\sigma}_v$ and $\hat{\sigma}_d$. The standard choice for orienting these elements in the hexagonal molecular

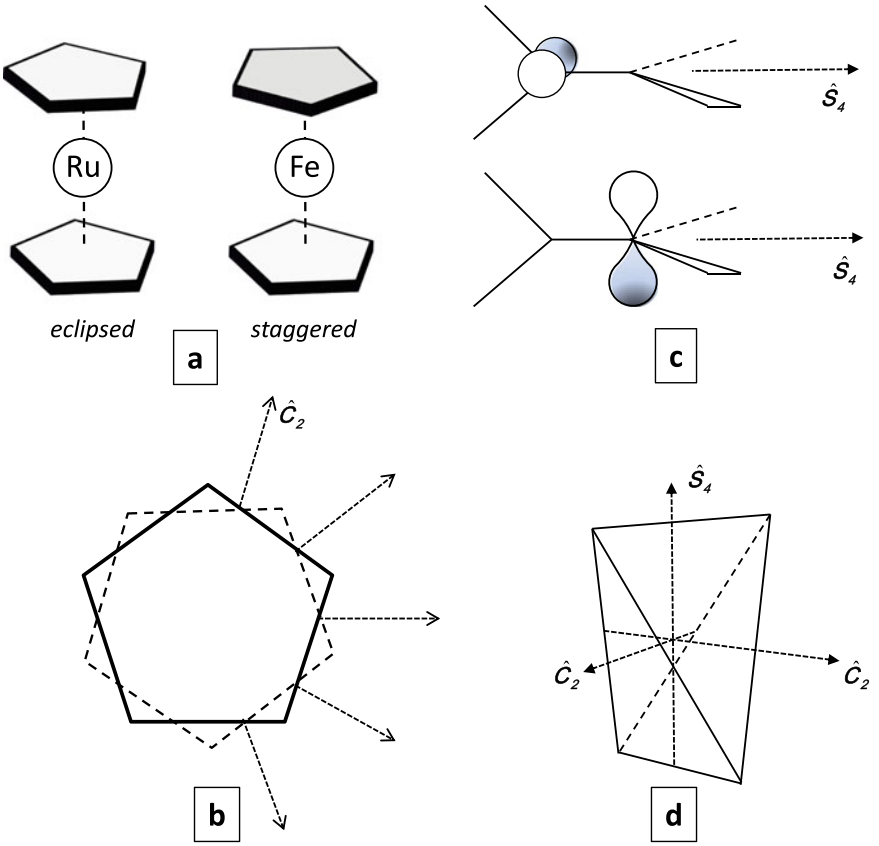


Fig. 3.9 Molecular realizations of cylindrical symmetry: **a** D_{5h} ruthenocene in an eclipsed pentagonal prismatic conformation and D_{5d} ferrocene in a staggered pentagonal antiprismatic conformation, **b** top view of the pentagonal antiprism with the position of the five twofold axes perpendicular to the pentagonal direction, **c** staggered conformation of triplet ethylene with D_{2d} symmetry, and horizontal $\hat{S}_4$-axis; (in the triplet the two carbon $2p$-orbitals shown are singly occupied), (**d**) alternative view of the D_{2d} symmetry of a bisphenoid, with vertical $\hat{S}_4$-axis

frame is shown in Fig. 3.10. This choice is conventional and may be changed, but note that the two symmetries are coupled, in that the $\hat{\sigma}_v$ planes contain the $\hat{C}_2'$-axes, and, similarly, the $\hat{\sigma}_d$-planes contain the $\hat{C}_2''$-axes.

The Rotating Cylinder

Other than geometrical distortions, symmetry breaking of a cylinder may also be realized by a dynamic effect, as in a *rotating cylinder*. In this case, only symmetry elements that will not change the direction of rotation are allowed. As a result, the

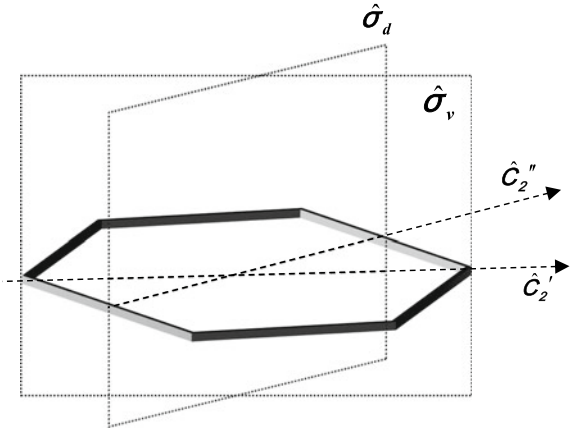


Fig. 3.10 Hexagonal D_{6h} symmetry of benzene: orientation of twofold symmetry elements. $\hat{C}_2'$ and $\hat{\sigma}_v$ pass through opposite atoms; $\hat{C}_2''$ and $\hat{\sigma}_d$ bisect opposite bonds

twofold rotation axes have to be removed and the cylindrical symmetry is reduced to that of the point group, $C_{\infty h}$. This is the symmetry of an *axial* vector or pseudovector. It corresponds to the spatial symmetry of a magnetic field. Note that this group is abelian, and so are its molecular subgroups, with symmetry C_{nh}. Again, the parity of n is important here. $C_{(2n+1)h}$ groups are cyclic. They have one generator which is a reflection axis, $\hat{S}_{2n+1}$, of order $4n+2$. For rotations of even order, two kinds of subgroups arise: the C_{2nh} groups and the S_{2n} groups. The latter are not to be confused with the symmetric groups but designate cyclic groups, generated by a $2n$-fold rotation–reflection axis. Figure 3.11 shows the stereographic projections for C_{2h} and S_4, and a molecular realization of C_{2h}. Note that the latter symmetry group is a combination of the three different kinds of binary symmetry elements of the point groups: a twofold rotation, the reflection plane, and the spatial inversion. It reminds us of the famous Euler identity $e^{i\pi} = -1$, which brings together three special numbers: the base of natural logarithms (e), a square-root of -1 (i), and the ratio of the circumference to the diameter of a circle (π).

The Twisted Cylinder

An alternative symmetry lowering of the cylinder is attained in the twisted cylinder. Here, all improper symmetry elements are broken, the remaining group being the rotational subgroup, D_∞ (see Fig. 3.12). The twist itself can be in two opposite directions, which are mirror images of each other. Molecules of this type may be formed by applying a twist to prisms or antiprisms. A simple example is ethane. In the eclipsed high energy conformation, this is a prism with D_{3h} symmetry, while in the staggered conformation it is an antiprism, D_{3d}. By rotating the two methyl groups with respect to each other in opposite senses about the threefold axis, we may interconvert these two conformations. In between the two extremal conformations, the frame adopts the largest common subgroup, which is D_3.

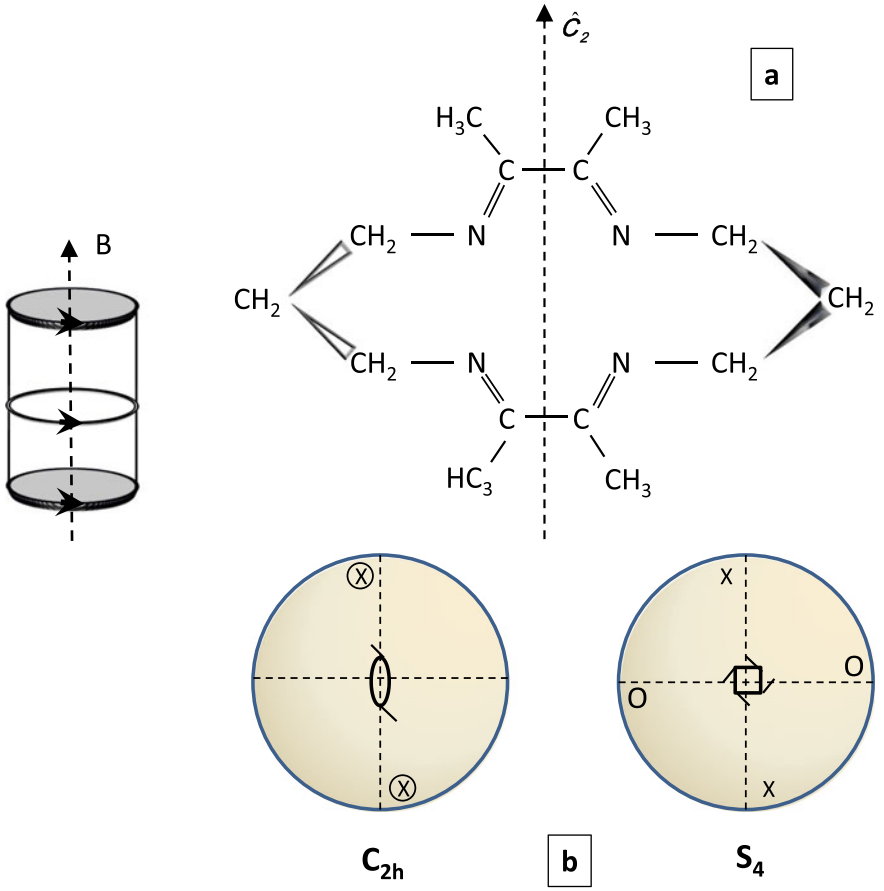


Fig. 3.11 The rotating cylinder has the symmetry of a magnetic field, B, along the cylinder axis. **a** shows a tetra-imine macrocycle, with the bridging CH_2 groups below and above the molecular plane; the resulting symmetry group is C_{2h}, **b** shows stereographic projections of C_{2h} and S_4

Cones

A cone can be considered as a deformation of a cylinder, in which the poles of the uniaxial direction are no longer equivalent. The symmetry group is reduced according to $C_{\infty v}$, where only the vertical symmetry planes remain. Conical symmetry is exemplified by hetero-nuclear diatomic molecules, but it is also the symmetry of a *polar* vector, such as a translation in a given direction, or a polarized medium or an electric field, etc. Conical molecules have C_{nv} symmetries, as was the case for the ammonia model. Again, the smallest trivial member of this series is C_{2v}, which is fully anisotropic. This is the point group of the water molecule.

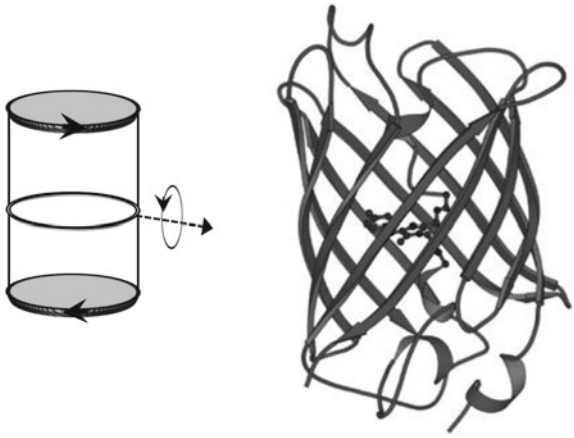


Fig. 3.12 The twisted cylinder. On the right is a cartoon of the green fluorescent protein (GFP). It adopts a β-barrel structure. Eleven β-sheet strands are wound in a helix around the central fluorophore. The approximate symmetry is D_{11}. In the barrel, a cleft is opened to provide interactions with the surrounding solvent

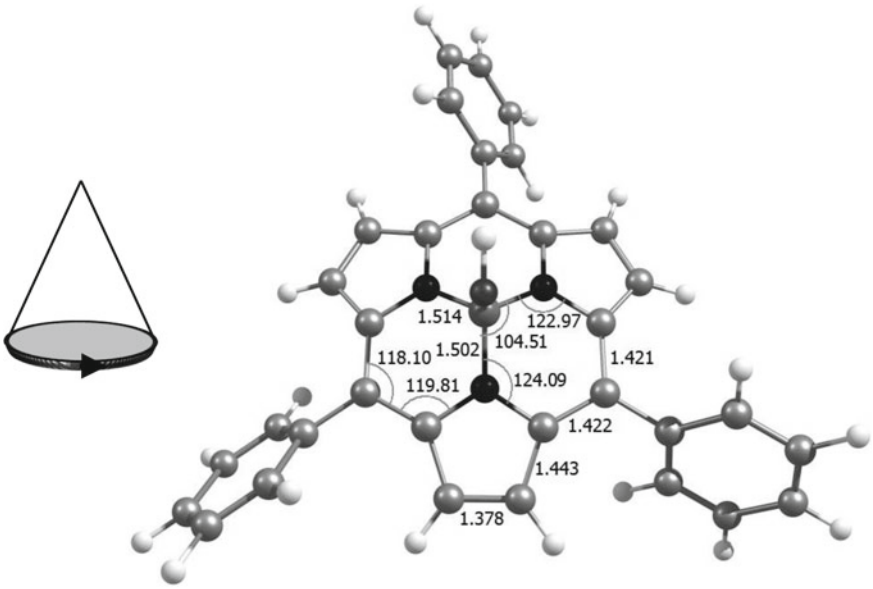


Fig. 3.13 The rotating cone. The molecule is a subporphyrin, and consists of a central boron in a tri-pyrrole macrocycle. The subporphyrin itself has the shape of a trigonal dome and exhibits C_{3v} symmetry. Three phenyl substituents at the meso-positions are arranged like a propeller and reduce the symmetry to C_3. A further symmetry lowering to C_1 is caused by an apical hydroxyl substituent at the boron position, with its hydrogen pointing in the direction of the upper phenyl group

The Rotating Cone

Adding a rotation to the cone will destroy the vertical symmetry planes, since reflection in these planes would alter the sense of the rotation. As a result, only rotations around the axis of the cone are retained, limiting the symmetry group to C_∞. Its molecular point groups are the cyclic groups, C_n. These symmetries are encountered in propeller-like molecules. An example of a subporphyrin [7] is shown in Fig. 3.13. The smallest non-trivial C_n group is found for $n = 2$. This is the symmetry of the Möbius strip, which may also be attained in Möbius-type annulenes.

3.8 Rotational Groups and Chiral Molecules

The symmetry operations that we have encountered are either *proper* or *improper*. Proper symmetry elements are rotations, also including the unit element. The improper rotations comprise planes of symmetry, rotation–reflection axes, and spatial inversion. All improper elements can be written as the product of spatial inversion and a proper rotation (see, e.g., Fig. 1.1). The difference between the two kinds of symmetry elements is that proper rotations can be carried out in real space, while improper elements require the inversion of space, and thus a mapping of every point onto its antipode. This can only be done in a virtual way by looking at the structure *via* a mirror. From a mathematical point of view, this difference is manifested in the sign of the determinant of the corresponding representation matrices in the (x, y, z) basis. For proper rotations, the determinant is equal to $+1$. For improper rotations, it is equal to -1. This minus sign comes from the representation matrix for the inversion center, which corresponds to minus the unit matrix.

$$\hat{\imath}\begin{pmatrix} x & y & z \end{pmatrix} = \begin{pmatrix} x & y & z \end{pmatrix}\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{3.34}$$

Since the determinant of a matrix product is the product of the determinants of the individual matrices, multiplication of proper rotations will yield again a proper rotation, and for this reason the proper rotations form a rotational group. In contrast, the product of improper rotations will square out the action of the spatial inversion, and thus yield a proper rotation. For this reason, improper rotations cannot form a subgroup, only a coset. Since the inversion matrix is proportional to the unit matrix, the result also implies that spatial inversion will commute with all symmetry elements.

In all the point groups with improper rotations, we shall thus always also have a rotational subgroup, like D_n in D_{nd} or D_{nh}, or T in T_d and T_h, etc. Moreover, this rotational subgroup is always a halving subgroup, i.e., its order is half the order of the full group. This can easily be demonstrated. Let H_{rot} be the rotational subgroup of G, and consider an improper symmetry element, $\hat{S}_i$, as coset generator. The coset $\hat{S}_i H_{\text{rot}}$ will contain only improper symmetry elements, and its order will be equal to $|H_{\text{rot}}|$.

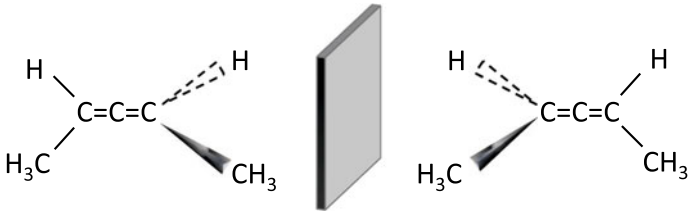


Fig. 3.14 Dimethyl substituted allene occurs in two chiral forms, which are the mirror image of each other, and cannot be superimposed; the point group of each form is C_2

Now is it possible that the group contains additional improper elements, outside this coset? Suppose we find such an element, say $\hat{S}_j$. Of course, the product $\hat{S}_i^{-1}\hat{S}_j$ is the combination of two improper elements, and thus must be a proper rotation, included in the rotational subgroup. Let us denote this element as $\hat{R}_z$. Hence, it follows that

$$\hat{S}_i\hat{R}_z = \hat{S}_i\hat{S}_i^{-1}\hat{S}_j = \hat{S}_j \tag{3.35}$$

This result confirms that S_j is included in the coset of $\hat{S}_i$, and thus implies that there is only one coset of improper rotations, covering half of the set of symmetry elements.

A group is a *direct product* of two subgroups, H_1 and H_2, if the operations of H_1 commute with the operations of H_2, and every operation of the group can be written uniquely as a product of an operation of H_1 and an operation of H_2. This may be denoted in general as

$$G = H_1 \times H_2 \tag{3.36}$$

This is certainly the case when a group is *centrosymmetric*, i.e., when it contains an inversion center. Since the inversion operation commutes with all operations, a centrosymmetric group can be written as the direct product: $C_i \times H_{\text{rot}}$, where $C_i = \{\hat{E}, \hat{\imath}\}$. Direct product groups are not limited to centrosymmetry, however. In the group D_{3h}, for example, the horizontal symmetry plane forms a separate conjugacy class, which means that it commutes with all the operations of the group. It thus generates a normal C_s subgroup. The full group D_{3h} can then be written as a direct product of this normal subgroup and the rotational subgroup:

$$D_{3h} = C_s \times D_3 \tag{3.37}$$

When the symmetry of a molecule is a purely rotational group, then the molecule does not coincide with its mirror image, and there will be two copies of it, which relate to each other as do right and left hands. This is illustrated in Fig. 3.14. Molecules with only rotational symmetry are *chiral*, meaning that the molecule and its reflection form optical antipodes. A synonym is *enantiomeric*, which literally means: on both sides of the mirror. By contrast, when a molecular point group contains any improper symmetry element, the molecule will be congruent to its mirror image,

and is *achiral*. Why does the absence of an improper symmetry element prevent the molecule's coinciding with its mirror image? Creating a mirror image of a structure involves a change of the parity of space. As we mentioned before, rotations and translations are not able to undo this change. Only an improper symmetry operation is able to revert the parity again in order to demonstrate the congruence between the molecule and its mirror image. Such symmetry elements are absent in molecules with only rotational point group symmetries, and, hence, they cannot be made to coincide with their mirror image. Molecular chirality is thus derived from an obvious and basic symmetry characteristic; nonetheless, it has far-reaching physical and chemical consequences. Chiral molecules are optically active, which means that the plane of polarization of linearly polarized light is rotated when passing through a medium with chiral molecules. The absorption coefficients of chiral molecules for left- and right-circularly polarized light are also different, giving rise to natural circular dichroism (CD) spectra. This will be illustrated in Sect. 6.8. The chemical consequences of chirality are of vital importance: living organisms are based on biochemical molecules with strict chirality.

3.9 Applications: Magnetic and Electric Fields

In many spectroscopic applications, external magnetic or electric fields are applied to a molecular sample. For magnetic fields, the perturbing influence of the field is known as the Zeeman effect. Electric fields give rise to the Stark effect. As we have seen, uniform magnetic and electric fields have symmetries $C_{\infty h}$ and $C_{\infty v}$, respectively. In this case, the symmetry group of the experiment will be constrained to these operations which leave the {molecule + field} combination invariant. It will thus consist only of symmetry elements that are common to both parts. These elements form the *intersection* of both symmetry groups. The elements of an intersection themselves form a group, which is the largest common subgroup of both symmetry groups. This can be written as follows:

$$\begin{aligned} \text{magnetic field} &: H = G \cap C_{\infty h} \\ \text{electric field} &: H = G \cap C_{\infty v} \end{aligned} \tag{3.38}$$

This intersection group will depend on the orientation of the field in the molecular frame. In Table 3.6, we work out an example of a tetrahedral molecule. The top row lists the symmetry elements of T_d. The fields can be oriented along several directions. The highest symmetry positions are along the fourfold or threefold axes. A lower symmetry position is within or perpendicular to a symmetry plane, or finally along

Table 3.6 Symmetries of a tetrahedral molecule in a uniform magnetic (B) or electric (E) field

	$\hat{E}$	$8\hat{C}_3$	$3\hat{C}_2$	$6\hat{S}_4$	$6\hat{\sigma}_d$	$T_d \cap C_{\infty h}$
$B \parallel \hat{S}_4$	$\hat{E}$		$\hat{C}_2$	$2\hat{S}_4$		S_4
$B \parallel \hat{C}_3$	$\hat{E}$	$2\hat{C}_3$				C_3
$B \perp \hat{\sigma}_d$	$\hat{E}$				$\hat{\sigma}_d$	C_s
						$T_d \cap C_{\infty v}$
$E \parallel \hat{S}_4$	$\hat{E}$		$\hat{C}_2$		$2\hat{\sigma}_d$	C_{2v}
$E \parallel \hat{C}_3$	$\hat{E}$	$2\hat{C}_3$			$3\hat{\sigma}_d$	C_{3v}
$E \in \hat{\sigma}_d$	$\hat{E}$				$\hat{\sigma}_d$	C_s

an arbitrary direction with no symmetry at all. In Appendix B, we list representative intersection groups for several point groups and orientations. Note that in the case of a magnetic field, the resulting intersection group is always abelian. This is of course a consequence of $C_{\infty h}$ being abelian.

3.10 Problems

3.1 The multiplication table of a set of elements is given below. Does this set form a group?

	A	**B**	**C**	**D**
A	C	D	A	B
B	D	C	B	A
C	A	B	C	D
D	B	A	D	C

3.2 The elements of the S_3 permutation group may be defined in an active sense, with symmetry elements that remain fixed, as follows: consider three boxes, labeled as 1, 2, and 3. A configuration is a distribution of the three atomic labels A, B, C over these three boxes. The starting configuration is A in 1, B in 2, and C in 3. Now a permutation operation (12) means: swap the contents of boxes 1 and 2. Likewise a 3-cycle such as (123) signifies: the content of box 1 is transferred to box 2, the content of box 2 goes to box 3, and the content of box 3 is moved to box 1. In this way, symmetry elements are associated to fixed permutation operators: $\hat{C}_3$ corresponds to (123), $\hat{\sigma}_1$ corresponds to (23), etc. Using this definition, construct the multiplication table for the permutational group and show it is isomorphic to C_{3v}.

3.3 Consider the set of cosets of a normal subgroup. Multiplication of two cosets $g_i H$ and $g_j H$ covers all possible products of type $g_i h_x g_j h_y$. Show that for the cosets of a normal subgroup multiplication of two cosets obeys closure, i.e., yields a coset. Also verify the other group properties. The group formed by the

cosets of a normal subgroup may thus be considered as the result of the division of a group by one of its normal subgroups.

3.4 Use molecular ball and stick models to construct examples of molecules that have a reflection plane as the only symmetry element. Similarly for a center of inversion, and for a twofold axis. In each case find the solution with the smallest number of atoms! Explain your reasoning. What is the smallest molecule with no symmetry at all?

3.5 The 2D analogue of a polyhedron is a polygon. In a regular polygon, all vertices, edges, and angles between adjacent edges are identical. A 2D plane can be tessellated in identical regular polygons, which then form a covering of the plane. In how many ways can this be performed?

3.6 Why is the order of a rotational axis of a polyhedral object always an integer?

3.7 Prove that a halving subgroup is always a normal subgroup.

3.8 Determine the point group of the structures in Fig. 3.15.

a. soccer ball
b. tennis ball
c. basketball
d. trefoil knot
e. rhombic dodecahedron
f. parallelepiped
g. pentadiamond
h. adamantane
i. cyclohexane, chair
j. cyclohexane, boat

3.9 Which one of the structures in Fig. 3.15 is chiral?

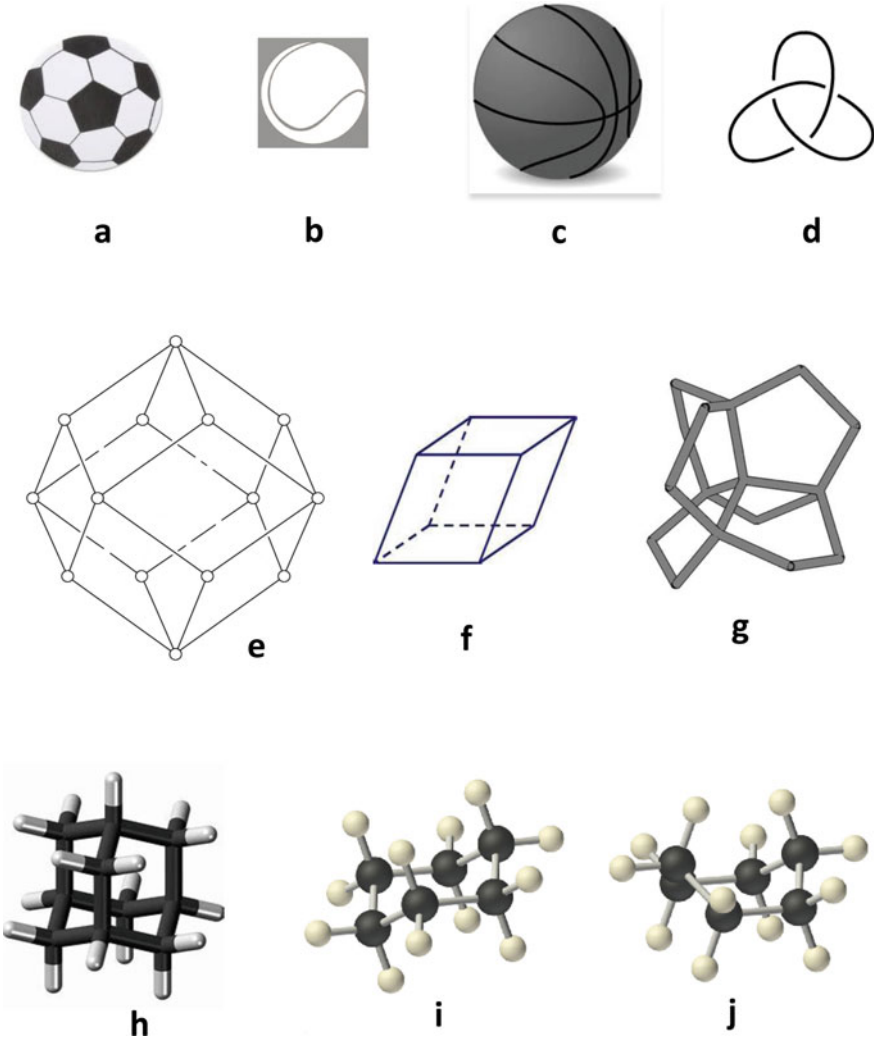


Fig. 3.15 Study Problem 3.8

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Chapter 4
Representations



Abstract Having made acquaintance with the basic properties of groups, we now turn our attention to the structure of the matrices which represent the group action in a function space. It turns out that there exists only a limited set of standard patterns. These are called the irreducible representations. They form the principal mathematical concept on which this monograph is based, and much care is devoted to acquire a gradual understanding of what this concept really means. Then the character theorem, matrix theorems, and projection operators are introduced. The concepts of subduction and induction relate the representations of subgroups and those of their parent group. The chapter also offers a detailed group-theoretical analysis of three chemical applications: the tetrahedral hybridization of carbon, the molecular vibrations of PF_5 and UF_6, and the electronic structure and magnetic properties of conjugated hydrocarbons, according to the Hückel and London models.

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4.1 Symmetry-Adapted Linear Combinations of Hydrogen Orbitals in Ammonia

From now on we shall no longer be working with the nuclei, but with electronic orbital functions that are anchored on the nuclei. As an example we take the $1s$ orbitals on the hydrogen atoms in ammonia. The $|1s_\mathrm{A}\rangle$ is defined in Eq. (4.1), where $\mathbf{R}_\mathrm{A}$ denotes the position vector of atom A with respect to the Cartesian origin; a_0 is the Bohr radius (0.529 Å) which is the atomic unit of length.

$$|1s_\mathrm{A}\rangle = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2} \exp\left(-\frac{|\mathbf{r}-\mathbf{R}_\mathrm{A}|}{a_0}\right) \tag{4.1}$$

Following Sect. 1.2, the transformation of this function by the threefold axis is given by

$$\hat{C}_3|1s_\mathrm{A}\rangle = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2} \exp\left(-\frac{\left|\left[\hat{C}_3^{-1}\mathbf{r}\right]-\mathbf{R}_\mathrm{A}\right|}{a_0}\right) \tag{4.2}$$

The distance between the electron at position $\hat{C}_3^{-1}\mathbf{r}$ and nucleus A is equal to the distance between the electron at position $\mathbf{r}$ and nucleus B. This can be established by working out the distances as a function of the Cartesian coordinates, but a straightforward demonstration is based on the fact that the distance does not change if we rotate *both* nuclei and electrons (Fig. 4.1).

$$\left|\left[\hat{C}_3^{-1}\mathbf{r}\right]-\mathbf{R}_\mathrm{A}\right| = \hat{\mathcal{Q}}_3\left|\left[\hat{C}_3^{-1}\mathbf{r}\right]-\mathbf{R}_\mathrm{A}\right| = \left|\left[\hat{\mathcal{Q}}_3\hat{C}_3^{-1}\mathbf{r}\right]-\hat{\mathcal{Q}}_3\mathbf{R}_\mathrm{A}\right| = |\mathbf{r}-\mathbf{R}_\mathrm{B}| \tag{4.3}$$

Here, we have denoted the bodily rotation of the entire molecule as $\hat{\mathcal{Q}}_3$, in order to indicate that its action also involves the nuclei, as opposed to $\hat{C}_3$, which is reserved for electrons. In the expression of Eq. (4.1), $\mathbf{R}_\mathrm{A}$ is replaced by $\mathbf{R}_\mathrm{B}$, or:

$$\hat{C}_3|1s_\mathrm{A}\rangle = |1s_\mathrm{B}\rangle \tag{4.4}$$

This confirms the active view, propagated from the beginning, as applied to the functions. We rotate the $|1s_\mathrm{A}\rangle$ orbital itself counterclockwise over 120°. The result is equal to the $|1s_\mathrm{B}\rangle$ orbital. We now put the three components of the function space together in a row vector:

$$|\mathbf{f}\rangle = \left(|1s_\mathrm{A}\rangle\ |1s_\mathrm{B}\rangle\ |1s_\mathrm{C}\rangle\right) \tag{4.5}$$

The action of the operator in the function space now reads as follows:

$$\hat{C}_3|\mathbf{f}\rangle = |\mathbf{f}\rangle\mathbb{D}(C_3) \tag{4.6}$$

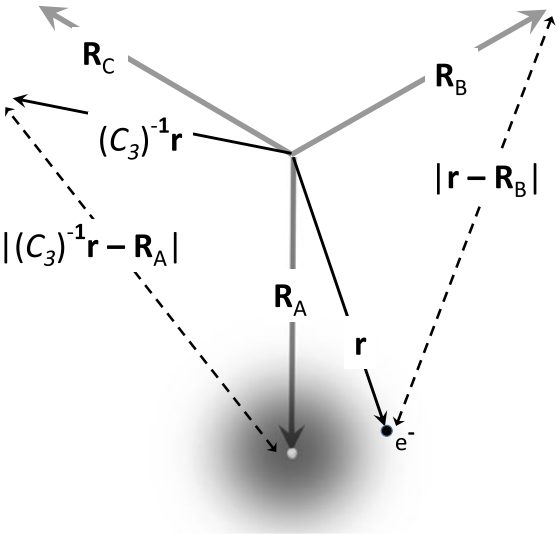


Fig. 4.1 Top view of the NH_3 structure; vector diagram of a threefold rotation of an electron in the $1s_A$ orbital

where the representation matrix is given by

$$\mathbb{D}(C_3) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \tag{4.7}$$

While the function space is clearly invariant, i.e., it transforms into itself under the rotation, the individual components are not: they undergo a cyclic permutation. Our objective is to find *symmetry-adapted linear combinations* (SALCs) that are invariant under the operator, except, possibly, for a phase factor. The combinations we are looking for are thus nothing other than eigenfunctions of the symmetry operator, in the same way as solutions of the Schrödinger equation are eigenfunctions of the Hamiltonian. Unlike the Hamiltonian eigenfunctions, however, which will usually consist of linear combinations in an infinite Hilbert space, the present exercise is carried out in a space of three functions only, since this space is already closed under the operator. We shall solve this symmetry eigenvalue problem in a purely algebraic way. Let $|\psi_m\rangle$ be a SALC:

$$|\psi_m\rangle = \sum_{\mathrm{X=A,B,C}} c_{\mathrm{X}} |1s_{\mathrm{X}}\rangle \tag{4.8}$$

which we shall again write as the product of a row vector and a column vector:

$$|\psi_m\rangle = \left(|1s_{\mathrm{A}}\rangle \; |1s_{\mathrm{B}}\rangle \; |1s_{\mathrm{C}}\rangle \right) \begin{pmatrix} c_{\mathrm{A}} \\ c_{\mathrm{B}} \\ c_{\mathrm{C}} \end{pmatrix} \tag{4.9}$$

The transformation of this function is then given by

$$\hat{C}_3|\psi_m\rangle = |\mathbf{f}\rangle \begin{pmatrix} 0\,0\,1 \\ 1\,0\,0 \\ 0\,1\,0 \end{pmatrix} \begin{pmatrix} c_\mathrm{A} \\ c_\mathrm{B} \\ c_\mathrm{C} \end{pmatrix} \tag{4.10}$$

We now require this function to be an eigenfunction of the threefold rotation operator with eigenvalue λ:

$$\hat{C}_3|\psi_m\rangle = \lambda|\psi_m\rangle \tag{4.11}$$

Combining Eqs. (4.10) and (4.11), we see that the function is an eigenfunction if the product of the $\mathbb{D}$ matrix with the column vector of the coefficients returns the coefficient column, multiplied by the eigenvalue λ, i.e.,

$$\begin{pmatrix} 0\,0\,1 \\ 1\,0\,0 \\ 0\,1\,0 \end{pmatrix} \begin{pmatrix} c_\mathrm{A} \\ c_\mathrm{B} \\ c_\mathrm{C} \end{pmatrix} = \lambda \begin{pmatrix} c_\mathrm{A} \\ c_\mathrm{B} \\ c_\mathrm{C} \end{pmatrix} \tag{4.12}$$

This equation can also be rewritten as

$$\begin{pmatrix} -\lambda & 0 & 1 \\ 1 & -\lambda & 0 \\ 0 & 1 & -\lambda \end{pmatrix} \begin{pmatrix} c_\mathrm{A} \\ c_\mathrm{B} \\ c_\mathrm{C} \end{pmatrix} = \mathbf{0} \tag{4.13}$$

Equation (4.13) forms a homogeneous system of equations in the three unknowns. It will have solutions only if the matrix preceding the column vector of the unknowns has determinant zero. This requirement is written as

$$|\mathbb{D}(C_3) - \lambda\mathbb{I}| = 0 \tag{4.14}$$

Here, the vertical bars denote the determinant. Equation (4.14) is called the *secular equation*. It has the form of a simple cubic equation in the eigenvalue λ.

$$-\lambda^3 + 1 = 0 \tag{4.15}$$

This equation is the Euler equation. It has three roots:

$$\lambda_m = \exp\left(\frac{2m\pi i}{3}\right) \tag{4.16}$$

where m can take the values $(-1, 0, +1)$. What we have just performed is a matrix diagonalization of the representation matrix. We obtain at once not one but three eigenvalues. The eigenfunction corresponding to a given root can now be found by introducing this λ value in the system of equations, Eq. (4.13). Since the system is homogeneous, the three unknown coefficients can be determined only up to a

constant factor. We find the absolute values of these vector coefficients by invoking a normalization condition that requires the vectors to be of unit length. The simplified normalization condition, neglecting overlap integrals, reads

$$|c_{\mathrm{A}}|^2 + |c_{\mathrm{B}}|^2 + |c_{\mathrm{C}}|^2 = 1 \tag{4.17}$$

In this way we obtain three SALCs, each characterized by a different eigenvalue for the symmetry operator:

$$\begin{aligned}
|\psi_0\rangle &= \frac{1}{\sqrt{3}}\left(|1s_{\mathrm{A}}\rangle + |1s_{\mathrm{B}}\rangle + |1s_{\mathrm{C}}\rangle\right) \\
|\psi_{+1}\rangle &= \frac{1}{\sqrt{3}}\left(|1s_{\mathrm{A}}\rangle + \overline{\epsilon}|1s_{\mathrm{B}}\rangle + \epsilon|1s_{\mathrm{C}}\rangle\right) \\
|\psi_{-1}\rangle &= \frac{1}{\sqrt{3}}\left(|1s_{\mathrm{A}}\rangle + \epsilon|1s_{\mathrm{B}}\rangle + \overline{\epsilon}|1s_{\mathrm{C}}\rangle\right)
\end{aligned} \tag{4.18}$$

where $\epsilon = \exp 2\pi i/3$. The set of corresponding eigenvalues is called the spectrum of the operator. It will be evident that this spectrum consists of the cubic roots of 1, since operating with $\hat{C}_3$ three times in succession is equivalent to applying the identity operator.

$$\hat{C}_3^3|\psi_m\rangle = \lambda^3|\psi_m\rangle = \hat{E}|\psi_m\rangle \tag{4.19}$$

For any operator, one thus can find eigenfunctions by simply diagonalizing the corresponding representation matrix. Our objective is more ambitious, however. We want to obtain functions that are not only adapted to a single symmetry element but to the group as a whole. This really amounts to finding SALCs for the set of group generators, since adaptation to the generators implies that the function is adapted to any combination of generators and, hence, to the whole group. So, in the case of C_{3v}, we have to examine the behavior of the $|\psi_m\rangle$ functions under a vertical symmetry plane as well, say $\hat{\sigma}_1$. This plane will leave $|1s_{\mathrm{A}}\rangle$ unchanged and will interchange $|1s_{\mathrm{B}}\rangle$ and $|1s_{\mathrm{C}}\rangle$. Its effect on the trigonal eigenfunctions is thus given by

$$\begin{aligned}
\hat{\sigma}_1|\psi_0\rangle &= |\psi_0\rangle \\
\hat{\sigma}_1|\psi_{+1}\rangle &= |\psi_{-1}\rangle \\
\hat{\sigma}_1|\psi_{-1}\rangle &= |\psi_{+1}\rangle
\end{aligned} \tag{4.20}$$

What does this result tell us? The SALC $|\psi_0\rangle$ is simultaneously an eigenfunction of both $\hat{C}_3$ and $\hat{\sigma}_1$; hence, it forms in itself a one-dimensional function space that is completely adapted to the full group. This symmetry characteristic will be denoted by the totally symmetric representation A_1. However, the other two SALCs are transformed into each other. We can easily turn them into eigenfunctions of $\hat{\sigma}_1$, and in this way obtain alternative eigenfunctions, one which is symmetric under reflection, and one which is anti-symmetric. These will be labeled as x and y, respectively, since their symmetry under reflection mimics the symmetries of p_x and p_y:

$$|\psi_x\rangle = \frac{1}{\sqrt{2}}\left(|\psi_{+1}\rangle + |\psi_{-1}\rangle\right) = \frac{1}{\sqrt{6}}\left(2|1s_A\rangle - |1s_B\rangle - |1s_C\rangle\right)$$
$$|\psi_y\rangle = \frac{i}{\sqrt{2}}\left(|\psi_{+1}\rangle - |\psi_{-1}\rangle\right) = \frac{1}{\sqrt{2}}\left(|1s_B\rangle - |1s_C\rangle\right) \tag{4.21}$$

A schematic drawing of these eigenfunctions is shown in Fig. 4.2. The downside of this symmetry adaptation is that it has destroyed the diagonalization along the trigonal axis. Indeed, one has

$$\hat{R}\left(|\psi_x\rangle\ |\psi_y\rangle\right) = \left(|\psi_x\rangle\ |\psi_y\rangle\right)\begin{pmatrix}\cos(2\pi/3) & -\sin(2\pi/3)\\ \sin(2\pi/3) & \cos(2\pi/3)\end{pmatrix} \tag{4.22}$$

Hence, it is impossible to resolve the function space formed by $|\psi_{\pm 1}\rangle$ into simultaneous eigenfunctions of both generators. The action of the symmetry group ties these functions together into a two-dimensional space which is thus *irreducible*. This symmetry characteristic is denoted by the degenerate irreducible representation E. What we have learned form this simple example is the following. The construction of SALCs of a symmetry group is based on simultaneous diagonalization of the representation matrices of the group generators. This will resolve the function space into separate blocks, which may consist of one function, or which may form a subspace

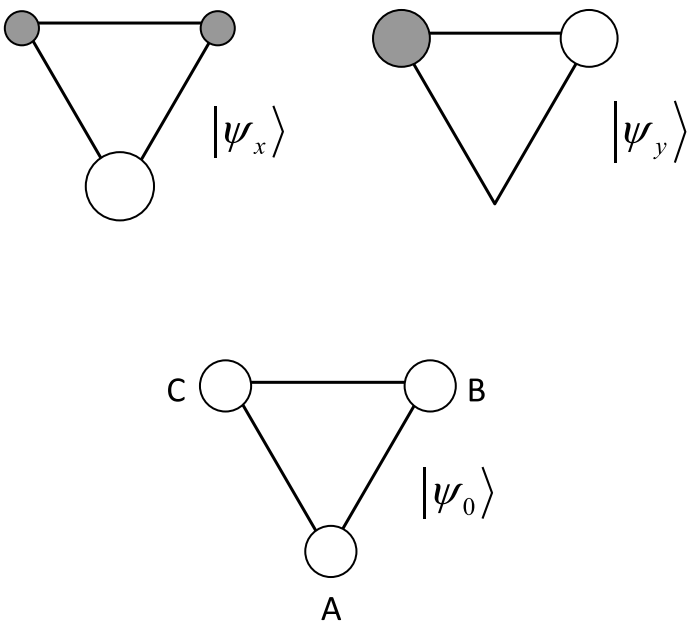


Fig. 4.2 Hydrogen SALCs in ammonia. The size of the circles is proportional to the eigenfunction coefficients. $|\psi_0\rangle$ transforms as the totally symmetric irrep A_1; $|\psi_x\rangle$ and $|\psi_y\rangle$ are components of the degenerate E representation

that cannot be further reduced. The results are functions that transform as irreducible representations (irreps). Why is such a resolution important? The irreducible subspaces, into which the function space has been separated, are invariant under the actions of the actual symmetry group. This means that there are no operators that will send SALCs with one irreducible symmetry over into SALCs with a different irreducible symmetry. This will also imply that the eigenenergies associated with these irreducible blocks will not be related. We will return to this in much more detail in Sect. 5.2. The algebraic treatment also provides an insight into the meaning of degeneracy. The two components of the E irrep are locked in the same function space, because it was not possible to diagonalize the representation matrices for both generators simultaneously. If two operators commute, it is always possible to find solutions that are simultaneous eigenfunctions of both. However, the two generators of C_{3v} do not commute, as this group is not abelian. The fact that the corresponding representation matrices also do not commute explains why it is impossible to block-diagonalize the E irrep.

4.2 Character Theorems

When examining a function space from a symmetry point of view, we note that there are two basic questions to be asked:

1. What are the symmetry ingredients of the function space; in other words, which irreps describe the symmetry of this space?
2. What do the corresponding SALCs look like?

The present section on characters deals with the first question and provides an elegant description of the symmetries of function spaces. In the subsequent sections, matrix theorems are used for the construction of projection operators that will carry out the job of obtaining the suitable SALCs. The intuitive algebraic approach that we have demonstrated in the previous section has been formalized by Schur [1], Frobenius [2] and others into a fully fledged character theory, which reveals which irreps a given group can sustain, and how to analyze the irreducible contents of a given function space. Characters are nothing other than the trace (that is the sum of the diagonal elements) of a representation matrix. They will be represented as $\chi(R)$:

$$\chi(R) = \sum_i D_{ii}(R) \tag{4.23}$$

If the functional basis of a representation is transformed by a unitary transformation, the trace does not change, as can easily be demonstrated. Define $|\mathbf{f}'\rangle = |\mathbf{f}\rangle\mathbb{U}$. Then the corresponding representation matrices, $\mathbb{D}'(R)$, also undergo a unitary transformation:

$$\begin{aligned}\hat{R}\,|\mathbf{f}'\rangle &= \hat{R}\,|\mathbf{f}\rangle\mathbb{U} = |\mathbf{f}\rangle\mathbb{D}(R)\mathbb{U} \\ &= |\mathbf{f}'\rangle\mathbb{U}^{-1}\mathbb{D}(R)\mathbb{U} \\ &= |\mathbf{f}'\rangle\mathbb{D}'(R)\end{aligned} \tag{4.24}$$

from which it follows:

$$\mathbb{D}'(R) = \mathbb{U}^{-1}\mathbb{D}(R)\mathbb{U} \tag{4.25}$$

or, for $\mathbb{U}$ unitary:

$$\begin{aligned}D'_{ij}(R) &= \sum_{kl} \overline{U}^{T}_{ik} D_{kl}(R) U_{lj} \\ &= \sum_{kl} \overline{U}_{ki} D_{kl}(R) U_{lj}\end{aligned} \tag{4.26}$$

The invariance of the character then follows from the orthogonality of the rows of the unitary matrix:

$$\begin{aligned}\chi'(R) &= \sum_{i} D'_{ii}(R) \\ &= \sum_{kl} D_{kl} \left(\sum_{i} \overline{U}_{ki} U_{li} \right) \\ &= \sum_{kl} D_{kl}(R)\delta_{kl} \\ &= \sum_{k} D_{kk}(R) \\ &= \chi(R)\end{aligned} \tag{4.27}$$

Hence, sets of characters literally characterize representations, since they are immune to the effects of unitary transformations, such as occur in Eq. (4.21) between the complex functions $|\psi_{+1}\rangle$, $|\psi_{-1}\rangle$ and the real functions $|\psi_x\rangle$, $|\psi_y\rangle$. The characters for the irreps are brought together in a character table. Here, the conjugacy class concept comes in very handy. Indeed, since all elements belonging to the same class are similarity transforms, their representation matrices will be unitary transforms, and, hence, will all have the same character. We can thus group elements together in classes. In Table 4.1 we show the character table for C_{3v} as can be obtained by algebraic techniques such as the one we used in the previous section. We recognize at once the characters for the totally symmetric A_1 and the twofold degenerate E irrep. In addition, there is another one-dimensional irrep, A_2, which is symmetric under the threefold axis and anti-symmetric under the reflection planes.

Table 4.1 Character table for the group C_{3v}, and reducible characters of the hydrogen 1s functions $\chi(1s)$ and hydrogen bends $\chi(\Delta\phi)$. The $\chi(1s)$ row is equal to the sum of the A_1 and E rows, and the $\chi(\Delta\phi)$ row is equal to the sum of the A_2 and E rows

C_{3v}	$\hat{E}$	$2\hat{C}_3$	$3\hat{\sigma}_v$	$\langle\chi\|\chi\rangle$
A_1	1	1	1	6
A_2	1	1	−1	6
E	2	−1	0	6
$\chi(1s)$	3	0	1	12
$\chi(\Delta\phi)$	3	0	−1	12

Let us denote an irrep as Γ_i, and the string of characters, arranged in a row over the full group, in a Dirac form as $|\chi^{\Gamma_i}\rangle$. The norm of this string[1] will then be denoted as a bracket, i.e.,

$$\langle\chi^{\Gamma_i}|\chi^{\Gamma_i}\rangle = \sum_{R\in G} \overline{\chi}^{\Gamma_i}(R)\chi^{\Gamma_i}(R) \tag{4.28}$$

Since the matrices are unitary we could also replace the complex-conjugate character by the character of the inverse element:

$$\overline{\chi}^{\Gamma_i}(R) = \chi^{\Gamma_i}(R^{-1}) \tag{4.29}$$

The character strings obey the following character theorem:

Theorem 4.1 *The norm of the character string is equal to the order of the group if and only if the characters refer to an irreducible representation. The scalar product of two character strings of different irreps is equal to zero.*

This theorem can be expressed as follows:

$$\langle\chi^{\Gamma_i}|\chi^{\Gamma_j}\rangle = \sum_{R\in G} \overline{\chi}^{\Gamma_i}(R)\chi^{\Gamma_j}(R) = \delta_{ij}|G| \tag{4.30}$$

where Γ_i and Γ_j refer to irreducible representations, and $|G|$ is the order of the group. This theorem provides an elegant and simple solution for determining the irrep content of a function space. One first determines the characters of the representation matrices, $\chi(R)$. They will be equal to the sums of the traces of the individual irreducible symmetry blocks. This can be expressed as follows:

$$\chi(R) = \sum_i c_i \chi^{\Gamma_i}(R) \tag{4.31}$$

[1] Since elements in the same class have the same character, we can also simplify the expression to a summation over all classes, provided that we then multiply each term by the number of elements in the class under consideration.

Here, the important quantities are the c_i coefficients. These are integers that tell how many times a given irrep Γ_i is contained in the function space. They can easily be calculated by using the character theorems. To find a given coefficient, say c_k, all one has to do is to evaluate the scalar product of a the corresponding irreducible character, $\Gamma_k(R)$, and the reducible character, $\chi(R)$, and then divide by the group order.

$$c_k = \frac{1}{|G|}\langle \chi^{\Gamma_k} | \chi \rangle = \frac{1}{|G|} \sum_i c_i \langle \chi^{\Gamma_k} | \chi^{\Gamma_i} \rangle \tag{4.32}$$

It is also clear that the norm of the reducible character will be a multiple of the group order, since:

$$\langle \chi | \chi \rangle = \sum_{ij} c_i c_j \langle \chi^{\Gamma_i} | \chi^{\Gamma_j} \rangle = \sum_i c_i^2 |G| \tag{4.33}$$

This procedure may seem quite complicated but it is in fact very simple. Let us demonstrate this for the previous example, the set of the three 1*s*-orbitals on the hydrogens in NH_3. We first determine the reducible character of this set (see $\chi(1s)$ in Table 4.1). We do not have to do this for all six elements of C_{3v} but only for one representative of each class, since conjugate elements have the same characters. For the unit element, the representation matrix is of course the 3×3 unit matrix, and its trace is equal to three, being the dimension of the set. For the other elements, we do not need to know the full matrix representation; indeed, *we need only the elements on the diagonal.* Now a diagonal entry in a representation matrix can differ from zero only if a component function is turned into itself, or at least into a fraction of itself. The threefold axis moves all three 1*s*-orbitals around and hence will have zeros only on the diagonal, thus: $\chi(C_3) = 0$ (see Eq. (4.7)). For the reflection planes one element is stabilized while the other two are interchanged, hence, there is only one non-zero diagonal element, which will be equal to $+1$ since the orbital is not affected; hence, $\chi(\sigma_1) = +1$. The norm of the character string is given by

$$\langle \chi | \chi \rangle = 3^2 + 2 \times 0 + 3 \times 1^2 = 12 = 2|C_{3v}| \tag{4.34}$$

Since the norm is twice the order of the group, we know for certain that the symmetry of the function space is reducible. In fact, we also predict that it will reduce to two irreps, which each occur once, since the sum of the squares of the multiplicities in Eq. (4.33) is equal to 2. These coefficients can now be obtained by calculating the character brackets, which yields the previous result: $c_{A_1} = 1, c_{A_2} = 0, c_E = 1$. From Table 4.1, one can also verify that the sum of χ^{A_1} and χ^E equals $\chi(1s)$. This is sometimes also expressed in a formal way as

$$\Gamma = A_1 + E \tag{4.35}$$

As a further example we can take the set of the two in-plane $2p$-orbitals on nitrogen, p_x, p_y. Here, the character for the threefold axis is

$$\chi^{p_x,p_y}(C_3) = 2\cos(2\pi/3) = -1 \tag{4.36}$$

Under the $\hat{\sigma}_1$ plane of symmetry, one of the orbitals is anti-symmetric, while the other is symmetric, and so the trace of the matrix will vanish. The characters of this p-set are thus precisely the ones for the E irrep, and we can therefore state that the set $\{p_x, p_y\}$ transforms irreducibly as the E representation. As a final example let us take an alternative set, consisting of three counterclockwise displacement vectors of the hydrogen atoms, along a tangent to the (imaginery) circumscribed circle through the hydrogens, as indicated in Fig. 4.3. In this example the function space is the set of three bendings: $\Delta\phi_A$, $\Delta\phi_B$, $\Delta\phi_C$. We must realize that in this case the function on which we are operating is actually the distortion itself, and to rotate this distortion in an active sense means to take the distortion vector and displace it to the next nucleus. Hence, nuclei are left immobile: only the distortions are moved. One could easily reconcile this view with the previous orbital rotations by thinking of the distortions as little tangent p-orbitals, with positive and negative lobes corresponding to the head and tail of the vectors, respectively. Hence as an example one has

$$\hat{C}_3\Delta\phi_A = \Delta\phi_B \tag{4.37}$$

Since all distortions are cyclically permuted the character under this generator will be zero. On the other hand, for the reflection plane, one has

$$\begin{aligned}
\hat{\sigma}_1\Delta\phi_A &= -\Delta\phi_A \\
\hat{\sigma}_1\Delta\phi_B &= -\Delta\phi_C \\
\hat{\sigma}_1\Delta\phi_C &= -\Delta\phi_B
\end{aligned} \tag{4.38}$$

In this case, the only contribution to the trace comes from the distortion of A, which is mapped onto minus itself; hence, $\chi(\sigma_1) = -1$. The resulting character string also has norm 12, but in this case it decomposes into $A_2 + E$. As shown in Fig. 4.3, the A_2 component is the sum of the three distortions, which corresponds to a bodily rotation of the molecule around its vertical axis. The E components are given by

$$\begin{aligned}
Q_{E_x} &= \frac{1}{\sqrt{2}}(\Delta\phi_C - \Delta\phi_B) \\
Q_{E_y} &= \frac{1}{\sqrt{6}}(2\Delta\phi_A - \Delta\phi_B - \Delta\phi_C)
\end{aligned} \tag{4.39}$$

Perhaps the most surprising result is that the symmetry eigenfunctions of the C_{3v} group can be only of three different types: A_1, A_2, and E. In fact, this is another result derived by Schur.

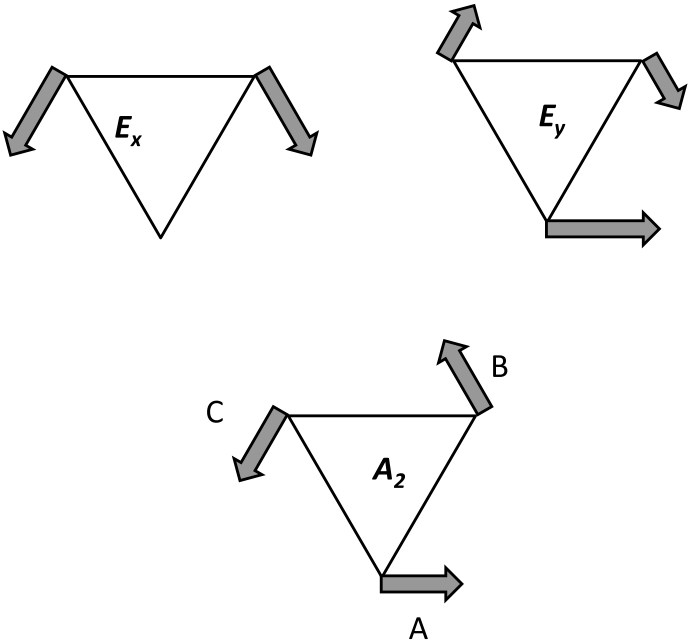


Fig. 4.3 Symmetry coordinates for the in-plane bending of the hydrogen atoms in ammonia. The length of the arrows is proportional to the SALC coefficients

Theorem 4.2 *The number of irreps in a group is equal to the number of conjugacy classes.*

The fact that there are only a few canonical ways of representing a symmetry group can be rationalized in a general way as follows. Consider a molecule with symmetry group G and a function that is localized on an arbitrary point in the molecule, i.e., a point which is not lying on an axis of symmetry, nor in a reflection plane, nor in the inversion center or the center of a rotation-reflection axis. We shall denote this function as $|f_E\rangle$ since it is stabilized by the unit element, and only by the unit element. Any other element will move this function to some other point that is a copy of the original one. This we write as

$$\hat{R}\,|f_E\rangle = |f_R\rangle \tag{4.40}$$

In this way the entire group will generate a set of $|G|$ functions, which will all be different. Indeed, suppose that, for $\hat{R} \neq \hat{S}$, the two corresponding functions would be the same, then the product operation $\hat{S}^{-1}\hat{R}$ would map $|f_E\rangle$ onto itself. This would contradict the assumption that $|f_E\rangle$ be stabilized only by the unit element. Furthermore, the closure of the group also guarantees that this set of functions forms an invariant function space. This function space transforms according to a reducible representation which is called the *regular* representation, Γ_{reg}. This representation

describes the most general function basis that one can consider, since it is based on functions that have no symmetry whatsoever. Now let us determine the symmetry ingredients of this space, using the standard character procedure. The character of the regular representation will be equal to $|G|$ for the unit element, and zero for all other elements since none of the functions is stabilized

$$\chi^{\Gamma_{reg}}(R) = |G|\delta_{ER} \tag{4.41}$$

Inserting this result into the expression for the multiplicity coefficients yields

$$c_k = \frac{1}{|G|}\langle\chi^{\Gamma_k}|\chi^{\Gamma_{reg}}\rangle = \overline{\chi}^{\Gamma_k}(E) = \dim(\Gamma_k) \tag{4.42}$$

Hence every irrep occurs as many times as its own dimension. The sum of all these irreducible blocks must yield the regular representation. Thus one has

$$\sum_k c_k \dim(\Gamma_k) = \sum_k [\dim(\Gamma_k)]^2 = |G| \tag{4.43}$$

This result tells us that, in a group of order $|G|$, there are exactly $|G|$ independent vectors. Rewriting these vectors in the form of SALCs will exhaust all possible symmetries that can be realized in this group. If a group is abelian, every class is a singleton, and hence the number of classes will be equal to the order of the group. In this case, Eq. (4.43) can be fulfilled only if all irreps are one-dimensional. Hence, in an abelian group all irreps are one-dimensional.

4.3 Character Tables

In Appendix A, we reproduce the character tables for the point groups, following the standard form which was introduced by Mulliken [3]. The top row of the table lists the conjugacy classes. In some cases the designations of the symmetry operations could be ambiguous, and additional labels are added, such as h, v, and d for *horizontal*, *vertical*, and *dihedral*, respectively (see also e.g., Fig. 3.10). In the final column of the tables we list some simple functions, which transform according to the corresponding irreps. Irreps are denoted by letters that are related to their degeneracy. A and B stand for one-dimensional irreps, which are symmetric or anti-symmetric with respect to some principal symmetry element. E and T are used for two- and three-dimensional irreps, respectively. Sometimes in physics textbooks T is replaced by F. This alphabetical order is then continued for the fourfold- and fivefold-degenerate irreps in the icosahedral symmetry, which are denoted as G and H. Further subscripts are added to distinguish symmetry characteristics with respect to secondary symmetry elements. Best known are the g and u subscripts which distinguish between even (*gerade*) and odd (*ungerade*) symmetries with respect to spatial inversion. Primes

or double primes are used to distinguish symmetric versus anti-symmetric behavior with respect to a horizontal symmetry plane in groups such as $D_{(2n+1)h}$ or $C_{(2n+1)h}$. In addition, numerical indices can appear as subscripts, such as in A_1, A_2 in the C_{3v} group. It should be clear that this labeling is somewhat ad hoc, and one should consult the actual tables in order to find out the precise meaning of the symbols used.

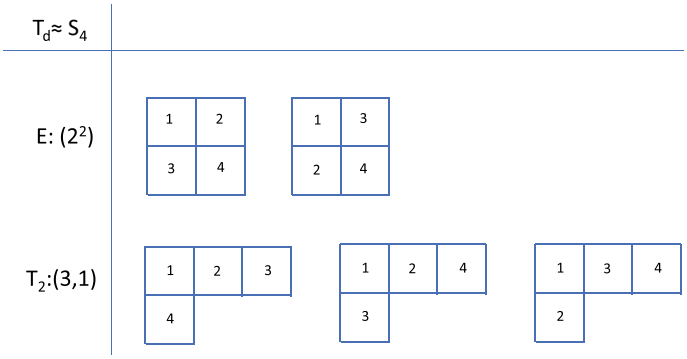
Some point groups, *viz.* the cyclic groups C_n, $C_{(2n+1)h}$, S_{2n} and also T and T_h, have irreps with complex characters. In these cases, for an irrep Γ_k with complex characters, there will always be a complementary irrep with a complex-conjugate character string, which is denoted as $\overline{\Gamma}_k$. Hence one has

$$\chi^{\overline{\Gamma}_k}(R) = \overline{\chi}^{\Gamma_k}(R) \tag{4.44}$$

Note that if Γ_k fulfills the character theorems, then $\overline{\Gamma}_k$ will also, and hence it, too, must be irreducible. The two irreps are said to be complex conjugates. They are orthogonal to each other, hence there is no point group operation that can turn a function belonging to Γ_k into a function belonging to $\overline{\Gamma}_k$. For this reason, we also should denote them by two separate labels. However, in the absence of external magnetic fields, symmetry is not restricted to spatial symmetry, but also includes time-reversal symmetry. As we have seen in Sect. 2.4, this symmetry will precisely turn functions into their complex conjugates, and thus also interchange the corresponding conjugate irreps. Complex-conjugate irreps thus remain degenerate under time reversal, and for this reason, they are usually indicated by means of a brace.

Appendix A also provides the character tables for the symmetric groups of permutations of three, four, and five elements. Permutation operators are grouped in classes, characterized by a common cycle index. The cycle index corresponds to the lengths of the permutational cycles. As an example the index $(1)^2(3)$ in S_5 refers to a permutation of two identities (1-cycles) and one 3-cycle. The sum of the individual cycle lengths for S_n equals n. Also the number of classes must be equal to all possible sums of n elements. The irreps of the symmetric groups are denoted as partitions of n elements and refer to Young tableaux [4].

Young Tableaux
A Young tableau is a practical graphical way to denote the partition of a permutation. For the group S_n, n building blocks are arranged in rows, such that a subsequent row can never be larger than a foregoing one. The dimension of an irrep equals the number of ways in which the numbers $1, \ldots, n$ can be inserted in the tableau in such a way that the numbers in the tableau strictly increase from left to right, and from top to bottom.



The diagram shows the tableaux for the partitions (2^2) and (3, 1) for the group S_4, which correspond to the irreps E and T_2 in the isomorphic group T_d. The number of allowed distributions of the numbers 1 to 4 over the tableaux matches the respective degeneracies 2 and 3 of the irreps.

4.4 Matrix Theorem

Determining the symmetry contents of a function space is only a first step. We would also like to know what are the SALCs that correspond to the different irreps. To carry out this task we have to work with the representation matrices themselves. In the group C_{3v} the matrices for the one-dimensional irreps A_1 and A_2 are trivial, since these are simply equal to the corresponding characters. For the E irrep, we need to determine the generator matrices and perform the proper multiplications in order to generate all $\mathbb{D}^E(R)$ for the whole group. These matrices are already available from Table 3.2 for the standard basis of the p_x, p_y orbitals. An important theorem, known as the *Great Orthogonality Theorem* (GOT), is due to Schur.

Theorem 4.3 *Let Ω and Ω' be two irreducible representations of a group G, and consider vectors formed by taking elements $\{ij\}$ and $\{kl\}$ from the respective representation matrices for every element of the group, then these vectors will be orthogonal to each other, and their squared norm is equal to the order of the group, divided by the dimension of the irrep:*

$$\sum_{R\in G} \overline{D}^{\Omega}_{ij}(R) D^{\Omega'}_{kl}(R) = \frac{|G|}{\dim(\Omega)} \delta_{\Omega,\Omega'} \delta_{ik} \delta_{jl} \tag{4.45}$$

The theorem thus proceeds as follows: take a given entry ij in the representation matrix of the irrep Ω for every R and order these elements to form a vector of length $|G|$. Do the same with another entry, kl, for a different representation, Ω', and also

Table 4.2 Complete set of matrix element strings for the group C_{3v}

C_{3v}	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_1$	$\hat{\sigma}_2$	$\hat{\sigma}_3$	$\langle D_{ij}^{\Omega} \vert D_{ij}^{\Omega} \rangle$
A_1	1	1	1	1	1	1	6
A_2	1	1	1	-1	-1	-1	6
E_{11}	1	$-\frac{1}{2}$	$-\frac{1}{2}$	1	$-\frac{1}{2}$	$-\frac{1}{2}$	3
E_{21}	0	$+\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{2}$	0	$-\frac{\sqrt{3}}{2}$	$+\frac{\sqrt{3}}{2}$	3
E_{12}	0	$-\frac{\sqrt{3}}{2}$	$+\frac{\sqrt{3}}{2}$	0	$-\frac{\sqrt{3}}{2}$	$+\frac{\sqrt{3}}{2}$	3
E_{22}	1	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$+\frac{1}{2}$	$+\frac{1}{2}$	3

arrange these to form a vector. Then take the scalar product of these two vectors, bearing in mind that, in this process, the complex conjugate of one of them should be taken (which one does not matter since the scalar product will always be real). The theorem states that this scalar product will be zero unless the same irrep is taken, and in this irrep the same row and column index are selected. In that case, the scalar product yields the norm of the vector which is equal to $|G|/\dim(\Omega)$.

Let us apply this to C_{3v}. For this group the total number of $\{\Omega, i, j\}$ combinations that can be formed according to the GOT procedure is equal to 6. These are listed in Table 4.2, based on the matrices in Table 3.2. In general, this number will always be equal to the order of the group, since, for every irrep, the number of $\{i, j\}$ combinations is equal to the squared dimension of that irrep, and, according to Eq. (4.43), the sum of these squares is equal to $|G|$. The strings in Table 4.2 thus form a set of six linearly independent vectors. This is in accord with our earlier finding that the set of arbitrary functions that form the most general function space for a group has dimension $|G|$. The GOT thus offers the complete list of coefficients, from which SALCs may be constructed. How this can be done is shown in the next section. Note that the trace theorem in the previous section is a direct consequence of the GOT which is obtained by taking diagonal matrix entries ii and kk and summing over i and k.

$$\begin{aligned}
\langle \chi^{\Omega} | \chi^{\Omega'} \rangle &= \sum_{i,k} \sum_{R \in G} \overline{D}_{ii}^{\Omega}(R) D_{kk}^{\Omega'}(R) \\
&= \frac{|G|}{\dim(\Omega)} \delta_{\Omega,\Omega'} \sum_{ik} \delta_{ik} \\
&= \frac{|G|}{\dim(\Omega)} \delta_{\Omega,\Omega'} \dim(\Omega) \\
&= \delta_{\Omega,\Omega'} |G| \qquad (4.46)
\end{aligned}$$

4.5 Projection Operators

We recapitulate what we have so far: a group G has been identified and a function space $|\mathbf{f}\rangle$ was constructed which is invariant under the action of the group. Next, the characters were determined for each conjugacy class and arranged in a character string, $|\chi\rangle$, which was mapped onto the irreducible characters in the table. Non-zero brackets determined which irreps are present in the function space. Now, the final step is to carry out the actual symmetry adaptation and to obtain the resulting SALCs, say $|\Phi_i^\Omega\rangle$. The SALCs are characterized by two indices: the upper index, Ω which stands for the irrep, and the lower index, i. This latter index is sometimes called the subrepresentation and also contains a specific piece of information. It determines the component of the irrep, i.e., it refers to a particular column in the irreducible representation matrix that describes the transformation of this component:

$$\hat{R}|\Phi_i^\Omega\rangle = \sum_j |\Phi_j^\Omega\rangle D_{ji}^\Omega(R) \tag{4.47}$$

This symbol of course makes sense only if we do not limit ourselves to the characters, but also determine the representation matrices for all degenerate irreps of the group. These will depend on the choice of a particular canonical basis set. Tabular material containing suitable sets of irrep matrices is rather sparse. Strategies to develop practical basis sets will be presented later in Sect. 5.3 and Appendix D. Now we construct a projector, $\hat{P}$, based on the available matrices:

$$\hat{P}_{kl}^{\Omega'} = \frac{\dim(\Omega')}{|G|} \sum_{R\in G} \overline{D}_{kl}^{\Omega'}(R)\hat{R} \tag{4.48}$$

Let us apply this projector onto the SALC $|\Phi_i^\Omega\rangle$. This requires the combination of Eqs. (4.47) and (4.48) and exploits the full GOT potential:

$$\begin{aligned}
\hat{P}_{kl}^{\Omega'}|\Phi_i^\Omega\rangle &= \frac{\dim(\Omega')}{|G|} \sum_{R\in G}\sum_j \overline{D}_{kl}^{\Omega'}(R) D_{ji}^\Omega(R)|\Phi_j^\Omega\rangle \\
&= \sum_j \delta_{\Omega',\Omega}\delta_{kj}\delta_{li}|\Phi_j^\Omega\rangle \\
&= |\Phi_k^{\Omega'}\rangle\delta_{\Omega',\Omega}\delta_{li}
\end{aligned} \tag{4.49}$$

The action of the projector entails a twofold selection, both at the level of the representation and of the subrepresentation, indicated by the two Kronecker deltas. First, it compares the irreps of the operator and of the SALC. If they do not match, then the SALC is simply destroyed. Second, the δ_{li} selection rule comes into play—under the 'protection', as it were, of the first Kronecker delta, which assures that the second selection rule will matter only when we are already inside the same irrep. This second rule compares the l index of the projection operator with the i index of the target

and will annihilate the target unless they are the same; it therefore selectively picks out the SALC which transforms exactly as the lth component of the Ω' irrep. Third, instead of delivering as result this particular component, the projector also has a built-in ability to act as a ladder operator and turn the lth component so obtained into a kth component. If we do not want this ladder aspect, we simply use the diagonal projection operator with $k = l$. Let us illustrate this for the Q_{Ex} hydrogen bending mode in Fig. 4.3. If we want to obtain from this the Q_{Ey} component, we should use a projection operator that recognizes the x and replaces it by y; hence, it must belong to the E irrep, its row index should be 2, and its column index 1. This indeed generates the y component of Eq. (4.39).

$$\begin{aligned}\hat{P}^E_{21} Q_{E_x} &= \frac{1}{3}\frac{\sqrt{3}}{2}\left[\hat{C}_3 - \hat{C}_3^2 - \hat{\sigma}_2 + \hat{\sigma}_3\right] \frac{1}{\sqrt{2}}(\Delta\phi_C - \Delta\phi_\mathrm{B}) \\ &= \frac{1}{2\sqrt{6}}[(\Delta\phi_\mathrm{A} - \Delta\phi_C) - (\Delta\phi_\mathrm{B} - \Delta\phi_\mathrm{A}) \\ &\quad - (-\Delta\phi_\mathrm{A} + \Delta\phi_\mathrm{B}) + (-\Delta\phi_C + \Delta\phi_\mathrm{A})] \\ &= \frac{1}{\sqrt{6}}(2\Delta\phi_\mathrm{A} - \Delta\phi_\mathrm{B} - \Delta\phi_C) \\ &= Q_{E_y}\end{aligned} \tag{4.50}$$

For a given irrep, the number of projectors that can be constructed is equal to the number of all possible $\{k, l\}$ combinations, which equals $\dim(\Omega')^2$. By varying the row index, k, one can obtain all the components of the invariance space of a given irrep. This is demonstrated by acting on the operators with an element $\hat{S}$.

$$\begin{aligned}\hat{S}\hat{P}^{\Omega'}_{kl} &= \frac{\dim(\Omega')}{|G|}\sum_{R\in G}\sum_j \overline{D}^{\Omega'}_{kl}(R)\hat{S}\hat{R} \\ &= \frac{\dim(\Omega')}{|G|}\sum_{T\in G}\overline{D}^{\Omega'}_{kl}(S^{-1}T)\hat{T} \\ &= \frac{\dim(\Omega')}{|G|}\sum_{T\in G}\left(\sum_m \overline{D}^{\Omega'}_{km}(S^{-1})\overline{D}^{\Omega'}_{ml}(T)\right)\hat{T} \\ &= \frac{\dim(\Omega')}{|G|}\sum_m D^{\Omega'}_{mk}(S)\left(\sum_{T\in G}\overline{D}^{\Omega'}_{ml}(T)\hat{T}\right) \\ &= \sum_m D^{\Omega'}_{mk}(S)\hat{P}^{\Omega'}_{ml}\end{aligned} \tag{4.51}$$

In this derivation we have used the substitution $\hat{S}\hat{R} = \hat{T}$. The result shows that the set of projectors with fixed index l forms a complete basis set for the Ω' irrep. On the other hand, by changing the column index, l, we have access to different sets of SALCs with the same symmetry. This applies only when the function space has

multiplicities, c_Γ, greater than one. This can be illustrated for the $\{|f_R\rangle\}$ function space, transforming as the regular representation, which has the maximal degree of freedom. Two projectors with the same k index, but different l indices, will project out two functions that are linearly independent, as the following overlap calculation shows

$$\begin{aligned}
\langle \hat{P}^{\Omega}_{kl} f_E | \hat{P}^{\Omega}_{kl'} f_E \rangle &= \frac{\dim(\Omega)^2}{|G|^2} \langle \sum_R \overline{D}^{\Omega}_{kl}(R) \hat{R} f_E | \sum_S \overline{D}^{\Omega}_{kl'}(S) \hat{S} f_E \rangle \\
&= \frac{\dim(\Omega)^2}{|G|^2} \sum_{R,S} D^{\Omega}_{kl}(R) \overline{D}^{\Omega}_{kl'}(S) \langle f_R | f_S \rangle \\
&= \frac{\dim(\Omega)^2}{|G|^2} \sum_{R,S} D^{\Omega}_{kl}(R) \overline{D}^{\Omega}_{kl'}(S) \delta_{R,S} \\
&= \frac{\dim(\Omega)^2}{|G|^2} \sum_{R} D^{\Omega}_{kl}(R) \overline{D}^{\Omega}_{kl'}(R) \\
&= \frac{\dim(\Omega)}{|G|} \delta_{l,l'}
\end{aligned} \tag{4.52}$$

Hence, if the multiplicity is greater than one, an additional label preceding the irrep label should be introduced in order to distinguish SALCs with the same symmetry, and, by varying the l index of the projector, all these can be projected out. Note that the maximal invariance space of a symmetry group is bound to be the regular representation; hence, multiplicities of an invariant function space cannot exceed the dimensions of the irreps, and thus will always be covered by the variation of the l index. If the multiplicity is smaller than $\dim(\Omega)$, variation of l will give rise to redundancies.

The action of the projector on an arbitrary function can be written as

$$\hat{P}^{\Gamma}_{kl} |f_x\rangle = S^{\Gamma l}_x |\Phi^{\Gamma}_k\rangle \tag{4.53}$$

Hence, the projector takes out of the function an irreducible part which transforms as the $|\Phi^{\Gamma}_k\rangle$ SALC, multiplied by an overlap factor, $S^{\Gamma l}_x$, which indicates the extent to which the $|\Phi^{\Gamma}_l\rangle$ SALC was present in this function. A very concise formulation of this result can be achieved by the use of the Dirac notation. In this notation, the projector is written as

$$\hat{P}^{\Gamma}_{kl} = |\Phi^{\Gamma}_k\rangle\langle\Phi^{\Gamma}_l| \tag{4.54}$$

In the ket-bra combination, all the aspects of the projector come together. Let us apply this to our function

$$\begin{aligned}
\hat{P}^{\Gamma}_{kl} |f_x\rangle &= |\Phi^{\Gamma}_k\rangle\langle\Phi^{\Gamma}_l| \, |f_x\rangle \\
&= |\Phi^{\Gamma}_k\rangle\langle\Phi^{\Gamma}_l | f_x\rangle
\end{aligned} \tag{4.55}$$

where the convention is followed that the juxtaposition of two vertical lines is contracted to one. Comparing Eqs. (4.54) and (4.55), one can identify the bracket:

$$S_x^{\Gamma l} = \langle \Phi_l^\Gamma | f_x \rangle \tag{4.56}$$

When the projection operator acts (on the left) upon a function $|f_x\rangle$ it forms a bracket, which is the overlap factor measuring how much of the $|\Phi_l^\Gamma\rangle$ SALC is present in the target. This is the 'recognition' part of the projection. It then returns, as a result, the desired SALC $|\Phi_k^\Gamma\rangle$ multiplied by the overlap factor. This is the ladder aspect. As an example, consider the action of the $\hat{P}^E$ projection operators on the $|1s_\mathrm{A}\rangle$ orbital in ammonia. One has

$$\begin{aligned}
\hat{P}_{11}|1s_\mathrm{A}\rangle &= \frac{1}{3}\left[2|1s_\mathrm{A}\rangle - |1s_\mathrm{B}\rangle - |1s_C\rangle\right] = \frac{\sqrt{2}}{\sqrt{3}}|\psi_x\rangle \\
\hat{P}_{21}|1s_\mathrm{A}\rangle &= \frac{\sqrt{2}}{\sqrt{3}}|\psi_y\rangle \\
\hat{P}_{12}|1s_\mathrm{A}\rangle &= 0 \\
\hat{P}_{22}|1s_\mathrm{A}\rangle &= 0
\end{aligned} \tag{4.57}$$

Note that the bracket $\langle \psi_y | 1s_\mathrm{A}\rangle$ vanishes, because $|1s_\mathrm{A}\rangle$ does not occur in the $|\psi_y\rangle$ target, and this gives rise to the zeros in Eq. 4.57. If one wants to avoid the cumbersome construction of the irreducible representation matrices, one can construct trace projectors by putting $k = l$ and summing over all k.

$$\begin{aligned}
\sum_k \hat{P}_{kk}^\Gamma &= \frac{\dim(\Gamma)}{|G|} \sum_k \sum_R \overline{D}_{kk}^\Gamma(R)\hat{R} \\
&= \frac{\dim(\Gamma)}{|G|} \sum_R \overline{\chi}^\Gamma(R)\hat{R}
\end{aligned} \tag{4.58}$$

In this case, only the character tables are needed in order to construct such projectors. They will certainly destroy all parts of the function space that do not belong to the irrep Γ, but, on the other hand, one loses the additional information in the subrepresentation. As we will see in the subsequent chapters these little auxiliary indices are nonetheless valuable. A further remarkable property of a projector is that if it is applied twice with inverted kl indices, one again obtains a projector.

$$\begin{aligned}
\hat{P}^{\Gamma}_{lk}\hat{P}^{\Gamma}_{kl} &= \frac{\dim(\Gamma)^2}{|G|^2}\sum_{RS}\overline{D}^{\Gamma}_{lk}(R)\overline{D}^{\Gamma}_{kl}(S)\hat{R}\hat{S} \\
&= \frac{\dim(\Gamma)^2}{|G|^2}\sum_{RT}\overline{D}^{\Gamma}_{lk}(R)\overline{D}^{\Gamma}_{kl}(R^{-1}T)\hat{T} \\
&= \frac{\dim(\Gamma)^2}{|G|^2}\sum_{T}\sum_{m}\sum_{R}\overline{D}^{\Gamma}_{lk}(R)\overline{D}^{\Gamma}_{km}(R^{-1})\overline{D}^{\Gamma}_{ml}(T)\ \hat{T} \\
&= \frac{\dim(\Gamma)^2}{|G|^2}\sum_{T}\sum_{m}\left(\sum_{R}\overline{D}^{\Gamma}_{lk}(R)D^{\Gamma}_{mk}(R)\right)\overline{D}^{\Gamma}_{ml}(T)\ \hat{T} \\
&= \frac{\dim(\Gamma)}{|G|}\sum_{T}\left(\sum_{m}\delta_{m,l}\overline{D}^{\Gamma}_{ml}(T)\right)\hat{T} \\
&= \hat{P}^{\Gamma}_{ll}
\end{aligned} \tag{4.59}$$

This also implies that the diagonal operator $\hat{P}_{kk}$ is *idempotent*, i.e., applying it twice gives exactly the same result as applying it once:

$$\hat{P}^{\Gamma}_{kk}\hat{P}^{\Gamma}_{kk} = \hat{P}^{\Gamma}_{kk} \tag{4.60}$$

Finally, summing over all diagonal projectors gives rise to the unit element:

$$\sum_{\Gamma}\sum_{i}\hat{P}^{\Gamma}_{ii} = \hat{E} \tag{4.61}$$

The proof is as follows:

$$\begin{aligned}
\sum_{\Gamma}\sum_{i}\hat{P}^{\Gamma}_{ii} &= \frac{1}{|G|}\sum_{\Gamma}\dim(\Gamma)\sum_{i}\sum_{R}\overline{D}^{\Gamma}_{ii}(R)\hat{R} \\
&= \frac{1}{|G|}\sum_{\Gamma}\dim(\Gamma)\sum_{R}\overline{\chi}^{\Gamma}(R)\hat{R} \\
&= \frac{1}{|G|}\sum_{R}\left(\sum_{\Gamma}\dim(\Gamma)\chi^{\Gamma}(R)\right)\hat{R} \\
&= \sum_{R}\delta_{R,E}\hat{R} \\
&= \hat{E}
\end{aligned} \tag{4.62}$$

Here, we have made use of the fact that the sum over all characters, multiplied by the dimension of the irrep, is the character of the regular representation, and this vanishes for all R except for the unit element, where it is equal to $|G|$ (see Eq. (4.41)). In Dirac terminology this reads

$$\sum_{\Gamma}\sum_{i}|\Phi_i^{\Gamma}\rangle\langle\Phi_i^{\Gamma}| = 1 \tag{4.63}$$

This relation is also known as the *closure* relation. It is frequently applied in the context of the crystal field theory of the lanthanides.

4.6 Subduction and Induction

Many applications are concerned with the reduction of symmetry by external or internal perturbations. *Subduction* corresponds to the lowering of a symmetry group G to one of its subgroups, H, and is denoted by $G \downarrow H$. It can consist of a chain of consecutive symmetry lowerings, following a path of descent in symmetry down the genealogical tree of the group. In physics a typical form of *external* symmetry breaking is through the application of a uniform magnetic or electric field. It leads to a subgroup which is the intersection of the molecular point group and the axial or polar symmetry of the applied field, as discussed in Sect. 3.9. In chemistry a common approach to external symmetry breaking is to substitute one or more atoms or atomic groups by homologs, or to interchange sites, which may give rise to different stereoisomers. *Internal* symmetry breaking is more subtle, and may arise as a consequence of the Jahn−Teller effect. In this case the presence of a degenerate electronic state in the high-symmetry conformation of the molecule may provoke a spontaneous geometric distortion of the nuclear frame, leading to a lower symmetry in which the degeneracy is removed. This effect involves the coupling of representations and will be discussed in Sect. 6.6.

Induction is the reverse process of subduction. Here we start from local properties of an atom or a group of atoms that are bound together, the symmetry characteristic of which are described by a subgroup H. The purpose of induction is then to determine which symmetries will result in a covering group G when these sites become the building blocks of a larger symmetric system. Typically, this technique is useful to construct molecular orbitals from atomic basis orbitals on equivalent sites or to derive collective motions of a cluster starting from individual displacement of the individual atoms.

4.6.1 Subduction

Subduction defines what will happen to the irreps of G when the symmetry is reduced to H. This can easily be decided on the basis of the character theorem. We simply have to determine the character of the representation in the subgroup. The procedure consists of three steps. One first identifies the correspondence between the elements of G and the elements of H. Then the characters of the irrep in G are transcribed to the characters for the corresponding operations in the subgroup. Thirdly, the

Table 4.3 Subduction of T_{1u} in $O_h \downarrow D_{3d}$

O_h	$\hat{E}$	$8\hat{C}_3$	$6\hat{C}_2$	$6\hat{C}_4$	$3\hat{C}_2$	$\hat{\imath}$	$6\hat{S}_4$	$8\hat{S}_6$	$3\hat{\sigma}_h$	$6\hat{\sigma}_d$
T_{1u}	3	0	−1	1	−1	−3	−1	0	1	1
D_{3d}	$\hat{E}$	$2\hat{C}_3$	$3\hat{C}_2$			$\hat{\imath}$		$2\hat{S}_6$		$3\hat{\sigma}_d$
T_{1u}	3	0	−1			−3		0		1
A_{2u}	1	1	−1			−1		−1		1
E_u	2	−1	0			−2		1		0

character string in the subgroup is reduced according to the standard procedure of the character theorem. Hence, let Γ denote an irrep of the parent group, and γ an irrep of the subgroup. The number of times that this subgroup representation occurs in the subduction $G \downarrow H$ is given by

$$c_\gamma(\Gamma G \downarrow H) = \frac{1}{|H|} \sum_{h \in H} \overline{\chi}^\gamma(h) \chi^\Gamma(h) \tag{4.64}$$

In Table 4.3 we follow as an example the fate of the octahedral T_{1u} irrep for the subduction $O_h \downarrow D_{3d}$. The first step of the procedure is to tie the symmetry elements of H to the appropriate classes of G. For instance, the octahedron has two conjugacy classes of $\hat{C}_2$ axes. The $6\,\hat{C}_2$ class collects the twofold axes which bisect the Cartesian directions, while the $3\,\hat{C}_2$ class is made up of the $\hat{C}_4^2$ axes along the Cartesian directions. In the case of subduction to D_{3d}, the three axes perpendicular to the trigonal direction belong to the $6\,\hat{C}_2$ class. As the table indicates, in D_{3d} the threefold-degenerate representation becomes reducible by splitting into two trigonal irreps:

$$O_h \downarrow D_{3d} \ : T_{1u} \rightarrow A_{2u} + E_u \tag{4.65}$$

In some cases a subgroup can be reached via two different symmetry breakings. An example is the subduction $O_h \downarrow D_{2h}$. Here, two pathways for symmetry breaking are present. The three $\hat{C}_2$ axes of the orthorhombic symmetry are either based on the $3\,\hat{C}_2$ class, $\{\hat{C}_2^x, \hat{C}_2^y, \hat{C}_2^z\}$, or on a mixture of the two classes, as in $\{\hat{C}_2^z, \hat{C}_2^{xy}, \hat{C}_2^{\bar{x}y}\}$. We shall denote the latter group as D'_{2h}. Simple molecular examples of both are shown in Fig. 4.4. Tables 4.4 and 4.5 present the splitting of the T_{2g} irrep over these two subduction paths. The corresponding splitting schemes are as follows:

$$\begin{aligned} O_h \downarrow D_{2h} &: T_{2g} \rightarrow B_{1g} + B_{2g} + B_{3g} \\ O_h \downarrow D'_{2h} &: T_{2g} \rightarrow A_g + B_{2g} + B_{3g} \end{aligned} \tag{4.66}$$

Subduction tables are available in Appendix C.1.

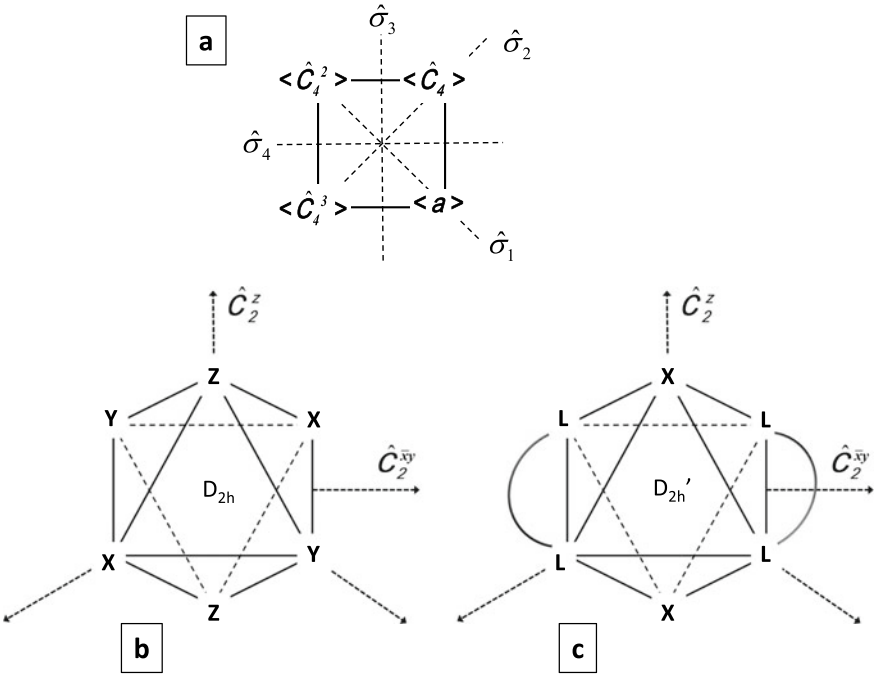


Fig. 4.4 **a** Generation of equivalent sites in a square, starting from $\langle a \rangle$, **b** substitutional symmetry lowering of $O_h \downarrow D_{2h}$ in the $MX_2Y_2Z_2$ complex isomer, **c** $O_h \downarrow D'_{2h}$ symmetry lowering by bidentate ligands in *trans*-$M(L\text{-}L)_2X_2$

Table 4.4 Subduction of T_{2g} under $O_h \downarrow D_{2h}$

O_h	$\hat{E}$	$3\hat{C}_2$			$\hat{i}$	$3\hat{\sigma}_h$		
	$\downarrow$	$\swarrow$	$\downarrow$	$\searrow$	$\downarrow$	$\swarrow$	$\downarrow$	$\searrow$
D_{2h}	$\hat{E}$	$\hat{C}_2^x$	$\hat{C}_2^y$	$\hat{C}_2^z$	$\hat{i}$	$\hat{\sigma}_{xy}$	$\hat{\sigma}_{xz}$	$\hat{\sigma}_{yz}$
T_{2g}	3	−1	−1	−1	3	−1	−1	−1
B_{1g}	1	−1	−1	1	1	1	−1	−1
B_{2g}	1	−1	1	−1	1	−1	1	−1
B_{3g}	1	1	−1	−1	1	−1	−1	1

Table 4.5 Subduction of T_{2g} under $O_h \downarrow D'_{2h}$

O_h	$\hat{E}$	$6\hat{C}_2$		$3\hat{C}_2$	$\hat{i}$	$3\hat{\sigma}_h$	$6\hat{\sigma}_d$	
	$\downarrow$	$\downarrow$	$\searrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\searrow$
D'_{2h}	$\hat{E}$	$\hat{C}_2^x$	$\hat{C}_2^y$	$\hat{C}_2^z$	$\hat{i}$	$\hat{\sigma}_{xy}$	$\hat{\sigma}_{xz}$	$\hat{\sigma}_{yz}$
T_{2g}	3	1	1	−1	3	−1	1	1
A_g	1	1	1	1	1	1	1	1
B_{2g}	1	−1	1	−1	1	−1	1	−1
B_{3g}	1	1	−1	−1	1	−1	−1	1

4.6.2 Induction

The opposite process to subduction is induction. Here, we start from an irrep in a subgroup H. By coset expansion, this subgroup is put on an orbit inside a higher symmetry group. This leads to an extension of the function space and will generate irreps of the parent group. The outcome of the induction is determined by the reciprocity theorem due to Frobenius.

Theorem 4.4 *The number of times that a given irrep* Γ *of a parent group* G *occurs in the induction* $H \uparrow G$ *of a subgroup irrep* γ *is equal to the number of times that* γ *is present in the subduction* $G \downarrow H$ *of that irrep* Γ.

It is worthwhile to study the proof in detail, since it introduces the important concept of the *ground* representation [5, 6]. Before starting the proof itself, we must introduce the concept of the induction space. This proceeds in three steps. The first step is to generate all the cosets of H in G. This provides us with the set of equivalent sites and their local symmetry groups. The second step involves the construction of the ground or positional representation. This representation refers to the collection of sites. Indeed, as all these sites are replicas of one another, the action of G will permute them among themselves, so they form a representation of the group. Finally, in the third step, the sites are 'dressed' with 'objects', such as atomic orbitals, dipole moments, and displacement vectors. At this stage, the intermediate step of the positional representation pays off, as it provides full control of the symmetry properties of the induction space. So to begin with, consider a polyhedral molecular cluster or a complex, consisting of equivalent sites. Typically, these sites could be the atoms or groups of atoms in a network covering a hollow cage, or ligands in a metal complex. In the case of ammonia, the sites are simply the three hydrogen atoms. Usually, the site group is of type C_{nv}. In the case of ammonia it is simply C_s. We choose site $\langle a \rangle$ as the starting site, which is stabilized by the subgroup H_a. The group G is now expanded in cosets of this subgroup, with coset representatives $\hat{g}_\kappa$:

$$G = \sum_\kappa \hat{g}_\kappa H_a \tag{4.67}$$

As we have seen in the previous chapter, the coset representatives each address a copy of site $\langle a \rangle$, which we shall label as $\langle \kappa \rangle$. The site group that stabilizes this site is isomorphic to H_a and is denoted by H_κ. We thus have the following mappings:

$$\begin{aligned} \langle \kappa \rangle &= \hat{g}_\kappa \langle a \rangle \\ \hat{g}_\kappa^{-1} \langle \kappa \rangle &= \langle a \rangle \\ H_\kappa &= \hat{g}_\kappa H_a \hat{g}_\kappa^{-1} \end{aligned} \tag{4.68}$$

The mapping of the stabilizer, $H_a \rightarrow H_\kappa$, is recognized as a similarity transformation of the whole subgroup. Two different sites can share the same site group. As an example, in a square pyramidal complex, with parent group C_{4v} and site groups C_s,

two ligands, *trans* to each other, have the same site group. With reference to the square in Fig. 4.4a, the cosets may be generated as follows:

$$C_{4v} = \sum_{k=0}^{3} \hat{C}_4^k \{\hat{E}, \hat{\sigma}_1\} = \{\hat{E}, \hat{\sigma}_1\} + \{\hat{C}_4, \hat{\sigma}_4\} + \{\hat{C}_4^2, \hat{\sigma}_2\} + \{\hat{C}_4^3, \hat{\sigma}_3\} \tag{4.69}$$

Hence, the four sites of the square are denoted by the coset generators as $\langle a \rangle$, $\langle \hat{C}_4 \rangle$, $\langle \hat{C}_4^2 \rangle$, and $\langle \hat{C}_4^3 \rangle$. The operators of the group will permute equivalent sites; we say they act transitively on the cosets. This means that the cosets themselves form a basis for a representation of G. This representation has the form of a permutation matrix and is denoted as $\mathbb{P}(g)$. One has

$$\hat{g}\left(\hat{g}_\kappa H_a\right) = \sum_\lambda P_{\lambda\kappa}(g)\hat{g}_\lambda H_a \tag{4.70}$$

with

$$\begin{aligned} P_{\lambda\kappa}(g) &= 1 \text{ if } \hat{g}\left(\hat{g}_\kappa H_\mathrm{a}\right) = \hat{g}_\lambda H_a \\ P_{\lambda\kappa}(g) &= 0 \text{ if } \hat{g}\left(\hat{g}_\kappa H_\mathrm{a}\right) \neq \hat{g}_\lambda H_a \end{aligned} \tag{4.71}$$

This permutational representation is called the *ground* representation. It describes the transformation of the coset space. The dimension of this coset space is $|G|/|H|$. As each coset corresponds to a site, this matrix represents the permutation of the positions of the sites. For this reason it is also called the *positional* representation. Indeed, Eq. (4.70) could equally well be written as

$$\hat{g}\langle\kappa\rangle = \sum_\lambda P_{\lambda\kappa}(g)\langle\lambda\rangle \tag{4.72}$$

For the λ-value, which marks the position of the non-zero element in the κth column of the matrix $\mathbb{P}$, the product $\hat{g}_\lambda^{-1}\hat{g}\hat{g}_\kappa$ will be an element of H_a. We call this the *subelement* of $\hat{g}$ in H_a. The subelement of a given group element $\hat{g}$ may be seen as the mapping of $\hat{g}$ onto an element of the subgroup, by the concerted action of a $\lambda\kappa$ pair for which $P_{\lambda\kappa}(g) = 1$. One indeed verifies for $P_{\lambda\kappa}(g) = 1$:

$$\begin{aligned} \hat{g}\langle\kappa\rangle &= \langle\lambda\rangle \\ \hat{g}\hat{g}_\kappa\langle a\rangle &= \hat{g}_\lambda\langle a\rangle \\ \hat{g}_\lambda^{-1}\hat{g}\hat{g}_\kappa\langle a\rangle &= \langle a\rangle \end{aligned} \tag{4.73}$$

As an example, for the case of the C_{4v} pyramidal complex, the matrices of the positional representation are listed in Table 4.6. If $\hat{g} = \hat{g}_\kappa\hat{h}\hat{g}_\kappa^{-1}$, the diagonal element will be non-zero: $P_{\kappa\kappa}(g) = 1$. The following sum rules will thus hold, as can be verified from Table 4.6:

Table 4.6 Ground or positional representation of the four equatorial ligand sites in a square pyramidal complex; the sites are ordered as in Fig. 4.4a

$$\mathbb{P}(E) = \begin{pmatrix} 1&0&0&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1 \end{pmatrix} \qquad \mathbb{P}(\sigma_1) = \begin{pmatrix} 1&0&0&0\\ 0&0&0&1\\ 0&0&1&0\\ 0&1&0&0 \end{pmatrix}$$

$$\mathbb{P}(C_4) = \begin{pmatrix} 0&0&0&1\\ 1&0&0&0\\ 0&1&0&0\\ 0&0&1&0 \end{pmatrix} \qquad \mathbb{P}(\sigma_2) = \begin{pmatrix} 0&0&1&0\\ 0&1&0&0\\ 1&0&0&0\\ 0&0&0&1 \end{pmatrix}$$

$$\mathbb{P}(C_4^2) = \begin{pmatrix} 0&0&1&0\\ 0&0&0&1\\ 1&0&0&0\\ 0&1&0&0 \end{pmatrix} \qquad \mathbb{P}(\sigma_3) = \begin{pmatrix} 0&0&0&1\\ 0&0&1&0\\ 0&1&0&0\\ 1&0&0&0 \end{pmatrix}$$

$$\mathbb{P}(C_4^3) = \begin{pmatrix} 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\\ 1&0&0&0 \end{pmatrix} \qquad \mathbb{P}(\sigma_4) = \begin{pmatrix} 0&1&0&0\\ 1&0&0&0\\ 0&0&0&1\\ 0&0&1&0 \end{pmatrix}$$

$$\forall \hat{h} \in H_a \rightarrow \sum_{\kappa} P_{\kappa\kappa}(\hat{g}_\kappa \hat{h} \hat{g}_\kappa^{-1}) = \frac{|G|}{|H|}$$

$$\forall \kappa \in \left(1, 2, \ldots, \frac{|G|}{|H|}\right) \rightarrow \sum_{\hat{h} \in H} P_{\kappa\kappa}(\hat{g}_\kappa \hat{h} \hat{g}_\kappa^{-1}) = |H| \tag{4.74}$$

As an example, for $h = E$, the matrix $\mathbb{P}(E)$ is the unit matrix and its trace is equal to 4, which corresponds to the quotient of the group orders, in accordance with Eq. (4.74). For $\hat{h} = \hat{\sigma}_1$, again only four diagonal elements are non-zero:

$$\begin{aligned} \kappa = 1: &\ P_{11}(E\sigma_1 E) = P_{11}(\sigma_1) = 1\\ \kappa = 2: &\ P_{22}(C_4\sigma_1 C_4^3) = P_{22}(\sigma_2) = 1\\ \kappa = 3: &\ P_{33}(C_4^2\sigma_1 C_4^2) = P_{33}(\sigma_1) = 1\\ \kappa = 4: &\ P_{44}(C_4^3\sigma_1 C_4) = P_{44}(\sigma_2) = 1 \end{aligned} \tag{4.75}$$

Finally in the third step, we attach to each site a functional basis, which is represented by the irrep γ, with component labeling m_γ. So the first site $\langle a \rangle$ will be hosting the functions $|\gamma m_\gamma : a\rangle$, with transformation properties:

$$\hat{h}_a |\gamma m_\gamma; a\rangle = \sum_{m'_\gamma} |\gamma m'_\gamma; a\rangle D^{\gamma}_{m'_\gamma m_\gamma}(h_a) \tag{4.76}$$

The coset generators will once again take this functional basis around in an orbit which visits all the equivalent sites. Local basis sets are thus defined as

$$|\gamma m_\gamma; \kappa\rangle = \hat{g}_\kappa |\gamma m_\gamma; a\rangle \tag{4.77}$$

The *total induction space* is the sum of all these basis sets on the different sites.

We are now ready to start the proof. The total induction space is invariant under the operations of the group G. Our careful construction of this space, based on the coset partition, implies that the action of any operator of the group on any basis function is fully defined. One has

$$\begin{aligned}
\hat{g}|\gamma m_\gamma; \kappa\rangle &= \hat{g}\hat{g}_\kappa |\gamma m_\gamma; a\rangle \\
&= \left(\sum_\lambda P_{\lambda\kappa}(g) \hat{g}_\lambda \hat{g}_\lambda^{-1} \right) \hat{g}\hat{g}_\kappa |\gamma m_\gamma; a\rangle \\
&= \sum_\lambda P_{\lambda\kappa}(g) \hat{g}_\lambda \left[\hat{g}_\lambda^{-1} \hat{g} \hat{g}_\kappa \right] |\gamma m_\gamma; a\rangle
\end{aligned} \tag{4.78}$$

Note that the insertion of a summation over coset representatives in the second line of this equation is perfectly legitimate, since it simply corresponds to the unit operation. Indeed, one has

$$\sum_\lambda P_{\lambda\kappa}(g) \hat{g}_\lambda \hat{g}_\lambda^{-1} = \sum_\lambda P_{\lambda\kappa}(g) \hat{E} = \hat{E} \tag{4.79}$$

where we used the property that any column of a permutation matrix contains exactly one non-zero element, which is equal to 1. In the final line of Eq. (4.78) we highlighted the appearance of the subelement of $\hat{g}$ by placing it between square brackets. This subelement belongs to H_a, under the protection of the $P_{\lambda\kappa}(g)$ prefactor, which will be non-zero only for values of $\hat{g}$ for which this is indeed the case. We can thus introduce the representation matrix for the local on-site transformations:

$$\begin{aligned}
\hat{g}|\gamma m_\gamma; \kappa\rangle &= \sum_\lambda \sum_{m'_\gamma} P_{\lambda\kappa}(g) \hat{g}_\lambda |\gamma m'_\gamma; a\rangle D^\gamma_{m'_\gamma m_\gamma}(g_\lambda^{-1} g g_\kappa) \\
&= \sum_\lambda \sum_{m'_\gamma} P_{\lambda\kappa}(g) |\gamma m'_\gamma; \lambda\rangle D^\gamma_{m'_\gamma m_\gamma}(g_\lambda^{-1} g g_\kappa)
\end{aligned} \tag{4.80}$$

This result provides the matrix transformation, which shows how the basis functions of the total induction space are transformed under the operations of G. We shall denote this matrix as $\mathbb{D}^{H\uparrow G}$. The structure of this matrix is based on the permutational structure of the ground representation, but the zeros are replaced by small zero blocks of dimension $\dim(\gamma) \times \dim(\gamma)$, and, instead of the ones, the $\mathbb{D}^\gamma(g_\lambda^{-1} g g_\kappa)$ matrices are inserted. Our aim is now to determine all the irreps of the induction space. Hence we only need to determine the trace of the matrix representation of the induction space. The diagonal element of this matrix are given by $P_{\kappa\kappa}(g) D^\gamma_{m_\gamma m_\gamma}(\hat{g}_\kappa^{-1} \hat{g} \hat{g}_\kappa)$.

From these we can calculate at once the frequency of a given Γ irrep of the main group, using the character theorem:

$$\begin{aligned} c_\Gamma(\gamma H \uparrow G) &= \frac{1}{|G|}\sum_{g\in G}\overline{\chi}^\Gamma(g)\mathrm{Tr}\left[\mathbb{D}^{H\uparrow G}(g)\right] \\ &= \frac{1}{|G|}\sum_{g\in G}\overline{\chi}^\Gamma(g)\sum_\kappa P_{\kappa\kappa}(g)\chi^\gamma(g_\kappa^{-1}gg_\kappa) \end{aligned} \tag{4.81}$$

The only elements, $\hat{g}$, that will be allowed in the summation over κ are the ones where $(\hat{g}_\kappa^{-1}\hat{g}\hat{g}_\kappa) \in H_a$. For other elements $P_{\kappa\kappa}(g)$ will be zero. Let us denote the subelement as $\hat{h}$, which allows us to express $\hat{g}$ as

$$\hat{g} = \hat{g}_\kappa\hat{h}\hat{g}_\kappa^{-1} \tag{4.82}$$

Introducing this substitution in Eq. (4.81) yields

$$c_\Gamma(\gamma H \uparrow G) = \frac{1}{|G|}\sum_{h\in H}\sum_\kappa \overline{\chi}^\Gamma(g_\kappa h g_\kappa^{-1})\chi^\gamma(h)P_{\kappa\kappa}(g_\kappa h g_\kappa^{-1}) \tag{4.83}$$

The first character in this equation belongs to the full group, and is the same for all elements of a conjugacy class, hence

$$\chi^\Gamma(g_\kappa h g_\kappa^{-1}) = \chi^\Gamma(h) \tag{4.84}$$

Substituting the result of Eq. (4.84) and the sum rule in Eq. (4.74) in the character expression finally gives

$$\begin{aligned} c_\Gamma(\gamma H \uparrow G) &= \frac{1}{|G|}\sum_{h\in H}\sum_\kappa \overline{\chi}^\Gamma(h)\chi^\gamma(h)P_{\kappa\kappa}(g_\kappa h g_\kappa^{-1}) \\ &= \frac{1}{|G|}\sum_{h\in H}\overline{\chi}^\Gamma(h)\chi^\gamma(h)\left(\sum_\kappa P_{\kappa\kappa}(g_\kappa h g_\kappa^{-1})\right) \\ &= \frac{1}{|H|}\sum_{h\in H}\overline{\chi}^\Gamma(h)\chi^\gamma(h) \\ &= c_\gamma(\Gamma G \downarrow H) \end{aligned} \tag{4.85}$$

which concludes the proof. Armed with the subduction tables, we can now read these at once in the opposite sense and obtain the corresponding induction frequencies. As a simple example, consider a hydrogen atom in ammonia. The site symmetry is C_s, and the subduction $(C_{3v} \downarrow C_s)$ reads

Table 4.7 Relationship between the character tables of D_{3h} and its C_{nv} sitegroups

$\mathbf{G}: D_{3h}$	$\hat{E}$	$2\hat{C}_3$	$3\hat{C}_2$	$\hat{\sigma}_h$	$2\hat{S}_3$	$3\hat{\sigma}_v$	
$\Gamma: A_1'$	1	1	1	1	1	1	
$\Gamma: A_2'$	1	1	−1	1	1	−1	R_z
$\Gamma: E'$	2	−1	0	2	−1	0	(x, y)
$\Gamma: A_1''$	1	1	1	−1	−1	−1	
$\Gamma: A_2''$	1	1	−1	−1	−1	1	z
$\Gamma: E''$	2	−1	0	−2	1	0	(R_x, R_y)
$\mathbf{H}: C_{3v}$	$\hat{E}$	$2\hat{C}_3$				$3\hat{\sigma}_v$	
$\gamma: a_1$	1	1				1	
$\gamma: a_2$	1	1				−1	
$\gamma: e$	2	−1				0	
$\mathbf{H}: C_{2v}$	$\hat{E}$		$\hat{C}_2$	$\hat{\sigma}_v^{xy}$		$\hat{\sigma}_v^{xz}$	
$\gamma': a_1$	1		1	1		1	
$\gamma': a_2$	1		1	−1		−1	
$\gamma': b_1$	1		−1	1		−1	
$\gamma': b_1$	1		−1	−1		1	

$$\begin{aligned} A_1 &\to a \\ A_2 &\to b \\ E &\to a + b \end{aligned} \tag{4.86}$$

Using reciprocity, we can thus immediately infer the SALC symmetries of the hydrogen basis functions by selecting the C_{3v} irreps that subduce a in the case of the $1s$ orbitals, and b in the case of the bending coordinates:

$$\begin{aligned} \text{Hydrogen} \quad 1s : aC_s \uparrow C_{3v} &= A_1 + E \\ \text{Hydrogen} \quad \Delta\phi : bC_s \uparrow C_{3v} &= A_2 + E \end{aligned} \tag{4.87}$$

A more elaborate example is offered by the symmetries of the vibrations of Phosphorus pentafluoride, PF_5, with a trigonal bipyramidal structure. As always, we start by providing a picture of the molecular frame, indicating the coordinate system. Axial Fluorines on the z-axis occupy C_{3v} sites, while equatorial Fluorines in the xy-plane occupy C_{2v} sites. As each atom can move in three directions, there are 18 degrees of freedom in total. The center of the trigonal bipyramid has full D_{3h} symmetry. The translational modes of the Phosphorus atom transform as p_x, p_y, p_z. Their symmetry is $E' \oplus A_2''$, as can directly be read off from the character table of Appendix A. In order to determine the vibrational degrees of freedom of the Fluorine atoms we now use induction. The first step is to define the correspondence between the symmetry elements in D_{3h} and the site subgroups, C_{3v} and C_{2v}. The result is shown in Table 4.7.

The second step is to determine from this table the subduction relations. One has for the subduction $D_{3h} \downarrow C_{3v}$:

$$\begin{array}{ll} A_1' \to a_1 & A_1'' \to a_2 \\ A_2' \to a_2 & A_2'' \to a_1 \\ E' \to e & E'' \to e \end{array} \tag{4.88}$$

Similarly for the subduction $D_{3h} \downarrow C_{2v}$:

$$\begin{array}{ll} A_1' \to a_1 & A_1'' \to a_2 \\ A_2' \to b_1 & A_2'' \to b_2 \\ E' \to a_1 + b_1 & E'' \to a_2 + b_2 \end{array} \tag{4.89}$$

We are now ready to determine the symmetries of the Fluorine displacements, making use of the Frobenius induction theorem. Each Fluorine has three degrees of freedom: one displacement along the bond direction, and two displacements perpendicular to the bond direction. These are also denoted as σ and π types, resp. These labels refer to the cylindrical characteristics of a diatomic bond: σ basis functions, be it orbitals or displacement vectors, are axially symmetric, while π functions have nodal planes containing the bond axis. Clearly the σ displacements correspond to bond stretchings, while the π modes are bendings. In Table 4.8 we present the D_{3h} symmetries that result from the stretchings and bendings of the five Fluorine atoms. Note that in the C_{2v} site group of the equatorial sites, the π modes are further resolved into in-plane and out-of-plane components, which are respectively symmetric and anti-symmetric with respect to the common equatorial plane. The 18 degrees of freedom of the PF_5 molecule, including the P atom, are then given by

$$\Gamma = 2A_1' + A_2' + 4E' + 3A_2'' + 2E'' \tag{4.90}$$

This set contains six spurious modes, three translations and three rotations. Their symmetries are specified in the final column of Table 4.7: $A_2'' \oplus E'$ for the translations, and $A_2' \oplus E''$ for the rotations. Subtracting these from the list in Eq. (4.90), then yields the 12 vibrational modes, five stretching modes transforming as $2A_1' \oplus A_2'' \oplus E'$, and seven bending modes, transforming as $A_2'' \oplus 2E' \oplus E''$. The two A_2'' modes are depicted in Fig. 4.5.

In Appendix C.2 induction schemes for σ, π, and δ orbital basis sets on C_{nv} sites of polyhedral complexes are to be found, including the D_{3h} group. Here we also provide the induction for an atom with C_s^v symmetry, in the xz-plane in between the axial and equatorial sites. In addition to the Frobenius theorem, there is also a stronger result for induction theory based on the concept of a fiber bundle. This requires the coupling of representations and will be considered in Sect. 6.9.

Table 4.8 Displacement symmetries of the Fluorine atoms in PF_5, determined by induction from the axial and equatorial sitegroups; σ-modes are bond stretchings, π-modes are bendings

		γ	Γ
ax	σ	a_1	$A_1' \oplus A_2''$
ax	π	e	$E' \oplus E''$
eq	σ	a_1	$A_1' \oplus E'$
eq	π_h	b_1	$A_2' \oplus E'$
	π_v	b_2	$A_2'' \oplus E''$

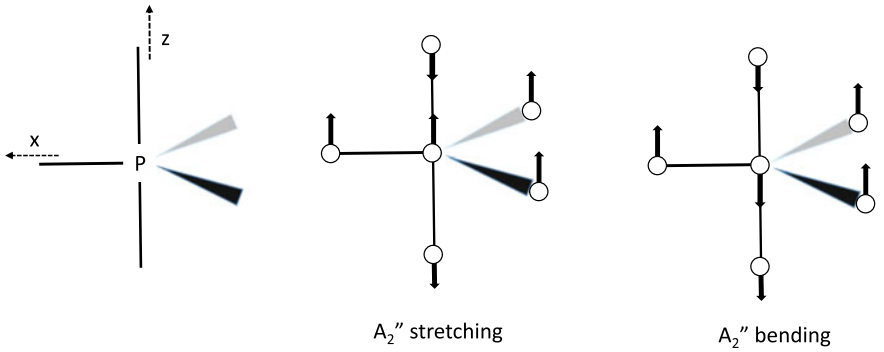


Fig. 4.5 Phosphorus pentafluoride: D_{3h} coordinate system and A_2'' vibrational modes

4.7 Application: The sp^3 Hybridization of Carbon

Methane is the prototype of the saturated aliphatic hydrocarbons. The four hydrogen atoms occupy the corners of a regular tetrahedron, as in Fig. 4.6. Their site symmetry is C_{3v}. The $1s$ atomic orbital on hydrogen is totally symmetric in the site group. The symmetries of the corresponding SALCs can of course be obtained by the standard character procedure, as for the case of ammonia, but we might as well directly obtain them by induction. Using the $\Gamma(\sigma C_{3v} \uparrow T_d)$ entry in Appendix C.2, one has

$$\Gamma\left(a_1 C_{3v} \uparrow T_d\right) = A_1 + T_2 \tag{4.91}$$

The corresponding SALCs can easily be projected:

$$\begin{aligned}\left(|A_1\rangle\ |T_{2x}\rangle\ |T_{2y}\rangle\ |T_{2z}\rangle\right) = \\ \left(|1s_A\rangle\ |1s_B\rangle\ |1s_C\rangle\ |1s_D\rangle\right) \times \mathbb{T}\end{aligned} \tag{4.92}$$

with

$$\mathbb{T} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix} \tag{4.93}$$

SALCs are normalized to unity, neglecting overlap between the sites. The matrix $\mathbb{T}$ transforms the localized orbitals on the sites to delocalized molecular orbitals with irreducible symmetry characteristics. The inverse matrix $\mathbb{T}^{-1}$ fulfills the opposite role, and localizes the molecular orbital set back on the atomic sites.

The valence shell of the central carbon atom contains four orbitals, which incidentally also transform as $A_1 + T_2$. The precise correspondence is as follows:

$$\begin{aligned} |2\,s\rangle &\leftrightarrow A_1 \\ |2\,p_x\rangle &\leftrightarrow T_{2x} \\ |2\,p_y\rangle &\leftrightarrow T_{2y} \\ |2\,p_z\rangle &\leftrightarrow T_{2z} \end{aligned} \tag{4.94}$$

Hence, we can match the central valence shell with the hydrogen SALCs. In fact, this correspondence provides a simple pictorial method for obtaining the SALCs immediately. The weighting coefficients for a given SALC are simply taken as proportional to the local amplitude of the central $2s$ or $2p$ function, with the same symmetry, as is illustrated in Fig. 4.6. In this way one obtains a SALC that has the same nodal characteristics, and thus the same symmetry, as the central orbital. Note that this procedure also aligns the phases of the peripheral and central orbitals.

Starting from on-site localized atomic orbitals, we have thus transformed these into SALCs using the $\mathbb{T}$ matrix, and then found a perfect matching with the central valence orbitals on carbon. What would now be the effect of applying the inverse transformation, $\mathbb{T}^{-1}$, not to the hydrogen SALCs but to the central carbon orbitals? This yields an interesting result. The inverse matrix, which transforms delocalized SALCs back into localized orbitals, reshapes the carbon valence orbitals, by projecting out linear combinations, of mixed or *hybrid* character, which are maximally directed to a single site of the tetrahedron. These are the ubiquitous sp^3 hybrids of Pauling, which we can label with the site labels A, B, C, and D:

$$\begin{aligned} &\left(|sp^3_{\mathrm{A}}\rangle\ |sp^3_{\mathrm{B}}\rangle\ |sp^3_{C}\rangle\ |sp^3_{D}\rangle\right) = \\ &\qquad \left(|2s\rangle\ |2p_x\rangle\ |2p_y\rangle\ |2p_z\rangle\right)\frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \end{pmatrix} \end{aligned} \tag{4.95}$$

The components of the valence shell of carbon being a scalar ($2s$) and a vector ($2p$), the tetrahedron is the optimal geometry that provides four valence sites, which together transform precisely as scalar and vector. The alternative high-symmetry

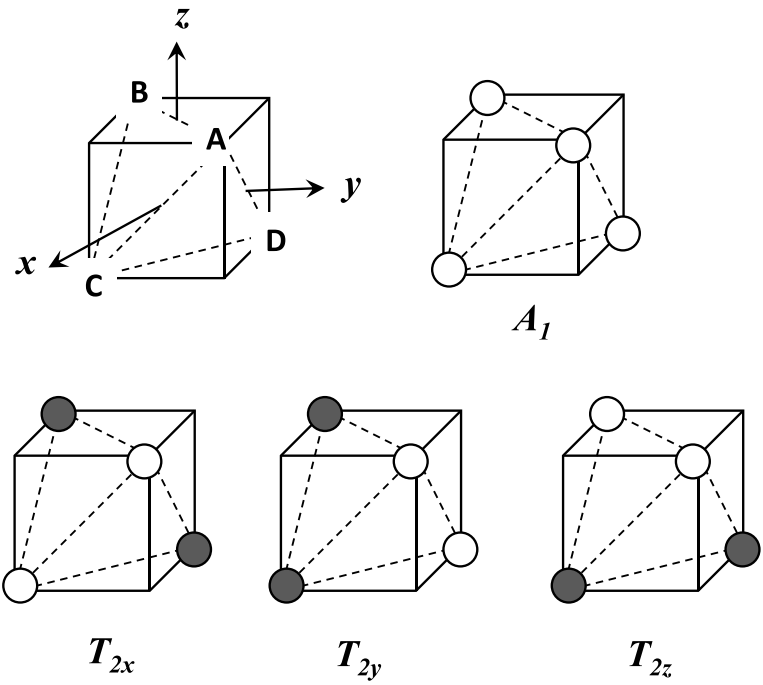


Fig. 4.6 Tetrahedral SALCs in methane. The open circles are positive and the filled circles are negative. The T_2 orbitals match the sign pattern of central p-orbitals

four-site structure is the square, but, from the tables in Appendix C.2, the induction from the C_{2v} sites in a square-planar structure yields

$$\Gamma(a_1 C_{2v} \uparrow D_{4h}) = A_{1g} + B_{2g} + E_u \tag{4.96}$$

This matches the symmetry of sp^2d hybrids. It is thus not suitable for carbon, but indeed describes the valence structure of square-planar transition-metal complexes where d-orbitals are involved in the bonding.

4.8 Application: The Vibrations of UF_6

Previously we determined the vibrational symmetries of a trigonal bipyramidal molecule PF_5. Here we continue the study of vibrations by looking at the actual expressions for normal modes, which involves the construction of SALC's. As a sample case we take the octahedral complex, UF_6, which can be brought into the gas phase, and which has been studied in great detail, since it is the carrier of Uranium in the gas diffusion process for enrichment of nuclear fuel. The atoms are labeled as

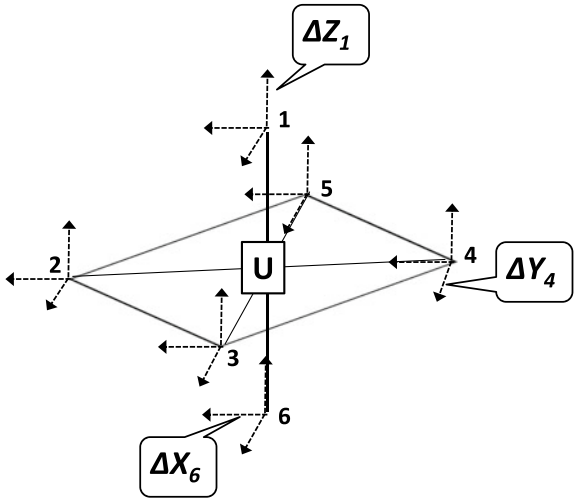


Fig. 4.7 Ligand numbering, and displacement coordinates for UF_6

in Fig. 4.7, and on each atom we define a local coordinate system, which parallels the central system. In the notation adopted, ΔY_2 is a variable for a displacement of atom 2 over a distance Δ in the positive y-direction. Remember that symmetry operations do not move the atoms, but the distortions. A symmetry operation such as $\hat{C}_4^z$ will transform this ΔY_2 into $-\Delta X_3$. The seven atoms give rise to 21 distortions, which include six spurious modes, corresponding to three translations and three rotations. The seven atoms form two different orbits: the orbit containing the six fluoride ligands and the one-atom orbit of the central Uranium atom. The displacements of one ligand can further be separated into a radial or σ-mode, and two tangential or π-modes, which, in the C_{4v} site group, transform as a_1 and e, respectively. Altogether, the distortion space thus contains three different basis sets: the central atom, the ligand σ-modes, and the ligand π-modes. For each of these the symmetry content may be determined by directly applying the character theorem, or—for the case of the ligands—by using induction. The three displacements of the Uranium atom transform as the T_{1u} irrep of the central translation mode. The ligand inductions are as follows:

$$\begin{aligned} \mathrm{F}\sigma &: \Gamma\,(a_1 C_{4v} \uparrow O_h) = A_{1g} + E_g + T_{1u} \\ \mathrm{F}\pi &: \Gamma\,(e C_{4v} \uparrow O_h) = T_{1g} + T_{2g} + T_{1u} + T_{2u} \end{aligned} \tag{4.97}$$

We can now determine the symmetry-adapted coordinates by applying the projection operators, but the results can be written down almost immediately, by again using the criterion of overlap with central symmetry functions. The A_{1g}, T_{1u}, and $E_g + T_{2g}$ SALCs reflect the nodal patterns of central s, p, and d functions respectively. The T_{1g} mode corresponds to the rotation, and evidently will consist of tangential displacements of ligands in the equator perpendicular to the rotation axis. Finally, the T_{2u} is a buckling mode, which has the symmetry of central f orbitals, *viz.* $f_{z(x^2-y^2)}$, and its

Table 4.9 Symmetry coordinates for UF_6

U $T_{1u}\,p_x$	ΔX_0
U $T_{1u}\,p_y$	ΔY_0
U $T_{1u}\,p_z$	ΔZ_0
Fσ $A_{1g}\,s$	$1/\sqrt{6}\,(\Delta Z_1 + \Delta X_2 + \Delta Y_3 - \Delta X_4 - \Delta Y_5 - \Delta Z_6)$
Fσ $E_g\,d_{z^2}$	$1/\sqrt{12}\,(2\Delta Z_1 - \Delta X_2 - \Delta Y_3 + \Delta X_4 + \Delta Y_5 - 2\Delta Z_6)$
Fσ $E_g\,d_{x^2-y^2}$	$1/2\,(\Delta X_2 - \Delta Y_3 - \Delta X_4 + \Delta Y_5)$
Fσ $T_{1u}\,p_x$	$1/\sqrt{2}\,(\Delta X_2 + \Delta X_4)$
Fσ $T_{1u}\,p_y$	$1/\sqrt{2}\,(\Delta Y_3 + \Delta Y_5)$
Fσ $T_{1u}\,p_z$	$1/\sqrt{2}\,(\Delta Z_1 + \Delta Z_6)$
Fπ $T_{1g}\,x$	$1/2\,(-\Delta Y_1 + \Delta Z_3 - \Delta Z_5 + \Delta Y_6)$
Fπ $T_{1g}\,y$	$1/2\,(\Delta X_1 - \Delta Z_2 - \Delta X_6 + \Delta Z_4)$
Fπ $T_{1g}\,z$	$1/2\,(\Delta Y_2 - \Delta X_3 - \Delta Y_4 + \Delta X_5)$
Fπ $T_{2g}\,d_{yz}$	$1/2\,(\Delta Y_1 + \Delta Z_3 - \Delta Z_5 - \Delta Y_6)$
Fπ $T_{2g}\,d_{xz}$	$1/2\,(\Delta X_1 + \Delta Z_2 - \Delta X_6 - \Delta Z_4)$
Fπ $T_{2g}\,d_{xy}$	$1/2\,(\Delta Y_2 + \Delta X_3 - \Delta Y_4 - \Delta X_5)$
Fπ $T_{1u}\,p_x$	$1/2\,(\Delta X_1 + \Delta X_3 + \Delta X_5 + \Delta X_6)$
Fπ $T_{1u}\,p_y$	$1/2\,(\Delta Y_1 + \Delta Y_2 + \Delta Y_4 + \Delta Y_6)$
Fπ $T_{1u}\,p_z$	$1/2\,(\Delta Z_2 + \Delta Z_3 + \Delta Z_4 + \Delta Z_5)$
Fπ $T_{2u}\,f_{x(y^2-z^2)}$	$1/2\,(-\Delta X_1 + \Delta X_3 + \Delta X_5 - \Delta X_6)$
Fπ $T_{2u}\,f_{y(z^2-x^2)}$	$1/2\,(\Delta Y_1 - \Delta Y_2 - \Delta Y_4 + \Delta Y_6)$
Fπ $T_{2u}\,f_{z(x^2-y^2)}$	$1/2\,(\Delta Z_2 - \Delta Z_3 + \Delta Z_4 - \Delta Z_5)$

cyclic permutations $f_{x(y^2-z^2)}$ and $f_{y(z^2-x^2)}$. In Table 4.9 we list all 21 symmetry coordinates by category, using labels that refer to the central harmonic functions. A short overview of the classical mechanics framework to determine molecular vibrations is presented in the box below.

Vibrational spectroscopy

A set of N vibrating atoms is characterized by 3N Cartesian displacement coordinates. Classical mechanics is applied to find the vibrational modes of the system. Let **S** be a 3N-dimensional vector of the atomic displacements. The kinetic energy is given by

$$T = \frac{1}{2}\sum_{i=1}^{3N} m_i \left(\frac{\mathrm{d}S_i}{\mathrm{d}t}\right)^2$$

Here, m_i is the atomic mass associated with the S_i coordinate. The kinetic energy can be reduced to a uniform scalar product by mass weighting the coordinates, i.e., by multiplying the S coordinates with the square root of the atomic mass of the displaced atom. We shall denote these as the vector $\mathbf{Q}$. Hence: $Q_i = \sqrt{m_i} S_i$:

$$T = \frac{1}{2} \sum_i \left(\frac{\mathrm{d}Q_i}{\mathrm{d}t} \right)^2 = \frac{1}{2} \sum_i \dot{Q}_i^2$$

where the dot over Q implies the time derivative. The potential energy at a minimal point of the energy surface is approximated as

$$V = \frac{1}{2} \sum_{i,j} V_{ij} Q_i Q_j$$

with V_{ij} an element of $V(\mathbf{Q})$ in the mass-weighted coordinates:

$$V_{ij} = \frac{\partial^2 V}{\partial Q_i \partial Q_j}$$

These derivatives are the elements of the Hessian matrix, $\mathbb{V}$, which is symmetric about the diagonal. The kinetic and potential energies are combined to form the Lagrangian, $L = T - V$. The equation of motion is given by

$$\frac{\partial L}{\partial Q_k} = \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{Q}_k}$$

The partial derivatives in this equation are given by

$$\frac{\partial L}{\partial Q_k} = -\frac{\partial V}{\partial Q_k} = -V_{kk} Q_k - \sum_{i \neq k} V_{ki} Q_i = -\sum_i V_{ki} Q_i$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{Q}_k} = \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial T}{\partial \dot{Q}_k} = \frac{\mathrm{d}}{\mathrm{d}t} \dot{Q}_k = \ddot{Q}_k$$

In the harmonic approximation the coordinates oscillate with an angular frequency ω; hence: $Q_k = Q_k^{\max} \cos \omega t$. The second derivative is then given by

$$\ddot{Q}_k = -\omega^2 Q_k = -(2\pi\nu)^2 Q_k$$

where ν is the vibrational frequency in Hertz. The equation of motion then is turned into a set of homogeneous linear equations:

$$\forall k \ : \ \sum_i \left(V_{ki} - \delta_{ki}\omega^2\right) Q_i = 0$$

This set of equations is solved in the standard way by diagonalizing the Hessian matrix, as

$$|\mathbb{V} - \omega^2 \mathbb{I}| = 0$$

The eigenvalues of the secular equation yield the frequencies of the normal modes, which are usually expressed as wavenumbers, $\overline{\nu}$, preferentially in reciprocal centimeters, cm^{-1} by dividing the frequency by the speed of light, c.

$$\overline{\nu} = \frac{\nu}{c} = \frac{\omega}{2\pi c}$$

The corresponding eigenfunctions are the actual normal modes.

When symmetry is present, it is useful to replace the original basis set of atomic displacements with the SALC basis. The vector **S** thus henceforth consists of the 21 symmetry coordinates of Table 4.9. In this basis, the kinetic energy is given by

$$T = \frac{M}{2} \sum_{i=1}^{3} \left(\frac{\mathrm{d}S_i}{\mathrm{d}t}\right)^2 + \frac{m}{2} \sum_{i=4}^{21} \left(\frac{\mathrm{d}S_i}{\mathrm{d}t}\right)^2 \tag{4.98}$$

Here, M is the atomic mass of Uranium, and m is the atomic mass of Fluorine. As a result of rewriting the displacement basis in SALC format the Hessian matrix will be block diagonal over the irreps of the group, and, within each irrep, over the individual components of the irrep. Moreover, the blocks are independent of the components. All this illustrates the power of symmetry, and the reasons for it will be explained in detail in the next chapter. As an immediate consequence, symmetry coordinates, which belong to irreps that occur only once, are exact normal modes of the Hessian. Five irreps fulfill this criterion: the T_{1g} mode, which corresponds to the overall rotations, and the vibrational modes: $A_{1g} + E_g + T_{2g} + T_{2u}$. Only the T_{1u} irrep gives rise to a triple multiplicity. In this case, the actual normal modes will depend on the matrix elements in the Hessian.

In Table 4.10 we present the experimental results [7] for $U^{238}F_6$, as compared with the Hessian eigenvalues, based on extensive relativistic calculations [8]. To further illustrate the tools, that are provided by the SALC basis, we analyze in more detail the three $T_{1u}z$-components, which we shall abbreviate as follows:

$$\begin{aligned} Q_{\mathrm{U}} &= \sqrt{M}\,\Delta Z_0 \\ Q_\sigma &= \frac{\sqrt{m}}{\sqrt{2}} \left(\Delta Z_1 + \Delta Z_6\right) \\ Q_\pi &= \frac{\sqrt{m}}{2} \left(\Delta Z_2 + \Delta Z_3 + \Delta Z_4 + \Delta Z_5\right) \end{aligned} \tag{4.99}$$

Table 4.10 Vibrational spectrum of UF_6

Symmetry	Type	$\overline{\nu}\,(\mathrm{cm}^{-1})$	Calc.	Technique
$\nu_1(A_{1g})$	Breathing	667	669	Raman (very strong)
$\nu_2(E_g)$	Stretching	533	534	Raman (weak)
$\nu_3(T_{1u})$	Stretching	626	624	IR
$\nu_4(T_{1u})$	Bending	186	181	IR
$\nu_5(T_{2g})$	Bending	202	191	Raman (weak)
$\nu_6(T_{2u})$	Buckling	142	140	Overtone

This space is still reducible, since it includes the translation in the z-direction. The translation coordinate corresponds to the displacement of the center of mass in the z-direction. It is given by $\sum_i m_i \Delta Z_i$, which can be expressed as follows:

$$\begin{aligned} T_z &= M\Delta Z_0 + m\,(\Delta Z_1 + \Delta Z_2 + \Delta Z_3 + \Delta Z_4 + \Delta Z_5 + \Delta Z_6) \\ &= \sqrt{M}\,Q_{\mathrm{U}} + \sqrt{2\,m}\,Q_\sigma + \sqrt{4\,m}\,Q_\pi \end{aligned} \tag{4.100}$$

We can remove this degree of freedom from the function space by a standard orthogonalization procedure. One option is to construct first a pure stretching mode, which does not involve the Q_π coordinate. This mode is denoted by $|\sigma_1\rangle$. The remainder of the function space, which is orthogonal both to the translation and to this pure stretching mode, is then denoted by $|\pi_1\rangle$. Normalizing these modes with respect to mass-weighted coordinates yields

$$\begin{aligned} |\sigma_1\rangle &= \frac{-\sqrt{2\,m}\,Q_{\mathrm{U}} + \sqrt{M}\,Q_\sigma}{\sqrt{M+2m}} \\ |\pi_1\rangle &= \frac{-\sqrt{4mM}\,Q_{\mathrm{U}} - m\sqrt{8}\,Q_\sigma + (M+2m)\,Q_\pi}{\sqrt{(M+2m)(M+6m)}} \end{aligned} \tag{4.101}$$

An alternative option would be to construct a pure bending mode, based on the tangent motions. Let us denote this by $|\pi_2\rangle$. The remainder is then denoted by $|\sigma_2\rangle$.

$$\begin{aligned} |\sigma_2\rangle &= \frac{-\sqrt{2mM}\,Q_{\mathrm{U}} + (M+4m)\,Q_\sigma - m\sqrt{8}\,Q_\pi}{\sqrt{(M+4m)(M+6m)}} \\ |\pi_2\rangle &= \frac{-\sqrt{4m}\,Q_{\mathrm{U}} + \sqrt{M}\,Q_\pi}{\sqrt{M+4m}} \end{aligned} \tag{4.102}$$

In Fig. 4.8 we present both choices of bases. The angle, α, between both basis sets is defined by

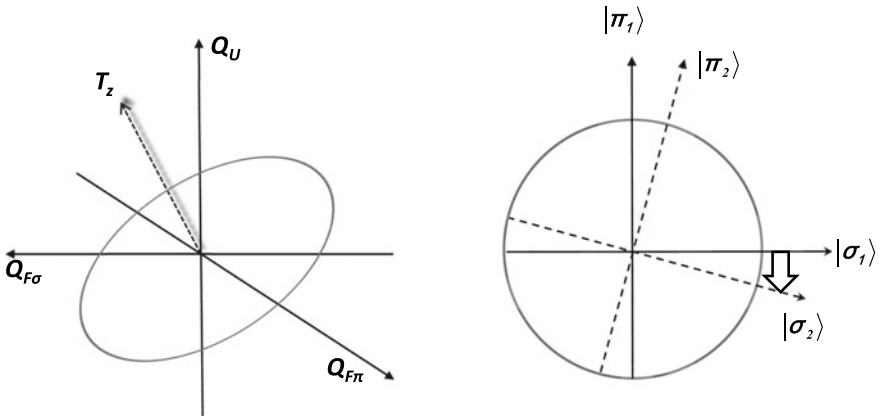


Fig. 4.8 T_{1u} distortion space for UF_6, with coordinates as defined in Eq. (4.99); T_z is the translation of mass. The circle, perpendicular to this direction, is the space of vibrational stretching and bending, with coordinates defined in Eqs. (4.101) and (4.102). The angle $\langle\sigma_2|\sigma_1\rangle$ is $-10.5°$

$$\cos\alpha = \sqrt{\frac{M(M+6m)}{(M+2m)(M+4m)}} \tag{4.103}$$

In the case of UF_6 ($m = 18.998$, $M = 238.050$) this angle is $-10.5°$. The actual eigenmodes are found by setting up the Hessian in one of these coordinate sets and diagonalizing it. This Hessian matrix is symmetric, and thus contains three independent parameters: the two diagonal elements, and the single off-diagonal element. On the other hand, there are only two observable eigenfrequencies: the sum of the eigenvalues is equal to the trace of the matrix, and their product is equal to its determinant. This leaves still one degree of freedom, which is associated with the *composition* of the normal mode, *viz.*, the angle of rotation in the diagram. It is important to realize that this composition also shows up in observables, albeit not the eigenfrequencies, but a variety of other properties, such as the intensities of the vibrational transition, isotope shifts and isotope splittings, or electron diffraction amplitudes. For most octahedral complexes, as in the case of UF_6, the rotation angle for the actual T_{1u} eigenmodes lies in the interval $[0, \alpha]$. This means that the modes may approximately be assigned as a stretching and a bending mode. In the spectrum their frequencies are denoted as ν_3 and ν_4, respectively. The isotope effect of the radioactive nucleus U^{235}, as distinct from U^{238}, is absent for all modes, except for the T_{1u} modes, since these involve the displacement of Uranium. The strongest effect is expected for the stretching vibration, ν_3, since it involves the largest displacement of the central atom. The pure stretching mode, $|\sigma_1\rangle$, can be expressed in terms of the displacements along the z-direction as

$$|\sigma_1\rangle = \sqrt{\frac{2mM}{M+2m}}\left(-\Delta Z_0 + \frac{\Delta Z_1 + \Delta Z_6}{2}\right) \tag{4.104}$$

This is precisely the anti-symmetric mode for a triatomic F–U–F oscillator. The square root preceding the modes corresponds to a mass weighting by the reduced mass, μ, for such an oscillator.

$$\mu = \left(\frac{1}{M} + \frac{1}{2m}\right)^{-1} = \frac{2mM}{M+2m} \tag{4.105}$$

Substitution of U^{238} by the U^{235} isotope will reduce this effective mass by a factor 0.9982. The frequency is accordingly increased by the square root of this factor. This gives an increase of frequency of 0.56 cm^{-1}, which is close to the experimental value [9] of 0.60 cm^{-1}. This confirms the dominant stretching character of the ν_3 mode.

What have we learned from this example? The Hessian matrix is block diagonal over the irreps of the point group, and, as a result, the normal modes are characterized by symmetry labels. These labels are characteristics of the exact vibrational wave function. Vibrational IR and Raman spectra are directly linked to these labels. This is due to the presence of symmetry-based selection rules, which will be explained in detail in Chap. 6. In the long run the relevance of symmetry as an exact quantum property will be vastly more important than the temporary gain in computational time for evaluation and diagonalization of the Hessian matrix.

4.9 Application: Hückel Theory

4.9.1 Introduction

The Hückel model for the chemist (or the analogous tight-binding model for the condensed-matter physicist) is an extremely simplified molecular orbital model [10], which nevertheless continues to play an important role in our understanding of electronic structures and properties. It emphasizes the molecule–graph analogy and uses what is now regarded as spectral graph theory [11, 12] in order to obtain molecular orbitals. Its strength comes from the fact that in spite of the approximations involved, it incorporates the essential topological and symmetry aspects of electronic structures, and, as we keep repeating, these are simple but exact properties of complex molecular quantum systems.

Hückel Hamiltonian
Consider molecular systems, where each atom or node carries one atomic orbital, say $|\phi_i\rangle$. Molecular orbitals will be denoted as $|\Phi_k\rangle$:

$$|\Phi_k\rangle = \sum_i c_i^k |\phi_i\rangle$$

The Hamiltonian matrix in Hückel theory is given by

$$\langle\phi_i|\mathcal{H}|\phi_j\rangle = \alpha\delta_{ij} + \beta A_{ij}$$

or in operator form:

$$\mathcal{H} = \sum_i \alpha|\phi_i\rangle\langle\phi_i| + \sum_{i\neq j} \beta A_{ij}|\phi_i\rangle\langle\phi_j|$$

with

- α: on-site interaction element (so-called Coulomb integral)
- β: inter-site hopping or resonance integral
- $\mathbb{A}$: adjacency matrix of the molecular graph. ($A_{ij} = 1$ if nodes i and j are neighboring, otherwise $A_{ij} = 0$).

The α-parameter corresponds to the zero-point of energy, and thus has only a symbolic significance in homogeneous systems. However, in hetero-atomic systems, it is important to differentiate the atoms. As an example, the Coulomb integral for nitrogen will be more negative than the one for carbon, because the heavier nitrogen nucleus exerts a greater attraction on the electrons. The β-parameter represents the bonding interaction between neighboring atoms and thus will be negative. The Hückel method is especially suited for the study of conjugated hydrocarbons. Figure 4.9 shows the spectrum for the prototypal ethylene, C_2H_4, and for butadiene, C_4H_6, as an example of a small conjugated chain molecule. The Hamiltonian eigenvalues and eigenfunctions for the π-bond in ethylene are given by

$$\begin{aligned} E(\pi) &= \alpha + \beta \rightarrow |\Phi_\pi\rangle = \frac{1}{\sqrt{2}}\left(|p_1\rangle + |p_2\rangle\right) \\ E(\pi^*) &= \alpha - \beta \rightarrow |\Phi_{\pi^*}\rangle = \frac{1}{\sqrt{2}}\left(|p_1\rangle - |p_2\rangle\right) \end{aligned} \tag{4.106}$$

For trans-butadiene the molecular-symmetry group is C_{2h}, and the four atomic p_z orbitals transform as $2A_u + 2B_g$. The Hückel matrix thus can be separated in a A_u and a B_g block. The spectrum is described by

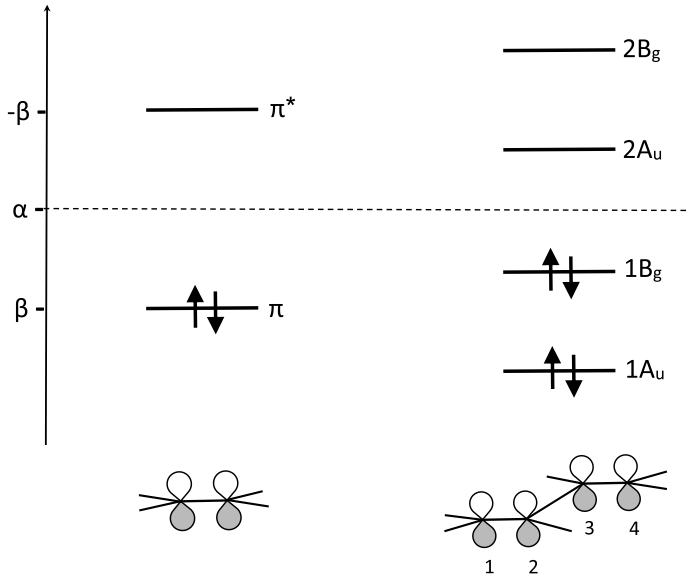


Fig. 4.9 Hückel orbital energy spectrum of ethylene and butadiene

$$
\begin{aligned}
E(1A_u) &= \alpha + \varphi\beta \rightarrow |\Phi_{1A_u}\rangle = \frac{1}{\sqrt{2+2\varphi^2}}\left(|p_1\rangle + \varphi|p_2\rangle + \varphi|p_3\rangle + |p_4\rangle\right)\\
E(1B_g) &= \alpha + \varphi^{-1}\beta \rightarrow |\Phi_{1B_g}\rangle = \frac{1}{\sqrt{2+2\varphi^2}}\left(\varphi|p_1\rangle + |p_2\rangle - |p_3\rangle - \varphi|p_4\rangle\right)\\
E(2A_u) &= \alpha - \varphi^{-1}\beta \rightarrow |\Phi_{2A_u}\rangle = \frac{1}{\sqrt{2+2\varphi^2}}\left(\varphi|p_1\rangle - |p_2\rangle - |p_3\rangle + \varphi|p_4\rangle\right)\\
E(2B_g) &= \alpha - \varphi\beta \rightarrow |\Phi_{2B_g}\rangle = \frac{1}{\sqrt{2+2\varphi^2}}\left(|p_1\rangle - \varphi|p_2\rangle + \varphi|p_3\rangle - |p_4\rangle\right)
\end{aligned}
\tag{4.107}
$$

Here φ is the golden ratio, $(1+\sqrt{5})/2$, and normalization does not take into account inter-site overlap integrals. Cis-butadiene is characterized by the same Hückel Hamiltonian, but its point group is C_{2v}. From the eigenfunctions one can determine π-contributions to several properties such as the on-site atomic population, q_r, and the inter-site π-bond order, p_{rs}. The population of atom r, and the bond order of the bond between atoms r and s are given by

$$
\begin{aligned}
q_r &= \sum_k n_k |c_r^k|^2\\
p_{rs} &= \sum_k n_k \overline{c_r^k} c_s^k
\end{aligned}
\tag{4.108}
$$

Here, the index k runs over the eigenfunctions: n_k is the occupation number (0, 1, or 2) of the kth eigenlevel, and c_r^k is the coefficient of the $|\phi_r\rangle$ atomic orbital for the normalized eigenfunction. The atomic populations are simply the densities or weights at the atomic sites and may vary between 0 and 2. The neutral atom has a population of one p_z-electron, sites with $q_r < 1$ are cationic, with $q_r > 1$ anionic. The bond order adopts the form of a correlation coefficient between two sites. The π-bond order for a full π-bond in ethylene is equal to 1. In butadiene, the order of the central bond is 0.477, versus 0.894 for the terminal bonds. This difference is reflected in the bond lengths of 1.46 Å and 1.35 Å, respectively. In the case of 1D conjugated chains molecular symmetry does not seem to be of great importance as there is only one essential binary symmetry element in the center of the chain. Surprisingly though, the actual symmetry of the Hückel Hamiltonian for the linear chain is much higher, as will be discussed in Study Problem 4.13 at the end of this chapter.

4.9.2 Cyclic Polyenes

Cyclic polyenes, also known as *annulenes*, are hydrocarbon rings, C_nH_n. Each carbon atom contributes one p_z-orbital, perpendicular to the plane of the ring, which gives rise to conjugated π-bonding. The prototype is the aromatic molecule benzene. The adjacency matrix has the form of a *circulant* matrix. This is a matrix where each row is rotated one element to the right relative to the preceding row. Because each atom is linked to only two neighbors each row contains only two elements. These are arranged left and right of the diagonal, which is characteristic for a chain, but with additional non-zero elements in the upper right and lower left corners, where both ends of the chain meet. For benzene it is given by

$$\mathbb{A} = \begin{pmatrix} 0\,1\,0\,0\,0\,1 \\ 1\,0\,1\,0\,0\,0 \\ 0\,1\,0\,1\,0\,0 \\ 0\,0\,1\,0\,1\,0 \\ 0\,0\,0\,1\,0\,1 \\ 1\,0\,0\,0\,1\,0 \end{pmatrix} \tag{4.109}$$

The symmetry of an N-atom ring is D_{Nh}, but in practice the cyclic group C_N is sufficient to solve the eigenvalue problem. Atoms are numbered from 0 to $N-1$. The cyclic projection operator,[2] $\hat{P}_k$, is given by

$$\hat{P}_k = \frac{1}{N} \sum_{j=0}^{N-1} \exp\left(2\pi i \frac{jk}{N}\right) \hat{C}_N^j \tag{4.110}$$

[2] Equation (4.110) is based on $D^k_{1,1}(C_N) = \exp(-2\pi ik/N)$. See also: Eq. (4.18).

Projectors are characterized by an integer k in a periodic interval. We may choose the range $]-N/2, +N/2]$ as the standard interval. The total number of integers in this interval is N. Keeping in mind the active view, where the rotation axis will rotate all the orbitals one step further in a counterclockwise way, we now act with the projection operator on the starting orbital, $|\phi_0\rangle$.

$$\hat{P}_k|\phi_0\rangle = \frac{1}{N}\sum_{j=0}^{N-1} \exp(2\pi i \frac{jk}{N})|\phi_j\rangle \tag{4.111}$$

The result is an unnormalized SALC. Neglecting overlap between adjacent atoms, we obtain the normalized SALC:

$$|\Phi_k\rangle = \sqrt{N}\,\hat{P}_k|\phi_0\rangle \tag{4.112}$$

The transformation properties of this SALC under the rotation axis are characterized as

$$\begin{aligned}\hat{C}_N|\Phi_k\rangle &= \frac{1}{\sqrt{N}}\sum_{j=0}^{N-1} \exp(2\pi i \frac{jk}{N})|\phi_{j+1}\rangle \\ &= \frac{1}{\sqrt{N}}\sum_{j=0}^{N-1} \exp(2\pi i \frac{(j-1)k}{N})|\phi_j\rangle \\ &= \exp(-2\pi i \frac{k}{N})|\Phi_k\rangle \end{aligned} \tag{4.113}$$

Applying this symmetry element N times is identical to the unit operation, and raises the exponential factor in this expression to the Nth power.

$$\left(\exp(-2\pi i \frac{k}{N})\right)^N = \exp(-2\pi i\, k) = 1 \tag{4.114}$$

Each integer value of k in the periodic interval $[-N/2, +N/2]$ thus characterizes a different SALC. The corresponding energy eigenvalues are also easily extracted:

$$\begin{aligned} E_k &= \langle\Phi_k|\mathcal{H}|\Phi_k\rangle \\ &= \frac{1}{N}\sum_{j,j'=0}^{N-1} \exp(-2\pi i \frac{k(j-j')}{N})\langle\phi_j|\mathcal{H}|\phi_{j'}\rangle \\ &= \frac{1}{N}\sum_{j=0}^{N-1}\Big(\alpha + \beta\big[\exp(-2\pi i k/N) + \exp(+2\pi i k/N)\big]\Big) \\ &= \alpha + 2\beta\cos(2\pi k/N) \end{aligned} \tag{4.115}$$

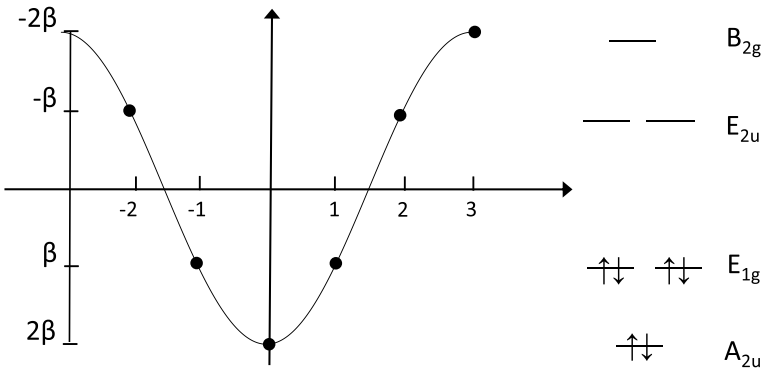


Fig. 4.10 Hückel orbital energy spectrum of benzene, as a function of the index k, with allowed values $0, \pm 1, \pm 2, 3$

The energies are thus seen to form N discrete levels, which are points on a cosine curve, as shown in Fig. 4.10. Except for $k = 0$ and, in the case of N even, $k = N/2$, all levels E_k and E_{-k} are twofold degenerate. Closed-shell structures thus will be realized for $N = 4n + 2$, which is the famous Hückel condition for aromaticity. The bond order in benzene is uniformly equal to 2/3, and all bond lengths are the same. The cyclic labels can be expanded to the full irrep designations of the D_{6h} symmetry group for benzene. The atomic p_z-orbitals transform as b_1 in the C_{2v} site group. In accord with the conventions for the D_{6h} point group symmetry, as pictured in Fig. 3.10, this site group is based on operators of type $\hat{C}_2'$ and $\hat{\sigma}_v$. The induced irrep of the six atomic orbitals then becomes

$$\Gamma(b_1 C_{2v} \uparrow D_{6h}) = A_{2u} + E_{1g} + E_{2u} + B_{2g} \tag{4.116}$$

Since each irrep occurs only once there is a one-to-one correlation between these irreps and the cycle index k, which can be retrieved from the $D_{6h} \downarrow C_6$ subduction rules:

$$\begin{aligned} A_{2u} &\longrightarrow A \ (k = 0) \\ E_{1g} &\longrightarrow E_1 \ (k = \pm 1) \\ E_{2u} &\longrightarrow E_2 \ (k = \pm 2) \\ B_{2g} &\longrightarrow B \ (k = 3) \end{aligned} \tag{4.117}$$

The diagram in Fig. 4.10 has also a broader significance, as it corresponds to a discrete representation of the continuous band structure in 1D extended chains. This connection will be discussed in detail in Chap. 8.

4.9.3 Zeeman Effect in Cyclic Polyenes

We will now engage in a more elaborate application of Hückel theory, which demonstrates the power of the model. The purpose is to determine the energy shifts of the eigenvalues when an annulene is brought into a uniform magnetic field, **B**. This field is independent of position and time. It can be defined as the 'curl' (or rotation) of a vector potential **A** and, in terms of a position vector **r** from a given origin, the relevant relations are as follows:

$$\begin{aligned} \mathbf{B} &= \boldsymbol{\nabla} \wedge \mathbf{A} \\ \mathbf{A} &= \frac{1}{2}\mathbf{B} \wedge \mathbf{r} \end{aligned} \tag{4.118}$$

This implies that the divergence of the vector potential is zero, hence **A** and $\boldsymbol{\nabla}$ commute: $[\boldsymbol{\nabla}, \mathbf{A}] = \mathbf{0}$. The introduction of the magnetic field will add an extra term to the kinetic energy operator, which becomes

$$\begin{aligned} T &= \frac{1}{2m}\left(\frac{\hbar}{i}\boldsymbol{\nabla} + \mathrm{e}\mathbf{A}\right)^2 \\ &= -\frac{h^2}{8\pi^2 m}\left(\boldsymbol{\nabla} + i\frac{\mathrm{e}}{\hbar}\mathbf{A}\right)^2 \\ &= -\frac{h^2}{8\pi^2 m}\left(\Delta + i\frac{\mathrm{e}}{\hbar}\mathbf{A}\cdot\boldsymbol{\nabla} + i\frac{\mathrm{e}}{\hbar}\boldsymbol{\nabla}\cdot\mathbf{A} - \frac{\mathrm{e}^2}{\hbar^2}A^2\right) \\ &= -\frac{h^2}{8\pi^2 m}\left(\Delta + 2i\frac{\mathrm{e}}{\hbar}\mathbf{A}\cdot\boldsymbol{\nabla} - \frac{\mathrm{e}^2}{\hbar^2}A^2\right) \end{aligned} \tag{4.119}$$

where we have taken into account that the 'del' (or nabla) operator and the vector potential commute. The electron charge is $-\mathrm{e}$. London proposed that the atomic basis functions should be multiplied by a phase factor, which explicitly depends on the vector potential [13]. In this London gauge the atomic orbitals are rewritten as

$$|\chi_j\rangle = \exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \mathbf{r})|\phi_j\rangle \tag{4.120}$$

where $\mathbf{A}_j$ is the vector potential at the position of the jth atom. The effect of this phase factor is to move the origin of the vector potential from an arbitrary origin to the local position of atom j. The action of the del operator and Laplacian on this gauge is given by

$$\begin{aligned} \boldsymbol{\nabla}\exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \mathbf{r}) &= \exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \mathbf{r})\left(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j + \boldsymbol{\nabla}\right) \\ \Delta\exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \mathbf{r}) &= \exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \mathbf{r})\left(-\frac{\mathrm{e}^2}{\hbar^2}A_j^2 - 2i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \boldsymbol{\nabla} + \Delta\right) \end{aligned} \tag{4.121}$$

Combining this result with Eqs. (4.119) and (4.120) yields

$$T|\chi_j\rangle = -\frac{h^2}{8\pi^2 m}\exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j\cdot\mathbf{r})\times \left[\Delta + 2i\frac{\mathrm{e}}{\hbar}\left(\mathbf{A}-\mathbf{A}_j\right)\cdot\boldsymbol{\nabla} - \frac{\mathrm{e}^2}{\hbar^2}(\mathbf{A}-\mathbf{A}_j)\cdot(\mathbf{A}-\mathbf{A}_j)\right]|\phi_j\rangle \tag{4.122}$$

The first term in the brackets is the usual kinetic energy term while the second term produces the orbital Zeeman effect. The third term describes the second-order interactions, corresponding to the atomic contribution to the susceptibility. The second term can easily be converted into the more familiar form of the Zeeman operator as follows:

$$\begin{aligned}\frac{\mathrm{e}}{m}\left(\mathbf{A}-\mathbf{A}_j\right)\cdot\left(\frac{\hbar}{i}\boldsymbol{\nabla}\right) &= \frac{\mathrm{e}}{2m}\mathbf{B}\cdot\left[\left(\mathbf{r}-\mathbf{R}_j\right)\wedge\mathbf{p}\right]\\ &= \frac{\mathrm{e}}{2m}\boldsymbol{l}\cdot\mathbf{B}\\ &= -\boldsymbol{m}\cdot\mathbf{B}\end{aligned} \tag{4.123}$$

Here, $\mathbf{p}$ is the momentum operator of the electron in atom j, $\boldsymbol{l}$ is the corresponding angular momentum operator, and $\boldsymbol{m}$ is the magnetic dipole operator. These operators are related by

$$\boldsymbol{m} = -\frac{\mathrm{e}}{2m}\boldsymbol{l} = -\frac{\mu_\mathrm{B}}{\hbar}\boldsymbol{l} \tag{4.124}$$

Here μ_B is the Bohr magneton. Angular momentum is thus expressed in units of $\hbar$, and the magnetic moment in units of the Bohr magneton. The basis atomic orbitals will be eigenfunctions of the first two operators. So to first order the London basis orbitals are eigenfunctions of the total Hamiltonian. Moreover, for a p_z-orbital the Zeeman effect for a magnetic field along the z-axis vanishes. As a result the on-site parameter α will be independent of the London gauge:

$$\langle\chi_j|\mathcal{H}|\chi_j\rangle = \alpha\langle\chi_j|\chi_j\rangle = \alpha \tag{4.125}$$

However the inter-site integrals, which depend on the potential energy, V, are influenced by the gauge factors:

$$\langle\chi_i|V|\chi_j\rangle = \langle\phi_i|V\exp(i\frac{\mathrm{e}}{\hbar}(\mathbf{A}_i-\mathbf{A}_j)\cdot\mathbf{r})|\phi_j\rangle \tag{4.126}$$

At this point London introduced an important approximation by replacing the variable position vector in this equation by the position vector (relative to the arbitrary origin) of the center of the bond between the two atoms:

$$\mathbf{r} = \left(\mathbf{R}_i+\mathbf{R}_j\right)/2 \tag{4.127}$$

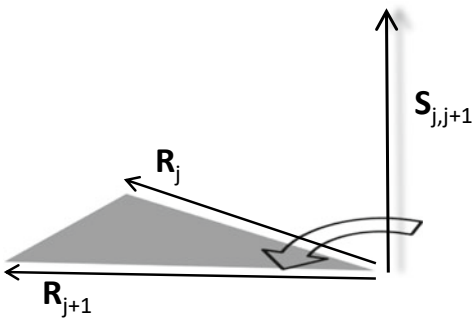


Fig. 4.11 Triangular surface vector: $\mathbf{S}_{j,j+1} = \frac{1}{2}\,\mathbf{R}_j \wedge \mathbf{R}_{j+1}$

In this approximation the phase factor is turned into a constant which can be removed from the brackets. One has

$$\begin{aligned}\frac{1}{2}(\mathbf{A}_i - \mathbf{A}_j)\cdot\left(\mathbf{R}_i + \mathbf{R}_j\right) &= {}^1\!/_4\left[(\mathbf{B}\wedge\mathbf{R}_i)\cdot\mathbf{R}_j - (\mathbf{B}\wedge\mathbf{R}_j)\cdot\mathbf{R}_i\right]\\ &= {}^1\!/_2\,\mathbf{B}\cdot(\mathbf{R}_i\wedge\mathbf{R}_j)\\ &= \mathbf{B}\cdot\mathbf{S}_{ij}\end{aligned} \tag{4.128}$$

Here $\mathbf{S}_{ij}$ is the directed area of the triangle formed by the position vectors of the atoms i and j from the origin, as shown in Fig. 4.11. The orientation of the vector $\mathbf{S}_{ij}$ follows the right thumb rule. So if the atom numbers increase counterclockwise, this vector will be oriented along the positive z-direction. One has also

$$\mathbf{S}_{ij} = -\mathbf{S}_{ji} \tag{4.129}$$

The interaction elements in the Hückel matrix are thus replaced by

$$\begin{aligned}\langle\chi_i|V|\chi_j\rangle &= \exp(i\frac{\mathrm{e}}{\hbar}\mathbf{B}\cdot\mathbf{S}_{ij})\beta\\ \langle\chi_j|V|\chi_i\rangle &= \exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{B}\cdot\mathbf{S}_{ij})\beta\end{aligned} \tag{4.130}$$

We further define a vector $\mathbf{S}$ as

$$\sum_{j=0}^{N-1}\mathbf{S}_{j,j+1} = \mathbf{S} \tag{4.131}$$

The magnitude of this vector is equal to the area of the polygon. Because of cyclic symmetry, we can also write

$$\mathbf{S}_{j,j+1} = \frac{1}{N}\mathbf{S} \tag{4.132}$$

The action of the symmetry operators on the London gauge is as follows:

$$\hat{C}_N \exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \mathbf{r}) = \exp\left[-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_j \cdot \left(\hat{C}_N^{-1}\mathbf{r}\right)\right]$$
$$= \exp(-i\frac{\mathrm{e}}{\hbar}\mathbf{A}_{j+1} \cdot \mathbf{r}) \tag{4.133}$$

which may easily be proven by writing out $\mathbf{A}_j$ and $\hat{C}_N^{-1}\mathbf{r}$ in full. Hence, the rotation axis will perform a cyclic permutation of the $|\chi_j\rangle$ kets, exactly in the same way as for the $|\phi_j\rangle$ kets. The magnetic field reduces the D_{nh} symmetry of the regular polygon to C_{nh} (see Appendix B), so the cyclic symmetry is conserved and thus the projection operators of Eq. (4.110) remain valid, and so do the eigenfunctions. We shall denote these as $|\Psi_k\rangle$. One has

$$|\Psi_k\rangle = \frac{1}{\sqrt{N}}\sum_{j=0}^{N-1}\exp(2\pi i\frac{jk}{N})|\chi_j\rangle \tag{4.134}$$

The corresponding eigenvalues can be worked out in the same way as in the absence of the field (see Eq. (4.115)).

$$\begin{aligned}
E_k &= \langle\Psi_k|\mathcal{H}|\Psi_k\rangle \\
&= \frac{1}{N}\sum_{j,j'=0}^{N-1}\exp(2\pi i\frac{k(-j+j')}{N})\langle\chi_j|\mathcal{H}|\chi_{j'}\rangle \\
&= \alpha + \frac{1}{N}\sum_{j=0}^{N-1}\beta\left[\exp\left(-\frac{2\pi ik}{N} + i\frac{\mathrm{e}}{\hbar}\mathbf{B}\cdot\mathbf{S}_{j,j-1}\right) + \exp\left(\frac{2\pi ik}{N} + i\frac{\mathrm{e}}{\hbar}\mathbf{B}\cdot\mathbf{S}_{j,j+1}\right)\right] \\
&= \alpha + 2\beta\cos\left[\frac{2\pi}{N}(k + \frac{\mathrm{e}}{\hbar}\mathbf{B}\cdot\mathbf{S})\right]
\end{aligned} \tag{4.135}$$

The result is of the same form as in the absence of the field, except for a shift of the quantum number k under the influence of the magnetic field. The magnitude of this shift, $\mathrm{e}/\hbar\ \mathbf{B}\cdot\mathbf{S}$, is equal to the magnetic flux through the area of the ring, multiplied by the constant $\mathrm{e}/\hbar$. As a result of this shift, the energy levels which were originally degenerate now display a Zeeman splitting. For $\mathbf{B}\cdot\mathbf{S} > 0$, the Zeeman contribution adds to k in the energy expression. This implies that the points on the k axis in Fig. 4.10 are displaced to the right. The roots with $k = 1, 2$ thus increase in energy, while their counterparts, $k = -1, -2$, become lower in energy. Likewise, the root at the bottom ($k = 0$) will increase in energy, while the root at the top, $k = 3$, will decrease, but the changes in these extremal points are only of second order.

This simplified model is at the basis of a whole corpus of electromagnetic studies of conjugated polyenes, involving, *inter alia*, the calculation of magnetic susceptibilities, current densities, ring currents, and chemical shifts in nuclear magnetic resonance (NMR). From the point of view of symmetry, it is to be noted that the magnetic field has removed all degeneracies. The time-reversal symmetry is indeed no longer valid. However, if one reverses the momenta, $k \to -k$, and at the same time

switches the magnetic field, $\mathbf{B} \to -\mathbf{B}$, the energies are still invariant. This operation is no longer an invariance operation of one measurement though, but a comparison between two separate experiments, with opposite fields.

4.9.4 Polyhedral Hückel Systems of Equivalent Atoms

The polygonal system of the annulenes can be extended to polyhedral systems of equivalent atoms. Atoms are equivalent if the symmetry group of the molecule—or more generally the automorphism group of the molecular graph—acts transitively on the set of atomic nodes. So for any pair of atoms, $\langle k\rangle\langle l\rangle$, there is a symmetry element that will map $\langle k\rangle$ onto $\langle l\rangle$, and then of course there is always an inverse element that will map $\langle l\rangle$ onto $\langle k\rangle$. The solution of the Hamiltonian matrix for such a system can almost entirely be performed by symmetry arguments. This was first demonstrated in 1896 by Frobenius in his derivation of the 'group determinant' [14]. Almost a century later it was rediscovered independently by Samuel [15], who called this the method of the group lattice. The first step consists in the construction of SALCs, using the projection operators on the atomic orbital on site $\langle a\rangle$.

$$|\Phi_{ik}^{\Omega}\rangle = \hat{P}_{ik}^{\Omega}|\phi_{\mathrm{A}}\rangle = \frac{\dim(\Omega)}{|G|}\sum_{R\in G}\overline{D}_{ik}^{\Omega}(R)\hat{R}|\phi_{\mathrm{A}}\rangle \tag{4.136}$$

These functions are not yet normalized. This can be done later. Let us first consider a general matrix element.

$$\begin{aligned}
\langle\Phi_{ik}^{\Omega}|\mathcal{H}|\Phi_{jl}^{\Omega'}\rangle &= \frac{\dim(\Omega)^2}{|G|^2}\sum_{R,S}D_{ik}^{\Omega}(R)\overline{D}_{jl}^{\Omega'}(S)\langle\hat{R}\phi_{\mathrm{A}}|\mathcal{H}|\hat{S}\phi_{\mathrm{A}}\rangle \\
&= \frac{\dim(\Omega)^2}{|G|^2}\sum_{R,Q}D_{ik}^{\Omega}(R)\overline{D}_{jl}^{\Omega'}(RQ)\langle\hat{R}\phi_{\mathrm{A}}|\mathcal{H}|\hat{RQ}\phi_{\mathrm{A}}\rangle \\
&= \frac{\dim(\Omega)^2}{|G|^2}\sum_{R,Q}\sum_{m}D_{ik}^{\Omega}(R)\overline{D}_{jm}^{\Omega'}(R)\overline{D}_{ml}^{\Omega'}(Q)\langle\phi_{\mathrm{A}}|\hat{R}^{-1}\mathcal{H}\hat{R}|\hat{Q}\phi_{\mathrm{A}}\rangle \\
&= \frac{\dim(\Omega)}{|G|}\delta_{\Omega\Omega'}\delta_{ij}\delta_{km}\sum_{Q}\overline{D}_{ml}^{\Omega'}(Q)\langle\phi_{\mathrm{A}}|\hat{R}^{-1}\mathcal{H}\hat{R}|\hat{Q}\phi_{\mathrm{A}}\rangle \\
&= \frac{\dim(\Omega)}{|G|}\delta_{\Omega\Omega'}\delta_{ij}\sum_{Q}\overline{D}_{kl}^{\Omega}(Q)\langle\phi_{\mathrm{A}}|\mathcal{H}|\hat{Q}\phi_{\mathrm{A}}\rangle
\end{aligned} \tag{4.137}$$

The by-now experienced reader will have recognized in the second line of this derivation the use of a substitution, $\hat{S} \to \hat{R}\hat{Q}$, as well as the invariance of the Hamiltonian under the symmetry transformation in the third line. Let us now use this equation to normalize the SALCs. This can be done by simply setting the Hamiltonian equal to

unity. Adopting the Hückel approximation, which neglects all overlaps between the sites, we obtain

$$\langle \Phi_{ik}^{\Omega} | \Phi_{ik}^{\Omega} \rangle = \frac{\dim(\Omega)}{|G|} \sum_{Q} \overline{D}_{kk}^{\Omega}(Q) \delta_{Q,E} = \frac{\dim(\Omega)}{|G|} \tag{4.138}$$

Hence, the normalized SALCs should be redefined as

$$|\Phi_{ik}^{\Omega}\rangle = \sqrt{\frac{\dim(\Omega)}{|G|}} \sum_{R \in G} \overline{D}_{ik}^{\Omega}(R) \hat{R} |\phi_{\mathrm{A}}\rangle \tag{4.139}$$

The matrix elements are accordingly simplified to

$$\langle \Phi_{ik}^{\Omega} | \mathcal{H} | \Phi_{il}^{\Omega} \rangle = \sum_{Q} \overline{D}_{kl}^{\Omega}(Q) \langle \phi_{\mathrm{A}} | \mathcal{H} | \hat{Q} \phi_{\mathrm{A}} \rangle \tag{4.140}$$

The Hückel approximation considers only hopping between nearest neighbors, hence the sum in this expression will be limited to only those operators that link the starting site to its neighbors. We thus do not need all the irrep matrices, but only a few of them. As an example, the cube is a trivalent polyhedron, i.e., each atom has three bonds to its neighbors. The neighbors of site $\langle a \rangle$ can be reached by $\hat{C}_4, \hat{C}_4^{-1}, \hat{C}_2^{xy}$, where the twofold axis belongs to the $6\hat{C}_2'$ class. We thus need to know the irrep matrices for these three elements only. In principle, according to the above derivation, for each component of each irrep we can, by varying the index k, construct a number of projection operations which is equal to $\dim(\Omega)$. As we have already discussed, we shall effectively need all these only when the induced representation of the atoms is equal to the regular representation. This is the case if the number of atoms is equal to the order of the group, i.e., when there are no symmetry elements 'going through' the atoms, so that their site group is the trivial C_1. For octahedral symmetry, this is a polyhedron with 48 vertices. It belongs to the family of the Archimedean solids and is known as the *great rhombicuboctahedron*. The highest Archimedean solid is a polyhedron with 120 equivalent vertices, which is known as the *great rhombicosidodecahedron, or Archimedene*.[3] The representation of its vertices is the regular representation of the group I_h. In the case where the number of atoms is less, and thus the site symmetry is higher, the projection operators will give rise to redundancies. To avoid these we can make use of the Frobenius reciprocity theorem. The number of times a given irrep will occur in the induction is exactly equal to the number of times it subduces the totally symmetric irrep at the site group. The number of projection operations should thus also be restricted to this number. This can be achieved when the $\mathbb{D}^{\Omega}$ matrices are constructed in such a way that they are block diagonal in the stabilizing sitegroup of site $\langle a \rangle$. In that case the k indices of the projection operators should then be chosen in such a way that they correspond

[3] Calculations predict that a C_{120} molecular realization of this solid should exist [16, 17].

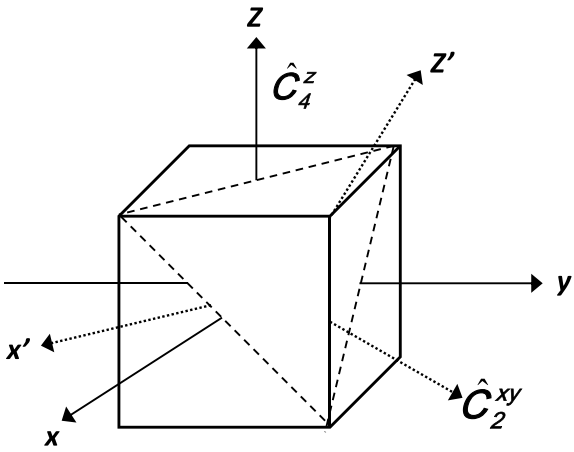


Fig. 4.12 Orientation in a cube of a tetragonal (x, y, z) and trigonal (x', y', z') system. The symmetry elements $\hat{C}_4^z$, $(\hat{C}_4^z)^{-1}$, and $\hat{C}_2^{xy}$ take the trigonal site to its nearest neighbors

to components that are totally symmetric in that sitegroup. For more elaborate treatments, see [14, 15]. As an example, consider the cube. The site group is C_{3v} and induction tells us that the function space, spanned by the 8 atoms, is given by

$$\Gamma(a_1 C_{3v} \uparrow Oh) = A_{1g} + T_{2g} + T_{1u} + A_{2u} \tag{4.141}$$

All these irreps occur only once; hence, when we choose a trigonal symmetry adaptation for constructing representation matrices, for all these matrices there will be exactly one index k that labels the component which is totally symmetric in C_{3v}. For this value, the matrix elements read

$$\langle \Phi_{ik}^{\Omega} | \mathcal{H} | \Phi_{ik}^{\Omega} \rangle = \beta \left[\overline{D}_{kk}^{\Omega}(C_4) + \overline{D}_{kk}^{\Omega}(C_4^{-1}) + \overline{D}_{kk}^{\Omega}(C_2) \right] \tag{4.142}$$

To obtain these eigenvalues we thus have to mold the representation matrices for O_h in a trigonal setting. For the one-dimensional irreps, A_{1g} and A_{2u}, this is trivial, since the matrix elements in Eq. (4.142) are simply the characters. We thus obtain

$$\begin{aligned} E(A_{1g}) &= \beta \left[\chi^{A_{1g}}(C_4) + \chi^{A_{1g}}(C_4^{-1}) + \chi^{A_{1g}}(C_2) \right] = 3\beta \\ E(A_{2u}) &= \beta \left[\chi^{A_{2u}}(C_4) + \chi^{A_{2u}}(C_4^{-1}) + \chi^{A_{2u}}(C_2) \right] = -3\beta \end{aligned} \tag{4.143}$$

For the T_{1u} irrep, we can use the set of the p-orbitals. In the standard Cartesian orientation, this set is adapted to the tetragonal site symmetry. In Fig. 4.12 we illustrate an alternative trigonal basis (see also Fig. 3.6d). The transformation between the two sets is given by

$$
\begin{aligned}
|p'_z\rangle &= \frac{1}{\sqrt{3}}\left(|p_x\rangle + |p_y\rangle + |p_z\rangle\right) \\
|p'_x\rangle &= \frac{1}{\sqrt{2}}\left(|p_x\rangle - |p_y\rangle\right) \\
|p'_y\rangle &= \frac{1}{\sqrt{6}}\left(|p_x\rangle + |p_y\rangle - 2|p_z\rangle\right)
\end{aligned} \tag{4.144}
$$

Here, the first component points in the threefold direction and thus is adapted to the C_{3v} site symmetry. We require only the diagonal matrix elements for this component. They can easily be obtained by expressing these elements in the standard canonical set:

$$
\begin{aligned}
D_{z'z'}(C_4) &= \langle p'_z|\hat{C}_4|p'_z\rangle \\
&= 1/3\left(\langle p_x| + \langle p_y| + \langle p_z|\right)\hat{C}_4\left(|p_x\rangle + |p_y\rangle + |p_z\rangle\right) \\
&= 1/3\left(\langle p_x| + \langle p_y| + \langle p_z|\right)\left(|p_y\rangle - |p_x\rangle + |p_z\rangle\right) = 1/3 \\
D_{z'z'}(C_4^{-1}) &= \langle p'_z|\hat{C}_4^{-1}|p'_z\rangle \\
&= 1/3\left(\langle p_x| + \langle p_y| + |p_z\rangle\right)\hat{C}_4^{-1}\left(|p_x\rangle + |p_y\rangle + |p_z\rangle\right) \\
&= 1/3\left(\langle p_x| + \langle p_y| + \langle p_z|\right)\left(-|p_y\rangle + |p_x\rangle + |p_z\rangle\right) = 1/3 \\
D_{z'z'}(C'_2) &= \langle p'_z|\hat{C}'_2|p'_z\rangle \\
&= 1/3\left(\langle p_x| + \langle p_y| + \langle p_z|\right)\hat{C}'_2\left(|p_x\rangle + |p_y\rangle + |p_z\rangle\right) \\
&= 1/3\left(\langle p_x| + \langle p_y| + \langle p_z|\right)\left(|p_y\rangle + |p_x\rangle - |p_z\rangle\right) = 1/3
\end{aligned} \tag{4.145}
$$

An entirely similar derivation can be made for the T_{2g} irrep, using the d-orbital set. The results are

$$
\begin{aligned}
E(T_{1u}) &= \beta \\
E(T_{2g}) &= -\beta.
\end{aligned} \tag{4.146}
$$

4.9.5 Triphenylmethyl Radical and Hidden Symmetry

As a final application we discuss an example of a molecular radical, where more symmetry is present than meets the eye. The triphenylmethyl radical, $C_{19}H_{15}$, is a planar, conjugated, hydrocarbon-radical, with 19 π-electrons. The molecular point group for the planar configuration is D_{3h}, but, since all valence $2p_z$-orbitals are antisymmetric with respect to the horizontal symmetry plane, the relevant symmetry of the valence shell is only C_{3v} as seen from Fig. 4.13. The molecular-symmetry group distributes the 19 atoms over five trigonal orbits of atoms that, under C_{3v}, can solely be permuted with partners in the same orbit.

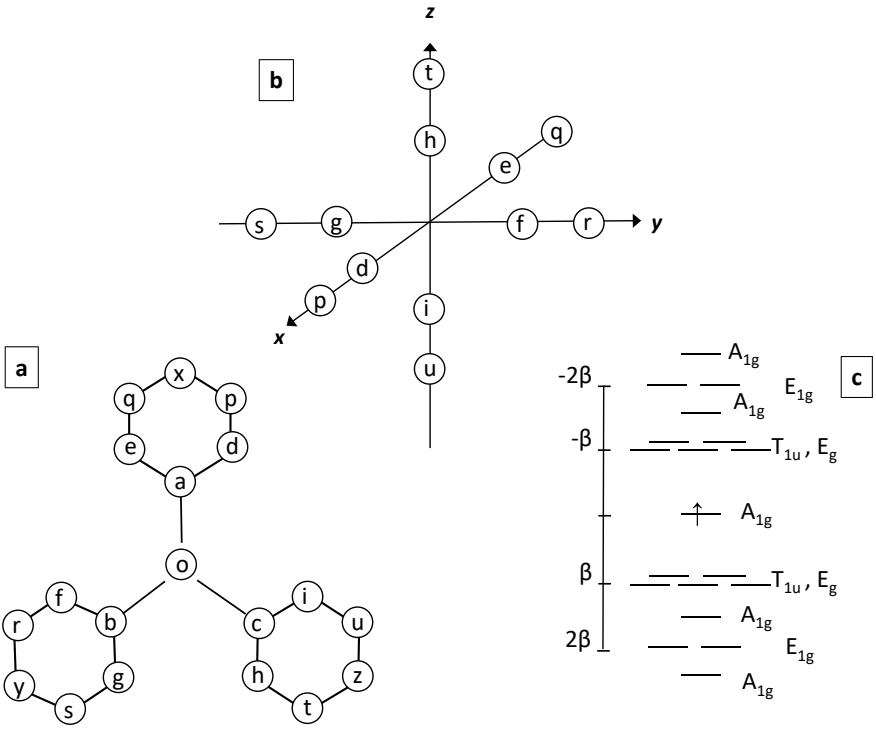


Fig. 4.13 **a** Molecular graph of triphenylmethyl, **b** correspondence between ortho and meta sites and octahedral C_{4v} positions, **c** Hückel spectrum

1. The central atom {o}.
2. The three atoms that are adjacent to o: $\{a, b, c\}$.
3. The six atoms in the *ortho* positions: $\{d, e, f, g, h, i\}$
4. The six atoms in the *meta* positions: $\{p, q, r, s, t, u\}$.
5. The three atoms in the *para* positions: $\{x, y, z\}$.

The separate rotation of a single phenyl group by 180° around its twofold direction will not change the connectivity of the graph. Yet this cannot be achieved by elements of the point group. It is, however, a legitimate symmetry operation as far as the graph is concerned, since it preserves the connectivity. The resulting automorphism group is thus larger than the point group, and in fact is isomorphic to O_h [18]. The three phenyl groups can be associated with the three Cartesian directions of this octahedral group. The six atoms in the ortho orbit can be formally associated with the six corners of this octahedron, each connected to a meta position (Fig. 4.10). This implies that the ortho and meta atoms occupy C_{4v} sites. The three-atom orbits correspond to the three tetragonal directions in the octahedron. The site group which leaves such a tetragonal direction invariant is not of the conical C_{nv} type, but D_{4h}. These correspondences allow identification of all the permutations. As examples, the $\hat{C}_4$ symmetry element

through the upper phenyl group, and the $\hat{S}_6$ rotation-reflection extension along the threefold direction of the trigonal axis of the molecular frame, permute the atoms as follows:

$$\begin{aligned}\hat{C}_4 &\to (o)(a)(d)(p)(x)(q)(e)(b,c)(y,z)(f,h,g,i)(r,t,s,u)\\ \hat{S}_6 &\to (o)(a,c,b)(x,z,y)(d,i,f,e,h,g)(p,u,r,q,t,s)\end{aligned} \tag{4.147}$$

A horizontal coordinate plane, $\hat{\sigma}_h$, of the octahedron has the effect of flipping one single phenyl group around. We can at once determine the irreps of the different orbits by induction:

$$\begin{aligned}\{a,b,c\} &: \Gamma(a_{1g}D_{4h} \uparrow O_h) = A_{1g} + E_g\\ \{x,y,z\} &: \Gamma(a_{1g}D_{4h} \uparrow O_h) = A_{1g} + E_g\\ \{d,e,f,g,h,i\} &: \Gamma(a_1C_{4v} \uparrow O_h) = A_{1g} + E_g + T_{1u}\\ \{p,q,r,s,t,u\} &: \Gamma(a_1C_{4v} \uparrow O_h) = A_{1g} + E_g + T_{1u}\end{aligned} \tag{4.148}$$

The central atom is invariant in O_h and thus transforms as A_{1g}. The total induced representation of the function space thus is given by

$$\Gamma = 5A_{1g} + 4E_g + 2T_{1u} \tag{4.149}$$

The 19-dimensional Hückel matrix thus will be resolved into five blocks, one of dimension 5, two identical blocks of dimension 4, and three identical blocks of dimension 2. In Table 4.11 we display the blocks for each irrep, and the corresponding SALCs for one component. The corresponding secular equations are

$$\begin{aligned}A_{1g} &: \lambda(\lambda^4 - 8\lambda^2 + 13)^2 = 0\\ E_g &: \lambda^4 - 5\lambda^2 + 4 = 0\\ T_{1u} &: \lambda^2 - 1 = 0\end{aligned} \tag{4.150}$$

Symmetry has taken us to a point where still quintic, quartic, and quadratic secular equations must be solved. However, a closer look at these equations shows that they can easily be solved. Apparently, a further symmetry principle is present, which leads to simple analytical solutions of the secular equations. Triphenylmethyl is an *alternant* hydrocarbon. In an alternant atoms can be given two different colors, in such a way that all bonds are between atoms of different colors; hence, no atoms of the same color are adjacent. A graph with this property is *bipartite*,[4] and its eigenvalue spectrum obeys the celebrated Coulson−Rushbrooke theorem [19].

Theorem 4.5 *The eigenvalues of an alternant are symmetrically distributed about the zero energy level. The corresponding eigenfunctions also show a mirror relationship, except for a difference of sign (only) in every other atomic orbital*

[4] Note that in fact a molecular graph will always be bipartite unless it contains one or more odd-membered rings.

Table 4.11 SALCs and Hückel matrices, in units of β, for triphenylmethyl. For the degenerate irreps., only one component is given

A_{1g}					
$\|o\rangle$	$-\lambda$	$\sqrt{3}$	0	0	0
$1/\sqrt{3}\,(\|a\rangle + \|b\rangle + \|c\rangle)$	$\sqrt{3}$	$-\lambda$	0	$\sqrt{2}$	0
$1/\sqrt{3}\,(\|x\rangle + \|y\rangle + \|z\rangle)$	0	0	$-\lambda$	0	$\sqrt{2}$
$1/\sqrt{6}\,(\|d\rangle + \|e\rangle + \|f\rangle + \|g\rangle + \|h\rangle + \|i\rangle)$	0	$\sqrt{2}$	0	$-\lambda$	1
$1/\sqrt{6}\,(\|p\rangle + \|q\rangle + \|r\rangle + \|s\rangle + \|t\rangle + \|u\rangle)$	0	0	$\sqrt{2}$	1	$-\lambda$
E_g					
$1/\sqrt{2}\,(\|b\rangle - \|c\rangle)$	$-\lambda$	0	$\sqrt{2}$	0	
$1/\sqrt{2}\,(\|y\rangle - \|z\rangle)$	0	$-\lambda$	0	$\sqrt{2}$	
$1/2\,(\|f\rangle + \|g\rangle - \|h\rangle - \|i\rangle)$	$\sqrt{2}$	0	$-\lambda$	1	
$1/2\,(\|r\rangle + \|s\rangle - \|t\rangle - \|u\rangle)$	0	$\sqrt{2}$	1	$-\lambda$	
T_{1u}					
$1/\sqrt{2}\,(\|d\rangle - \|e\rangle)$	$-\lambda$	1			
$1/\sqrt{2}\,(\|p\rangle - \|q\rangle)$	1	$-\lambda$			

coefficient. The total charge density at any carbon atom in the neutral alternant hydrocarbon equals unity.

Since triphenylmethyl is an odd alternant, there should be at least one eigenvalue at energy zero. This root will be necessary of A_{1g} symmetry, since this is the only irrep which occurs an odd number of times. All other roots occur in pairs of opposite energies. This is confirmed by the secular equations in Eq. (4.150), where the A_{1g} equation indeed has a root at $\lambda = 0$, and all remaining equations contain only even powers of λ. The roots are then easily determined.

$$\begin{aligned} A_{1g} &: \lambda = 0, \pm\sqrt{4 \pm \sqrt{3}} \\ E_g &: \lambda = \pm 1, \pm 2 \\ T_{1u} &: \lambda = \pm 1 \end{aligned} \tag{4.151}$$

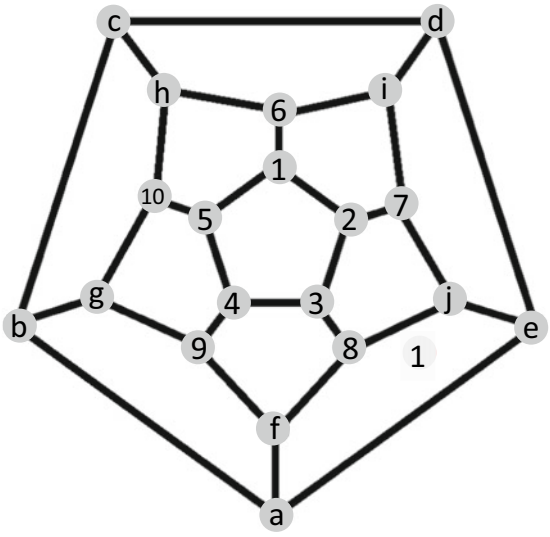
The spectrum contains unexpected fivefold degeneracies at $E = \pm\beta$, where the T_{1u} and E_g levels coincide. This degeneracy is considered *accidental*, to the extent that it does not correspond to a single irrep of the automorphism group of the graph. However, the fivefold degeneracy can easily be rationalized as follows: $E = \beta$ is the eigenenergy of the degenerate highest occupied molecular orbital (HOMO) in an isolated phenyl-ring. The three rings thus give rise to six orbitals with this energy. The symmetry of this orbital space in O_h is equal to $A_{1g} + E_g + T_{1u}$. Of these only the A_{1g} combination is of the right symmetry to interact with the central atom. For the five others, there can be no communication between the phenyl rings, since the

channel via the central atom is open only to A_{1g} symmetries. As a result, for these solutions, there is no overlap between the rings, and five isolated phenyl solutions persist at $E = \beta$. A similar argument applies to the level at $E = -\beta$ level, which stems from the phenyl lowest unoccupied orbital (LUMO).

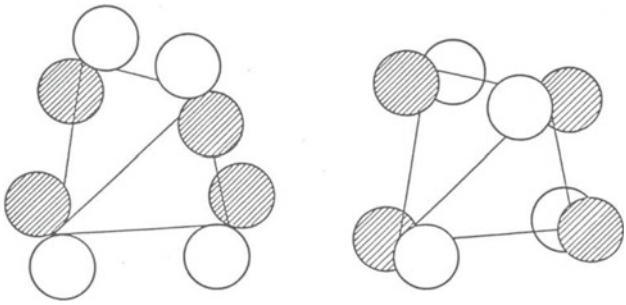
Many features thus come together in triphenylmethyl. Besides the C_{3v} molecular point group, the Hückel matrix obeys an additional or hidden O_h symmetry. This is a typical feature of the nearest-neighbor approximation, which requires only that symmetry operations should preserve the connections with the nearest neighbors. This precisely complies with the definition of the automorphism group of the graph. Furthermore, the special bipartite properties of the graph further impose constraints on the spectrum, which in this case lead to a complete reduction of the secular equations. Finally, an unexpected additional degeneracy manifests itself, which is related to the composite nature of the molecule being considered, with several fragments bound to one central atom.

4.10 Problems

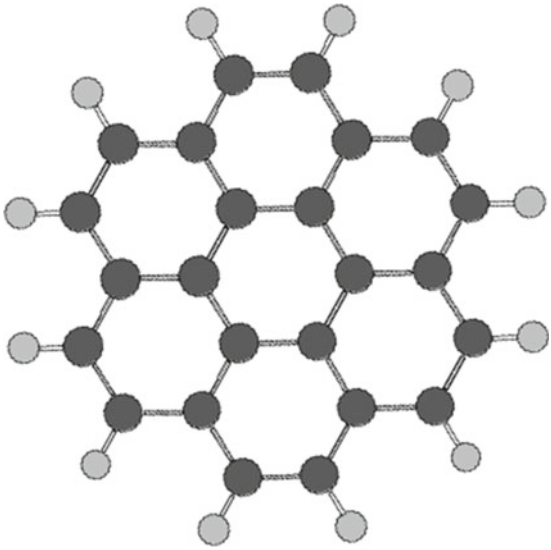
4.1 A Schlegel diagram of a polyhedron is a projection into a plane. The figure below shows the Schlegel diagram for a dodecahedron. As was explained in the preceding chapter (See Fig. 3.8a) a dodecahedron contains five cubes, e.g., the cube based on the nodes $\langle 1, 3, 9, 10, a, c, i, j\rangle$. Symmetry elements of I_h will permute these cubes. Construct the set of the five cubes and determine the irreps of this set. Can you obtain this result by induction?



4.2 Consider the set of 8 tangential π-orbitals on a tetrahedron. Derive the irreducible representations in the T_d point group. How would you label the canonical symmetry of the combinations which are shown in the figure below?



4.3 Consider the set of perpendicular p_z-orbitals in the polyaromatic planar molecule coronene, shown below. This set gives rise to Molecular Orbitals of a_{1u} and b_{1g} symmetry. Can you draw both of these orbitals? (Use the standard orientation of the central benzene frame, as shown in Fig. 3.10)



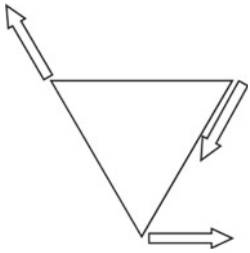
4.4 In dodecaborane, $B_{12}H_{12}^{2-}$, the twelve boron atoms occupy the vertices of an icosahedral cage. Determine the irreducible representations of the set of their tangential π-orbitals.

4.5 Consider a degenerate manifold, transforming as the irrep Γ_i. The appropriate projector was applied to the basis to yield one component of the degeneracy space. Which other projectors do you need to use to generate all the remaining components?

4.6 Consider the expression for a totally symmetric projection operator and demonstrate that it is indeed totally symmetric under any group element.

4.7 Construct the three sp^2 hybrids in the xy plane.

4.8 The picture below shows the distortion of an equilateral triangle. Determine the representation of this distortion.

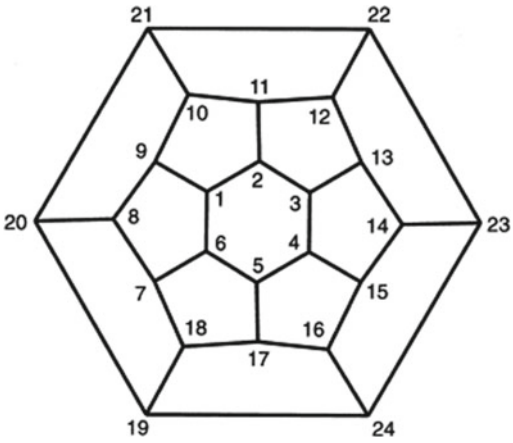


4.9 Construct the character table for a cyclic group of order 5.

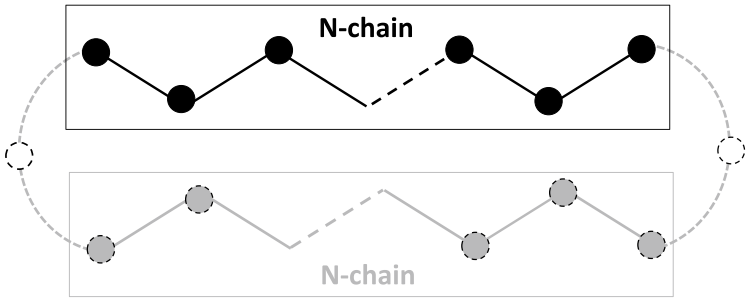
4.10 In a molecule with D_{6h} symmetry, the choice of $\hat{C}_2'$ and $\hat{C}_2''$ directions is arbitrary but, once the twofold directions are chosen, the $\hat{\sigma}_v$ and $\hat{\sigma}_v$ reflection planes are also fixed, as was indicated in Fig. 3.10. Show that these spatial relationships between the axes and the planes is in line with the character table for D_{6h}.

4.11 Consider a molecule with D_{3h} symmetry and apply a distortion mode which transforms according to the a_2'' irrep. A distortion along this coordinate will break the original symmetry to a subgroup of D_{3h}. Which subgroup? Try to generalize this example to a general rule.

4.12 Below is shown the Schlegel diagram for the C_{24} fullerene. It is a cage structure with hexagonal faces at top and bottom, capping a crown of twelve pentagons around its waist. The valence shell of this structure is formed by a set of 24 radial p_σ-orbitals, one on each carbon. Determine the point group of this molecule and the irreps describing the valence shell. (Hint: divide the set of 24 orbitals into two separate orbits.)



4.13 Determine the Hückel spectrum and eigenvalues of a linear chain with N atoms. Hint: incorporate the chain in a cyclic $(2N + 2)$-polyene, consisting of two N-chains with 2 bridging atoms, as shown in the figure. The eigenstates of the chain correspond to the cyclic states with nodes at the bridging atoms.



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Chapter 5
What has Quantum Chemistry Got to Do with It?



Abstract The time has come to see how the concept of irreducible representations ties in with quantum chemistry. After a brief introduction to the pre-quantum principles of symmetry, it will be shown that eigenfunctions of the Hamiltonian are also eigenfunctions of the symmetry operators that commute with the Hamiltonian. The concept of a degeneracy is further analyzed, and it is shown how the degenerate components can be characterized by canonical symmetry-relationships. The final section will then provide a detailed account of the symmetry operations that leave the Hamiltonian invariant.

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5.1 The Prequantum Era

Nature around us is full of disorder and chaos, yet it also offers intriguing examples of perfect order and symmetry. Ever since prehistoric times, man has been admiring the circular geometry of a full moon or the perfectly flat surface of a calm sea. Crystals offer another example of almost ideal symmetrical shapes, and it is no surprise that early recognition of the important role of symmetry in physics was based on the study of the properties of crystals. Two pioneers of the pre-quantum era stand out for their important conjectures: Franz Neumann and Pierre Curie [1].

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Theorem 5.1 *Neumann's principle states that the symmetry elements of any physical property of a crystal must include all the symmetry elements of the point group of the crystal.*

We can apply this directly to a molecule such as ammonia. Ammonia carries a permanent dipole moment, μ_z, which is oriented along the threefold axis. In-plane components of the dipole moment are strictly forbidden. This is in agreement with the Neumann principle. A dipole moment corresponds to a displacement of charge and the only displacement which does not destroy the molecular point group is along the z-direction. Hence, this is the only direction which is compatible with the Neumann principle.

Pierre Curie realized that this principle is not limited to crystals, but applies to physical phenomena in general, and not only to isolated systems, but also to systems subject to external perturbations. His proposition is known as the principle of *dissymmetry*.

Theorem 5.2 *The symmetry of a phenomenon is the maximal symmetry compatible with the existence of the phenomenon. In order for a phenomenon to exist, it is necessary that certain elements of symmetry are absent: dissymmetry creates the phenomenon.*

Curie understood that under stress, or in the presence of external electric or magnetic fields, the symmetry of a system is changed. The Neumann principle still applies but should no longer be based on the symmetry of the isolated crystal, but on that of the combined system of crystal and external field, as we have considered in Sect. 3.9. In the case of ammonia, application of an electric field has the $C_{\infty v}$ symmetry of a polar vector. The symmetry which results from the superposition of the field with the molecular point group C_{3v} depends on the orientation (see Appendix B). In the coordinate frame of Fig. 3.1 one has

$$\begin{aligned} z &: C_{\infty v} \cap C_{3v} = C_{3v} \\ x &: C_{\infty v} \cap C_{3v} = C_s \\ y &: C_{\infty v} \cap C_{3v} = C_1 \end{aligned} \tag{5.1}$$

If the field is oriented along the z-direction it will keep the C_{3v} symmetry of the molecule. This is also in line with the existence of a permanent dipole in the z-direction according to the Neumann principle, as a charge dipole has the same symmetry as an electric field. If an external field is applied in the x-direction the symmetry is reduced to the reflection group, $C_s = \{\hat{E}, \hat{\sigma}_1\}$. In the presence of such a field displacement of charge in the x-direction is compatible with the extended symmetry principle, which means that ammonia can acquire an *induced* dipole moment in the x-direction. However, a field along x cannot induce a dipole moment in the y-direction, since the σ_1 reflection plane is incompatible with the displacement of charge across the plane of symmetry.

The symmetry principles of Neumann and Curie can be recast in the language of irreducible representations. The requirement that physical properties be invariant

under the symmetry elements of the point group simply means that they should transform as the totally symmetric irrep. For the dipole moment, the components transform as follows:

$$\begin{aligned} \Gamma(\mu_z) &= a_1 \\ \Gamma(\mu_x, \mu_y) &= e \end{aligned} \tag{5.2}$$

Hence, only the z-component is compatible with the existence of a dipole moment.

The principles of Neumann and Curie are of course based on classical physics. They remain valid for quantum systems, but do not include typical quantum phenomena, such as the existence of electronic degeneracy, or transitions between quantum states. A proper quantum description of molecular symmetry is thus required.

5.2 The Schrödinger Equation

According to quantum mechanics, the stationary states of a molecule are described by the eigenfunctions of the Hamiltonian, $\mathcal{H}$, corresponding to a quantized set of eigenvalues.

$$\mathcal{H}|\Psi_j^k\rangle = E_k|\Psi_j^k\rangle \tag{5.3}$$

Here, E_k is a fixed eigenvalue, and $|\Psi_j^k\rangle$ is an associated eigenfunction. The index j takes into account the possibility that several eigenfunctions may be associated with the same eigenvalue. In this case the set of these functions forms an eigenspace: $\{|\Psi_j^k\rangle\}_{j=1\ldots n}$, where n denotes the dimension of this space, and the E_k eigenvalue is said to be n-fold *degenerate*. As usual, the eigenspace will be taken to be orthonormal. The Hamiltonian expresses the kinematics of the electrons in the frame of the nuclei, subject to Coulomb forces. We shall study this in detail in Sect. 5.4. For the moment all we need to know is that the operators of the molecular point group leave $\mathcal{H}$ invariant.

$$\forall \hat{R} \in G \rightarrow \ [\hat{R}, \mathcal{H}] = 0 \tag{5.4}$$

Now applying $\hat{R}$ to the Schrödinger equation yields

$$\hat{R}\mathcal{H}|\Psi_j^k\rangle = \mathcal{H}\hat{R}|\Psi_j^k\rangle = E_k\hat{R}|\Psi_j^k\rangle \tag{5.5}$$

Here, we have made use of the commutation relation in Eq. (5.4) and the property that $\hat{R}$ as a linear operator will not affect the constant eigenvalue. The equation signifies that, if $|\Psi_j^k\rangle$ is an eigenfunction, the transformed function, $\hat{R}|\Psi_j^k\rangle$, will also be an eigenfunction *with the same eigenvalue.* This is an important result, which ties quantum mechanics and group theory together, and is essentially the reason why group theory can be applied to chemistry! Now, there are two possibilities, depending on the degeneracy.

1. The electronic state is non-degenerate ($n = 1$). In this case the transformed eigenfunction must necessarily be proportional to the original one. Since the transformation will not change normalization, the proportionality constant must be an unimodular number:

$$\forall \hat{R} \in G : \hat{R}|\Psi^k\rangle = \exp(i\kappa_R)|\Psi^k\rangle \tag{5.6}$$

   Here, we have dropped the index j since there is only one eigenfunction in this case. This equation indicates that this eigenfunction must transform as a non-degenerate irrep of the point group, say Γ_k, with

$$\exp(i\kappa_R) = \chi^{\Gamma_k}(R) \tag{5.7}$$

2. The electronic state is degenerate ($n > 1$). In this case the transformed function may not be proportional to the original one, but in any case it is also an eigenfunction of H with the same eigenvalue. This means that it must be mapped onto a linear combination of the components of the eigenspace; hence, the eigenspace itself must form a function space that is invariant under G.

$$R|\Psi_j^k\rangle = \sum_{i=1}^{n} D_{ij}(R)|\Psi_i^k\rangle \tag{5.8}$$

   Here, the eigenfunctions have again been arranged as a row vector, and the coefficients are gathered in a transformation matrix, $\mathbb{D}(R)$. There are now two possibilities:

   - The matrix representation is an irrep of the point group.
     In this case the electronic degeneracy is equal to the dimension of an irrep of the point group.
   - The matrix representation is reducible.
     In this case the eigenspace can always be separated into irreducible blocks, using projection operators.

This list of possibilities shows that eigenfunctions of the Hamiltonian will also be (or can be made to be) eigenfunctions of the symmetry group of the Hamiltonian. When there is a perfect match between the eigenfunctions and an irrep, the presence of degeneracy or non-degeneracy can directly be attributed to the symmetry of the eigenstates. The remaining possibility that the eigenspace may consist of several irreducible blocks could be described as a case of 'accidental degeneracy', in the sense that symmetry cannot explain the fact that stationary states are degenerate. When this happens, it could mean that the symmetry of the system exceeds the apparent spatial symmetry group. This case is referred to as 'hidden symmetry'. A special case of this is Kramers' degeneracy, which is treated in Sect. 7.6. Or it could be that a simplified model Hamiltonian was used, such as, e.g., the nearest-neighbor Hamiltonian in Hückel theory, which may give rise to additional degeneracies, as we

have explained for the example of triphenylmethyl in the previous chapter. With the exceptions of these possibilities, experience tells—in the words of Griffith [2]—that 'accidents don't happen, at least in that part of physics which is understood'.

The significance of these observations can hardly be overestimated. The derivation refers to the properties of the exact Hamiltonian and its eigenfunctions. In actual calculations one must usually introduce approximate Hamiltonians and eigenspaces. However, there is at least one simple way to endow these approximate eigenfunctions with an exact property, i.e., by making sure that they are eigenfunctions of the symmetry group of the exact Hamiltonian. The great success of semi-empirical theories, such as ligand-field theory for transition-metal and lanthanide complexes, is in fact due to this consistent use of atomic and molecular symmetry. But also in computational chemistry the effort of symmetrizing the orbital basis pays off in a considerable gain of computation time and it facilitates the assignment of spectroscopic data.

5.3 How to Structure a Degenerate Space

The eigenfunctions of an n-fold degenerate state are defined up to unitary equivalence, which means that any complex linear combination of eigenfunctions is again an eigenfunction. It is convenient to define a standard basis, which brings some structure in this degenerate space. The tool that can be used for this is furnished by the following lemma, due to Schur.

Theorem 5.3 *If a matrix commutes with all the matrices of an irreducible representation, the matrix must be a multiple of the unit matrix.*

$$\forall \hat{R} \in G \ : \ \mathbb{U}\,\mathbb{D}(R) = \mathbb{D}(R)\,\mathbb{U} \ \rightarrow \ \mathbb{U} \sim \mathbb{I} \tag{5.9}$$

This theorem implies that if we combine the basis functions of a space in such a way that their representation matrices of the group generators coincide with a canonical choice, then the entire basis set will be completely fixed, up to a global phase factor. Indeed, if a basis is reached which complies with a standard choice of representation matrices, the theorem tells us that this basis will be uniquely defined, except for an overall phase change. This remaining phase freedom is external to the symmetry group. A basis set that complies with a specified set of representation matrices is called a *canonical* basis. A convenient strategy for defining a canonical basis is founded on a *splitting field*. In this case the basis is chosen in such a way that it is diagonal with respect to a particular generator, usually a principal axis of rotation. The components then appear as eigenfunctions of the splitting field. In order for the splitting to be unequivocal, all eigenvalues should be different. The splitting field then 'recognizes' each component by its individual eigenvalue. As an example consider the twofold degenerate E-state in an octahedron. As a splitting field we take the $\hat{C}_4$ axis along the z-direction. The standard components, which

are recognized individually by this field, are denoted as $|E\theta\rangle$ and $|E\epsilon\rangle$. They are respectively symmetric and anti-symmetric with respect to the rotation axis.

$$\begin{aligned}\hat{C}_4^z|E\theta\rangle &= |E\theta\rangle \\ \hat{C}_4^z|E\epsilon\rangle &= -|E\epsilon\rangle\end{aligned} \tag{5.10}$$

In octahedral transition-metal complexes the d-orbitals, which transform as the E irrep, are d_{z^2} and $d_{x^2-y^2}$. They are seen to match $|E\theta\rangle$ and $|E\epsilon\rangle$, respectively. The conventional choice of the real d-orbitals is thus based on a tetragonal splitting field. Once the splitting field has been applied, the eigenspace is already nearly fixed; the only freedoms that remain are the *relative* phases of the components. To freeze these phases one can sometimes use an extra ladder operator, that moves from one component to the other, thereby imposing a phase convention. For the E-state, one uses a threefold axis to connect the two components, and the corresponding representation matrix is copied from the standard canonical choice for a trigonal matrix, as constructed in Sect. 3.1:

$$\hat{C}_3\left(|E\theta\rangle\ |E\epsilon\rangle\right) = \left(|E\theta\rangle\ |E\epsilon\rangle\right)\begin{pmatrix}-1/2 & -\sqrt{3}/2\\ \sqrt{3}/2 & -1/2\end{pmatrix} \tag{5.11}$$

This operator connects the two components and, therefore, if we require that the matrix be of the specified form, then the relative phase freedom is lifted. With such a connecting element, one can indeed easily construct a proper ladder operator:

$$\begin{aligned}\frac{2}{\sqrt{3}}\left[\hat{C}_3 + \frac{1}{2}\hat{E}\right]|E\theta\rangle &= |E\epsilon\rangle \\ \frac{2}{\sqrt{3}}\left[\hat{C}_3 + \frac{1}{2}\hat{E}\right]|E\epsilon\rangle &= -|E\theta\rangle\end{aligned} \tag{5.12}$$

In Appendix D we list standard conventions that are frequently used to define canonical basis sets for degenerate irreps.

For the octahedral T_1 state the standard basis complies with the transformation properties of the real p-orbitals and is marked as $|T_{1x}\rangle, |T_{1y}\rangle, |T_{1z}\rangle$. If we diagonalize this set under a fourfold splitting field one obtains the complex p-orbitals. In applications where real functions are preferred, the $\hat{C}_4$ axis is represented as

$$\hat{C}_4\left(|T_{1x}\rangle\ |T_{1y}\rangle\ |T_{1z}\rangle\right) = \left(|T_{1x}\rangle\ |T_{1y}\rangle\ |T_{1z}\rangle\right)\begin{pmatrix}0 & -1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{pmatrix} \tag{5.13}$$

Here, $|T_{1z}\rangle$ is recognized as a totally symmetric eigenfunction, and it is uniquely defined by this eigenvalue, since the other eigenvalues are $\pm i$. The splitting field is

thus operating only partially; nonetheless, it uniquely picks the z-component. Then the $\hat{C}_3$ axis is a perfect ladder operator, effecting a cyclic permutation of z to x, and further to y.

5.4 The Molecular Symmetry Group

So far, the symmetry of the Hamiltonian was defined as the set of all operations that leave the Hamiltonian invariant. This invariance group was assumed to coincide with the point group of the nuclear frame of the molecule, but it is now time to provide a clear explanation of this connection. This section relies on the definition of the symmetry groups of non-rigid molecules proposed by Longuet-Higgins [3]. We start by writing down in explicit form the Schrödinger Hamiltonian for a molecule.

$$\mathcal{H} = T_e + T_N + V_{NN} + V_{eN} + V_{ee} \tag{5.14}$$

The T operators are the kinetic energy operators for electrons and nuclei:

$$\begin{aligned} T_e &= -\frac{\hbar^2}{2m_e} \sum_i \left(\frac{\partial^2}{\partial x_i^2} + \frac{\partial^2}{\partial y_i^2} + \frac{\partial^2}{\partial z_i^2} \right) \\ T_N &= -\frac{\hbar^2}{2} \sum_n \frac{1}{m_n} \left(\frac{\partial^2}{\partial X_n^2} + \frac{\partial^2}{\partial Y_n^2} + \frac{\partial^2}{\partial Z_n^2} \right) \end{aligned} \tag{5.15}$$

The V operators are the Coulomb interactions between the particles.

$$\begin{aligned} V_{NN} &= \sum_{n<m} \frac{Z_n Z_m \mathrm{e}^2}{4\pi\epsilon_0 |\mathbf{R}_n - \mathbf{R}_m|} \\ V_{eN} &= -\sum_{i,n} \frac{Z_n \mathrm{e}^2}{4\pi\epsilon_0 |\mathbf{R}_n - \mathbf{r}_i|} \\ V_{ee} &= \sum_{i<j} \frac{\mathrm{e}^2}{4\pi\epsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} \end{aligned} \tag{5.16}$$

The Hamiltonian may contain additional terms describing coupling between orbital and spin momenta. Longuet-Higgins stated that this full Hamiltonian must be invariant under the following types of transformations:

1. Any permutation of the positions and spins of the electrons.
2. Any rotation of the positions and spins of all particles (electrons and nuclei) about any axis through the center of mass.
3. Any overall translation in space.
4. The reversal of all-particle momenta and spins.

5. The simultaneous inversion of the positions of all particles in the center of mass.
6. Any permutation of the positions and spins of any set of identical nuclei.

The complete group of the Hamiltonian is the combination of all these possible symmetries. This derivation is directly evident from the mathematical form of the Hamiltonian and expresses fundamental properties of molecular space and time. Yet it took 40 years, from Schrödinger to Longuet-Higgins, to obtain a clear definition of the molecular-symmetry group. Three kinds of symmetries may be identified:

- Space symmetries. Space is uniform, isotropic, and has inversion symmetry. This is clear from the fact that the kinetic energy Laplacian operators are trace operators of second derivatives; hence, they will be invariant under a sign change of all coordinates, and isotropic under rotations. All potential-energy operators depend only on relative distances between particles, and thus will not change under the isometries: translations, rotations, or inversion.
- Time symmetry. The reversal of all momenta and spins is nothing other than the time-reversal operator, which we introduced earlier. As long as no external magnetic fields are present, time is reversible.
- Elementary particles of the same kind are indistinguishable. The remaining symmetries do not refer to space or time but are *collective* symmetries: they refer to the permutational symmetry of a set of particles. All electrons are the same and thus the Hamiltonian does not change when we permute electron labels. This symmetry becomes apparent only when multi-electronic wavefunctions are considered, and these will be treated in the next chapter. Likewise, identical nuclei can be permuted without changing the Hamiltonian. This is reflected in the potential-energy terms which consist of sums over all pairwise interactions. Permutations of particle labels will change only the order of the terms in these summations.

As an example, in Fig. 5.1 we return to our favorite ammonia molecule, and list all nuclear permutations, with and without the all-particle inversion operator, that leave the full Hamiltonian invariant. Nuclear permutations are defined here in the same way as in Sect. 3.3. A permutation such as (ABC) means that the letters A, B, and C are replaced by B, C, and A, respectively [4]. The overall inversion operator, $\hat{E}^*$, inverts the positions of all particles through a common inversion center, which can conveniently be chosen in the mass origin. In total 12 combinations of such operations are found which together form a group that is isomorphic to D_{3h}. How is this related to our previous C_{3v} point group? At this point it is very important to recall that the state of a molecule is not only determined by its Hamiltonian but also, and to an equal extent, by the boundary conditions. The eigenvalue equation is a differential equation which has a very extensive set of mathematical solutions, but not all these solutions are also acceptable states of the physical system. The role of the boundary conditions is to define constraints which filter out physically unacceptable states of the system. In most cases these constraints also lead to the quantization of the energies.

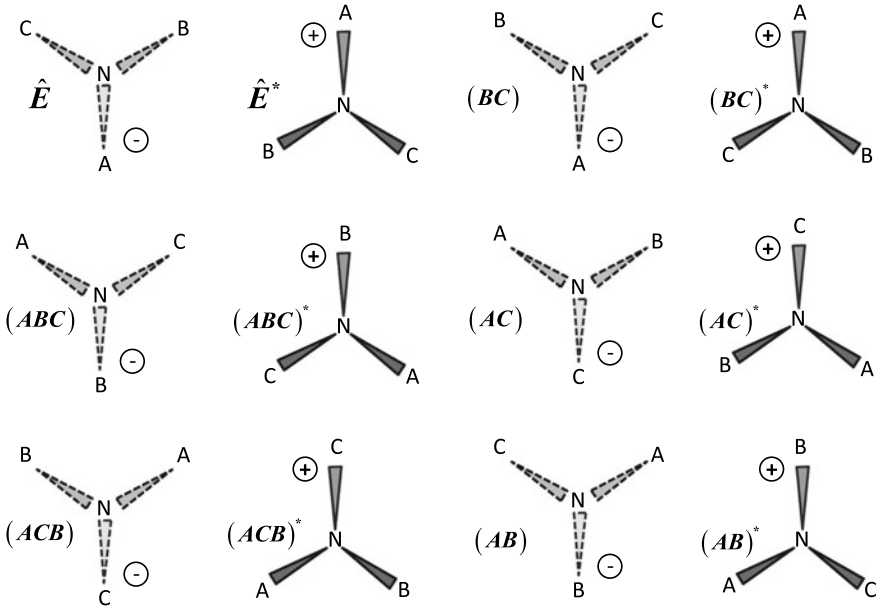


Fig. 5.1 Dynamic symmetry group of ammonia, with permutations of nuclei, and inversion of all particles (indicated by an asterisk). The plus (minus) sign indicates the position of an electron above (below) the plane of the hydrogen atoms

From our present perspective the invariance group should not only leave the Hamiltonian invariant, but also *it should not alter the boundary conditions*. Now, in most quantum-chemical applications a very stringent boundary condition is offered by the Born−Oppenheimer approximation. This simply states that the nuclei are considered immobile. It seriously restricts the Longuet-Higgins list. Indeed, we should retain only those operations that leave all nuclei in their rest positions. Applying this to ammonia, of the twelve operations we retain only those that do not affect the starting structure, i.e., only those combinations of permutations, permutation inversion, and overall rotations that, as a net result, keep the nuclei fixed in space. Of the twelve operations of the symmetry group in Fig. 5.1 only six turn out to be compatible with the condition of fixed nuclei, and these are precisely the six operations of the C_{3v} point group. To illustrate this, consider the permutation of nuclei B and C followed by the inversion of all particles, yielding the structure, marked $(BC)^*$ in Fig. 5.1. In this case, the ammonia molecule has been turned upside down. The nuclei can be moved back to their original positions by rotating the whole molecule by 180° about an axis which is perpendicular to the $\hat{\sigma}_1$ reflection plane. The net result is that the electron, marked by the little circle in the figure, has been reflected in this plane, as depicted in Fig. 5.2. This operation thus leaves the nuclei in place and only the positions of the electrons are changed. This is precisely the definition of symmetry operators which we have been using all along. The operators are displacing the

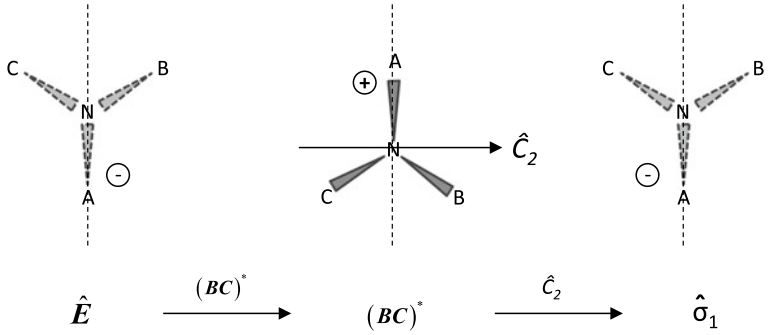


Fig. 5.2 Point group symmetry of ammonia: the $\hat{\sigma}_1$ symmetry plane is recovered as the consecutive application of the dynamic symmetry element $(BC)^*$, followed by an overall rotation about a twofold axis, perpendicular to $\hat{\sigma}_1$

Table 5.1 Embedding of the C_{3v} point group in the Longuet-Higgins group. The symmetry elements of the point group act on the electrons. They are identified as the product of nuclear permutations, inversion of all particles (star operation), and bodily rotations of all particles ($\hat{\mathcal{Q}}$ operators) along particular directions

C_{3v}	Rotation	×	Inversion	×	Permutation
$\hat{E}$	$\hat{E}$		$\hat{E}$		$\hat{E}$
$\hat{C}_3$	$\hat{\mathcal{Q}}_3^z$		$\hat{E}$		(ABC)
$\hat{C}_3^2$	$\left(\hat{\mathcal{Q}}_3^z\right)^{-1}$		$\hat{E}$		(ACB)
$\hat{\sigma}_1$	$\hat{\mathcal{Q}}_2^{\perp\sigma_1}$		$\hat{E}^*$		(A)(BC)
$\hat{\sigma}_2$	$\hat{\mathcal{Q}}_2^{\perp\sigma_2}$		$\hat{E}^*$		(B)(AC)
$\hat{\sigma}_3$	$\hat{\mathcal{Q}}_2^{\perp\sigma_3}$		$\hat{E}^*$		(C)(AB)

electrons, and in this way lead to transformations of the electronic wavefunctions. The complete embedding of C_{3v} in the Longuet-Higgins group is given in Table 5.1. By contrast, an operation such as (A)(BC), not followed by spatial inversion of all particles, gives rise to an alternative arrangement of the nuclei, which cannot be brought into coincidence with the original positions by mere spatial rotations. As a result, this operation is not compatible with the Born−Oppenheimer boundary conditions. On the other hand, going beyond the Born−Oppenheimer approximation, one may consider the dynamic states of ammonia, which correspond to the tunneling of the nitrogen through the triangle of the hydrogens. For the tunneling states the all-particle inversion operator, $\hat{E}^*$, is also a symmetry element and the symmetry group of the non-rigid ammonia thus attains the full D_{3h} Longuet-Higgins group.

5.5 Problems

5.1 Prove that the electron-repulsion operator, V_{ee}, is invariant under a rotation around the z-axis.

5.2 Construct a splitting field and ladder operators for the canonical components of the icosahedral irreps in Appendix D.

5.3 Derive the permutation-inversion group for CH_3BF_2 (methyl-boron-difluoride), under the assumption that the methyl group is almost freely rotating. This means that the result of a permutation inversion can be rotated back by a bodily rotation to a rotamer of the original structure. Determine the point group which is isomorphic to the resulting dynamic symmetry group.

5.4 The barrier to rotation of the cyclopentadienyl rings in $Fe(C_5H_5)_2$ (ferrocene, see Fig. 3.9a), measured in the gas phase, is only one kcal/mol. Construct a dynamic symmetry group for this molecule.

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Chapter 6
Interactions



Abstract In quantum mechanics, the observable phenomena are interactions, expressed as matrix elements of operators in a function space. These spaces and operators are like communicating vessels, reality is neither the operation nor the representation, but the interaction. The evaluation of the corresponding matrix elements requires the coupling of representations and can be factorized into an intrinsic scalar quantity that contains the physics of the interaction, and a tensorial coupling coefficient that contains its symmetry. This factorization is first illustrated for the case of overlap integrals, where the operator is just the unit operator, and then extended to the case of non-trivial operators, such as the Hamiltonian, and electric and magnetic dipole operators. The Wigner–Eckart theorem is introduced, together with the symmetry selection rules, both at the level of representations and subrepresentations. The results are applied to chemical reaction theory, and to the theory of the Jahn–Teller effect. Selection rules are illustrated for linear and circular dichroism. Finally, the polyhedral Euler theorem is introduced and applied to valence-bond theory for clusters.

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6.1 Overlap Integrals

Operations and representations are merely theoretical constructs. What is actually observed are the *interactions*. In quantum mechanics, interactions are expressed as matrix elements of operators in a function space. When the operator is the unit operator, the matrix elements are just overlap integrals. These are the simplest form of interactions.

We start our analysis by examining symmetry selection rules for overlap integrals. Consider the overlap integral between the ith-component of a function space which transforms according to the irrep Γ, and the kth-component of another function space transforming as Γ'. The overlap integral, S_{ki}, is a scalar quantity and thus must be invariant under the action of linear symmetry operators acting on the functions.

$$S_{ki} = \langle \phi_k^{\Gamma'} | \psi_i^{\Gamma} \rangle = \hat{R} \langle \phi_k^{\Gamma'} | \psi_i^{\Gamma} \rangle \tag{6.1}$$

An integral being an infinite sum, the operator can be brought inside the bracket and then transform the bra and ket parts directly.

$$\hat{R} \langle \phi_k^{\Gamma'} | \psi_i^{\Gamma} \rangle = \langle \hat{R} \phi_k^{\Gamma'} | \hat{R} \psi_i^{\Gamma} \rangle = \sum_{jl} \overline{D}_{lk}^{\Gamma'}(R) D_{ji}^{\Gamma}(R) \langle \phi_l^{\Gamma'} | \psi_j^{\Gamma} \rangle \tag{6.2}$$

By summing over all $\hat{R} \in G$ and dividing by the group order, one obtains a form to which the GOT can be applied.

$$\begin{aligned} \langle \phi_k^{\Gamma'} | \psi_i^{\Gamma} \rangle &= \frac{1}{|G|} \sum_{R \in G} \hat{R} \langle \phi_k^{\Gamma'} | \psi_i^{\Gamma} \rangle \\ &= \frac{1}{|G|} \sum_{jl} \left(\sum_R \overline{D}_{lk}^{\Gamma'}(R) D_{ji}^{\Gamma}(R) \right) \langle \phi_l^{\Gamma'} | \psi_j^{\Gamma} \rangle \\ &= \delta_{\Gamma'\Gamma} \delta_{ik} \frac{1}{\dim(\Gamma)} \sum_j \langle \phi_j^{\Gamma'} | \psi_j^{\Gamma} \rangle \end{aligned} \tag{6.3}$$

We now rewrite this result in terms of elements of the overlap matrix $\mathbb{S}$:

$$S_{ki} = \delta_{\Gamma'\Gamma} \delta_{ik} \frac{1}{\dim(\Gamma)} \sum_j S_{jj} = \delta_{\Gamma'\Gamma} \delta_{ik} \frac{1}{\dim(\Gamma)} \mathrm{Tr}(\mathbb{S}) \tag{6.4}$$

This simple derivation yields three important results:

1. Overlap integrals between functions which transform according to different irreps are zero.
2. Overlap integrals between functions which belong to different components of the same irrep are zero.

3. Overlap integrals between functions with the same symmetry properties, i.e. transforming as the *same component* of the *same irrep*, are independent of the component choice provided that both components are normalized.

These results clearly illustrate the importance of the GOT. It not only provides a selection rule at the level of the irreps but also at the level of the components. Of course, the latter selection rule will work only if we ensured that the symmetry adaptation of the basis set has been carried out at the component level, as was explained in Sect. 5.3.

A further consequence is that SALCs on peripheral atom sites can quite often easily be derived from central symmetry-adapted orbitals. One simply has to make sure that the SALCs have the same nodal characteristics as the central functions, so as to guarantee maximal overlap. This is well illustrated in Fig. 4.6.

6.2 The Coupling of Representations

Overlap integrals are scalar products of only two factors: a bra and a ket. A general matrix element however involves three factors: it is an integral of the outer product of a bra, an operator, and a ket. This means that three irreps are combined. The evaluation of such a triad is based on the *coupling* of irreps. This concept refers to the formation of a product space. The simplest example is the formation of a two-electron wavefunction, obtained by multiplying two one-electron functions. This section will be devoted entirely to the formation of such product spaces.

Consider two sets of orbitals, transforming as the irreps Γ_a and Γ_b, respectively, each occupied by one electron. A two-electron wavefunction with electron 1 in the γ_a-component of the first set, and electron 2 in the γ_b-component of the second set is written as a simple product function: $|\Gamma_a\gamma_a(1)\rangle|\Gamma_b\gamma_b(2)\rangle$. Clearly, since the one-electron function spaces are invariants of the group, their product space is invariant, too. Now the question is to determine the symmetry of this new space. The recipe to find this symmetry can safely be entrusted to the character theorem: first, determine the character string for the product basis, and then carry out the reduction according to the character theorem. Symmetry operators are all-electron operators affecting all particles together; hence, the effect of a symmetry operation on a ket product is to transform both kets simultaneously.

$$\hat{R}\Big(|\Gamma_a\gamma_a(1)\rangle|\Gamma_b\gamma_b(2)\rangle\Big) = \sum_{\gamma'_a}\sum_{\gamma'_b} D^{\Gamma_a}_{\gamma'_a\gamma_a}(R)D^{\Gamma_b}_{\gamma'_b\gamma_b}(R)\ \ |\Gamma_a\gamma'_a(1)\rangle|\Gamma_b\gamma'_b(2)\rangle \quad (6.5)$$

The transformation of the product functions is thus expressed by a supermatrix, each element of which is a product of two matrix elements for the individual orbital transformations. The trace of this supermatrix is given by

Table 6.1 Direct product of $E_g \times T_{2g}$ in O_h symmetry

O_h	$\hat{E}$	$8\hat{C}_3$	$6\hat{C}_2$	$6\hat{C}_4$	$3\hat{C}_2$	$\hat{i}$	$6\hat{S}_4$	$8\hat{S}_6$	$3\hat{\sigma}_h$	$6\hat{\sigma}_d$
E_g	2	–1	0	0	2	2	0	–1	2	0
T_{2g}	3	0	1	–1	–1	3	–1	0	–1	1
$E_g \times T_{2g}$	6	0	0	0	–2	6	0	0	–2	0
T_{1g}	3	0	–1	1	–1	3	1	0	–1	–1
T_{2g}	3	0	1	–1	–1	3	–1	0	–1	1

$$\begin{aligned}\chi^{\Gamma_a \times \Gamma_b}(R) &= \sum_{\gamma_a \gamma_b} D^{\Gamma_a}_{\gamma_a \gamma_a}(R) D^{\Gamma_b}_{\gamma_b \gamma_b}(R) \\ &= \chi^{\Gamma_a}(R) \chi^{\Gamma_b}(R) \end{aligned} \tag{6.6}$$

This is a gratifying result. The character of a product space is simply the product of the characters of the factor spaces. Accordingly, the symmetry of the product space is identified as the *direct product* of the orbital irreps, and is denoted as $\Gamma_a \times \Gamma_b$. If both irreps are degenerate, the direct product will be reducible. Let c_Γ be the number of times that the irrep Γ occurs in the direct product:

$$\Gamma_a \times \Gamma_b = \sum_\Gamma c_\Gamma \Gamma \tag{6.7}$$

By straightforward application of the character theorem, one obtains:

$$\begin{aligned} c_\Gamma &= \frac{1}{|G|} \sum_R \overline{\chi}^\Gamma(R) \chi^{\Gamma_a \times \Gamma_b}(R) \\ &= \frac{1}{|G|} \sum_R \overline{\chi}^\Gamma(R) \chi^{\Gamma_a}(R) \chi^{\Gamma_b}(R) \end{aligned} \tag{6.8}$$

Here we have, for the first time, a formula with a triad of irreps. This will form the basis for the symmetry evaluation of general matrix elements, which will be performed in Sect. 6.5. The c_Γ coefficients are obtained by performing product manipulations on the character tables. As an example, Table 6.1 illustrates the reduction of the $E_g \times T_{2g}$ product in O_h, as given in Eq. (6.9).

$$E_g \times T_{2g} = T_{1g} + T_{2g} \tag{6.9}$$

The product of the characters on the left is equal to the sum of the characters on the right. Appendix E lists direct products for several point groups. In the case of centrosymmetric groups, only the results for the rotational subgroup are given. In this case, inversion symmetry adds to each irrep the labels g or u, and the outcome

of the product of these labels obeys the following selection rule:

$$\begin{aligned} g \times g &= g \\ u \times u &= g \\ g \times u &= u \times g = u \end{aligned} \tag{6.10}$$

This rule is the basis of the Laporte selection rule in spectroscopy. Similarly, for $D_{(2n+1)h}$ groups, the unique horizontal reflection plane $\hat{\sigma}_h$ adds to the irreps of the rotational subgroup the prime and double prime labels for symmetric and anti-symmetric reflection symmetry, respectively. The outcome of the products of these labels is entirely similar to the selection rules in Eq. (6.10): products of symmetric irreps are symmetric, as are products of anti-symmetric irreps, while mixed products are always anti-symmetric.

Let us now proceed with the two-electron problem and address the next question: after having determined which symmetry species are present, we now want to know the explicit form of the corresponding two-electron wavefunctions, i.e., we must construct the SALCs. This construction does not pose any new problems; the projection operators that were introduced in Sect. 4.5 will do the job perfectly well. Some notation is important here. The product function will be written as

$$|\Gamma\gamma(1,2)\rangle = \sum_{\gamma_a} \sum_{\gamma_b} c^{\Gamma\gamma}_{\gamma_a\gamma_b} |\Gamma_a\gamma_a(1)\rangle |\Gamma_b\gamma_b(2)\rangle \tag{6.11}$$

The combination coefficient is itself identified as a matrix element, by multiplying left and right with the one-electron bra functions and using orthonormality of the basis orbitals.

$$\begin{aligned} c^{\Gamma}_{\gamma_a\gamma_b} &= \langle \Gamma_a\gamma_a(1)\, \Gamma_b\gamma_b(2) | \Gamma\gamma(1,2)\rangle \\ &\equiv \langle \Gamma_a\gamma_a\Gamma_b\gamma_b | \Gamma\gamma\rangle \end{aligned} \tag{6.12}$$

This coefficient is known as a Clebsch–Gordan (CG) coupling coefficient and denoted by the 3Γ bracket $\langle \Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle$. It indicates how the orbital irreps Γ_a and Γ_b have to be combined to yield a product ket that transforms as $|\Gamma\gamma\rangle$. The CG-coefficients can be determined by using projection operators. The results are listed in Appendix F.

Note on the use of the coupling tables in Appendix F
The left column in the table lists all products of Γ_a and Γ_b ket functions. The $|\Gamma\gamma\rangle$ column lists the corresponding CGCs. They correspond to the notation $\langle\Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle$.

	$\|\Gamma\gamma\rangle$
$\|\Gamma_a\gamma_a\rangle\|\Gamma_b\gamma_b\rangle$	$\langle\Gamma_a\gamma_a\Gamma_b\gamma_b\|\Gamma\gamma\rangle$

- Centrosymmetric groups: CGCs identical to rotational subgroup for all products allowed under the selection rule for inversion (see Eq. (6.10)).
- $D_{(2n+1)h}$ groups: CGC identical to D_{2n+1} subgroup for all products allowed by the selection rule for the horizontal plane.
- C_{nv} groups: CGC identical to isomorphic D_n groups.

General closed formulas are available to calculate CGCs, but it is often possible to obtain these results by a simpler procedure. We illustrate this for the components of the T_{1g} two-electron state, obtained in Eq. (6.9). The z-component of this state is the only component that is totally symmetric under the $\hat{C}_4$ splitting field. It is clear that this symmetry can be obtained only by multiplying the $|e_g\epsilon\rangle$ and $|t_{2g}\zeta\rangle$-components, since these are both anti-symmetric and thus will form a symmetric product. From here on, we will adopt for the product functions the usual notation of small letters for the orbitals and capital letters for the coupled states. Hence:

$$|T_{1g}z\rangle = |e_g\epsilon\rangle|t_{2g}\zeta\rangle \tag{6.13}$$

The coupling coefficient $\langle E_g\epsilon T_{2g}\zeta|T_{1g}z\rangle$ is thus equal to 1. The x and y-components may then immediately be obtained by applying the cyclic $\hat{C}_3$ generator. As an example for the x-component,

$$\begin{aligned}
|T_{1g}x\rangle &= \hat{C}_3|T_{1g}z\rangle \\
&= \left(\hat{C}_3|e_g\epsilon\rangle\right)\left(\hat{C}_3|t_{2g}\zeta\rangle\right) \\
&= \left(-\frac{\sqrt{3}}{2}|e_g\theta\rangle - \frac{1}{2}|e_g\epsilon\rangle\right)|t_{2g}\xi\rangle \\
&= -\frac{\sqrt{3}}{2}|e_g\theta\rangle|t_{2g}\xi\rangle - \frac{1}{2}|e_g\epsilon\rangle|t_{2g}\xi\rangle
\end{aligned} \tag{6.14}$$

Thus: $\langle\theta\xi|x\rangle = -\sqrt{3}/2$; $\langle\epsilon\xi|x\rangle = -1/2$. The resulting coupling coefficients are shown in Table 6.2.

Table 6.2 The coupling coefficients for the direct product $E_g \times T_{2g}$ in O_h symmetry

$E_g \times T_{2g}$	T_{1g}			T_{2g}		
	x	y	z	ξ	η	ζ
$\|\theta\rangle\|\xi\rangle$	$-\frac{\sqrt{3}}{2}$	0	0	$-\frac{1}{2}$	0	0
$\|\theta\rangle\|\eta\rangle$	0	$\frac{\sqrt{3}}{2}$	0	0	$-\frac{1}{2}$	0
$\|\theta\rangle\|\zeta\rangle$	0	0	0	0	0	1
$\|\epsilon\rangle\|\xi\rangle$	$-\frac{1}{2}$	0	0	$\frac{\sqrt{3}}{2}$	0	0
$\|\epsilon\rangle\|\eta\rangle$	0	$-\frac{1}{2}$	0	0	$-\frac{\sqrt{3}}{2}$	0
$\|\epsilon\rangle\|\zeta\rangle$	0	0	1	0	0	0

6.3 Symmetry Properties of the Coupling Coefficients

The CG-coefficients in the finite point groups stem from Wigner's celebrated coupling coefficients for the spherical symmetry group. Wigner proposed reformulating these coefficients in terms of more primitive $3j$ symbols, which contain, in a uniform way, the permutational properties of the spherical coupling coefficients [1]. Several attempts have been made to define similar 3Γ symbols for the point group, but this requires the introduction of quite detailed phase conventions, which limits the efficiency of this formalism [2, 3]. We shall therefore not engage in a further factorization of the coupling coefficients, but express the important symmetry properties of the couplings at the level of the brackets. Two guidelines will thereby be used: when dealing with coupling coefficients, it is important to bear in mind that the coupling is based on the formation of a product, as we have illustrated in the preceding section, and, secondly, that we should treat the coupling coefficients as far as possible as ordinary brackets.

A direct consequence of the latter viewpoint is that the rules for complex conjugation of brackets apply

$$\overline{\langle \Gamma_a \gamma_a \Gamma_b \gamma_b | \Gamma \gamma \rangle} = \langle \Gamma \gamma | \Gamma_a \gamma_a \Gamma_b \gamma_b \rangle \tag{6.15}$$

Being expansion coefficients of SALCs, the coupling coefficients also obey two orthogonality rules. Column-wise orthonormality results from the orthonormal properties of the coupled states.

$$\sum_{\gamma_a \gamma_b} \langle \Gamma' \gamma' | \Gamma_a \gamma_a \Gamma_b \gamma_b \rangle \langle \Gamma_a \gamma_a \Gamma_b \gamma_b | \Gamma \gamma \rangle = \delta_{\Gamma\Gamma'} \delta_{\gamma\gamma'} \tag{6.16}$$

In addition, the scalar products along rows are orthonormal, because of the orthonormal properties of the basic kets. Note that the summation runs over the irreps of the entire product space: $\Gamma \in \Gamma_a \times \Gamma_b$.

$$\sum_{\Gamma\gamma} \langle \Gamma_a \gamma'_a \Gamma_b \gamma'_b | \Gamma\gamma \rangle \langle \Gamma\gamma | \Gamma_a \gamma_a \Gamma_b \gamma_b \rangle = \delta_{\gamma'_a \gamma_a} \delta_{\gamma'_b \gamma_b} \tag{6.17}$$

The permutational properties of the CG-coefficients refer to interchange of the bra and ket irreps. If Γ_a and Γ_b are not equivalent, their ordering will not affect the symmetry of the coupled state, since the factors in the direct product commute:

$$\Gamma_a \times \Gamma_b = \Gamma_b \times \Gamma_a \tag{6.18}$$

We can therefore define the coupling coefficients in such a way that interchange of the coupled irreps leaves the coefficient invariant:

$$\Gamma_a \neq \Gamma_b : \ \langle \Gamma_a \gamma_a \Gamma_b \gamma_b | \Gamma\gamma \rangle \equiv \langle \Gamma_b \gamma_b \Gamma_a \gamma_a | \Gamma\gamma \rangle \tag{6.19}$$

If the coupled electrons are equivalent, i.e., belong *to the same shell*, the situation is different. In this case, $\Gamma_a = \Gamma_b$, and the direct product becomes a *direct square*. For non-equivalent irreps, exchange of the components γ_a and γ_b was not possible, because they refer to different irreps. However, when they are components of the same irrep, this exchange is an important symmetry of the product space. Indeed, the squared space can be split into two separate blocks, one block which contains product functions that are symmetric under exchange of the component labels and one block which is anti-symmetric. This implies that there are two linearly independent sets of direct square coupling coefficients, which are either symmetric or anti-symmetric under exchange of the labels, i.e., $\langle \Gamma_a \gamma_a \Gamma_a \gamma_b | \Gamma\gamma \rangle = \pm \langle \Gamma_a \gamma_b \Gamma_a \gamma_a | \Gamma\gamma \rangle$. The symmetrized part of the direct square is denoted as $[\Gamma_a]^2$. For $n = \dim(\Gamma_a)$, the dimension of this subspace is equal to the number of symmetric combinations:

$$\dim([\Gamma_a]^2) = \sum_{\gamma_a} 1 + \sum_{\gamma_a < \gamma_b} 1 = n + n(n-1)/2 = n(n+1)/2 \tag{6.20}$$

On the other hand, if the coupling coefficients are anti-symmetric under exchange of the labels, the coupled state belongs to the anti-symmetrized direct square, denoted as $\{\Gamma_a\}^2$. This product space is restricted to combinations with $\gamma_a \neq \gamma_b$; its dimension is equal to $n(n-1)/2$. The characters for either part of the square can be determined separately. For the character of the $\{\Gamma_a\}^2$ part, the derivation runs as follows: one first applies a symmetry operator to an arbitrary anti-symmetric function. The ket product $|\Gamma_a \gamma_a(1)\rangle |\Gamma_a \gamma_b(2)\rangle$ will be abbreviated here as $\gamma_a(1)\gamma_b(2)$.

$$
\begin{aligned}
&\hat{R}\left(\gamma_a(1)\gamma_b(2)-\gamma_b(1)\gamma_a(2)\right)\\
&=\sum_{\gamma_a'\gamma_b'}\left(\gamma_a'(1)\gamma_b'(2)-\gamma_b'(1)\gamma_a'(2)\right)D^{\Gamma_a}_{\gamma_a'\gamma_a}(R)D^{\Gamma_a}_{\gamma_b'\gamma_b}(R)\\
&=\sum_{\gamma_a'\gamma_b'}\gamma_a'(1)\gamma_b'(2)\left(D^{\Gamma_a}_{\gamma_a'\gamma_a}(R)D^{\Gamma_a}_{\gamma_b'\gamma_b}(R)-D^{\Gamma_a}_{\gamma_a'\gamma_b}(R)D^{\Gamma_a}_{\gamma_b'\gamma_a}(R)\right)\\
&=\frac{1}{2}\sum_{\gamma_a'\gamma_b'}\left(\gamma_a'(1)\gamma_b'(2)-\gamma_b'(1)\gamma_a'(2)\right)\left(D^{\Gamma_a}_{\gamma_a'\gamma_a}(R)D^{\Gamma_a}_{\gamma_b'\gamma_b}(R)-D^{\Gamma_a}_{\gamma_a'\gamma_b}(R)D^{\Gamma_a}_{\gamma_b'\gamma_a}(R)\right)
\end{aligned}
\tag{6.21}
$$

Taking the trace then yields

$$
\begin{aligned}
\chi^{\{\Gamma_a\}^2}(R)&=\frac{1}{2}\sum_{\gamma_a\gamma_b}\left(D^{\Gamma_a}_{\gamma_a\gamma_a}(R)D^{\Gamma_a}_{\gamma_b\gamma_b}(R)-D^{\Gamma_a}_{\gamma_a\gamma_b}(R)D^{\Gamma_a}_{\gamma_b\gamma_a}(R)\right)\\
&=\frac{1}{2}\left(\left(\chi^{\Gamma_a}(R)\right)^2-\sum_{\gamma_a}D^{\Gamma_a}_{\gamma_a\gamma_a}(R^2)\right)\\
&=\frac{1}{2}\left(\left(\chi^{\Gamma_a}(R)\right)^2-\chi^{\Gamma_a}(R^2)\right)
\end{aligned}
\tag{6.22}
$$

The trace for the symmetrized product is then found by subtracting the trace in Eq. (6.22) from the total trace for the direct product.

$$
\begin{aligned}
\chi^{[\Gamma_a]^2}(R)&=\chi^{\Gamma_a{}^2}(R)-\chi^{\{\Gamma_a\}^2}(R)\\
&=\frac{1}{2}\left(\left(\chi^{\Gamma_a}(R)\right)^2+\chi^{\Gamma_a}(R^2)\right)
\end{aligned}
\tag{6.23}
$$

In Table 6.3, these quantities are given for the direct product $H_g \times H_g$ in icosahedral symmetry. The product resolution is as follows:

$$
H_g \times H_g = \left[A_g + G_g + 2H_g\right] + \left\{T_{1g} + T_{2g} + G_g\right\} \tag{6.24}
$$

Note that this product contains one totally symmetric irrep, notably in the symmetrized part. In general, for irreps with real characters the totally symmetric irrep,

Table 6.3 Direct square $H_g \times H_g$ in I_h symmetry

I_h	$\hat{E}$	$12\hat{C}_5$	$12\hat{C}_5^2$	$20\hat{C}_3$	$15\hat{C}_2$	$\hat{i}$	$12\hat{S}_{10}$	$12\hat{S}_{10}^3$	$20\hat{S}_6$	$15\hat{\sigma}$
χ^{H_g}	5	0	0	–1	1	5	0	0	–1	1
$\chi^{H_g{}^2}$	25	0	0	1	1	25	0	0	1	1
$\chi^{H_g}(R^2)$	5	0	0	–1	5	5	0	0	–1	5
$\chi^{[H_g]^2}$	15	0	0	0	3	15	0	0	0	3
$\chi^{\{H_g\}^2}$	10	0	0	1	–2	10	0	0	1	–2

Γ_0, appears in a direct square only once. This can easily be derived from Eq. (6.8). First note that, for a general direct product $\Gamma_a \times \Gamma_b$, consisting of irreps with *real characters*, one has

$$\begin{aligned}\langle \chi^{\Gamma_0} | \chi^{\Gamma_a \times \Gamma_b} \rangle &= \sum_R \overline{\chi}^{\Gamma_0}(R) \chi^{\Gamma_a}(R) \chi^{\Gamma_b}(R) \\ &= \sum_R \chi^{\Gamma_a}(R) \chi^{\Gamma_b}(R) \\ &= |G| \delta_{\Gamma_a \Gamma_b} \end{aligned} \tag{6.25}$$

Hence, only a direct square contains the totally symmetric irrep, Γ_0, exactly once. To determine if this belongs to the symmetric or anti-symmetric part, the following intermediate result is needed:

$$\begin{aligned}\sum_R \chi^{\Gamma_a}(R^2) &= \sum_R \sum_i \left[\mathbb{D}^{\Gamma_a}(R) \times \mathbb{D}^{\Gamma_a}(R) \right]_{ii} \\ &= \sum_R \sum_i \sum_j D^{\Gamma_a}_{ij}(R) D^{\Gamma_a}_{ji}(R) \\ &= \frac{|G|}{\dim(\Gamma_a)} \sum_i \sum_j \delta_{ij} = |G| \end{aligned} \tag{6.26}$$

In order to arrive at this result, we have made use of the GOT, on the assumption that the $\mathbb{D}$ matrices are real. Combining the results of Eqs. (6.25) and (6.26) with Eq. (6.23) then leads to the conclusion that the unique totally symmetric irrep belongs to the symmetrized part of the direct square:

$$\frac{1}{|G|} \langle \chi^{\Gamma_0} | \chi^{[\Gamma_a]^2} \rangle = \frac{1}{2|G|} \sum_R \chi^{\Gamma_0}(R) \left[\chi^{\Gamma_a{}^2}(R) + \chi^{\Gamma_a}(R^2) \right] = 1 \tag{6.27}$$

For irreps with real characters, but transformation matrices that cannot be all real, the unique totally symmetric product appears in the anti-symmetrized part. This is the case for spin representations, which will be dealt with in Chap. 7. In summary, as far as complex-conjugation properties are concerned, we have three kinds of irreps:

1. Irreps with real characters, and for which all $\mathbb{D}(R)$ transformation matrices can be put in real form. In this case: $\Gamma_0 \in [\Gamma_a]^2$.
2. Irreps with real characters, but which cannot be represented by transformation matrices that are all real. In this case, $\Gamma_0 \in \{\Gamma_a\}^2$.
3. Irreps with complex characters. In this case, there is always a complex-conjugate irrep, and $\Gamma_0 \in \Gamma \times \overline{\Gamma}$.

Equation (6.24) further exemplifies a case of *product multiplicity*. This is when an irrep occurs more than once in the decomposition of a direct product. Both the H_g and G_g irreps appear twice in the direct product $H_g \times H_g$. In the point groups,

product multiplicity is quite rare. It occurs only in the icosahedral group for the products $G \times H$ and $H \times H$, as well as for spin representations in cubic and icosahedral symmetries. Product multiplicity means that there are different coupling schemes for arriving at the product states. Each of these 'channels' corresponds to a separate set of CG coupling coefficients. There are several ways of obtaining linearly independent sets of coupling coefficients. For the separation of the two G_g irreps in Eq. (6.24) symmetrization of the product space is sufficient, since the symmetrized and anti-symmetrized parts contain one G_g. This strategy does not work for the two H_g irreps, which are both the result of symmetrized coupling. In this case, more elaborate splitting schemes have been constructed, based *inter alia* on higher symmetries [4, 5].

Last but not least, we should consider the relationship between coupling coefficients where irreps from bra and ket parts are interchanged. We shall limit the discussion here to the simplified case in which all ingredients of the coupling are taken to be real. A case with complex irreps will be treated in Chap. 7. Consider two related couplings: $\Gamma_a \times \Gamma_b = \Gamma$ and $\Gamma \times \Gamma_b = \Gamma_a$. The corresponding expansion coefficients are scalar matrix elements and are thus invariant under the group action. By importing the group action inside the brackets, as we have frequently done before, we obtain a set of equations in the CG-coefficients

$$\begin{aligned}
\langle \Gamma_a \gamma_a \Gamma_b \gamma_b | \Gamma \gamma \rangle &= \sum_{\gamma'_a \gamma'_b \gamma} \left[\frac{1}{|G|} \sum_R \overline{D}^{\Gamma_a}_{\gamma'_a \gamma_a}(R) \overline{D}^{\Gamma_b}_{\gamma'_b \gamma_b}(R) D^{\Gamma}_{\gamma' \gamma}(R) \right] \langle \Gamma_a \gamma'_a \Gamma_b \gamma'_b | \Gamma \gamma' \rangle \\
\langle \Gamma \gamma \Gamma_b \gamma_b | \Gamma_a \gamma_a \rangle &= \sum_{\gamma'_a \gamma'_b \gamma} \left[\frac{1}{|G|} \sum_R D^{\Gamma_a}_{\gamma'_a \gamma_a}(R) \overline{D}^{\Gamma_b}_{\gamma'_b \gamma_b}(R) \overline{D}^{\Gamma}_{\gamma' \gamma}(R) \right] \langle \Gamma \gamma' \Gamma_b \gamma'_b | \Gamma_a \gamma'_a \rangle
\end{aligned} \tag{6.28}$$

These equations form a system of homogeneous linear equations from which the coupling coefficients can be obtained. At present, we shall use this result only in the simplified case where all components have been chosen to be real, so that both CGCs in Eq. (6.28) obey the same system of equations.[1] From this, it follows that the corresponding coupling coefficients will be proportional to each other, independent of the components; hence:

$$\langle \Gamma_a \gamma_a \Gamma_b \gamma_b | \Gamma \gamma \rangle = x \langle \Gamma \gamma \Gamma_b \gamma_b | \Gamma_a \gamma_a \rangle \tag{6.29}$$

The proportionality constant can be determined by summing the square of the coefficients over all components and using the normalization result from Eq. (6.16).

[1] The general case with complex irreps is exemplified for the coupling of spin representations in Sect. 7.4.

$$\sum_{\gamma_a\gamma_b\gamma} |\langle \Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle|^2 = x^2 \sum_{\gamma_a\gamma_b\gamma} |\langle \Gamma\gamma\Gamma_b\gamma_b|\Gamma_a\gamma_a\rangle|^2$$
$$\dim(\Gamma) = x^2 \dim(\Gamma_a) \tag{6.30}$$

The permutation of irreps between bra and ket in the CG-coefficients thus requires a uniform dimensional renormalization:

$$[\dim(\Gamma)]^{-1/2}\,\langle \Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle = \pm\,[\dim(\Gamma_a)]^{-1/2}\,\langle \Gamma\gamma\Gamma_b\gamma_b|\Gamma_a\gamma_a\rangle \tag{6.31}$$

The renormalization leaves a phase factor undetermined. This phase factor is the same for the entire coupling table, and thus can be chosen arbitrarily. As an example, in the group O (see Appendix F), the coefficients $\langle T_2\xi T_1 x|E\theta\rangle$ and $\langle E\theta T_1 x|T_2\xi\rangle$ are related as follows:

$$\frac{1}{\sqrt{2}}\langle T_2\xi T_1 x|E\theta\rangle = \frac{1}{\sqrt{3}}\langle E\theta T_1 x|T_2\xi\rangle = -\frac{1}{2} \tag{6.32}$$

Here the conventional phase was chosen to be $+1$.

6.4 Pauli Exchange-Symmetry and Slater Determinants

When constructing multi-electron wavefunctions by coupling individual electrons, one must take into account the collective symmetry of the Hamiltonian: $\mathcal{H}$ is invariant under *any permutation of the positions and spins of all electrons*. This requirement reflects the fundamental indistinguishability of electrons. The permutation operation for a state of two coupled electrons is given by

$$|\Gamma\gamma(1,2)\rangle = \sum_{\gamma_a}\sum_{\gamma_b}\langle \Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle|\Gamma_a\gamma_a(1)\rangle|\Gamma_b\gamma_b(2)\rangle$$
$$\hat{P}_{12}|\Gamma\gamma(1,2)\rangle = |\Gamma\gamma(2,1)\rangle = \sum_{\gamma_a}\sum_{\gamma_b}\langle \Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle|\Gamma_a\gamma_a(2)\rangle|\Gamma_b\gamma_b(1)\rangle \tag{6.33}$$

Here $\hat{P}_{12}$ is the operator that permutes the positions of the two electrons. It commutes with the spatial symmetry operators. The $|\Gamma\gamma(1,2)\rangle$ and $|\Gamma\gamma(2,1)\rangle$ states thus have exactly the same spatial symmetries and are described by the same coupling coefficients. But are these functions the same or not? In fact, there are two possible outcomes: (i) if the two electrons are occupying different shells, the $|\Gamma\gamma(1,2)\rangle$ and $|\Gamma\gamma(2,1)\rangle$ states will be linearly independent functions, (ii) but if the two electrons occupy substates of the same shell, $|\Gamma\gamma(1,2)\rangle$ and $|\Gamma\gamma(2,1)\rangle$ are the same function, except possibly for a phase factor. First, we deal with the case of non-equivalent

electrons, this is when the two electrons occupy different shells.[2] In this case, neither $|\Gamma\gamma(1,2)\rangle$ nor $|\Gamma\gamma(2,1)\rangle$ are eigenfunctions of the permutation operator and a symmetry adaptation is in order. The permutation operator is the generator of the symmetric group, S_2, which has only two irreps, one symmetric and one anti-symmetric. Symmetry adaptation to S_2 is achieved by the usual projection method:

$$|\Gamma\gamma\rangle = \frac{1}{\sqrt{2}}\Big(|\Gamma\gamma(1,2)\rangle + (-1)^p|\Gamma\gamma(1,2)\rangle\Big) \tag{6.34}$$

where p is the parity of the permutation symmetry: symmetric for p even, anti-symmetric for p odd. Now according to the Pauli principle, one has:

Theorem 6.1 *The total wavefunction is anti-symmetric with respect to exchange of any pair of electrons. For an n-electron system, the total wavefunction changes sign under the odd permutations of the symmetric group, S_n, i.e., under permutations that consist of an odd number of transpositions of two elements, and remains invariant under even permutations.*

Until now we only considered the spatial orbital part of the wavefunction, but the requirement concerns permutations both of positions and spins and applies to the *total* wavefunction. This also involves a spin part, which should be multiplied by the orbital part. Anticipating the results of Chap. 7, we here provide the spin functions for a two-electron system. Spin functions are characterized by a spin quantum number, S, and a component, M_S, in the range $\{-S, -S+1, -S+2, ..., S-1, S\}$. The total number of components, hence the dimension of the spin-space for a given S, is equal to $2S+1$. This number is called the spin multiplicity. For a two-electron system, S can be 0 or 1; hence, there is one singlet state, and there are three components belonging to a triplet state. In a $|SM_S\rangle$ notation, they are given by

$$\begin{aligned}
|0,0\rangle &= \frac{1}{\sqrt{2}}\left[\alpha(1)\beta(2) - \beta(1)\alpha(2)\right] \\
|1,1\rangle &= \alpha(1)\alpha(2) \\
|1,0\rangle &= \frac{1}{\sqrt{2}}\left[\alpha(1)\beta(2) + \beta(1)\alpha(2)\right] \\
|1,-1\rangle &= \beta(1)\beta(2)
\end{aligned} \tag{6.35}$$

These functions also exhibit permutation symmetry: the triplet functions are symmetric under exchange of the two particles, while the singlet function is anti-symmetric under such an exchange. The total wavefunction can thus always be put in line with the Pauli principle by combining coupled orbital states with spin states of opposite permutation symmetry. For $M_S = 0$, the total wavefunction is denoted as $|\Gamma\gamma(S, M_S)\rangle$

[2] This does not necessarily imply that the irreps of these shells are different. It might be that there are two separate Γ_a levels, each occupied by one electron.

$$
\begin{aligned}
|\Gamma\gamma(0,0)\rangle &= \frac{1}{2}\sum_{\gamma_a}\sum_{\gamma_b}\langle\Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle \\
&\quad\times\Big[|\Gamma_a\gamma_a(1)\rangle|\Gamma_b\gamma_b(2)\rangle + |\Gamma_a\gamma_a(2)\rangle|\Gamma_b\gamma_b(1)\rangle\Big] \\
&\quad\times\big[\alpha(1)\beta(2) - \beta(1)\alpha(2)\big] \\
|\Gamma\gamma(1,0)\rangle &= \frac{1}{2}\sum_{\gamma_a}\sum_{\gamma_b}\langle\Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle \\
&\quad\times\Big[|\Gamma_a\gamma_a(1)\rangle|\Gamma_b\gamma_b(2)\rangle - |\Gamma_a\gamma_a(2)\rangle|\Gamma_b\gamma_b(1)\rangle\Big] \\
&\quad\times\big[\alpha(1)\beta(2) + \beta(1)\alpha(2)\big]
\end{aligned}
\tag{6.36}
$$

The signs of the spatial and spin parts in this equation are inversely correlated: if the spatial part is symmetric under exchange, the spin part is the anti-symmetric singlet, and if the spatial part is anti-symmetric, the spin part is the symmetric triplet. As a result the total waveform changes sign under the combined permutation of the positions and spins, as required by the Pauli principle. As proposed by Slater, a convenient format to express this total function is the determinantal form:

$$
|\Gamma\gamma(S,0)\rangle = \frac{1}{2}\sum_{\gamma_a}\sum_{\gamma_b}\langle\Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle\left\{\begin{vmatrix}|\Gamma_a\gamma_a\alpha(1)\rangle & |\Gamma_b\gamma_b\beta(1)\rangle\\ |\Gamma_a\gamma_a\alpha(2)\rangle & |\Gamma_b\gamma_b\beta(2)\rangle\end{vmatrix} - (-1)^S\begin{vmatrix}|\Gamma_a\gamma_a\beta(1)\rangle & |\Gamma_b\gamma_b\alpha(1)\rangle\\ |\Gamma_a\gamma_a\beta(2)\rangle & |\Gamma_b\gamma_b\alpha(2)\rangle\end{vmatrix}\right\}
\tag{6.37}
$$

or in shorthand notation, which includes normalization of the determinants:

$$
|\Gamma\gamma(S,0)\rangle = \frac{1}{\sqrt{2}}\sum_{\gamma_a}\sum_{\gamma_b}\langle\Gamma_a\gamma_a\Gamma_b\gamma_b|\Gamma\gamma\rangle\big\{|(\Gamma_a\gamma_a\alpha)(\Gamma_b\gamma_b\beta)| - (-1)^S|(\Gamma_a\gamma_a\beta)(\Gamma_b\gamma_b\alpha)|\big\}
\tag{6.38}
$$

As an example for the coupling of e_g and t_{2g} electrons, we can construct construct four states: ${}^1T_{1g}$, ${}^1T_{2g}$, ${}^3T_{1g}$, ${}^3T_{2g}$. This set of four states, totalling 24 wavefunctions, forms a *manifold*, representing all the coupled states resulting from the $(t_{2g})^1(e_g)^1$ configuration. The dimension of the manifold is equal to the product of the six possible t_{2g} substates (including spin), and the four possible e_g substates. All states can be written as linear combinations of Slater determinants. As an example, for the $|{}^1T_{1g}z\rangle$ state, one writes:

$$
\begin{aligned}
|{}^1T_{1g}z\rangle &= \frac{1}{\sqrt{2}}\Big(|\epsilon(1)\rangle|\zeta(2)\rangle + |\epsilon(2)\rangle|\zeta(1)\rangle\Big)\frac{1}{\sqrt{2}}\left[\alpha(1)\beta(2) - \beta(1)\alpha(2)\right] \\
&= \frac{1}{\sqrt{2}}\,|(\epsilon\alpha)(\zeta\beta)| - \frac{1}{\sqrt{2}}\,|(\epsilon\beta)(\zeta\alpha)|
\end{aligned}
\tag{6.39}
$$

When the coupling concerns two equivalent electrons, *i.e.* electrons that belong to *the same irreducible shell*, permutation symmetry and spatial symmetry are not independent. Indeed, in this case, symmetry adaptation of the coupled states automatically implies symmetry adaptation to the permutation operator as well. This is because the spatial symmetry operations induce permutations of the component

labels, and thus of the position of the electrons in the shell. The coupled state may now be written as:

$$|\Gamma\gamma(1,2)\rangle = \sum_{\gamma_a\gamma_b}\langle\gamma_a\gamma_b|\Gamma\gamma\rangle|\gamma_a(1)\rangle|\gamma_b(2)\rangle = \sum_{\gamma_a}\langle\gamma_a\gamma_a|\Gamma\gamma\rangle|\gamma_a(1)\rangle|\gamma_a(2)\rangle + \sum_{\gamma_a<\gamma_b}\Big[\langle\gamma_a\gamma_b|\Gamma\gamma\rangle|\gamma_a(1)\rangle|\gamma_b(2)\rangle + \langle\gamma_b\gamma_a|\Gamma\gamma\rangle|\gamma_b(1)\rangle|\gamma_a(2)\rangle\Big] \tag{6.40}$$

where the first sum is over all diagonal terms, while the second sum runs over all pairs of different component labels. Now, if the product representation belongs to the symmetrized square, $\Gamma \in [\Gamma_a]^2$, this result simplifies to:

$$|\Gamma\gamma(1,2)\rangle = \sum_{\gamma_a}\langle\gamma_a\gamma_a|\Gamma\gamma\rangle|\gamma_a(1)\rangle|\gamma_a(2)\rangle + \sum_{\gamma_a<\gamma_b}\langle\gamma_a\gamma_b|\Gamma\gamma\rangle\Big(|\gamma_a(1)\rangle|\gamma_b(2)\rangle + |\gamma_b(1)\rangle|\gamma_a(2)\rangle\Big) \tag{6.41}$$

Hence, this function is symmetric under the $\hat{P}_{12}$ operator. It also obeys the normalization condition of Eq. (6.16). It should always be multiplied by an anti-symmetric singlet spin-function in order to obey the Pauli exclusion principle. On the other hand, if the product representation belongs to the anti-symmetrized square, $\Gamma \in \{\Gamma_a\}^2$, the coupled state is given by:

$$|\Gamma\gamma(1,2)\rangle = \sum_{\gamma_a<\gamma_b}\langle\gamma_a\gamma_b|\Gamma\gamma\rangle\Big(|\gamma_a(1)\rangle|\gamma_b(2)\rangle - |\gamma_b(1)\rangle|\gamma_a(2)\rangle\Big) \tag{6.42}$$

This function is anti-symmetric under the $\hat{P}_{12}$ operator, and should be multiplied by a symmetric triplet spin-function in order to obey the Pauli principle. In summary, for the spatial part:

$$\begin{aligned}\Gamma \in [\Gamma_a]^2 &\rightarrow |\Gamma\gamma\rangle = |\Gamma\gamma(1,2)\rangle = |\Gamma\gamma(2,1)\rangle \\ \Gamma \in \{\Gamma_a\}^2 &\rightarrow |\Gamma\gamma\rangle = |\Gamma\gamma(1,2)\rangle = -|\Gamma\gamma(2,1)\rangle\end{aligned} \tag{6.43}$$

As an example, for the $(e_g)^2$ configuration the allowed states are:

$$e_g \times e_g = [A_{1g} + E_g] + \{A_{2g}\} \Rightarrow {}^1A_{1g} + {}^1E_g + {}^3A_{2g} \tag{6.44}$$

For equivalent electrons the Pauli principle thus really does function as an exclusion principle, since the coupled symmetry states are either triplets or singlets. The dimension of the manifold is given by the binomial coefficient, where q is the number of equivalent substates (including spin), and n is the number of electrons:

$$\binom{q}{n} = \frac{q!}{n!(q-n)!} \tag{6.45}$$

For the $(e_g)^2$ problem, one has $n = 2$ and $q = 4$; there are thus six two-electron states in this configuration (see Eq. (6.44)).

As a special result, we examine the symmetry of the maximal spin-multiplicity ground state of a system with a half-filled shell. The shell consists of the components $|f_1\rangle ... |f_n\rangle$, transforming according to the irrep Γ_a, and each will be occupied by one electron with α spin. The ground state corresponds to a single determinant:

$$\begin{aligned} |\Psi\rangle = |f_1\alpha ... f_2\alpha| &= \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \left[f_1\alpha(\sigma_1) ... f_n\alpha(\sigma_n) \right] \\ &= \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \left[f_{\sigma_1}\alpha(1) ... f_{\sigma_n}\alpha(n) \right] \end{aligned} \tag{6.46}$$

Here, $\sigma \in S_n$ is an element of the permutation group of the n electron labels, and $\mathrm{sgn}(\sigma)$ is its parity.[3] Equation (6.46) indicates that this permutation can equally well be applied to the component labels, since the determinant is invariant under matrix transposition. We can now calculate the matrix element in the symmetry operator:

$$\begin{aligned} \langle \Psi | \hat{R} | \Psi \rangle &= \frac{1}{n!} \sum_{\sigma, \pi \in S_n} \mathrm{sgn}(\sigma)\mathrm{sgn}(\pi) \left[\langle f_{\sigma_1}\alpha(1) | \hat{R} | f_{\pi_1}\alpha(1) \rangle ... \langle f_{\sigma_n}\alpha(n) | \hat{R} | f_{\pi_n}\alpha(n) \rangle \right] \\ &= \frac{1}{n!} \sum_{\sigma, \pi \in S_n} \mathrm{sgn}(\sigma)\mathrm{sgn}(\pi) \left[D^{\Gamma_a}_{\sigma_1\pi_1}(R) ... D^{\Gamma_a}_{\sigma_n\pi_n}(R) \right] \\ &= \sum_{\lambda \in S_n} \mathrm{sgn}(\lambda) \left[D^{\Gamma_a}_{1\lambda_1}(R) ... D^{\Gamma_a}_{n\lambda_n}(R) \right] \\ &= \det(\mathbb{D}^{\Gamma_a}) \end{aligned} \tag{6.47}$$

In this equation we have used the result from Eq. 2.9, which identified matrix elements over symmetry operators as elements of the representation matrix. The double summation over permutations covers the permutation group twice and could be reduced to a single sum. The result indicates that the spatial symmetry of the half-filled shell ground state transforms as the determinant of the irrep of the shell. This is also called the *determinantal* representation. For the e_g shell the determinantal irrep is A_{2g}. The shell ground state is thus a $^3A_{2g}$.

[3] Any permutation can be expressed as a sequence of transpositions of two elements. If the total number of transpositions is even, $\mathrm{sgn}(\sigma) = +1$; if it is odd, $\mathrm{sgn}(\sigma) = -1$. See also Sect. 3.3.

6.5 Matrix Elements and the Wigner–Eckart Theorem

A general interaction element is a bracket around an operator. Each of the three ingredients, bra, ket, and operator, can be put in symmetry-adapted form, so that they transform according to a given irrep. Moreover, provided that the symmetry adaptation is done properly, not only the irrep itself but also the subrepresentation is well defined. Altogether, the matrix element will thus be characterized by six symmetry labels, as $\langle\psi^{\Omega}_{\omega}|O^{\Lambda}_{\lambda}|\phi^{\Gamma}_{\gamma}\rangle$. The labels imply that the symmetry behavior of each of these ingredients is fully known:

$$\begin{aligned}\hat{R}\langle\psi^{\Omega}_{\omega}| &= \sum_{\omega'}\overline{D}^{\Omega}_{\omega'\omega}(R)\langle\psi^{\Omega}_{\omega'}|\\ \hat{R}|\phi^{\Gamma}_{\gamma}\rangle &= \sum_{\gamma'}D^{\Gamma}_{\gamma'\gamma}(R)|\phi^{\Gamma}_{\gamma'}\rangle\\ \hat{R}|O^{\Lambda}_{\lambda}|R^{-1} &= \sum_{\lambda'}D^{\Lambda}_{\lambda'\lambda}(R)|O^{\Lambda}_{\lambda'}|\end{aligned} \tag{6.48}$$

Note that the general form of the operator $|O^{\Lambda}_{\lambda}|$ refers to a component of an irreducible set. Such a set of operators is usually referred to as a *tensor operator*. Obvious examples are the components of the electric or magnetic dipole-moment operators. The Wigner–Eckart theorem introduces a symmetry factorization, which simplifies the evaluation of matrix elements.

Theorem 6.2 *A matrix element, involving a tensor operator, may be factorized into a product of an intrinsic scalar part and an appropriate* 3Γ *coupling coefficient.*

The scalar constant is denoted by $\langle\psi^{\Omega}||O^{\Lambda}||\phi^{\Gamma}\rangle$, and is called the *reduced matrix element*.

$$\langle\psi^{\Omega}_{\omega}|O^{\Lambda}_{\lambda}|\phi^{\Gamma}_{\gamma}\rangle = \langle\psi^{\Omega}||O^{\Lambda}||\phi^{\Gamma}\rangle\langle\Omega\omega|\Lambda\lambda\Gamma\gamma\rangle \tag{6.49}$$

To prove this theorem, one first considers the coupling of two ingredients of the matrix element and then compares the result with the third one. We thus first consider the coupling of the operator and the ket. The transformation of their product does indeed correspond to the supermatrix which is due to the direct product $\Lambda \times \Gamma$.

$$\hat{R}|O^{\Lambda}_{\lambda}|\phi^{\Gamma}_{\gamma}\rangle = \hat{R}|O^{\Lambda}_{\lambda}|\hat{R}^{-1}\hat{R}|\phi^{\Gamma}_{\gamma}\rangle = \sum_{\lambda'}\sum_{\gamma'}D^{\Lambda}_{\lambda'\lambda}(R)D^{\Gamma}_{\gamma'\gamma}(R)|O^{\Lambda}_{\lambda'}|\phi^{\Gamma}_{\gamma'}\rangle \tag{6.50}$$

This means that we couple the tensor operator and the ket to form product entities, say with irrep Π and component π:

$$|(O\phi)^{\Pi}_{\pi}\rangle = \sum_{\lambda'}\sum_{\gamma'}|O^{\Lambda}_{\lambda'}|\phi^{\Gamma}_{\gamma'}\rangle\langle\Lambda\lambda'\Gamma\gamma'|\Pi\pi\rangle \tag{6.51}$$

We now invert this equation, using the unitary properties of the coupling coefficients, to yield

$$|O_\lambda^\Lambda|\phi_\gamma^\Gamma\rangle = \sum_{\Pi'}\sum_{\pi'}|(O\phi)_{\pi'}^{\Pi'}\rangle\langle\Pi'\pi'|\Lambda\lambda\Gamma\gamma\rangle \tag{6.52}$$

Then we combine this expression with the bra.

$$\langle\psi_\omega^\Omega|O_\lambda^\Lambda|\phi_\gamma^\Gamma\rangle = \sum_{\Pi'}\sum_{\pi'}\langle\psi_\omega^\Omega|(O\phi)_{\pi'}^{\Pi'}\rangle\langle\Pi'\pi'|\Lambda\lambda\Gamma\gamma\rangle \tag{6.53}$$

The matrix elements on the right-hand side are now in fact reduced to an overlap integral where the direct product irreps are compared with the irrep of the bra. Hence, the selection rules for the overlap integrals apply

$$\langle\Omega\omega|(O\phi)_{\pi'}^{\Pi'}\rangle = \delta_{\Omega,\Pi'}\delta_{\omega\pi'}\frac{1}{\dim(\Omega)}\sum_{\omega'}\langle\Omega\omega'|(O\phi)_{\omega'}^{\Omega'}\rangle \equiv \langle\psi^\Omega||O^\Lambda||\phi^\Gamma\rangle \tag{6.54}$$

The trace summation in this equation is identified as a scalar interaction constant, which is represented by the reduced matrix element.

Combination of Eqs. (6.53) and (6.54) then yields the Wigner–Eckart theorem of Eq. (6.49), where the total interaction is the product of a scalar interaction constant and a CG coupling coefficient. The former refers to the interaction itself, the latter extracts the transformation properties. In case of product multiplicity, there will be one reduced matrix element for every coupling channel, and the matrix element is decomposed into a sum over the channels. Note that the coupling coefficient in the theorem, $\langle\Omega\omega|\Lambda\lambda\Gamma\gamma\rangle$ is the complex conjugate of the coupling coefficients found in Appendix F, of type: $\langle\Lambda\lambda\Gamma\gamma|\Omega\omega\rangle$. The Wigner–Eckart theorem or matrix-element theorem is at the heart of most chemical applications of group theory. It provides an elegant method for separating interactions into an intrinsic part and a part that depends only on the symmetry of the problem under consideration.

An important consequence of the matrix element theorem concerns the definition of selection rules. An interaction will be forbidden if the corresponding coupling coefficient in the Wigner–Eckart theorem is zero. The conditions that control the zero values of the coupling coefficients are called *triangular conditions*, since they involve the combination of three irreps. Two kinds of triangular conditions must be taken into account:

1. Selectivity at the level of representations: an interaction element is forbidden if the coupling of the three irreps involved is zero, i.e., if the direct product of the operator and ket parts does not include the irrep of the bra.

$$\Omega \notin \Lambda \times \Gamma \tag{6.55}$$

   The triad of the three irreps may also be seen as a triple direct product, $\overline{\Omega} \times \Lambda \times \Gamma$, where the bra irrep appears in its complex-conjugate form. Equation (6.8) can

now also be read as the character overlap between the totally symmetric irrep and the triple product. Accordingly, the selection rule of Eq. (6.55) can also be reformulated as follows: an interaction will be forbidden if the triple product of the irreps does not contain the totally symmetric irrep.

$$\Gamma_0 \notin \overline{\Omega} \times \Lambda \times \Gamma \tag{6.56}$$

2. Selectivity at the level of subrepresentations: subrepresentations that are defined in a splitting field must obey the triangular conditions for the subduced irreps in the corresponding subgroup.

6.6 Application: The Jahn–Teller Effect

In 1937, Jahn and Teller made the claim that degenerate states of molecules are intrinsically unstable [6–8].

Theorem 6.3 *Non-linear molecules in a spatially degenerate electronic state are subject to spontaneous symmetry-breaking forces that distort the molecule to a geometry of lower symmetry, where the degeneracy is removed.*

The theorem is based on a perturbation of the Hamiltonian by small displacements of the nuclei. A high-symmetry geometry is chosen as the origin, and the nuclear displacements are described by normal modes which transform as irreps of the point group. The nuclear positions are parameters in the electronic Hamiltonian. One has, to second-order:

$$\begin{aligned} \mathcal{H} &= \mathcal{H}_0 + \mathcal{H}' \\ \mathcal{H}' &= \sum_{Q_{\Gamma\gamma}} \left(\frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right)_0 Q_{\Gamma\gamma} + \frac{1}{2} \sum_{Q_{\Gamma\gamma}} \sum_{Q_{\Gamma'\gamma'}} \left(\frac{\partial^2 \mathcal{H}}{\partial Q_{\Gamma\gamma} \partial Q_{\Gamma'\gamma'}} \right)_0 Q_{\Gamma\gamma} Q_{\Gamma'\gamma'} \end{aligned} \tag{6.57}$$

The partial derivatives with respect to the normal modes will affect only the electrostatic V_{Ne} term in the Hamiltonian. These operators are thus electrostatic one-electron operators. At the coordinate origin, the electronic state is degenerate, and is described by a set of wavefunctions, $|\Gamma_a \gamma_a\rangle$, where Γ_a is a degenerate irrep. The energies as functions of the coordinates are obtained by diagonalizing the Hamiltonian matrix, $\mathbb{H}$, with elements:

$$H_{\gamma_a \gamma_b} = \langle \Gamma_a \gamma_a | \mathcal{H} | \Gamma_a \gamma_b \rangle = E_0 \delta_{\gamma_a \gamma_b} + \langle \Gamma_a \gamma_a | \mathcal{H}' | \Gamma_a \gamma_b \rangle \tag{6.58}$$

The matrix in $\mathcal{H}'$ is also called the Jahn–Teller (JT) matrix. The linear terms in this matrix are of type:

$$\left\langle \Gamma_a \gamma_a \left| \left(\frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right)_0 Q_{\Gamma\gamma} \right| \Gamma_a \gamma_b \right\rangle = \left\langle \Gamma_a \gamma_a \left| \frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right| \Gamma_a \gamma_b \right\rangle_0 Q_{\Gamma\gamma} \tag{6.59}$$

We have used the fact that the integration in this matrix element runs over electronic coordinates, and does not affect the nuclear coordinates. The Wigner–Eckart theorem can be applied to derive the selection rules. Since the Hamiltonian is invariant under the elements of the symmetry group, the transformation properties of the operator part in this matrix element will be determined by the partial derivatives, $\partial/\partial Q_{\Gamma\gamma}$. As we have seen in Sect. 1.3, a partial derivative in a variable has the same transformation properties as the variable itself.[4] The operator part is thus given by

$$\left\langle \Gamma_a \gamma_a \left| \frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right| \Gamma_a \gamma_b \right\rangle_0 = \langle \Gamma_a || \Gamma || \Gamma_a \rangle \langle \Gamma_a \gamma_a | \Gamma \gamma \Gamma_a \gamma_b \rangle \tag{6.60}$$

The coupling coefficient on the right-hand side of Eq. (6.60) restricts the symmetry of the nuclear displacements to the direct square of the irrep of the electronic wavefunction. This selection rule is made even more stringent by time-reversal symmetry. The Hamiltonian is based on displacement of nuclear charges, and not on momenta, so as an operator, it is time-even or real.[5] For spatially degenerate irreps, which are of the first kind, i.e., can be represented by real functions, JT matrix elements can thus be chosen to be entirely real, which implies

$$\langle \Gamma_a \gamma_a | \mathcal{H}' | \Gamma_a \gamma_b \rangle = \langle \Gamma_a \gamma_b | \mathcal{H}' | \Gamma_a \gamma_a \rangle \tag{6.61}$$

Combining this result with Eq. (6.60) implies that the coupling coefficients, to first-order, should obey

$$\langle \Gamma_a \gamma_a | \Gamma \gamma \Gamma_a \gamma_b \rangle = \langle \Gamma_a \gamma_b | \Gamma \gamma \Gamma_a \gamma_a \rangle \tag{6.62}$$

In view of Eq. (6.31), this condition can be rewritten as

$$\langle \Gamma_a \gamma_a \Gamma_a \gamma_b | \Gamma \gamma \rangle = \langle \Gamma_a \gamma_b \Gamma_a \gamma_a | \Gamma \gamma \rangle \tag{6.63}$$

The JT distortion modes are thus restricted to the symmetrized square of the degenerate irrep of the electronic state, minus the totally symmetric modes, since these cannot lower the symmetry:

$$\Gamma \in \Big([\Gamma_a \times \Gamma_a] - \Gamma_0 \Big) \tag{6.64}$$

Modes that obey this selection rule, are said to be *JT active*. The evaluation of the second-order matrix elements requires two steps. One first couples the two distortion modes to a composite tensor operator: $|\Omega\omega|$.

[4] For complex variables, variable and derivative have complex-conjugate transformation properties.

[5] The general time-reversal selection rules are discussed in Sect. 7.6.

$$\left|\frac{\partial^2\mathcal{H}}{\partial Q_{\Gamma\gamma}\partial Q_{\Gamma'\gamma'}}\right| = \sum_{\Omega\omega} |\Omega\omega\rangle\langle\Omega\omega|\Gamma\gamma\Gamma'\gamma'\rangle \tag{6.65}$$

The second-order matrix element then becomes

$$\left\langle \Gamma_a\gamma_a \left|\frac{\partial^2\mathcal{H}}{\partial Q_{\Gamma\gamma}\partial Q_{\Gamma'\gamma'}}\right| \Gamma_a\gamma_b \right\rangle_0 = \sum_{\Omega\omega} \langle\Gamma_a||\Omega||\Gamma_a\rangle\langle\Omega\omega|\Gamma\gamma\Gamma'\gamma'\rangle\langle\Gamma_a\gamma_a|\Omega\omega\Gamma_a\gamma_b\rangle \tag{6.66}$$

The second-order elements thus are related to a product of two 3Γ symbols.[6] A special element arises when Ω is totally symmetric. In this case, the coupling coefficients are given by

$$\begin{aligned}\langle\Gamma_0|\Gamma\gamma\Gamma'\gamma'\rangle &= \frac{1}{\sqrt{\dim(\Gamma)}}\delta_{\Gamma\Gamma'}\delta_{\gamma\gamma'}\\ \langle\Gamma_a\gamma_a|\Gamma_0\Gamma_a\gamma_b\rangle &= \delta_{\gamma_a\gamma_b}\end{aligned} \tag{6.67}$$

The second-order expressions then are reduced to a diagonal matrix element:

$$\begin{aligned}&\langle\Gamma_a||\Gamma_0||\Gamma_a\rangle\,\langle\Gamma_0|\Gamma\gamma\Gamma'\gamma'\rangle\langle\Gamma_a\gamma_b|\Gamma_0\Gamma_a\gamma_b\rangle\\ &= \frac{1}{\sqrt{\dim(\Gamma)}}\,\langle\Gamma_a||\Gamma_0||\Gamma_a\rangle\,\delta_{\Gamma\Gamma'}\delta_{\gamma\gamma'}\delta_{\gamma_a\gamma_b} = K_\Gamma\ \delta_{\Gamma\Gamma'}\delta_{\gamma\gamma'}\delta_{\gamma_a\gamma_b}\end{aligned} \tag{6.68}$$

K_Γ in this equation is the harmonic force-constant. It gives rise to a constant diagonal term which provides an attractive potential around the minimum and keeps the surface bound at larger distances from the origin. The general expression for the potential-energy surface then becomes

$$E_k(Q) = E_0 + \sum_\Gamma \frac{1}{2}K_\Gamma\left(\sum_\gamma Q^2_{\Gamma_\gamma}\right) + \varepsilon_k(Q) \tag{6.69}$$

Here, $\varepsilon_k(Q)$ represents the kth root of the Hamiltonian matrix. This equation describes a surface with multiple sheets, one for each root, which cross in the high-symmetry origin. In its simplest form, the Hamiltonian can be restricted to the linear terms only. In the second-order approximation, non-totally symmetric second-order terms will also be included.

The prototype of the JT surface is the celebrated *Mexican hat* potential, which describes the effect of the twofold-degenerate cubic or trigonal E state. A typical example is the ^2E_g ground state of octahedral Cu^{2+} complexes, with $(t_{2g})^6(e_g)^3$

[6] Such combinations can be cast in a higher order symbol, known as 6Γ symbol, by analogy with the $6j$ coupling coefficients in atomic spectroscopy.

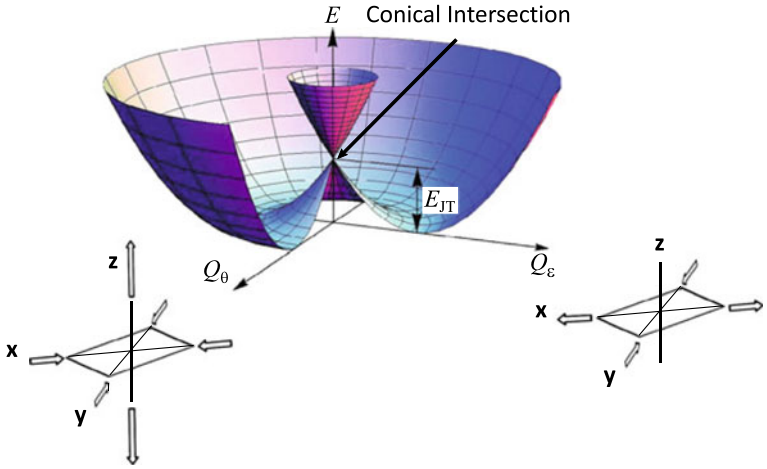


Fig. 6.1 The Mexican hat potential-energy surface of the $E \times e$ linear JT problem. The nuclear displacement coordinates are the tetragonal elongation, Q_θ, and the orthorhombic in-plane distortion, Q_ϵ

configuration. The JT-active mode in this case is restricted to an e_g mode, corresponding to the symmetrized square.

$$\left[E_g \times E_g\right] - A_{1g} = E_g \tag{6.70}$$

This distortion mode consists of the tetragonal and orthorhombic stretchings, which we already encountered as vibrational modes of UF_6, and are depicted in Fig. 6.1. By use of the appropriate $\langle Ei|Ej\,Ek\rangle$ coupling coefficients the JT matrix can easily be derived. The force element is usually defined as

$$\sqrt{2}F_E = \left\langle E||\frac{\partial \mathcal{H}}{\partial Q}||E\right\rangle \tag{6.71}$$

Using the $E \times E$ coupling coefficients in Appendix F for D_3 or O_h, the matrix then becomes

$$\mathbb{H} = \left(E_0 + \frac{1}{2}K_E\left(Q_\theta^2 + Q_\epsilon^2\right)\right)\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} + F_E\begin{bmatrix}-Q_\theta & Q_\epsilon\\ Q_\epsilon & Q_\theta\end{bmatrix} \tag{6.72}$$

In the literature, frequent use is made of an alternative sign convention for the coupling coefficients [9]. This convention is based on the formation of the quadrupole tensor elements $x^2 - y^2, xy$ from the dipole vector components x, y:

$$\mathbb{H}' = F_E\begin{bmatrix}Q_{x^2-y^2} & Q_{xy}\\ Q_{xy} & -Q_{x^2-y^2}\end{bmatrix}$$

These are precisely the coupling coefficients that appear in the $E_1 \times E_1 = E_2$ coupling table for the group D_5, where the electronic states and the coupling modes transform as E_1 and E_2, respectively. In the following, we will stick to the trigonal and cubic coupling scheme of Eq. (6.72). It can be changed into the sign pattern of the alternative convention by simultaneously changing the sign of F_E and replacing Q_ϵ by $-Q_\epsilon$. To diagonalize this Hamiltonian, it is convenient to introduce cylindrical coordinates $\{\rho, \varphi\}$:

$$\begin{aligned} Q_\theta &= \rho \cos\varphi \\ Q_\epsilon &= \rho \sin\varphi \end{aligned} \tag{6.73}$$

Then the secular equation of the force element matrix in Eq. (6.72) becomes

$$\varepsilon_k^2 - F_E^2 \rho^2 \cos^2\varphi - F_E^2 \rho^2 \sin^2\varphi = 0 \tag{6.74}$$

Two roots are found, which are independent of the angular coordinate. The corresponding eigenfunctions are

$$\begin{aligned} \epsilon_1 = -F_E\rho \;&\longrightarrow\; |\psi_1\rangle = \cos\frac{\varphi}{2}|E\theta\rangle - \sin\frac{\varphi}{2}|E\epsilon\rangle \\ \epsilon_2 = F_E\rho \;&\longrightarrow\; |\psi_2\rangle = \sin\frac{\varphi}{2}|E\theta\rangle + \cos\frac{\varphi}{2}|E\epsilon\rangle \end{aligned} \tag{6.75}$$

The surface consists of two sheets and exhibits rotational symmetry.

$$\begin{aligned} E_\pm &= E_0 + \frac{1}{2}K_E\rho^2 \pm F_E\rho \\ &= E_0 + \frac{1}{2}K_E\left(Q_\theta^2 + Q_\epsilon^2\right) \pm F_E\sqrt{Q_\theta^2 + Q_\epsilon^2} \end{aligned} \tag{6.76}$$

A cross section of this surface looks like a two-well potential, with two displaced parabolæ. The depth of the well is called the JT stabilization energy:

$$E_{JT} = -\frac{F_E^2}{2K_E} \tag{6.77}$$

In the 2D space of the active modes, these parabolæ revolve around the center, giving rise to the Mexican hat appearance. At the origin this surface has the shape of a conical intersection, indicating that the high-symmetry point is unstable, and will spontaneously relax to the circular trough surrounding the degeneracy [10]. The distorted system in the trough orbits around the origin. This motion is a *pseudo*-rotation, i.e., it is not a rotation of the molecular frame, but a gradual redistribution of the distortions between the Cartesian directions. The distortions as a function of the rotation angle are shown in Table 6.4, for ϵ_1 the lower root, with $F_E > 0$. The table is calculated by the following system of linear equations:

Table 6.4 Distortions of the octahedral frame as a function of the angle of revolution on the Jahn–Teller surface (in units of ρ)

ϕ	0°	30°	60°	90°	120°	150°	180°
ΔZ	0.816	0.707	0.408	0	–0.408	–0.707	–0.816
ΔX	–0.408	0	0.408	0.707	0.816	0.707	0.408
ΔY	–0.408	–0.707	–0.816	–0.707	–0.408	0	0.408

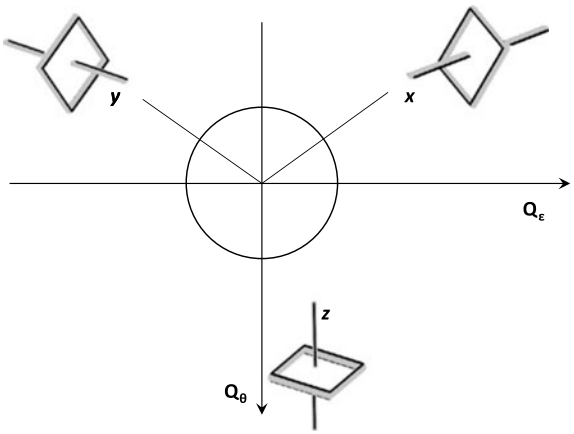


Fig. 6.2 Rotation of the distortion in the trough of the Mexican hat. Along the Q_θ-coordinate, the complex is elongated along its z-axis. Rotation around the center in the direction of Q_ϵ will shorten the z-axis and increase the x-axis. At an angle of revolution of 120°, a tetragonally elongated structure is again found, but this time with the elongation along the x-direction, and similarly at 240°, with the elongation along the y-axis

$$
\begin{aligned}
Q_a &= \frac{1}{\sqrt{3}}(\Delta Z + \Delta X + \Delta Y) = 0 \\
Q_\theta &= \frac{1}{\sqrt{6}}(2\Delta Z - \Delta X - \Delta Y) = \rho\cos\phi \\
Q_\epsilon &= \frac{1}{\sqrt{2}}(\Delta X - \Delta Y) = \rho\sin\phi
\end{aligned}
\tag{6.78}
$$

The turning points are illustrated in Fig. 6.2. The starting point is at $\varphi = 0$, in the direction of the Q_θ mode. In this mode, the z-axis is elongated, and the xy-plane is contracted. A counterclockwise rotation activates the orthorhombic Q_ϵ mode, while the tetragonal Q_θ mode is receding. This introduces a difference between the x- and y-axes: the distortion in the x-direction becomes more pronounced, while the y-axis contracts further. At the same time, the elongation of the z-axis diminishes. At an angle of 60°, the x- and z-axes are both elongated to an equal extent, giving rise to a weak xz-plane and a strong y-axis. At the 90° point, the Q_θ contribution vanishes, and the distortion is orthorhombic, with a short y-axis, an elongated x-axis, and

an undistorted z-axis. At an angle of 120°, we reach the point where the x-axis is weak, and the perpendicular yz-plane is strong. We thus have regained an elongated tetragonal configuration, but the elongation has been rotated from the z-axis to the x-axis. Continuing now at 240°, we shall have traveled another third of the trough and reoriented the tetragonal axis along the y-direction.

We can also follow the wavevector along the trough. If it is assumed that $F_E > 0$, the lower eigenfunction will be the $|\psi_1\rangle$ eigenfunction of Eq. (6.75). In the starting elongated tetragonal configuration, the ground state coincides with the $|E\theta\rangle$ basis function. By the time we have reached the orthorhombic configuration at $\varphi = 90°$, the ground state has rotated in the opposite direction by only half that angle and equals $1/\sqrt{2}\,(|E\theta\rangle - |E\epsilon\rangle)$. At 180°, we reach a structure which is tetragonally compressed along the z-direction. Accordingly, the Q_θ mode has changed sign, in contrast to the eigenfunction, where $|E\theta\rangle$ is replaced by $-|E\epsilon\rangle$. Finally, when completing a full rotation, the eigenfunction becomes $-|E\theta\rangle$. It thus takes two rotations over 720° to return to the starting point in function space. The observation of this remarkable rotational symmetry is an unexpected feature, which is not related to the point group but is due to the linear dependence of the JT Hamiltonian on the distortion coordinates in the vicinity of the degenerate origin. To describe this symmetry, we first reformulate the force element Hamiltonian in Eq. (6.72) in Dirac notation:

$$\mathcal{H}' = F_E\Big(Q_\theta\big[-|E\theta\rangle\langle E\theta| + |E\epsilon\rangle\langle E\epsilon|\big] + Q_\epsilon\big[|E\theta\rangle\langle E\epsilon| + |E\epsilon\rangle\langle E\theta|\big]\Big) \tag{6.79}$$

The operator, corresponding to a rotation in coordinate space, is given by

$$\mathcal{L} = \frac{\partial}{\partial\varphi} = \frac{\partial Q_\epsilon}{\partial\varphi}\frac{\partial}{\partial Q_\epsilon} + \frac{\partial Q_\theta}{\partial\varphi}\frac{\partial}{\partial Q_\theta} = Q_\theta\frac{\partial}{\partial Q_\epsilon} - Q_\epsilon\frac{\partial}{\partial Q_\theta} \tag{6.80}$$

The partial derivatives were obtained from Eq. (6.73). The commutator of this operator with the Hamiltonian is

$$[\mathcal{L}, \mathcal{H}'] = F_E\Big(Q_\epsilon\big[|E\theta\rangle\langle E\theta| - |E\epsilon\rangle\langle E\epsilon|\big] + Q_\theta\big[|E\theta\rangle\langle E\epsilon| + |E\epsilon\rangle\langle E\theta|\big]\Big) \tag{6.81}$$

Surprisingly, this commutator does not vanish. This is an important observation, which directly points to the vibrational-electronic or *vibronic* coupling between the distortion modes and the electronic wavevector. When the system rotates around the origin in coordinate space, not only are the coordinates changing but also the wavevector is rotating simultaneously, so we must also provide an angular momentum operator for a rotation in the function space [11]. We can construct this by analogy with Eq. (6.80), but with an important amendment: as we have argued while discussing Fig. 6.2, the coordinates rotate in the opposite sense. The angular variation of the wavefunction thus reads

$$\mathcal{S}|\psi_1\rangle = -\frac{\partial}{\partial\phi}|\psi_1\rangle = \frac{1}{2}\Big(\sin\frac{\phi}{2}|E\theta\rangle + \cos\frac{\phi}{2}|E\epsilon\rangle\Big)$$
$$\mathcal{S} = \frac{1}{2}\Big(-|E\theta\rangle\langle E\epsilon| + |E\epsilon\rangle\langle E\theta|\Big) \tag{6.82}$$

The factor $1/2$ in this result indicates that the wavevector rotates at half speed as compared to the coordinates. While the coordinate and wavefunction rotations do not commute with the Hamiltonian on their own, the total rotation operator $\mathcal{J} = \mathcal{L} + \mathcal{S}$ indeed does

$$[\mathcal{J}, \mathcal{H}'] = [\mathcal{L}, \mathcal{H}'] + [\mathcal{S}, \mathcal{H}'] = 0 \tag{6.83}$$

As the reader will have noticed, we have made use of the standard spectroscopic symbols for orbital angular momentum and spin momentum. The relation between the electronic components and the distortion coordinates is indeed somewhat analogous to the relationship between spin and spatial position of electrons. Spin–orbit coupling aligns the orientation of the spins with the electronic movement in space. In a similar fashion, vibronic coupling aligns the electronic eigenvectors with the distortions of the nuclear frame as it vibrates. Hence, two spaces are involved: the coordinate space, formed by Q_θ, Q_ϵ, and the function space, formed by the components $|\theta\rangle$, $|\epsilon\rangle$. The Hamiltonian in Eq. (6.72) couples both spaces. As shown in Eq. (6.72), the eigenvector is reoriented as the system rotates around the surface pivot, but at half speed, and in opposite direction. The latter aspect is reflected in the minus sign for the spin operator in Eq. (6.82). An alternative formulation for the coupling Hamiltonian exists with a plus sign instead. Ultimately, this depends on phase conventions for the coupling coefficients.

To form the vibronic wavefunction, describing the dynamics of the Mexican hat system, the electronic state has to be combined with nuclear wavefunctions. If the JT effect is pronounced, the vibronic levels take the form of a radial oscillator, describing transverse oscillations in the bottom of the through, and pseudo-rotational levels, describing the longitudinal motion along the bottom of the trough. The total vibronic wavefunction should of course be single-valued after a full turn around the trough, which takes the system back to the starting point. Hence, since the electronic part changes sign after a full turn, the vibrational part should also show a compensating sign change. This is indeed the case: the pseudo-rotational levels are characterized by half-integral angular momentum [12].

6.7 Application: Pseudo-Jahn–Teller Interactions

Pseudo-JT interactions (PJT) refer to the second-order vibronic coupling between electronic states which are separated by a gap [13]. In this section we describe the case of a non-degenerate ground state, $|\psi_{\Sigma\sigma}\rangle$, which is coupled to an excited manifold. The Hamiltonian is identical to the expression in Eq. (6.57). Application of the selection rules shows that the diagonal contribution, $\langle\psi_{\Sigma\sigma}|\mathcal{H}|\psi_{\Sigma\sigma}\rangle$, is limited to the totally

symmetric operator associated with the harmonic restoring-force, as in Eq. (6.69). Perturbation theory further provides interactions between the ground and excited states. These interactions are usually limited to first-order contributions, which give rise to a quadratic coordinate dependence. Hence one has, to second order in the displacements:

$$\begin{aligned} E(Q) = E_0 + \sum_{\Gamma} \frac{1}{2} K_{\Gamma} \left(\sum_{\gamma} Q_{\Gamma_\gamma}^2 \right) \\ + \sum_{\Lambda\lambda} \sum_{\Gamma\gamma} \frac{\left| \langle \psi_{\Lambda\lambda} \left| \frac{\partial \mathcal{H}}{\partial Q_{\Gamma_\gamma}} \right| \psi_{\Sigma\sigma} \rangle_0 \right|^2}{E_0 - E_\Lambda} Q_{\Gamma_\gamma}^2 \end{aligned} \tag{6.84}$$

where we have used the property that the Hamiltonian matrix is hermitian. The selection rule in this process resides with the matrix elements in the enumerator of the bilinear term. The vibronic operator must couple ground and excited states; hence, it is required that their triple direct product contains the totally symmetric irrep:

$$\Gamma_0 \in \overline{\Gamma}_\Lambda \times \Gamma \times \Sigma \tag{6.85}$$

Applying the Wigner–Eckart theorem to the matrix element yields

$$\langle \psi_{\Lambda\lambda} \left| \frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right| \psi_{\Sigma\sigma} \rangle_0 = \langle \Lambda\lambda | \Gamma\gamma\Sigma\sigma \rangle \langle \Lambda || \Gamma || \Sigma \rangle \tag{6.86}$$

The sum over the λ-components of the excited state, transforming as the Λ irrep, can be simplified by using the orthonormality property of the coupling coefficients from Eq. (6.17).

$$\begin{aligned} \sum_{\lambda \in \Lambda} \frac{\left| \langle \psi_{\Lambda\lambda} \left| \frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right| \psi_{\Sigma\sigma} \rangle_0 \right|^2}{E_0 - E_\Lambda} &= \sum_{\lambda \in \Lambda} \frac{|\langle \Lambda\lambda | \Gamma\gamma\Sigma\sigma \rangle \langle \Lambda || \Gamma || \Sigma \rangle|^2}{E_0 - E_\Lambda} \\ &= \frac{|\langle \Lambda || \Gamma || \Sigma \rangle|^2}{E_0 - E_\Lambda} \end{aligned} \tag{6.87}$$

Note that, in case of a non-degenerate ground state, the product $\Gamma \times \Sigma$ yields only one irrep, since the norm of the product character string equals the order of the group.

$$\langle \chi^{\Gamma \times \Sigma} | \chi^{\Gamma \times \Sigma} \rangle = \langle \chi^{\Gamma} \chi^{\Sigma} | \chi^{\Gamma} \chi^{\Sigma} \rangle = \langle \chi^{\Gamma} | \chi^{\Gamma} \rangle = |G| \tag{6.88}$$

The summation over $\lambda \in \Lambda$ thus covers the entire product space of $\Gamma \times \Sigma$. Combining the sum rule of Eq. (6.87) with the total expression for the PJT, one finds

$$E(Q) = E_0 + \sum_{\Gamma} \left\{ \frac{1}{2} K_\Gamma + \sum_{\Lambda} \frac{|\langle \Lambda || \Gamma || \Sigma \rangle|^2}{E_0 - E_\Lambda} \right\} \left(\sum_{\gamma} Q_{\Gamma\gamma}^2 \right) \tag{6.89}$$

Hence, when the ground state is non-degenerate, the first-order dependence of the energy on symmetry-lowering displacement vanishes, and the second-order term contains two contributions: the diagonal harmonic force constant, which is always positive, and the bilinear relaxation term, which is always negative. If the excited states are close in energy to the ground state, and if the vibronic coupling is strong, the relaxation term may be dominant, and a second-order symmetry-breaking effect will result. This is known as the pseudo-JT effect. There are two main applications of this effect: in geometry optimization, and in reaction dynamics.

In reaction dynamics, the PJT may be responsible for stereoselectivity, because of the selection rules for vibronic coupling matrix elements. Via these relaxation matrix elements the Wigner–Eckart theorem is at the basis of the Woodward–Hoffmann rules [14]. We shall not discuss these rules in general, but consider a few simple illustrations, related to electrocyclic reactions [15]. The ring closure of *cis*-butadiene is illustrated in Fig. 6.3. The relevant occupied orbitals of the reagent are the two bonding π-bonds of butadiene, with the same nodal pattern as was demonstrated for the trans-isomer in Eq. 4.107. On the right are the occupied bonding orbitals in

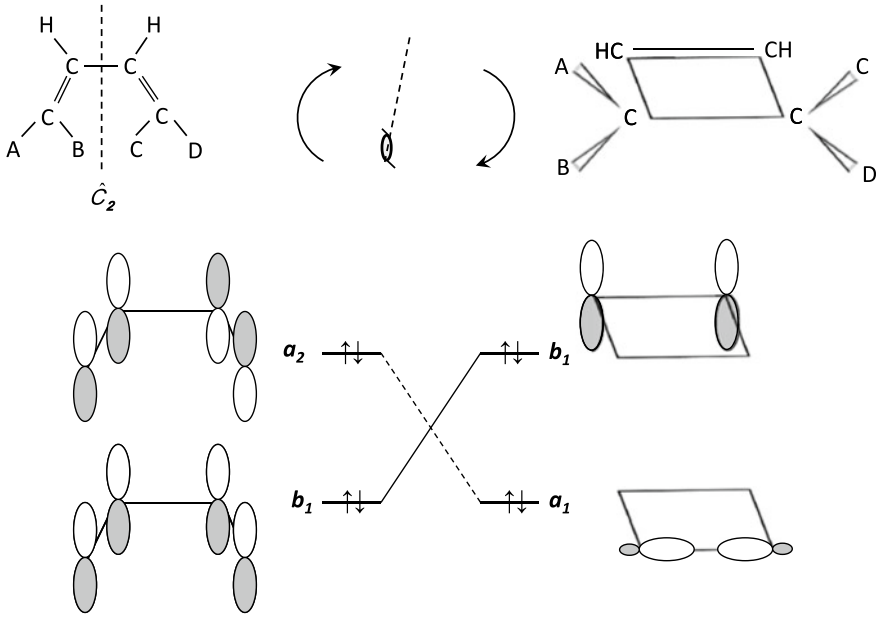


Fig. 6.3 Ring closure of *cis*-butadiene to *cyclo*-butene. In C_{2v} symmetry, there is a symmetry mismatch between the a_2 and a_1 occupied orbitals. Vibronic orbital coupling requires a concerted mechanism, based on a conrotatory ring closure, which conserves only the $\hat{C}_2$-axis

the product: a central π-bond and a newly formed σ-bond. In order for the reaction to occur, there has to be a seamless correlation between the bonds in the reagent and the product. However, as the diagram shows, in the common C_{2v} point group there is a mismatch between the symmetries left and right. For the correlation to take place, the reaction coordinate has to reduce the symmetry so that the a_2-orbital can interchange with an a_1-orbital. This interchange is taking place via a PJT mechanism which couples the a_2 occupied orbital to an a_1 virtual orbital in the reagent. As the reaction coordinate proceeds, this coupling is intensified and leads to an interchange of both. The relevant matrix element is thus an orbital vibronic coupling element:

$$\langle a_2 \left| \frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right| a_1 \rangle \neq 0 \tag{6.90}$$

Hence, a distortion coordinate is required which transforms as $a_2 \times a_1 = a_2$. The coordinate with this symmetry is the one that destroys the symmetry planes but keeps the twofold axis. This is typically a *conrotatory* reaction, where the extremal carbon atoms rotate simultaneously in the same sense, to form the σ-bond. Ring closure of substituted butadienes thus follows a conrotatory reaction stereochemistry, at least if the reaction is concerted.

This ring-closure selection rule is further confirmed by the closure reaction for the cis-1,3,5 hexatriene to 1,3-cyclohexadiene, as illustrated in the Walsh diagram of Fig. 6.4. Here, a b_1-orbital has to interchange with a virtual orbital of a_1 symmetry. The selection thus takes place at the level of the orbital matrix element:

$$\langle b_1 \left| \frac{\partial \mathcal{H}}{\partial Q_{\Gamma\gamma}} \right| a_1 \rangle \neq 0 \tag{6.91}$$

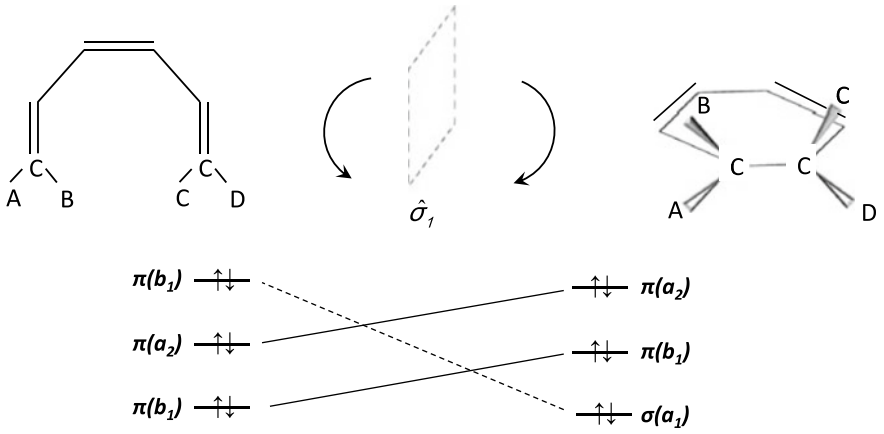


Fig. 6.4 Ring closure of *cis*-1,3,5-hexatriene to *cyclo*-hexadiene. In C_{2v} symmetry, there is a symmetry mismatch between the b_1 and a_1 occupied orbitals. Vibronic orbital coupling requires a concerted mechanism, based on a disrotatory ring closure, which conserves only the $\hat{\sigma}_1$ plane

Clearly, the distortion coordinate should now be of $b_1 \times a_1 = b_1$ symmetry, and this corresponds to the *disrotatory* mode, which destroys the $\hat{C}_2$-axis but keeps the vertical reflection plane.

6.8 Application: Linear and Circular Dichroism

Selection rules are of primary importance in spectroscopy, where they provide direct evidence concerning the nature of excited states. As an application, we study the linear and circular dichroism of tris-chelate transition-metal complexes [16]. The prototype is a divalent ruthenium complex with three 2, 2′-bipyridyl ligands, which is an important chromophore for energy conversion. In this section, we shall describe the charge transfer and intra-ligand transitions of this type of complex. The linear dichroism (LD) spectrum measures the absorption of the chromophore under plane-polarized incident light for different orientations of the polarization with respect to the molecular frame. This requires that the molecules should be embedded in an oriented phase, such as a crystalline host. Circular dichroism (CD) measures the difference in absorption between left and right circularly polarized light. Since this is based on the intrinsic helicity of the molecule, it can be performed in a non-oriented medium, such as a solution.

As always, we start the treatment by making a simple sketch of the structure in Fig. 6.5. Two sets of Cartesian axes are relevant. In the usual octahedral coordinate system, the x, y, z-axes coincide with the metal-ligand bond directions, assuming that the ligator atoms form a perfect octahedron. In addition, in Fig. 3.6 of Chap. 3,

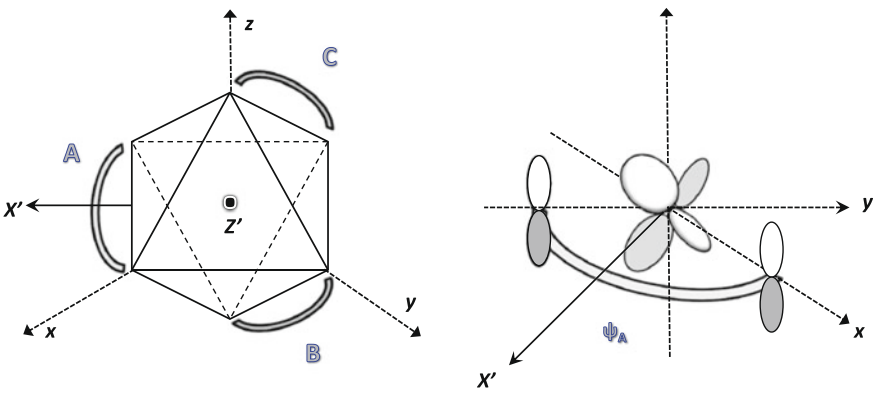


Fig. 6.5 Δ-enantiomer of tris-chelate octahedral complex of D_3 symmetry. The xyz-coordinate system passes through the ligator atoms; the primed coordinate system has the z'-direction along the threefold axis, and x' on a twofold axis through ligand A. The ligand orbital shown on the right is of ψ-type: it is anti-symmetric under a rotation about the twofold axis through the ligator bridge. The ψ orbital on ligand A interacts with the $\frac{1}{\sqrt{2}}\left(d_{xz} - d_{yz}\right)$-combination on the metal

Table 6.5 Symmetry-adapted zeroth-order metal and ligand orbital functions

t_{2g}-orbitals		Dipole moments	
$\|a_1\rangle =$	$\frac{1}{\sqrt{3}}\left(d_{xy}+d_{xz}+d_{yz}\right)$	$a_2:$	$\mu_{z'}$
$\|e_\theta\rangle =$	$\frac{1}{\sqrt{6}}\left(-2d_{xy}+d_{xz}+d_{yz}\right)$	$e_\theta:$	$\mu_{x'}$
$\|e_\epsilon\rangle =$	$\frac{1}{\sqrt{2}}\left(d_{xz}-d_{yz}\right)$	$e_\epsilon:$	$\mu_{y'}$
ψ-orbitals		χ-orbitals	
$\|a_2\rangle =$	$\frac{1}{\sqrt{3}}\left(\|\psi_A\rangle+\|\psi_B\rangle+\|\psi_C\rangle\right)$	$\|a_1\rangle =$	$\frac{1}{\sqrt{3}}\left(\|\chi_A\rangle+\|\chi_B\rangle+\|\chi_C\rangle\right)$
$\|e_\theta\rangle =$	$\frac{1}{\sqrt{2}}\left(\|\psi_C\rangle-\|\psi_B\rangle\right)$	$\|e_\theta\rangle =$	$\frac{1}{\sqrt{6}}\left(2\|\chi_A\rangle-\|\chi_B\rangle-\|\chi_C\rangle\right)$
$\|e_\epsilon\rangle =$	$\frac{1}{\sqrt{6}}\left(2\|\psi_A\rangle-\|\psi_B\rangle-\|\psi_C\rangle\right)$	$\|e_\epsilon\rangle =$	$\frac{1}{\sqrt{2}}\left(\|\chi_B\rangle-\|\chi_C\rangle\right)$

a primed x', y', z'-coordinate system was introduced, which is adapted to the tris-chelate geometry. The z'-axis is along the threefold direction, and the x'-axis is oriented along a twofold axis, coinciding with the bisector of the positive x and the negative y-axes. Next, we determine the point group, which in the present case is D_3. This is a rotational group, which implies that the molecule is chiral. The figure shows the Δ-enantiomer.[7] Thirdly, we define the functional basis. The relevant orbitals are the metal t_{2g} orbitals, which are fully occupied in the Ru^{2+} ground state, and the frontier orbitals on the ligand. For conjugated bidentate ligands, such as 2, 2′-bipyridyl (bipy) or 2,4-pentanedionate (also named 'acetylacetonate', $acac^-$), the frontier orbitals are of π-type. The essential parts of these orbitals are the contributions on the ligator atoms. These are either symmetric or anti-symmetric with respect to the twofold axis through the bidentate ligand, as shown in Fig. 6.5. Following Orgel [17], we denote them as χ- or ψ-type, respectively. The standard techniques of characters and projection operators yield SALCs for all these basis sets. The results are shown in Table 6.5. For the e-irrep, the components are labeled as e_θ and e_ϵ, following the standard canonical format. As a splitting field, we use the twofold axis along x'. Finally, we also include in the table the symmetries of the transition dipoles, which are the operators for the optical transitions. This completes the groundwork for the symmetry analysis.

6.8.1 Linear Dichroism

The linear dichroism is associated with the metal-to-ligand charge-transfer (CT) transitions [18]. Dipole-allowed transitions between the orbitals are governed by the appropriate D_3 coupling coefficients. However, since both donor and acceptor orbitals, as well as the transition operators, each involve two irreps, several symmetry-independent coupling channels are possible. As is often the case in transition-metal

[7] In tris-chelate complexes, Δ refers to a right-handed (*dextro*) helix. A left-handed helix (*lævo*) is denoted as Λ.

spectroscopy, it is not sufficient to identify the reduced matrix elements; for a deeper understanding, a further development of the model is often required to compare the reduced matrix elements. In the case of the CT bands, the model of Day and Sanders offers just that little extra [19]. According to this simple model, a charge-transfer (CT) transition between metal and ligand gains intensity when the relevant metal and ligand orbitals interact.

We first calculate the interaction terms between the metal and isolated ligand orbitals. The bipy ligand has low-lying unoccupied levels of ψ-character, which form π-acceptor interactions with the metal t_{2g} orbitals. Let H_π represent the elementary interaction between a ligand ψ orbital and a metal t_{2g} orbital, directed towards one ligator. The allowed interactions are then obtained by cyclic permutation:

$$\begin{aligned} H_\pi &= \langle d_{xz}|\mathcal{H}|\psi^A\rangle = -\langle d_{yz}|\mathcal{H}|\psi^A\rangle \\ &= \langle d_{xy}|\mathcal{H}|\psi^B\rangle = -\langle d_{xz}|\mathcal{H}|\psi^B\rangle \\ &= \langle d_{yz}|\mathcal{H}|\psi^C\rangle = -\langle d_{xy}|\mathcal{H}|\psi^C\rangle \end{aligned} \tag{6.92}$$

In order to apply the model of Day and Sanders, we now consider the CT transition between the ligand orbital on A and the t_{2g} combination that interacts with it. As shown in Fig. 6.5, the ψ^A-acceptor orbital is anti-symmetric with respect to the $\hat{C}_2^{x'}$-axis and anti-symmetric in the xy-plane. The only matching t_{2g} combination on the metal is the $|e_\epsilon(t_{2g})\rangle$-component (see Table 6.5). In the local C_{2v} symmetry, $|\psi_A\rangle$ and $|e_\epsilon(t_{2g})\rangle$ both transform as b_2 (taking the horizontal plane as the local $\hat{\sigma}_1$). Their interaction element is expressed as

$$\langle e_\epsilon(t_{2g})|\mathcal{H}|\psi^A\rangle = \frac{1}{\sqrt{2}}\langle d_{xz} - d_{yz}|\mathcal{H}|\psi^A\rangle = \sqrt{2}H_\pi \tag{6.93}$$

We now consider the transition dipole moment between these orbitals along the x' direction, with $\mu_{x'} = -\mathrm{e}x'$. In C_{2v} symmetry, this component transforms as a_1, while $\mu_{y'}$ and $\mu_{z'}$ are anti-symmetric with respect to the $\hat{C}_2^{x'}$-axis. According to the Wigner–Eckart theorem, a transition dipole between two b_2 orbitals must transform as the direct product $b_2 \times b_2 = a_1$; hence, only the x'-component will be dipole-allowed. In a perturbative approach, which takes into account the symmetry-allowed interaction between the metal and ligand orbitals, one has

$$\mu(e_\epsilon(t_{2g}) \to \psi^A) = \langle e_\epsilon(t_{2g})|\mu_{x'}|\psi^A\rangle - \frac{\langle e_\epsilon(t_{2g})|\mathcal{H}|\psi^A\rangle}{E_\psi - E_{t_{2g}}}\langle \psi^A|\mu_{x'}|\psi^A\rangle \tag{6.94}$$

In this expression, the first term is the *contact* term between the zeroth-order orbitals. The second term is the *transfer* term, arising from the interaction between the donor and acceptor orbitals. In the simplified model of Day and Sanders, this term is the dominant contribution. The transfer-dipole matrix element in Eq. (6.94) is

approximated as the dipole length of the transferred charge, which we will represent as μ_A.

$$\langle \psi^A | \mu_{x'} | \psi^A \rangle = -\mathrm{e} \langle \psi^A | x' | \psi^A \rangle \approx -\mathrm{e} |\mathbf{R}_A| = -\mathrm{e}\rho \equiv \mu_A \tag{6.95}$$

where $\mathbf{R}_A$ is the radius vector from the origin to the center of ligand A, with length ρ. Since the three ligands are equivalent, we further write

$$\mu_A = \mu_B = \mu_C \equiv \mu^{\perp} \tag{6.96}$$

The three vectors of the ligand positions can be expressed in a row notation for the primed x', y', z'-coordinate system as

$$\begin{aligned} \boldsymbol{R}_A &= \rho\,(1, 0, 0) \\ \boldsymbol{R}_B &= \rho \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0 \right) \\ \boldsymbol{R}_C &= \rho \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0 \right) \end{aligned} \tag{6.97}$$

The transfer term for the transition to a single ligand then becomes

$$\boldsymbol{\mu}(e_\epsilon(t_{2g}) \rightarrow \psi^A) = -\mathrm{e}\kappa \mathbf{R}_A = \kappa \boldsymbol{\mu}_A \tag{6.98}$$

where the parameter κ is an overlap factor which indicates what fraction of the charge is actually transferred:

$$\kappa = -\frac{\langle e_\epsilon(t_{2g}) | \mathcal{H} | \psi^A \rangle}{E_\psi - E_{t_{2g}}} = -\frac{\sqrt{2} H_\pi}{E_\psi - E_{t_{2g}}} \tag{6.99}$$

Note that the transfer term is always polarized in the direction of the transferred charge. This parametrization can now be extended to the transfer terms for the relevant trigonal transitions. The Hamiltonian operator is of course totally symmetric, so allowed interactions can take place only between orbitals with the same symmetry, and are independent of the component; hence:

$$\begin{aligned} \langle e_\epsilon(t_{2g}) | \mathcal{H} | e_\epsilon(\psi) \rangle &= \sqrt{3} H_\pi \\ \langle e_\theta(t_{2g}) | \mathcal{H} | e_\theta(\psi) \rangle &= \sqrt{3} H_\pi \end{aligned} \tag{6.100}$$

Symmetry prevents interaction between the $a_1(t_{2g})$ and $a_2(\psi)$ orbitals. The metal–ligand π acceptor interaction will thus stabilize the e-component of the t_{2g} shell while leaving the a_1-orbital in place, as shown in the simple orbital-energy diagram in the left panel of Fig. 6.6. We can now calculate the transfer term for the $e \rightarrow e$ and $e \rightarrow a_2$ orbital transitions. In each case, only one component needs to be calculated.

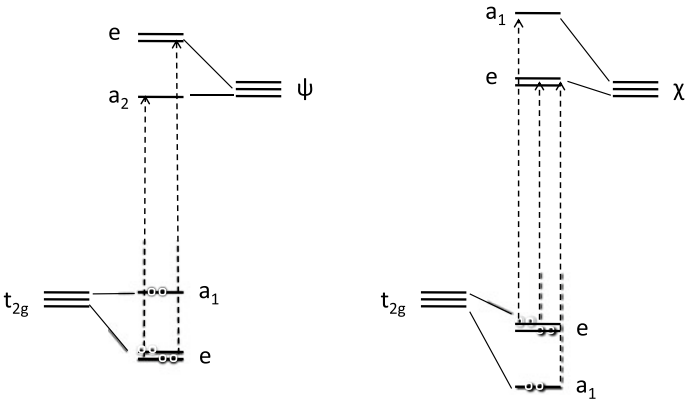


Fig. 6.6 Allowed CT transitions from the t_{2g} shell to ψ- or χ-type ligand acceptor orbitals for tris-chelate complexes with D_3 symmetry

The interaction element in this case is obtained from Eq. (6.100) and the transfer fraction reads

$$-\frac{\sqrt{3}H_{\pi}}{E_{\psi}-E_{t_{2g}}}=\sqrt{\frac{3}{2}}\kappa \tag{6.101}$$

The transfer-dipole element is given by

$$\frac{1}{6}\left\langle 2\psi^{A}-\psi^{B}-\psi^{C}\,|\boldsymbol{\mu}|\,2\psi^{A}-\psi^{B}-\psi^{C}\right\rangle=\frac{1}{6}\left[4\boldsymbol{\mu}_{A}+\boldsymbol{\mu}_{B}+\boldsymbol{\mu}_{C}\right]=\frac{1}{2}\boldsymbol{\mu}_{A} \tag{6.102}$$

Here, we have made use of the fact that the sum of the three dipole vectors vanishes. The effective transfer term thus becomes

$$\boldsymbol{\mu}(e_{\epsilon}(t_{2g})\rightarrow e_{\epsilon}(\psi))=\sqrt{\frac{3}{8}}\kappa\boldsymbol{\mu}_{A} \tag{6.103}$$

In the Wigner–Eckart formalism, this matrix element is written as

$$\langle e_{\epsilon}(t_{2g})|\mu_{x'}|e_{\epsilon}(\psi)\rangle=\langle E\epsilon|E\theta E\epsilon\rangle\langle e(t_{2g})||e(\mu)||e(\psi)\rangle \tag{6.104}$$

The coupling coefficient in this equation is equal to $1/\sqrt{2}$. We can thus identify the reduced matrix element as

$$\langle e(t_{2g})||e(\mu)||e(\psi)\rangle=\frac{\sqrt{3}}{2}\kappa\mu_{A} \tag{6.105}$$

All other $e\rightarrow e$ transfer terms can then be obtained by simply varying the coupling coefficients. We give one more example of a transition that requires an operator

which is $\mu_{y'}$ polarized:

$$\begin{aligned}\langle e_\theta(t_{2g})|\boldsymbol{\mu}|e_\epsilon(\psi)\rangle &= \frac{1}{\sqrt{8}}\kappa\left\langle\psi^C-\psi^B\middle|\boldsymbol{\mu}\middle|2\psi^A-\psi^B-\psi^C\right\rangle\\ &= \frac{1}{\sqrt{8}}\kappa\left(\boldsymbol{\mu}_B-\boldsymbol{\mu}_C\right)\end{aligned} \tag{6.106}$$

The vector $\boldsymbol{\mu}_B-\boldsymbol{\mu}_C$ in this expression is directed in the $\mu_{y'}$ direction, as required by the selection rule. Moreover, the length of this vector is $\sqrt{3}\mu^\perp$

$$\left(\boldsymbol{\mu}_B-\boldsymbol{\mu}_C\right)\cdot\left(\boldsymbol{\mu}_B-\boldsymbol{\mu}_C\right)=2\left(\mu^\perp\right)^2-2\boldsymbol{\mu}_B\cdot\boldsymbol{\mu}_C=3\left(\mu^\perp\right)^2 \tag{6.107}$$

Hence, the transfer-dipole length for this y'-polarized transition also measures $\sqrt{3/8}\kappa$, which is exactly the same as for the x'-polarized transition, given in Eq. (6.103). This is expected since the corresponding coupling coefficients, $\langle E\epsilon|E\theta E\epsilon\rangle$ and $\langle E\theta|E\epsilon E\epsilon\rangle$, are equal.

Using the transfer model, we can also express the reduced matrix elements for the $e\rightarrow a_2$ channel. Even though there is no overlap between these orbitals, they do give rise to a transfer-term intensity. Orbital interaction does indeed delocalize the $e(t_{2g})$ orbitals over the ligands. The dipole operators, centered on the complex origin, will then couple the $e(\psi)$ and $a_2(\psi)$ ligand-centered orbitals. Hence, we write

$$\begin{aligned}\boldsymbol{\mu}(e_\epsilon(t_{2g})\rightarrow a_2(\psi)) &= -\frac{\langle e_\epsilon(t_{2g})|\mathcal{H}|e_\epsilon(\psi)\rangle}{E_\psi-E_{t_{2g}}}\langle e_\epsilon(\psi)|\boldsymbol{\mu}|a_2(\psi)\rangle\\ &= \sqrt{\frac{3}{2}}\kappa\langle e_\epsilon(\psi)|\boldsymbol{\mu}|a_2(\psi)\rangle\end{aligned} \tag{6.108}$$

The dipole matrix element in this expression can easily be evaluated:

$$\begin{aligned}\langle e_\epsilon(\psi)|\boldsymbol{\mu}|a_2(\psi)\rangle &= \frac{1}{3\sqrt{2}}\left\langle 2\psi^A-\psi^B-\psi^C\,|\boldsymbol{\mu}|\,\psi^A+\psi^B+\psi^C\right\rangle\\ &= \frac{1}{3\sqrt{2}}\left(2\boldsymbol{\mu}_A-\boldsymbol{\mu}_B-\boldsymbol{\mu}_C\right)\\ &= \frac{1}{\sqrt{2}}\boldsymbol{\mu}_A\end{aligned} \tag{6.109}$$

The total transfer term is obtained by combining Eqs. (6.108) and (6.109):

$$\boldsymbol{\mu}(e_\epsilon(t_{2g})\rightarrow a_2(\psi))=\frac{\sqrt{3}}{2}\kappa\boldsymbol{\mu}_A \tag{6.110}$$

A final task is to calculate the transition moments between the corresponding multi-electronic states based on the orbital transition moments obtained. In the tris-chelate complex under consideration, a ${}^1A_1\rightarrow{}^1E$ state transition can be associated with

each allowed orbital-transition. The $^1A_1$ corresponds to the closed-shell ground state, based on the $(t_{2g})^6$ configuration. Both the $e \rightarrow a_2$ and $e \rightarrow e$ transitions will give rise to a twofold-degenerate ^1E state. As an example, the θ states are written in determinantal notation as follows, where we write only the orbitals that are singly occupied:

$$
\begin{aligned}
|^1E_\theta(e \rightarrow a_2)\rangle &= \frac{1}{\sqrt{2}}\Big[|(e_\epsilon(t_{2g})\alpha)(a_2(\psi)\beta)| - |(e_\epsilon(t_{2g})\beta)(a_2(\psi)\alpha)|\Big] \\
|^1E_\theta(e \rightarrow e)\rangle &= \frac{1}{2}\Big[|(e_\theta(t_{2g})\alpha)(e_\theta(\psi)\beta)| - |(e_\theta(t_{2g})\beta)(e_\theta(\psi)\alpha)| \\
&\quad -|(e_\epsilon(t_{2g})\alpha)(e_\epsilon(\psi)\beta)| + |(e_\epsilon(t_{2g})\beta)(e_\epsilon(\psi)\alpha)|\Big] \qquad (6.111)
\end{aligned}
$$

The resulting state transition moments are then expressed in terms of orbital transition moments as

$$
\begin{aligned}
\langle^1A_1|\boldsymbol{\mu}|^1E_\theta(e \rightarrow a_2)\rangle &= \sqrt{2}\langle e_\epsilon(t_2g)|\boldsymbol{\mu}|a_2(\psi)\rangle = \sqrt{3/2}\kappa \\
\langle^1A_1|\boldsymbol{\mu}|^1E_\theta(e \rightarrow e)\rangle &= 2\langle e_\theta(t_2g)|\boldsymbol{\mu}|e_\theta(\psi)\rangle = \sqrt{3/2}\kappa \qquad (6.112)
\end{aligned}
$$

On the other hand, the $a_1(t_{2g})$ orbital does not delocalize over the ligands. As a result, there can be no transfer term associated with transitions from this orbital. One expects only a weak contact term. The lowest transition corresponds to $a_1(t_{2g}) \rightarrow a_2(\psi)$. The only non-zero coupling coefficient for this transition is $\langle A_1|A_2A_2\rangle$. This transition will thus be dipole allowed under $\mu_{z'}$. Polarized absorption spectra are in line with this analysis: the spectral onset of the CT region is characterized by a weak absorption band in parallel polarization, followed by two strong absorption bands in perpendicular polarization. This assignment is based on the assumption that the vertical Franck–Condon excitations reach delocalized charge-transfer states. At least in the case of $Ru(bipy)_3^{2+}$, this is supported by detailed spectral measurements [20]. An entirely similar analysis can be performed in the case when the ligand orbital is of χ-type. The transition moments are collected in Table 6.6. In this case, the ligand and metal part both transform as $a_1 + e$ (see Table 6.5). As a result, three transitions are found to carry transfer-term intensity, as indicated in Fig. 6.6.

Table 6.6 Transfer-term contributions to $^1A_1 \rightarrow {}^1E$ CT transitions, with ψ and χ acceptor orbitals

ψ ligand orbitals		χ ligand orbitals	
$^1E(a_1 \rightarrow e(\psi))$	0	$^1E(a_1 \rightarrow e(\chi))$	$\sqrt{2}\kappa'$
$^1E(e \rightarrow a_2(\psi))$	$\sqrt{\frac{3}{2}}\kappa$	$^1E(e \rightarrow a_1(\chi))$	$\frac{1}{\sqrt{2}}\kappa'$
$^1E(e \rightarrow e(\psi))$	$\sqrt{\frac{3}{2}}\kappa$	$^1E(e \rightarrow e(\chi))$	$\frac{1}{\sqrt{2}}\kappa'$

6.8.2 Circular Dichroism

Circular dichroism (CD) is the difference between absorption of left and right circularly polarizes light.

$$\Delta A = A_l - A_r \tag{6.113}$$

The tris-chelate compounds are chiral compounds, with an apparent helical structure, which gives rise to their circular-dichroic properties. Here, we will focus on the symmetry selection rules that determine the CD signal. The CT transitions that we have just discussed cannot be responsible for the primary CD strength, since they are in-plane polarized, and thus do not carry intrinsic helicity. Instead, the prominent peaks in the CD spectrum are observed at higher energies and are associated with the intra-ligand $\pi\pi^*$-transitions. These transitions take place between occupied and virtual ligand-centered orbitals which are of opposite signature, and hence are of type $\psi \rightarrow \chi$ or vice versa. Such transitions are long-axis polarized, i.e., the transition dipole moment is oriented along the ligand bridge as shown in Fig. 6.7.

We designate these dipole moments as $\boldsymbol{\mu}_A^{\parallel}$, $\boldsymbol{\mu}_B^{\parallel}$, $\boldsymbol{\mu}_C^{\parallel}$. These vectors can be expressed in a row notation for the primed x', y', z'-coordinate system as follows:

$$\begin{aligned}
\boldsymbol{\mu}_A^{\parallel} &= \mu^{\parallel}\left(0, \frac{1}{\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{3}}\right) \\
\boldsymbol{\mu}_B^{\parallel} &= \mu^{\parallel}\left(-\frac{1}{2}, -\frac{1}{2\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{3}}\right) \\
\boldsymbol{\mu}_C^{\parallel} &= \mu^{\parallel}\left(\frac{1}{2}, -\frac{1}{2\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{3}}\right)
\end{aligned} \tag{6.114}$$

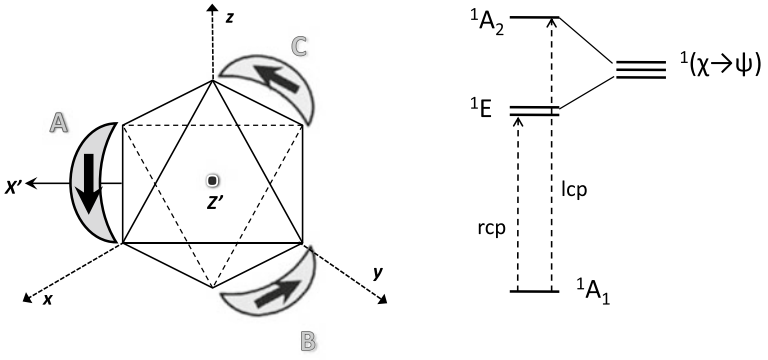


Fig. 6.7 Allowed intra-ligand transitions from χ- to ψ-type ligand orbitals for tris-chelate complexes with D_3 symmetry, in the Δ conformation. The circular dichroism for the right-handed helix in the diagram has a lower right-circularly polarized (rcp) band and an upper left-circularly polarized (lcp) band. This gives the CD spectrum the appearance of the first derivative of a Gaussian curve, with a negative part at longer wavelength and a positive part at shorter wavelength

The scalar products between these orientations in units of $\mu_{\parallel}^2$ are equal to $1/2$, which corresponds to angles of 60°. Each of the three transitions gives rise to an excited state. In D_3 symmetry these states transform as $A_2 + E$. The composition of these *exciton* states[8] is as follows:

$$
\begin{aligned}
|^1A_2\rangle &= \frac{1}{\sqrt{3}}\left[(\chi_A \to \psi_A)^1 + (\chi_B \to \psi_B)^1 + (\chi_C \to \psi_C)^1\right] \\
|^1E_\theta\rangle &= \frac{1}{\sqrt{2}}\left[-(\chi_B \to \psi_B)^1 + (\chi_C \to \psi_C)^1\right] \\
|^1E_\epsilon\rangle &= \frac{1}{\sqrt{6}}\left[2(\chi_A \to \psi_A)^1 - (\chi_B \to \psi_B)^1 - (\chi_C \to \psi_C)^1\right]
\end{aligned}
\tag{6.115}
$$

Here, the notation refers to a singlet orbital transition, which can be written in determinantal form as

$$
(\chi_A \to \psi_A)^1 = \frac{1}{\sqrt{2}}\Big[|(\chi_A\alpha)(\psi_A\beta)| - |(\chi_A\beta)(\psi_A\alpha)|\Big] \tag{6.116}
$$

To first approximation, the metal center is not taking part in the electronic properties, but merely serves as a structural template which keeps the ligands in place. Distant interactions between the three transitions can be described by a simple exciton-coupling model. In this model, the interaction between transitions is approximated by the electrostatic interaction potential between the corresponding transition dipoles. This potential is given by

$$
V_{ij} = \frac{1}{4\pi\epsilon_0}\left(\frac{\boldsymbol{\mu}^i \cdot \boldsymbol{\mu}^j}{R_{ij}^3} - \frac{3\left(\boldsymbol{\mu}^i \cdot \mathbf{R}_{ij}\right)\left(\boldsymbol{\mu}^j \cdot \mathbf{R}_{ij}\right)}{R_{ij}^5}\right) \tag{6.117}
$$

where R_{ij} is the distance between the dipoles, and $\mathbf{R}_{ij} = \mathbf{R}_j - \mathbf{R}_i$. The length of the distance vector is thus $\sqrt{3}\rho$, where ρ is the length of the individual vectors. The energies of the exciton states are then given by

$$
\begin{aligned}
\langle^1A_2|V|^1A_2\rangle &= \frac{\left(\mu^{\parallel}\right)^2}{4\pi\epsilon_0\rho^3}\frac{1}{6\sqrt{3}} \\
\langle^1E|V|^1E\rangle &= -\frac{\left(\mu^{\parallel}\right)^2}{4\pi\epsilon_0\sqrt{3}\rho^3}\frac{1}{12\sqrt{3}}
\end{aligned}
\tag{6.118}
$$

The $^1A_2$ state thus goes up in energy twice as much as the ^1E state goes down, thus keeping the barycenter energy at the zeroth-order position. Now, in order to determine the CD strength, we need for the two states both the electric and the

[8] The excitation creates an electron–hole pair, which can move from one ligand to another. This is called an exciton.

magnetic transition dipoles from the ground state. The electric dipoles are easily obtained by combining the state vectors:

$$\begin{aligned}
\boldsymbol{\mu}(^1A_1 \rightarrow {}^1\ A_2) &= \frac{1}{\sqrt{3}}\left(\boldsymbol{\mu}_A^{\parallel} + \boldsymbol{\mu}_B^{\parallel} + \boldsymbol{\mu}_C^{\parallel}\right) = \sqrt{2}\mu^{\parallel}(0,0,1) \\
\boldsymbol{\mu}(^1A_1 \rightarrow {}^1\ E_\epsilon) &= \frac{1}{\sqrt{6}}\left(2\boldsymbol{\mu}_A^{\parallel} - \boldsymbol{\mu}_B^{\parallel} - \boldsymbol{\mu}_C^{\parallel}\right) = \frac{1}{\sqrt{2}}\mu^{\parallel}(0,1,0) \\
\boldsymbol{\mu}(^1A_1 \rightarrow {}^1\ E_\theta) &= \frac{1}{\sqrt{2}}\left(-\boldsymbol{\mu}_B^{\parallel} + \boldsymbol{\mu}_C^{\parallel}\right) = \frac{1}{\sqrt{2}}\mu^{\parallel}(1,0,0)
\end{aligned} \tag{6.119}$$

The calculation of the magnetic transition dipoles requires a preamble. The magnetic moment was already defined in Eq. 4.124 of Chap. 4. By explicitly writing the angular momentum operator in terms of the linear momentum operator as $\boldsymbol{r} \times \boldsymbol{p}$, one obtains

$$\boldsymbol{m} = -\frac{\mathrm{e}}{2m}\boldsymbol{l} = -\frac{\mathrm{e}}{2m}\boldsymbol{r} \times \boldsymbol{p} \tag{6.120}$$

The commutator of the one-electron Hamiltonian with the position operator is given by

$$[\mathcal{H}, \boldsymbol{r}] = \left[\left(\frac{\boldsymbol{p}\cdot\boldsymbol{p}}{2m} + V(\boldsymbol{r})\right), \boldsymbol{r}\right] = -\frac{i\hbar}{m}\boldsymbol{p} \tag{6.121}$$

Here, we used the Heisenberg commutator relation between the conjugate position and momentum operators: $[x, p_x] = i\hbar$. The magnetic moment matrix element of the intra-ligand transition with respect to the common origin of the coordinate system is given by

$$\mathbf{m}_A = \langle\psi_A|\boldsymbol{m}|\chi_A\rangle = -\frac{\mathrm{e}}{2m}\langle\psi_A|(\mathbf{R}_A + \boldsymbol{r}) \times \boldsymbol{p}|\chi_A\rangle = -\frac{\mathrm{e}}{2m}\mathbf{R}_A \times \langle\psi_A|\boldsymbol{p}|\chi_A\rangle \tag{6.122}$$

where it was assumed that the chromophore has no intrinsic magnetic transition moment. The momentum matrix element in this equation can now be evaluated with the help of Eq. (6.121):

$$\begin{aligned}
\langle\psi_A|\boldsymbol{p}|\chi_A\rangle &= \frac{im}{\hbar}\langle\psi_A|\mathcal{H}\boldsymbol{r} - \boldsymbol{r}\mathcal{H}|\chi_A\rangle \\
&= \frac{im}{\hbar}\left(\langle\mathcal{H}\psi_A|\boldsymbol{r}|\chi_A\rangle - \langle\psi_A|\boldsymbol{r}\mathcal{H}|\chi_A\rangle\right) \\
&= \frac{im}{\hbar}\left(E_\psi - E_\chi\right)\langle\psi_A|\boldsymbol{r}|\chi_A\rangle \\
&= 2\pi i m\nu\langle\psi_A|\boldsymbol{r}|\chi_A\rangle
\end{aligned} \tag{6.123}$$

Here, ν is the frequency of the intra-ligand transition. The combination of this result with Eq. (6.122) yields

$$\begin{aligned}
\mathbf{m}_A &= i\pi\nu\left(\mathbf{R}_A \times \boldsymbol{\mu}_A^{\parallel}\right) = i\pi\nu\rho\mu^{\parallel}\left(0, -\frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \\
\mathbf{m}_B &= i\pi\nu\left(\mathbf{R}_B \times \boldsymbol{\mu}_B^{\parallel}\right) = i\pi\nu\rho\mu^{\parallel}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{3}}\right) \\
\mathbf{m}_C &= i\pi\nu\left(\mathbf{R}_C \times \boldsymbol{\mu}_C^{\parallel}\right) = i\pi\nu\rho\mu^{\parallel}\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{3}}\right)
\end{aligned} \tag{6.124}$$

As we indicated the above formalism applies to chromophores that have no intrinsic magnetic moment.

$$\begin{aligned}
\mathbf{m}({}^1A_1 \rightarrow {}^1A_2) &= \frac{1}{\sqrt{3}}\left(\mathbf{m}_A^{\parallel} + \mathbf{m}_B^{\parallel} + \mathbf{m}_C^{\parallel}\right) = i\pi\nu\rho\mu^{\parallel}(0,0,1) \\
\mathbf{m}({}^1A_1 \rightarrow {}^1E_\epsilon) &= \frac{1}{\sqrt{6}}\left(2\mathbf{m}_A^{\parallel} - \mathbf{m}_B^{\parallel} - \mathbf{m}_C^{\parallel}\right) = -i\pi\nu\rho\mu^{\parallel}(0,1,0) \\
\mathbf{m}({}^1A_1 \rightarrow {}^1E_\theta) &= \frac{1}{\sqrt{2}}\left(-\mathbf{m}_B^{\parallel} + \mathbf{m}_C^{\parallel}\right) = -i\pi\nu\rho\mu^{\parallel}(1,0,0)
\end{aligned} \tag{6.125}$$

A transition will be characterized by a helical displacement of the electron if the magnetic and electric transition dipoles are aligned. This is reflected in the Rosenfeld equation for the CD intensity or *rotatory strength*, $\mathcal{R}_{a\rightarrow j}$, for a transition from a ground state a to an excited state j in a collection of randomly oriented molecules:

$$\mathcal{R}_{a\rightarrow j} = \mathrm{Im}\left\{\langle a|\boldsymbol{\mu}|j\rangle \cdot \langle j|\boldsymbol{m}|a\rangle\right\} \tag{6.126}$$

Straightforward application to the exciton bands yields for a Δ-trischelate:

$$\begin{aligned}
\mathcal{R}({}^1A_1 \rightarrow {}^1A_2) &= \sqrt{2}\pi\nu\rho\left(\mu^{\parallel}\right)^2 \\
\mathcal{R}({}^1A_1 \rightarrow {}^1E_\epsilon) &= -\frac{1}{\sqrt{2}}\pi\nu\rho\left(\mu^{\parallel}\right)^2 \\
\mathcal{R}({}^1A_1 \rightarrow {}^1E_\theta) &= -\frac{1}{\sqrt{2}}\pi\nu\rho\left(\mu^{\parallel}\right)^2
\end{aligned} \tag{6.127}$$

The out-of-plane polarized transition to the ${}^1A_2$ state, which lies at higher energy, has a positive CD signal, while the in-plane polarized transition to the lower ${}^1E$ state has a negative CD signal. The latter transition consists of two components along the two in-plane directions. Summing over the three components in Eq. (6.127), shows that the total rotatory strength, for randomly oriented molecules, is exactly zero. This is a general sum rule for CD spectra. If one now takes the spectrum of the chiral antipode, the Λ tris-chelate complex, the spectra are exactly the same but the signs are reversed. Mirror image in actual geometry thus becomes reflection symmetry in the spectrum.

6.9 Induction Revisited: The Fiber Bundle

In Chap. 4, we left induction after the proof of the Frobenius reciprocity theorem. In that proof, the important concept of the positional representation was introduced. This described the permutation of the sites under the action of the group elements. Further, we defined local functions on the sites which transformed as irreps of the site symmetry. As an example, if we want to describe the displacement of a cluster atom in a polyhedron, two local functions are required: a totally symmetric one for the radial displacement and a twofold-degenerate one for the tangential displacements. In cylindrical symmetry, these are labeled σ and π, respectively. The *mechanical* representation, i.e., the representation of the cluster displacements, is then the sum of the two induced representations:

$$\Gamma_{\text{mech}} = \Gamma(\sigma H \uparrow G) + \Gamma(\pi H \uparrow G) \tag{6.128}$$

As an example using the induction tables in Appendix C.2 for an octahedron, we have

$$\Gamma_{\text{mech}} = \left(A_{1g} + E_g + T_{1u}\right) + (T_{1g} + T_{2g} + T_{1u} + T_{2u}) \tag{6.129}$$

This is precisely the set of Fluorine displacements that we constructed in Sect. 4.8 in order to describe the vibrational modes of UF_6. One remarkable result of induction theory is that the mechanical representation can also be obtained as the direct product of the positional representation and the translational representation, T_{1u}; this is the representation of the three displacements of the center of the cluster.

$$\begin{aligned}\Gamma_{\text{mech}} &= T_{1u} \times \left(A_{1g} + E_g + T_{1u}\right) \\ &= T_{1u} + (T_{1u} + T_{2u}) + \left(A_{1g} + E_g + T_{1g} + T_{2g}\right)\end{aligned} \tag{6.130}$$

It is as if the displacements of the central point of the octahedron were relocated to every ligand site. The elementary function space of the displacements of the central atom, which transforms as the translational irrep, T_{1u}, is called the *standard fiber*. This fiber is attached to every site of the cluster, and the set of these fibers is the *fiber bundle*, as represented in Fig. 6.8. The action of the group interconnects fibers of the bundle by permuting them. The following induction theorem holds:

Theorem 6.4 *Consider a standard fiber, consisting of a function space that is invariant under the action of the group. In a cluster of equivalent sites, we can form a fiber bundle by associating this standard fiber with every site position. The induced representation of the fiber bundle is then the direct product of the irrep of the standard fiber with the positional representation.*

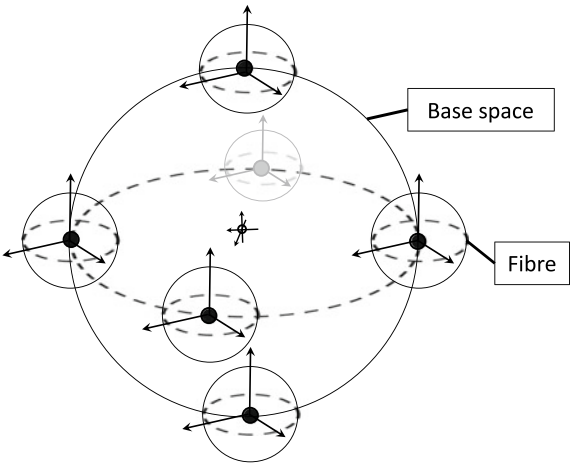


Fig. 6.8 Pictorial representation of a fiber bundle: the sphere corresponds to the base space, limited here to six discrete points at the corners of an octahedron. To each of these points, a vector space is attached of three translations, forming the standard fiber. The set of six fibers, attached to the base space, forms the fiber bundle

For $\mathcal{V}$ being the representation of the standard fiber, T_{1u} in our example, and $\mathcal{P}$ the positional representation of the set of equivalent sites in the molecule, one has for the induced representation:

$$\Gamma(\{\mathcal{V}\}H \uparrow G) = \mathcal{V} \times \mathcal{P}(H \uparrow G) \tag{6.131}$$

For a proof of this theorem, we refer to the literature [21, 22]. The theorem is not only applicable to molecular vibrations but also directly in line with the LCAO method in molecular quantum chemistry. In this method, the molecular orbitals (MOs) are constructed from atomic basis sets that are defined on the constituent atoms. An atomic basis set, such as $3d$ or $4f$, corresponds to a fiber, emanating, as it were, from the atomic center. Usually, such basis sets obey spherical symmetry, since they are defined for the isolated atoms. As such, they are also invariant under the molecular point group [23]. As an example, a set of $4f$ polarisation functions on a chloride ligand in a $RhCl_6^{3-}$ complex is itself adapted to octahedral symmetry as $a_{2u} + t_{1u} + t_{2u}$. This representational sum thus corresponds to $\mathcal{V}$. In the C_{4v} site symmetry these irreps subduce: $a_1 + b_1 + b_2 + 2e$. According to the theorem, the LCAOs based on the $4f$ orbitals thus will transform as

$$\begin{aligned}\Gamma(\{a_1 + b_1 + b_2 + 2e\}\, C_{4v} \uparrow O_h) &= (a_{2u} + t_{1u} + t_{2u}) \times (a_{1g} + e_g + t_{1u}) \\ &= a_{1g} + a_{2g} + 2e_g + 2t_{1g} + 3t_{2g} + a_{2u} + e_u + 3t_{1u} + 3t_{2u}\end{aligned} \tag{6.132}$$

In this LCAO space, several irreps occur multiple times, but they can all be distinguished by the specific direct product from which they originated. In the following section, we will apply the theorem to obtain relationships between polyhedral orbitals in clusters. These relations will be based on Euler polyhedral formula, and its subsequent group-theoretical corollaries.

6.10 Application: Bonding Schemes for Polyhedra

Leonhard Euler dominated the mathematics of the eighteenth century. One of his famous discoveries was the polyhedral theorem, which marks the beginning of topology. A polyhedron has three structural elements: vertices, edges, and faces.[9] The numbers of these will be represented as v, e, and f, respectively. Then, for a polyhedron, the following theorem holds:

Theorem 6.5 *In a convex polyhedron the alternating sum of the numbers of vertices, edges, and faces is always equal to 2.*

$$v - e + f = 2. \tag{6.133}$$

As an example, in a cube one has $v = 8, e = 12, f = 6$, and hence $8 - 12 + 6 = 2$. The 2 in the right-hand side of Eq. (6.133) is called the Euler invariant. It is a *topological* characteristic. Topology draws attention to properties of surfaces, which are not affected when the surfaces are stretched or deformed, as one can do with objects made of rubber or clay. Topology is thus not concerned with regular shapes, and in this sense seems to be completely outside our subject of symmetry; yet, as we intend to show in this section, there is in fact a deep connection, which also carries over to molecular properties. The surface to which the 2 in the theorem refers is the surface of a sphere. A convex polyhedron is indeed a polyhedron which can be embedded or mapped on the surface of a sphere. Group theory, and in particular the induction of representations, provides the tools to understand this invariant. To this end, each of the terms in the Euler equation is replaced by an induced representation, which is based on the particular nature of the corresponding structural element. In Fig. 6.9, we illustrate the results for the case of the tetrahedron.

- The vertices, being zero-dimensional points, form a set of nodes, $\{\langle u \rangle\}$, which are permuted under the symmetry operations of the polyhedron. The representation of this set is the positional representation, $\Gamma_\sigma(v)$. The σ here refers to the fact that the sites themselves transform as totally symmetric objects in the site group. If the cluster contains several orbits, the induced representation is of course the sum of the individual positional representations. In Fig. 6.9, the vertex representation of a tetrahedron is $A_1 + T_2$. In Sect. 4.7, we have already encountered these irreps, when discussing the sp^3 hybridization of carbon.

$$\Gamma_\sigma(v) = \Gamma(a_1 C_{3v} \uparrow T_d) = A_1 + T_2 \tag{6.134}$$

- The edges are one-dimensional lines. They form a set of ordered pairs, $\{\langle u, v \rangle\}$. Each of these can be thought of as an arrow, directed along the edge. The symmetry operations will interchange these arrows, but may also change their sense. The corresponding representation is labeled as $\Gamma_\parallel(e)$. This symbol indicates that the basic objects on the edge sites are not symmetric points but directed arrows,

[9] In geometry, a *vertex* is a point where two or more lines meet.

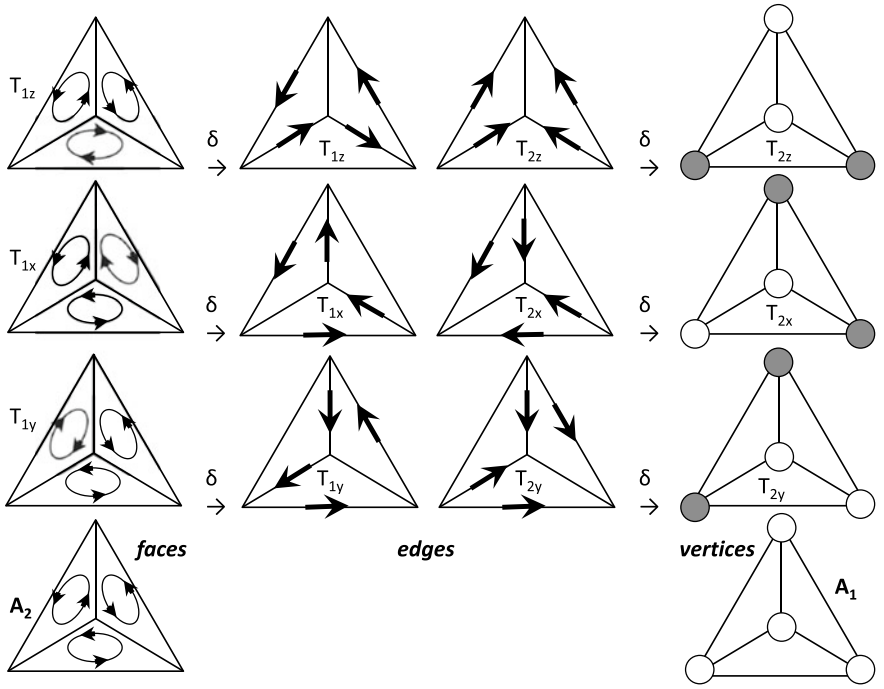


Fig. 6.9 Face, edge, and vertex SALCs for a tetrahedron. The δ symbol denotes taking the boundary, from faces to edges, and from edges to vertices (see text). The two topological invariants are the A_2 face term and the A_1 vertex term

parallel to the edge. The site group through the center of an edge has maximal symmetry C_{2v} and, in this site group, the arrows transform as the b_1 irrep, which is symmetric under reflection in a plane containing the edge and anti-symmetric under the symmetry plane perpendicular to the edge. For a tetrahedron there are six edge vectors, transforming as $T_1 + T_2$.

$$\Gamma_{\parallel}(e) = \Gamma(b_1 C_{2v} \uparrow T_d) = T_1 + T_2 \tag{6.135}$$

- The faces may be represented as closed chains of nodes, which are bordering a polyhedral face, $\{\langle u, v, w, ...\rangle\}$. The sequence forms a circulation around the face, in a particular sense (going from $\langle u\rangle$ to $\langle w\rangle$ over $\langle v\rangle$, etc.). The set of face rotations forms the basis for the face representation, which is denoted as $\Gamma_{\circlearrowleft}(f)$. In a polyhedron the maximal site group of a face is C_{nv}, and in this site group the face rotation transforms as the rotation around the $\hat{C}_n$-axis, i.e., it is symmetric under the axis and anti-symmetric with respect to the vertical $\hat{\sigma}_v$ planes, which invert the sense of rotation. For the tetrahedron, the face circulations transform as $A_2 + T_1$, as shown in Fig. 6.9.

$$\Gamma_{\circlearrowleft} = \Gamma(a_2 C_{3v} \uparrow T_d) = A_2 + T_1 \tag{6.136}$$

The following theorem [24] applies:

Theorem 6.6 *The alternating sum of induced representations of the vertex nodes, edge arrows, and face rotations, is equal to the sum of the totally symmetric representation,* Γ_0*, and the pseudo-scalar representation,* Γ_ϵ*. The latter representation is symmetric under proper symmetry elements and anti-symmetric under improper symmetry elements.*

$$\Gamma_\sigma(v) - \Gamma_\parallel(e) + \Gamma_{\circlearrowleft}(f) = \Gamma_0 + \Gamma_\epsilon \tag{6.137}$$

The Euler theorem may be considered as the dimensional form of this theorem, which states that the alternating sum of the characters of the induced representations under the unit element, $\hat{E}$, is equal to 2, but the present theorem extends this character equality to all the operations of the group. The theorem silently implies that irreps can be added and subtracted. In the example of the tetrahedron, the theorem is expressed as

$$\Gamma_\sigma(v) - \Gamma_\parallel(e) + \Gamma_{\circlearrowleft}(f) = (A_1 + T_2) - (T_1 + T_2) + (A_2 + T_2) = A_1 + A_2 \tag{6.138}$$

A straightforward interpretation of the theorem is possible in terms of fluid flow on the surface of a polyhedron.[10] Suppose observers are positioned on the vertices, edge centers, and face centers, and register the local fluid flow. When the incoming and outgoing currents at a node are not in balance, the observers located on these nodes will report piling up or depletion of the local fluid level. This is the scalar property represented by the vertex term. The corresponding connection between edge flow and vertex density is expressed by the *boundary operation*, indicated by δ in Fig. 6.9. Taking the boundary of an edge arrow means replacing the arrow by the difference of two vertex-localized scalars: a positive one (indicated by a white circle in the figure) at the node to which the arrow's head is pointing, and a negative one (indicated by a black circle) at the node facing the arrow's tail. This projection from edge to vertex will not change the symmetry. Hence, in this way, the boundary of the T_2 edge irrep is the T_2 vertex SALC, as illustrated in the figure. Similarly, observers in face centers will notice the net current that is circulating around the face. Such a circular current through the edges does not give rise to changes at the nodes (indeed the incoming flow at a node is also leaving again), but is observable from the center of the face around which the current is circulating. The boundaries of circular currents around face centers are thus chains of arrows on the edges, which again conserve the symmetry. In Fig. 6.9, the boundary of the T_1 face term is thus the T_1 edge term. Clearly, the sum of the vertex and face observations should account for all currents going through the edges, except for two additional terms which escape edge observations. These are the two Euler invariants: the totally symmetric Γ_0-component corresponds to a uniform change of fluid amplitude at all vertex basins. This does not give rise

[10] This flow description provides a simple pictorial illustration of the abstract homology theory. The standard reference is Hilton and Wylie [25].

to edge currents, since it creates no gradients over the edges. The other is the Γ_ϵ-component. It corresponds to a simultaneous rotation around all faces in the same sense. Again, such rotor flows do not create net flows through the edges, because two opposite currents are flowing through every edge. The Euler invariant thus points to two invariant characteristic modes of the sphere. They are not boundaries of a mode at a higher level, nor are they bounded by a mode at a lower level. The phenomena, that these two terms describe, might also be referred to in a topological context as the electric and magnetic *monopoles*.

Because of this connection to density and current, this theorem may be applied in various ways to describe chemical bonding, frontier orbital structure, and vibrational properties. The applications of this theorem can be greatly extended by introducing fiber representations, as shown below.

Taking the Dual

To take the dual of a polyhedron is to replace vertices by faces and vice versa, as was already mentioned in Sect. 3.7 in relation to the Platonic solids. The dual has the same number of edges as the original, but every edge is rotated 90°. Hence, the relations between v^D, e^D, f^D for the dual and v, e, f for the original are:

$$\begin{aligned} v^D &= f \\ e^D &= e \\ f^D &= v \end{aligned} \tag{6.139}$$

As a result, the Euler formula is invariant under the dual operation.

$$v - e + f = v^D - e^D + f^D = 2 \tag{6.140}$$

A similar invariance holds for the symmetry extension, but in this case, 'to take the dual' corresponds to multiplying all terms by the pseudo-scalar irrep Γ_ϵ. The terms are then changed as follows:

$$\begin{aligned} \Gamma_\sigma(v) \times \Gamma_\epsilon &= \Gamma_\circlearrowleft(v) = \Gamma_\circlearrowleft(f^D) \\ \Gamma_\parallel(e) \times \Gamma_\epsilon &= \Gamma_\perp(e) = \Gamma_\parallel(e^D) \\ \Gamma_\circlearrowleft(f) \times \Gamma_\epsilon &= \Gamma_\sigma(f) = \Gamma_\sigma(v^D) \\ (\Gamma_0 + \Gamma_\epsilon) \times \Gamma_\epsilon &= \Gamma_0 + \Gamma_\epsilon \end{aligned} \tag{6.141}$$

Hence, if the theorem holds for the original, it also holds for the dual.

$$\big(\Gamma_\sigma(v) - \Gamma_\parallel(e) + \Gamma_\circlearrowleft(f)\big) \times \Gamma_\epsilon = \Gamma_\sigma(v^D) - \Gamma_\parallel(e^D) + \Gamma_\circlearrowleft(f^D) = \Gamma_0 + \Gamma_\epsilon \tag{6.142}$$

Note especially the fiber modification of the edge term. The maximal local symmetry of an edge is C_{2v}. The arrow along the edge transforms as b_1, while the pseudo-scalar irrep in C_{2v} is a_2. The product $b_1 \times a_2$ produces b_2, which is precisely

the symmetry of an arrow, tangent to the surface of the polyhedron, but directed perpendicular to the edge. Multiplication with the pseudo-scalar irrep thus has the effect of rotating the edges through 90°. In Eq. (6.141), the resulting representation is denoted as $\Gamma_\perp(e)$.

Deltahedra

Deltahedra are polyhedra that consist entirely of triangular faces. Three of the Platonic solids are deltahedra: the tetrahedron, the octahedron, and the icosahedron. In a convex deltahedron, the bond stretches (i.e., stretchings of the edges) span precisely the representation of the internal vibrations. In other words, a convex deltahedron cannot vibrate if it is made of rigid rods. This is the Cauchy theorem:

Theorem 6.7 *Convex polyhedra in three dimensions with congruent corresponding faces must be congruent to each other. In consequence, if a polyhedron is made up of triangles with rigid rods, the angles between the triangular faces are fixed.*

This result can be cast in the language of induced representations. The stretchings of the edges correspond to scalar changes of edge lengths and transform as σ-type objects, and hence will correspond to $\Gamma_\sigma(e)$. On the other hand, the internal vibrations span the mechanical representation, which can be written as a bundle of the translation, minus the spurious modes of translation and rotation. The symmetries of these will be denoted as Γ_T and Γ_R, respectively. One thus has

$$\text{Deltahedron: } \Gamma_\sigma(v) \times \Gamma_T - \Gamma_T - \Gamma_R = \Gamma_\sigma(e) \tag{6.143}$$

Trivalent Polyhedra

The dual of a deltahedron is a trivalent polyhedron, meaning that every vertex is connected to three nearest neighbors. The fullerene networks of carbon are usually trivalent polyhedra. This reflects the sp^2 hybridization of carbon, which can form three σ-bonds. Also in this case several specialized forms of the Euler symmetry theorem can be formulated. We may start from Eq. (6.143) and replace vertices by faces. The edge terms remain the same since they are totally symmetric under the local symmetries of the edges. Rotations of the edges by 90° will thus not affect these terms.

$$\text{Trivalent: } \Gamma_\sigma(f) \times \Gamma_T - \Gamma_T - \Gamma_R = \Gamma_\sigma(e) \tag{6.144}$$

Furthermore, by multiplying the vertices in a trivalent polyhedron by three, we have accounted for all the edges twice, since each edge is linked to two vertices, hence:

$$\text{Trivalent: } 3v = 2e \tag{6.145}$$

The $3v$ in this formula suggests once again taking the fiber representation $\Gamma_\sigma(v) \times \Gamma_T$. In doing so, we have considered on each vertex one σ and two π objects. Hence, this is not only the mechanical representation with three displacements on each vertex, but it is equally well the symmetry of a set of sp^2 hybrids on every vertex, directed

along the three edges. Along each edge, the hybrids at either end can be combined in a local bonding and anti-bonding combination. The corresponding induced representations are, respectively: $\Gamma_\sigma(e)$ and $\Gamma_\parallel(e)$; hence, the symmetry extension of Eq. (6.145) reads

$$\text{Trivalent: } \Gamma_\sigma(v) \times \Gamma_T = \Gamma_\sigma(e) + \Gamma_\parallel(e). \tag{6.146}$$

6.10.1 Edge Bonding in Trivalent Polyhedra

The understanding of the bonding schemes in polyhedra is based on the correct identification of the local hybridization scheme on the constituent fragments. Trivalent polyhedra are often *electron-precise*: this means that the fragment has three electrons in three orbitals, which are available for cluster bonding and give rise to edge-localized σ-bonds. Such is the case for the methyne fragment, CH, forming *polyhedranes*, but equally well for the *isolobal* [26] organo-transition-metal fragments such as $M(CO)_3$, where M is a d^9 metal such as Co, Rh or Ir. Figure 6.10 shows the bonding pattern based on such electron-precise fragments. As indicated before, the orbital basis corresponds to the fiber representation $\Gamma_\sigma(v) \times \Gamma_T$, and contains $3n$ orbitals. Local interactions along the edges will split this orbital basis into an occupied σ-bonding half and a virtual σ-anti-bonding counterpart, transforming as $\Gamma_\sigma(e)$ and $\Gamma_\parallel(e)$, respectively. This is precisely the result of Eq. (6.146). Now, for each half, a more detailed pattern can be discerned [27]. For the anti-bonding orbitals, the general theorem, Eq. (6.137), can be applied directly. The result is illustrated in

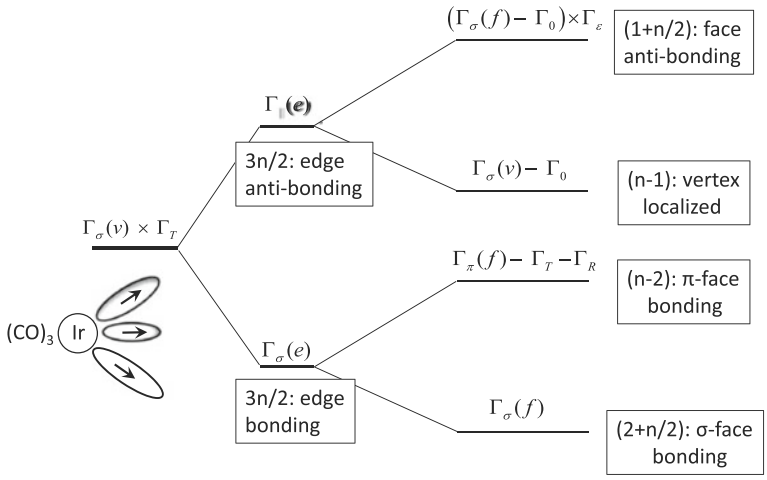


Fig. 6.10 Edge bonding in electron-precise trivalent cages. The valence shell splits into an occupied set of localized edge-bonds, and a matching virtual set of edge-anti-bonds. The sets may be further differentiated by use of the symmetry theorems

Fig. 6.10. By this theorem, the $3n/2$ edge anti-bonds are split into two subsets containing $(1 + n/2)$ and $(n - 1)$ orbitals. The former, higher lying, subset transforms as $\Gamma_{\circlearrowleft} - \Gamma_{\epsilon}$. These terms correspond to circulations around the faces, which means that these levels will be highly anti-bonding. In fact, they are always at the top of the skeletal spectrum. Note that the pseudo-scalar term, Γ_{ϵ}, does not take part. This is because a uniform circulation around all faces in the same sense has no contribution on the edges. Below this is a subset of weakly anti-bonding orbitals, transforming as $\Gamma_{\sigma}(v) - \Gamma_0$. These orbitals are more localized on the vertices. The Γ_0 term is not included since this is the totally symmetric molecular orbital which is completely bonding, and thus will appear in the lower half of the diagram.

Furthermore, the edge-bonding half can be analyzed with the help of Eq. (6.144). The $3n/2$ edge bonds split into two subsets of dimension $(n - 2)$ and $(2 + n/2)$. This analysis involves the fiber representation $\Gamma_{\sigma}(f) \times \Gamma_T$, which can be decomposed into a radial σ- and tangential π-part. The σ-part corresponds to cylindrically symmetric bonds around the faces, and will thus be strongly bonding. For the π-part, the face terms contain a nodal plane through the faces, and thus will be less bonding.

6.10.2 *Frontier Orbitals in Leapfrog Fullerenes*

Fullerenes are trivalent polyhedra of carbon, consisting of hexagons and pentagons. The following relations hold

$$\begin{aligned} v - e + f &= 2 \\ 3v &= 2e \\ f_5 + f_6 &= f \\ 5f_5 + 6f_6 &= 3v \end{aligned} \tag{6.147}$$

The first two relationships are from Eqs. (6.133) and (6.145). The third expresses that the total number of the faces is the sum of the number of pentagons (f_5), and hexagons (f_6). The final equation indicates that by counting the hexagons six times, and the pentagons five times, we have counted all vertices three times, since every vertex is at the junction of three faces. Even though there are fewer equations, here, than unknowns, it can easily be seen by manipulation of Eq. (6.147) that the only value that the number of pentagons, f_5, can take on is 12. Hence, the smallest fullerene is the dodecahedron C_{20}, which only consists of pentagons. Also note that the number of atoms in a fullerene must be even, since $3v$ must be divisible by 2, as e is an integer. Taking the *leapfrog*, L, of a primitive fullerene, P, is an operation of cage expansion, which yields a fullerene with three times as many atoms [28]. This procedure is described by the following rule:

$$L = \text{Dual (Omnicap } P) \tag{6.148}$$

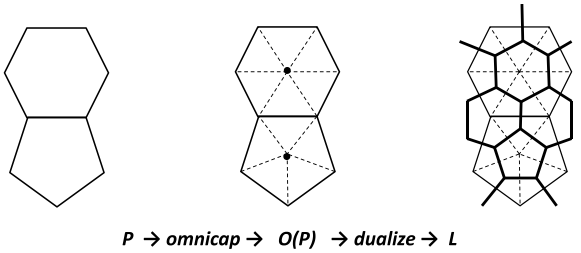


Fig. 6.11 The leapfrog extension consists of two operations: first, place an extra atom in the centers of all the polygons (middle panel), then, take the dual. The result is indicated by the solid lines in the right panel

It involves two operations, which are carried out consecutively, as illustrated in Fig. 6.11. One first places an extra *capping* atom on all pentagons and hexagons. This leads to a cage which consists only of triangles, and this is a deltahedron. By taking the dual, one restores a trivalent cage. As can be seen, all vertices of the primitive have been turned into hexagons, while the original pentagons and hexagons are recovered, but in a rotational stagger. The edges of the primitive are also recovered, but rotated 90°. In summary, the leapfrog operation inserts six vertices in the hexagons of P, and five vertices in the pentagons, which, according to the final expression in Eq. (6.147), multiplies the number of atoms by 3. The first and best-known leapfrog is Buckminsterfullerene, C_{60}, which is the leapfrog of the dodecahedron itself. Each carbon atom contributes, besides the sp^2 orbitals, which build the σ-frame, one radial p_r-orbital. These orbitals form π-bonds which control the frontier orbitals of fullerenes. In the case of the leapfrog, this frontier MO region is always characterized by six low-lying almost non-bonding orbitals, which, moreover, always transform as $\Gamma_T + \Gamma_R$. This can be explained with the help of the Euler rules [29].

As in the case of C_{60}, all leapfrogs can be considered to be truncations of the primitive fullerenes, in the sense that all the faces of the primitive have become isolated *islands*, surrounded by rings of hexagons. With every bond of the primitive is associated a perpendicular bond, which always forms a *bridge* between these islands. Based on this neat bond separation, two canonical valence-bond frames can be constructed for the leapfrog, which in a sense are the extremes of a correlation diagram, with the actual bonding somewhere in between. These bond schemes are known as the Fries and the Clar structures. The Fries structure is an extreme case where all *bridges* are isolated π-bonds. The induced representation for these bonds corresponds to $\Gamma_\sigma(e^P)$. On the other hand, in the Clar structures the bonding is completely redistributed to aromatic sextets on the hexagonal and pentagonal islands. The corresponding representation is the fiber bundle $\Gamma_\sigma(f^P) \times \Gamma_T$. We now compare the representations of both bonding schemes, using the symmetry theorems. We start with the main theorem, applied to the primitive P, and multiply left and right with Γ_R.

$$\Gamma_\sigma(v^P) \times \Gamma_R - \Gamma_\parallel(e^P) \times \Gamma_R + \Gamma_\circlearrowleft(f^P) \times \Gamma_R = \Gamma_T + \Gamma_R \tag{6.149}$$

Since $\Gamma_R = \Gamma_T \times \Gamma_\epsilon$, we could already simplify the face term to a form which precisely corresponds to the Clar representation:

$$\Gamma_{\circlearrowleft}(f^P) \times \Gamma_R = \Gamma_\sigma(f^P) \times \Gamma_T = \Gamma_{\text{Clar}} \tag{6.150}$$

The vertex term can be expressed with the help of Eq. (6.146):

$$\text{Trivalent: } \Gamma_\sigma(v^P) \times \Gamma_R = \Gamma_\sigma(e^P) \times \Gamma_\epsilon + \Gamma_\parallel(e^P) \times \Gamma_\epsilon = \Gamma_{\circlearrowleft}(e^P) + \Gamma_\perp(e^P) \tag{6.151}$$

where the pseudo-scalar irrep turns a σ-object into a circular current, and rotates the parallel edge current over 90°. Note that this is applied in the primitive cage, to the edges of P only. To complete the derivation one final fiber bundle is needed, which applies to all convex polyhedra:

$$\Gamma_\sigma(e) \times \Gamma_T = \Gamma_\sigma(e) + \Gamma_\parallel(e) + \Gamma_\perp(e) \tag{6.152}$$

This result is based on the C_{2v} site-symmetry of an edge. The translation in this site has a radial σ-component of a_1 symmetry, and two tangential π-components of $b_1 + b_2$ symmetry. The fiber bundle will thus correspond to the induction of $a_1 + b_1 + b_2$, which is precisely the meaning of the three terms on the right-hand side of Eq. (6.152). This expression may be transformed in two steps to the term which is required in the derivation. One first changes the substrate of the fiber from $\Gamma_\sigma(e)$ to $\Gamma_\parallel(e)$. This associates the edges with b_1 objects, and combination with $a_1 + b_1 + b_2$ will thus yield $b_1 + a_1 + a_2$, or:

$$\Gamma_\parallel(e) \times \Gamma_T = \Gamma_\parallel(e) + \Gamma_\sigma(e) + \Gamma_{\circlearrowleft}(e) \tag{6.153}$$

Finally, multiply this result by Γ_ϵ:

$$\Gamma_\parallel(e) \times \Gamma_R = \Gamma_\perp(e) + \Gamma_{\circlearrowleft}(e) + \Gamma_\sigma(e) \tag{6.154}$$

We now combine Eqs. (6.151) and (6.154), and find

$$\text{Trivalent: } \Gamma_\parallel(e^P) \times \Gamma_R - \Gamma_\sigma(v^P) \times \Gamma_R = \Gamma_\sigma(e^P) = \Gamma_{\text{Fries}} \tag{6.155}$$

This is precisely the representation of the Fries bonds. We can thus compare the Fries and Clar structures in a general leapfrog, and find from Eq. (6.149):

$$\Gamma_{\text{Clar}} - \Gamma_{\text{Fries}} = \Gamma_T + \Gamma_R \tag{6.156}$$

The Clar structure thus has six extra bonding orbitals as compared with the Fries structure. When both bonding schemes are correlated, as illustrated in the Walsh diagram of Fig. 6.12, this sextet must correlate with the anti-bonding half of the Fries structure. It will thus be placed on top of the Clar band, and actually be nearly non-bonding, forming six low-lying virtual orbitals, which explains the electron deficiency of the leapfrog fullerenes. Moreover, as the derivation shows, they transform exactly as rotations and translations.

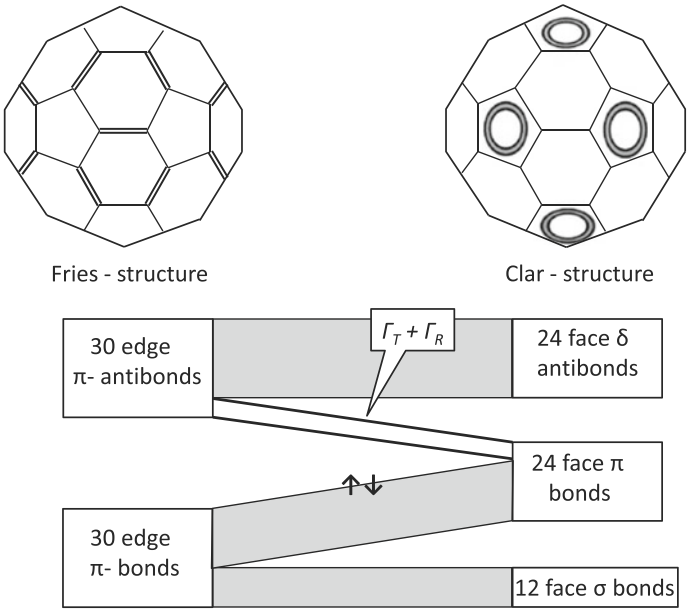


Fig. 6.12 Correlation diagram for C_{60}. The Fries and Clar structures are bonding extremes, where double bonds are either localized on the 30 bonds between the pentagons (Fries), or form isolated aromatic sextets on the twelve pentagons. The true conjugation scheme is found in between, and is characterized by six unoccupied levels, which are anti-bonding in the Fries structure and bonding in the Clar structure, and which transform as rotations and translations. Buckminsterfullerene has low-lying LUMO and LUMO+1 levels of $t_{1u}(\Gamma_T)$ and $t_{1g}(\Gamma_R)$ symmetry

6.11 Problems

6.1 A three-electron wavefunction in an octahedron is given by

$$\Psi = |(t_{1u}x\alpha)(t_{1u}y\alpha)(t_{1u}z\alpha)| \tag{6.157}$$

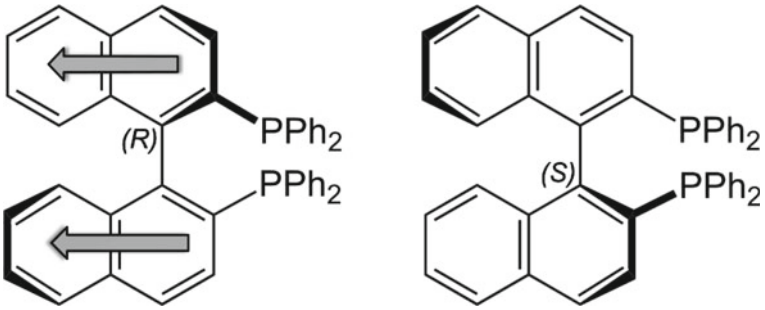
The vertical bars denote a Slater determinant. Determine the symmetry of this function, starting from parent two-electron coupled states, to which the third electron is coupled. Make use of the coupling coefficients in Appendix F.

6.2 Write the Jahn–Teller matrix for a threefold degenerate T_{1u} level in an octahedral molecule. How many reduced matrix elements are needed?

6.3 Do you expect octahedral e_g orbitals to show a magnetic dipole moment?

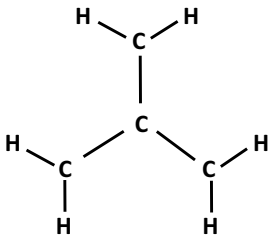
6.4 Binaphthyl consists of two linked naphthalene molecules. The dihedral angle between the two naphthyl planes is around 70°, and can be stabilized by bulky substituents on the naphthyl units, as indicated below for the case of 2,2'-di-biphenylphosphine-1,1'-binaphthyl. A circular dichroism signal is detected in the UV region, corresponding to the long-axis polarized transitions of the naph-

thyl units (indicated by the arrows in the figure). Construct the appropriate exciton states and determine the CD profile of the two enantiomers of binaphthyl.



6.5 A diradical is a molecule with two *open* orbitals, each containing one electron. Consider as an example twisted ethylene (D_{2d} symmetry, see Fig. 3.9). The HOMO is a degenerate e-orbital, occupied by two electrons. Construct the e^2 diradical states for this molecule, and determine their symmetries.

6.6 Planar trimethylenemethane (TMM), C_4H_6, is a diradical with trigonal symmetry. Determine the Hückel spectrum for the four carbon p_z-orbitals perpendicular to the plane of the molecule. The HOMO in D_{3h} has e'' symmetry and is also occupied by two electrons. Determine the corresponding diradical states, and compare with the results for twisted ethylene. How would you describe the valence bond structure of this molecule?



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Chapter 7
Spherical Symmetry and Spins



Abstract A brief excursion is made into the concept of the continuous groups, with the example of the rotation groups. The purpose is to familiarize the reader with the concept of electron spin. The coupling of spins is discussed. Applications are taken from crystal-field theory and electron spin resonance.

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7.1 The Spherical Symmetry Group

In Chap. 3, we introduced the symmetry group of the sphere, and found that all proper and improper rotations which leave the center of the sphere invariant can be represented by orthogonal matrices. These form the Lie group $O(3)$, and its rotational subgroup $SO(3)$. In the present chapter, we approach this representation from the opposite perspective: we start from the most general form of $SO(3)$ matrices, and try to find out with which rotations it can be associated. The construction of $SO(3)$ matrices is based on Lie algebra. We will not evocate it here (see [1]), but directly state the result: any orthogonal matrix in $SO(3)$ can be parametrized by four parameters:

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n_x, n_y, n_z, α, subject to the condition:

$$n_x^2 + n_y^2 + n_z^2 = 1 \tag{7.1}$$

The $SO(3)$ matrix for the row vector $(|x\rangle\ |y\rangle\ |z\rangle)$ is given by

$$\mathbb{O}(R) = \begin{pmatrix} 1 - 2(n_y^2 + n_z^2)\gamma & -n_z \sin\alpha + 2n_x n_y \gamma & n_y \sin\alpha + 2n_z n_x \gamma \\ n_z \sin\alpha + 2n_x n_y \gamma & 1 - 2(n_z^2 + n_x^2)\gamma & -n_x \sin\alpha + 2n_y n_z \gamma \\ -n_y \sin\alpha + 2n_z n_x \gamma & n_x \sin\alpha + 2n_y n_z \gamma & 1 - 2(n_x^2 + n_y^2)\gamma \end{pmatrix} \tag{7.2}$$

with: $\gamma = \sin^2(\alpha/2)$. These four parameters are identified as the rotation $\hat{R}(\alpha, n_x, n_y, n_z)$: α is the rotational angle, and n_x, n_y, n_z are three direction cosines for the orientation of the pole of the rotation axis in the Cartesian frame. The latter are normalized as $n_x^2 + n_y^2 + n_z^2 = 1$. Because of this relationship the rotational pole is defined by only two angles, θ and ϕ, as depicted in Fig. 7.1.

$$\begin{aligned} n_x &= \sin\theta\cos\phi \\ n_y &= \sin\theta\sin\phi \\ n_z &= \cos\theta \end{aligned} \tag{7.3}$$

Hence, in total to describe any rotation, only three angles are required, a result which was known to Euler [2]. The determinant of the matrix in Eq. (7.2) is $+1$, as expected. In (n_x, n_y, n_z) space, any rotation $\hat{R} \in G$ has two poles. These are the points on the unit sphere that are invariant under the rotation. These points are at $\pm\mathbf{n}$, and thus are mutual antipodes. We recall that a rotation over a positive angle is viewed from the pole as counterclockwise. In the antipodal pole, this rotation will be observed as clockwise. Rotations about antipodal poles and over opposite angles thus produce the same effect. Hence, a sign change of all parameters leaves the matrix invariant. Moreover, since a rotation through an angle of 2π is equivalent to the unit element, the rotation angle can also be specified as $\alpha - 2\pi$. In total, in the range $[-2\pi, +2\pi]$, there are thus four equivalent sets of parameters that give rise to the same matrix $\mathbb{O}(R)$.

$$\begin{gathered} \hat{R}(\alpha, n_x, n_y, n_z) \\ \hat{R}(-\alpha, -n_x, -n_y, -n_z) \\ \hat{R}(-2\pi + \alpha, n_x, n_y, n_z) \\ \hat{R}(2\pi - \alpha, -n_x, -n_y, -n_z) \end{gathered} \tag{7.4}$$

As mentioned before, the difference between rotations which give rise to the same $SO(3)$ only involves the path that is taken, not the resulting transformation. The transformations of the standard vector form the fundamental irrep of spherical symmetry. All other irreps can be constructed by taking direct products of this vector. In particular, the *spherical harmonic functions* can be constructed by taking fully symmetrized powers of the vector. The symmetrized direct square of the p-functions yields a six-dimensional function space with components: $\{z^2, x^2, y^2, yz, xz, xy\}$. This space is

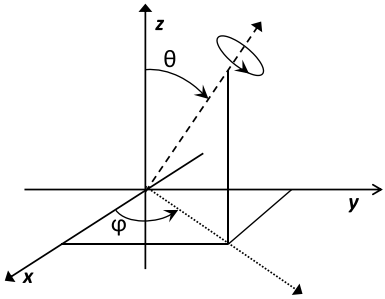


Fig. 7.1 The dashed line represents an axis of rotation, at an angle θ from the positive z-axis, and an angle ϕ in the x, y-plane, measured counterclockwise from the positive x-direction

not orthonormal: the components are not normalized, and the first three components do overlap. In fact, the space is reducible, since the sum of the squares $z^2 + x^2 + y^2$ is a radial function, which is totally symmetric under rotations. Taking out this root leaves five components, which are irreducible and correspond to the five d orbitals, shown in Table 7.1. This result parallels the cubic $[T_{1u}]^2 = A_{1g} + E_g + T_{2g}$ coupling [3].

When extending these results to the nth power of the p-irrep, symmetrization will be governed by the irreducible representations of the corresponding S_n permutation group. The f-orbitals may be generated by the third power of the p-irrep. Full symmetrization of the three components generates 10 functions: $\{z^3, x^3, y^3, z^2x, z^2y, x^2z, x^2y, y^2z, y^2x, xyz\}$, which in cubic symmetry transforms as $A_{2u} + 2T_{1u} + T_{2u}$. Again, this space is reducible and contains a p and an f-subspace. The reduction is based on the removal of the totally symmetric trace. Indeed, the combination $z(x^2 + y^2 + z^2)$ and its cyclic permutations reduce to the fundamental p-vector. The remainder of the space is irreducible and corresponds to the seven f-functions, listed in Table 7.1. Further explorations of the spherical symmetry group opens the book of angular momentum arithmetic and the underlying theory of Lie groups (see, e.g., [1, 4, 5]). This is outside the present scope. We shall restrict the treatment to indicating the subduction rules, which describe the decomposition of the spherical irreps in point-group symmetries. To obtain these rules, we have to derive the character of function space of the spherical harmonics, $\{Y_{\ell m_\ell}\}$, for $m_\ell = -\ell, -\ell + 1, \ldots, \ell - 1, \ell$, under the proper and improper rotations of the point group. We shall start by considering the proper ones first. It is sufficient to limit the treatment to rotations around the z-axis, since on a sphere all directions are equivalent. Rotations around z will affect only the angular coordinate, ϕ, in the equatorial plane, and leave the azimuthal coordinate, θ, unchanged. The ϕ-dependence of the spherical harmonics is given by

$$\Phi_{m_\ell}(\phi) = \frac{1}{\sqrt{2\pi}} \exp(i m_\ell \phi) \tag{7.5}$$

A rotation $\hat{C}_\alpha$ about the z-direction affects this function in the following way:

$$\hat{C}_\alpha \Phi_{m_\ell}(\phi) = \Phi_{m_\ell}(\phi - \alpha) = \exp(-i m_\ell \alpha) \Phi_{m_\ell}(\phi) \tag{7.6}$$

Table 7.1 Complex and cubic real forms of the spherical harmonics, for $\ell = 0, 1, 2, 3$. The constants N_ℓ are the common normalizing factors over the θ and ϕ coordinates

ℓ	N_ℓ	$\|LM\rangle$	$\|\Gamma\gamma\rangle$
s	$\sqrt{\frac{1}{4\pi}}$	$\|0\,0\rangle = 1$	$\|A_{1g}\rangle = 1$
p	$\sqrt{\frac{3}{4\pi}}\ r^{-1}$	$\|1+1\rangle = -\frac{1}{\sqrt{2}}(x+iy)$	$\|T_{1u}x\rangle = x$
		$\|1-1\rangle = \frac{1}{\sqrt{2}}(x-iy)$	$\|T_{1u}y\rangle = y$
		$\|1\,0\rangle = z$	$\|T_{1u}z\rangle = z$
d	$\sqrt{\frac{15}{8\pi}}\ r^{-2}$	$\|2+2\rangle = \frac{1}{2}(x+iy)^2$	$\|E_g\theta\rangle = \frac{1}{\sqrt{6}}(3z^2 - r^2)$
		$\|2-2\rangle = \frac{1}{2}(x-iy)^2$	$\|E_g\epsilon\rangle = \frac{1}{\sqrt{2}}(x^2 - y^2)$
		$\|2+1\rangle = -(x+iy)z$	$\|T_{2g}\xi\rangle = \sqrt{2}yz$
		$\|2-1\rangle = (x-iy)z$	$\|T_{2g}\eta\rangle = \sqrt{2}xz$
		$\|2\,0\rangle = \frac{1}{\sqrt{6}}(3z^2 - r^2)$	$\|T_{2g}\zeta\rangle = \sqrt{2}xy$
f	$\sqrt{\frac{35}{8\pi}}\ r^{-3}$	$\|3+3\rangle = -\frac{1}{2\sqrt{2}}(x+iy)^3$	$\|A_{2u}\rangle = \sqrt{6}xyz$
		$\|3-3\rangle = \frac{1}{2\sqrt{2}}(x-iy)^3$	$\|T_{1u}x\rangle = \frac{1}{\sqrt{10}}x(5x^2 - 3r^2)$
		$\|3+2\rangle = \frac{\sqrt{3}}{2}z(x+iy)^2$	$\|T_{1u}y\rangle = \frac{1}{\sqrt{10}}y(5y^2 - 3r^2)$
		$\|3-2\rangle = \frac{\sqrt{3}}{2}z(x-iy)^2$	$\|T_{1u}z\rangle = \frac{1}{\sqrt{10}}z(5z^2 - 3r^2)$
		$\|3+1\rangle = -\frac{\sqrt{3}}{2\sqrt{10}}(x+iy)(5z^2 - r^2)$	$\|T_{2u}\xi\rangle = \sqrt{\frac{3}{2}}x(z^2 - y^2)$
		$\|3-1\rangle = \frac{\sqrt{3}}{2\sqrt{10}}(x-iy)(5z^2 - r^2)$	$\|T_{2u}\eta\rangle = \sqrt{\frac{3}{2}}y(x^2 - z^2)$
		$\|3\,0\rangle = \frac{1}{\sqrt{10}}z(5z^2 - 3r^2)$	$\|T_{2u}\zeta\rangle = \sqrt{\frac{3}{2}}z(y^2 - x^2)$

The trace over the entire function space is then given by:

$$\chi^\ell(C_\alpha) = \sum_{m_\ell=-\ell}^{\ell} \exp(-im_\ell\alpha) = \frac{\sin(\ell + 1/2)\alpha}{\sin(\alpha/2)} \tag{7.7}$$

To obtain this result use was made of the following sum rule, which is obtained by carrying out a straightforward division:

$$\sum_{n=0}^{N-1} r^n = \frac{1 - r^N}{1 - r} \tag{7.8}$$

As an example, for $\ell = 1$, the rotation matrix corresponds to the matrix $\mathbb{O}(R)$ in Eq. (7.2). Its trace is given by

$$\chi^p(C_\alpha) = 3 - 4\sin^2(\alpha/2) = \frac{\sin(3\alpha/2)}{\sin(\alpha/2)} \tag{7.9}$$

Improper rotations can always be written as the product of a proper rotation and the inversion operation. The resulting character is then given by the product of the rotational character in Eq. (7.7), times the parity of the spherical harmonics, which is given by

$$\hat{\imath}\ Y_{\ell m_\ell} = (-1)^\ell Y_{\ell m_\ell} \tag{7.10}$$

Finally, we also remind that the definition of the spherical harmonics in the standard phase convention[1] implies complex conjugation, as

$$\overline{Y}_{\ell m_\ell} = (-1)^{m_\ell} Y_{\ell -m_\ell} \tag{7.11}$$

In Appendix A.2, the character tables for the groups $O(3)$ and $SO(3)$ are given. The subduction relations are listed in Appendix C.1.

7.2 Application: Crystal-Field Potentials

Model treatments of transition-metal and lanthanide complexes are essentially based on the d^n or f^n open-shell states of the central metal atom, which are perturbed by the electrostatic field of the surrounding ligating groups.

$$\mathcal{V}_{CF} = \sum_L \frac{\mathrm{e}^2 Z_L}{4\pi\epsilon_0 |\mathbf{R}_L - \mathbf{r}|} \tag{7.12}$$

This term describes the electrostatic repulsion between an electron residing in the metal orbital at a position $\mathbf{r}$ and a negatively charged ligand, with charge $-\mathrm{e}Z_L$, at a position $\mathbf{R}_L$. In crystal-field theory, the electrostatic field of the surroundings is written as an expansion in spherical harmonic operators, as these elements will be evaluated in the d or f function space of the central metal atom. Series expansion for distances $r < R_L$ yields

$$\mathcal{V}_{CF} = \sum_L \frac{\mathrm{e}^2 Z_L}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \sum_{m_\ell=-\ell}^{m_\ell=+\ell} \frac{4\pi}{2\ell+1} \frac{r^\ell}{R^{\ell+1}} Y_{\ell m_\ell}(\theta,\phi) \overline{Y}_{\ell m_\ell}(\theta_L,\phi_L) \tag{7.13}$$

This potential is invariant under the symmetry properties of the metal complex. As a result, the operator part reduces to the totally symmetric components of the spherical harmonics. Moreover, interactions with d electrons imply that ℓ must be limited to

[1] The standard phase convention is imposed by the choice for positive normalization constants for the shift operator

four, and to six for f electrons. In the case of an octahedral field, the subduction relations for spherical harmonics (see Appendix C.1) indicate that a totally symmetric A_{1g} component can be subduced only from $\ell = 4$ and $\ell = 6$. Filling in the angular positions of the ligands in an octahedron for d-electrons then yields

$$\mathcal{V}_{O_h} = \frac{6e^2 Z_L}{\sqrt{4\pi}\epsilon_0 R_L} Y_{00} + \frac{e^2 Z_L r^4}{\sqrt{4\pi}\epsilon_0 R_L^5} \frac{\sqrt{35}}{6\sqrt{2}} \left(Y_{44} + Y_{4-4} + \frac{\sqrt{14}}{\sqrt{5}} Y_{40} \right) \tag{7.14}$$

This is the famous crystal-field operator for an octahedron, which splits the d-shell into e_g and t_{2g} subshells. The crystal-field interaction is usually parametrized by the crystal-field parameter, $10Dq$, which corresponds to the splitting of the e_g and t_{2g} orbitals. The term in brackets here is the octahedral invariant of rank 4. In the multipole expansion, this corresponds to a *hexadecapole* operator. In normalized form, it reads

$$|4\ A_{1g}| = \frac{1}{2\sqrt{3}} \left(\frac{\sqrt{5}}{\sqrt{2}} (Y_{44} + Y_{4-4}) + \sqrt{7} Y_{40} \right) \tag{7.15}$$

Here, the notation between vertical bars indicates that this is an operator. We shall now derive this expression with the aid of the coupling coefficients. Since the d orbitals transform as $e_g + t_{2g}$, the squares of the d-orbitals yield two totally symmetric results, which may be abbreviated as follows:

$$\begin{aligned} |A_{1g}(e \times e)| &= \frac{1}{\sqrt{2}} \left(\theta^2 + \epsilon^2 \right) \\ |A_{1g}(t_2 \times t_2)| &= \frac{1}{\sqrt{3}} \left(\xi^2 + \eta^2 + \zeta^2 \right) \end{aligned} \tag{7.16}$$

These two invariants are at the origin of two spherical operators: one corresponds to the constant scalar $|0\ A_{1g}|$, and the other to $|4\ A_{1g}|$. The former invariant may be obtained by taking the norm of the entire d-manifold, which can be expressed in the A_{1g} functions of Eq. (7.16) as follows:

$$\begin{aligned} |0\ A_{1g}| &= \frac{1}{\sqrt{5}} \left(\theta^2 + \epsilon^2 + \xi^2 + \eta^2 + \zeta^2 \right) \\ &= \frac{1}{\sqrt{5}} \left(\sqrt{2} |A_{1g}(e \times e)| + \sqrt{3} |A_{1g}(t_2 \times t_2)| \right) \end{aligned} \tag{7.17}$$

The $|4\ A_{1g}|$ invariant must be orthonormal to this result, and thus will be given by

$$|4\ A_{1g}| = \frac{1}{\sqrt{5}} \left(\sqrt{3} |A_{1g}(e \times e)| - \sqrt{2} |A_{1g}(t_2 \times t_2)| \right) \tag{7.18}$$

We can use this result to write the functional form of the hexadecapolar invariant by combining the squares of the d-functions:

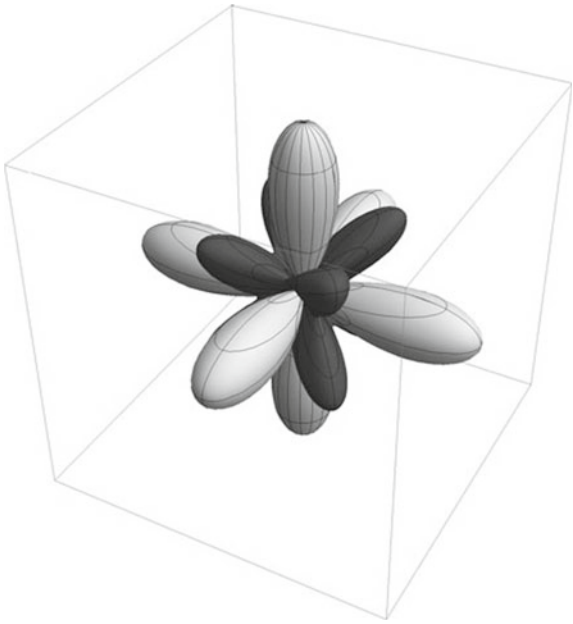


Fig. 7.2 The octahedral crystal-field potential, corresponding to the $|4A_1|$ hexadecapole. Gray and black refer to positive and negative values, respectively

$$
\begin{aligned}
|4\ A_{1g}| &= \frac{1}{\sqrt{5}}\left[\frac{\sqrt{3}}{\sqrt{2}}\left(\theta^2+\epsilon^2\right)-\frac{\sqrt{2}}{\sqrt{3}}\left(\xi^2+\eta^2+\zeta^2\right)\right] \\
&\sim \left[z^4+x^4+y^4-3\left(x^2y^2+x^2z^2+y^2z^2\right)\right] \qquad (7.19)
\end{aligned}
$$

This function corresponds precisely to the crystal-field operator of Eq. (7.14). Figure 7.2 shows this invariant. It is clearly a function that mimics the octahedral symmetry. Moreover, it reflects the multipole character of the crystal-field potential. The potential is repulsive along the coordinate axes where the ligands are, and of opposite sign along the threefold directions, in between the ligands, corresponding to the vertices of the cube. In fact, Eq. (7.19) provides a direct route to the crystal-field splitting. The coefficient in front of θ^2 and ϵ^2 in this equation is proportional to the interaction of these e_g-orbitals with the crystal-field operator, and similarly for the coefficient in front of the t_{2g} orbitals. The ratio of these coefficients can be reduced to

$$
\frac{\frac{\sqrt{3}}{\sqrt{2}}}{-\frac{\sqrt{2}}{\sqrt{3}}}=\frac{3}{-2} \qquad (7.20)
$$

which reflects the crystal-field splitting of the d-shell, where the e_g level is raised by $6Dq$, while the t_{2g} level is lowered by $4Dq$, so that the barycenter is conserved and that the total splitting remains $10Dq$.

Typically, the first invariant multipole corresponds to the charge distribution of the Platonic solids of the given symmetry: hence, the $L = 4$ term describes the

octahedron, and its dual the cube, while the next invariant, belonging to the $L = 6$ multipoles, represents the dominant term for the charge distribution in the 12-vertex Archimedean solid, known as the cuboctahedron, with a ligand at every edge center of the octahedron. Entirely similar relations exist for the icosahedral group. The first icosahedral invariant, with $L = 6$, is the dominant term for the icosahedron and its dual dodecahedron. The next term, with $L = 10$, is the leading multipole for the truncated dodecahedron, alias the 'buckyball' [6, 7].

7.3 Spinors and Spinor Groups

A spinor is an entity with two components which lives in a complex two-dimensional space. Norm-preserving transformations of this space are 2×2 unitary matrices. Together they form the unitary group $U(2)$. Here, we will restrict this group to the special unitary group $SU(2)$, of unitary matrices with determinant $+1$. There is an intricate relationship between the group $SO(3)$ of orthogonal transformations in three-dimensional Euclidean space, and the group $SU(2)$ of unitary transformations of two-dimensional space. Hence, whenever we consider a rotation in ordinary space, there is a corresponding unitary transformation in a spinor space. It means ordinary space is not so ordinary after all: it has a structure, based on more primitive constituents. The purpose of this section will be to clarify this structure. To this end, we seek an explicit connection between the rotations of a vector and the transformations of the underlying spinor.

Like $SO(3)$ the group $SU(2)$ group has three degrees of freedom, or parameters (see Problem 2.1). The corresponding generators can be identified with the elements of a traceless Hermitian 2×2 matrix. For a full description, we must refer to the construction of the Lie algebra of $SU(2)$. Here instead we will limit ourselves to the construction of a general 2×2 Hermitian matrix in the base space of the two spin components, $|\alpha\rangle$, $|\beta\rangle$. Unitary transformations of this basis induce transformations of the Hermitian matrix, which in turn can be expressed as orthogonal transformations of the three matrix parameters. In this way, a connection is found between spinors and vectors. The strategy is thus to set up a general Hamiltonian matrix in a space with two components and verify that it has spherical symmetry. The general interaction Hamiltonian in a space of only two components is a 2×2 Hermitian matrix, $\mathbb{H}$. We can take this matrix to be traceless since the trace will not introduce an interaction inside the spin space but will simply shift the barycenter of the two levels with respect to an external reference. In its most general form, such a traceless Hermitian matrix will thus contain three independent real parameters, which we shall label as x, y, z.

$$\mathbb{H} = \begin{pmatrix} z & x - iy \\ x + iy & -z \end{pmatrix} \tag{7.21}$$

We have purposely chosen the Cartesian labels for the three independent parameters, since, later on, a connection will be established between the 2D complex interaction

space and the real 3D vector space. An example of such a Hamiltonian is the Zeeman interaction of an isolated spin in a magnetic field:

$$\mathcal{H} = 2.0023\ \frac{\mu_B}{\hbar}\ \mathbf{B}\cdot\mathbf{S} \tag{7.22}$$

Here, $\mathbf{S}$ is the operator for the spin momentum, expressed in units of $\hbar$. The x, y, z parameters in this case are proportional to the magnitude of the magnetic field, $\mathbf{B}$, in the three Cartesian directions.

However, for now, we shall not yet make use of this spatial connotation and just continue to consider x, y, z as the general variables of the Hamiltonian matrix. The two components of the space will be denoted as the spin functions $|\alpha\rangle$, $|\beta\rangle$ which together form a spinor. The corresponding interaction operator can then be expressed as

$$\begin{aligned}\mathcal{H} = z\,(|\alpha\rangle\langle\alpha| - |\beta\rangle\langle\beta|) + x\,(|\alpha\rangle\langle\beta| + |\beta\rangle\langle\alpha|)\\ -iy\,(|\alpha\rangle\langle\beta| - |\beta\rangle\langle\alpha|)\end{aligned} \tag{7.23}$$

This result can also be recast in matrix form as

$$\mathcal{H} = \begin{pmatrix} |\alpha\rangle & |\beta\rangle \end{pmatrix} \mathbb{H} \begin{pmatrix} \langle\alpha| \\ \langle\beta| \end{pmatrix} \tag{7.24}$$

or, inversely, as

$$\mathbb{H} = \begin{pmatrix} \langle\alpha| \\ \langle\beta| \end{pmatrix} \mathcal{H} \begin{pmatrix} |\alpha\rangle & |\beta\rangle \end{pmatrix} \tag{7.25}$$

To establish the connection between the spinor and the vector, we now need to verify how transformations in the spinor are manifested as transformations in the vector. Consider a finite unitary transformation of the spinor. The transformation belongs to the unitary group, $U(2)$, and, as we have seen, the determinant of this matrix is unimodular. We consider the special case, however, when the determinant is $+1$. Such matrices form the special unitary group, $SU(2)$. The most general form of an $SU(2)$ matrix involves two complex parameters, say a and b, subject to the condition that their squared norm, $|a|^2 + |b|^2$, equals unity. These parameters are also known as the Cayley−Klein parameters. One has

$$\mathbb{U} = \begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix} \tag{7.26}$$

The operation $\hat{R}$ transforms the spinor as follows:

$$\left(|\alpha'\rangle\ |\beta'\rangle\right) = \hat{R}\left(|\alpha\rangle\ |\beta\rangle\right) = \left(|\alpha\rangle\ |\beta\rangle\right)\mathbb{U}(R) \tag{7.27}$$

In order to apply the transformation to the interaction operator, we must also consider the effect of $\hat{R}$ on the column of bra-functions. This simply requires the inverse of the matrix, which, for a unitary matrix, is nothing other than its complex conjugate transposed.

$$\begin{pmatrix}\langle\alpha'|\\ \langle\beta'|\end{pmatrix} = \hat{R}\begin{pmatrix}\langle\alpha|\\ \langle\beta|\end{pmatrix} = \overline{\mathbb{U}}^T(R)\begin{pmatrix}\langle\alpha|\\ \langle\beta|\end{pmatrix} \tag{7.28}$$

The transformation of the spinor thus changes the interaction matrix as follows:

$$\begin{aligned}\mathbb{H}' &= \begin{pmatrix}\langle\alpha'|\\ \langle\beta'|\end{pmatrix}\mathcal{H}\left(|\alpha'><|\beta'>\right)\\ &= \overline{\mathbb{U}}^T\begin{pmatrix}\langle\alpha|\\ \langle\beta|\end{pmatrix}\mathcal{H}\left(|\alpha><|\beta>\right)\mathbb{U}\\ &= \overline{\mathbb{U}}^T \times \mathbb{H} \times \mathbb{U}\end{aligned} \tag{7.29}$$

The transformed Hamiltonian matrix is defined by a new set of parameters (x', y', z').

$$\mathbb{H}' = \begin{pmatrix} z' & x' - iy' \\ x' + iy' & -z' \end{pmatrix} \tag{7.30}$$

In this way, the transformation of the spinor $(|\alpha\rangle\,|\beta\rangle) \rightarrow (|\alpha'\rangle\,|\beta'\rangle)$ *induces* a transformation of the vector $(x\ y\ z) \rightarrow (x'\ y'\ z')$. In the vector space, this transformation is described by a matrix $\mathbb{O}(R)$. This matrix may be constructed by combining the previous two equations. One has

$$(x'\ y'\ z') = \hat{R}(x\ y\ z) = (x\ y\ x)\mathbb{O}(R) \tag{7.31}$$

where the transformation matrix is given by

$$\mathbb{O}(R) = \begin{pmatrix} \frac{1}{2}\left(a^2 + \overline{a}^2 - b^2 - \overline{b}^2\right) & -\frac{i}{2}\left(a^2 - \overline{a}^2 + b^2 - \overline{b}^2\right) & -ab - \overline{ab} \\ \frac{i}{2}\left(a^2 - \overline{a}^2 - b^2 + \overline{b}^2\right) & \frac{1}{2}\left(a^2 + \overline{a}^2 + b^2 + \overline{b}^2\right) & -i\left(ab - \overline{ab}\right) \\ a\overline{b} + \overline{a}b & -i\left(a\overline{b} - \overline{a}b\right) & |a|^2 - |b|^2 \end{pmatrix} \tag{7.32}$$

It can be shown that this matrix is an orthogonal transformation with determinant equal to unity. Hence, it belongs to the $SO(3)$ group. As a result, it will leave the squared length of the vector invariant:

$$x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2 \tag{7.33}$$

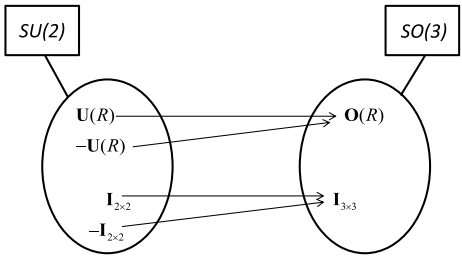


Fig. 7.3 Homomorphism between $SU(2)$ and $SO(3)$

This conservation of length is the property that confirms the previous identification of the interaction matrix elements with a 3-vector, and relates it to ordinary space. In fact, by identifying the rotation matrices in Eqs. (7.2) and (7.32), we may determine the Cayley−Klein parameters. Two solutions with opposite signs are possible:

$$\begin{aligned} a &= \left(\cos\frac{\alpha}{2} - in_z \sin\frac{\alpha}{2}\right) \\ b &= \left(-n_y \sin\frac{\alpha}{2} - in_x \sin\frac{\alpha}{2}\right) \end{aligned} \tag{7.34}$$

or:

$$\begin{aligned} a &= -\left(\cos\frac{\alpha}{2} - in_z \sin\frac{\alpha}{2}\right) \\ b &= -\left(-n_y \sin\frac{\alpha}{2} - in_x \sin\frac{\alpha}{2}\right) \end{aligned} \tag{7.35}$$

As this equation shows, the mapping $\mathbb{O}(R)$ is not an isomorphism but a homomorphism (see Fig. 7.3). Indeed, the elements of the matrix $\mathbb{O}$ are bilinear in the a and b parameters; hence, an overall sign change of the two Cayley−Klein parameters will give the same rotational matrix. The mapping between $SU(2)$ and $SO(3)$ is a two-to-one mapping. Each element of the rotation group in 3D space is the image of two elements in $SU(2)$. For this reason, $SU(2)$ is also called a covering group of $SO(3)$. The unit element in $SO(3)$ is the image of the identity matrix in $SU(2)$ and minus the identity matrix. This homomorphism also appears when we check the parameter list of Eq. (7.4), which leaves the rotation matrix of the vector unchanged. The overall sign change of the rotation angle and the directional cosines to $\hat{R}(-\alpha, -\mathbf{n})$, leaves the Cayley−Klein parameters unchanged. By contrast, the combinations $\hat{R}(2\pi - \alpha, -\mathbf{n})$ and $\hat{R}(-2\pi + \alpha, \mathbf{n})$ change the signs of both Cayley−Klein parameters.

7.4 The Coupling of Spins

Whenever a vector is rotating in space, we can view this as the result of a complex transformation which is taking place in a more primitive underlying spinor. In a sense, a vector is formed by 'squaring a spinor'. In fact, as we already indicated in Chap. 6,

coupling of two spins gives rise to a triplet state, which represents indeed the three vector components, as well as a singlet which is the totally symmetric length scalar. The coupling coefficients, which describe these binary entities, can be obtained from the matrix elements in the Hamiltonian. By applying the Wigner–Eckart theorem, with K the reduced matrix element, the interaction matrix in Eq. (7.21) yields

$$\begin{aligned}
\langle\alpha|\mathcal{H}_z|\alpha\rangle &= K\langle\alpha|z\alpha\rangle = 1\\
\langle\beta|\mathcal{H}_z|\beta\rangle &= K\langle\beta|z\beta\rangle = -1\\
\langle\alpha|\mathcal{H}_x|\beta\rangle &= K\langle\alpha|x\beta\rangle = 1\\
\langle\beta|\mathcal{H}_x|\alpha\rangle &= K\langle\beta|x\alpha\rangle = 1\\
\langle\alpha|\mathcal{H}_y|\beta\rangle &= K\langle\alpha|y\beta\rangle = -i\\
\langle\beta|\mathcal{H}_y|\alpha\rangle &= K\langle\beta|y\alpha\rangle = i
\end{aligned} \tag{7.36}$$

These coefficients can also be reversed to describe the coupling between two spinors to yield a vector. This requires that the ket spin in the coefficients is relocated to the bra part. In Sect. 6.3, we indicated that for real functions such a shift does not change the coupling, except for a renormalization factor. The transposition of the spin functions is more delicate, since the bra functions transform as the complex conjugate of the ket functions, as indicated in Eq. (7.28). In order to establish the equivalences between transformations of bra and ket spins, one must identify the basis transformation that turns the matrix $\mathbb{U}$ into its complex conjugate. This transformation is readily achieved by replacing $|\alpha\rangle$ by $|\beta\rangle$, and $|\beta\rangle$ by $-|\alpha\rangle$. The interchange corresponds to the following matrix transformation:

$$\left(|\alpha\rangle\ |\beta\rangle\right)\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix} \tag{7.37}$$

This matrix gives rise to the mapping of $\mathbb{U}$ onto its complex conjugate:

$$\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\begin{pmatrix}a & b\\ -\overline{b} & \overline{a}\end{pmatrix}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix} = \begin{pmatrix}\overline{a} & \overline{b}\\ -b & a\end{pmatrix} \tag{7.38}$$

For this reason, the matrix in Eq. (7.37) is also called the *conjugating matrix*. The conjugating matrix is defined up to an arbitrary phase. We have taken here the standard phase choice. The conjugating relationships between the two spins can now be used to transfer the spin functions in the coupling coefficients from the ket to the bra part. An α spin in the ket part becomes a β spin in the bra part, while a β spin in the ket becomes a $-\alpha$ spin in the bra part.

The corresponding coupling coefficients are summarized in Table 7.2. They indicate how two spins can be coupled to a vector. Here, we also express the vector in complex form, as

Table 7.2 The coupling coefficients for the direct product of two spins

	$S=0$	$S=1$					
		$\|+1\rangle$	$\|0\rangle$	$\|-1\rangle$	$\|x\rangle$	$\|y\rangle$	$\|z\rangle$
$\|\alpha\rangle\|\alpha\rangle$	0	1	0	0	$-1/\sqrt{2}$	$i/\sqrt{2}$	0
$\|\alpha\rangle\|\beta\rangle$	$1/\sqrt{2}$	0	$1/\sqrt{2}$	0	0	0	$1/\sqrt{2}$
$\|\beta\rangle\|\alpha\rangle$	$-1/\sqrt{2}$	0	$1/\sqrt{2}$	0	0	0	$1/\sqrt{2}$
$\|\beta\rangle\|\beta\rangle$	0	0	0	1	$1/\sqrt{2}$	$i/\sqrt{2}$	0

$$
\begin{aligned}
|+1\rangle &= -\frac{1}{\sqrt{2}}\left(|x\rangle + i|y\rangle\right) \\
|0\rangle &= |z\rangle \\
|-1\rangle &= \frac{1}{\sqrt{2}}\left(|x\rangle - i|y\rangle\right)
\end{aligned}
\tag{7.39}
$$

This transformation is in accordance with the Condon−Shortley phase conventions for the spherical basis functions [8]. In fact, our initial Hamiltonian matrix in Eq. (7.21) was constructed in this way. The resulting vector corresponds to the triplet spin functions, which we used in Sect. 6.4. The total spinor product space has dimension 4. The remainder after extraction of the three triplet functions corresponds to the spin singlet, which is invariant and transforms as a scalar. Spinors are thus the fundamental building blocks of 3D space. Their transformation properties were known to Rodrigues as early as 1840. It was some 90 years before Pauli realized that elementary particles, such as electrons, had properties that could be described as internal spin states. In spin–orbit coupling, the internal spin degrees of the electron are coupled to its external momentum, and this can be based on the embedding of $SO(3)$ in $SU(2)$.

7.5 Double Groups

The spinor basis enables us to obtain a two-dimensional matrix representation of the point-group operations. Let us first limit ourselves to the relationships for the proper rotations. The counterclockwise rotation of the vector over an angle α with the pole at the positive z-axis is given by the matrix:

$$
\hat{R}\left(|x\rangle\ |y\rangle\ |z\rangle\right) = \left(|x\rangle\ |y\rangle\ |z\rangle\right)\begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}
\tag{7.40}
$$

According to Eqs. (7.34) and (7.35), the corresponding rotation matrix in the spinor basis is determined up to a sign by

$$\hat{R}\left(|\alpha\rangle\ |\beta\rangle\right) = \pm\left(|\alpha\rangle\ |\beta\rangle\right)\begin{pmatrix}\exp(\frac{-i\alpha}{2}) & 0\\ & \exp(\frac{i\alpha}{2})\end{pmatrix} \tag{7.41}$$

How does one deal with this ambiguity of sign? A plausible way is to use a continuity argument [9]. If we approach the neighborhood of the unit element for both spinor and vector by letting α decrease to zero, we should converge to the unit matrix and, hence, take the + sign in Eq. (7.41), with $\alpha = 0$. Now let the rotation angle increase continuously from 0 to 2π. While the matrix $\mathbb{O}(R)$ is periodic in α, and passes again to the unit matrix, the spinor matrix becomes minus the unit matrix. Continuing the path in parameter space and increasing the angle to 4π will drive the vector rotation once again over the same interval, while the spinor rotation finally completes its path and reaches the unit element again. So, the difference between $\mathbb{U}$ and $-\mathbb{U}$ can be interpreted as a rotation over a full angle of 2π. From the topological point of view, the path that we have described corresponds to a full circle in the four-parameter space of $SU(2)$, and the rotation over 2π connects a point in this space to its antipode. In this space the $SO(3)$ operations may be identified as the set of straight lines connecting antipodal points.

Our real interest at present are molecular Hamiltonians that are characterized by a point group G. However, as compared with the Hamiltonian considered in Chap. 5, we should also include *spin–orbit coupling* operators. These will be invariant only under concerted transformations of the orbital and spin parts. The homomorphism between $SO(3)$ and $SU(2)$ provides a straightforward *algebraic* way to construct the spinor group associated with the point group. For each operation, $\hat{R} \in G$, a matrix $\mathbb{O}(R)$ is defined which offers a faithful representation of G and, in turn, gives rise to two spinor matrices, $\pm\mathbb{U}(R)$, which describe the spinor transformation. The set of these matrices forms a group, which contains twice as many elements as G, and thus is called the *double group*, denoted by G^*. While this construction is algebraic, the definition of a *geometrical* link between both groups is much less straightforward and unavoidably involves the introduction of phase conventions. Bethe introduced a formal symmetry operation, R, which corresponds to a rotation over 2π. Subsequently, we shall replace this by the aleph symbol $\aleph$, in order to avoid confusion. This operation is fictitious to the extent that the poles of this rotation are left undefined. It can be multiplied with every operator in the group and thus leads to an actual doubling of the number of symmetry elements. Nonetheless, the double group is not the direct product of G with the group $\{\hat{E}, \aleph\}$. Indeed, G is not a subgroup of G^*, because it is no longer closed: applying a $\hat{C}_n$ axis in G n times will not lead to the unit element but to $\aleph$.

For the actual construction of the double group as a group of operators, we need a convention to connect the spatial operators to the spinor matrices. As we have seen in Sect. 7.2, the four possible parametric descriptions of a given rotation yield two different choices for the Cayley−Klein parameters. Hence, our convention should define how to characterize unequivocally the parameters of a rotation. It will consist of two criteria: the rotation angle must be positive, and the pole from which the rotation is seen as counterclockwise should belong to the *Northern hemisphere* in the n_x, n_y, n_z parameter space. This is the hemisphere above the equatorial plane, i.e., with $n_z > 0$. In the (n_x, n_y)-plane, we include the half-circle of points with positive

Table 7.3 Representation matrices for the spinor basis in D_2^*

$\mathbb{D}(E) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\mathbb{D}(\aleph) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
$\mathbb{D}(C_2^x) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$	$\mathbb{D}(\aleph C_2^x) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$
$\mathbb{D}(C_2^y) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$\mathbb{D}(\aleph C_2^y) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
$\mathbb{D}(C_2^z) = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$	$\mathbb{D}(\aleph C_2^z) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$

n_x-value, i.e. with $n_z = 0, n_x > 0$, and also the point with $n_y = 1, n_x = 0$ and $n_z = 0$. The rotational parameters $(\alpha, \mathbf{n})$ are thus chosen in such a way that α is positive, i.e., counterclockwise, and that the vector $\mathbf{n}$ points to the Northern hemisphere. This eliminates three of the four equivalent parameter choices of Eq. (7.4). The only remaining description is then inserted into Eq. (7.34) to determine the Cayley−Klein parameters.

As a straightforward example, we take the double group of D_2. The standard drawing puts the twofold rotational axes in the Northern hemisphere, and the corresponding spinor matrices are easily obtained from Eq. (7.34). The results are given in Table 7.3. For each operation of G, there are two operations in the double group: $\hat{R}$ and $\aleph\hat{R}$. Note that the Bethe operation, $\aleph$, commutes with every element of the group. Armed with this set of matrices, one can easily construct the multiplication table; as an example:

$$\mathbb{D}(C_{2y}) \times \mathbb{D}(C_{2x}) = \mathbb{D}(\aleph C_{2z}) \tag{7.42}$$

From the multiplication table, one can obtain the conjugacy classes. The double group, D_2^*, has five classes; hence, it will contain one extra irrep, as compared with the parent group. The character table is shown in Table 7.4.

The irreps are of two different kinds. The *orbital* irreps are not changed under $\aleph$, and thus retain the characters of the single group. The other kind are the *spin* irreps which are anti-symmetric under $\aleph$. In D_2^*, there is one spin representation, denoted as $E_{1/2}$. Direct product tables may also be constructed. For the spin representation, its direct square is given by

$$E_{1/2} \times E_{1/2} = \{A\} + [B_1 + B_2 + B_3] \tag{7.43}$$

Note that the totally symmetric component in this case belongs to the anti-symmetrized part of the direct square of the spinor irreps. Relevant tabular information concerning the double groups and spinor irreps is gathered in Appendix G. As a further example, in the D_3 symmetry group, the three $\hat{C}_2$ operators, which

Table 7.4 Character table for the double group D_2^*

	$\hat{E}$	$\aleph$	$\begin{pmatrix}\hat{C}_2^z\\ \aleph\hat{C}_2^z\end{pmatrix}$	$\begin{pmatrix}\hat{C}_2^y\\ \aleph\hat{C}_2^y\end{pmatrix}$	$\begin{pmatrix}\hat{C}_2^x\\ \aleph\hat{C}_2^x\end{pmatrix}$
A	1	1	1	1	1
B_1	1	1	1	−1	−1
B_2	1	1	−1	1	−1
B_3	1	1	−1	−1	1
$E_{1/2}$	2	−2	0	0	0

bisect the chelating ligand (*cf.* Fig. 6.5), all lie in the (x', y') plane. In order to obey the conventions we have to take the poles of these axes in the positive semicircle with $x' > 0$. The $(\alpha, n_{x'}, n_{y'}, n_{z'})$ labeling in the primed coordinate system is thus as follows:

$$\begin{aligned}\hat{C_2^A}\ &(180°,\ 1,\ \ 0,\ \ 0)\\ \hat{C_2^B}\ &(180°,\ \tfrac{1}{2},\ -\tfrac{\sqrt{3}}{2},\ 0)\\ \hat{C_2^C}\ &(180°,\ \tfrac{1}{2},\ \ \tfrac{\sqrt{3}}{2},\ 0)\end{aligned} \tag{7.44}$$

The corresponding Cayley−Klein parameters are then determined as

$$\begin{aligned}\hat{C_2^A}\ &(a = 0, b = -i)\\ \hat{C_2^B}\ &(a = 0, b = \tfrac{\sqrt{3}}{2} - \tfrac{i}{2})\\ \hat{C_2^C}\ &(a = 0, b = -\tfrac{\sqrt{3}}{2} - \tfrac{i}{2})\end{aligned} \tag{7.45}$$

In Tables 7.5 and 7.6, we provide the corresponding matrices and the character table. The ρ designations in the latter table refer to Kramers doublets and will be explained in the subsequent section. The class structure of the double group in connection to the point group is determined by the Opechowski theorem [10].

Theorem 7.1 *If a set of proper rotations, $\hat{C}_n$, forms a class in the single group G, then it will give rise to two separate classes in the double group, corresponding to conjugacy classes $\{\hat{C}_n\}$ and $\{\aleph\hat{C}_n\}$. The case of $n = 2$ is exceptional: when $n = 2$ ánd there is another twofold axis in G perpendicular to this $\hat{C}_2$, both this $\hat{C}_2$ and $\aleph\hat{C}_2$ will belong to the same class.*

Table 7.5 Representation matrices for the spinor basis in D_3^*. $\omega = \exp\frac{i\pi}{6}$

$$\mathbb{D}(E) = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \qquad \mathbb{D}(C_2^A) = \begin{pmatrix}0 & -i\\ -i & 0\end{pmatrix}$$

$$\mathbb{D}(C_3) = \begin{pmatrix}\overline{\omega}^2 & 0\\ 0 & \omega^2\end{pmatrix} \qquad \mathbb{D}(C_2^B) = \begin{pmatrix}0 & \overline{\omega}\\ -\omega & 0\end{pmatrix}$$

$$\mathbb{D}(C_3^2) = \begin{pmatrix}-\omega^2 & 0\\ 0 & -\overline{\omega}^2\end{pmatrix} \qquad \mathbb{D}(C_2^C) = \begin{pmatrix}0 & -\omega\\ \overline{\omega} & 0\end{pmatrix}$$

Table 7.6 Character table for the double group D_3^*

		$\hat{E}$	$\aleph$	$\begin{pmatrix}\hat{C}_3\\ \aleph\hat{C}_3^2\end{pmatrix}$	$\begin{pmatrix}\aleph\hat{C}_3\\ \hat{C}_3^2\end{pmatrix}$	$\begin{pmatrix}\hat{C}_2^A\\ \aleph\hat{C}_2^B\\ \aleph\hat{C}_2^C\end{pmatrix}$	$\begin{pmatrix}\aleph\hat{C}_2^A\\ \hat{C}_2^B\\ \hat{C}_2^C\end{pmatrix}$
$E_{1/2}$		2	-2	1	-1	0	0
$E_{3/2}$	ρ_1	1	-1	-1	1	i	$-i$
	ρ_2	1	-1	-1	1	$-i$	i

The theorem can easily be demonstrated in an algebraic way. If a symmetry element $\hat{A}$ is conjugate to $\hat{B}$, the element $\aleph\hat{A}$ will be conjugate to $\aleph\hat{B}$, since the Bethe operation commutes with all symmetry elements:

$$\hat{A} = \hat{X}\hat{B}\hat{X}^{-1} \;\rightarrow\; \aleph\hat{A} = \hat{X}\aleph\hat{B}\hat{X}^{-1} \tag{7.46}$$

Since multiplication by $\aleph$ corresponds to multiplication by $-\mathbb{I}$, a symmetry element $\hat{A}$ and its double-group partner $\aleph\hat{A}$ will have characters that differ by sign. Unless their characters are zero, they cannot belong to the same class since symmetry elements in the same class must have the same character. This explains the first rule of the theorem.[2]

Exceptions can exist when the character is zero. The character of the spinor irrep is given by

$$\chi^{\text{spin}} = a + \overline{a} = \pm 2\cos\frac{\alpha}{2} \tag{7.47}$$

This character can be zero only for $\alpha = \pm\pi$, and, hence, for binary rotations, with $n = 2$. To examine whether or not the matrix for a binary rotation can be class-conjugated to minus itself, we may limit ourselves to the study of one orientation of the rotation axis, say $\hat{C}_2^z$. Indeed, in $SU(2)$ any orientation can always be transformed backwards to this standard choice by a unitary transformation. The problem thus reduces to finding a spinor operation $\hat{X}$, represented by a matrix $\mathbb{X}$ with Cayley−Klein parameters a, b, which transforms $\mathbb{D}(C_2^z)$ into minus itself:

$$\hat{X}\hat{C}_2^z\hat{X}^{-1} = \aleph\hat{C}_2^z \tag{7.48}$$

or, in terms of the spinor matrices:

$$\begin{pmatrix} a & b \\ -\overline{b} & \overline{a}\end{pmatrix}\begin{pmatrix} -i & 0 \\ 0 & i\end{pmatrix}\begin{pmatrix} \overline{a} & -b \\ \overline{b} & a\end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i\end{pmatrix} \tag{7.49}$$

This expression reduces to a set of two equations:

[2] Note that not all elements in the same class will be oriented in the Northern hemisphere, as is clear from Tables 7.4 and 7.6.

$$\begin{aligned} |a|^2 - |b|^2 &= -1 \\ ab &= 0 \end{aligned} \tag{7.50}$$

These equations can be solved only when a is equal to zero. This implies that both the real and complex parts of this parameter must be equal to zero; hence:

$$\begin{aligned} \Re(a) &= \cos \alpha/2 = 0 \;\rightarrow\; \alpha = \pi \mod 2\pi \\ \Im(a) &= -n_z \sin \alpha/2 = 0 \;\rightarrow\; n_z = 0 \end{aligned} \tag{7.51}$$

This result identifies the $\hat{X}$ operator as a twofold rotation in a plane perpendicular to the z-direction. Hence, a twofold rotation axis, $\hat{C}_2$, and its double group extension $\aleph\hat{C}_2$ will belong to the same class, only if the group contains additional binary elements in a plane perpendicular to this axis. Otherwise, the first rule will apply. The exception referred to in this theorem is illustrated by the D_2^* class structure as shown in Table 7.4.

So far, we have been concerned only with proper rotations, because in the spinor basis improper rotations are not defined. Since the electron spin is associated with an internal 'spinning' of the electron around its axis, the electron spin is assigned an intrinsic positive parity. As we have seen before in Sect. 3.8, all improper rotations can be written as the product of a proper rotation and an inversion; therefore, whenever an improper rotation acts on a spinor, we simply take the matrix representation for the proper factor in this improper rotation.

7.6 Kramers Degeneracy

In 1930, Kramers showed that, in the presence of an arbitrary electrostatic field, all states with an odd number of spins must still have even degeneracy (adapted from [11]).

Theorem 7.2 *The energy levels of a system that contains an odd number of spin-$\frac{1}{2}$ particles are at least doubly degenerate in the absence of an external magnetic field.*

This theorem reflects the influence of time-reversal symmetry. Already in Chap. 2, we showed that the time-reversal operator, ϑ, is an anti-linear operator. It will turn any spatial wavefunction into its complex conjugate. Applying it twice in succession will return the original wavefunction and, hence, we may write for spatial functions:

$$\text{vectors}: \quad \vartheta^2 = +1 \tag{7.52}$$

The time-reversal properties of a spinor are different, however. Here we can use the same argument as in Sect. 7.4, *viz.* in Eq. (7.38), by requiring that the time reversal of the spinor components will lead to complex conjugation of their transformation properties. This implies that time reversal must turn $|\alpha\rangle$ into $|\beta\rangle$, and vice versa, but

with a sign difference:

$$\begin{aligned} \vartheta|\alpha\rangle &= |\beta\rangle \\ \vartheta|\beta\rangle &= -|\alpha\rangle \end{aligned} \tag{7.53}$$

Hence, for spinors, applying time reversal twice leads to a sign change:

$$\text{spinors}: \quad \vartheta^2 = -1 \tag{7.54}$$

Note that the action of time reversal on the spin functions precisely corresponds to the $\hat{C}_2^y$ operator, and thus is represented by $\mathbb{D}(C_2^y)$. This result may be generalized, in the sense that time reversal can be represented as the product of complex conjugation, denoted as K, and a unitary operator acting on the components of a function space, which we shall denote by the unitary matrix $\mathbb{U}$. We thus write: $\vartheta = \mathbb{U}\ K$. When this operator is applied twice it must return the same state, except possibly for a phase factor, say $\exp(i\kappa)$. Following Wigner, we now show that the two cases $\vartheta^2 = \pm 1$ are in fact the only possibilities. Hence, the phase factor can be only either +1 (time-even state) or -1 (time-odd state) [12]. Taking time reversal twice, we have

$$\vartheta^2 = \mathbb{U}K\mathbb{U}K = \mathbb{U} \times \overline{\mathbb{U}} = \exp(i\kappa)\mathbb{I} \tag{7.55}$$

Since $\mathbb{U}$ is unitary, we have $\mathbb{U} \times \overline{\mathbb{U}}^T = \mathbb{I}$. Comparing this result with the previous expression, it follows that:

$$\overline{\mathbb{U}} = \exp(i\kappa)\overline{\mathbb{U}}^T \tag{7.56}$$

Taking the transpose of both matrices in this equation will not affect the phase factor:

$$\overline{\mathbb{U}}^T = \exp(i\kappa)\overline{\mathbb{U}} \tag{7.57}$$

Combining Eqs. (7.56) and (7.57) yields

$$\overline{\mathbb{U}} = \exp(2i\kappa)\overline{\mathbb{U}} \tag{7.58}$$

which implies that $\exp(i\kappa) = \pm 1$.

We shall now examine the effects of these two kinds of time-reversal symmetry on quantum systems under time-even Hamiltonians, i.e., in the absence of external magnetic fields.

- $\boldsymbol{\vartheta^2 = +1}$. In the case of a positive sign, it is always possible to write states that are time-invariant. Consider a state that is described by a complex wavefunction, $|\Psi\rangle$, which is an eigenfunction of a time-even Hamiltonian. The time-reversed function, $\vartheta|\Psi\rangle$, will thus also be an eigenfunction with the same eigenenergy. The two functions will either coincide or be linearly independent. The latter case leads to a state which is at least twofold-degenerate. Both components of this

degeneracy may transform as the complex-conjugate irreps of point groups such as appear for the cyclic groups, or the T_h group. When $\vartheta^2 = +1$ it is always possible to recombine these two degenerate states into two linear combinations, which are invariant under time reversal. It is indeed sufficient to project the real and imaginary parts of these functions:

$$\begin{aligned} |\phi\rangle &= \frac{1}{\sqrt{2}}\left(|\Psi\rangle + \vartheta|\Psi\rangle\right) \\ |\chi\rangle &= \frac{i}{\sqrt{2}}\left(|\Psi\rangle - \vartheta|\Psi\rangle\right) \end{aligned} \tag{7.59}$$

It can easily be demonstrated that the Hamiltonian in this new basis is real:

$$\langle\phi|\mathcal{H}|\chi\rangle = \overline{\langle\vartheta\phi|\vartheta(\mathcal{H}\chi)\rangle} = \overline{\langle\vartheta\phi|\vartheta\mathcal{H}\vartheta|\vartheta\chi\rangle} = \overline{\langle\phi|\mathcal{H}|\chi\rangle} \tag{7.60}$$

Hence, in this case, it is always possible to rewrite the basis in such a way that the Hamiltonian matrix is completely real, and the states behave in all respects as a real twofold-degenerate irrep, which can be split by symmetry-lowering electrostatic fields. In particular, such states will be subject to Jahn−Teller distortions.

- $\boldsymbol{\vartheta^2 = -1}$. In the case of a negative sign, it is impossible to obtain states that are time-invariant. This can be shown as follows. We start again with two states that are each other's time inverse ($|\Psi\rangle$ and $\vartheta|\Psi\rangle$) and first show that these states must be linearly independent:

$$\langle\Psi|\vartheta\Psi\rangle = \overline{\langle\vartheta\Psi|\vartheta^2\Psi\rangle} = -\overline{\langle\vartheta\Psi|\Psi\rangle} = -\langle\Psi|\vartheta\Psi\rangle = 0 \tag{7.61}$$

In contrast to the previous case, all attempts to construct a linear combination of these basis states that is invariant under time reversal, fail. Indeed, suppose $|X\rangle$ is a linear combination with coefficients a and b, such that $\vartheta|X\rangle = |X\rangle$. Then we have

$$\begin{aligned} |X\rangle &= a|\Psi\rangle + b\vartheta|\Psi\rangle \\ \vartheta|X\rangle &= \overline{a}\vartheta|\Psi\rangle - \overline{b}|\Psi\rangle \equiv a|\Psi\rangle + b\vartheta|\Psi\rangle \end{aligned} \tag{7.62}$$

Since the two kets are linearly independent, their respective coefficients must coincide, and this implies: $a = b = 0$. Hence, it is not possible to remove the degeneracy by time-even external fields. In particular, these states will not be subject to the JT effect.

7.6.1 Time-Reversal Selection Rules

The argument used in Eq. (7.60) can be generalized to describe selection rules that depend on time reversal [13]. We first introduce two parities, τ and η, which describe the time-dependence of the state and of the Hamiltonian:

$$\begin{aligned} \vartheta^2 &= (-1)^\tau \hat{E} \\ \vartheta \mathcal{H} \vartheta^{-1} &= (-1)^\eta \mathcal{H} \end{aligned} \tag{7.63}$$

The first label, τ, indicates the parity of the state functions, as we have just introduced in this section. The second label, η, indicates whether the Hamiltonian is time-even or time-odd. Time-even interactions are typically interactions associated with the electrostatic potential, such as the Jahn−Teller and the Stark effect. Time-odd interactions are electrodynamic in nature, the most common one being the Zeeman interaction. We shall now study a function space that is invariant under time reversal, and transforms according to a degenerate irrep Γ with all characters real. As we have seen, this can be either an orbital or a spinor irrep. If $|\phi\rangle$ is an element of this space, so is $\vartheta|\phi\rangle$. Now, instead of considering matrix elements of type $\langle\phi|\mathcal{H}|\chi\rangle$, we shall replace the bra-functions by their time-reversed partners. The interaction element will then be of type $\langle\vartheta\phi|\mathcal{H}|\chi\rangle$. This may seem awkward, but in fact, it does not lead to inconsistencies. In the case of orbital irreps, basis functions will either be real or may be arranged in complex-conjugate pairs, which are mutually time inverses. For spinor irreps, we can always write the basis functions in time-reversal pairs, such as the α and β spins.

The Hamiltonian for physical interactions must be Hermitian; hence, the bracket will be equal to the complex-conjugate inverted bracket:

$$\langle\vartheta\phi|\mathcal{H}|\chi\rangle = \overline{\langle\chi|\mathcal{H}|\vartheta\phi\rangle} \tag{7.64}$$

Complex conjugation of the bracket can also be achieved by time reversal. But, as an operator, time reversal enters into the bracket and operates on the components.

$$\overline{\langle\chi|\mathcal{H}|\vartheta\phi\rangle} = \langle\vartheta\chi|\vartheta(\mathcal{H}\vartheta\phi)\rangle = \langle\vartheta\chi|\vartheta\mathcal{H}\vartheta^{-1}|\vartheta^2\phi\rangle = (-1)^{\tau+\eta}\langle\vartheta\chi|\mathcal{H}|\phi\rangle \tag{7.65}$$

By combining these results we may thus write

$$\langle\vartheta\phi|\mathcal{H}|\chi\rangle = \frac{1}{2}\left[\langle\vartheta\phi|\mathcal{H}|\chi\rangle + (-1)^{\tau+\eta}\langle\vartheta\chi|\mathcal{H}|\phi\rangle\right] \tag{7.66}$$

The transformations of a time-reversed bra, $\langle\vartheta f_i|$, and its original ket, $|f_i\rangle$, are described by exactly the same matrices, because complex conjugation is applied twice:

$$\hat{R}\langle\vartheta f_i| = \sum_j D^{\Gamma}_{ji}(R)\langle\vartheta f_j|$$
$$\hat{R}|f_i\rangle = \sum_j D^{\Gamma}_{ji}(R)|f_j\rangle \tag{7.67}$$

This implies that the matrix elements in Eq. (7.66) are described by CG coupling coefficients which are symmetric with respect to exchange of the bra and ket parts. This result leads to the following selection rules:

1. If the Hamiltonian and the system have the same parity under time reversal, interaction can take place only with the symmetrized square of the irrep, $[\Gamma]^2$.
2. If the Hamiltonian and the system have opposite parity under time reversal, the only allowed interaction elements must belong to the anti-symmetrized square of the irrep, $\{\Gamma\}^2$.

As a case in point, the JT Hamiltonian for orbital systems is limited to the symmetrized square, whereas the Zeeman Hamiltonian will only arise if the anti-symmetrized square contains the symmetries of an axial field. For systems with an odd number of electrons, which therefore transform according to spinor representations, the selection rules are exactly opposite.

7.7 Application: Spin Hamiltonian for the Octahedral Quartet State

The octahedral double group contains a four-dimensional spin representation, which is commonly denoted as the Γ_8 quartet, or as $G_{3/2}$, or as U' in Griffith's notation. The direct square of this irrep is given by

$$\Gamma_8 \times \Gamma_8 = \left[A_{2g} + 2T_{1g} + T_{2g}\right] + \{A_{1g} + E_g + T_{2g}\} \tag{7.68}$$

According to the time-reversal selection rules, time-odd interactions with a magnetic field will be based on the symmetrized square. The spin-operator of the Zeeman Hamiltonian transforms as T_{1g}, which is indeed included in the symmetrized square. The present case is, however, special since the T_{1g} irrep occurs twice in the product. The multiplicity separation cannot be achieved on the basis of symmetrization, since both T_{1g} irreps appear in the symmetrized part. One way to distinguish the two products is through the subduction process from spherical symmetry. The addition rules of angular momenta give rise to

$$3/2 \times 3/2 = [p + f] + \{s + d\} \tag{7.69}$$

On these grounds, the two T_{1g} interactions can be distinguished on the basis of a different spherical parentage, corresponding to p or f coupling. We shall return

to this point in a moment. For a systematic treatment of this problem, we start by setting up a suitable function space. The spherical $S = 3/2$ spin-quartet level subduces directly the octahedral Γ_8. We can thus use the quartet spin functions as symmetry bases. The components of $S = 3/2$ can be obtained by a fully symmetrized product of the basic spinor:

$$\begin{aligned}
|3/2 + 3/2\rangle &= \alpha_1\alpha_2\alpha_3 \\
|3/2 + 1/2\rangle &= \frac{1}{\sqrt{3}}(\alpha_1\alpha_2\beta_3 + \alpha_1\beta_2\alpha_3 + \beta_1\alpha_2\alpha_3) \\
|3/2 - 1/2\rangle &= \frac{1}{\sqrt{3}}(\alpha_1\beta_2\beta_3 + \beta_1\alpha_2\beta_3 + \beta_1\beta_2\alpha_3) \\
|3/2 - 3/2\rangle &= \beta_1\beta_2\beta_3
\end{aligned} \tag{7.70}$$

In this expression, the fundamental $\{|\alpha\rangle, |\beta\rangle\}$ spinor resembles a quark state: three quarks are coupled together to form the quartet result. The use of the quartet spin bases does not mean that our Γ_8 really corresponds to a quartet spin. It only means that we can introduce a fictitious spin operator, $\widetilde{\mathbf{S}}$, which acts on the Γ_8 components in the same way as the real spin momentum would act on the components of a spin quartet. The transformations of the Γ_8 spinor under the elements of the group O^* may be obtained by combining the transformation matrices for the fundamental $(|\alpha\rangle|\beta\rangle)$ spins with the quartet coupling scheme in Eq. (7.70). In the group O^*, this irrep is denoted as Γ_6. In Table 7.7, the results are shown for two generators of the octahedral group. These matrices can be taken as the canonical basis relationships which define the components of the Γ_8. In Griffith's notation, these four components are defined in the following way:

Table 7.7 Representation matrices for the spinor basis in O^*

$$\mathbb{D}^{\Gamma_6}(C_4^z) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix} \quad \mathbb{D}^{\Gamma_6}(C_3^{xyz}) = \frac{1}{2}\begin{pmatrix} 1-i & -1-i \\ 1-i & 1+i \end{pmatrix}$$

$$\mathbb{D}^{\Gamma_8}(\hat{C}_4^z) = \frac{1}{\sqrt{2}}\begin{pmatrix} -1-i & 0 & 0 & 0 \\ 0 & 1-i & 0 & 0 \\ 0 & 0 & 1+i & 0 \\ 0 & 0 & 0 & -1+i \end{pmatrix}$$

$$\mathbb{D}^{\Gamma_8}(\hat{C}_3^{xyz}) = \frac{1}{4}\begin{pmatrix} -1-i & \sqrt{3}(-1+i) & \sqrt{3}(1+i) & 1-i \\ \sqrt{3}(-1-i) & -1+i & -1-i & \sqrt{3}(-1+i) \\ \sqrt{3}(-1-i) & 1-i & -1-i & \sqrt{3}(1-i) \\ -1-i & \sqrt{3}(1-i) & \sqrt{3}(1+i) & -1+i \end{pmatrix}$$

$$
\begin{aligned}
|U'\kappa\rangle &\sim |\frac{3}{2}+\frac{3}{2}\rangle \\
|U'\lambda\rangle &\sim |\frac{3}{2}+\frac{1}{2}\rangle \\
|U'\mu\rangle &\sim |\frac{3}{2}-\frac{1}{2}\rangle \\
|U'\nu\rangle &\sim |\frac{3}{2}-\frac{3}{2}\rangle
\end{aligned}
\tag{7.71}
$$

Knowing the symmetries of the components we can now turn to the coupling coefficients that describe their interactions. The coupling coefficients that we need are determined by the Zeeman Hamiltonian, which can be written as

$$
\mathcal{H}_{Ze} = \frac{\mu_B}{\hbar}\,\mathbf{B}\cdot(\mathbf{L}+2.0023\;\mathbf{S}) \tag{7.72}
$$

The electronic part of this operator contains the orbital angular momentum and the spin operator. In octahedral symmetry, the overall electronic operator transforms as T_{1g}. Applying the Wigner−Eckart theorem to the interaction elements in this operator yields

$$
\begin{aligned}
\frac{\mu_B}{\hbar}\,\langle\Gamma_8\, i|\,(\mathbf{L}+2.0023\;\mathbf{S})_k\,|\Gamma_8\, j\rangle = {} & \langle\Gamma_8||T_1||\Gamma_8\rangle_a\langle\Gamma_8\, i|T_1\, k\,\Gamma_8\, j\rangle_a + \\
& \langle\Gamma_8||T_1||\Gamma_8\rangle_b\langle\Gamma_8\, i|T_1\, k\,\Gamma_8\, j\rangle_b
\end{aligned}
\tag{7.73}
$$

Here, k denotes the Cartesian component of the momentum operator. Here, we have introduced the extra labels a and b in order to distinguish that there are two coupling channels. The coupling coefficients that are required are of type $\langle\Gamma_{8\,k}|\Gamma_{8\,i}\;T_{1\,j}\rangle$, while the coefficients, as given in Appendix G, are of type $\langle\Gamma_{8\,i}\;T_{1\,j}|\Gamma_{8\,k}\rangle$ and describe the spin–orbit coupling coefficients for the spin–orbit levels of a ${}^4T_1$ state. However, since all coefficients in the table are real, turning them around does not make any difference. Hence, we can directly use the spin–orbit tables to obtain the Zeeman matrix. Only one of the coupling channels is seen to link the κ and ν levels. In spherical symmetry, this requires a jump of 3 spin units, which can be bridged only by an $\ell = 3$ operator. The coupling coefficients of this channel are thus of spherical octupole parentage ($\ell = 3$), while the other set is of dipole parentage ($\ell = 1$). We shall parametrize the corresponding reduced matrix elements as J_f and J_p, respectively. The constant $\mu_B/\hbar$, as well as a common factor of $1/\sqrt{15}$, are also absorbed into these parameters. The electronic operator will be represented as $|T_i|$ and the Zeeman Hamiltonian is then recast in complex form as

$$
\begin{aligned}
\mathcal{H}_{Ze} &= B_x|T_x| + B_y|T_y| + B_z|T_z| \\
&= B_z|T_0| - \frac{1}{\sqrt{2}}(B_x - i\,B_y)|T_{+1}| + \frac{1}{\sqrt{2}}(B_x + i\,B_y)|T_{-1}|
\end{aligned}
\tag{7.74}
$$

This expression follows the convention of Eq. (7.39). The operator part is defined by

$$\begin{aligned} |T_0| &= |T_z| \\ |T_{+1}| &= -\frac{1}{\sqrt{2}}(|T_x| + i|T_y|) \\ |T_{-1}| &= \frac{1}{\sqrt{2}}(|T_x| - i|T_y|) \end{aligned} \tag{7.75}$$

The elements of the interaction matrix are then given by

$$\begin{aligned} H_{ij} &= B_z\Big(J_p\langle\Gamma_8 i|T_0\ \Gamma_8 j\rangle_p + J_f\langle\Gamma_8 i|T_0\ \Gamma_8 j\rangle_f\Big) \\ &-\frac{1}{\sqrt{2}}(B_x - iB_y)\Big(J_p\langle\Gamma_8 i|T_{+1}\ \Gamma_8 j\rangle_p + J_f\langle\Gamma_8 i|T_{+1}\ \Gamma_8 j\rangle_f\Big) \\ &+\frac{1}{\sqrt{2}}(B_x + iB_y)\Big(J_p\langle\Gamma_8 i|T_{-1}\ \Gamma_8 j\rangle_p\Big) + J_f\langle\Gamma_8 i|T_{-1}\ \Gamma_8 j\rangle_f\Big) \end{aligned} \tag{7.76}$$

The resulting interaction matrix is given in Table 7.8. Since the Zeeman interaction leads to a splitting of the levels which conserves the barycenter, the secular equation does not contain odd powers in the energy.

$$aE^4 + bE^2 + c = 0 \tag{7.77}$$

Table 7.8 Spin Hamiltonian matrix for the octahedral quartet irrep. A common factor of $1/\sqrt{15}$ is absorbed into the J-parameters

B_z	$\|\Gamma_8\kappa\rangle$	$\|\Gamma_8\lambda\rangle$	$\|\Gamma_8\mu\rangle$	$\|\Gamma_8\nu\rangle$
$\langle\Gamma_8\kappa\|$	$(3J_p - J_f)$			
$\langle\Gamma_8\lambda\|$		$(J_p + 3J_f)$		
$\langle\Gamma_8\mu\|$			$-(J_p + 3J_f)$	
$\langle\Gamma_8\nu\|$				$-(3J_p - J_f)$
$B_x - iB_y$	$\|\Gamma_8\kappa\rangle$	$\|\Gamma_8\lambda\rangle$	$\|\Gamma_8\mu\rangle$	$\|\Gamma_8\nu\rangle$
$\langle\Gamma_8\kappa\|$		$\sqrt{3}(J_p + \frac{1}{2}J_f)$		
$\langle\Gamma_8\lambda\|$			$(2J_p - \frac{3}{2}J_f)$	
$\langle\Gamma_8\mu\|$				$\sqrt{3}(J_p + \frac{1}{2}J_f)$
$\langle\Gamma_8\nu\|$	$-\frac{5}{2}J_f$			
$B_x + iB_y$	$\|\Gamma_8\kappa\rangle$	$\|\Gamma_8\lambda\rangle$	$\|\Gamma_8\mu\rangle$	$\|\Gamma_8\nu\rangle$
$\langle\Gamma_8\kappa\|$				$-\frac{5}{2}J_f$
$\langle\Gamma_8\lambda\|$	$\sqrt{3}(J_p + \frac{1}{2}J_f)$			
$\langle\Gamma_8\mu\|$		$(2J_p - \frac{3}{2}J_f)$		
$\langle\Gamma_8\nu\|$			$\sqrt{3}(J_p + \frac{1}{2}J_f)$	

The parameters are identified as

$$\begin{aligned} a &= 1 \\ b &= -10\big(B_x^2 + B_y^2 + B_z^2\big)\big(J_p^2 + J_f^2\big) \\ c &= \big(B_x^4 + B_y^4 + B_z^4\big)\big(9J_p^4 + 48J_p^3 J_f + 46J_p^2 J_f^2 - 48J_p J_f^3 + 9J_f^4\big) \\ &\quad + \big(B_x^2 B_y^2 + B_x^2 B_z^2 + B_y^2 B_z^2\big)\big(18J_p^4 - 144J_p^3 J_f + 32J_p^2 J_f^2 + 24J_p J_f^3 + 63J_f^4\big) \end{aligned} \tag{7.78}$$

The parameter b in this expression is isotropic, i.e. it does not depend on the orientation of the magnetic field in the octahedron. On the other hand, the parameter c contains an anisotropic contribution. It depends on the orientation of the magnetic field in the octahedron, but symmetry-equivalent orientations must, of course, yield the same splitting. This means that c is certainly an octahedral invariant and may thus be written as the sum of the familiar scalar $L = 0$ and hexadecapolar $L = 4$ cubic invariants that we derived in Sect. 7.2. We thus write

$$c = c_1\big(B_x^2 + B_y^2 + B_z^2\big)^2 + c_2\big(B_x^4 + B_y^4 + B_z^4 - 3B_x^2 B_y^2 - 3B_x^2 B_z^2 - 3B_y^2 B_z^2\big) \tag{7.79}$$

The coefficients in this equation are identified as

$$\begin{aligned} c_1 &= 9J_p^4 + 34J_p^2 J_f^2 - 24J_p J_f^3 + 18J_f^4 \\ c_2 &= J_f(4J_p + 2J_f)^2(3J_p - 9/4J_f) \end{aligned} \tag{7.80}$$

The c_2 coefficient is of special interest, since it controls the only octahedral term in the linear Zeeman effect. If this coefficient vanishes, the splitting will be completely isotropic, and does not depend on the orientation of the magnetic field in the octahedron. There are three possible isotropies [14].

- For $J_f = 0$, the spin operator is strictly dipolar, and the Zeeman Hamiltonian will induce a regular splitting of the quartet level, which is proportional to the spin quantum number, M_S. This case is illustrated in the left-hand panel of Fig. 7.4. Such cases will arise for an octahedral ${}^4A_1$ state, and also for the quartet spin–orbit level of a ${}^2T_1$ state.
- For $4J_p + 2J_f = 0$, the matrix splits into two separate 2×2 blocks, which have the same eigenvalues. The splitting pattern is thus as in the central panel of Fig. 7.4. Such a case can occur for a ${}^2E$ state. The orbital part of this state has no angular momentum, since the corresponding operator is not included in the direct square: $T_1 \notin E \times E$. As a result, the magnetic moment of such a state is due only to the doublet spin part. Such a state behaves as a *pseudo-doublet*.
- Finally, for $3J_p - 9/4J_f = 0$, the splitting again resembles the Zeeman splitting of a regular spherical quartet, but the spin labels have been interchanged, as compared with the standard quartet splitting: the $\pm 3/2$ levels become the inner levels of the split manifold, while the $\pm 1/2$ levels form the outer levels. This *inverted quartet* behavior is observed for ${}^2A_2$ and ${}^2T_2$ states. The orbital part in these states reverses

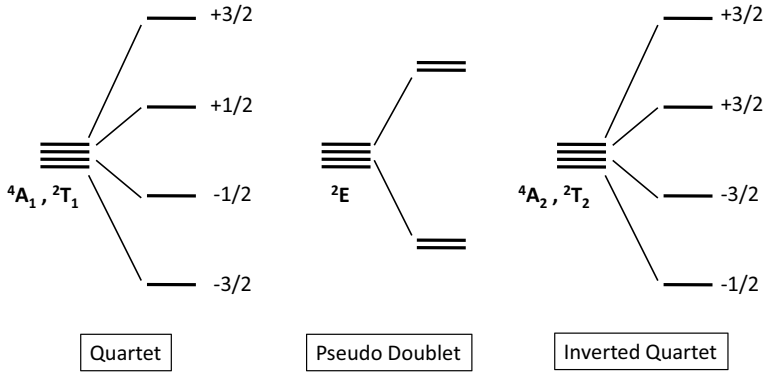


Fig. 7.4 Possible isotropic Zeeman splittings of the octahedral Γ_8 spinor irrep

the assignment of the fictitious spin levels, as can be seen, for instance, from the $A_2 \times \Gamma_8$ coupling table in Appendix G.

To conclude, we present the eigenenergies in the notation of Satten [15], who used parameters g_1 and g_2. The reduced matrix elements are expressed as

$$\begin{aligned} J_p &= \frac{\mu_B}{2\hbar}\,\frac{g_1 + 9g_2}{10} \\ J_f &= \frac{\mu_B}{2\hbar}\,\frac{3(g_1 - g_2)}{10} \end{aligned} \tag{7.81}$$

Furthermore, the magnetic field is represented by directional cosines as $B_z = Bn_z$, $B_x = Bn_x$, $B_y = Bn_y$. The four eigenvalues then become

$$E = \pm\frac{\mu_B}{2\hbar}B\times \left[\frac{1}{2}(g_1^2 + 9g_2^2) \pm \frac{1}{4\sqrt{2}}(g_1 + 3g_2)\Big(9(g_1 - 9g_2)(g_1 - g_2)\mathcal{F} - (g_1^2 - 42g_1g_2 + 9g_2^2)\Big)^{1/2}\right]^{1/2} \tag{7.82}$$

The three isotropic cases are reflected by the zeroes of the three factors preceding the function $\mathcal{F} = (n_x^4 + n_y^4 + n_z^4)$.

7.8 Problems

7.1 Find a relationship between the crystal-field potentials of an octahedron and a cube.

7.2 The product of two rotations is a rotation. Obtain an expression for the Cayley−Klein parameters of the product as a function of the parameters of its

factors. Is the product commutative? The $SU(2)$ matrices may also be identified as normalized *quaternions* [2].

7.3 Work out the group multiplication table for the D_3^* double group, and derive the class structure.

7.4 Consider a set of three eigenlevels transforming as $A_1 + E$ in D_3 symmetry. A general matrix for the interaction between the states can be written as

$\mathcal{H}$	$\|A_1\rangle$	$\|E_x\rangle$	$\|E_y\rangle$
$\langle A_1\|$	-2Δ	a+ib	c+id
$\langle E_x\|$	a-ib	Δ	e-if
$\langle E_y\|$	c-id	e+if	Δ

Introduce a fictitious spin operator $\widetilde{\mathbf{S}}$ which recognizes these states as the components of a triplet spin, $\widetilde{S} = 1$, and consider the spin Hamiltonian:

$$\mathcal{H}_{Ze} = \frac{\mu_B}{\hbar}\left[g_{\parallel} B_z \widetilde{S}_z + g_{\perp}\left(B_x \widetilde{S}_x + B_y \widetilde{S}_y\right)\right] \tag{7.83}$$

Express the a, ..., f parameters as a function of the two g-parameters.

7.5 In octahedral symmetry the fictitious $\widetilde{\mathbf{S}}$ operator follows the T_1 irrep. Its third symmetrized power transforms as the components of the f-harmonics, and also subduces a T_1 irrep, as indicated in Table 7.1. Rewrite the p- and f-parts of the $|T_i|$ operators for the Γ_8 quartet state as a spin Hamiltonian of the fictitious spin.

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Chapter 8
Line Groups and Plane Groups

Abstract This chapter offers an introduction to the group theory of one-dimensional and two-dimensional extended structures. Elementary band structures for glide planes and screw axes are derived. Applications include poly-acetylene and carbon nanotubes.

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8.1 Translational Symmetry Along a Line

Translational symmetry appears when a structure exhibits a long-range order. Ideally, for a 1D crystal, we can think of an infinite molecular chain of equidistant atoms. Let the atoms be arranged along the z-axis. The atomic z-coordinates are given by

$$\text{atom } j \ : \ z_j = j\ a \ (j \in \mathbb{Z}) \tag{8.1}$$

Here, a is the distance between neighboring atoms. The entire chain is divided into elementary cells, containing exactly one atom, and with a length equal to a. These cells are called unit cells. We now define the translational symmetry operator $\hat{T}_a$ in the active sense in line with the standard conventions of the previous chapters as the displacement of every atom over the distance a. An atomic point with coordinate z_j

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is then displaced to the position $z_j + a = z_{j+1}$. For an infinite chain, this operation will map every atom onto an identical neighboring atom, and as a result, the entire chain is translationally invariant. An equivalent way to express this is to state *that the unit cell of a translationally invariant lattice is periodic*: when you leave it to the right you enter it to the left. While the $\hat{T}_a$ operator has characteristics of a generator of a cyclic group, it is not as it cannot generate the unit element, nor can it generate its inverse. To complete the translational symmetry of the linear lattice, we thus have to add the inverse translation $\hat{T}_{-a}$. Nonetheless, cyclic groups prove to provide a useful analogy, as will be shown in a moment.

From here on, we will follow the Koster–Seitz notation for lattice symmetries. A translation operator over an interval a is written as a combined symbol of a point group unit operation, and a translation part over one unit distance:

$$\hat{T}_a = (E|1) \tag{8.2}$$

In this notation, the elements of the translational group, with $t, s \in \mathbb{Z}$, are written as

$$\begin{aligned} \left(\hat{T}_a\right)^t &= (E|t) \\ \left(\hat{T}_a\right)^t \left(\hat{T}_a\right)^s &= (E|t+s) \\ \left(\hat{T}_a\right)^t \left(\hat{T}_a\right)^{-t} &= (E|0) \end{aligned} \tag{8.3}$$

As these results indicate the translation group is isomorphic to the set of the integer numbers under addition, which implies that it is an abelian group. Its irreps are thus one-dimensional, and since the group T is also unitary, its characters are unimodular constants. Hence, with $\kappa \in \mathbb{R}$, we can write

$$\mathbb{D}(E|t) = (\exp i\kappa)^t = \exp i\kappa t = \exp(ikta) \tag{8.4}$$

Here, we have expressed $\kappa = ka$, where the k parameter refers to the momentum $p = \hbar k$ of a free wave. The resulting character table of the translation group **T** is presented in Table 8.1. There is an infinity of irreps, with $k \in \mathbb{R}$. Note that all irreps occur in conjugate pairs, which are related by time-reversal symmetry, except for $k = 0$.

Table 8.1 Character table for the group **T**

T	$(E\|0)$	$(E\|t)$	$(E\|-t)$
Γ_k	1	$\exp(ikta)$	$\exp(-ikta)$
Γ_{-k}	1	$\exp(-ikta)$	$\exp(ikta)$

8.2 Band Structures

Before turning to the actual line groups, a brief introduction to band theory of solids is in order. Consider a chain of hydrogen atoms, where every atom contributes one isotropic s-orbital to the function space. Let these atomic orbitals be denoted as $|\phi_j\rangle$, as in Fig. 8.1. In line with the standard definition, provided in Sect. 4.1, the translation operator displaces the orbitals, thus:

$$\hat{T}_a|\phi_j\rangle = |\phi_{j+1}\rangle \tag{8.5}$$

The atomic orbital basis is now turned into SALC's, using standard projection techniques:

$$\begin{aligned} |\Phi_k\rangle &= \sum_{j\in\mathbb{Z}} \overline{D}^{\Gamma_k}(E|j)|\phi_0\rangle \\ &= \sum_j \exp(-ikz_j)|\phi_j\rangle \end{aligned} \tag{8.6}$$

This function is indeed adapted to translational symmetry, as can be established by acting with the translation operator:

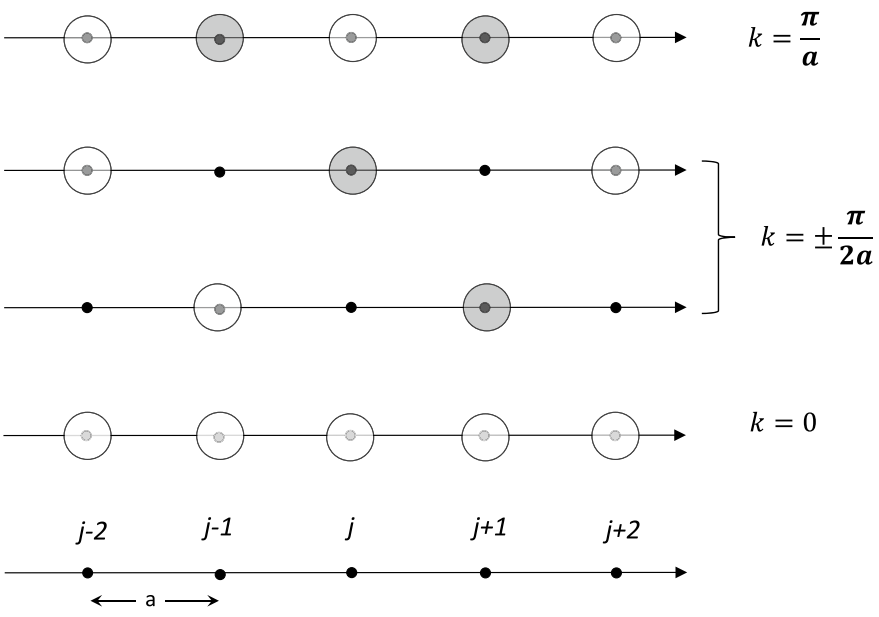


Fig. 8.1 Linear chain of hydrogen atoms: Bloch waves for $k = 0, \pm\pi/2a, \pi/a$

$$\begin{aligned}\hat{T}_a|\Phi_k\rangle &= \sum_j \exp(-ikz_j)\hat{T}_a|\phi_j\rangle \\ &= \sum_j \exp(-ikz_j)|\phi_{j+1}\rangle \\ &= \sum_j \exp\left(-ikz_{j-1}\right)|\phi_j\rangle \\ &= \exp(ika)\sum_j \exp(-ikz_j)|\phi_j\rangle \\ &= \exp(ika)|\Phi_k\rangle \end{aligned} \tag{8.7}$$

Hence, the function $|\Phi_k\rangle$ is an eigenfunction.[1] This exponent thus corresponds to the representational character of the translation operator. Clearly, these functions are projections of the wavefunction of a free particle moving along the z-line, with linear momentum $p = \hbar k$. However, unlike the free particle, the momentum of the particle hopping along a discrete lattice is constrained: replacing the k parameter by $k + {}^{2\pi}/_{a}$ will return the same wavefunction. Hence, the k parameter is periodic. The entire function space is parametrized in an interval with length ${}^{2\pi}/_{a}$. This interval is usually delineated as

$$-\frac{\pi}{a} < k \leq +\frac{\pi}{a} \tag{8.8}$$

This interval is called the first Brillouin zone of k space. Note that the wavefunction consists of an infinite sum which cannot be normalized. A useful tool for practical calculations is to apply the Born–von Karman cyclic boundary conditions. In this approach the chain is divided into long identical segments of N atoms and it is assumed that the wavepattern is repetitive: this means that the coefficient of the $(N + 1)$th atom in the chain is the same as the coefficient of the $j = 1$ atom. It is thus as if the end of the segment is connected to its beginning, as in a cycle. By taking the limit of the cyclic expressions for $N \to \infty$ the full translational symmetry can be attained. The cyclic boundary assumption puts further constraints on the k parameter, as it must be commensurate with the length of the segment, i.e.,:

$$\exp\left(ik(N+1)a\right) = \exp(ika) \tag{8.9}$$

This implies $\exp(ikNa) = 1$. It means that k is no longer a continuous real parameter, but is limited to a set of discrete points in the standard interval, defined as

[1] This wavefunction corresponds to the projection of a running wave with wavevector k and is called a Bloch wave. Note that a negative sign in the exponential coefficients in Eq. (8.5) of the translation operator goes together with a positive sign in the exponential of the translational character in Eq. (8.7), and vice versa. This is a consequence of the complex conjugate form of the transformation matrix in the projection operator. Usually, Bloch waves are written with k instead of $-k$, but then the translational character is $\exp(-ika)$. The same relationship is found for the cyclic polyenes, and also in Eq. 4.18.

$$-\frac{\pi}{a} < \frac{2\pi n}{Na} \leq \frac{\pi}{a} \tag{8.10}$$

The number n is restricted to

$$\begin{aligned} N\text{even} : n &= -\frac{N}{2}+1, ..., \frac{N}{2} \\ N\text{odd} : n &= -\frac{N-1}{2}, ..., \frac{N-1}{2} \end{aligned} \tag{8.11}$$

As a result, the interval is discretized in N points, exactly the number of atoms in the interval. The advantage of the Born–von Karman approach is that the wavefunction for the segment can be normalized:

$$|\Phi_k(N)\rangle = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} \exp(-ikz_j)|\phi_j\rangle \tag{8.12}$$

Its energy can be evaluated as a function of N. In the tight-binding approximation, one has

$$\begin{aligned} \epsilon_k &= \frac{1}{N} \sum_{j=1}^{N} \sum_{j'=1}^{N} \langle \phi_j | H | \phi_{j'} \rangle \\ &= \frac{1}{N} \sum_{j=1}^{N} \sum_{j'=1}^{N} \beta \left(\delta_{j+1,j'} + \delta_{j-1,j'} \right) \exp\left(\frac{2\pi i n (j-j')}{N} \right) \\ &= 2\beta \cos\left(\frac{2\pi n}{N} \right) \end{aligned} \tag{8.13}$$

From here the energy for the infinite lattice can be obtained by taking the limit for $N \to \infty$:

$$\epsilon_k = \lim_{N\to\infty} 2\beta \cos\left(\frac{2\pi n}{N} \right) = 2\beta \cos(ka) \tag{8.14}$$

Energy diagrams for solids are represented in the first Brillouin zone of k-space as shown in Fig. 8.2. As the β parameter is a negative quantity the energy reaches a minimum of $-2|\beta|$ at the origin $k = 0$. The wavefunction at this point is the fully bonding combination where all the orbitals are in phase. The maximum of the energy $2|\beta|$ is reached at the top of the band at $k = \pi/a$. At this point, the wavefunction is fully anti-bonding. The profile that we have obtained is of course identical to what was obtained in Chap. 4 when studying the tight-binding model for cyclic polyenes. The difference is of course that in the cyclic case only a discrete set of eigenvalues was obtained, while, for the infinite lattice, a continuous band is formed. The variation of the energy as a function of k is called the dispersion of the band. We also understand why extension of k beyond the boundaries of the standard interval does not introduce

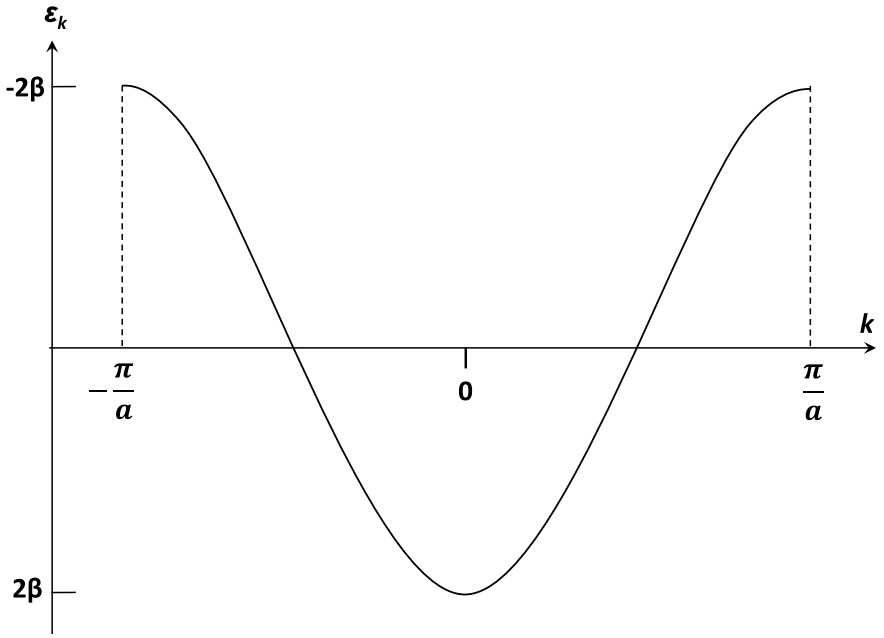


Fig. 8.2 1D crystal: band dispersion in the first Brillouin zone

new eigenvalues. Outside the first Brillouin zone, the phase of the wavefunction changes more rapidly than the period of the lattice. The wavefunction is not able to follow this fast variation as it can only probe the phase at the atomic positions. Hence, if the phase changes twice over the distance between two neighboring atoms, the atomic orbitals will be in phase, as if there was no phase change between them, and we are back in the first Brillouin zone. Finally, note that the dispersion is symmetric in $k = 0$. This reflects of course the time inversion symmetry of the band in the absence of external magnetic fields. The density of states (DOS) represents the distribution of states over the energy scale. Again the cyclic boundary conditions offer a convenient way to calculate this. Let Δn be the number of discrete k-points between k and $k + \Delta k$:

$$\Delta n = \frac{aN}{2\pi}\Delta k \tag{8.15}$$

Likewise, the spread of energy associated with the Δk interval is given by

$$\Delta\epsilon_k = \epsilon_{k+\Delta k} - \epsilon_k \tag{8.16}$$

The DOS is then defined as

$$\rho(\epsilon) = \lim_{N\to\infty} \frac{1}{N}\frac{|\Delta n|}{|\Delta\epsilon|} = \frac{a}{2\pi}\left|\frac{dk}{d\epsilon}\right| = \frac{a}{2\pi}\frac{1}{\left|\frac{d\epsilon}{dk}\right|} \tag{8.17}$$

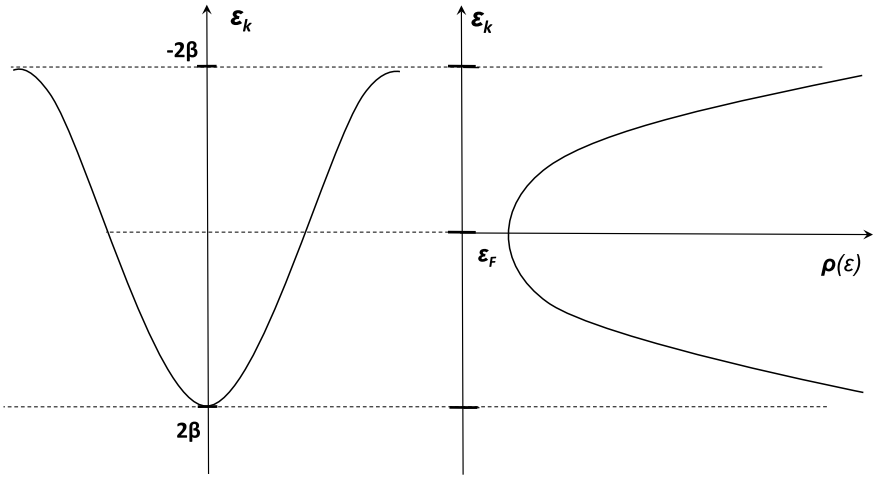


Fig. 8.3 Relation between dispersion, ϵ_k, and Density of States, $\rho(\epsilon)$; ϵ_F is the Fermi energy

The DOS is thus inversely proportional to the derivative of the dispersion function, given in Eq. (8.14). One thus has

$$\begin{aligned} \frac{d\epsilon}{dk} &= -2\beta a \sin(ka) \\ \rho(\epsilon) &= \frac{1}{4\pi|\beta||sin(ka)|} = \frac{1}{2\pi\sqrt{4\beta^2 - \epsilon^2}} \end{aligned} \tag{8.18}$$

A plot of this density is shown in the Fig. 8.3. The density is at a minimum for $k = \pi/2a$, where the slope of the dispersion function is maximal. In contrast, both at the origin and at the top the DOS tends to infinity, as the slope of the dispersion function approaches zero. These points are known as Van Hove singularities. Finally, we introduce the concept of the Fermi level. First note that the integral of the DOS over the entire band is normalized to unity, as can easily be shown:

$$\int_{-\infty}^{+\infty} \rho(\epsilon) d\epsilon = \int_{-\infty}^{+\infty} \frac{1}{N} \frac{dn}{d\epsilon} d\epsilon = \frac{1}{N} N = 1 \tag{8.19}$$

In general, this integral is equal to the number of basis functions per unit cell. The Fermi level is then defined by

$$\int_{-\infty}^{\epsilon_F} \rho(\epsilon) d\epsilon = m/2 \tag{8.20}$$

where m is the number of electrons per unit cell. In the case of the hydrogen chain, with one orbital and one electron per unit cell, m = 1. If we populate the band, starting

at the fully bonding MO at $k = 0$ and accommodate two electrons with opposite spins per level, the Fermi level will be reached exactly halfway between the fully bonding and the fully anti-bonding levels, at $k = \pm\pi/2a$. In the MO picture, the Fermi level thus consists of two non-bonding orbitals, corresponding to a bipartite partition of the lattice, as shown in Fig. 8.1.

8.3 Line Groups

So far, we have treated the chain essentially as a one-dimensional entity. However, the monomers making up the chain are three-dimensional fragments and display point group symmetry characteristics. The line groups, **L**, combine these characteristics with translational symmetry. Elements of **L** are denoted in general as $(R|f + t)$. Here: $t \in \mathbb{Z}$ and $0 < f \leq 1$. These symbols are defined by their action on a position vector **r**:

$$(R|f + t)\mathbf{r} = \hat{R}\mathbf{r} + (f + t)\mathbf{a} \tag{8.21}$$

The elements form a group, with a composition rule and a definition of the inverse:

$$\begin{aligned}(R_2|f_2 + t_2) \times (R_1|f_1 + t_1) &= (R_2R_1|f_2 + t_2 + R_2f_1 + R_2t_1)\\ (R|f + t)^{-1} &= (R^{-1}| - R^{-1}f - R^{-1}t)\end{aligned} \tag{8.22}$$

The $(R|f + t)$ operations cover four different kinds of group elements:

1. *Translations*: $(E|t)$, with $t = 0, \pm1, \pm2, \ldots$
2. *Point group operations*: $(R|0)$ with $\hat{R}$ elements of cylindrical symmetry groups
3. *Glide planes*: $(\sigma_v|1/2)$
4. *Screw axes*: $(C_\alpha|f_\alpha)$.

The pure translation operators commute with all the elements of the group and form a normal subgroup, $\mathbf{T} \in \mathbf{L}$. The point group operations refer to individual symmetry characteristics of the monomers. Since these are embedded in a chain, cubic or icosahedral symmetries are excluded, and the point group operations are limited to the so-called axial or cylindrical symmetry groups, presented in Chap. 3. Glide planes or screw axes however do not refer to symmetries of individual monomers, but describe the arrangement of the equivalent individual monomers in unit cells containing several monomers. An example of a glide plane is shown in Fig. 8.4, with the structure of gutta percha, which is trans-poly-isoprene [2]. The unit cell here contains two isoprene units. Mapping between the monomers is achieved by a translation over half a unit cell followed by reflection in a vertical symmetry plane, as indicated in the figure. The parameter f in the symbol, $(R|f + t)$, indicates a translation over a fraction of the unit cell ($f \leq 1$). Clearly, for a glide plane, this fraction is 1/2, as the square of the glide plane is a pure translation.

The screw axis $(C_\alpha|f_\alpha)$ refers to a helical arrangement of the monomers in a chain. Going from one bead to the next corresponds to a rotation over an angle α

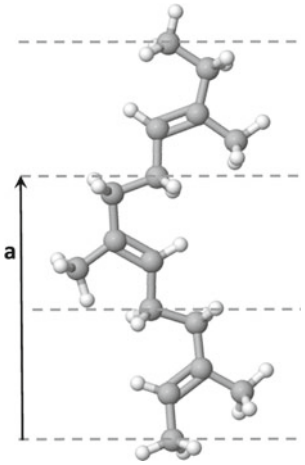


Fig. 8.4 Polymer crystal structure of gutta percha (trans-poly-isoprene); **a** is the length of the unit cell, consisting of two monomers, connected via a vertical glide plane

combined with a translation over a fraction f_α of the unit distance. This is shown in Fig. 8.5 for a sixfold rotation. In this case, rotation over a full circle produces a unit translation. The angle of rotation is here a crucial parameter. If the fraction f_α is an irrational number, the helix and the translational lattice are non-commensurate, and the chain will never attain a perfect translation periodicity, no matter how many full rotations are executed. In non-commensurate crystals, the translational symmetry is, at least formally, broken, and the order is quasi-crystalline. On the other hand, if $\alpha = {}^{2r\pi}/_{q}$, applying the screw axis q times will correspond to r full rotations, and provide perfect translational symmetry. In case the fractional translation is $f_\alpha = 1/q$, a full unit length is reached after applying the screw axis q times. In the example of Fig. 8.5, we illustrate the straightforward case, with α is one sixth of a full angle, and six turns of the helix yield one translational unit. This axis is thus denoted as $(C_6|1/6)$. In the following, we will only be concerned with commensurate line groups. As the translation group is a subgroup of the line group, we may expand the line group in cosets, the generators of which are all remaining group elements:

$$\mathbf{L} = \sum_{i=1}^{|\mathbf{P}|} (R_i|f_i)\mathbf{T} \tag{8.23}$$

Here the sum runs over all point group symmetries, glide planes, and screw axes, that are present in **L**. Moreover, one has

$$(R|f)(E|t) = (R|f + Rt) = (E|Rt)(R|f) \tag{8.24}$$

Note that this relation is not a proper commutation, as the symmetry elements $\hat{R}$ may invert the direction of translation. However, the equation does prove that left and right cosets of **T** are identical. The translation group is thus an invariant or normal

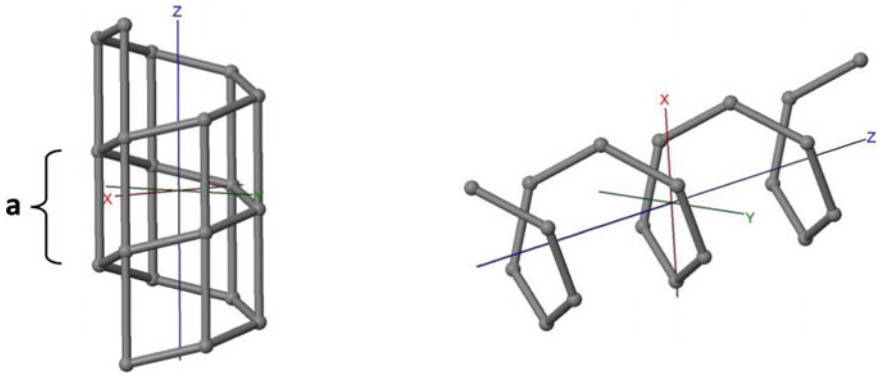


Fig. 8.5 Axial and lateral views of a helical polymer chain with a right-handed sixfold screw axis; the z-direction corresponds to the translation direction, the x-direction coincides with a twofold rotation axis, perpendicular to the helical axis

subgroup. The set of the cosets of a normal subgroup has the group property, as we already explored in Problem 3.3. This group of cosets is denoted as the factor group, **P**, with dimension |**P**|. Please bear in mind that an element of this factor group is not a coset generator, but a whole coset, as explained in the box.

Factor group
A normal or invariant subgroup N of parent group G remains unchanged under similarity transformations:

$$\forall g \in G \;\&\; N \subset G : \; gNg^{-1} = N$$

A normal subgroup is thus mapped onto itself by the elements of the group. The definition implies that a normal subgroup can only contain entire classes. An example is $D_{2h} \subset O_h$ in Table 4.4, which contains all three elements of the $3C_2$ and $3\sigma_h$ classes. Now consider the coset expansion of G over N with coset generators g_i:

$$G = \sum_{i=1}^{|G|/|N|} g_i N$$

The cosets themselves, $\{g_i N\}_i$, have the group property:

1. Unit element: $\hat{E}\, N$
2. Closure:

$$g_i N \, g_j N = g_i g_j N g_j^{-1} g_j N = g_i g_j N N = g_i g_j N = g_k N$$

3. Associative since G is associative

4. inverse element:

$$(g_i N)^{-1} = N g_i^{-1} = g_i^{-1} g_i N g_i^{-1} = g_i^{-1} N$$

Note that this result is only valid for normal subgroups. The resulting group of cosets is the factor group or quotient group G/N. It may formally be considered as the result of a division of G by one of its 'factors' N. In the present application, the parent group is the line group **L**, the normal subgroup is the group of pure translations, **T**, and the factor group is called the isogonal point group, **P**, thus:

$$\mathbf{P} \sim \frac{\mathbf{L}}{\mathbf{T}}$$

When a line group does not contain glide planes or screw axes, the coset generators all belong to the point group of the monomer. This implies that, in this case, the isogonal point group is identical to the point group of the monomer. These line groups are called *symmorphic*. Line groups have been categorized into 11 families, based on possible symmetries of the isogonal point group. For a detailed overview of these families, we refer to the textbook [1]. Instead, here we will analyze in detail two case studies, one based on a glide plane, and one on a screw axis [4]. This will provide practical experience to deal with these essential translational elements of the line groups, and their irreps.

8.4 Applications

8.4.1 Line Group with a Glide Plane

Group Structure

In Fig. 8.6, we present the symmetry points of a line which exhibits a glide plane, $(\sigma_v|1/2)$. The unit cell contains two monomers, A and B, which are mapped onto each other by translation over one half of the unit cell, combined with a reflection in a σ_v-plane. In addition, we have assumed that a twofold symmetry axis runs through each monomer. This axis is perpendicular to the z-direction and reverses the sense of the translations. This symmetry element is denoted as $(U|0)$. Here the symbol U, from the German 'Umklapp' which signifies to fold over, is used commonly for the perpendicular C_2-axes, as opposed to an ordinary C_2-axis which is aligned with the z-axis. The glide plane is perpendicular to the U-elements. The unit cell contains not one but two U operations, one through A and one through B. However, these belong

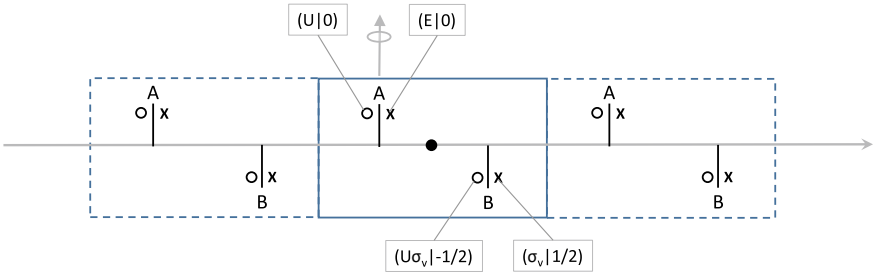


Fig. 8.6 Symmetry points in the unit cell with $\mathbf{L}\bar{1}c$ symmetry; the black dot represents an inversion center

Table 8.2 Multiplication table for the isogonal point group of $\mathbf{L}\bar{1}c$

$\mathbf{L}\bar{1}c$	$(E\|0)$	$(U\|0)$	$(\sigma_v\|1/2)$	$(U\sigma_v\|1/2)$
$(E\|0$	$(E\|0)$	$(U\|0)$	$(\sigma_v\|1/2)$	$(U\sigma_v\|1/2)$
$(U\|0)$	$(U\|0)$	$(E\|0)$	$(U\sigma_v\|1/2)$	$(\sigma_v\|1/2)$
$(\sigma_v\|1/2)$	$(\sigma_v\|1/2)$	$(U\sigma_v\|1/2)$	$(E\|0)$	$(U\|0)$
$(U\sigma_v\|1/2)$	$(U\sigma_v\|1/2)$	$(\sigma_v\|1/2)$	$(U\|0)$	$(E\|0)$

to the same coset (see Problem 8.2). Following the notation of Vujicic et al. [5], a line group with a glide plane is denoted as $\mathbf{L}1c$. Addition of the U element leads to line group $\mathbf{L}\bar{1}c$, with twice as many elements:

$$\mathbf{L}\bar{1}c = \mathbf{L}1c + (U|0)\mathbf{L}1c \tag{8.25}$$

The isogonal point group is obtained by coset expansion over the group of pure translations:

$$\mathbf{P} = (E|0)\mathbf{T} + (\sigma_v|1/2)\mathbf{T} + (U|0)\mathbf{T} + (U\sigma_v|1/2)\mathbf{T} \tag{8.26}$$

The isogonal point group thus contains four cosets. Its multiplication table is presented in Table 8.2. The derivation of the products in this table differs from our experience with the point groups and requires to take into account that these elements stand for entire cosets. This requires to work out the products, modulo an element of the group $\mathbf{T}$. A few examples of the calculus with $(R|f + t)$ operations are given below:

$$
\begin{aligned}
(U|0)(\sigma_v|1/2) &= (U\sigma_v|-1/2) \\
(U|0)(\sigma_v|1/2)(U|0) &= (\sigma_v|1/2)^{-1} = (\sigma_v|-1/2) \\
&= (\sigma_v|1/2)(E|-1) \in (\sigma_v|1/2)\mathbf{T} \\
(\sigma_v U|1/2) &= (\sigma_v|1/2)(U|0) = (U|0)(\sigma_v|-1/2)(U|0)(U|0) \\
&= (U\sigma_v|-1/2) = (U\sigma_v|1/2)(E|-1) \in (U\sigma_v|1/2)\mathbf{T} \\
(U\sigma_v|1/2)(U\sigma_v|1/2) &= (U|0)(\sigma_v|-1/2)(U|0)(\sigma_v|-1/2) \\
&= (\sigma_v|1/2)(\sigma_v|-1/2) = (E|0) \\
(U|0)(E|t)(U|0) &= (E|-t)
\end{aligned}
\tag{8.27}
$$

As these equations show a similarity transformation under the Umklapp operation changes the sign of the translational part. Translations thus do not commute with all operations of the isogonal point group. However, as a subgroup, $\mathbf{T}$ commutes. The isogonal point group is an Abelian group with two generators. It is isomorphic to the C_{2h} point group, with the following correspondence:

$$
\begin{aligned}
(U|0) &\leftrightarrow \hat{C}_2 \\
(\sigma_v|1/2) &\leftrightarrow \sigma_h \\
(U\sigma_v|1/2) &\leftrightarrow \hat{i} \sim \hat{S}_2
\end{aligned}
\tag{8.28}
$$

Vujicic et al. [3] denote this isogonal point group as D_{1d}, rather than C_{2h}, the reason being that the C_{2h} symbol usually refers to a C_2-axis along the z-direction, with a horizontal σ_{xy}-plane.

Band Theory

We develop here a simplified band structure model, which can be applied to explain the electronic structure in polyacetylene. Polyacetylene forms long chains of sp^2 carbon atoms, as shown in Fig. 8.7. Two bond isomers are conceivable, which—in analogy to the Kekule forms in benzene—might form a delocalized state, that would make polyacetylene an organic conductor. Unit cells of the delocalized structure contain two carbon atoms and exhibit a zigzag structure which points to the presence of a glide plane. Note that this glide plane also connects the two bond isomers, making it an important symmetry to understand the conducting properties of polyacetylene. In polyacetylene further binary symmetry elements, including horizontal symmetry planes, are present as well, but the treatment can conveniently be limited to our $\mathbf{L}\bar{1}c$ group, which covers all the essential features. Each atom has a $2p_\perp$ orbital, carrying one electron. These orbitals form the π-type valence shell of the polymer. The dual orbital basis is denoted as $\{|\phi_j\rangle, |\chi_j\rangle\}$, connected by the glide plane:

$$
\begin{aligned}
(\sigma_v|1/2)|\phi_j\rangle &= |\chi_j\rangle \\
(\sigma_v|1/2)|\chi_j\rangle &= |\psi_{j+1}\rangle
\end{aligned}
\tag{8.29}
$$

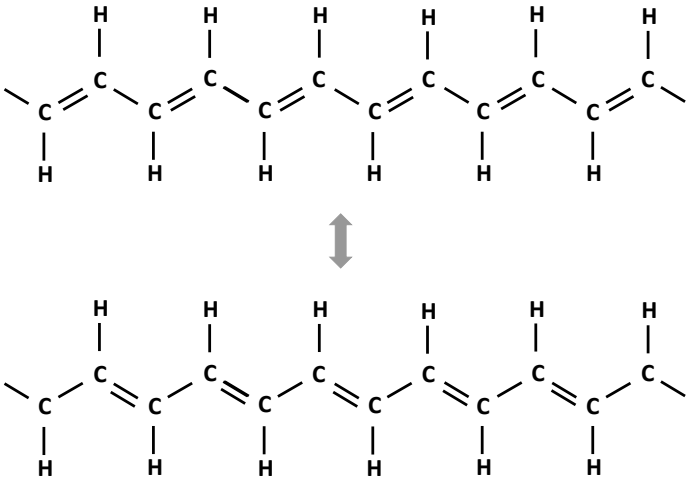


Fig. 8.7 Kekule structures for polyacetylene

where the ϕ and χ orbitals are localized on the A and B sites respectively. Translational symmetry is then introduced by transforming the orbital basis into Bloch waves[2]:

$$|\Phi_k\rangle = \frac{1}{\sqrt{N}} \sum_j e^{ikja} |\phi_j\rangle$$
$$|X_k\rangle = \frac{e^{\frac{ika}{2}}}{\sqrt{N}} \sum_j e^{ikja} |\chi_j\rangle \tag{8.30}$$

As before, the range of the first Brillouin zone is defined by Eq. (8.8). Besides of the formal normalization factor, an additional overall phase factor is introduced for the $|X_k\rangle$ Bloch wave. This factor anticipates the translational character associated with the glide plane. Alternatively, one could leave out this extra feature, and see it appearing when working out the eigenfunctions of the tight-binding Hamiltonian. The translational symmetry of the Bloch waves is described by the wavevector k. Only waves with the same k value can interact. Hence, the tight-binding Hamiltonian reduces to 2×2 interaction matrices, of the following type:

$$\begin{pmatrix} 0 & 2\beta \cos \frac{ka}{2} \\ 2\beta \cos \frac{ka}{2} & 0 \end{pmatrix} \tag{8.31}$$

[2] We follow here the traditional expressions for Bloch waves, keeping in mind that this definition implies that they transform as Γ_{-k}.

The diagonal elements of type $\langle\psi_j|\mathcal{H}|\psi_{j\pm1}\rangle$ vanish since the atoms are not nearest neighbors. The off-diagonal elements are evaluated as follows:

$$\begin{aligned}\langle\Phi_k|\mathcal{H}|X_k\rangle &= \frac{1}{N}\sum_{j,j'} e^{\frac{ika}{2}} e^{ika(j'-j)}\langle\phi_j|\mathcal{H}|\chi_{j'}\rangle \\ &= \frac{1}{N}\sum_{j\,j'} \beta e^{\frac{ika}{2}}\left(\delta_{j',j}+\delta_{j',j-1}\right) e^{ika(j'-j)} \\ &= \frac{1}{N}\sum_{j} \beta\left(e^{\frac{ika}{2}}+e^{\frac{-ika}{2}}\right) \\ &= 2\beta\cos\frac{ka}{2} \end{aligned} \tag{8.32}$$

As this equation demonstrates the nearest neighbors of a given ϕ orbital are the χ orbitals in the same unit cell and in the preceding unit cell. Diagonalization of the matrix gives rise to two bands, a bonding one $|\Psi^I\rangle$, and an anti-bonding one $|\Psi^{II}\rangle$:

$$\begin{aligned} E^I = 2\beta\cos\frac{ka}{2} \;&\rightarrow\; |\Psi^I\rangle = \frac{1}{\sqrt{2}}\left(|\Phi_k\rangle + |X_k\rangle\right) \\ E^{II} = -2\beta\cos\frac{ka}{2} \;&\rightarrow\; |\Psi^{II}\rangle = \frac{1}{\sqrt{2}}\left(|\Phi_k\rangle - |X_k\rangle\right) \end{aligned} \tag{8.33}$$

These bands meet at $E = 0$, which corresponds to the Fermi level, as shown in the Fig. 8.8. Comparison of Figs. 8.2 and 8.8 reveals several interesting observations. As the unit cell now contains two atoms, the reciprocal lattice unit vector shrinks, or in other words, the dispersion of the first energy band in the kinked chain is only half that for the linear chain. As a result the Fermi level is reached only at the zone boundary, where before it was already reached for $k = \pi/2a$. The overall result is that the band structure of the linear chain is folded back when reaching the Fermi level. Since the dimension of the Brillouin zone is the reciprocal of the dimension of the unit cell, a larger unit cell will lead to a more compressed Brillouin zone, but which is compensated by the appearance of additional bands, which account for the increased orbital basis of the larger unit cell. As the two bands meet at the Fermi point, the question arises whether the Fermi point exhibits orbital degeneracy or not. It is here where the glide plane symmetry is a reliable tool to analyze what is going on.

Irreducible Representations

To find out the irreducible representations of the line group $\mathbf{L}1c$ one must act with the glide plane on the wavefunctions. The $2p_\perp$ orbital basis is entirely anti-symmetric with respect to σ_v. In order not to be distracted by this intrinsic anti-symmetry, we simplify the problem to a zigzag chain of hydrogen atoms and apply the tight-binding approximation. One has for such nodeless basis functions:

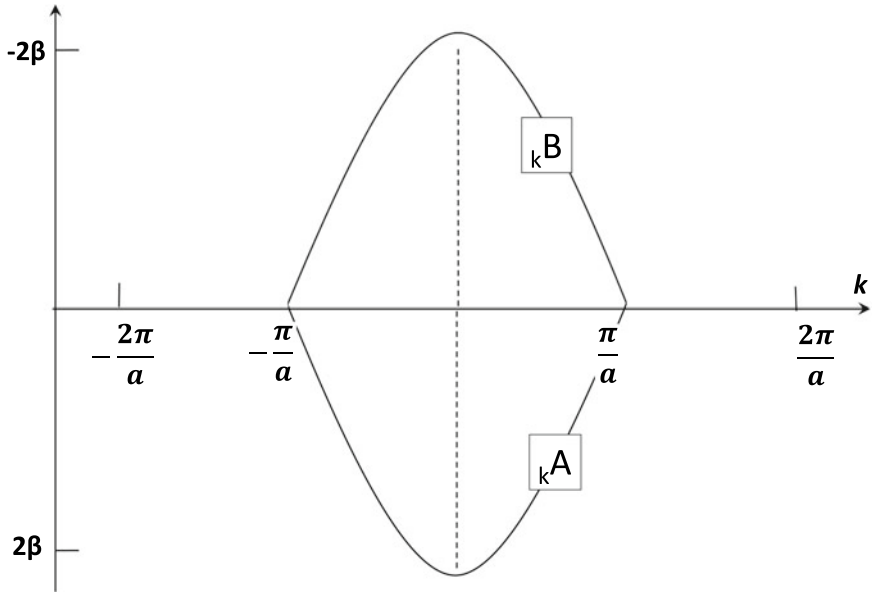


Fig. 8.8 Band structure of valence shell in trans-polyacetylene; the Fermi level is at $k = \pi/a$

$$
\begin{aligned}
(\sigma_v|1/2)|\Phi_k\rangle &= \frac{1}{\sqrt{N}} \sum_j e^{ikja} |\chi_j\rangle \\
&= e^{-\frac{ika}{2}} |X_k\rangle \\
(\sigma_v|1/2)|X_k\rangle &= \frac{e^{\frac{ika}{2}}}{\sqrt{N}} \sum_j e^{ikja} |\phi_{j+1}\rangle \\
&= e^{-\frac{ika}{2}} |\Phi_k\rangle
\end{aligned}
\tag{8.34}
$$

Combining this result with the waveforms in Eq. (8.33) then yields

$$
\begin{aligned}
(\sigma_v|1/2)|\Psi^I\rangle &= e^{-\frac{ika}{2}} |\Psi^I\rangle \\
(\sigma_v|1/2)|\Psi^{II}\rangle &= -e^{-\frac{ika}{2}} |\Psi^{II}\rangle
\end{aligned}
\tag{8.35}
$$

Both eigenstates are thus eigenfunctions of the glide plane. The corresponding character contains on the one hand the translational factor associated with translation over half the unit cell, and a plus or minus sign indicating that the wavefunction is symmetric or anti-symmetric with respect to the reflection itself. The plus or minus sign refers to the irreducible representations of the isogonal point group, in case the group C_s. The resulting character table is shown in Table 8.3. The symbols A and B are used for symmetric and anti-symmetric representations, and they receive a subscript k which denotes the symmetry of the Bloch waves under the elements of

Table 8.3 Character table for the group $\mathbf{L}1c$

$\mathbf{L}1c$	$(E\|0)$	$(E\|t)$	$(\sigma_v\|1/2)$
${}_kA$	1	$\exp(ikta)$	$\exp(\frac{ika}{2})$
${}_kB$	1	$\exp(ikta)$	$-\exp(\frac{ika}{2})$

the translational group $\mathbf{T}$. At the Fermi level, lower and upper bands are, respectively, symmetric and anti-symmetric with respect to the glide plane. As these states are degenerate at the Fermi level, distortion of the lattice may exhibit a first-order coupling to the electronic manifold, not unlike the Jahn–Teller effect in point groups. This phenomenon is known as Peierls distortion. Symmetry dictates that the interaction is governed by the direct product of the interacting bands:

$$ {}_kA \times {}_kB = {}_kB \tag{8.36}$$

Vibrations of the lattice, which are called phonons in solid state theory, need thus be anti-symmetric with respect to the glide plane. These motions will break the conjugation and create an alternating pattern of single and double bonds, reminiscent of the Kekule forms of benzene. As a result, a gap will open between the two bands, and—depending on the size of this gap—create an isolator.

Adding the Umklapp symmetry to the Line group will not greatly influence the irreducible characteristics. The effect of the $(U|0)$ operator is to invert the direction of the translations. This will correspond to taking the complex conjugate of the Bloch waves. As a result, symmetry adaptation of the basis vectors to the Umklapp operation involves the transformation of the translational waves into standing waves, corresponding to $\cos(kja)$, $\sin(kja)$. This will not affect the character under the glide plane as the translational waves with opposite k vector are equisymmetric in the isogonal point group.

8.4.2 Line Group with a Screw Axis

For the application of a screw axis, we consider the simple case of a threefold right-handed screw axis, $(C_3|1/3)$, as illustrated in the Fig. 8.9. The Line group is denoted as $\mathbf{L}3_1$. The coset decomposition reads

$$ \mathbf{L}3_1 = (E|0)\mathbf{T} + (C_3|1/3)\mathbf{T} + (C_3^2|2/3)\mathbf{T} \tag{8.37}$$

The isogonal point group contains three cosets, generated by the screw axis, and is isomorphic to the rotational group C_3. Addition of the Umklapp expands the line group $\mathbf{L}n_1$ to a group $\mathbf{L}n_12$, which doubles the number of cosets. The isogonal point group is then a group of six elements, which is isomorphic to D_3. It is represented in Fig. 8.9. The group multiplication table is displayed in Table 8.4. Below some of these products are worked out in detail to illustrate how the calculus works.

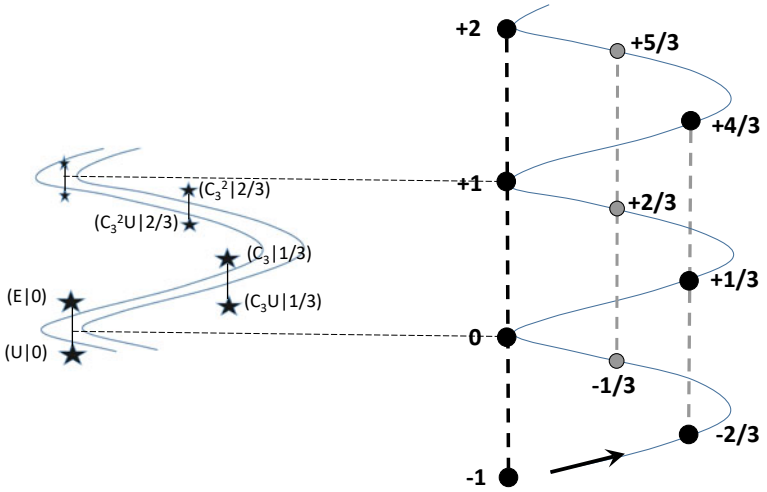


Fig. 8.9 Symmetry points in the unit cell of a right-handed helix with a threefold screw axis, and twofold $(U|0)$-axis through atom 0; the unit cell between atoms 0 and 1 contains six symmetry points as labeled in the left panel: these cover the six cosets of D_3

Table 8.4 Multiplication table for the isogonal point group of $\mathbf{L}3_1$

$\mathbf{L}3_1$	$(E\|0)$	$(C_3\|\frac{1}{3})$	$(C_3^2\|\frac{2}{3})$	$(U\|0)$	$(C_3U\|\frac{1}{3})$	$(C_3^2U\|\frac{2}{3})$
$(E\|0$	$(E\|0)$	$(C_3\|\frac{1}{3})$	$(C_3^2\|\frac{2}{3})$	$(U\|0)$	$(C_3U\|\frac{1}{3})$	$(C_3^2U\|\frac{2}{3})$
$(C_3\|\frac{1}{3})$	$(C_3\|\frac{1}{3})$	$(C_3^2\|\frac{2}{3})$	$(E\|0)$	$(C_3U\|\frac{1}{3})$	$(C_3^2U\|\frac{2}{3})$	$(U\|0)$
$(C_3^2\|\frac{2}{3})$	$(C_3^2\|\frac{2}{3})$	$(E\|0)$	$(C_3\|\frac{1}{3})$	$(C_3^2U\|\frac{2}{3})$	$(U\|0)$	$(C_3U\|\frac{1}{3})$
$(U\|0)$	$(U\|0)$	$(C_3^2U\|\frac{2}{3})$	$(C_3U\|\frac{1}{3})$	$(E\|0)$	$(C_3^2\|\frac{2}{3})$	$(C_3\|\frac{1}{3})$
$(C_3U\|\frac{1}{3})$	$(C_3U\|\frac{1}{3})$	$(U\|0)$	$(C_3^2U\|\frac{2}{3})$	$(C_3\|\frac{1}{3})$	$(E\|0)$	$(C_3^2\|\frac{2}{3})$
$(C_3^2U\|\frac{2}{3})$	$(C_3^2U\|\frac{2}{3})$	$(C_3U\|\frac{1}{3})$	$(U\|0)$	$(C_3^2\|\frac{2}{3})$	$(C_3\|\frac{1}{3})$	$(E\|0)$

$$
\begin{aligned}
(C_3|\frac{1}{3})(U|0) &= (C_3U|\frac{1}{3}) \\
(U|0)(C_3|\frac{1}{3})(U|0) &= ((C_3)^{-1}|-\frac{1}{3}) = (C_3^2|\frac{2}{3})(E|-1) \\
(U|0)(C_3|\frac{1}{3}) &= (U|0)(C_3|\frac{1}{3})(U|0)(U|0) = (C_3^{-1}|-\frac{1}{3})(U|0) \\
&= (C_3^2|\frac{2}{3})(E|-1)(U|0) = (C_3^2|\frac{2}{3})(U|0)(E|+1) \in (C_3^2U|\frac{2}{3}) \\
(C_3U|\frac{1}{3})(C_3U|\frac{1}{3}) &= (C_3|\frac{1}{3})(C_3^{-1}|-\frac{1}{3}) = (E|0)
\end{aligned}
\tag{8.38}
$$

Band Theory

Consider an elementary orbital basis set consisting of three $1s$ orbitals per helical unit cell. Following the same procedure as for the glide plane, we start by constructing Bloch waves for the three consecutive basis orbitals in the unit cell, introducing an appropriate phase factor for fractional translation along the screw axis:

$$\begin{aligned} |\Phi_k\rangle &= \frac{1}{\sqrt{N}} \sum_j e^{ikja} |\phi_j\rangle \\ |X_k\rangle &= \frac{e^{\frac{ika}{3}}}{\sqrt{N}} \sum_j e^{ikja} |\chi_j\rangle \\ |\Psi_k\rangle &= \frac{e^{\frac{2ika}{3}}}{\sqrt{N}} \sum_j e^{ikja} |\psi_j\rangle \end{aligned} \tag{8.39}$$

All these states have the same translational symmetry, expressed by the k momentum. For a given k, there are thus only three states that can interact, giving rise to a 3×3 Hamiltonian. In the tight-binding approximation, with the orbitals ordered as (Φ, X, Ψ), this reads

$$\begin{pmatrix} 0 & \beta e^{\frac{ika}{3}} & \beta e^{-\frac{ika}{3}} \\ \beta e^{-\frac{ika}{3}} & 0 & \beta e^{\frac{ika}{3}} \\ \beta e^{\frac{ika}{3}} & \beta e^{-\frac{ika}{3}} & 0 \end{pmatrix} \tag{8.40}$$

As an example we work out the matrix element between Φ and Ψ:

$$\begin{aligned} \langle \Phi_k | \mathcal{H} | \Psi_k \rangle &= \frac{e^{\frac{2ika}{3}}}{N} \sum_{j,j'} e^{ika(j'-j)} \langle \phi_j | \mathcal{H} | \psi_{j'} \rangle \\ &= \frac{e^{\frac{2ika}{3}}}{N} \sum_{j\, j'} \beta \delta_{j',j-1} e^{ika(j'-j)} \\ &= \beta e^{-\frac{ika}{3}} \end{aligned} \tag{8.41}$$

The secular equation reads

$$E^3 - 3\beta^2 E - 2\beta^3 \cos ka \tag{8.42}$$

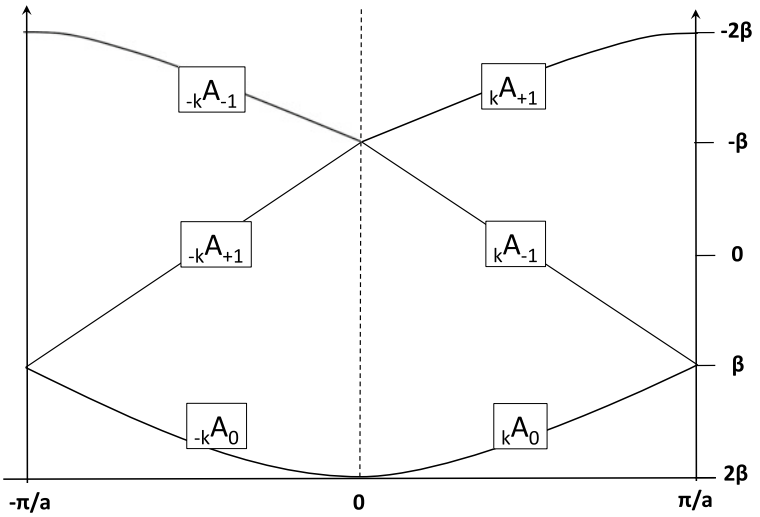


Fig. 8.10 Band structure of helix with threefold screw axis

Its solution generates three bands:

$$
\begin{aligned}
E^I &= 2\beta\cos\frac{ka}{3} \rightarrow |\Omega_k^I\rangle = \frac{1}{\sqrt{3}}\left(|\Phi_k\rangle + |X_k\rangle + |\Psi_k\rangle\right) \\
E^{II} &= 2\beta\cos(\frac{ka}{3} - \frac{2\pi}{3}) \rightarrow |\Omega_k^{II}\rangle = \frac{1}{\sqrt{3}}\left(|\Phi_k\rangle + \epsilon^*|X_k\rangle + \epsilon|\Psi_k\rangle\right) \\
E^{III} &= -2\beta\cos(\frac{ka}{3} + \frac{2\pi}{3}) \rightarrow |\Omega_k^{III}\rangle = \frac{1}{\sqrt{3}}\left(|\Phi_k\rangle + \epsilon|X_k\rangle + \epsilon^*|\Psi_k\rangle\right)
\end{aligned}
\tag{8.43}
$$

with $\epsilon = \exp(2\pi i/3)$. The resulting energy profile is shown schematically in the Fig. 8.10. In the present case, the dispersion depends on $k/3$. The profile may thus be understood as a triple fold of the original Brillouin zone in Fig. 8.2. In case all atoms contribute one electron, the Fermi level will be found halfway the middle band at $k = \pi/2a$.

Irreducible Representations

Knowing the eigenfunctions, the irreducible symmetry characteristics can easily be determined. For the $\mathbf{L}3_1$ helical symmetry group, the isogonal point group is isomorphic to the rotational group C_3, which is Abelian and has three irreps. These are readily associated with the three bands that have been identified. The action of the screw axis on the Bloch waves is given by

$$
\begin{aligned}
(C_3|\frac{1}{3})|\Phi_k\rangle &= \frac{1}{\sqrt{N}}\sum_j e^{ikja}|\chi_j\rangle = e^{-ika/3}|X_k\rangle \\
(C_3|\frac{1}{3})|X_k\rangle &= \frac{e^{\frac{ika}{3}}}{\sqrt{N}}\sum_j e^{ikja}|\psi_j\rangle = e^{-ika/3}|\Psi_k\rangle \\
(C_3|\frac{1}{3})|\Psi_k\rangle &= \frac{e^{\frac{2ika}{3}}}{\sqrt{N}}\sum_j e^{ikja}|\phi_{j+1}\rangle = e^{-ika/3}|\Phi_k\rangle
\end{aligned}
\tag{8.44}
$$

Armed with these expressions, one can now identify the symmetries of the eigenstates.

$$
\begin{aligned}
(C_3|\frac{1}{3})|\Omega_k^I\rangle &= e^{-ika/3}|\Omega_k^I\rangle \\
(C_3|\frac{1}{3})|\Omega_k^{II}\rangle &= \frac{e^{-\frac{ika}{3}}}{\sqrt{3}}\Big(|X_k\rangle + \epsilon^*|\Psi_k\rangle + \epsilon|\Phi_k\rangle\Big) = \epsilon\, e^{-\frac{ika}{3}}|\Omega_k^{II}\rangle \\
(C_3|\frac{1}{3})|\Omega_k^{III}\rangle &= \frac{e^{-\frac{ika}{3}}}{\sqrt{3}}\Big(|X_k\rangle + \epsilon|\Psi_k\rangle + \epsilon^*|\Phi_k\rangle\Big) = \epsilon^*\, e^{-\frac{ika}{3}}|\Omega_k^{III}\rangle
\end{aligned}
\tag{8.45}
$$

The symmetry character now consists of two parts: the characteristic helical phase factor, $\exp(-ika/3)$, and the rotational character, 1, ϵ, ϵ^*, of the trigonal point group. The corresponding character table is produced in Table 8.5. Note that the two irreps ${}_kA_1$ and ${}_kA_{-1}$ do not form conjugate pairs. Indeed conjugation of the point group character, $\epsilon \to \epsilon^*$, is not accompanied by a simultaneous phase change of the k wavevector. Hence, these states are not related by time reversal, and thus they do not form degenerate pairs. The pairs that are degenerate for time-even interactions are the pairs $[{}_kA_m] \leftrightarrow \left[{}_{-k}A_{-m}\right]$. In the k-interval $[0, \pi/a]$, we thus assign the three bands as shown in the figure.

Now consider the effect of the Umklapp operator. By inverting the direction of translation, this operator will connect states that are conjugate under time reversal, exactly as in the case of the glide plane. Special degeneracy points occur at the zone reflections at $k = 0$ and $k = \pi/a$. We will study these in detail, making use of the explicit wavefunctions. To start with, consider a twofold axis through the $|\phi_0\rangle$ orbital

Table 8.5 Character table for the group $\mathbf{L}3_1$

$\mathbf{L}c$	$(E\|0)$	$(E\|t)$	$(C_3\|1/3)$	$(C_3^2\|2/3)$
${}_kA_0$	1	$\exp(ikta)$	$\exp(\frac{ika}{3})$	$\exp(\frac{2ika}{3})$
${}_kA_1$	1	$\exp(ikta)$	$\epsilon\exp(\frac{ika}{3})$	$\epsilon^*\exp(\frac{2ika}{3})$
${}_kA_{-1}$	1	$\exp(ikta)$	$\epsilon^*\exp(\frac{ika}{3})$	$\epsilon\exp(\frac{2ika}{3})$

in the origin of the unit cell, with $j = 0$. The action of the operator on the orbitals of ϕ type is thus simply to invert j. In contrast the orbital $|\chi_0\rangle$, one third of a helical turn forward from $|\phi_0\rangle$ will be rotated backwards to the orbital $|\psi_{-1}\rangle$, of the unit cell with $j = -1$, or in general:

$$\begin{aligned}
(U|0)|\phi_j\rangle &= |\phi_{-j}\rangle \\
(U|0)|\chi_j\rangle &= (C_3^{-1}|-\frac{1}{3})(U|0)|\phi_j\rangle = (C_3^{-1}|-\frac{1}{3})|\phi_{-j}\rangle = |\psi_{-(j+1)}\rangle \\
(U|0)|\psi_j\rangle &= (C_3^{-2}|-\frac{2}{3})(U|0)|\phi_j\rangle = (C_3^{-2}|-\frac{2}{3})|\phi_{-j}\rangle = |\chi_{-(j+1)}\rangle
\end{aligned} \tag{8.46}$$

These orbital symmetries will affect the Bloch waves. Focusing on the waves at $k = \pi/a$, with

$$\begin{aligned}
|\Phi_{\pi/a}\rangle &= \frac{1}{\sqrt{N}}\sum_j (-1)^j |\phi_j\rangle \\
|X_{\pi/a}\rangle &= \frac{e^{\frac{\pi i}{3}}}{\sqrt{N}}\sum_j (-1)^j |\chi_j\rangle \\
|\Psi_{\pi/a}\rangle &= \frac{e^{\frac{2\pi i}{3}}}{\sqrt{N}}\sum_j (-1)^j |\psi_j\rangle
\end{aligned} \tag{8.47}$$

Combining these waves with the symmetry operations in Eq. (8.46), one obtains

$$\begin{aligned}
(U|0)|\Phi_{\pi/a}\rangle &= |\Phi_{\pi/a}\rangle \\
(U|0)|X_{\pi/a}\rangle &= \frac{-e^{\frac{\pi i}{3}}}{\sqrt{N}}\sum_j (-1)^j |\psi_j\rangle = e^{\frac{2\pi i}{3}}|\Psi_{\pi/a}\rangle \\
(U|0)|\Psi_{\pi/a}\rangle &= \frac{-e^{\frac{2\pi i}{3}}}{\sqrt{N}}\sum_j (-1)^j |\chi_j\rangle = e^{-\frac{2\pi i}{3}}|X_{\pi/a}\rangle
\end{aligned} \tag{8.48}$$

With these expressions, we can now evaluate the effect of the Umklapp operation on the first and second eigenstates. They become degenerate with eigenvalue $E = \beta$ at the zone boundary.

$$\begin{aligned}
(U|0)|\Omega^I_{\pi/a}\rangle &= \frac{1}{\sqrt{3}}\Big[|\Phi_{\pi/a}\rangle + \epsilon|\Psi_{\pi/a}\rangle + \epsilon^*|X_{\pi/a}\rangle\Big] = |\Omega^{II}_{\pi/a}\rangle \\
(U|0)|\Omega^{II}_{\pi/a}\rangle &= \frac{1}{\sqrt{3}}\Big[|\Phi_{\pi/a}\rangle + |\Psi_{\pi/a}\rangle + |X_{\pi/a}\rangle\Big] = |\Omega^I_{\pi/a}\rangle
\end{aligned} \tag{8.49}$$

The Umklapp operation thus simply interchanges the two states at the degeneracy point, indicating that this degeneracy is caused by the action of the isogonal group. Accordingly, this state is indicated in the diagram as an E state, as in the isomorphic

D_3 point group. Note that this derivation was based on the use of a perpendicular twofold axis through the first atom of unit cell $j = 0$ of the helix. This choice has yielded a symmetry matrix which adopts a simple permutational form. Other choices would of course produce unitary transformations of this matrix, but not affect the symmetry character, as all these C_2-axes are conjugate through similarity transformations.

In summary, we especially highlight the importance of the isogonal point group, which provides the link between the line and point groups:

- The isogonal point group forms the core of the line group, and connects it to the axial point groups.
- In the absence of glide planes or screw axes, the isogonal point group is isomorphous to the point group of the monomer. This is clear because, in that case, the only coset generators are the point group operations of the monomer symmetry group. Line groups with this property are called symmmorphic.
- In case the line group contains elements that reverse the direction of the z-axis, such as a horizontal symmetry plane, or a U-axis, the set of all elements that conserve the direction, is a halving subgroup of $\mathbf{P}$, denoted as $\mathbf{P}^+$, the other half being the set of elements that reverse the direction.
- The character of an irrep of a Line group covers three aspects: the point group character of the isogonal group, the fractional translation due to glide planes or screw axes, and the translational character of the integral translations.

8.5 Plane Groups: The Graphene Lattice

The final application of this chapter concerns the line symmetries of some carbon nanotubes, which are formed by cutting out straps of the graphene honeycomb lattice and enrolling them into infinite seamless cylinders. This construction can be viewed as a projection of the translations of a two-dimensional lattice onto a line group with a cyclic symmetry. This urges us to insert first this penultimate section on the symmetry of the two-dimensional hexagonal lattice, which takes our understanding of line groups to the next level of plane groups. It provides us with a direct practical experience with plane groups and puts us in the front row to observe how a plane can fold into a line, and what that does to the Brillouin zone.

Translation Group in Two Dimensions

The Bravais lattice of graphene is a hexagonal mesh, formed by sp^2 hybridized carbon. The lattice has translational symmetry in two dimensions. In Fig. 8.11, we define two unit vectors, $\{\mathbf{a}_1, \mathbf{a_2}\}$, spanning an angle of 60°, and forming a rhombic unit cell. As the area of this unit cell covers exactly one hexagon, one can as well choose elementary hexagons as unit cells, simply by reordering the appropriate fragments of the rhombic cell. We also define a Cartesian (x, y)-coordinate frame, as shown in the figure:

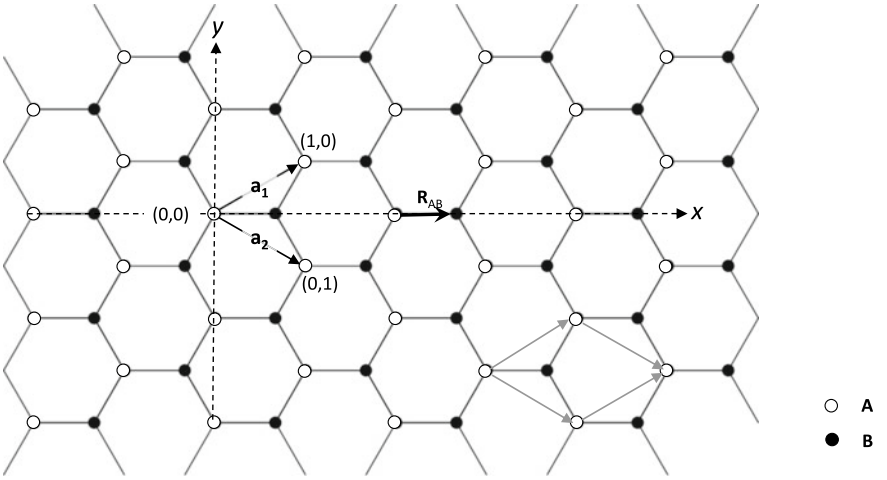


Fig. 8.11 Hexagonal graphene lattice: $\mathbf{a}_1$ and $\mathbf{a}_2$ are the elementary translations which define a rhombic unit cell; lattice points are labeled by (n_1, n_2) indices, $\mathbf{R}_{AB}$ is the elementary bond vector from A to B along the x-direction

$$\mathbf{a}_1 = a\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$
$$\mathbf{a}_2 = a\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right) \tag{8.50}$$

Here a is the length of the unit translation. The unit cell contains two atoms, labeled as A and B. In the figure, these atoms are indicated by alternating white and black circles. R_{AB} is the elementary bond length. We define a vector $\mathbf{R}_{AB}$ along the x-direction as

$$R_{AB} = \frac{a}{\sqrt{3}}(1, 0) \tag{8.51}$$

The carbon–carbon bond in graphene is approximately 1.42 Å. In the rhombic unit cell, the $\mathbf{R}_{AB}$ vector does not have translation symmetry. As the unit cell consists of two atoms, each will generate a translation lattice. We now introduce a number system n_1, n_2 to assign the lattice cells. The lattice vectors of the A and B atoms are then generated by

$$\mathbf{r}^A_{n_1,n_2} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 = a\left(\frac{\sqrt{3}(n_1+n_2)}{2}, \frac{n_1-n_2}{2}\right)$$
$$\mathbf{r}^B_{n_1,n_2} = \mathbf{r}_A + \mathbf{R}_{AB} = a\left(\frac{1}{\sqrt{3}} + \frac{\sqrt{3}(n_1+n_2)}{2}, \frac{n_1-n_2}{2}\right) \tag{8.52}$$

Note that the B atom with indices (n_1, n_2) is in the same unit cell as the A atom, displaced by one bond length along the x-axis. Atoms of type A are only bound to atoms of type B and vice versa. Such a lattice is called bipartite. The core feature of plane groups is the two-dimensional translational symmetry. The translation operations of the planar Bravais lattice are direct products of translations along the directions of the two unit vectors. Moreover these translations commute. The group $\mathbf{T}$ is thus a direct product group: $\mathbf{T} = \mathbf{T}_1 \times \mathbf{T}_2$, with

$$\hat{T}_{s,t}\ \mathbf{r} = \hat{T}_1^s \hat{T}_2^t\ \mathbf{r} = (E|s,t)\mathbf{r} = \mathbf{r} + s\mathbf{a}_1 + t\mathbf{a}_2 \tag{8.53}$$

As before, the corresponding SALCs that are adapted to the translation symmetry take the form of Bloch waves, but taking into account that separate phase factors are now required for $\mathbf{T}_1$ and $\mathbf{T}_2$. One has for both atom types:

$$\begin{aligned} |\psi_{\mathbf{k}}^A\rangle &= \frac{1}{\sqrt{N}} \sum_{n_1} \sum_{n_2} \exp(ik_1 n_1 a) \exp(ik_2 n_2 a) |\phi_{n_1,n_2}^A\rangle \\ |\psi_{\mathbf{k}}^B\rangle &= \frac{\exp(i\mathbf{k}\cdot\mathbf{R}_{AB})}{\sqrt{N}} \sum_{n_1,n_2} \exp(ik_1 n_1 a) \exp(ik_2 n_2 a) |\phi_{n_1,n_2}^B\rangle \end{aligned} \tag{8.54}$$

Here B atoms are displaced in the unit cell over a distance $\mathbf{R}_{AB}$ relative to $\mathbf{r}_(n_1, n_2)$. Both k-coefficients are components of a wavevector $\mathbf{k}$. According to Eq. (8.54) the scalar product of this wavevector with the position vector is given by

$$\mathbf{k} \cdot \mathbf{r}_{n_1,n_2} = k_1 n_1 a + k_2 n_2 a \tag{8.55}$$

In order to determine the Brillouin zones, one cannot simply match the orientation of the wavevector components with the directions in the unit cell. The problem is indeed that, at least for the case of the hexagonal lattice, the unit vectors of the lattice are not orthogonal to each other but at 60°. This complicates the scalar product. One could of course opt for a rectangular unit cell, based on the Cartesian frame, but this would contain four atoms, and thus give rise to unnecessary zone folding. Instead, this problem is readily solved by defining the momentum vector in a $(\mathbf{b}_1, \mathbf{b}_2)$ basis which obeys the following orthogonality relations:

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{i,j} \tag{8.56}$$

We can orient these vectors in the Cartesian frame, by combining Eqs. (8.50) and (8.56):

$$\begin{aligned} \mathbf{b}_1 &= \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, 1\right) \\ \mathbf{b}_2 &= \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -1\right) \end{aligned} \tag{8.57}$$

Momentum space is thus described by a two-component vector in the $(\mathbf{b}_1, \mathbf{b}_2)$ unit vectors, as

$$\mathbf{k} = \kappa_1 \mathbf{b}_1 + \kappa_2 \mathbf{b}_2 \tag{8.58}$$

The connection between the κ and k coefficients is obtained by substituting this result in Eq. (8.55):

$$\begin{aligned} \mathbf{k} \cdot \mathbf{r}_{n_1,n_2} &= (\kappa_1 \mathbf{b}_1 + \kappa_2 \mathbf{b}_2)\,(n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2) \\ &= 2\pi\,(\kappa_1 n_1 + \kappa_2 n_2) = k_1 n_1 a + k_2 n_2 a \end{aligned} \tag{8.59}$$

from which follows:

$$\begin{aligned} k_1 &= \frac{2\pi\kappa_1}{a} \\ k_2 &= \frac{2\pi\kappa_2}{a} \end{aligned} \tag{8.60}$$

The $\mathbf{b}$ vectors also define a lattice, which is called the reciprocal lattice, as it relates to the real Bravais lattice, such as position and momentum space relate to each other. This lattice is also periodic, since translations do not affect the phase factors of the Bloch waves:

$$\exp\left(i\,(\mathbf{k} + m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2) \cdot (n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2)\right) = \exp\left(i \mathbf{k} \cdot (n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2)\right) \tag{8.61}$$

with $m_1, m_2 \in \mathbb{Z}$. This implies that the first Brillouin zone corresponds to the unit cell in the reciprocal lattice, as it contains all the essential wavevectors. It is now practical to rewrite the reciprocal lattice coordinates also in a Cartesian frame:

$$\mathbf{k} = \kappa_1 \mathbf{b}_1 + \kappa_2 \mathbf{b}_2 \equiv k_x(1, 0) + k_y(0.1) \tag{8.62}$$

and accordingly:

$$\begin{aligned} k_x &= \frac{2\pi}{a\sqrt{3}}(\kappa_1 + \kappa_2) \\ k_y &= \frac{2\pi}{a}(\kappa_1 - \kappa_2) \end{aligned} \tag{8.63}$$

Equation (8.59) then becomes

$$\begin{aligned} \mathbf{k} \cdot \mathbf{r}_{n_1,n_2} &= 2\pi\,(\kappa_1 n_1 + \kappa_2 n_2) \\ &= 2\pi \left((\kappa_1 + \kappa_2)(\frac{n_1 + n_2}{2}) + (\kappa_1 - \kappa_2)(\frac{n_1 - n_2}{2}) \right) \\ &= a\sqrt{3} k_x (\frac{n_1 + n_2}{2}) + a k_y (\frac{n_1 - n_2}{2}) \end{aligned} \tag{8.64}$$

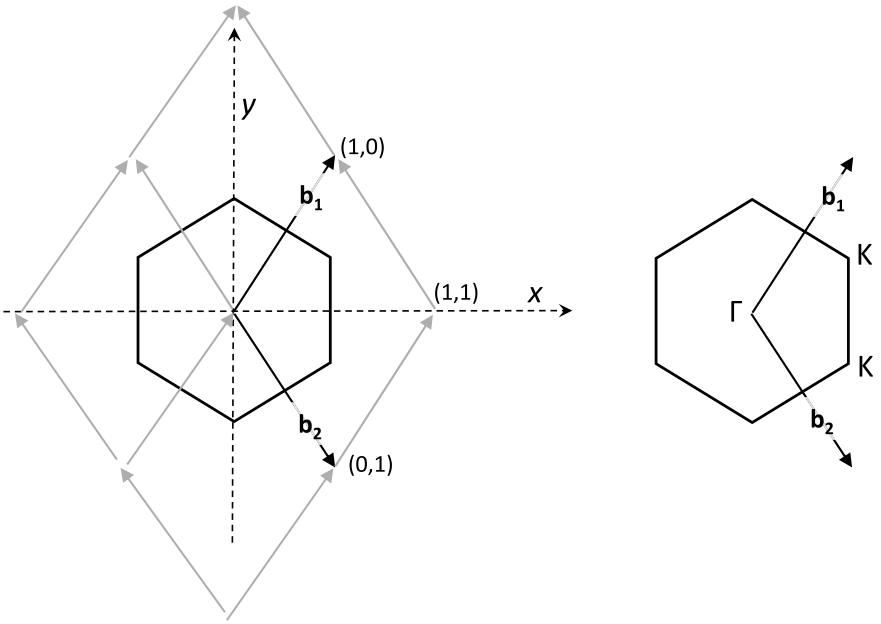


Fig. 8.12 Reciprocal lattice for hexagonal graphene: $\mathbf{b}_1$ and $\mathbf{b}_2$ are the elementary translations which define a rhombic unit cell; the central hexagon is the Wigner–Seitz unit cell, corresponding to the first Brillouin zone

In Fig. 8.12, we present the reciprocal lattice, and the first Brillouin zone. Integer values for κ_1 and κ_2 address lattice points. At these points, the product in Eq. (8.59) is an integer multiple of 2π. Since the Brillouin zone is periodic, its unit cell can be delineated in several ways. Remember that already in the case of the one-dimensional translation group, we choose to position the Brillouin zone in the interval $]-\pi/a, \pi/a]$ rather than $]0, 2\pi/a]$, because this diagram depicts more clearly the time-reversal symmetry between $-k$ and $+k$ points. A similar procedure is adopted for the presentation of a symmetrized unit cell in the 2D case. This is known as the Wigner–Seitz unit cell. It is constructed by drawing lines from the space origin to all its nearest neighbors. In the case of graphene, a given origin has six nearest neighbors, corresponding to $(\mathbf{b}_1, \mathbf{b}_2)$ points with coordinates $(\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1)$. For each of these lines, one draws the perpendicular bisector. These bisectors delineate the primitive Wigner–Seitz unit cell. This cell is a regular hexagon, which in the Cartesian frame is turned 90°, as compared to the direct space unit cell. The zone boundaries along the horizontal x-direction are $\pm 1/2\,(\mathbf{b}_1 + \mathbf{b}_2)$, which yields

$$-\frac{2\pi}{a\sqrt{3}} < k_x \leq \frac{2\pi}{a\sqrt{3}} \tag{8.65}$$

The boundaries in the vertical direction are obtained as $\pm 1/3(\mathbf{b}_1 - \mathbf{b}_2)$

$$-\frac{4\pi}{3a} < k_y \leq \frac{4\pi}{3a} \tag{8.66}$$

Several special points in the diagram are to be noted. They are symmetry points of the plane group. We will come back to these after the derivation of the tight-binding results.

Tight Binding Model for Graphene

The translation SALC's for the A and B atoms were given previously in Eq. (8.54). Translation symmetry dictates that only SALC's with the same $\mathbf{k}$ wavevector can interact. The Hamiltonian matrix is thus limited to 2×2 matrices over $|\psi^A_{\mathbf{k}}\rangle, \psi^B_{\mathbf{k}}\rangle$. Furthermore, as the unit cell is bipartite, all nearest neighbors of A atoms are B atoms, and vice versa. As a result in the tight binding approximation the diagonal matrix elements vanish. The off-diagonal elements are given by

$$\begin{aligned}\langle\psi^A_{\mathbf{k}}|\mathcal{H}|\psi^B_{\mathbf{k}}\rangle &= \frac{1}{N}\exp((i\mathbf{k}\cdot\mathbf{R}_{AB})\times \\ \sum_{n_1,n_2}\sum_{n_1',n_2'}&\exp\left[i\mathbf{k}\cdot\left((n_1'-n_1)\mathbf{a}_1+(n_2'-n_2)\mathbf{a}_2\right)\right]\langle\phi^A_{n_1,n_2}|\mathcal{H}|\phi^B_{n_1',n_2'}\rangle\end{aligned} \tag{8.67}$$

The atom A at position (n_1, n_2) has only three neighbors. Their position vectors are displayed in Fig. 8.13. As a result, the summation over the primed positions reduces to only three terms:

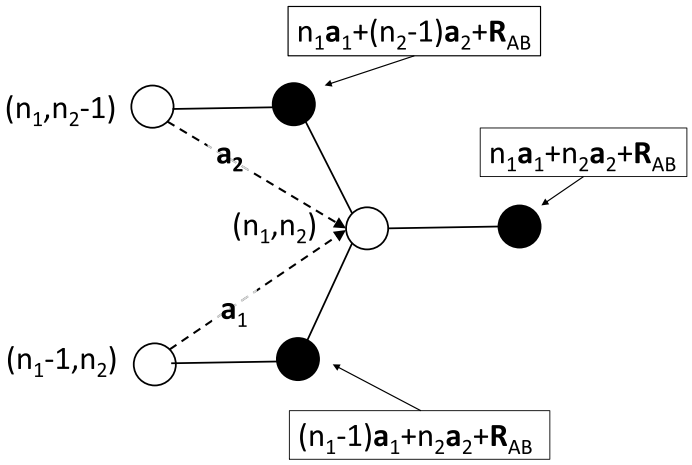


Fig. 8.13 Coordinates of nearest neighbor B atoms (black) to a given A atom (white) with address (n_1, n_2)

$$\langle \psi_{\mathbf{k}}^{A} | \mathcal{H} | \psi_{\mathbf{k}}^{B} \rangle = \frac{\beta}{N} \exp(i\mathbf{k} \cdot \mathbf{R}_{AB}) \sum_{n_1, n_2} \left[1 + \exp\left(-i\mathbf{k} \cdot \mathbf{a}_1\right) + \exp\left(-i\mathbf{k} \cdot \mathbf{a}_2\right) \right] \tag{8.68}$$

Converting these expressions to Cartesian form, using Eq. (8.64), the matrix element finally can be rewritten as

$$\langle \psi_{\mathbf{k}}^{A} | \mathcal{H} | \psi_{\mathbf{k}}^{B} \rangle = \beta \exp\left(\frac{ik_x a}{\sqrt{3}}\right) \left[1 + 2 \exp\left(-\frac{i\sqrt{3}k_x a}{2}\right) \cos\left(\frac{k_y a}{2}\right) \right] \tag{8.69}$$

The energy roots resulting from this matrix are then given by

$$\epsilon_{\mathbf{k}} = \pm\beta \left[1 + 4\cos\left(\frac{\sqrt{3}k_x a}{2}\right) \cos\left(\frac{k_y a}{2}\right) + 4\cos^2\left(\frac{k_y a}{2}\right) \right]^{1/2} \tag{8.70}$$

As each carbon contributes one p_z orbital, there are two valence electrons per unit cell. The lower energy root then forms the valence band, while the upper root forms the conduction band, which is an exact mirror image of the lower bands. Both bands are touching at the K points in the Brillouin zone. The band structure is shown in Fig. 8.14. The point in the center of the zone, with $k_x = k_y = 0$, is called the Γ point. It corresponds to total bonding, with an energy of 3β. The upper points of the band are at the vertices of the hexagonal unit cell. These are the **K** points, at the trigonal vertex sites of the hexagon. Since each of these is shared between three neighboring zones, the total number of **K** points amounts to 2. The coordinates of the two **K** in the figure are $k_x = 2\pi/a\sqrt{3}, k_y = \pm 2\pi/3a$. Each **K** point is characterized by two zero eigenvalues, as the upper and lower bands form a conical crossing in these sites. In total, there are thus exactly four non-bonding orbitals in the first Brillouin zone. Finally, also noteworthy are the orthorhombic edge centers, denoted as the points **M**.

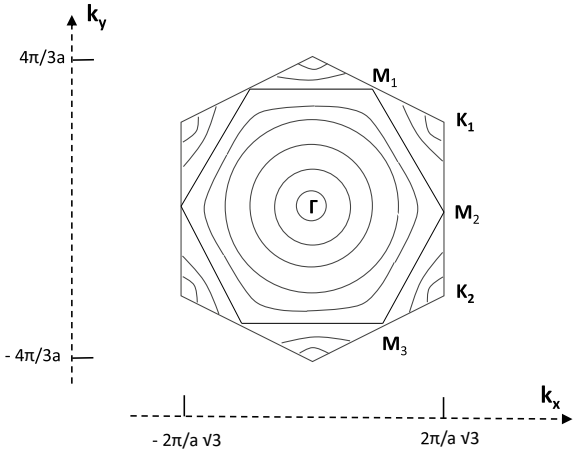


Fig. 8.14 Energy surface of the valence band for graphene in the first Brillouin zone; the central Γ point has minimal energy of 3β, the two trigonal **K** points are at zero energy, and the three orthorhombic **M** points at the hexagonal edges are connected by an equipotential inscribed hexagonal line with energy β

Each of these points is shared between two neighboring zones, so there are three **M** sites in total, as indicated in the figure. The eigenvalue on these sites is exactly equal to β, and moreover, an equipotential line $\epsilon = \beta$ connects these points, forming an inscribed hexagon.

Eigenvectors and Irreducible Representations

Our treatment is limited to the symmorphic plane group, generated by the central hexagonal symmetry axis, and the six in-plane twofold rotations. In Fig. 8.15 are shown both sets of symmetries. The hexagonal unit cell is characterized by a central hexagonal axis, threefold symmetry axes at the vertices, and twofold symmetry axes at the edges. The $\hat{C}_3$ and $\hat{C}_2$-axes belong to the translational cosets of the sixfold axis: This is demonstrated in the figure for the case of the $\hat{C}_3$-axis. The right-handed threefold rotation moves the black-starred point to the white-starred point. This can be achieved by first translating the point over a vertical distance $-\mathbf{a}_1 + \mathbf{a}_2$ to the gray-starred point, followed by the central C_6^2 rotation.

$$\hat{C}_3 = \hat{C}_6^2(E|-1,+1) \tag{8.71}$$

Hence: $\hat{C}_3 \in \hat{C}_6^2\mathbf{T}$. On the right of Fig. 8.15 are shown two of the six twofold in-plane rotations of the hexagon. These are genuine Umklapp operations of the unit cell. Each of these rotations generates a separate coset of the plane group. The resulting plane group is denoted as the $\mathbf{P}_{622}$ group. It consists of twelve cosets:

$$\mathbf{P}_{622} = \sum_{n=1,6} \hat{C}_6^n \times \mathbf{T} + \sum_{n=1,6} \hat{U}\hat{C}_6^n \times \mathbf{T} \tag{8.72}$$

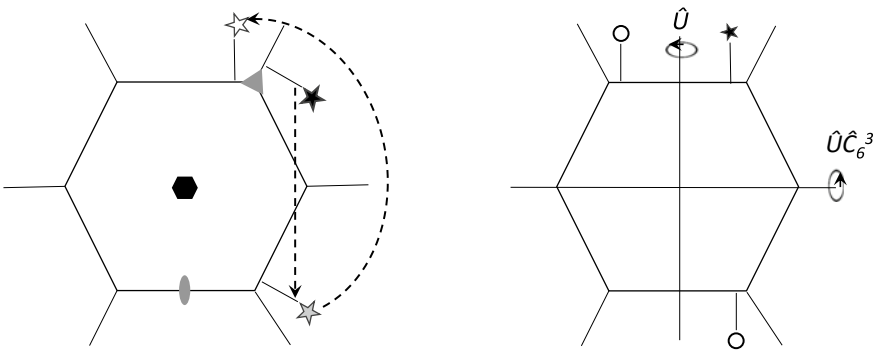


Fig. 8.15 Plane group elements of the hexagonal lattice: on the left are shown the six-, three-, and twofold rotation axes perpendicular to the plane, on the right two in-plane Umklapp operations: $\hat{U}$ through edges and $\hat{U}\hat{C}_6^3$ through vertices

Table 8.6 Coordinates of face, vertex and edge points in the Brillouin zone

	k_x	k_y
Γ	0	0
$\mathbf{K}_1$	$\frac{2\pi}{a\sqrt{3}}$	$\frac{\pi}{3a}$
$\mathbf{K}_2$	$\frac{2\pi}{a\sqrt{3}}$	$-\frac{\pi}{3a}$
$\mathbf{M}_1$	$\frac{\pi}{a\sqrt{3}}$	$\frac{\pi}{a}$
$\mathbf{M}_2$	$\frac{2\pi}{a\sqrt{3}}$	0
$\mathbf{M}_3$	$\frac{\pi}{a\sqrt{3}}$	$-\frac{\pi}{a}$

The corresponding isogonal point group is isomorphic to the D_6 point group. Adding symmetry planes will enlarge this group further to D_{6h}, but our analysis can conveniently be limited to $\mathbf{P}_{622}$. The focus is on the Γ, K and M special points in the Brillouin zone. These points are indicated in Fig. 8.14 and their coordinates are provided in Table 8.6. We first derive the eigenvector spaces in these points and then examine their symmetry behavior. In this way, we construct special irreducible representations of the plane group. For simplicity, we will consider a hexagonal lattice of nodeless s-type orbitals.

The Γ point at the origin is the totally bonding combination of all atomic sites and therefore transforms according to the totally symmetric scalar irrep. As for K_1 and K_2 points, since each point has two zero eigenvalues, the total basis covers four wavevectors. The tight-binding Hamiltonian relates the coefficient of an atom on site A to the sum of the coefficients of the surrounding B sites. Referring to Fig. 8.13 a line of the Hückel matrix equation thus reads

$$c^B_{n_1,n_2} + c^B_{n_1-1,n_2} + c^B_{n_1,n_2-1} = \lambda c^A_{n_1,n_2} \tag{8.73}$$

where λ is the eigenvalue. A non-bonding eigenvalue requires that the sum of the coefficients of the surrounding B atoms is also equal to zero. For a bipartite lattice, the construction of an eigenvector based on this rule is straightforward: one surrounds a given A atom by the (unnormalized) sequence of the roots of the Euler equation $x^3 = 1 : 1, \epsilon, \epsilon^*$, with $\epsilon = \exp(2\pi i/3)$. This sequence of roots forms a rotation around the atom A. The sum of these roots is zero, and hence the combination is non-bonding. Moreover, by putting all rotations in the same sense, one obtains at once one complete eigenfunction, as seen in Fig. 8.16. In the figure, the eigenvector consists of right-handed rotation around A, and is denoted[3] as $|\psi^B_r\rangle$. By taking the complex conjugate of this eigenfunction, one reverts the sense of rotation and in this way generates its orthogonal counterpart. Finally, to obtain the remaining two non-bonding combinations, one repeats the same procedure, with now vanishing coefficients on B and rotations on A. Returning to the expressions for the Bloch waves in Eq. (8.54), we verify that the eigenvectors for the K_1 point are $|\psi^A_l\rangle$, $|\psi^B_r\rangle$. The K_2 point is

[3] Note that the rotational sequence of the roots around the center of the hexagons is in the opposite sense, so for $|\psi^B_r\rangle$ the rotation around the hexagonal face is left-handed.

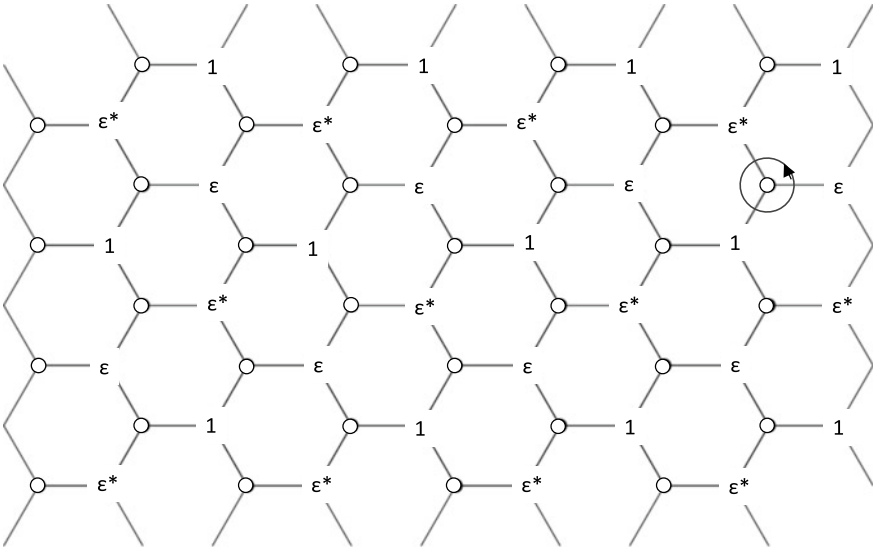


Fig. 8.16 Non-bonding eigenfunction on the hexagonal lattice: each A atom is surrounded by a right-handed rotation of the sequence of cubic roots of 1: $1 \to \epsilon \to \epsilon*$, with $\epsilon = \exp(2\pi i/3)$

exactly opposite: $|\psi_r^A\rangle$, $|\psi_l^B\rangle$. The sixfold rotation axis will interchange A and B but will keep the handedness of the rotation. In contrast, the Umklapp operations will also interchange A and B, but in the same time switch the handedness. In this way, the four eigenvectors sustain one fourfold degenerate irrep. In order to construct the corresponding transformation matrices, one must fix the relative phase of the eigenvectors. It suffices to pick one hexagon and specify the coefficients of the respective eigenvectors in this hexagon. In Fig. 8.17, we show our canonical hexagons.

Now order these components in a row vector as

$$|\mathbf{f}\rangle = \left(|\psi_r^A\rangle, |\psi_r^B\rangle, |\psi_l^A\rangle, |\psi_l^B\rangle\right)$$

Applying the generators of the plane group to this function space, using the canonical representations in Fig. 8.17 then yields

$$\begin{aligned} \hat{C}_6|\mathbf{f}\rangle &= |\mathbf{f}\rangle \begin{pmatrix} 0 & \epsilon^* & 0 & 0 \\ \epsilon^* & 0 & 0 & 0 \\ 0 & 0 & 0 & \epsilon \\ 0 & 0 & \epsilon & 0 \end{pmatrix} \\ \hat{U}|\mathbf{f}\rangle &= |\mathbf{f}\rangle \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \end{aligned} \tag{8.74}$$

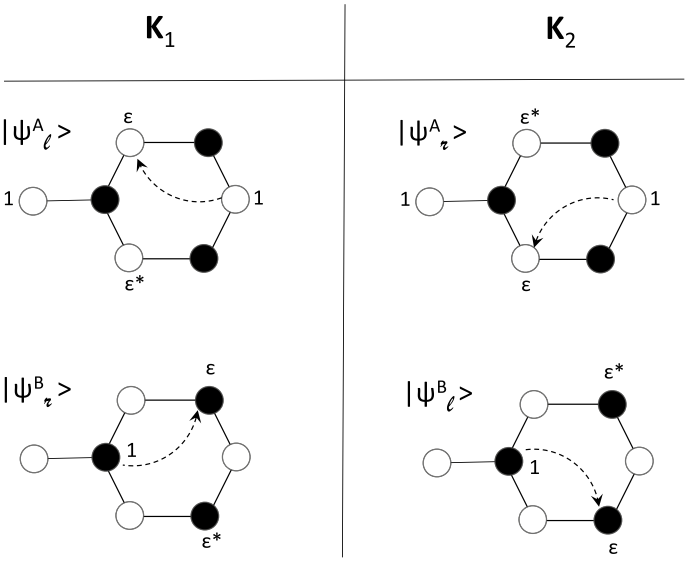


Fig. 8.17 The four non-bonding eigenfunctions of the hexagonal lattice: a combination such as A, l means that density is on the A atom sites and that a left-handed rotation of the A atoms surrounding a given B site orders the trigonal roots in the sequence $(1, \epsilon, \epsilon^*)$, as indicated by the dashed arrows, etc.; note that the rotation of the roots around the center of the hexagon is always in the opposite sense

While both operations switch A and B, the sixfold rotation keeps the handedness while the Umklapp operation reverses the handedness. As a result, the four components are intrinsically tied together and form one fourfold degenerate irrep.[4]

Finally, for the three M points, indicated in Fig. 8.14, the wavevectors are given in Table 8.6. The corresponding eigenfunctions can then be worked out with the help of the expressions in Eqs. (8.54), (8.64), and (8.69). As an example for the $\mathbf{M}_1$ point, one has

$$\begin{aligned}\sqrt{N}|\psi^A_{M_1}\rangle &= \sum_{n_1,n_2} \exp\left(\frac{i\sqrt{3}k_x a}{2}(n_1+n_2)+\frac{ik_y a}{2}(n_1-n_2)\right)|\phi^A_{n_1,n_2}\rangle \\ &= \sum_{n_1,n_2} \exp\left(i\pi n_1\right)|\phi^A_{n_1,n_2}\rangle = \sum_{n_1,n_2}(-1)^{n_1}|\phi^A_{n_1,n_2}\rangle \\ \sqrt{N}|\psi^B_{M_1}\rangle &= \exp\left(\frac{i\pi}{3}\right)\sum_{n_1,n_2}(-1)^{n_1}|\phi^B_{n_1,n_2}\rangle \end{aligned} \tag{8.75}$$

[4] According to the extensive symmetry tables, provided by Milosevic et al. [6], this irrep is labeled as the four-dimensional irrep Q_1. This irrep can be viewed as a vectorial extension of the two-dimensional irrep Q_0 which corresponds to the irrep formed by the two K points themselves in the Brillouin zone.

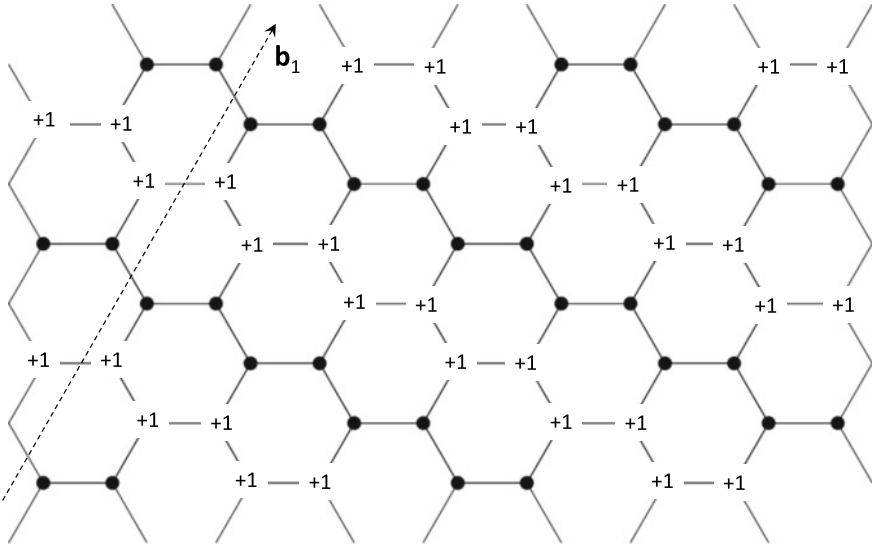


Fig. 8.18 Schematic presentation of the eigenfunction at M_1. At the black atoms, the eigenvector coefficients are all equal to -1. The wavefunction thus consists of an alternation of $+1$ and -1, which form a standing wave directed along the $\mathbf{b}_1$ vector

The (unnormalized) eigenvector of the lower root, $\epsilon_k = \beta$, of the Hamiltonian then reads

$$|\Psi_{M_1}\rangle = |\psi^A_{M_1}\rangle + \exp(\frac{-i\pi}{3})|\psi^B_{M_1}\rangle = \sum_{n_1,n_2}(-1)^{n_1}\left(|\phi^A_{n_1,n_2}\rangle + |\phi^B_{n_1,n_2}\rangle\right) \tag{8.76}$$

The distribution of the coefficients over the hexagonal lattice is shown in Fig. 8.18. The figure clearly shows stripes of plus and minus one, corresponding to a wave which propagates in the direction of the $\mathbf{b}_1$ translation vector. Similar waves are formed for the other M points: for M_2 the direction of propagation is along the x-axis, corresponding to $\mathbf{b}_1 + \mathbf{b}_2$, and for M_3, it follows the direction of $\mathbf{b}_2$. In summary:

$$\begin{aligned}
|\Psi_{M_1}\rangle &= \sum_{n_1,n_2}(-1)^{n_1}\left(|\phi^A_{n_1,n_2}\rangle + |\phi^B_{n_1,n_2}\rangle\right)\\
|\Psi_{M_2}\rangle &= \sum_{n_1,n_2}(-1)^{n_1+n_2}\left(|\phi^A_{n_1,n_2}\rangle + |\phi^B_{n_1,n_2}\rangle\right)\\
|\Psi_{M_3}\rangle &= \sum_{n_1,n_2}(-1)^{n_2}\left(|\phi^A_{n_1,n_2}\rangle + |\phi^B_{n_1,n_2}\rangle\right)
\end{aligned} \tag{8.77}$$

The three eigenfunctions thus correspond to standing waves in three directions at 60° apart. Clearly, a sixfold rotation will introduce a cyclic permutation of these waves. One has

$$\hat{C}_6\left(|\Psi_{M_1}\rangle|\Psi_{M_2}\rangle|\Psi_{M_3}\rangle\right) = \left(|\Psi_{M_1}\rangle|\Psi_{M_2}\rangle|\Psi_{M_3}\rangle\right)\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{pmatrix} \tag{8.78}$$

The negative sign in the final matrix element is because the sixfold rotation of $\mathbf{M}_1$ projects the point M_1 to the point $\mathbf{M}_3$ in the neighboring Brillouin zone, which is one translation away. Finally, the Umklapp operation will leave one wave invariant while interchanging the other two. Again the triplet of orthorhombic M points forms an irreducible space, corresponding to a triple degenerate irrep. All special points thus form irreducible representations of the plane group. Apart from these points however points inside the Brillouin zone may realize even higher degeneracies, as the plane group also allows for sixfold and even 12-fold degenerate irreps. This reflects the hexagonal symmetry of the Wigner–Seitz unit cell.

8.6 Application: Nanotubes

Nanotubes are exquisite molecular structures with intriguing properties. They can be obtained on paper by cutting out stripes of the graphene sheet. In Fig. 8.19 is shown the construction of an armchair nanotube. Here C_h is called the chiral vector. It connects two points on the stripe which meet when the tube is closed and forms the

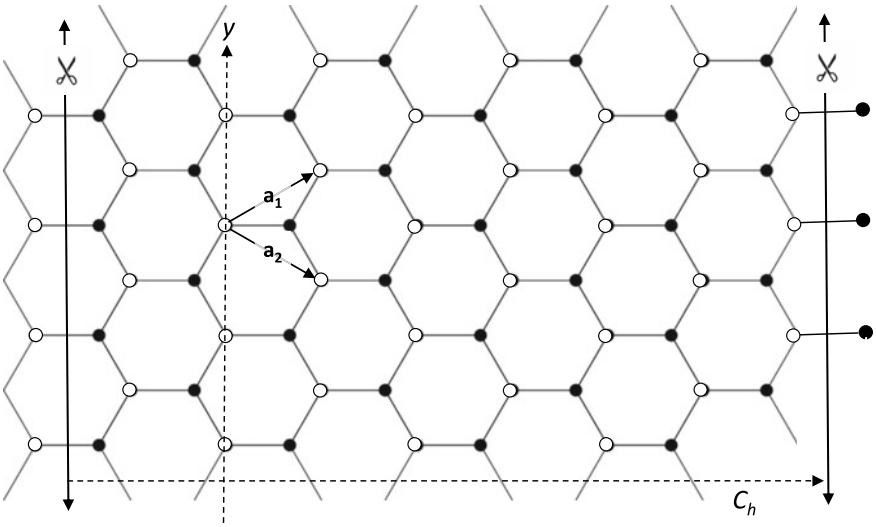


Fig. 8.19 Paper construction of a nanotube: cut the graphene plane along the vertical lines and roll up the stripe; C_h is the chiral vector which connects the points that are glued together. The figure illustrates a (4, 4) armchair tube, with $C_h = 4\mathbf{a}_1 + 4\mathbf{a}_2$

circumference of the tube. The hexagonal coordinates (n_1, n_2) of the chiral vector denote the type of tube:

$$C_h = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 \tag{8.79}$$

High-symmetry cases are the armchair tubes, with $n_1 = n_2$ as shown in the figure, and the zigzag tubes, with $n_2 = 0$. The two-dimensional translation group of the plane is reduced to a Line group oriented along the cylinder axis. In the perpendicular dimension, a rotational or skew-rotational axis replaces the translational symmetry. The reduction of this dimension to a finite cyclic symmetry literally provides a molecular realization of the Born–von Karman cyclic boundary conditions. This will clearly show up in the band structure of the nanotube, as we will demonstrate in a moment.

Line Group of Armchair Nanotube

Group theory is now applied to the (4, 4) armchair nanotube. In Fig. 8.20 we show the unit cell. A modal point inside this unit cell has 32 equivalent points, generated by symmetry. This implies the existence of a 32-dimensional isogonal point group. The 16 black stars in the figure denote points that are reached by rotations. The obvious rotational symmetry element is a fourfold rotation axis, $(C_4|0)$, oriented along the spine of the tube. But in fact, this element itself is the square of a skew axis, $(C_8|1/2)$. Indeed, the unit cell contains not four but eight hexagons in total, forming the orbit of this skew axis. As a result, the isogonal point group is non-symmorphic, and this is the case for all nanotubes. The skew axis reaches 8 of the 16 black stars, the remaining 8 are generated by Umklapp operations. These are twofold axes perpendicular to the tube direction. There are eight of these in total. The resulting isogonal point group is isomorphic to the dihedral group, D_8. The white stars in the figure are 16 further copies that are obtained by adding reflection planes. The principal planes are the horizontal symmetry plane $(\sigma_h|0)$ and the vertical symmetry plane $(\sigma_v|0)$.

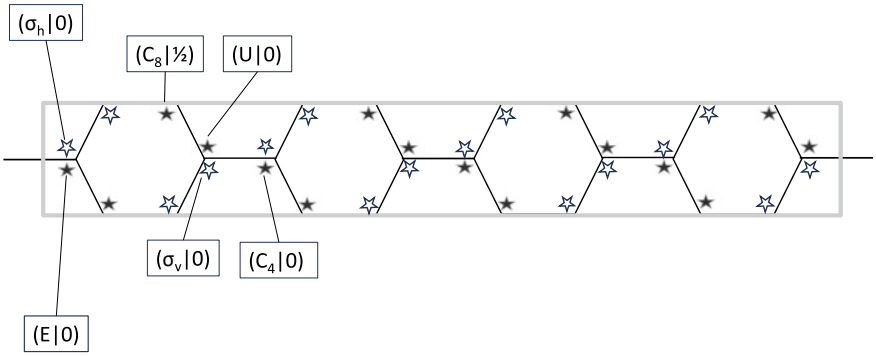


Fig. 8.20 Unit cell of the (4, 4) armchair nanotube, with assignments for some generators of the $\mathbf{L}8_4/mcm$ line group

The eightfold axis lies in the vertical plane and is perpendicular to the horizontal plane. One has

$$(\sigma_h|0) = (\sigma_v|0)(U|0) \tag{8.80}$$

These planes extend the isogonal group to a group which is isomorphic to the 32-dimensional prismatic point group D_{8h}. The general symbol for the line groups of $(n.n)$ armchair tubes, as well as $(n, 0)$ zigzag tubes, is $\mathbf{L}(2n)_n/mcm$, and the isogonal point group is D_{2nh}:

$$\mathbf{L}(2n)_n/mcm = D_{2nh} \times \mathbf{T} \tag{8.81}$$

The precise relationship between the symmetry points in Fig. 8.20 and the D_{8h} point group is the topic of Problem 8.5.

The irreps of the line group were obtained by Damnjanovic et al. [7]. They are represented in Table 8.7. Irreps are classified by k and m quantum numbers: k is the momentum of the wave in the translational y-direction, while m is the angular momentum under the principal $\hat{C}_n$ rotation axis, along the x-direction. The cyclization of this direction replaces the $\hat{T}_x$ translation of the Plane Group by a finite Born–von Karman condition. For $k = 0$, the irreps coincide with the irreps of the prismatic group D_{8h} as given in Appendix A. The G symbol refers to a four-dimensional irrep. The associated representation matrices in the table are 4×4 matrices, as follows:

Table 8.7 Irreducible representations for the line group $\mathbf{L}(2n)_{n/mcm}$: $m \in (0, n)$ denotes that m is an integer with $0 < m < n$; irreps with $m = n/2$ only occur for even n

	m	$(C_{2n}\|a/2)$	$(C_n\|0)$	$(U\|0)$	$(\sigma_v\|0)$
${}_0A_m^+$	0,n	$e^{im\pi/n}$	$+1$	$+1$	$+1$
${}_0A_m^-$	0,n	$e^{im\pi/n}$	$+1$	-1	$+1$
${}_0B_m^+$	0,n	$e^{im\pi/n}$	$+1$	$+1$	-1
${}_0B_m^-$	0,n	$e^{im\pi/n}$	$+1$	-1	-1
${}_0E_m^+$	$(0, n)$	$\begin{pmatrix} e^{im\pi/n} & 0 \\ 0 & e^{-im\pi/n} \end{pmatrix}$	$\begin{pmatrix} e^{2im\pi/n} & 0 \\ 0 & e^{-2im\pi/n} \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
${}_0E_m^-$	$(0, n)$	$\begin{pmatrix} e^{im\pi/n} & 0 \\ 0 & e^{-im\pi/n} \end{pmatrix}$	$\begin{pmatrix} e^{2im\pi/n} & 0 \\ 0 & e^{-2im\pi/n} \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
${}_kE_m^{(A)}$	$0, n$	$\begin{pmatrix} e^{i(m\pi/n+ka/2)} & 0 \\ 0 & e^{i(m\pi/n-ka/2} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
${}_kE_m^{(B)}$	$0, n$	$\begin{pmatrix} e^{i(m\pi/n+ka/2)} & 0 \\ 0 & e^{i(m\pi/n-ka/2} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
${}_{\pi/a}E_{n/2}^A$	$n/2$	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
${}_{\pi/a}E_{n/2}^B$	$n/2$	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
${}_kG_m$	(0,n)	$\mathbb{K}_4(\frac{k}{2}, \frac{m}{2})$	$\mathbb{M}_4(m)$	$\mathbb{O}_4$	$\mathbb{V}_4$

$$\mathbb{K}_4(\frac{k}{2},\frac{m}{2}) = \begin{pmatrix} e^{i(m\pi/n+ka/2)} & 0 & 0 & 0 \\ 0 & e^{i(-m\pi/n+ka/2)} & 0 & 0 \\ 0 & 0 & e^{i(m\pi/n-ka/2)} & 0 \\ 0 & 0 & 0 & e^{i(-m\pi/n-ka/2)} \end{pmatrix}$$
$$\mathbb{M}_4(m) = \begin{pmatrix} e^{im\pi/n} & 0 & 0 & 0 \\ 0 & e^{-im\pi/n} & 0 & 0 \\ 0 & 0 & e^{im\pi/n} & 0 \\ 0 & 0 & 0 & e^{-im\pi/n} \end{pmatrix}$$
$$\mathbb{O}_4 = \begin{pmatrix} 0&0&0&1 \\ 0&0&1&0 \\ 0&1&0&0 \\ 1&0&0&0 \end{pmatrix}$$
$$\mathbb{V}_4 = \begin{pmatrix} 0&1&0&0 \\ 1&0&0&0 \\ 0&0&0&1 \\ 0&0&1&0 \end{pmatrix} \tag{8.82}$$

Band Structure

As the nanotube lattice is a stripe with cyclic boundary conditions of the graphene lattice, the general eigenvalue expression in Eq. (8.70) applies. In the nanotube, the graphene k_y-coordinate is preserved as the translational momentum k. The k_x-coordinate is no longer continuous, as it must obey a cyclic boundary condition which requires discrete integer values of the angular momentum. The reduction from plane to line is summarized in Table 8.8, and graphically represented in Fig. 8.21. The Brillouin zone of the nanotube consists of a set of vertical lines in a rectangular reciprocal unit cell, which has the same area as the hexagonal zone of graphene. The irreducible representation of the line group is clearly resolved. A modal point in a periodic rectangle has four copies, with coordinates $(\pm k, \pm m)$. These points are lying on the dashed vertical lines and form function spaces which transform as the four-dimensional ${}_kG_m$ irreps. In contrast points in the center of the rectangle, and on the edge centers are non-degenerate and transform as the one-dimensional A and B irreps. This is the case for the Γ ground level in the center, and the $\mathbf{M}_1$ point at $m = n$. The twofold degenerate ${}_0E_m$ irreps correspond to discrete $\pm m$-functions at either side of the origin along the k_x-direction. For $0 < k < \pi/a$, the central vertical line through the origin, as well as the vertical line at the rectangular edge, consist of $\pm k$ pairs which transform as the ${}_kE_m$ irreps, with $m = 0$ and $m = n$ resp. The edge line contains the Fermi points of the graphite sheet, $\mathbf{K}_1$ and $\mathbf{K}_2$, with symmetries ${}_{2\pi/3a}E_n^A$ and ${}_{2\pi/3a}E_n^B$. For even values of n, as is the case for the $(4, 4)$ nanotube, two additional $\mathbf{M}_2$, $\mathbf{M}_3$ points are found at the upper rectangular edge, as

Table 8.8 Transition from graphene plane to nanotube line for armchair and zigzag nanotubes: the translation in the x-direction is replaced by discrete angular momenta, due to cyclic boundary conditions

	Topology	Momenta	Boundaries
Graphene	Hexagon	(k_x, k_y)	$-\frac{2\pi}{\sqrt{3}a} < k_x \leq +\frac{2\pi}{a\sqrt{3}}$ $-\frac{4\pi}{3a} < k_y \leq \frac{4\pi}{3a}$
Nanotube	Rectangle	$k_x = \frac{2m\pi}{\sqrt{3}a}$	$-n < m \leq n$
		$k_y = k$	$-\frac{\pi}{a} < k \leq \frac{\pi}{a}$

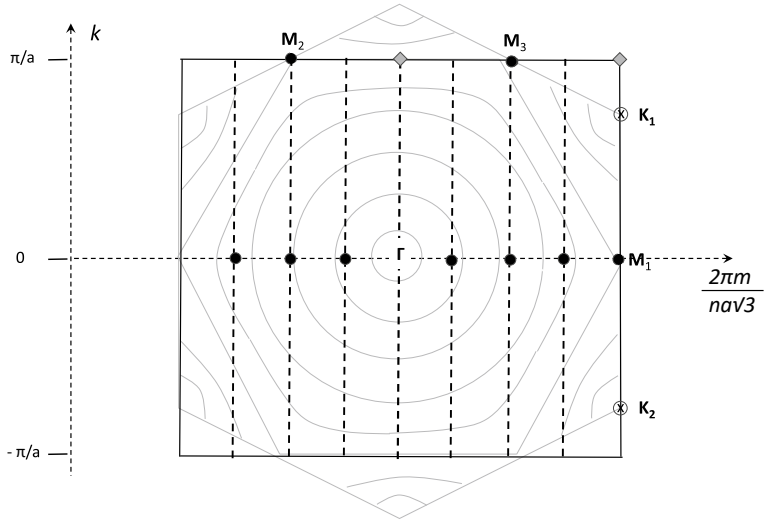


Fig. 8.21 Brillouin line intervals of the (4, 4) armchair nanotube in the graphene reciprocal space; the zone consists of a set of vertical lines in the (k, m) diagram

indicated in the figure. These transform according to the exceptional irrep ${}_{\pi/a}E_{n/2}$, also listed in Table 8.7. Their translational period corresponds to $3a$. Also exceptional are the end points of the ${}_{k}E_{m}$ vertical lines, which are indicated by diamonds. Using the standard expressions of the graphene tight-binding model with the appropriate values for k_x, k_y derivation of the unnormalized eigenfunctions in the diamond points is straightforward. One obtains for the points on the upper zone boundary:

$$\begin{aligned}
\text{edge: } m = 0 \quad (0, \tfrac{\pi}{a}) \quad |\psi_0^A\rangle &= \sum_{n_1,n_2} (i)^{n_1-n_2} |\phi^A_{n1,n_2}\rangle \\
|\psi_0^B\rangle &= \sum_{n_1,n_2} (i)^{n_1-n_2} |\phi^B_{n1,n_2}\rangle \\
|\Psi_0\rangle &= |\psi_0^A\rangle + |\psi_0^B\rangle \\
\text{vertex: } m = n \ (\tfrac{2\pi}{a\sqrt{3}}, \tfrac{\pi}{a}) \quad |\psi_n^A\rangle &= \sum_{n_1,n_2} (-1)^{n_1+n_2} (i)^{n_1-n_2} |\phi^A_{n1,n_2}\rangle \\
|\psi_n^B\rangle &= \epsilon \sum_{n_1,n_2} (-1)^{n_1+n_2} (i)^{n_1-n_2} |\phi^B_{n1,n_2}\rangle \\
|\Psi_n\rangle &= |\psi_n^A\rangle + \epsilon^* |\psi_n^B\rangle
\end{aligned} \tag{8.83}$$

When inserting these eigenfunction coefficients at the corresponding atomic positions in the armchair unit cell, it is verified that $|\Psi_0\rangle$ and $|\Psi_n\rangle$ are complex conjugate. Their symmetry behavior is thus exactly the same as the complex conjugate partners of the ${}_{k}E_{m}$ eigenfunctions on the central and lateral vertical lines, with the special feature

however that they are not paired by a sign change of the momentum k, but by the switch of the angular momentum m from 0 to n. The symmetry of the two diamond states may thus be denoted as the ${}_{\pi/a}E_{\{0,n\}}$ irrep. In summary, the Brillouin zone, shown in Fig. 8.21, turns out to be a true map of the entire irreducible representation structure of the line group.

8.7 Problems

1. Consider a commensurate helix, generated by an n-fold screw axis. The cylindrical coordinates of successive helical points are given by

$$(R, \phi_t, z_t) = (\hat{C}_n|\frac{1}{n})^t(R, \phi_0, z_0) \tag{8.84}$$

   Let R be the radius of the helix, and H the helix step, i.e., the total increase of z in a single turn. Derive the parameter expressions for the cylindrical coordinates. The inclination of the helix is given by

$$\chi = \arctan \frac{H}{2\pi R} \tag{8.85}$$

   Note however that every helical arrangement can be drawn as a right-handed as well as a left-handed minimal helix, connecting all the points. The latter is generated by

$$(R, \phi_t, z_t) = (C_m^{-1}|\frac{1}{n})^t(R, \phi_0, z_0) \tag{8.86}$$

   Here $\hat{C}_m^{-1}$ is a left-handed rotation over an angle $-2\pi/m$, reaching the same point as the right-handed rotation $\hat{C}_n$ over an angle $2\pi/n$, hence: $m = n/(n-1)$. Determine the inclination χ' associated with the left helix, and calculate the sum: $(\tan\chi)^{-1} + (\tan\chi')^{-1}$.
   In view of this ambiguity, how would you decide whether a helix is right-handed or left-handed?
2. In Fig. 8.6, the unit cell contains not one but two $\hat{U}$ symmetry axes. Prove that these belong to the same coset. Idem for the inversion center: there is one in the center and one at the edge boundary of the unit cell, but they belong to the same coset.
3. Figure 8.15 shows that the hexagonal Bravais lattice has at the edge centers a twofold rotational axis perpendicular to the molecular plane. Show that this symmetry operation belongs to a coset of the sixfold rotation axis.
4. Figure 8.18 represents the wavefunction at the $\mathbf{M}_1$ point of graphene. The associated energy is equal to one unit of β. Can you obtain this eigenvalue on the basis of the orbital picture of a single unit cell?

5. Figure 8.20 shows generating elements of the armchair $\mathbf{L}8_4/mcm$ line group. The corresponding isogonal point group contains 32 symmetry elements and is isomorphic to D_{8h} point group, listed in Appendix A. Identify the correspondence between the elements of both groups.

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Appendix A
Character Tables

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Character tables were introduced to chemistry through the pioneering work of Robert Mulliken [1]. The book on "Chemical applications of Group Theory", by F. Albert Cotton, has been instrumental in disseminating their use in chemistry [2]. Atkins, Child, and Phillips [3] produced a handy pamphlet of the point group character tables.[1]

[1] In the tables the columns on the right list representative coordinate functions which transform according to the corresponding irrep. The symbols R_x, R_y, R_z stand for rotations about the Cartesian directions.

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A.1 Finite Point Groups

C_1 and the Binary Groups: C_s, C_i, C_2

C_1	$\hat{E}$
A	1

C_s	$\hat{E}$	$\hat{\sigma}_h$		
A'	1	1	x, y, R_z	x^2, y^2, z^2, xy
A''	1	-1	z, R_x, R_y	yz, xz

C_i	$\hat{E}$	$\hat{\imath}$		
A_g	1	1	R_x, R_y, R_z	$x^2, y^2, z^2, yz, xz, xy$
A_u	1	-1	x, y, z	

C_2	$\hat{E}$	$\hat{C}_2^z$		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

The Cyclic Groups: C_n ($n = 3, 4, 5, 6, 7, 8$)

C_3	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\epsilon = \exp(2\pi i/3)$	
A	1	1	1	z, R_z	$x^2 + y^2, z^2$
E	1	ϵ	$\overline{\epsilon}$	$(x, y)(R_x, R_y)$	$(x^2 - y^2, xy)(yz, xz)$
	1	$\overline{\epsilon}$	ϵ		

C_4	$\hat{E}$	$\hat{C}_4$	$\hat{C}_2$	$\hat{C}_4^3$		
A	1	1	1	1	z, R_z	$x^2 + y^2, z^2$
B	1	-1	1	-1		$x^2 - y^2, xy$
E	1	i	-1	$-i$	$(x, y)(R_x, R_y)$	(yz, xz)
	1	$-i$	-1	i		

C_5	$\hat{E}$	$\hat{C}_5$	$\hat{C}_5^2$	$\hat{C}_5^3$	$\hat{C}_5^4$	$\epsilon = \exp(2\pi i/5)$	
A	1	1	1	1	1	z, R_z	$x^2 + y^2, z^2$
E_1	1	ϵ	ϵ^2	$\bar{\epsilon}^2$	$\bar{\epsilon}$	$(x, y)(R_x, R_y)$	(yz, xz)
	1	$\bar{\epsilon}$	$\bar{\epsilon}^2$	ϵ^2	ϵ		
E_2	1	ϵ^2	$\bar{\epsilon}$	ϵ	$\bar{\epsilon}^2$		$(x^2 - y^2, xy)$
	1	$\bar{\epsilon}^2$	ϵ^2	$\bar{\epsilon}$	ϵ^2		

C_6	$\hat{E}$	$\hat{C}_6$	$\hat{C}_3$	$\hat{C}_2$	$\hat{C}_3^2$	$\hat{C}_6^5$	$\epsilon = \exp(2\pi i/6)$	
A	1	1	1	1	1	1	z, R_z	$x^2 + y^2, z^2$
B	1	-1	1	-1	1	-1		
E_1	1	ϵ	$-\bar{\epsilon}$	-1	$-\epsilon$	$\bar{\epsilon}$	$(x, y)(R_x, R_y)$	(yz, xz)
	1	$\bar{\epsilon}$	$-\epsilon$	-1	$-\bar{\epsilon}$	ϵ		
E_2	1	$-\bar{\epsilon}$	$-\epsilon$	1	$-\bar{\epsilon}$	$-\epsilon$		$(x^2 - y^2, xy)$
	1	$-\epsilon$	$-\bar{\epsilon}$	1	$-\epsilon$	$-\bar{\epsilon}$		

C_7	$\hat{E}$	$\hat{C}_7$	$\hat{C}_7^2$	$\hat{C}_7^3$	$\hat{C}_7^4$	$\hat{C}_7^5$	$\hat{C}_7^6$	$\epsilon = \exp(2\pi i/7)$	
A	1	1	1	1	1	1	1	z, R_z	$x^2 + y^2, z^2$
E_1	1	ϵ	ϵ^2	ϵ^3	$\bar{\epsilon}^3$	$\bar{\epsilon}^2$	$\bar{\epsilon}$	$(x, y)(R_x, R_y)$	(yz, xz)
	1	$\bar{\epsilon}$	$\bar{\epsilon}^2$	$\bar{\epsilon}^3$	ϵ^3	ϵ^2	ϵ		
E_2	1	ϵ^2	$\bar{\epsilon}^3$	$\bar{\epsilon}$	ϵ	ϵ^3	$\bar{\epsilon}^2$		$(x^2 - y^2, xy)$
	1	$\bar{\epsilon}^2$	ϵ^3	ϵ	$\bar{\epsilon}$	$\bar{\epsilon}^3$	ϵ^2		
E_3	1	ϵ^3	$\bar{\epsilon}$	ϵ^2	$\bar{\epsilon}^2$	ϵ	$\bar{\epsilon}^3$		$[x(x^2 - 3y^2),$
	1	$\bar{\epsilon}^3$	ϵ	$\bar{\epsilon}^2$	ϵ^2	$\bar{\epsilon}$	ϵ^3		$y(3x^2 - y^2)]$

C_8	$\hat{E}$	$\hat{C}_8$	$\hat{C}_4$	$\hat{C}_2$	$\hat{C}_4^3$	$\hat{C}_8^3$	$\hat{C}_8^5$	$\hat{C}_8^7$	$\epsilon = \exp(2\pi i/8)$	
A	1	1	1	1	1	1	1	1	z, R_z	$x^2 + y^2, z^2$
B	1	-1	1	1	1	-1	-1	-1		
E_1	1	ϵ	i	-1	$-i$	$-\bar{\epsilon}$	$-\epsilon$	$\bar{\epsilon}$	$(x, y)(R_x, R_y)$	(yz, xz)
	1	$\bar{\epsilon}$	$-i$	-1	i	$-\epsilon$	$-\bar{\epsilon}$	ϵ		
E_2	1	i	-1	1	-1	$-i$	i	$-i$		$(x^2 - y^2, xy)$
	1	$-i$	-1	1	-1	i	$-i$	i		
E_3	1	$-\epsilon$	i	-1	$-i$	$\bar{\epsilon}$	ϵ	$-\bar{\epsilon}$		$[x(x^2 - 3y^2),$
	1	$-\bar{\epsilon}$	$-i$	-1	i	ϵ	$\bar{\epsilon}$	$-\epsilon$		$y(3x^2 - y^2)]$

The Dihedral Groups: D_n ($n = 2, 3, 4, 5, 6$)

D_2	$\hat{E}$	$\hat{C}_2^z$	$\hat{C}_2^y$	$\hat{C}_2^x$		
A	1	1	1	1		x^2, y^2, z^2
B_1	1	1	-1	-1	z, R_z	xy
B_2	1	-1	1	-1	y, R_y	xz
B_3	1	-1	-1	1	x, R_x	yz

D_3	$\hat{E}$	$2\hat{C}_3$	$3\hat{C}_2$		
A_1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	-1	z, R_z	
E	2	-1	0	$(x, y)(R_x, R_y)$	$(xz, yz)(x^2 - y^2, xy)$

D_4	$\hat{E}$	$2\hat{C}_4$	$\hat{C}_2(=\hat{C}_4^2)$	$2\hat{C}_2'$	$2\hat{C}_2''$		
A_1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	1	-1	-1	z, R_z	
B_1	1	-1	1	1	-1		$x^2 - y^2$
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)

D_5	$\hat{E}$	$2\hat{C}_5$	$2\hat{C}_5^2$	$5\hat{C}_2'$		
A_1	1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	1	-1	z, R_z	
E_1	2	$2\cos(2\pi/5)$	$2\cos(4\pi/5)$	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	$2\cos(4\pi/5)$	$2\cos(2\pi/5)$	0		$(x^2 - y^2, xy)$

D_6	$\hat{E}$	$2\hat{C}_6$	$2\hat{C}_3$	$\hat{C}_2$	$3\hat{C}_2'$	$3\hat{C}_2''$		
A_1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	1	1	-1	-1	z, R_z	
B_1	1	-1	1	-1	1	-1		$x(x^2 - 3y^2)$
B_2	1	-1	1	-1	-1	1		$y(3x^2 - y^2)$
E_1	2	1	-1	-2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	-1	-1	2	0	0		$(x^2 - y^2, xy)$

The Conical Groups: C_{nv} ($n = 2, 3, 4, 5, 6$)

C_{2v}	$\hat{E}$	$\hat{C}_2^z$	$\hat{\sigma}_v^{xz}$	$\hat{\sigma}_v^{yz}$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	−1	−1	R_z	xy
B_1	1	−1	1	−1	x, R_y	xz
B_2	1	−1	−1	1	y, R_x	yz

C_{3v}	$\hat{E}$	$2\hat{C}_3$	$3\hat{\sigma}_v$		
A_1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	−1	R_z	
E	2	−1	0	$(x, y)(R_x, R_y)$	$(x^2 - y^2, xy)(xz, yz)$

C_{4v}	$\hat{E}$	$2\hat{C}_4$	$\hat{C}_2$	$2\hat{\sigma}_v$	$2\hat{\sigma}_d$		
A_1	1	1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	1	−1	−1	R_z	
B_1	1	−1	1	1	−1		$x^2 - y^2$
B_2	1	−1	1	−1	1		xy
E	2	0	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)

C_{5v}	$\hat{E}$	$2\hat{C}_5$	$2\hat{C}_5^2$	$5\hat{\sigma}_v$		
A_1	1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	1	−1	R_z	
E_1	2	$2\cos(2\pi/5)$	$2\cos(4\pi/5)$	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	$2\cos(4\pi/5)$	$2\cos(2\pi/5)$	0		$(x^2 - y^2, xy)$

C_{6v}	$\hat{E}$	$2\hat{C}_6$	$2\hat{C}_3$	$\hat{C}_2$	$3\hat{\sigma}_v$	$3\hat{\sigma}_d$		
A_1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2	1	1	1	1	−1	−1	z, R_z	
B_1	1	−1	1	−1	1	−1		$x(x^2 - 3y^2)$
B_2	1	−1	1	−1	−1	1		$y(3x^2 - y^2)$
E_1	2	1	−1	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)
E_2	2	−1	−1	2	0	0		$(x^2 - y^2, xy)$

The C_{nh} Groups ($n = 2, 3, 4, 5, 6$)

C_{2h}	$\hat{E}$	$\hat{C}_2^z$	$\hat{\imath}$	$\hat{\sigma}_h$		
A_g	1	1	1	1	R_z	x^2, y^2, z^2, xy
B_g	1	-1	1	-1	R_x, R_y	xz, yz
A_u	1	1	-1	-1	z	
B_u	1	-1	-1	1	x, y	

C_{3h}	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_h$	$\hat{S}_3$	$\hat{S}_3^2$	$\epsilon = \exp(2\pi i/3)$	
A'	1	1	1	1	1	1	R_z	$x^2 + y^2, z^2$
E'	1	ϵ	$\bar{\epsilon}$	1	ϵ	$\bar{\epsilon}$	(x, y)	$(x^2 - y^2, xy)$
	1	$\bar{\epsilon}$	ϵ	1	$\bar{\epsilon}$	ϵ		
A''	1	1	1	-1	-1	-1	z	
E''	1	ϵ	$\bar{\epsilon}$	-1	$-\epsilon$	$-\bar{\epsilon}$	(R_x, R_y)	(xz, yz)
	1	$\bar{\epsilon}$	ϵ	-1	$-\bar{\epsilon}$	$-\epsilon$		

C_{4h}	$\hat{E}$	$\hat{C}_4$	$\hat{C}_2$	$\hat{C}_4^3$	$\hat{\imath}$	$\hat{S}_4^3$	$\hat{\sigma}_h$	$\hat{S}_4$		
A_g	1	1	1	1	1	1	1	1	R_z	$x^2 + y^2, z^2$
B_g	1	-1	1	-1	1	-1	1	-1		$x^2 - y^2, xy$
E_g	1	i	-1	$-i$	1	i	-1	$-i$	(R_x, R_y)	(yz, xz)
	1	$-i$	-1	i	1	$-i$	-1	i		
A_u	1	1	1	1	-1	-1	-1	-1	R_z	$x^2 + y^2, z^2$
B_u	1	-1	1	-1	-1	1	-1	1		$x^2 - y^2, xy$
E_u	1	i	-1	$-i$	-1	$-i$	1	i	(x, y)	
	1	$-i$	-1	i	-1	i	1	$-i$		

C_{5h}	$\hat{E}$	$\hat{C}_5$	$\hat{C}_5^2$	$\hat{C}_5^3$	$\hat{C}_5^4$	$\hat{\sigma}_h$	$\hat{S}_5$	$\hat{S}_5^7$	$\hat{S}_5^3$	$\hat{S}_5^9$	$\epsilon = \exp(2\pi i/5)$	
A'	1	1	1	1	1	1	1	1	1	1	R_z	x^2+y^2, z^2
E_1'	1	ϵ	ϵ^2	$\bar{\epsilon}^2$	$\bar{\epsilon}$	1	ϵ	ϵ^2	$\bar{\epsilon}^2$	$\bar{\epsilon}$	(x, y)	
	1	$\bar{\epsilon}$	$\bar{\epsilon}^2$	ϵ^2	ϵ	1	$\bar{\epsilon}$	$\bar{\epsilon}^2$	ϵ^2	ϵ		
E_2'	1	ϵ^2	$\bar{\epsilon}$	ϵ	$\bar{\epsilon}^2$	1	ϵ^2	$\bar{\epsilon}$	ϵ	$\bar{\epsilon}^2$		(x^2-y^2, xy)
	1	$\bar{\epsilon}^2$	ϵ^2	$\bar{\epsilon}$	ϵ^2	1	$\bar{\epsilon}^2$	ϵ^2	$\bar{\epsilon}$	ϵ^2		
A''	1	1	1	1	1	−1	−1	−1	−1	−1	z	
E_1''	1	ϵ	ϵ^2	$\bar{\epsilon}^2$	$\bar{\epsilon}$	−1	$-\epsilon$	$-\epsilon^2$	$-\bar{\epsilon}^2$	$-\bar{\epsilon}$	(R_x, R_y)	(xz, yz)
	1	$\bar{\epsilon}$	$\bar{\epsilon}^2$	ϵ^2	ϵ	−1	$-\bar{\epsilon}$	$-\bar{\epsilon}^2$	$-\epsilon^2$	$-\epsilon$		
E_2''	1	ϵ^2	$\bar{\epsilon}$	ϵ	$\bar{\epsilon}^2$	−1	$-\epsilon^2$	$-\bar{\epsilon}$	$-\epsilon$	$-\bar{\epsilon}^2$		
	1	$\bar{\epsilon}^2$	ϵ^2	$\bar{\epsilon}$	ϵ^2	−1	$-\bar{\epsilon}^2$	$-\epsilon^2$	$-\bar{\epsilon}$	$-\epsilon^2$		

C_{6h}	$\hat{E}$	$\hat{C}_6$	$\hat{C}_3$	$\hat{C}_2$	$\hat{C}_3^2$	$\hat{C}_6^5$	$\hat{\imath}$	$\hat{S}_3^5$	$\hat{S}_6^5$	$\hat{\sigma}_h$	$\hat{S}_6$	$\hat{S}_3$	$\epsilon = \exp(2\pi i/6)$	
A_g	1	1	1	1	1	1	1	1	1	1	1	1	R_z	x^2+y^2, z^2
B_g	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1		
E_{1g}	1	ϵ	$-\bar{\epsilon}$	−1	$-\epsilon$	$\bar{\epsilon}$	1	ϵ	$-\bar{\epsilon}$	−1	$-\epsilon$	$\bar{\epsilon}$	$(R_x, R_y)(yz, xz)$	
	1	$\bar{\epsilon}$	$-\epsilon$	−1	$-\bar{\epsilon}$	ϵ	1	$\bar{\epsilon}$	$-\epsilon$	−1	$-\bar{\epsilon}$	ϵ		
E_{2g}	1	$-\bar{\epsilon}$	$-\epsilon$	1	$-\bar{\epsilon}$	$-\epsilon$	1	$-\bar{\epsilon}$	$-\epsilon$	1	$-\bar{\epsilon}$	$-\epsilon$		
	1	$-\epsilon$	$-\bar{\epsilon}$	1	$-\epsilon$	$-\bar{\epsilon}$	1	$-\epsilon$	$-\bar{\epsilon}$	1	$-\epsilon$	$-\bar{\epsilon}$		
A_u	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1	z	x^2+y^2, z^2
B_u	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1		
E_{1u}	1	ϵ	$-\bar{\epsilon}$	−1	$-\epsilon$	$\bar{\epsilon}$	−1	$-\epsilon$	$\bar{\epsilon}$	1	ϵ	$-\bar{\epsilon}$	(x, y)	
	1	$\bar{\epsilon}$	$-\epsilon$	−1	$-\bar{\epsilon}$	ϵ	−1	$-\bar{\epsilon}$	ϵ	1	$\bar{\epsilon}$	$-\epsilon$		
E_{2u}	1	$-\bar{\epsilon}$	$-\epsilon$	1	$-\bar{\epsilon}$	$-\epsilon$	−1	$\bar{\epsilon}$	ϵ	−1	$\bar{\epsilon}$	ϵ		
	1	$-\epsilon$	$-\bar{\epsilon}$	1	$-\epsilon$	$-\bar{\epsilon}$	−1	ϵ	$\bar{\epsilon}$	−1	ϵ	$\bar{\epsilon}$		

The Rotation–Reflection Groups: S_{2n} ($n = 2, 3, 4$)

S_4	$\hat{E}$	$\hat{S}_4$	$\hat{C}_2$	$\hat{S}_4^3$		
A	1	1	1	1	R_z	x^2+y^2, z^2
B	1	-1	1	-1	z	x^2-y^2, xy
E	1	i	-1	$-i$	$(x, y)(R_x, R_y)$	(xz, yz)
	1	$-i$	-1	i		

S_6	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\imath}$	$\hat{S}_6^5$	$\hat{S}_6$	$\epsilon = \exp(2\pi i/3)$	
A_g	1	1	1	1	1	1	R_z	x^2+y^2, z^2
E_g	1	ϵ	$\overline{\epsilon}$	1	ϵ	$\overline{\epsilon}$	(R_x, R_y)	$(x^2-y^2, xy)(yz, xz)$
	1	$\overline{\epsilon}$	ϵ	1	$\overline{\epsilon}$	ϵ		
A_u	1	1	1	-1	-1	-1	z	
E_u	1	ϵ	$\overline{\epsilon}$	-1	$-\epsilon$	$-\overline{\epsilon}$	(x, y)	
	1	$\overline{\epsilon}$	ϵ	-1	$-\overline{\epsilon}$	$-\epsilon$		

S_8	$\hat{E}$	$\hat{S}_8$	$\hat{C}_4$	$\hat{S}_8^3$	$\hat{C}_2$	$\hat{S}_8^5$	$\hat{C}_4^3$	$\hat{S}_8^7$	$\epsilon = \exp(2\pi i/8)$	
A	1	1	1	1	1	1	1	1	R_z	x^2+y^2, z^2
B	1	-1	1	-1	1	-1	1	-1	z	
E_1	1	ϵ	i	$-\overline{\epsilon}$	-1	$-\epsilon$	$-i$	$\overline{\epsilon}$	$(x, y)(R_x, R_y)$	
	1	$\overline{\epsilon}$	$-i$	$-\epsilon$	-1	$-\overline{\epsilon}$	i	ϵ		
E_2	1	i	-1	$-i$	1	i	-1	$-i$		(x^2-y^2, xy)
	1	$-i$	-1	i	1	$-i$	-1	i		
E_3	1	$-\overline{\epsilon}$	$-i$	ϵ	-1	$\overline{\epsilon}$	i	$-\epsilon$		(xz, yz)
	1	$-\epsilon$	i	$\overline{\epsilon}$	-1	ϵ	$-i$	$-\overline{\epsilon}$		

The Prismatic Groups: D_{nh} ($n = 2, 3, 4, 5, 6, 8$)

D_{2h}	$\hat{E}$	$\hat{C}_2^z$	$\hat{C}_2^y$	$\hat{C}_2^x$	$\hat{\imath}$	$\hat{\sigma}_{xy}$	$\hat{\sigma}_{xz}$	$\hat{\sigma}_{yz}$		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	−1	−1	1	1	−1	−1	R_z	xy
B_{2g}	1	−1	1	−1	1	−1	1	−1	R_y	xz
B_{3g}	1	−1	−1	1	1	−1	−1	1	R_x	yz
A_u	1	1	1	1	−1	−1	−1	−1		xyz
B_{1u}	1	1	−1	−1	−1	−1	1	1	z	
B_{2u}	1	−1	1	−1	−1	1	−1	1	y	
B_{3u}	1	−1	−1	1	−1	1	1	−1	x	

D_{3h}	$\hat{E}$	$2\hat{C}_3$	$3\hat{C}_2$	$\hat{\sigma}_h$	$2\hat{S}_3$	$3\hat{\sigma}_v$		
A_1'	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2'	1	1	−1	1	1	−1	R_z	
E'	2	−1	0	2	−1	0	(x, y)	$(x^2 - y^2, xy)$
A_1''	1	1	1	−1	−1	−1		
A_2''	1	1	−1	−1	−1	1	z	
E''	2	−1	0	−2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	$\hat{E}$	$2\hat{C}_4$	$\hat{C}_2$	$2\hat{C}_2'$	$2\hat{C}_2''$	$\hat{\imath}$	$2\hat{S}_4$	$\hat{\sigma}_h$	$2\hat{\sigma}_v$	$2\hat{\sigma}_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_{2g}	1	1	1	−1	−1	1	1	1	−1	−1	R_z	
B_{1g}	1	−1	1	1	−1	1	−1	1	1	−1		$x^2 - y^2$
B_{2g}	1	−1	1	−1	1	1	−1	1	−1	1		xy
E_g	2	0	−2	0	0	2	0	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)
A_{1u}	1	1	1	1	1	−1	−1	−1	−1	−1		
A_{2u}	1	1	1	−1	−1	−1	−1	−1	1	1	z	
B_{1u}	1	−1	1	1	−1	−1	1	−1	−1	1		
B_{2u}	1	−1	1	−1	1	−1	1	−1	1	−1		
E_u	2	0	−2	0	0	−2	0	2	0	0	(x, y)	

D_{5h}	$\hat{E}$	$2\hat{C}_5$	$2\hat{C}_5^2$	$5\hat{C}_2$	$\hat{\sigma}_h$	$2\hat{S}_5$	$2\hat{S}_5^3$	$5\hat{\sigma}_v$	$\alpha = \cos(2\pi/5)$	$\beta = \cos(4\pi/5)$
A_1'	1	1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A_2'	1	1	1	−1	1	1	1	−1	R_z	
E_1'	2	2α	2β	0	2	2α	2β	0	(x, y)	
E_2'	2	2β	2α	0	2	2β	2α	0		$(x^2 - y^2, xy)$
A_1''	1	1	1	1	−1	−1	−1	−1		
A_2''	1	1	1	−1	−1	−1	−1	1	z	
E_1''	2	2α	2β	0	−2	-2α	-2β	0	(R_x, R_y)	(xz, yz)
E_2''	2	2β	2α	0	−2	-2β	-2α	0		

D_{6h}	$\hat{E}$	$2\hat{C}_6$	$2\hat{C}_3$	$\hat{C}_2$	$3\hat{C}_2'$	$3\hat{C}_2''$	$\hat{\imath}$	$2\hat{S}_3$	$2\hat{S}_6$	$\hat{\sigma}_h$	$3\hat{\sigma}_d$	$3\hat{\sigma}_v$	
A_{1g}	1	1	1	1	1	1	1	1	1	1	1	1	$x^2 + y^2, z^2$
A_{2g}	1	1	1	1	−1	−1	1	1	1	1	−1	−1	R_z
B_{1g}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1	
B_{2g}	1	−1	1	−1	−1	1	1	−1	1	−1	−1	1	
E_{1g}	2	1	−1	−2	0	0	2	1	−1	−2	0	0	$(R_x, R_y)(xz, yz)$
E_{2g}	2	−1	−1	2	0	0	2	−1	−1	2	0	0	$(x^2 - y^2, xy)$
A_{1u}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1	
A_{2u}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1	z
B_{1u}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1	$x(x^2 - 3y^2)$
B_{2u}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1	$y(3x^2 - y^2)$
E_{1u}	2	1	−1	−2	0	0	−2	−1	1	2	0	0	(x, y)
E_{2u}	2	−1	−1	2	0	0	−2	1	1	−2	0	0	

D_{8h}	$\hat{E}$	$2\hat{C}_8$	$2\hat{C}_8^3$	$2\hat{C}_4$	$\hat{C}_2$	$4\hat{C}_2'$	$4\hat{C}_2''$	$\hat{\imath}$	$2\hat{S}_8^3$	$2\hat{S}_8$	$2\hat{S}_4$	$\hat{\sigma}_h$	$4\hat{\sigma}_v$	$4\hat{\sigma}_d$
A_{1g}	1	1	1	1	1	1	1	1	1	1	1	1	1	1
A_{2g}	1	1	1	1	1	−1	−1	1	1	1	1	1	−1	−1
B_{1g}	1	−1	−1	1	1	1	−1	1	−1	−1	1	1	1	−1
B_{2g}	1	−1	−1	1	1	−1	1	1	−1	−1	1	1	−1	1
E_{1g}	2	$\sqrt{2}$	$-\sqrt{2}$	0	−2	0	0	2	$\sqrt{2}$	$-\sqrt{2}$	0	−2	0	0
E_{2g}	2	0	0	−2	2	0	0	2	0	0	−2	2	0	0
E_{3g}	2	$-\sqrt{2}$	$\sqrt{2}$	0	−2	0	0	2	$-\sqrt{2}$	$\sqrt{2}$	0	−2	0	0
A_{1u}	1	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1	−1
A_{2u}	1	1	1	1	1	−1	−1	−1	−1	−1	−1	−1	1	1
B_{1u}	1	−1	−1	1	1	1	−1	−1	1	1	−1	−1	−1	1
B_{2u}	1	−1	−1	1	1	−1	1	−1	1	1	−1	−1	1	−1
E_{1u}	2	$\sqrt{2}$	$-\sqrt{2}$	0	−2	0	0	−2	$-\sqrt{2}$	$\sqrt{2}$	0	2	0	0
E_{2u}	2	0	0	−2	2	0	0	−2	0	0	2	−2	0	0
E_{3u}	2	$-\sqrt{2}$	$\sqrt{2}$	0	−2	0	0	−2	$\sqrt{2}$	$-\sqrt{2}$	0	2	0	0

The Anti-prismatic Groups: D_{nd} ($n = 2, 3, 4, 5, 6$)

D_{2d}	$\hat{E}$	$2\hat{S}_4$	$\hat{C}_2$	$2\hat{C}_2'$	$2\hat{\sigma}_d$		
A_1	1	1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	1	−1	−1	R_z	
B_1	1	−1	1	1	−1		$x^2 - y^2$
B_2	1	−1	1	−1	1		xy
E	2	0	−2	0	0	$(x, y)(R_x, R_y)$	(xz, yz)

D_{3d}	$\hat{E}$	$2\hat{C}_3$	$3\hat{C}_2$	$\hat{\imath}$	$2\hat{S}_6$	$3\hat{\sigma}_d$		
A_{1g}	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	−1	1	1	−1	R_z	
E_g	2	−1	0	2	−1	0	(R_x, R_y)	$(x^2-y^2, xy)(xz, yz)$
A_{1u}	1	1	1	−1	−1	−1		
A_{2u}	1	1	−1	−1	−1	1	z	
E_u	2	−1	0	−2	1	0	(x, y)	

D_{4d}	$\hat{E}$	$2\hat{S}_8$	$2\hat{C}_4$	$2\hat{S}_8^3$	$\hat{C}_2$	$4\hat{C}_2'$	$4\hat{\sigma}_d$		
A_1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	1	1	−1	−1	R_z	
B_1	1	−1	1	−1	1	1	−1		
B_2	1	−1	1	−1	1	−1	1	z	
E_1	2	$\sqrt{2}$	0	$-\sqrt{2}$	−2	−0	0	(x, y)	
E_2	2	0	−2	0	2	0	0		(x^2-y^2, xy)
E_3	2	$-\sqrt{2}$	0	$\sqrt{2}$	−2	0	0	(R_x, R_y)	(xz, yz)

D_{5d}	$\hat{E}$	$2\hat{C}_5$	$2\hat{C}_5^2$	$5\hat{C}_2$	$\hat{\imath}$	$2\hat{S}_{10}^3$	$2\hat{S}_{10}$	$5\hat{\sigma}_d$	$\alpha=\cos(2\pi/5)$	$\beta=\cos(4\pi/5)$
A_{1g}	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	−1	1	1	1	−1	R_z	
E_{1g}	2	2α	2β	0	2	2α	2β	0	(R_x, R_y)	(xz, yz)
E_{2g}	2	2β	2α	0	2	2β	2α	0		(x^2-y^2, xy)
A_{1u}	1	1	1	1	−1	−1	−1	−1		
A_{2u}	1	1	1	−1	−1	−1	−1	1	z	
E_{1u}	2	2α	2β	0	−2	-2α	-2β	0	(x, y)	
E_{2u}	2	2β	2α	0	−2	-2β	-2α	0		

D_{6d}	$\hat{E}$	$2\hat{S}_{12}$	$2\hat{C}_6$	$2\hat{S}_4$	$2\hat{C}_3$	$2\hat{S}_{12}^5$	$\hat{C}_2$	$6\hat{C}_2'$	$6\hat{\sigma}_d$		
A_1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_2	1	1	1	1	1	1	1	−1	−1	R_z	
B_1	1	−1	1	−1	1	−1	1	1	−1		
B_2	1	−1	1	−1	1	−1	1	−1	1	z	
E_1	2	$\sqrt{3}$	1	0	−1	$-\sqrt{3}$	−2	0	0	(x, y)	
E_2	2	1	−1	−2	−1	1	2	0	0		(x^2-y^2, xy)
E_3	2	0	−2	0	2	0	−2	0	0		
E_4	2	−1	−1	2	−1	−1	2	0	0		
E_5	2	$-\sqrt{3}$	1	0	−1	$\sqrt{3}$	−2	0	0	(R_x, R_y)	(xz, yz)

The Tetrahedral and Cubic Groups

T	$\hat{E}$	$4\hat{C}_3$	$4\hat{C}_3^2$	$3\hat{C}_2$	$\epsilon = \exp(2\pi i/3)$	
A	1	1	1	1		$x^2+y^2+z^2$
E {	1	ϵ	$\bar{\epsilon}$	1		$(2z^2-x^2-y^2, x^2-y^2)$
	1	$\bar{\epsilon}$	ϵ	1		
T	3	0	0	−1	$(R_x, R_y, R_z)(x, y, z)$	(xz, yx, xy)

T_d	$\hat{E}$	$8\hat{C}_3$	$3\hat{C}_2$	$6\hat{S}_4$	$6\hat{\sigma}_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	−1	−1		
E	2	−1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	−1	1	−1	(R_x, R_y, R_z)	
T_2	3	0	−1	−1	1	(x, y, z)	(xz, yz, xy)

T_h	$\hat{E}$	$4\hat{C}_3$	$4\hat{C}_3^2$	$3\hat{C}_2$	$\hat{\imath}$	$4\hat{S}_6^5$	$4\hat{S}_6$	$3\hat{\sigma}_h$	$\epsilon = \exp(2\pi i/3)$	
A_g	1	1	1	1	1	1	1	1		$x^2 + y^2 + z^2$
A_u	1	1	1	1	−1	−1	−1	−1		
E_g	1	ϵ	$\bar{\epsilon}$	1	1	ϵ	$\bar{\epsilon}$	1		$(2z^2 - x^2 - y^2, x^2 - y^2)$
	1	$\bar{\epsilon}$	ϵ	1	1	$\bar{\epsilon}$	ϵ	1		
E_u	1	ϵ	$\bar{\epsilon}$	1	−1	$-\epsilon$	$-\bar{\epsilon}$	−1		
	1	$\bar{\epsilon}$	ϵ	1	−1	$-\bar{\epsilon}$	$-\epsilon$	−1		
T_g	3	0	0	−1	3	0	0	−1	(R_x, R_y, R_z)	(xz, yz, xy)
T_u	3	0	0	−1	−3	0	0	1	(x, y, z)	

O	$\hat{E}$	$6\hat{C}_4$	$3\hat{C}_2 (= \hat{C}_4^2)$	$8\hat{C}_3$	$6\hat{C}_2$		
A_1	1	1	1	1	1		$x^2 + y^2 + z^2$
A_2	1	−1	1	1	−1		
E	2	0	2	−1	0		$(2z^2 - x^2 - y^2, x^2 - y^2)$
T_1	3	1	−1	0	−1	$(R_x, R_y, R_z)(x, y, z)$	
T_2	3	−1	−1	−1	1		(xz, yz, xy)

O_h	$\hat{E}$	$8\hat{C}_3$	$6\hat{C}_2$	$6\hat{C}_4$	$3\hat{C}_2$	$\hat{\imath}$	$6\hat{S}_4$	$8\hat{S}_6$	$3\hat{\sigma}_h$	$6\hat{\sigma}_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		$x^2 + y^2 + z^2$
A_{2g}	1	1	−1	−1	1	1	−1	1	1	−1		
E_g	2	−1	0	0	2	2	0	−1	2	0		$(2z^2 - x^2 - y^2, x^2 - y^2)$
T_{1g}	3	0	−1	1	−1	3	1	0	−1	−1	(R_x, R_y, R_z)	
T_{2g}	3	0	1	−1	−1	3	−1	0	−1	1		(xz, yz, xy)
A_{1u}	1	1	1	1	1	−1	−1	−1	−1	−1		
A_{2u}	1	1	−1	−1	1	−1	1	−1	−1	1		
E_u	2	−1	0	0	2	−2	0	1	−2	0		
T_{1u}	3	0	−1	1	−1	−3	−1	0	1	1	(x, y, z)	
T_{2u}	3	0	1	−1	−1	−3	1	0	1	−1		

The Icosahedral Groups

I	$\hat{E}$	$12\hat{C}_5$	$12\hat{C}_5^2$	$20\hat{C}_3$	$15\hat{C}_2$	$\phi = (1+\sqrt{5})/2$	
A	1	1	1	1	1		$x^2+y^2+z^2$
T_1	3	ϕ	$-\phi^{-1}$	0	-1	$(R_x, R_y, R_z)(x, y, z)$	
T_2	3	$-\phi^{-1}$	ϕ	0	-1		
G	4	-1	-1	1	0		
H	5	0	0	-1	1		$(2z^2-x^2-y^2, x^2-y^2,$ (xz, yz, xy)

I_h	$\hat{E}$	$12\hat{C}_5$	$12\hat{C}_5^2$	$20\hat{C}_3$	$15\hat{C}_2$	$\hat{\imath}$	$12\hat{S}_{10}$	$12\hat{S}_{10}^3$	$20\hat{S}_6$	$15\hat{\sigma}$	$\phi = (1+\sqrt{5})/2$
A_g	1	1	1	1	1	1	1	1	1	1	$x^2+y^2+z^2$
T_{1g}	3	ϕ	$-\phi^{-1}$	0	-1	3	$-\phi^{-1}$	ϕ	0	-1	(R_x, R_y, R_z)
T_{2g}	3	$-\phi^{-1}$	ϕ	0	-1	3	ϕ	$-\phi^{-1}$	0	-1	
G_g	4	-1	-1	1	0	4	-1	-1	1	0	
H_g	5	0	0	-1	1	5	0	0	-1	1	$(2z^2-x^2-y^2,$ $x^2-y^2,$ $xz, yz, xy)$
A_u	1	1	1	1	1	-1	-1	-1	-1	-1	
T_{1u}	3	ϕ	$-\phi^{-1}$	0	-1	-3	ϕ^{-1}	$-\phi$	0	1	(x, y, z)
T_{2u}	3	$-\phi^{-1}$	ϕ	0	-1	-3	$-\phi$	ϕ^{-1}	0	1	
G_u	4	-1	-1	1	0	-4	1	1	-1	0	
H_u	5	0	0	-1	1	-5	0	0	1	-1	

The Symmetric Groups

$\mathbf{S}_3 \sim C_{3v}$	1	3	2
	(1^3)	$(1, 2)$	(3)
(3)	1	1	1
$(2, 1)$	2	0	-1
(1^3)	1	-1	1

$\mathbf{S}_4 \sim T_d$	1	6	8	3	6
	(1^4)	$(1^2, 2)$	$(1, 3)$	(2^2)	(4)
(4)	1	1	1	1	1
$(3, 1)$	3	1	0	-1	-1
(2^2)	2	0	-1	2	0
$(2, 1^2)$	3	-1	0	-1	1
(1^4)	1	-1	1	1	-1

$\mathbf{S}_5$	1	10	20	15	30	20	24
	(1^5)	$(1^3, 2)$	$(1^2, 3)$	$(1, 2^2)$	$(1, 4)$	$(2, 3)$	(5)
(5)	1	1	1	1	1	1	1
$(4, 1)$	4	2	1	0	0	-1	-1
$(3, 2)$	5	1	-1	1	-1	1	0
$(3, 1^2)$	6	0	0	-2	0	0	1
$(2^2, 1)$	5	-1	-1	1	1	-1	0
$(2, 1^3)$	4	-2	1	0	0	1	-1
(1^5)	1	-1	1	1	-1	-1	1

A.2 Infinite Groups

Cylindrical Symmetry

C_∞	$\hat{E}$	$\hat{C}_2$	$\hat{C}_\phi$		
Σ	1	1	1	z	x^2+y^2, z^2
Π	1	−1	$\exp(i\phi)$	(x, y)	(xz, yz)
	1	−1	$\exp(-i\phi)$		
Δ	1	1	$\exp(2i\phi)$		(x^2-y^2, xy)
	1	1	$\exp(-2i\phi)$		
Φ	1	−1	$\exp(3i\phi)$		$\left[x(x^2-3y^2), y(3x^2-y^2)\right]$
	1	−1	$\exp(-3i\phi)$		

$C_{\infty v}$	$\hat{E}$	$\hat{C}_2$	$2\hat{C}_\phi$	$\infty\hat{\sigma}_v$		
Σ^+	1	1	1	1	z	x^2+y^2, z^2
Σ^-	1	1	1	−1	R_z	
Π	2	−2	$2\cos(\phi)$	0	$(x, y)(R_x, R_y)$	(xz, yz)
Δ	2	2	$2\cos(2\phi)$	0		(x^2-y^2, xy)
Φ	2	−2	$2\cos(3\phi)$	0		

$C_{\infty h}$	$\hat{E}$	$\hat{C}_2$	$\hat{C}_\phi$	$\hat{\imath}$	$\hat{S}_\phi$	$\hat{\sigma}_h$		
Σ_g	1	1	1	1	1	1		x^2+y^2, z^2
Π_g	1	−1	$\exp(i\phi)$	1	$-\exp(i\phi)$	−1		(xz, yz)
	1	−1	$\exp(-i\phi)$	1	$-\exp(-i\phi)$	−1		
Δ_g	1	1	$\exp(2i\phi)$	1	$\exp(2i\phi)$	1		(x^2-y^2, xy)
	1	1	$\exp(-2i\phi)$	1	$\exp(-2i\phi)$	1		
Φ_g	1	−1	$\exp(3i\phi)$	1	$-\exp(3i\phi)$	−1		
	1	−1	$\exp(-3i\phi)$	1	$-\exp(-3i\phi)$	−1		
Σ_u	1	1	1	−1	−1	−1	z	z^3
Π_u	1	−1	$\exp(i\phi)$	−1	$\exp(i\phi)$	1	(x, y)	(xz^2, yz^2)
	1	−1	$\exp(-i\phi)$	−1	$\exp(-i\phi)$	1		
Δ_u	1	1	$\exp(2i\phi)$	−1	$-\exp(2i\phi)$	−1		$\left((x^2-y^2)z, xyz\right)$
	1	1	$\exp(-2i\phi)$	−1	$-\exp(-2i\phi)$	−1		
Φ_u	1	−1	$\exp(3i\phi)$	−1	$\exp(3i\phi)$	1		
	1	−1	$\exp(-3i\phi)$	−1	$\exp(-3i\phi)$	1		

$D_{\infty h}$	$\hat{E}$	$\hat{C}_2^z$	$2\hat{C}_\phi$	$\infty\hat{\sigma}_v$	$\hat{\imath}$	$2\hat{S}_\phi$	$\infty\hat{C}_2^\perp$		
Σ_g^+	1	1	1	1	1	1	1		x^2+y^2, z^2
Σ_g^-	1	1	1	−1	1	1	−1	R_z	
Π_g	2	−2	$2\cos(\phi)$	0	2	$-2\cos(\phi)$	0	(R_x, R_y)	(xz, yz)
Δ_g	2	2	$2\cos(2\phi)$	0	2	$2\cos(2\phi)$	0		(x^2-y^2, xy)
Φ_g	2	−2	$2\cos(3\phi)$	0	2	$-2\cos(3\phi)$	0		
...									
Σ_u^+	1	1	1	1	−1	−1	−1	z	
Σ_u^-	1	1	1	−1	−1	−1	1		
Π_u	2	−2	$2\cos(\phi)$	0	−2	$2\cos(\phi)$	0	(x, y)	
Δ_u	2	2	$2\cos(2\phi)$	0	−2	$-2\cos(2\phi)$	0		
Φ_u	2	−2	$2\cos(3\phi)$	0	−2	$2\cos(3\phi)$	0		

Spherical Symmetry

$SO(3)$	$\hat{E}$	$\infty\hat{C}_\alpha^{\phi\theta}$		
S	1	1		$x^2+y^2+z^2$
P	3	$1+2\cos\alpha$	$(x, y, z)(R_x, R_y, R_z)$	
D	5	$1+2\cos\alpha+2\cos 2\alpha$		$(2z^2-x^2-y^2,$ $x^2-y^2, xz, yz, xy)$
...				
L	$2L+1$	$\frac{\sin(L+\frac{1}{2})\alpha}{sin\frac{1}{2}\alpha}$		

$O(3)$	$\hat{E}$	$\infty\hat{C}_\alpha^{\phi\theta}$	$\hat{\imath}$	$\infty\hat{\sigma}$	$\infty\hat{S}_\alpha^{\theta,\phi}$	
S_g	1	1	1	1	1	$x^2+y^2+z^2$
P_g	3	$1+2\cos\alpha$	3	-1	$1-2\cos\alpha$	(R_x, R_y, R_z)
D_g	5	$1+2\cos\alpha+2\cos 2\alpha$	5	1	$1-2\cos\alpha+2\cos 2\alpha$	d-orbitals
L_g	$2L+1$	$\frac{\sin(L+\frac{1}{2})\alpha}{sin\frac{1}{2}\alpha}$	$2L+1$	$(-1)^L$	$\frac{(-1)^L\cos(L+\frac{1}{2})\alpha}{cos\frac{1}{2}\alpha}$	
...						
S_u	1	1	-1	-1	-1	$x^2+y^2+z^2$
P_u	3	$1+2\cos\alpha$	-3	1	$-1+2\cos\alpha$	p-orbitals
D_u	5	$1+2\cos\alpha+2\cos 2\alpha$	-5	-1	$-1+2\cos\alpha-2\cos 2\alpha$	
L_u	$2L+1$	$\frac{\sin(L+\frac{1}{2})\alpha}{sin\frac{1}{2}\alpha}$	$-(2L+1)$	$(-1)^{L+1}$	$\frac{(-1)^{L+1}\cos(L+\frac{1}{2})\alpha}{cos\frac{1}{2}\alpha}$	

Appendix B
Symmetry Breaking by Uniform Linear Electric and Magnetic Fields

Contents

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B.2 Binary and Cylindrical Groups 279

B.1 Spherical Groups

G	B	E
T	C_3, C_2, C_1	C_3, C_2, C_1
T_d	S_4, C_3, C_2, C_s, C_1	C_{3v}, C_{2v}, C_s, C_1
T_h	S_6, C_{2h}, C_i	C_3, C_2, C_s, C_1
O	C_4, C_3, C_2, C_1	C_4, C_3, C_2, C_1
O_h	C_{4h}, S_6, C_{2h}, C_i	$C_{4v}, C_{3v}, C_{2v}, C_s, C_1$
I	C_5, C_3, C_2, C_1	C_5, C_3, C_2, C_1
I_h	S_{10}, S_6, C_{2h}, C_i	$C_{5v}, C_{3v}, C_{2v}, C_s, C_1$

B.2 Binary and Cylindrical Groups

The $\parallel$ notation refers to a field oriented along the principal cylindrical axis; in the $\perp$ direction several symmetry breakings are possible: C_2 symmetry implies that the

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field coincides with a $\hat{C}_2$ axis; a magnetic field perpendicular to a symmetry plane or an electric field in a symmetry plane will conserve at least C_s symmetry.

G	B		E	
	$\parallel$	$\perp$	$\parallel$	$\perp$
C_i	C_i		C_1	
C_s	C_1	C_s	C_s	C_1
C_2	C_2	C_1	C_2	C_1
C_n	C_n	C_1	C_n	C_1
D_n	C_n	C_2, C_1	C_n	C_2, C_1
C_{nv}	C_n	C_s, C_1	C_{nv}	C_s, C_1
C_{2nh}	C_{2nh}	C_i	C_{2n}	C_s
$C_{(2n+1)h}$	$C_{(2n+1)h}$	C_1	$C_{(2n+1)}$	C_s
S_{4n}	S_{4n}	C_1	C_{2n}	C_1
S_{4n+2}	S_{4n+2}	C_i	C_{2n+1}	C_1
D_{2nh}	C_{2nh}	C_{2h}, C_i	C_{2nv}	C_{2v}, C_s
$D_{(2n+1)h}$	$C_{(2n+1)h}$	C_2, C_s, C_1	$C_{(2n+1)v}$	C_2, C_s
D_{2nd}	S_{4n}	C_2, C_s, C_1	C_{2nv}	C_2, C_s, C_1
$D_{(2n+1)d}$	S_{4n+2}	C_{2h}, C_i	$C_{(2n+1)v}$	C_2, C_s, C_1

Appendix C
Subduction and Induction

Contents

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C.1 Subduction $G \downarrow H$

C_{6v}	C_6	$\frac{C_{3v}^v}{2\hat{C}_3, 3\hat{\sigma}_v}$	$\frac{C_{3v}^d}{2\hat{C}_3, 3\hat{\sigma}_d}$	C_{2v}
A_1	A	A_1	A_1	A_1
A_2	A	A_2	A_2	A_2
B_1	B	A_1	A_2	B_1
B_2	B	A_2	A_1	B_2
E_1	E_1	E	E	$B_1 + B_2$
E_2	E_2	E	E	$A_1 + A_2$

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D_{4h}	$\frac{D_{2d}}{\hat{C}_2^z,2\hat{C}_2',2\hat{\sigma}_d}$	C_{4h}	$\frac{C_{2v}^{zv}}{\hat{C}_2^z,2\hat{\sigma}_v}$	$\frac{C_{2v}^{zd}}{\hat{C}_2^z,2\hat{\sigma}_d}$	$\frac{C_{2v}'}{\hat{C}_2',\hat{\sigma}_h,\hat{\sigma}_v}$	$\frac{C_{2v}''}{\hat{C}_2'',\hat{\sigma}_h,\hat{\sigma}_d}$
A_{1g}	A_1	A_g	A_1	A_1	A_1	A_1
A_{2g}	A_2	A_g	A_2	A_2	B_1	B_1
B_{1g}	B_1	B_g	A_1	A_2	A_1	B_1
B_{2g}	B_2	B_g	A_2	A_1	B_1	A_1
E_g	E	E	B_1+B_2	B_1+B_2	A_2+B_2	A_2+B_2
A_{1u}	B_1	A_u	A_2	A_2	A_2	A_2
A_{2u}	B_2	A_u	A_1	A_1	B_2	B_2
B_{1u}	A_1	B_u	A_2	A_1	A_2	B_2
B_{2u}	A_2	B_u	A_1	A_2	B_2	A_2
E_u	E	E	B_1+B_2	B_1+B_2	A_1+B_1	A_1+B_1

D_{3d}	S_6	C_{3v}	C_{2h}
A_{1g}	A_g	A_1	A_g
A_{2g}	A_g	A_2	B_g
E_g	E_g	E	A_g+B_g
A_{1u}	A_u	A_2	A_u
A_{2u}	A_u	A_1	B_u
E_u	E_u	E	A_u+B_u

T_d	D_{2d}	C_{3v}	D_2	C_{2v}	S_4
A_1	A_1	A_1	A	A_1	A
A_2	B_1	A_2	A	A_2	B
E	A_1+B_1	E	$2A$	A_1+A_2	$A+B$
T_1	A_2+E	A_2+E	$B_1+B_2+B_3$	$A_2+B_1+B_2$	$A+E$
T_2	B_2+E	A_1+E	$B_1+B_2+B_3$	$A_1+B_1+B_2$	$B+E$

T_h	D_{2h}	S_6	C_{2v}	C_{2h}
A_g	A_g	A_g	A_1	A_g
A_u	A_u	A_u	A_2	A_u
E_g	$2A_g$	E_g	$2A_1$	$2A_g$
E_u	$2A_u$	E_u	$2A_2$	$2A_u$
T_g	$B_{1g} + B_{2g} + B_{3g}$	$A_g + E_g$	$A_2 + B_1 + B_2$	$A_g + 2B_g$
T_u	$B_{1u} + B_{2u} + B_{3u}$	$A_u + E_u$	$A_1 + B_1 + B_2$	$A_u + 2B_u$

O_h	T_d	D_{4h}	D_{3d}	T_h	D^z_{2h}	D'_{2h}
A_{1g}	A_1	A_{1g}	A_{1g}	A_g	A_g	A_g
A_{2g}	A_2	B_{1g}	A_{2g}	A_g	A_g	B_{1g}
E_g	E	$A_{1g} + B_{1g}$	E_g	E_g	$2A_g$	$A_g + B_{1g}$
T_{1g}	T_1	$A_{2g} + E_g$	$A_{2g} + E_g$	T_g	$B_{1g} + B_{2g} + B_{3g}$	$B_{1g} + B_{2g} + B_{3g}$
T_{2g}	T_2	$B_{2g} + E_g$	$A_{1g} + E_g$	T_g	$B_{1g} + B_{2g} + B_{3g}$	$A_g + B_{2g} + B_{3g}$
A_{1u}	A_2	A_{1u}	A_{1u}	A_u	A_u	A_u
A_{2u}	A_1	B_{1u}	A_{2u}	A_u	A_u	B_{1u}
E_u	E	$A_{1u} + B_{1u}$	E_u	E_u	$2A_u$	$A_u + B_{1u}$
T_{1u}	T_2	$A_{2u} + E_u$	$A_{2u} + E_u$	T_u	$B_{1u} + B_{2u} + B_{3u}$	$B_{1u} + B_{2u} + B_{3u}$
T_{2u}	T_1	$B_{2u} + E_u$	$A_{1u} + E_u$	T_u	$B_{1u} + B_{2u} + B_{3u}$	$A_u + B_{2u} + B_{3u}$

I_h	T_h	D_{5d}	D_{3d}	D_{2h}	C_{2v}
A_g	A_g	A_{1g}	A_{1g}	A_g	A_1
T_{1g}	T_g	$A_{2g} + E_{1g}$	$A_{2g} + E_g$	$B_{1g} + B_{2g} + B_{3g}$	$A_2 + B_1 + B_2$
T_{2g}	T_g	$A_{2g} + E_{2g}$	$A_{2g} + E_g$	$B_{1g} + B_{2g} + B_{3g}$	$A_2 + B_1 + B_2$
G_g	$A_g + T_g$	$E_{1g} + E_{2g}$	$A_{1g} + A_{2g} + E_g$	$A_g + B_{1g} + B_{2g} + B_{3g}$	$A_1 + A_2 + B_1 + B_2$
H_g	$E_g + T_g$	$A_{1g} + E_{1g} + E_{2g}$	$A_{1g} + 2E_g$	$2A_g + B_{1g} + B_{2g} + B_{3g}$	$2A_1 + A_2 + B_2 + B_3$
A_u	A_u	A_{1u}	A_{1u}	A_u	A_2
T_{1u}	T_u	$A_{2u} + E_{1u}$	$A_{2u} + E_u$	$B_{1u} + B_{2u} + B_{3u}$	$A_1 + B_1 + B_2$
T_{2u}	T_u	$A_{2u} + E_{2u}$	$A_{2u} + E_u$	$B_{1u} + B_{2u} + B_{3u}$	$A_1 + B_1 + B_2$
G_u	$A_u + T_u$	$E_{1u} + E_{2u}$	$A_{1u} + A_{2u} + E_u$	$A_u + B_{1u} + B_{2g} + B_{3g}$	$A_1 + A_2 + B_1 + B_2$
H_u	$E_u + T_u$	$A_{1u} + E_{1u} + E_{2u}$	$A_{1u} + 2E_u$	$2A_u + B_{1u} + B_{2g} + B_{3g}$	$A_1 + 2A_2 + B_2 + B_3$

$SO(3)$	I	O
ℓ		
0 (S)	A	A_1
1 (P)	T_1	T_1
2 (D)	H	$E + T_2$
3 (F)	$T_2 + G$	$A_2 + T_1 + T_2$
4 (G)	$G + H$	$A_1 + E + T_1 + T_2$
5	$T_1 + T_2 + H$	$E + 2T_1 + T_2$
6	$A + T_1 + G + H$	$A_1 + A_2 + E + T_1 + 2T_2$
7	$T_1 + T_2 + G + H$	$A_2 + E + 2T_1 + 2T_2$
8	$T_2 + G + 2H$	$A_1 + 2E + 2T_1 + 2T_2$
9	$T_1 + T_2 + 2G + H$	$A_1 + A_2 + E + 3T_1 + 2T_2$
10	$A + T_1 + T_2 + G + 2H$	$A_1 + A_2 + 2E + 2T_1 + 3T_2$
11	$2T_1 + T_2 + G + 2H$	$A_2 + 2E + 3T_1 + 3T_2$
12	$A + T_1 + T_2 + 2G + 2H$	$A_1 + \Gamma_{reg}$

C.2 Induction $H \uparrow G$

Ascent in symmetry tables have been provided by Boyle [4]. Fowler and Quinn have listed the irreps that are induced by σ-, π-, and δ-type orbitals on molecular sites [5]. These tables are reproduced below. They are useful for the construction of cluster orbitals. Γ_{reg} always denotes the regular representation. Γ_σ corresponds to the positional representation. The mechanical representation is the sum $\Gamma_\sigma + \Gamma_\pi$.

G	H	$\frac{\lvert G\rvert}{\lvert H\rvert}$	Γ_σ	Γ_π	Γ_δ
D_2	C_2^z	2	$A + B_1$	$2B_2 + 2B_3$	$2\Gamma_\sigma$
D_3	C_2	3	$A_1 + E$	$2A_2 + 2E$	$2\Gamma_\sigma$
D_4	C_4	2	$A_1 + A_2$	$2E$	$2B_1 + 2B_2$
	C_2'	4	$A_1 + B_1 + E$	$2A_2 + 2B_2 + 2E$	$2\Gamma_\sigma$
	C_2''	4	$A_2 + B_2 + E$	$2A_2 + 2B_1 + 2E$	$2\Gamma_\sigma$
D_5	C_5	2	$A_1 + A_2$	$2E_1$	$2E_2$
	C_2	4	$A_1 + E_1 + E_2$	$2A_2 + 2E_1 + 2E_2$	$2\Gamma_\sigma$
D_6	C_6	6	$A_1 + A_2$	$2E_1$	$2E_2$
	C_2'	4	$A_1 + B_1 + E_1 + E_2$	$2A_2 + 2B_2 + 2E_1 + 2E_2$	$2\Gamma_\sigma$
	C_2''	4	$A_1 + B_2 + E_1 + E_2$	$2A_2 + 2B_1 + 2E_1 + 2E_2$	$2\Gamma_\sigma$
C_{2v}	C_s^{xz}	2	$A_1 + B_1$	Γ_{reg}	Γ_{reg}
C_{3v}	C_s	3	$A_1 + E$	Γ_{reg}	Γ_{reg}
C_{4v}	C_s^v	4	$A_1 + B_1 + E$	Γ_{reg}	Γ_{reg}
	C_s^d	4	$A_1 + B_2 + E$	Γ_{reg}	Γ_{reg}
C_{5v}	C_s	5	$A_1 + E_1 + E_2$	Γ_{reg}	Γ_{reg}
C_{6v}	C_s^v	6	$A_1 + B_1 + E_1 + E_2$	Γ_{reg}	Γ_{reg}
	C_s^d	6	$A_1 + B_2 + E_1 + E_2$	Γ_{reg}	Γ_{reg}

G	H	$\frac{\lvert G\rvert}{\lvert H\rvert}$	Γ_σ	Γ_π	Γ_δ
C_{2h}	C_2	2	$A_g + A_u$	$2B_g + 2B_u$	Γ_σ
	C_s	2	$A_g + B_u$	Γ_{reg}	Γ_{reg}
C_{3h}	C_3	2	$A' + A''$	$E' + E''$	$E' + E''$
	C_s	3	$A' + E'$	Γ_{reg}	Γ_{reg}
C_{4h}	C_4	2	$A_g + A_u$	$E_g + E_u$	$2B_g + 2B_u$
	C_s	4	$A_g + B_u + E_u$	Γ_{reg}	Γ_{reg}
C_{5h}	C_5	2	$A' + A''$	$E'_1 + E''_1$	$E'_2 + E''_2$
	C_s	5	$A' + E'_1 + E'_2$	Γ_{reg}	Γ_{reg}
C_{6h}	C_6	2	$A_g + A_u$	$E_{1g} + E_{1u}$	$E_{2g} + E_{2u}$
	C_s	6	$A_g + B_u + E_{2g} + E_{1u}$	Γ_{reg}	Γ_{reg}
S_4	C_2	2	$A + B$	$2E$	$2\Gamma_\sigma$
S_6	C_3	2	$A_g + A_u$	$E_g + E_u$	$E_g + E_u$
D_{2h}	C^z_{2v}	2	$A_g + B_{1u}$	$B_{2g} + B_{2u} + B_{3g} + B_{3u}$	$A_g + A_u + B_{1g} + B_{1u}$
	C^{xy}_s	4	$A_g + B_{1g} + B_{2u} + B_{3u}$	Γ_{reg}	Γ_{reg}
D_{3h}	C_{3v}	2	$A'_1 + A''_2$	$E' + E''$	$E' + E''$
	C_{2v}	3	$A'_1 + E'$	$A'_1 + A''_2 + E' + E''$	$A'_1 + A''_1 + E' + E''$
	C_s	6	$A'_1 + A''_2 + E' + E''$	Γ_{reg}	Γ_{reg}
D_{4h}	C_{4v}	2	$A_{1g} + A_{2u}$	$E_g + E_u$	$B_{1g} + B_{1u} + B_{2g} + B_{2u}$
	C'_{2v}	4	$A_{1g} + B_{1g} + E_u$	$A_{2g} + A_{2u} + B_{2g} + B_{2u}$ $+ E_g + E_u$	$A_{1g} + A_{1u} + B_{1g} + B_{1u}$ $+ E_g + E_u$
	C''_{2v}	4	$A_{1g} + B_{2g} + E_u$	$A_{2g} + A_{2u} + B_{1g} + B_{1u}$ $+ E_g + E_u$	$A_{1g} + A_{1u} + B_{2g} + B_{2u}$ $+ E_g + E_u$
	C^h_s	8	$A_{1g} + A_{2g} + B_{1g}$ $+ B_{2g} + 2E_u$	Γ_{reg}	Γ_{reg}
	C^v_s	8	$A_{1g} + A_{2u} + B_{1g}$ $+ B_{2u} + E_g + E_u$	Γ_{reg}	Γ_{reg}
	C^d_s	8	$A_{1g} + A_{2u} + B_{1u}$ $+ B_{2g} + E_g + E_u$	Γ_{reg}	Γ_{reg}
D_{5h}	C_{5v}	2	$A'_1 + A''_2$	$E'_1 + E''_1$	$E'_2 + E''_2$
	C_{2v}	5	$A'_1 + E'_1 + E'_2$	$A'_2 + A''_2 + E'_1 + E''_1$ $+ E'_2 + E''_2$	$A'_1 + A''_1 + E'_1 + E''_1$ $+ E'_2 + E''_2$
	C^h_s	10	$A'_1 + A'_2 + 2E'_1 + 2E'_2$	Γ_{reg}	Γ_{reg}
	C^v_s	10	$A'_1 + A''_2 + E'_1 + E''_1$ $+ E'_2 + E''_2$	Γ_{reg}	Γ_{reg}

G	H	$\frac{\lvert G\rvert}{\lvert H\rvert}$	Γ_σ	Γ_π	Γ_δ
D_{6h}	C_{6v}	2	$A_{1g}+A_{2u}$	$E_{1g}+E_{1u}$	$E_{2g}+E_{2u}$
	C'_{2v}	6	$A_{1g}+B_{1u}+E_{1u}+E_{2g}$	$A_{2g}+A_{2u}+B_{2g}$ $+B_{2u}+E_{1g}+E_{1u}$ $+E_{2g}+E_{2u}$	$A_{1g}+A_{1u}+B_{1g}$ $+B_{1u}+E_{1g}+E_{1u}$ $+E_{2g}+E_{2u}$
	C''_{2v}	6	$A_{1g}+B_{2u}+E_{1u}+E_{2g}$	$A_{2g}+A_{2u}+B_{1g}$ $+B_{1u}+E_{1g}+E_{1u}$ $+E_{2g}+E_{2u}$	$A_{1g}+A_{1u}+B_{2g}$ $+B_{2u}+E_{1g}+E_{1u}$ $+E_{2g}+E_{2u}$
	C_s^h	12	$A_{1g}+A_{2g}+B_{1u}+B_{2u}$ $+2E_{1u}+2E_{2g}$	Γ_{reg}	Γ_{reg}
	C_s^v	12	$A_{1g}+A_{2u}+B_{1g}+B_{2u}$ $+E_{1g}+E_{1u}+E_{2g}+E_{2u}$	Γ_{reg}	Γ_{reg}
	C_s^d	12	$A_{1g}+A_{2u}+B_{1g}+B_{2u}$ $+E_{1g}+E_{1u}+E_{2g}+E_{2u}$	Γ_{reg}	Γ_{reg}
D_{2d}	C_{2v}^z	2	A_1+B_2	$2E$	$A_1+A_2+B_1+B_2$
	C'_2	4	A_1+B_1+E	$2A_2+2B_2+2E$	$2\Gamma_\sigma$
	C_s	4	A_1+B_2+E	Γ_{reg}	Γ_{reg}
D_{3d}	C_{3v}	2	$A_{1g}+A_{2u}$	E_g+E_u	E_g+E_u
	C_2	6	$A_{1g}+A_{1u}+E_g+E_u$	$2A_{2g}+2A_{2u}$ $+2E_g+2E_u$	$2\Gamma_\sigma$
	C_s	6	$A_{1g}+A_{2u}+E_g+E_u$	Γ_{reg}	Γ_{reg}
D_{4d}	C_{4v}	2	A_1+B_2	E_1+E_3	$2E_2$
	C'_2	8	$A_1+B_1+E_1+E_2+E_3$	$2A_2+2B_2+2E_1$ $+2E_2+2E_3$	$2\Gamma_\sigma$
	C_s	8	$A_1+B_2+E_1+E_2+E_3$	Γ_{reg}	Γ_{reg}
D_{5d}	C_{5v}	2	$A_{1g}+A_{2u}$	$E_{1g}+E_{1u}$	$E_{2g}+E_{2u}$
	C_2	10	$A_{1g}+A_{1u}+E_{1g}+E_{1u}$ $+E_{2g}+E_{2u}$	$2A_{2g}+2A_{2u}+2E_{1g}$ $+2E_{1u}+2E_{2g}+2E_{2u}$	$2\Gamma_\sigma$
	C_s	10	$A_{1g}+A_{2u}+E_{1g}+E_{1u}$ $+E_{2g}+E_{2u}$	Γ_{reg}	Γ_{reg}
D_{6d}	C_{6v}	2	A_1+B_2	E_1+E_5	E_2+E_4
	C'_2	12	$A_1+B_1+E_1+E_2$ $+E_3+E_4+E_5$	$2A_2+2B_2+2E_1$ $+2E_2+2E_3$ $+2E_4+2E_5$	$2\Gamma_\sigma$
	C_s	12	$A_1+B_2+E_1+E_2$ $+E_3+E_4+E_5$	Γ_{reg}	Γ_{reg}

G	H	$\frac{\lvert G\rvert}{\lvert H\rvert}$	Γ_σ	Γ_π	Γ_δ
T	C_3	4	$A+T$	$E+2T$	Γ_π
	C_2	6	$A+E+T$	$4T$	$2\Gamma_\sigma$
T_d	C_{3v}	4	A_1+T_2	$E+T_1+T_2$	Γ_π
	C_{2v}	6	A_1+E+T_2	$2T_1+2T_2$	A_1+A_2+2E $+T_1+T_2$
	C_s	12	$A_1+E+T_1+2T_2$	Γ_{reg}	Γ_{reg}
T_h	C_3	8	$A_g+A_u+T_g+T_u$	$E_g+E_u+2T_g+2T_u$	Γ_π
	C_{2v}	6	$A_g+E_g+T_u$	$2T_g+2T_u$	$A_g+A_u+E_g$ $+E_u+T_g+T_u$
	C_s	12	$A_g+E_g+T_g+2T_u$	Γ_{reg}	Γ_{reg}
O	C_4	6	A_1+E+T_1	$2T_1+2T_2$	$2A_2+2E+2T_2$
	C_3	8	$A_1+A_2+T_1+T_2$	$2E+2T_1+2T_2$	Γ_π
	C_2	12	$A_1+E+T_1+2T_2$	$2A_2+2E+4T_1+2T_2$	$2\Gamma_\sigma$
O_h	C_{4v}	6	$A_{1g}+E_g+T_{1u}$	$T_{1g}+T_{1u}+T_{2g}+T_{2u}$	$A_{2g}+A_{2u}+E_g$ $+E_u+T_{2g}+T_{2u}$
	C_{3v}	8	$A_{1g}+A_{2u}+T_{1u}+T_{2g}$	$E_g+E_u+T_{1g}+T_{1u}$ $+T_{2g}+T_{2u}$	Γ_π
	C_{2v}	12	$A_{1g}+E_g+T_{1u}$ $+T_{2g}+T_{2u}$	$A_{2g}+A_{2u}+E_g+E_u$ $+2T_{1g}+2T_{1u}$ $+T_{2g}+T_{2u}$	$A_{1g}+A_{1u}+E_g$ $+E_u+T_{1g}+T_{1u}$ $+2T_{2g}+2T_{2u}$
	C_s^h	24	$A_{1g}+A_{2g}+2E_g$ $+T_{1g}+2T_{1u}+T_{2g}+2T_{2u}$	Γ_{reg}	Γ_{reg}
	C_s^d	24	$A_{1g}+A_{2g}+E_g+E_u$ $+T_{1g}+2T_{1u}+2T_{2g}+T_{2u}$	Γ_{reg}	Γ_{reg}

G	H	$\frac{\lvert G\rvert}{\lvert H\rvert}$	Γ_σ	Γ_π	Γ_δ
I	C_5	12	$A+T_1+T_2+H$	$2T_1+2G+2H$	$2T_2+2G+2H$
	C_3	20	$A+T_1+T_2$ $+2G+H$	$2T_1+2T_2+2G+4H$	Γ_π
	C_2	30	$A+T_1+T_2$ $+2G+3H$	$4T_1+4T_2+4G+4H$	$2\Gamma_\sigma$
I_h	C_{5v}	12	$A_g+T_{1u}+T_{2u}+H_g$	$T_{1g}+T_{1u}+G_g+G_u$ $+H_g+H_u$	$T_{2g}+T_{2u}+G_g+G_u$ $+H_g+H_u$
	C_{3v}	20	$A_g+T_{1u}+T_{2u}$ $+G_g+G_u+H_g$	$T_{1g}+T_{1u}+T_{2g}+T_{2u}$ $+G_g+G_u+2H_g+2H_u$	Γ_π
	C_{2v}	30	$A_g+T_{1u}+T_{2u}+G_g$ $+G_u+2H_g+H_u$	$2T_{1g}+2T_{1u}+2T_{2g}+2T_{2u}$ $+2G_g+2G_u+2H_g+2H_u$	$A_g+A_u+T_{1g}+T_{1u}$ $+T_{2g}+T_{2u}+2G_g$ $+2G_u+3H_g+3H_u$
	C_s	60	$A_g+T_{1g}+2T_{1u}$ $+T_{2g}+2T_{2u}+2G_g$ $+2G_u+3H_g+2H_u$	Γ_{reg}	Γ_{reg}

Appendix D
Canonical-Basis Relationships

The importance of canonical-basis relationships was demonstrated by Griffith in his monumental work on the theory of transition-metal ions [6] (Fig. D.1). The icosahedral basis sets were defined by Boyle and Parker [7] (Fig. D.2).

D_3	$\mathbb{D}(C_3^z)$	$\mathbb{D}(C_2^x)$
$\lvert E_x\rangle, \lvert E_y\rangle$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

D_4	$\mathbb{D}(C_4^z)$	$\mathbb{D}(C_2^x)$
$\lvert E_x\rangle, \lvert E_y\rangle$	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

D_5	$\mathbb{D}(C_5^z)$	$\mathbb{D}(C_2^x)$
$\lvert E_{1x}\rangle, \lvert E_{1y}\rangle$	$\begin{pmatrix} \cos(2\pi/5) & -\sin(2\pi/5) \\ \sin(2\pi/5) & \cos(2\pi/5) \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
$\lvert E_{2x}\rangle, \lvert E_{2y}\rangle$	$\begin{pmatrix} \cos(4\pi/5) & -\sin(4\pi/5) \\ \sin(4\pi/5) & \cos(4\pi/5) \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

D_6	$\mathbb{D}(C_6^z)$	$\mathbb{D}(C_2^x)$
$\lvert E_{1x}\rangle, \lvert E_{1y}\rangle$	$\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
$\lvert E_{2x}\rangle, \lvert E_{2y}\rangle$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

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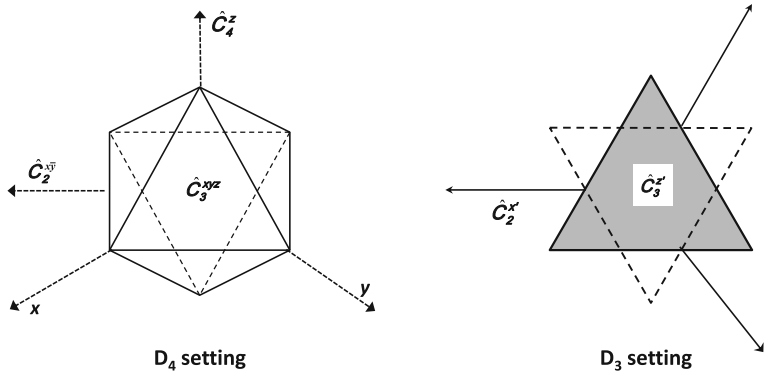
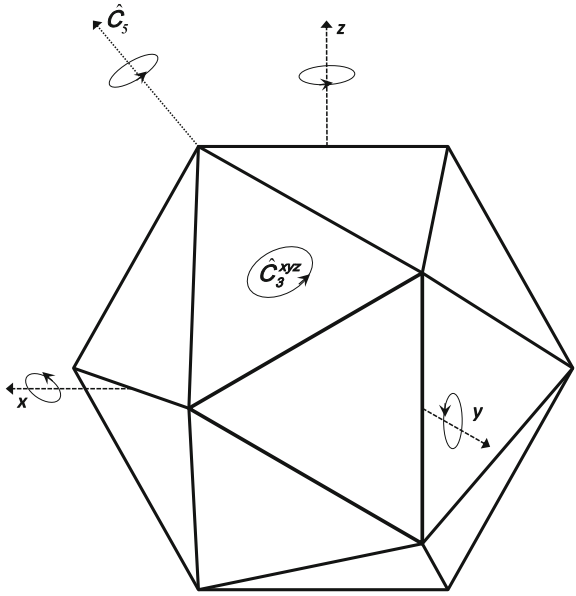


Fig. D.1 Octahedron with x, y, z-coordinates in D_4 and D_3 setting

Fig. D.2 Icosahedron with x, y, z-coordinates in D_2 setting



O (D_4 basis)	$\mathbb{D}(C_4^z)$	$\mathbb{D}(C_3^{xyz})$
$\|E_\theta\rangle, \|E_\epsilon\rangle$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$
$\|T_{1x}\rangle, \|T_{1y}\rangle, \|T_{1z}\rangle$	$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$
$\|T_{2\xi}\rangle, \|T_{2\eta}\rangle, \|T_{2\zeta}\rangle$	$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

Transformation to trigonal basis set (Fig. D.1):

$$|E_\theta\rangle = d_{z^2} = \frac{1}{\sqrt{3}}\left(-d_{x'^2-y'^2} - \sqrt{2}d_{y'z'}\right)$$

$$|E_\epsilon\rangle = d_{x^2-y^2} = \frac{1}{\sqrt{3}}\left(d_{x'y'} + \sqrt{2}d_{x'z'}\right)$$

$$|T_{1a}\rangle = \frac{1}{\sqrt{3}}\left(|T_{1x}\rangle + |T_{1y}\rangle + |T_{1z}\rangle\right) = p_{z'}$$

$$|T_{1\theta}\rangle = \frac{1}{\sqrt{2}}\left(|T_{1x}\rangle - |T_{1y}\rangle\right) = p_{x'}$$

$$|T_{1\epsilon}\rangle = \frac{1}{\sqrt{6}}\left(|T_{1x}\rangle + |T_{1y}\rangle - 2|T_{1z}\rangle\right) = p_{y'}$$

$$|T_{2a}\rangle = \frac{1}{\sqrt{3}}\left(|T_{2\xi}\rangle + |T_{2\eta}\rangle + |T_{2\zeta}\rangle\right) = d_{z'^2}$$

$$|T_{2\theta}\rangle = \frac{1}{\sqrt{6}}\left(|T_{2\xi}\rangle + |T_{2\eta}\rangle - 2|T_{2\zeta}\rangle\right) = \frac{1}{\sqrt{3}}\left(\sqrt{2}d_{x'^2-y'^2} - d_{y'z'}\right)$$

$$|T_{2\epsilon}\rangle = \frac{1}{\sqrt{2}}\left(|T_{2\eta}\rangle - |T_{2\xi}\rangle + |T_{2\zeta}\rangle\right) = \frac{1}{\sqrt{3}}\left(-\sqrt{2}d_{x'y'} + d_{x'z'}\right)$$

O (D_3 basis)	$\mathbb{D}(C_3^{z'})$	$\mathbb{D}(C_2^{x'})$
$\|E_\theta\rangle, \|E_\epsilon\rangle$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
$\|T_{1a}\rangle, \|T_{1\theta}\rangle, \|T_{1\epsilon}\rangle$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
$\|T_{2a}\rangle, \|T_{2\theta}\rangle, \|T_{2\epsilon}\rangle$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & +\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

I (D_2 basis)	$\mathbb{D}(C_5)$	$\mathbb{D}(C_3^{xyz})$	$\mathbb{D}(C_2^z)$
$\lvert T_{1x}\rangle, \lvert T_{1y}\rangle, \lvert T_{1z}\rangle$	$\frac{1}{2}\begin{pmatrix} 1 & -\phi & \phi^{-1} \\ \phi & \phi^{-1} & -1 \\ \phi^{-1} & 1 & \phi \end{pmatrix}$	$\begin{pmatrix} 0&0&1 \\ 1&0&0 \\ 0&1&0 \end{pmatrix}$	$\begin{pmatrix} -1&0&0 \\ 0&-1&0 \\ 0&0&1 \end{pmatrix}$
$\lvert T_{2x}\rangle, \lvert T_{2y}\rangle, \lvert T_{2z}\rangle$	$\frac{1}{2}\begin{pmatrix} 1 & \phi^{-1} & -\phi \\ -\phi^{-1} & -\phi & -1 \\ -\phi & 1 & -\phi^{-1} \end{pmatrix}$	$\begin{pmatrix} 0&0&1 \\ 1&0&0 \\ 0&1&0 \end{pmatrix}$	$\begin{pmatrix} -1&0&0 \\ 0&-1&0 \\ 0&0&1 \end{pmatrix}$
$\lvert G_a\rangle, \lvert G_x\rangle, \lvert G_y\rangle, \lvert G_z\rangle$	$\frac{1}{4}\begin{pmatrix} -1 & -\sqrt{5} & \sqrt{5} & \sqrt{5} \\ \sqrt{5} & -3 & -1 & -1 \\ \sqrt{5} & 1 & -1 & 3 \\ -\sqrt{5} & -1 & -3 & 1 \end{pmatrix}$	$\begin{pmatrix} 1&0&0&0 \\ 0&0&0&1 \\ 0&1&0&0 \\ 0&0&1&0 \end{pmatrix}$	$\begin{pmatrix} 1&0&0&0 \\ 0&-1&0&0 \\ 0&0&-1&0 \\ 0&0&0&1 \end{pmatrix}$

I	$\lvert H\theta\rangle, \lvert H\epsilon\rangle, \lvert H\xi\rangle, \lvert H\eta\rangle, \lvert H\zeta\rangle$		
	$\mathbb{D}(C_5)$	$\mathbb{D}(C_3^{xyz})$	$\mathbb{D}(C_2^z)$
	$\begin{pmatrix} -\frac{1}{4} & -\frac{\sqrt{3}}{4} & \frac{1}{\sqrt{8}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{8}} \\ -\frac{\sqrt{3}}{4} & \frac{1}{4} & -\frac{\sqrt{3}}{\sqrt{8}} & 0 & -\frac{\sqrt{3}}{\sqrt{8}} \\ -\frac{1}{\sqrt{8}} & \frac{\sqrt{3}}{\sqrt{8}} & 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{\sqrt{8}} & \frac{\sqrt{3}}{\sqrt{8}} & \frac{1}{2} & 0 & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0&0&0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0&0&0 \\ 0 & 0 & 0&0&1 \\ 0 & 0 & 1&0&0 \\ 0 & 0 & 0&1&0 \end{pmatrix}$	$\begin{pmatrix} 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&-1&0&0 \\ 0&0&0&-1&0 \\ 0&0&0&0&1 \end{pmatrix}$

It is important to note that, in the Boyle and Parker basis, the $\lvert H\theta\rangle$ and $\lvert H\epsilon\rangle$ components do not denote components that transform like the functions d_{z^2} and $d_{x^2-y^2}$, but refer to linear combinations of these:

$$\lvert H\theta\rangle = \sqrt{\frac{3}{8}}d_{z^2} + \sqrt{\frac{5}{8}}d_{x^2-y^2}$$
$$\lvert H\epsilon\rangle = -\sqrt{\frac{5}{8}}d_{z^2} + \sqrt{\frac{3}{8}}d_{x^2-y^2}$$

Griffith has presented the subduction of spherical $\lvert JM\rangle$ states to point-group canonical bases for the case of the octahedral group [6]. Similar tables for subduction to the icosahedral canonical basis have been published by Qiu and Ceulemans [8]. Extensive tables of bases in terms of spherical harmonics for several branching schemes are also provided by Butler [9].

Appendix E
Direct-Product Tables

Extensive direct-product tables are provided by Herzberg [10]. Anti-symmetrized and symmetrized parts of direct squares are indicated by braces and brackets, respectively.

D_3	A_1	A_2	E
A_1	A_1	A_2	E
A_2	A_2	A_1	E
E	E	E	$[A_1 + E] + \{A_2\}$

D_4	A_1	A_2	B_1	B_2	E
A_1	A_1	A_2	B_1	B_2	E
A_2	A_2	A_1	B_2	B_1	E
B_1	B_1	B_2	A_1	A_2	E
B_2	B_2	B_1	A_2	A_1	E
E	E	E	E	E	$[A_1 + B_1 + B_2] + \{A_2\}$

D_5	A_1	A_2	E_1	E_2
A_1	A_1	A_2	E_1	E_2
A_2	A_2	A_1	E_1	E_2
E_1	E_1	E_1	$[A_1 + E_2] + \{A_2\}$	$E_1 + E_2$
E_2	E_2	E_2	$E_1 + E_2$	$[A_1 + E_1] + \{A_2\}$

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D_6	A_1	A_2	B_1	B_2	E_1	E_2
A_1	A_1	A_2	B_1	B_2	E_1	E_2
A_2	A_2	A_1	B_2	B_1	E_1	E_2
B_1	B_1	B_2	A_1	A_2	E_2	E_1
B_2	B_2	B_1	A_2	A_1	E_2	E_1
E_1	E_1	E_1	E_2	E_2	$[A_1 + E_2] + \{A_2\}$	$B_1 + B_2 + E_1$
E_2	E_2	E_2	E_1	E_1	$B_1 + B_2 + E_1$	$[A_1 + E_2] + \{A_2\}$

T_d	A_1	A_2	E	T_1	T_2
A_1	A_1	A_2	E	T_1	T_2
A_2	A_2	A_1	E	T_2	T_1
E	E	E	$[A_1 + E] + \{A_2\}$	$T_1 + T_2$	$T_1 + T_2$
T_1	T_1	T_2	$T_1 + T_2$	$[A_1 + E + T_2] + \{T_1\}$	$A_2 + E + T_1 + T_2$
T_2	T_2	T_1	$T_1 + T_2$	$A_2 + E + T_1 + T_2$	$[A_1 + E + T_2] + \{T_1\}$

O	A_1	A_2	E	T_1	T_2
A_1	A_1	A_2	E	T_1	T_2
A_2	A_2	A_1	E	T_2	T_1
E	E	E	$[A_1 + E] + \{A_2\}$	$T_1 + T_2$	$T_1 + T_2$
T_1	T_1	T_2	$T_1 + T_2$	$[A_1 + E + T_2] + \{T_1\}$	$A_2 + E + T_1 + T_2$
T_2	T_2	T_1	$T_1 + T_2$	$A_2 + E + T_1 + T_2$	$[A_1 + E + T_2] + \{T_1\}$

I	A	T_1	T_2	G	H
A	A	T_1	T_2	G	H
T_1	T_1	$[A + H] + \{T_1\}$	$G + H$	$T_2 + G + H$	$T_1 + T_2 + G + H$
T_2	T_2	$G + H$	$[A + H] + \{T_2\}$	$T_1 + G + H$	$T_1 + T_2 + G + H$
G	G	$T_2 + G + H$	$T_1 + G + H$	$[A + G + H]$ $+\{T_1 + T_2\}$	$T_1 + T_2 + G + 2H$
H	H	$T_1 + T_2 + G$ $+H$	$T_1 + T_2 + G$ $+H$	$T_1 + T_2 + G$ $+2H$	$[A + G + 2H]$ $+\{T_1 + T_2 + G\}$

Appendix F
Coupling Coefficients

Coupling coefficients are denoted as 3Γ symbols: $\langle \Gamma_a \gamma_a \Gamma_b \gamma_b | \Gamma \gamma \rangle$. Their symmetry properties are given in Sect. 6.3. Octahedral coefficients have been listed by Griffith [6]. Icosahedral coefficients are taken from the work of Fowler and Ceulemans [11].

D_3		
$A_2 \times E$	E	
	x	y
a_2 x	0	-1
a_2 y	1	0

D_3				
$E \times E$	A_1	A_2	E	
	a_1	a_2	x	y
x x	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
y y	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
x y	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
y x	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

D_4				
$E \times E$	A_1	A_2	B_1	B_2
	a_1	a_2	b_1	b_2
x x	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
y y	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
x y	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
y x	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

D_5				
$E_1 \times E_1$	A_1	A_2	E_2	
	a_1	a_2	c	s
x x	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
y y	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
x y	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
y x	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

D_5				
$E_2 \times E_2$	A_1	A_2	E_1	
	a_1	a_2	x	y
c c	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
s s	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
c s	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
s c	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

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D_5

$E_1 \times E_2$	E_1		E_2	
	x	y	c	s
$x\ c$	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
$y\ s$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
$x\ s$	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
$y\ c$	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

D_6

$E_1 \times E_1$	A_1	A_2	E_2	
	a_1	a_2	c	s
$x\ x$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
$y\ y$	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
$x\ y$	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
$y\ x$	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

D_6

$E_2 \times E_2$	A_1	A_2	E_1	
	a_1	a_2	x	y
$c\ c$	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
$s\ s$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
$c\ s$	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
$s\ c$	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

D_6

$E_1 \times E_2$	B_1	B_2	E_1	
	b_1	b_2	x	y
$x\ c$	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
$y\ s$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
$x\ s$	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
$y\ c$	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

O

$A_2 \times E$	E	
	θ	ϵ
$a_2\ \theta$	0	-1
$a_2\ \epsilon$	1	0

O

$A_2 \times T_1$	T_2		
	ξ	η	ζ
$a_2\ x$	1	0	0
$a_2\ y$	0	1	0
$a_2\ z$	0	0	1

O

$A_2 \times T_2$	T_1		
	x	y	z
$a_2\ \xi$	1	0	0
$a_2\ \eta$	0	1	0
$a_2\ \zeta$	0	0	1

O

$E \times E$	A_1	A_2	E	
	a_1	a_2	θ	ϵ
$\theta\ \theta$	$\frac{1}{\sqrt{2}}$	0	$-\frac{1}{\sqrt{2}}$	0
$\epsilon\ \epsilon$	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	0
$\theta\ \epsilon$	0	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$
$\epsilon\ \theta$	0	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

O						
$E \times T_1$	T_1			T_2		
	x	y	z	ξ	η	ζ
θ x	$-\frac{1}{2}$	0	0	$-\frac{\sqrt{3}}{2}$	0	0
θ y	0	$-\frac{1}{2}$	0	0	$\frac{\sqrt{3}}{2}$	0
θ z	0	0	1	0	0	0
ϵ x	$\frac{\sqrt{3}}{2}$	0	0	$-\frac{1}{2}$	0	0
ϵ y	0	$-\frac{\sqrt{3}}{2}$	0	0	$-\frac{1}{2}$	0
ϵ z	0	0	0	0	0	1

O						
$E \times T_2$	T_1			T_2		
	x	y	z	ξ	η	ζ
θ ξ	$\frac{-\sqrt{3}}{2}$	0	0	$-\frac{1}{2}$	0	0
θ η	0	$\frac{\sqrt{3}}{2}$	0	0	$-\frac{1}{2}$	0
θ ζ	0	0	0	0	0	1
ϵ ξ	$-\frac{1}{2}$	0	0	$\frac{\sqrt{3}}{2}$	0	0
ϵ η	0	$-\frac{1}{2}$	0	0	$-\frac{\sqrt{3}}{2}$	0
ϵ ζ	0	0	1	0	0	0

O			
$T_1 \times T_1$ or $T_2 \times T_2$	A_1	E	
	a_1	θ	ϵ
x x ξ ξ	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{6}}$	$-\frac{1}{\sqrt{2}}$
y y η η	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{6}}$	$\frac{1}{\sqrt{2}}$
z z ζ ζ	$\frac{1}{\sqrt{3}}$	$-\frac{2}{\sqrt{6}}$	0

O			
$T_1 \times T_2$	A_2	E	
	a_2	θ	ϵ
x ξ	$\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{6}}$
y η	$\frac{1}{\sqrt{3}}$	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{6}}$
z ζ	$\frac{1}{\sqrt{3}}$	0	$\frac{2}{\sqrt{6}}$

O						
$T_1 \times T_1$ or $T_2 \times T_2$	T_1			T_2		
	x	y	z	ξ	η	ζ
x y ξ η	0	0	$-\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$
x z ξ ζ	0	$\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$	0
y x η ξ	0	0	$\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$
y z η ζ	$-\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$	0	0
z x ζ ξ	0	$-\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$	0
z y ζ η	$\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$	0	0

O						
$T_1 \times T_2$	T_1			T_2		
	x	y	z	ξ	η	ζ
x η	0	0	$-\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$
x ζ	0	$-\frac{1}{\sqrt{2}}$	0	0	$\frac{1}{\sqrt{2}}$	0
y ξ	0	0	$-\frac{1}{\sqrt{2}}$	0	0	$\frac{1}{\sqrt{2}}$
y ζ	$-\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$	0	0
z ξ	0	$-\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$	0
z η	$-\frac{1}{\sqrt{2}}$	0	0	$\frac{1}{\sqrt{2}}$	0	0

In the icosahedral tables, ϕ denotes the golden number $\left(1+\sqrt{5}\right)/2$, and $\alpha = 3\phi - 1, \beta = 3\phi^{-1} + 1$.

I

$T_1 \times T_1$	A	T_1			H				
		x	y	z	θ	ϵ	ξ	η	ζ
$x\ x$	$\frac{1}{\sqrt{3}}$	0	0	0	$\frac{\phi^{-1}}{2}$	$\frac{\phi^2}{2\sqrt{3}}$	0	0	0
$x\ y$	0	0	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$
$x\ z$	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$	0
$y\ x$	0	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$
$y\ y$	$\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{\phi}{2}$	$-\frac{\phi^{-2}}{2\sqrt{3}}$	0	0	0
$y\ z$	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$	0	0
$z\ x$	0	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$	0
$z\ y$	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$	0	0
$z\ z$	$\frac{1}{\sqrt{3}}$	0	0	0	$\frac{1}{2}$	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0

I

$T_1 \times T_2$	G				H				
	a	x	y	z	θ	ϵ	ξ	η	ζ
$x\ x$	$\frac{1}{\sqrt{3}}$	0	0	0	$\frac{1}{\sqrt{6}}$	$-\frac{1}{\sqrt{2}}$	0	0	0
$x\ y$	0	0	0	$-\frac{\phi^{-1}}{\sqrt{3}}$	0	0	0	0	$\frac{\phi}{\sqrt{3}}$
$x\ z$	0	0	$-\frac{\phi}{\sqrt{3}}$	0	0	0	0	$-\frac{\phi^{-1}}{\sqrt{3}}$	0
$y\ x$	0	0	0	$-\frac{\phi}{\sqrt{3}}$	0	0	0	0	$-\frac{\phi^{-1}}{\sqrt{3}}$
$y\ y$	$\frac{1}{\sqrt{3}}$	0	0	0	$\frac{1}{\sqrt{6}}$	$\frac{1}{\sqrt{2}}$	0	0	0
$y\ z$	0	$-\frac{\phi^{-1}}{\sqrt{3}}$	0	0	0	0	$\frac{\phi}{\sqrt{3}}$	0	0
$z\ x$	0	0	$-\frac{\phi^{-1}}{\sqrt{3}}$	0	0	0	0	$\frac{\phi}{\sqrt{3}}$	0
$z\ y$	0	$-\frac{\phi}{\sqrt{3}}$	0	0	0	0	$-\frac{\phi^{-1}}{\sqrt{3}}$	0	0
$z\ z$	$\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{\sqrt{2}}{\sqrt{3}}$	0	0	0	0

I

$T_1 \times G$	T_2			G					H				
	x	y	z	a	x	y	z	θ	ϵ	ξ	η	ζ
$x\ a$	$\frac{1}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0	0
$x\ x$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{\beta}{2\sqrt{6}}$	$-\frac{\phi}{2\sqrt{2}}$	0	0	0
$x\ y$	0	0	$-\frac{\phi}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^{-2}}{2\sqrt{3}}$
$x\ z$	0	$-\frac{\phi^{-1}}{2}$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^2}{2\sqrt{3}}$	0
$y\ a$	0	$\frac{1}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0
$y\ x$	0	0	$-\frac{\phi^{-1}}{2}$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^2}{2\sqrt{3}}$
$y\ y$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	$\frac{\alpha}{2\sqrt{6}}$	$-\frac{\phi^{-1}}{2\sqrt{2}}$	0	0	0
$y\ z$	$-\frac{\phi}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^{-2}}{2\sqrt{3}}$	0	0
$z\ a$	0	0	$\frac{1}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$
$z\ x$	0	$-\frac{\phi}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^{-2}}{2\sqrt{3}}$	0
$z\ y$	$-\frac{\phi^{-1}}{2}$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^2}{2\sqrt{3}}$	0	0
$z\ z$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{1}{2\sqrt{6}}$	$\frac{\sqrt{5}}{2\sqrt{2}}$	0	0	0

I

$T_2 \times G$	T_1			G				H				
	x	y	z	a	x	y	z	θ	ϵ	ξ	η	ζ
$x\ a$	$\frac{1}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0	0
$x\ x$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	$\frac{\alpha}{2\sqrt{6}}$	$\frac{\phi^{-1}}{2\sqrt{2}}$	0	0	0
$x\ y$	0	0	$-\frac{\phi^{-1}}{2}$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^2}{2\sqrt{3}}$
$x\ z$	0	$-\frac{\phi}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^{-2}}{2\sqrt{3}}$	0
$y\ a$	0	$\frac{1}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0
$y\ x$	0	0	$-\frac{\phi}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^{-2}}{2\sqrt{3}}$
$y\ y$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{\beta}{2\sqrt{6}}$	$\frac{\phi}{2\sqrt{2}}$	0	0	0
$y\ z$	$-\frac{\phi^{-1}}{2}$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^2}{2\sqrt{3}}$	0	0
$z\ a$	0	0	$\frac{1}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$
$z\ x$	0	$-\frac{\phi^{-1}}{2}$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^2}{2\sqrt{3}}$	0
$z\ y$	$-\frac{\phi}{2}$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{\phi^{-2}}{2\sqrt{3}}$	0	0
$z\ z$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{1}{2\sqrt{6}}$	$-\frac{\sqrt{5}}{2\sqrt{2}}$	0	0	0

I

$T_1 \times H$		T_1			T_2			G			
		x	y	z	x	y	z	a	x	y	z
x	θ	$\frac{\phi^{-1}\sqrt{3}}{2\sqrt{5}}$	0	0	$\frac{1}{\sqrt{10}}$	0	0	0	$\frac{\beta}{\sqrt{30}}$	0	0
x	ϵ	$\frac{\phi^2}{2\sqrt{5}}$	0	0	$-\frac{3}{\sqrt{10}}$	0	0	0	$\frac{\phi}{\sqrt{10}}$	0	0
x	ξ	0	0	0	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0
x	η	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	$-\frac{\phi^{-1}}{\sqrt{5}}$	0	0	0	$-\frac{\phi^2}{\sqrt{15}}$
x	ζ	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	$\frac{\phi}{\sqrt{5}}$	0	0	0	$-\frac{\phi^{-2}}{\sqrt{15}}$	0
y	θ	0	$-\frac{\phi\sqrt{3}}{2\sqrt{5}}$	0	0	$\frac{1}{\sqrt{10}}$	0	0	0	$-\frac{\alpha}{\sqrt{30}}$	0
y	ϵ	0	$-\frac{\phi^{-2}}{2\sqrt{5}}$	0	0	$\frac{3}{\sqrt{10}}$	0	0	0	$\frac{\phi^{-1}}{\sqrt{10}}$	0
y	ξ	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	$\frac{\phi}{\sqrt{5}}$	0	0	0	$-\frac{\phi^{-2}}{\sqrt{15}}$
y	η	0	0	0	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0
y	ζ	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	$-\frac{\phi^{-1}}{\sqrt{5}}$	0	0	0	$-\frac{\phi^2}{\sqrt{15}}$	0	0
z	θ	0	0	$\frac{\sqrt{3}}{2\sqrt{5}}$	0	0	$-\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	$\frac{1}{\sqrt{30}}$
z	ϵ	0	0	$-\frac{1}{2}$	0	0	0	0	0	0	$-\frac{1}{\sqrt{2}}$
z	ξ	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	$-\frac{\phi^{-1}}{\sqrt{5}}$	0	0	0	$-\frac{\phi^2}{\sqrt{15}}$	0
z	η	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	$\frac{\phi}{\sqrt{5}}$	0	0	0	$-\frac{\phi^{-2}}{\sqrt{15}}$	0	0
z	ζ	0	0	0	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0

I

$T_2 \times T_2$		A	T_2			H				
			x	y	z	θ	ϵ	x	y	z
x	x	$\frac{1}{\sqrt{3}}$	0	0	0	$\frac{\phi}{2}$	$-\frac{\phi^{-2}}{2\sqrt{3}}$	0	0	0
x	y	0	0	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$
x	z	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0
y	x	0	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$
y	y	$\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{\phi^{-1}}{2}$	$\frac{\phi^2}{2\sqrt{3}}$	0	0	0
y	z	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0	0
z	x	0	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0
z	y	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0	0
z	z	$\frac{1}{\sqrt{3}}$	0	0	0	$-\frac{1}{2}$	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0

I

$T_2 \times H$	T_1			T_2			G			
	x	y	z	x	y	z	a	x	y	z
$x\ \theta$	$\frac{1}{\sqrt{10}}$	0	0	$\frac{\phi\sqrt{3}}{2\sqrt{5}}$	0	0	0	$\frac{\alpha}{\sqrt{30}}$	0	0
$x\ \epsilon$	$-\frac{3}{\sqrt{10}}$	0	0	$-\frac{\phi^{-2}}{2\sqrt{5}}$	0	0	0	$\frac{\phi^{-1}}{\sqrt{10}}$	0	0
$x\ \xi$	0	0	0	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0
$x\ \eta$	0	0	$\frac{\phi}{\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	$\frac{\phi^{-2}}{\sqrt{15}}$
$x\ \zeta$	0	$-\frac{\phi^{-1}}{\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	$\frac{\phi^2}{\sqrt{15}}$	0
$y\ \theta$	0	$\frac{1}{\sqrt{10}}$	0	0	$-\frac{\phi^{-1}\sqrt{3}}{2\sqrt{5}}$	0	0	0	$-\frac{\beta}{\sqrt{30}}$	0
$y\ \epsilon$	0	$\frac{3}{\sqrt{10}}$	0	0	$\frac{\phi^2}{2\sqrt{5}}$	0	0	0	$\frac{\phi}{\sqrt{10}}$	0
$y\ \xi$	0	0	$-\frac{\phi^{-1}}{\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	$\frac{\phi^2}{\sqrt{15}}$
$y\ \eta$	0	0	0	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0
$y\ \zeta$	$\frac{\phi}{\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	$\frac{\phi^{-2}}{\sqrt{15}}$	0	0
$z\ \theta$	0	0	$-\frac{\sqrt{2}}{\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{2\sqrt{5}}$	0	0	0	$-\frac{1}{\sqrt{30}}$
$z\ \epsilon$	0	0	0	0	0	$-\frac{1}{2}$	0	0	0	$-\frac{1}{\sqrt{2}}$
$z\ \xi$	0	$\frac{\phi}{\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	$\frac{\phi^{-2}}{\sqrt{15}}$	
$z\ \eta$	$-\frac{\phi^{-1}}{\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	$\frac{\phi^2}{\sqrt{15}}$	0	0
$z\ \zeta$	0	0	0	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0

I

$T_1 \times H$	H				
	θ	ϵ	ξ	η	ζ
$x\ \theta$	0	0	$\frac{\phi^2}{2\sqrt{3}}$	0	0
$x\ \epsilon$	0	0	$-\frac{\phi^{-1}}{2}$	0	0
$x\ \xi$	$-\frac{\phi^2}{2\sqrt{3}}$	$\frac{\phi^{-1}}{2}$	0	0	0
$x\ \eta$	0	0	0	0	$\frac{1}{\sqrt{6}}$
$x\ \zeta$	0	0	0	$-\frac{1}{\sqrt{6}}$	0
$y\ \theta$	0	0	0	$-\frac{\phi^{-2}}{2\sqrt{3}}$	0
$y\ \epsilon$	0	0	0	$\frac{\phi}{2}$	0
$y\ \xi$	0	0	0	0	$-\frac{1}{\sqrt{6}}$
$y\ \eta$	$\frac{\phi^{-2}}{2\sqrt{3}}$	$-\frac{\phi}{2}$	0	0	0
$y\ \zeta$	0	0	$\frac{1}{\sqrt{6}}$	0	0
$z\ \theta$	0	0	0	0	$-\frac{\sqrt{5}}{2\sqrt{3}}$
$z\ \epsilon$	0	0	0	0	$-\frac{1}{2}$
$z\ \xi$	0	0	0	$\frac{1}{\sqrt{6}}$	0
$z\ \eta$	0	0	$-\frac{1}{\sqrt{6}}$	0	0
$z\ \zeta$	$\frac{\sqrt{5}}{2\sqrt{3}}$	$\frac{1}{2}$	0	0	0

I					
$T_2 \times H$	H				
	θ	ϵ	ξ	η	ζ
$x\ \theta$	0	0	$\frac{\phi^{-2}}{2\sqrt{3}}$	0	0
$x\ \epsilon$	0	0	$\frac{\phi}{2}$	0	0
$x\ \xi$	$-\frac{\phi^{-2}}{2\sqrt{3}}$	$-\frac{\phi}{2}$	0	0	0
$x\ \eta$	0	0	0	0	$\frac{1}{\sqrt{6}}$
$x\ \zeta$	0	0	0	$-\frac{1}{\sqrt{6}}$	0
$y\ \theta$	0	0	0	$-\frac{\phi^2}{2\sqrt{3}}$	0
$y\ \epsilon$	0	0	0	$-\frac{\phi^{-1}}{2}$	0
$y\ \xi$	0	0	0	0	$-\frac{1}{\sqrt{6}}$
$y\ \eta$	$\frac{\phi^2}{2\sqrt{3}}$	$\frac{\phi^{-1}}{2}$	0	0	0
$y\ \zeta$	0	0	$\frac{1}{\sqrt{6}}$	0	0
$z\ \theta$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$
$z\ \epsilon$	0	0	0	0	$-\frac{1}{2}$
$z\ \xi$	0	0	0	$\frac{1}{\sqrt{6}}$	0
$z\ \eta$	0	0	$-\frac{1}{\sqrt{6}}$	0	0
$z\ \zeta$	$-\frac{\sqrt{5}}{2\sqrt{3}}$	$\frac{1}{2}$	0	0	0

I							
$G \times G$	A	T_1			T_2		
		x	y	z	x	y	z
$a\ a$	$\frac{1}{2}$	0	0	0	0	0	0
$a\ x$	0	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0	0
$a\ y$	0	0	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0
$a\ z$	0	0	0	$\frac{1}{2}$	0	0	$\frac{1}{2}$
$x\ a$	0	$-\frac{1}{2}$	0	0	$-\frac{1}{2}$	0	0
$x\ x$	$\frac{1}{2}$	0	0	0	0	0	0
$x\ y$	0	0	0	$\frac{1}{2}$	0	0	$-\frac{1}{2}$
$x\ z$	0	0	$-\frac{1}{2}$	0	0	$\frac{1}{2}$	0
$y\ a$	0	0	$-\frac{1}{2}$	0	0	$-\frac{1}{2}$	0
$y\ x$	0	0	0	$-\frac{1}{2}$	0	0	$\frac{1}{2}$
$y\ y$	$\frac{1}{2}$	0	0	0	0	0	0
$y\ z$	0	$\frac{1}{2}$	0	0	$-\frac{1}{2}$	0	0
$z\ a$	0	0	0	$-\frac{1}{2}$	0	0	$-\frac{1}{2}$
$z\ x$	0	0	$\frac{1}{2}$	0	0	$-\frac{1}{2}$	0
$z\ y$	0	$-\frac{1}{2}$	0	0	$\frac{1}{2}$	0	0
$z\ z$	$\frac{1}{2}$	0	0	0	0	0	0

I

$G \times G$	G				H				
	a	x	y	z	θ	ϵ	ξ	η	ζ
$a\ a$	$\frac{\sqrt{3}}{2}$	0	0	0	0	0	0	0	0
$a\ x$	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0	0
$a\ y$	0	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0
$a\ z$	0	0	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$
$x\ a$	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0	0
$x\ x$	$-\frac{1}{2\sqrt{3}}$	0	0	0	$-\frac{1}{\sqrt{6}}$	$\frac{1}{\sqrt{2}}$	0	0	0
$x\ y$	0	0	0	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0	0	$-\frac{1}{2\sqrt{3}}$
$x\ z$	0	0	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0	0	$-\frac{1}{2\sqrt{3}}$	0
$y\ a$	0	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$	0
$y\ x$	0	0	0	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0	0	$-\frac{1}{2\sqrt{3}}$
$y\ y$	$-\frac{1}{2\sqrt{3}}$	0	0	0	$-\frac{1}{\sqrt{6}}$	$-\frac{1}{\sqrt{2}}$	0	0	0
$y\ z$	0	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0	0	$-\frac{1}{2\sqrt{3}}$	0	0
$z\ a$	0	0	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	0	$\frac{\sqrt{5}}{2\sqrt{3}}$
$z\ x$	0	0	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0	0	$-\frac{1}{2\sqrt{3}}$	0
$z\ y$	0	$-\frac{\sqrt{5}}{2\sqrt{3}}$	0	0	0	0	$-\frac{1}{2\sqrt{3}}$	0	0
$z\ z$	$-\frac{1}{2\sqrt{3}}$	0	0	0	$\frac{\sqrt{2}}{\sqrt{3}}$	0	0	0	0

I

$H \times H$	A	T_1			T_2		
		x	y	z	x	y	z
$\theta\ \theta$	$\frac{1}{\sqrt{5}}$	0	0	0	0	0	0
$\theta\ \epsilon$	0	0	0	0	0	0	0
$\theta\ \xi$	0	$\frac{\phi^2}{2\sqrt{5}}$	0	0	$\frac{\phi^{-2}}{2\sqrt{5}}$	0	0
$\theta\ \eta$	0	0	$-\frac{\phi^{-2}}{2\sqrt{5}}$	0	0	$-\frac{\phi^2}{2\sqrt{5}}$	0
$\theta\ \zeta$	0	0	0	$-\frac{1}{2}$	0	0	$\frac{1}{2}$
$\epsilon\ \theta$	0	0	0	0	0	0	0
$\epsilon\ \epsilon$	$\frac{1}{\sqrt{5}}$	0	0	0	0	0	0
$\epsilon\ \xi$	0	$-\frac{\phi^{-1}\sqrt{3}}{2\sqrt{5}}$	0	0	$\frac{\phi\sqrt{3}}{2\sqrt{5}}$	0	0
$\epsilon\ \eta$	0	0	$\frac{\phi\sqrt{3}}{2\sqrt{5}}$	0	0	$-\frac{\phi^{-1}\sqrt{3}}{2\sqrt{5}}$	0
$\epsilon\ \zeta$	0	0	0	$-\frac{\sqrt{3}}{2\sqrt{5}}$	0	0	$-\frac{\sqrt{3}}{2\sqrt{5}}$
$\xi\ \theta$	0	$-\frac{\phi^2}{2\sqrt{5}}$	0	0	$-\frac{\phi^{-2}}{2\sqrt{5}}$	0	0
$\xi\ \epsilon$	0	$\frac{\phi^{-1}\sqrt{3}}{2\sqrt{5}}$	0	0	$-\frac{\phi\sqrt{3}}{2\sqrt{5}}$	0	0
$\xi\ \xi$	$\frac{1}{\sqrt{5}}$	0	0	0	0	0	0
$\xi\ \eta$	0	0	0	$\frac{1}{\sqrt{10}}$	0	0	$\frac{1}{\sqrt{10}}$
$\xi\ \zeta$	0	0	$-\frac{1}{\sqrt{10}}$	0	0	$-\frac{1}{\sqrt{10}}$	0
$\eta\ \theta$	0	0	$\frac{\phi^{-2}}{2\sqrt{5}}$	0	0	$\frac{\phi^2}{2\sqrt{5}}$	0
$\eta\ \epsilon$	0	0	$-\frac{\phi\sqrt{3}}{2\sqrt{5}}$	0	0	$\frac{\phi^{-1}\sqrt{3}}{2\sqrt{5}}$	0
$\eta\ \xi$	0	0	0	$-\frac{1}{\sqrt{10}}$	0	0	$-\frac{1}{\sqrt{10}}$
$\eta\ \eta$	$\frac{1}{\sqrt{5}}$	0	0	0	0	0	0
$\eta\ \zeta$	0	$\frac{1}{\sqrt{10}}$	0	0	$\frac{1}{\sqrt{10}}$	0	0
$\zeta\ \theta$	0	0	0	$\frac{1}{2}$	0	0	$-\frac{1}{2}$
$\zeta\ \epsilon$	0	0	0	$\frac{\sqrt{3}}{2\sqrt{5}}$	0	0	$\frac{\sqrt{3}}{2\sqrt{5}}$
$\zeta\ \xi$	0	0	$\frac{1}{\sqrt{10}}$	0	0	$\frac{1}{\sqrt{10}}$	0
$\zeta\ \eta$	0	$-\frac{1}{\sqrt{10}}$	0	0	$-\frac{1}{\sqrt{10}}$	0	0
$\zeta\ \zeta$	$\frac{1}{\sqrt{5}}$	0	0	0	0	0	0

I								
$H \times H$	$[G]$				$\{G\}$			
	a	x	y	z	a	x	y	z
$\theta\ \theta$	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	0	0	0	0
$\theta\ \epsilon$	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0
$\theta\ \xi$	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	$\frac{\sqrt{3}}{2\sqrt{5}}$	0	0
$\theta\ \eta$	0	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	$-\frac{\sqrt{3}}{2\sqrt{5}}$	0
$\theta\ \zeta$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0
$\epsilon\ \theta$	0	0	0	0	$\frac{1}{\sqrt{2}}$	0	0	0
$\epsilon\ \epsilon$	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0	0	0	0	0
$\epsilon\ \xi$	0	$\frac{1}{2}$	0	0	0	$\frac{1}{2\sqrt{5}}$	0	0
$\epsilon\ \eta$	0	0	$-\frac{1}{2}$	0	0	0	$\frac{1}{2\sqrt{5}}$	0
$\epsilon\ \zeta$	0	0	0	0	0	0	0	$-\frac{1}{\sqrt{5}}$
$\xi\ \theta$	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	$-\frac{\sqrt{3}}{2\sqrt{5}}$	0	0
$\xi\ \epsilon$	0	$\frac{1}{2}$	0	0	0	$-\frac{1}{2\sqrt{5}}$	0	0
$\xi\ \xi$	$-\frac{\sqrt{2}}{\sqrt{15}}$	0	0	0	0	0	0	0
$\xi\ \eta$	0	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$
$\xi\ \zeta$	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0
$\eta\ \theta$	0	0	$-\frac{1}{2\sqrt{3}}$	0	0	0	$\frac{\sqrt{3}}{2\sqrt{5}}$	0
$\eta\ \epsilon$	0	0	$-\frac{1}{2}$	0	0	0	$-\frac{1}{2\sqrt{5}}$	0
$\eta\ \xi$	0	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$
$\eta\ \eta$	$-\frac{\sqrt{2}}{\sqrt{15}}$	0	0	0	0	0	0	0
$\eta\ \zeta$	0	$\frac{1}{\sqrt{6}}$	0	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0	0
$\zeta\ \theta$	0	0	0	$\frac{1}{\sqrt{3}}$	0	0	0	0
$\zeta\ \epsilon$	0	0	0	0	0	0	0	$\frac{1}{\sqrt{5}}$
$\zeta\ \xi$	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0
$\zeta\ \eta$	0	$\frac{1}{\sqrt{6}}$	0	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0
$\zeta\ \zeta$	$-\frac{\sqrt{2}}{\sqrt{15}}$	0	0	0	0	0	0	0

I

$H \times H$	H_a					H_b				
	θ	ϵ	ξ	η	ζ	θ	ϵ	ξ	η	ζ
$\theta\ \theta$	$\frac{\sqrt{3}}{2\sqrt{2}}$	0	0	0	0	0	$\frac{1}{2\sqrt{2}}$	0	0	0
$\theta\ \epsilon$	0	$-\frac{\sqrt{3}}{2\sqrt{2}}$	0	0	0	$\frac{1}{2\sqrt{2}}$	0	0	0	0
$\theta\ \xi$	0	0	$-\frac{1}{2\sqrt{6}}$	0	0	0	0	$\frac{\sqrt{3}}{2\sqrt{2}}$	0	0
$\theta\ \eta$	0	0	0	$-\frac{1}{2\sqrt{6}}$	0	0	0	0	$-\frac{\sqrt{3}}{2\sqrt{2}}$	0
$\theta\ \zeta$	0	0	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	0	0
$\epsilon\ \theta$	0	$-\frac{\sqrt{3}}{2\sqrt{2}}$	0	0	0	$\frac{1}{2\sqrt{2}}$	0	0	0	0
$\epsilon\ \epsilon$	$-\frac{\sqrt{3}}{2\sqrt{2}}$	0	0	0	0	0	$-\frac{1}{2\sqrt{2}}$	0	0	0
$\epsilon\ \xi$	0	0	$\frac{1}{2\sqrt{2}}$	0	0	0	0	$\frac{1}{2\sqrt{2}}$	0	0
$\epsilon\ \eta$	0	0	0	$-\frac{1}{2\sqrt{2}}$	0	0	0	0	$\frac{1}{2\sqrt{2}}$	0
$\epsilon\ \zeta$	0	0	0	0	0	0	0	0	0	$-\frac{1}{\sqrt{2}}$
$\xi\ \theta$	0	0	$-\frac{1}{2\sqrt{6}}$	0	0	0	0	$\frac{\sqrt{3}}{2\sqrt{2}}$	0	0
$\xi\ \epsilon$	0	0	$\frac{1}{2\sqrt{2}}$	0	0	0	0	$\frac{1}{2\sqrt{2}}$	0	0
$\xi\ \xi$	$-\frac{1}{2\sqrt{6}}$	$\frac{1}{2\sqrt{2}}$	0	0	0	$\frac{\sqrt{3}}{2\sqrt{2}}$	$\frac{1}{2\sqrt{2}}$	0	0	0
$\xi\ \eta$	0	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	0
$\xi\ \zeta$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	0	0
$\eta\ \theta$	0	0	0	$-\frac{1}{2\sqrt{6}}$	0	0	0	0	$-\frac{\sqrt{3}}{2\sqrt{2}}$	0
$\eta\ \epsilon$	0	0	0	$-\frac{1}{2\sqrt{2}}$	0	0	0	0	$\frac{\sqrt{1}}{2\sqrt{2}}$	0
$\eta\ \xi$	0	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	0
$\eta\ \eta$	$-\frac{1}{2\sqrt{6}}$	$-\frac{1}{2\sqrt{2}}$	0	0	0	$-\frac{\sqrt{3}}{2\sqrt{2}}$	$\frac{1}{2\sqrt{2}}$	0	0	0
$\eta\ \zeta$	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	0	0	0
$\zeta\ \theta$	0	0	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	0	0
$\zeta\ \epsilon$	0	0	0	0	0	0	0	0	0	$-\frac{1}{\sqrt{2}}$
$\zeta\ \xi$	0	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	0	0
$\zeta\ \eta$	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	0	0	0
$\zeta\ \zeta$	$\frac{1}{\sqrt{6}}$	0	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0

Appendix G
Spinor Representations

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Extensive character tables for double groups were provided by Herzberg. The $\aleph$ symbol in the present table corresponds to the Bethe rotation through an angle of 2π. Spin–orbit coupling coefficients for the icosahedral double group have been listed by Fowler and Ceulemans [12]. The notation ρ_1, ρ_2 for conjugate components follows Griffith [6]. The single-valued irreps in Appendix A also represent the double groups. The rotation through 2π leaves these irreps invariant. Their characters under $\hat{R}$ and $\aleph\hat{R}$ are thus the same.

G.1 Character Tables

D_2^*	$\hat{E}$	$\hat{\aleph}$	$2\hat{C}_2^z$	$2\hat{C}_2^y$	$2\hat{C}_2^x$
$E_{1/2}(\Gamma_5)$	2	-2	0	0	0

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D_3^*		$\hat{E}$	$\hat{\aleph}$	$2\hat{C}_3$	$2\hat{\aleph}\hat{C}_3$	$3\hat{C}_2$	$3\hat{\aleph}\hat{C}_2$
$E_{1/2}(\Gamma_4)$		2	−2	1	−1	0	0
$E_{3/2}$	ρ_1	1	−1	−1	1	i	$-i$
	ρ_2	1	−1	−1	1	$-i$	i

D_4^*	$\hat{E}$	$\hat{\aleph}$	$2\hat{C}_4$	$2\hat{\aleph}\hat{C}_4$	$2\hat{C}_2(=\hat{C}_4^2)$	$4\hat{C}_2'$	$4\hat{C}_2''$
$E_{1/2}(\Gamma_6)$	2	−2	$\sqrt{2}$	$-\sqrt{2}$	0	0	0
$E_{3/2}(\Gamma_7)$	2	−2	$-\sqrt{2}$	$\sqrt{2}$	0	0	0

D_5^*		$\hat{E}$	$\hat{\aleph}$	$2\hat{C}_5$	$2\hat{\aleph}\hat{C}_5$	$2\hat{C}_5^2$	$2\hat{\aleph}\hat{C}_5^2$	$5\hat{C}_2'$	$5\hat{\aleph}\hat{C}_2'$
$E_{1/2}$		2	−2	ϕ	$-\phi$	ϕ^{-1}	$-\phi^{-1}$	0	0
$E_{3/2}$		2	−2	$-\phi^{-1}$	ϕ^{-1}	$-\phi$	ϕ	0	0
$E_{5/2}$	ρ_1	1	−1	−1	1	1	−1	i	$-i$
	ρ_2	1	−1	−1	1	1	−1	$-i$	i

D_6^*	$\hat{E}$	$\hat{\aleph}$	$2\hat{C}_6$	$2\hat{\aleph}\hat{C}_6$	$2\hat{C}_3$	$2\hat{\aleph}\hat{C}_3$	$2\hat{C}_2$	$6\hat{C}_2'$	$6\hat{C}_2''$
$E_{1/2}(\Gamma_7)$	2	−2	$\sqrt{3}$	$-\sqrt{3}$	1	−1	0	0	0
$E_{5/2}(\Gamma_8)$	2	−2	$-\sqrt{3}$	$\sqrt{3}$	1	−1	0	0	0
$E_{3/2}(\Gamma_9)$	2	−2	0	0	−2	2	0	0	0

T^*		$\hat{E}$	$\hat{\aleph}$	$4\hat{C}_3$	$4\hat{\aleph}\hat{C}_3$	$4\hat{C}_3^2$	$4\hat{\aleph}\hat{C}_3^2$	$6\hat{C}_2$	$\epsilon = \exp(2\pi i/3)$
$E_{1/2}$		2	−2	1	−1	−1	1	0	
$G_{3/2}$	E''	2	−2	ϵ	$-\epsilon$	$-\bar{\epsilon}$	$\bar{\epsilon}$	0	
	E'''	2	−2	$\bar{\epsilon}$	$-\bar{\epsilon}$	$-\epsilon$	ϵ	0	

T_d^*	$\hat{E}$	$\hat{\aleph}$	$8\hat{C}_3$	$8\hat{\aleph}\hat{C}_3$	$6\hat{C}_2$	$6\hat{S}_4$	$6\hat{\aleph}\hat{S}_4$	$12\hat{\sigma}_d$
O^*	$\hat{E}$	$\hat{\aleph}$	$8\hat{C}_3$	$8\hat{\aleph}\hat{C}_3$	$6\hat{C}_2$	$6\hat{C}_4$	$6\hat{\aleph}\hat{C}_4$	$12\hat{C}_2'$
$E_{1/2}(\Gamma_6)$	2	-2	1	-1	0	$\sqrt{2}$	$-\sqrt{2}$	0
$E_{5/2}(\Gamma_7)$	2	-2	1	-1	0	$-\sqrt{2}$	$\sqrt{2}$	0
$G_{3/2}(\Gamma_8)$	4	-4	-1	1	0	0	0	0

I^*	$\hat{E}$	$\hat{\aleph}$	$12\hat{C}_5$	$12\hat{\aleph}\hat{C}_5$	$12\hat{C}_5^2$	$12\hat{\aleph}\hat{C}_5^2$	$20\hat{C}_3$	$20\hat{\aleph}\hat{C}_3$	$30\hat{C}_2$
$E_{1/2}(\Gamma_6)$	2	-2	ϕ	$-\phi$	ϕ^{-1}	$-\phi^{-1}$	1	-1	0
$E_{7/2}(\Gamma_7)$	2	-2	$-\phi^{-1}$	ϕ^{-1}	$-\phi$	ϕ	1	-1	0
$G_{3/2}(\Gamma_8)$	4	-4	1	-1	-1	1	-1	1	0
$I_{5/2}(\Gamma_9)$	6	-6	-1	1	1	-1	0	0	0

G.2 Subduction

$SO(3)$	I	O
j		
1/2	$E_{1/2}$	$E_{1/2}$
3/2	$G_{3/2}$	$G_{3/2}$
5/2	$I_{5/2}$	$E_{5/2} + G_{3/2}$
7/2	$E_{7/2} + I_{5/2}$	$E_{1/2} + E_{5/2} + G_{3/2}$
9/2	$G_{3/2} + I_{5/2}$	$E_{1/2} + 2G_{3/2}$
11/2	$E_{1/2} + G_{3/2} + I_{5/2}$	$E_{1/2} + E_{5/2} + 2G_{3/2}$
13/2	$E_{1/2} + E_{7/2} + G_{3/2} + I_{5/2}$	$E_{1/2} + 2E_{5/2} + 2G_{3/2}$
15/2	$G_{3/2} + 2I_{5/2}$	$E_{1/2} + E_{5/2} + 3G_{3/2}$
17/2	$E_{7/2} + G_{3/2} + 2I_{5/2}$	$2E_{1/2} + E_{5/2} + 3G_{3/2}$
19/2	$E_{1/2} + E_{7/2} + G_{3/2} + 2I_{5/2}$	$2E_{1/2} + 2E_{5/2} + 3G_{3/2}$

G.3 Canonical-Basis Relationships

D_3^*	$\mathbb{D}(C_3^z)$	$\mathbb{D}(C_2^x)$
$\|E_{1/2}\alpha\rangle, \|E_{1/2}\beta\rangle$	$\begin{pmatrix} \frac{1-i\sqrt{3}}{2} & 0 \\ 0 & \frac{1+i\sqrt{3}}{2} \end{pmatrix}$	$\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$
$\|E_{3/2}\rho_1\rangle$	$\begin{pmatrix} -1 \end{pmatrix}$	$\begin{pmatrix} i \end{pmatrix}$
$\|E_{3/2}\rho_2\rangle$	$\begin{pmatrix} -1 \end{pmatrix}$	$\begin{pmatrix} -i \end{pmatrix}$

D_4^*	$\mathbb{D}(C_4^z)$	$\mathbb{D}(C_2^x)$
$\|E_{1/2}\alpha\rangle, \|E_{1/2}\beta\rangle$	$\begin{pmatrix} \frac{1-i}{\sqrt{2}} & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$	$\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$
$\|E_{3/2}\alpha'\rangle, \|E_{3/2}\beta'\rangle$	$\begin{pmatrix} \frac{-1+i}{\sqrt{2}} & 0 \\ 0 & \frac{-1-i}{\sqrt{2}} \end{pmatrix}$	$\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$

The components of the fourfold-degenerate $G_{3/2}$ irrep in O^*, and I^* are labeled as κ, λ, μ and ν. For a quartet spin, these labels correspond to $M_S = +3/2, +1/2, -1/2$ and $-3/2$, respectively.

O^*	$\mathbb{D}(C_4^z)$	$\mathbb{D}(C_3^{xyz})$
$\|E_{1/2}\alpha\rangle, \|E_{1/2}\beta\rangle$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix}$	$\begin{pmatrix} \frac{1-i}{2} & \frac{-1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{pmatrix}$
$\|E_{5/2}\alpha'\rangle, \|E_{5/2}\beta'\rangle$	$\frac{1}{\sqrt{2}}\begin{pmatrix} -1+i & 0 \\ 0 & -1-i \end{pmatrix}$	$\begin{pmatrix} \frac{1-i}{2} & \frac{-1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{pmatrix}$

O^*		$\|G_{3/2}\kappa\rangle, \|G_{3/2}\lambda\rangle \|G_{3/2}\mu\rangle, \|G_{3/2}\nu\rangle$	
	$\mathbb{D}(C_4^z)$		$\mathbb{D}(C_3^{xyz})$
$\frac{1}{\sqrt{2}}$	$\begin{pmatrix} -1-i & 0 & 0 & 0 \\ 0 & 1-i & 0 & 0 \\ 0 & 0 & 1+i & 0 \\ 0 & 0 & 0 & -1+i \end{pmatrix}$	$\frac{1}{4}$	$\begin{pmatrix} -1-i & \sqrt{3}(-1+i) & \sqrt{3}(1+i) & 1-i \\ \sqrt{3}(-1-i) & -1+i & -1-i & \sqrt{3}(-1+i) \\ \sqrt{3}(-1-i) & 1-i & -1-i & \sqrt{3}(1-i) \\ -1-i & \sqrt{3}(1-i) & \sqrt{3}(1+i) & -1+i \end{pmatrix}$

I^*	$\mathbb{D}(C_5)$	$\mathbb{D}(C_3^{x,y,z})$
$\lvert E_{1/2}\alpha\rangle, \lvert E_{1/2}\beta\rangle$	$\frac{1}{2}\begin{pmatrix} \phi - i & -i\phi^{-1} \\ -i\phi^{-1} & \phi + i \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} 1-i & -1-i \\ 1-i & 1+i \end{pmatrix}$
$\lvert E_{7/2}\alpha'\rangle, \lvert E_{7/2}\beta'\rangle$	$\frac{1}{2}\begin{pmatrix} -\phi^{-1} - i & i\phi \\ i\phi & -\phi^{-1} + i \end{pmatrix}$	$\frac{1}{2}\begin{pmatrix} 1-i & -1-i \\ 1-i & 1+i \end{pmatrix}$

I^*	$\lvert G_{3/2}\kappa\rangle, \lvert G_{3/2}\lambda\rangle \lvert G_{3/2}\mu\rangle, \lvert G_{3/2}\nu\rangle$
	$\mathbb{D}(C_5)$
	$\frac{1}{8}\begin{pmatrix} -\phi^{-1}-i\phi^4 & -\sqrt{3}(2+i) & -\sqrt{3}(\phi^{-1}-i\phi^{-2}) & i\phi^{-3} \\ -\sqrt{3}(2+i) & 3+\phi-i(3-\phi^{-4}) & -i(4+\phi^{-3}) & -\sqrt{3}(\phi^{-1}+i\phi^{-2}) \\ -\sqrt{3}(\phi^{-1}-i\phi^{-2}) & -i(4+\phi^{-3}) & 3+\phi+i(3-\phi^{-4}) & \sqrt{3}(2-i) \\ i\phi^{-3} & -\sqrt{3}(\phi^{-1}+i\phi^{-2}) & \sqrt{3}(2-i) & -\phi^{-1}+i\phi^4 \end{pmatrix}$
	$\mathbb{D}(C_3^{x,y,z})$
	$\frac{1}{4}\begin{pmatrix} -1-i & \sqrt{3}(-1+i) & \sqrt{3}(1+i) & 1-i \\ \sqrt{3}(-1-i) & -1+i & -1-i & \sqrt{3}(-1+i) \\ \sqrt{3}(-1-i) & 1-i & -1-i & \sqrt{3}(1-i) \\ -1-i & \sqrt{3}(1-i) & \sqrt{3}(1+i) & -1+i \end{pmatrix}$

I^*	$I_{5/2}: \quad \lvert 5/2\rangle, \lvert 3/2\rangle \lvert 1/2\rangle, \lvert -1/2\rangle, \lvert -3/2\rangle, \lvert -5/2\rangle$
	$\mathbb{D}(C_5)$
	$\frac{1}{32}\begin{pmatrix} -7-i-\phi(10+5i) & -\sqrt{5}\phi(4-3i) & \sqrt{10}(\phi^{-3}+i\phi^2) & \sqrt{10}\phi^{-2}(2+i) & \sqrt{5}(\phi^{-3}-i\phi^{-4}) & -i\phi^{-5} \\ -\sqrt{5}\phi(4-3i) & -5-3i+6\phi^{-1}-7i\phi & -\sqrt{2}(2+i)(5\phi-2) & -\sqrt{2}(3+2\phi+i-3i\phi) & i(-8+7\phi) & \sqrt{5}(\phi^{-3}+i\phi^{-4}) \\ \sqrt{10}(\phi^{-3}+i\phi^2) & -\sqrt{2}(2+i)(5\phi-2) & 2\phi^3-2i\phi^2 & -2i(8+\phi) & -\sqrt{2}(3+2\phi-i+3i\phi) & -\sqrt{10}\phi^{-2}(2-i) \\ \sqrt{10}\phi^{-2}(2+i) & -\sqrt{2}(3+2\phi+i-3i\phi) & -2i(8+\phi) & 2\phi^3+2i\phi^2 & \sqrt{2}(2-i)(5\phi-2) & \sqrt{10}(\phi^{-3}-i\phi^2) \\ \sqrt{5}(\phi^{-3}-i\phi^{-4}) & i(-8+7\phi) & -\sqrt{2}(3+2\phi-i+3i\phi) & \sqrt{2}(2-i)(5\phi-2) & -5+3i+6\phi^{-1}+7i\phi & \sqrt{5}\phi(4+3i) \\ -i\phi^{-5} & \sqrt{5}(\phi^{-3}+i\phi^{-4}) & -\sqrt{10}\phi^{-2}(2-i) & \sqrt{10}(\phi^{-3}-i\phi^2) & \sqrt{5}\phi(4+3i) & -7+i-\phi(10-5i) \end{pmatrix}$
	$\mathbb{D}(C_3^{x,y,z})$
	$\frac{1}{8}\begin{pmatrix} -1+i & \sqrt{5}(1+i) & \sqrt{10}(1-i) & -\sqrt{10}(1+i) & -\sqrt{5}(1-i) & 1+i \\ -\sqrt{5}(1-i) & 3(1+i) & \sqrt{2}(1-i) & \sqrt{2}(1+i) & 3(1-i) & -\sqrt{5}(1+i) \\ -\sqrt{10}(1-i) & \sqrt{2}(1+i) & 2(-1+i) & 2(1+i) & -\sqrt{2}(1-i) & \sqrt{10}(1+i) \\ -\sqrt{10}(1-i) & -\sqrt{2}(1+i) & 2(-1+i) & -2(1+i) & -\sqrt{2}(1-i) & -\sqrt{10}(1+i) \\ -\sqrt{5}(1-i) & -3(1+i) & \sqrt{2}(1-i) & -\sqrt{2}(1+i) & 3(1-i) & \sqrt{5}(1+i) \\ -1+i & -\sqrt{5}(1+i) & \sqrt{10}(1-i) & \sqrt{10}(1+i) & -\sqrt{5}(1-i) & -1-i \end{pmatrix}$

G.4 Direct-Product Tables

D_3^*	$E_{1/2}$	ρ_1	ρ_2
A_1	$E_{1/2}$	ρ_1	ρ_2
A_2	$E_{1/2}$	ρ_2	ρ_1
E	$E_{1/2} + E_{3/2}$	$E_{1/2}$	$E_{1/2}$
$E_{1/2}$	$[A_2 + E] + \{A_1\}$	E	E
ρ_1	E	A_2	A_1
ρ_2	E	A_1	A_2

D_4^*	$E_{1/2}$	$E_{3/2}$
A_1	$E_{1/2}$	$E_{3/2}$
A_2	$E_{1/2}$	$E_{3/2}$
B_1	$E_{3/2}$	$E_{1/2}$
B_2	$E_{3/2}$	$E_{1/2}$
E	$E_{1/2} + E_{3/2}$	$E_{1/2} + E_{3/2}$
$E_{1/2}$	$[A_2 + E] + \{A_1\}$	$B_1 + B_2 + E$
$E_{3/2}$	$B_1 + B_2 + E$	$[A_2 + E] + \{A_1\}$

D_5^*	$E_{1/2}$	$E_{3/2}$	ρ_1	ρ_2
A_1	$E_{1/2}$	$E_{3/2}$	ρ_1	ρ_2
A_2	$E_{1/2}$	$E_{3/2}$	ρ_2	ρ_1
E_1	$E_{1/2} + E_{3/2}$	$E_{1/2} + E_{5/2}$	$E_{3/2}$	$E_{3/2}$
E_2	$E_{3/2} + E_{5/2}$	$E_{1/2} + E_{3/2}$	$E_{1/2}$	$E_{1/2}$
$E_{1/2}$	$[A_2 + E_1] + \{A_1\}$	$E_1 + E_2$	E_2	E_2
$E_{3/2}$	$E_1 + E_2$	$[A_2 + E_2] + \{A_1\}$	E_1	E_1
ρ_1	E_2	E_1	A_2	A_1
ρ_2	E_2	E_1	A_1	A_2

D_6^*	$E_{1/2}(\Gamma_7)$	$E_{5/2}(\Gamma_8)$	$E_{3/2}(\Gamma_9)$
A_1	$E_{1/2}$	$E_{5/2}$	$E_{3/2}$
A_2	$E_{1/2}$	$E_{5/2}$	$E_{3/2}$
B_1	$E_{5/2}$	$E_{1/2}$	$E_{3/2}$
B_2	$E_{5/2}$	$E_{1/2}$	$E_{3/2}$
E_1	$E_{1/2}+E_{3/2}$	$E_{5/2}+E_{3/2}$	$E_{1/2}+E_{5/2}$
E_2	$E_{5/2}+E_{3/2}$	$E_{1/2}+E_{3/2}$	$E_{1/2}+E_{5/2}$
$E_{1/2}$	$[A_2+E_1]+\{A_1\}$	$B_1+B_2+E_2$	E_1+E_2
$E_{5/2}$	$B_1+B_2+E_2$	$[A_2+E_1]+\{A_1\}$	E_1+E_2
$E_{3/2}$	E_1+E_2	E_1+E_2	$[A_2+B_1+B_2]+\{A_1\}$

O^*	$E_{1/2}$	$E_{5/2}$	$G_{3/2}$
A_1	$E_{1/2}$	$E_{5/2}$	$G_{3/2}$
A_2	$E_{5/2}$	$E_{1/2}$	$G_{3/2}$
E	$G_{3/2}$	$G_{3/2}$	$E_{1/2}+E_{5/2}+G_{3/2}$
T_1	$E_{1/2}+G_{3/2}$	$E_{5/2}+G_{3/2}$	$E_{1/2}+E_{5/2}+2G_{3/2}$
T_2	$E_{5/2}+G_{3/2}$	$E_{1/2}+G_{3/2}$	$E_{1/2}+E_{5/2}+2G_{3/2}$
$E_{1/2}$	$[T_1]+\{A_1\}$	A_2+T_2	$E+T_1+T_2$
$E_{5/2}$	A_2+T_2	$[T_1]+\{A_1\}$	$E+T_1+T_2$
$G_{3/2}$	$E+T_1+T_2$	$E+T_1+T_2$	$[A_2+2T_1+T_2]+\{A_1+E+T_2\}$

I^*	$E_{1/2}$	$E_{7/2}$	$G_{3/2}$	$I_{5/2}$
A	$E_{1/2}$	$E_{7/2}$	$G_{3/2}$	$I_{5/2}$
T_1	$E_{1/2}+G_{3/2}$	$I_{5/2}$	$E_{1/2}+G_{3/2}+I_{5/2}$	$E_{7/2}+G_{3/2}+2I_{5/2}$
T_2	$I_{5/2}$	$E_{7/2}+G_{3/2}$	$E_{7/2}+G_{3/2}+I_{5/2}$	$E_{1/2}+G_{3/2}+2I_{5/2}$
G	$E_{7/2}+I_{5/2}$	$E_{1/2}+I_{5/2}$	$G_{3/2}+2I_{5/2}$	$E_{1/2}+E_{7/2}+2G_{3/2}+2I_{5/2}$
H	$G_{3/2}+I_{5/2}$	$G_{3/2}+I_{5/2}$	$E_{1/2}+E_{7/2}+G_{3/2}+2I_{5/2}$	$E_{1/2}+E_{7/2}+2G_{3/2}+3I_{5/2}$
$E_{1/2}$	$[T_1]+\{A\}$	G	T_1+H	T_2+G+H
$E_{3/2}$	G	$[T_2]+\{A\}$	T_2+H	T_1+G+H
$G_{3/2}$	T_1+H	T_2+H	$[T_1+T_2+G]+\{A+H\}$	$T_1+T_2+2G+2H$
$I_{5/2}$	T_2+G+H	T_1+G+H	$T_1+T_2+2G+2H$	$[2T_1+2T_2+G+H]+\{A+G+2H\}$

G.5 Coupling Coefficients

O^*																
$A_2 \times G_{3/2}$		$G_{3/2}$				$E \times E_{1/2}$		$G_{3/2}$				$E \times E_{5/2}$		$G_{3/2}$		
		κ	λ	μ	ν			κ	λ	μ	ν			κ	λ	μ	ν
a_2	κ	0	0	1	0	θ	α	0	−1	0	0	θ	α'	0	0	0	−1
a_2	λ	0	0	0	−1	θ	β	0	0	1	0	θ	β'	1	0	0	0
a_2	μ	−1	0	0	0	ϵ	α	0	0	0	−1	ϵ	α'	0	1	0	0
a_2	ν	0	1	0	0	ϵ	β	1	0	0	0	ϵ	β'	0	0	−1	0

O^*										
$E \times G_{3/2}$		$E_{1/2}$		$E_{5/2}$			$G_{3/2}$			
		α	β	α'	β'	κ	λ	μ	ν
θ	κ	0	0	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	0	0
θ	λ	$\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0	0
θ	μ	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{1}{\sqrt{2}}$	0
θ	ν	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$
ϵ	κ	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$	0
ϵ	λ	0	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$
ϵ	μ	0	0	0	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0	0	0
ϵ	ν	$\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{1}{\sqrt{2}}$	0	0

Canonical complex T_1 and T_2 basis functions:

$$
\begin{aligned}
T_1: \quad |1\rangle &= \frac{1}{\sqrt{2}}\left[-|T_{1x}\rangle - i|T_{1y}\rangle\right] \\
|0\rangle &= |T_{1z}\rangle \\
|-1\rangle &= \frac{1}{\sqrt{2}}\left[|T_{1x}\rangle - i|T_{1y}\rangle\right]
\end{aligned}
$$

$$
\begin{aligned}
T_2: \quad |1\rangle &= \frac{1}{\sqrt{2}}\left[-|T_{2x}\rangle - i|T_{2y}\rangle\right] \\
|0\rangle &= |T_{2z}\rangle \\
|-1\rangle &= \frac{1}{\sqrt{2}}\left[|T_{2x}\rangle - i|T_{2y}\rangle\right]
\end{aligned}
$$

O^*									O^*								
$T_1 \times E_{1/2}$		$E_{1/2}$			$G_{3/2}$				$T_2 \times E_{1/2}$		$E_{5/2}$			$G_{3/2}$			
		α	β	κ	λ	μ	ν				α'	β'	κ	λ	μ	ν	
1	α	0	0	1	0	0	0		1	α'	0	0	0	0	1	0	
0	α	$\frac{1}{\sqrt{3}}$	0	0	$\frac{\sqrt{2}}{\sqrt{3}}$	0	0		0	α'	$\frac{1}{\sqrt{3}}$	0	0	0	0	$-\frac{\sqrt{2}}{\sqrt{3}}$	
−1	α	0	$\frac{\sqrt{2}}{\sqrt{3}}$	0	0	$\frac{1}{\sqrt{3}}$	0		−1	α'	0	$\frac{\sqrt{2}}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	0	0	0	
1	β	$-\frac{\sqrt{2}}{\sqrt{3}}$	0	0	$\frac{1}{\sqrt{3}}$	0	0		1	β'	$-\frac{\sqrt{2}}{\sqrt{3}}$	0	0	0	0	$-\frac{1}{\sqrt{3}}$	
0	β	0	$-\frac{1}{\sqrt{3}}$	0	0	$\frac{\sqrt{2}}{\sqrt{3}}$	0		0	β'	0	$-\frac{1}{\sqrt{3}}$	$-\frac{\sqrt{2}}{\sqrt{3}}$	0	0	0	
−1	β	0	0	0	0	0	1		−1	β'	0	0	0	1	0	0	

O^*		$T_1 \times G_{3/2}$													
		$E_{1/2}$		$E_{5/2}$		$G_{3/2}$				$G_{3/2}$					
		α	β	α'	β'	κ	λ	μ	ν	κ	λ	μ	ν		
1	κ	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	0	0	0	0	0	$\frac{\sqrt{5}}{\sqrt{6}}$		
0	κ	0	0	0	$-\frac{1}{\sqrt{3}}$	$\frac{\sqrt{3}}{\sqrt{5}}$	0	0	0	$-\frac{1}{\sqrt{15}}$	0	0	0		
−1	κ	$\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	$\frac{1}{\sqrt{10}}$	0	0		
1	λ	0	0	0	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	$-\frac{1}{\sqrt{10}}$	0	0	0		
0	λ	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$\frac{1}{\sqrt{15}}$	0	0	0	$\frac{\sqrt{3}}{\sqrt{5}}$	0	0		
−1	λ	0	$\frac{1}{\sqrt{6}}$	0	0	0	0	$\frac{2\sqrt{2}}{\sqrt{15}}$	0	0	0	$-\frac{\sqrt{3}}{\sqrt{10}}$	0		
1	μ	$\frac{1}{\sqrt{6}}$	0	0	0	0	$-\frac{2\sqrt{2}}{\sqrt{15}}$	0	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0		
0	μ	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$-\frac{1}{\sqrt{15}}$	0	0	0	$-\frac{\sqrt{3}}{\sqrt{5}}$	0		
−1	μ	0	0	$-\frac{1}{\sqrt{2}}$	0	0	0	0	$\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	$\frac{1}{\sqrt{10}}$		
1	ν	0	$\frac{1}{\sqrt{2}}$	0	0	0	0	$-\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	$-\frac{1}{\sqrt{10}}$	0		
0	ν	0	0	$-\frac{1}{\sqrt{3}}$	0	0	0	0	$-\frac{\sqrt{3}}{\sqrt{5}}$	0	0	0	$\frac{1}{\sqrt{15}}$		
−1	ν	0	0	0	$\frac{1}{\sqrt{6}}$	0	0	0	0	$-\frac{\sqrt{5}}{\sqrt{6}}$	0	0	0		

I^*		$T_1 \times E_{1/2}$								$T_2 \times E_{1/2}$					
		$E_{1/2}$		$G_{3/2}$						$I_{5/2}$					
		α	β	κ	λ	μ	ν			$\frac{5}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$
1	α	0	0	1	0	0	0	1	α'	0	$-\frac{1}{4}$	0	$\frac{\sqrt{5}}{2\sqrt{2}}$	0	$-\frac{\sqrt{5}}{4}$
0	α	$\frac{1}{\sqrt{3}}$	0	0	$\frac{\sqrt{2}}{\sqrt{3}}$	0	0	0	α'	$\frac{1}{2\sqrt{2}}$	0	$\frac{1}{2}$	0	$\frac{\sqrt{5}}{2\sqrt{2}}$	0
−1	α	0	$\frac{\sqrt{2}}{\sqrt{3}}$	0	0	$\frac{1}{\sqrt{3}}$	0	−1	α'	0	$\frac{\sqrt{5}}{4}$	0	$-\frac{1}{2\sqrt{2}}$	0	$-\frac{3}{4}$
1	β	$-\frac{\sqrt{2}}{\sqrt{3}}$	0	0	$\frac{1}{\sqrt{3}}$	0	0	1	β'	$\frac{3}{4}$	0	$\frac{1}{2\sqrt{2}}$	0	$-\frac{\sqrt{5}}{4}$	0
0	β	0	$-\frac{1}{\sqrt{3}}$	0	0	$\frac{\sqrt{2}}{\sqrt{3}}$	0	0	β'	0	$-\frac{\sqrt{5}}{2\sqrt{2}}$	0	$-\frac{1}{2}$	0	$-\frac{1}{2\sqrt{2}}$
−1	β	0	0	0	0	0	1	−1	β'	$\frac{\sqrt{5}}{4}$	0	$-\frac{\sqrt{5}}{2\sqrt{2}}$	0	$\frac{1}{4}$	0

I^*		$T_1 \times G_{3/2}$											
		$E_{1/2}$		$G_{3/2}$				$I_{5/2}$					
		α	β	κ	λ	μ	ν	$\frac{5}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$
1	κ	0	0	0	0	0	0	1	0	0	0	0	0
0	κ	0	0	$\frac{\sqrt{3}}{\sqrt{5}}$	0	0	0	0	$\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	0
-1	κ	$-\frac{1}{\sqrt{2}}$	0	0	$\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	0	$\frac{1}{\sqrt{10}}$	0	0	0
1	λ	0	0	$-\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	0	$\frac{\sqrt{3}}{\sqrt{5}}$	0	0	0	0
0	λ	$\frac{1}{\sqrt{3}}$	0	0	$\frac{1}{\sqrt{15}}$	0	0	0	0	$\frac{\sqrt{3}}{\sqrt{5}}$	0	0	0
-1	λ	0	$-\frac{1}{\sqrt{6}}$	0	0	$\frac{2\sqrt{2}}{\sqrt{15}}$	0	0	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0
1	μ	$-\frac{1}{\sqrt{6}}$	0	0	$-\frac{2\sqrt{2}}{\sqrt{15}}$	0	0	0	0	$\frac{\sqrt{3}}{\sqrt{10}}$	0	0	0
0	μ	0	$\frac{1}{\sqrt{3}}$	0	0	$-\frac{1}{\sqrt{15}}$	0	0	0	0	$\frac{\sqrt{3}}{\sqrt{5}}$	0	0
-1	μ	0	0	0	0	0	$\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	0	$\frac{\sqrt{3}}{\sqrt{5}}$	0
1	ν	0	$-\frac{1}{\sqrt{2}}$	0	0	$-\frac{\sqrt{2}}{\sqrt{5}}$	0	0	0	0	$\frac{1}{\sqrt{10}}$	0	0
0	ν	0	0	0	0	0	$-\frac{\sqrt{3}}{\sqrt{5}}$	0	0	0	0	$\frac{\sqrt{2}}{\sqrt{5}}$	0
-1	ν	0	0	0	0	0	0	0	0	0	0	0	1

Appendix H
Solutions to Problems

1.1 The diagram for the product $\hat{C}_2^z\ \hat{\imath}$ is the same as in Fig. 1.1, except for the intermediate point P_2, which should be denoted by a circle instead of a cross, since it is now below the gray disc. However, the end point P_3 remains the same, irrespective of the order of the operators. This implies that their commutator vanishes.

1.2 Represent the rotation of the coordinates by the rotational matrix $\mathbb{D}$, as given by

$$\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \mathbb{D} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$\begin{pmatrix} x_2 & y_2 \end{pmatrix} = \begin{pmatrix} x_1 & y_1 \end{pmatrix} \mathbb{D}^T = \begin{pmatrix} x_1 & y_1 \end{pmatrix} \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}$$

Express the sum $x_2^2 + y_2^2$ as the scalar product of the coordinate row with the coordinate column, and verify that this scalar product remains invariant under the matrix transformation.

1.3 Apply Eq. 1.9 twice:

$$\hat{R}\hat{S}f(P_1) = \hat{R}\left[\hat{S}f(P_1)\right] = \hat{R}\left[f(\hat{S}^{-1}P_1)\right] = f(\hat{S}^{-1}\hat{R}^{-1}P_1)$$

1.4 Apply the general rule that a displacement of the function corresponds to an opposite coordinate displacement. As a result of the transformation, the function acquires an additional phase factor.

$$\mathcal{T}_a\ e^{ikx} = e^{ik(x-a)} = e^{-ika}e^{ikx}$$

1.5 The action of a rotation about the z-axis can be expressed by a differential operator as

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$$\hat{O}(\alpha) = \cos\alpha \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right) + \sin\alpha \left(y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y} \right)$$

The unit element corresponds to $\alpha = 0$, hence:

$$\hat{E} = \hat{O}(0) = \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right)$$

The angular momentum operator is given by

$$\begin{aligned} \mathcal{L}_z &= x p_y - y p_x \\ &= \frac{\hbar}{i} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \\ &= -\frac{\hbar}{i} \lim_{\alpha \to 0} \frac{\hat{O}(\alpha) - \hat{E}}{\alpha} \end{aligned}$$

The angular momentum operator thus is proportional to an infinitesimal rotation in the neighborhood of the unit element.

2.1 One has

$$\mathbb{D}(R^{-1})\mathbb{D}(R) = \mathbb{D}(R)\mathbb{D}(R^{-1}) = \mathbb{D}(RR^{-1}) = \mathbb{I}$$

Thus the matrix representing the inverse operation multiplies left and right with $\mathbb{D}(R)$ to yield the unit matrix. This is a property which uniquely defines the inverse of $\mathbb{D}(R)$.

2.2 The condition that $\mathbb{C}$ be unitary gives rise to six equations:

$$\begin{aligned} 1 &= |a|^2 + |b|^2 \\ 1 &= |a|^2 + |c|^2 \\ 1 &= |b|^2 + |d|^2 \\ 1 &= |c|^2 + |d|^2 \\ 0 &= |ac| e^{i(\alpha-\gamma)} + |bd| e^{i(\beta-\delta)} \\ 0 &= |ab| e^{i(\alpha-\beta)} + |cd| e^{i(\gamma-\delta)} \end{aligned}$$

From these equations, it is clear that $|a| = |d|$ and $|b| = |c|$. The phase relationships may be reduced to

$$e^{i(\beta+\gamma)} = -e^{i(\alpha+\delta)}$$

With the help of these results, the four matrix entries can be rewritten as

$$
\begin{aligned}
|a|e^{i\alpha} &= |a|e^{i(\alpha+\delta)/2}e^{i(\alpha-\delta)/2} \\
|d|e^{i\delta} &= |a|e^{i(\alpha+\delta)/2}e^{-i(\alpha-\delta)/2} \\
|b|e^{i\beta} &= |b|e^{i(\alpha+\delta)/2}e^{i\left[\beta-\frac{\alpha+\delta}{2}\right]} \\
|c|e^{i\gamma} &= -|b|e^{i(\alpha+\delta)/2}e^{i\left[-\beta+\frac{\alpha+\delta}{2}\right]}
\end{aligned}
$$

The general $U(2)$ matrix may thus be rewritten as

$$
\mathbb{U} = e^{i(\alpha+\delta)/2}\begin{pmatrix} |a|e^{i(\alpha-\delta)/2} & |b|e^{i\left[\beta-\frac{\alpha+\delta}{2}\right]} \\ -|b|e^{i\left[-\beta+\frac{\alpha+\delta}{2}\right]} & |a|e^{-i(\alpha-\delta)/2} \end{pmatrix}
$$

with: $|a|^2 + |b|^2 = 1$. Note that a general phase factor has been taken out. The remaining matrix has determinant $+1$ and is called a special unitary matrix (see further in Chap. 7).

2.3 The relevant integrals are given by

$$
\begin{aligned}
\int_0^{2\pi} e^{-ik\phi}e^{ik\phi}d\phi &= [\phi]_0^{2\pi} = 2\pi \\
\int_0^{2\pi} e^{\pm 2ik\phi}d\phi &= \frac{1}{\pm 2ik}[e^{\pm 2ik\phi}]_0^{2\pi} = 0
\end{aligned}
$$

The normalized cyclic waves are thus given by

$$
|\pm k\rangle = \frac{1}{\sqrt{2\pi}}e^{\pm ik\phi}
$$

and these waves are orthogonal: $\langle -k|k\rangle = 0$.

2.4 The combination of transposition and complex conjugation is called the *adjoint* operation, indicated by a dagger. A Hermitian matrix is thus self-adjoint. An eigenfunction of this matrix, operating in a function space, may be expressed as a linear combination.

$$
|\psi_m\rangle = \sum_k = c_k|f_k\rangle
$$

We may arrange the expansion coefficients as a column vector $\mathbf{c}$. This is called the eigenvector. Its adjoint, $\mathbf{c}^\dagger$, is then the complex-conjugate row vector. The corresponding eigenvalue is denoted as E_m. Now start by writing the eigenvalue equation, and multiply left and right with the adjoint eigenvector.

$$
\begin{aligned}
\mathbb{H}\mathbf{c} &= E_m\mathbf{c} \\
\mathbf{c}^\dagger\mathbb{H}\mathbf{c} &= E_m\mathbf{c}^\dagger\mathbf{c}
\end{aligned}
$$

Now take the adjoint and use the self-adjoint property of $\mathbb{H}$:

$$\mathbf{c}^\dagger \mathbb{H}^\dagger \mathbf{c} = \overline{E}_m \mathbf{c}^\dagger \mathbf{c}$$
$$\mathbf{c}^\dagger \mathbb{H} \mathbf{c} = \overline{E}_m \mathbf{c}^\dagger \mathbf{c}$$

A comparison of both results shows that the eigenvalue must be equal to its complex conjugate, and hence be real. If $\mathbb{H}$ is skew-symmetric a similar argument shows that the eigenvalue must be imaginary.

3.1 The table is a valid multiplication table of a group which is isomorphic to D_2. The element C is the unit element. There are six ways to assign the three twofold axes to the letters A, B, D.

3.2 With $\hat{C}_3 \rightarrow (123)$ and $\hat{\sigma}_1 \rightarrow (23)$, one has

$$\hat{C}_3^2 = \hat{\sigma}_1 \hat{C}_3 \hat{\sigma}_1 \rightarrow \ (132)$$
$$\hat{\sigma}_2 = \hat{C}_3 \hat{\sigma}_1 \hat{C}_3^2 \rightarrow \ (13)$$
$$\hat{\sigma}_3 = \hat{C}_3 \hat{\sigma}_2 \hat{C}_3^2 \rightarrow \ (12)$$

Verify that these permutations strictly correspond to the conventions in the group multiplication table of C_{3v}, e.g., $\hat{C}_3 \hat{\sigma}_1$ is written as

$$\begin{array}{rc} & \langle \mathbf{1\ 2\ 3} \rangle \\ & \langle \mathrm{A\ B\ C} \rangle \\ (1)(23) & \Downarrow \\ & \langle \mathrm{A\ C\ B} \rangle \\ (123) & \Downarrow \\ & \langle \mathrm{B\ A\ C} \rangle \end{array}$$

The result is the permutation (12)(3), which corresponds to $\hat{\sigma}_3$ in accordance with the multiplication table for C_{3v}.

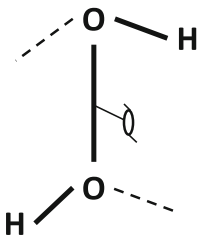
3.3 For H a normal subgroup, left and right cosets are the same, hence coset multiplication yields

$$\hat{g}_i H \times \hat{g}_j H = \hat{g}_i \hat{g}_j H \times H = \hat{g}_i \hat{g}_j H$$

Note that the product $H \times H$ is the multiplication table for the subgroup H. Every element appears in this table $|H|$ times, thus the ingredients of this product are the different elements of H, or: $H \times H = H$. The equation thus shows that the product of two cosets is again a coset. The other group properties are straightforward: the unit element is the subgroup H itself, the inverse of a coset is the coset formed by the inverse representative, and associativity is a property of the parent group G.

3.4 Any non-linear triatomic molecule with three different atoms has only C_s symmetry, e.g., a water molecule with one hydrogen replaced by deuterium. C_2 symmetry requires a non-planar tetra-atomic molecule, such as H_2O_2. In the free state, the dihedral angle of this molecule is almost a right angle (see Figure). To

realize C_i symmetry, one needs at least six atoms. Since three atoms are always coplanar, the smallest molecule with no symmetry at all has at least four atoms.



3.5 There are only three regular tesselations of the plane: triangles, squares, and hexagons.

3.6 The rotation generates points that are lying on a circle, perpendicular to the rotation. If the rotational angle is not a rational fraction of a full angle, every time the rotation is repeated a new point will be generated, until in the limit the whole circle is covered and we obtain a continuous body instead of a discrete polyhedron. To obtain an integer order, the additional requirement to be added is that the original point is retrieved after one full turn.

3.7 Consider a subgroup $H \subset G$, such that $|G|/|H| = 2$. Then the coset expansion of G will be limited to only two cosets.

$$G = H + \hat{g}\ H$$

Here $\hat{g}$ is a coset generator outside H. The subgroup is normal if the right and left cosets coincide. Since there is only one coset outside H, it is required that

$$\hat{g}\ H = H\ \hat{g}$$

Indeed, suppose that this equation does not hold. Then this can only mean that there are elements in H such that

$$\hat{h}_x \hat{g} = \hat{h}_y$$

But then the coset generator, $\hat{g}$, must be an element of H, which contradicts the starting assumption.

3.8 Soccer ball: I_h. Tennis ball: D_{2d}. Basketball: D_{2d}. Trefoil knot: D_3. Rhombic dodecahedron: O_h. Parallelepiped: D_{3d}. Pentadiamond: T_d. Adamantane: T_d. Cyclohexane chair: D_{3d}. Cyclohexane boat: D_{2h}.

3.9 The only chiral structure in the figure is the trefoil knot.

4.1 The site symmetry of a cube is T_h. The cube is an invariant of its site group, and transforms as a_g in T_h. The set of five cubes thus spans the induced representation: $aT_h \uparrow I_h$. Applying the Frobenius theorem to the subduction (see Appendix C.1), one obtains

$$aT_h \uparrow I_h = A_g + G_g$$

4.2 The irreps can be obtained from the induction table in Appendix C.2, as $\Gamma_\pi C_{3v} \uparrow T_d$:

$$\Gamma_\pi C_{3v} \uparrow T_d = E + T_1 + T_2$$

The SALCS shown span the tetrahedral E irrep, the one on the left is the E_θ component, the one on the right is the E_ϵ component. Note that they transform into each other by rotating all π-orbitals over 90° in the same sense [13].

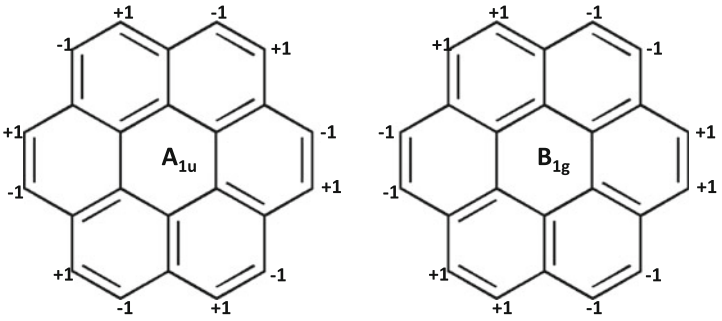
4.3 The 24 carbon atoms of coronene form three orbits: two orbits of six atoms, corresponding to the internal hexagon and to the six atoms on the outer ring that have bonds to the inner ring, and one orbit of the twelve remaining atoms. The elements of the six orbit occupy sites of C'_{2v} symmetry, based on $\hat{C}'_2$, $\hat{\sigma}_h$, $\hat{\sigma}_v$ in D_{6h}. The p_z orbitals on these sites transform as b_1, hence the induced irreps are as in the case of benzene:

$$b_1 C_{2v} \uparrow D_{6h} = B_{2g} + A_{2u} + E_{1g} + E_{2u}$$

The remaining 12-orbit connects carbon atoms with only C_s site symmetry, the p_z orbitals on these sites transforming as a''. The induced irreps read

$$a'' C_s \uparrow D_{6h} = B_{1g} + B_{2g} + A_{1u} + A_{2u} + 2E_{1g} + 2E_{2u}$$

The A_{1u} and B_{1g} irreps only appear in the 12-orbit, so we can infer that the molecular orbitals with this symmetry will entirely be localized on the 12-orbit. The SALCs can easily be constructed, as they should be anti-symmetric with respect to the $\hat{\sigma}_v$ planes in order not to hybridize with the SALCs based on the 6-orbits.



4.4 The tangential π-orbitals transform as Γ_π in the C_{5v} site group of I_h. According to Appendix C.2, one has

$$\Gamma_\pi\ C_{5v} \uparrow I_h = T_{1g} + T_{1u} + G_g + G_u + H_g + H_u$$

4.5 When the projector which generated the component is characterized as $\hat{P}^{\Gamma_i}_{kl}$ the other components may be found by varying the k index.

4.6 Act with an operator $\hat{S}$ on the projector, and carry out the substitution $\hat{R} = \hat{S}^{-1}\hat{T}$:

$$\begin{aligned}\hat{S}\ \hat{P}^{\Gamma_0}_{11} &= \hat{S}\frac{1}{|G|}\sum_R \hat{R} \\ &= \frac{1}{|G|}\sum_R \hat{S}\hat{R} = \frac{1}{|G|}\sum_T \hat{T} = \hat{P}^{\Gamma_0}_{11}\end{aligned}$$

4.7 Applying the inverse transformation to the SALCs of the hydrogens in ammonia, yields

$$\left(|sp^2_{\mathrm{A}}\rangle\ |sp^2_{\mathrm{B}}\rangle\ |sp^2_{C}\rangle\right) = \left(|2s\rangle\ |2p_x\rangle\ |2p_y\rangle\right)\begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

4.8 This mode transforms as E_y. It can be written as a linear combination of a radial and a tangent mode:

$$Q = -\frac{1}{\sqrt{2}}Q^{\mathrm{rad}}_y + \frac{1}{\sqrt{2}}Q^{\mathrm{tan}}_y$$

with:

$$\begin{aligned}Q^{\mathrm{rad}}_y &= \frac{1}{\sqrt{2}}\left(\Delta R_{\mathrm{B}} - \Delta R_{\mathrm{C}}\right) \\ Q^{\mathrm{tan}}_y &= \frac{1}{\sqrt{6}}R\left(2\Delta\phi_{\mathrm{A}} - \Delta\phi_{\mathrm{B}} - \Delta\phi_{\mathrm{C}}\right)\end{aligned}$$

This mode preserves the center of mass and is a genuine normal mode.

4.9 Since all irreps are one-dimensional, the characters can only consist of a phase factor:

$$\mathbb{D}(C_5) = e^{i\lambda}\mathbb{I}$$

The fifth power of the generator will yield the unit element, hence:

$$e^{5i\lambda} = 1$$

This is the Euler equation. Its solutions are the characters in the table of C_5, as given in Appendix A.

4.10 The product of inversion with a $\hat{C}_2$ axis must yield a reflection plane, perpendicular to this axis. As an example a product of type $\hat{\imath}\cdot\hat{C}'_2$ must yield a reflection plane of $\hat{\sigma}_d$ type, as this is perpendicular to the primed twofold axis. For the one-dimensional irreps of D_{6h}, one thus should have

$$\chi(\iota)\chi(C_2') = \chi(\sigma_d)$$

This is indeed verified to be the case.

4.11 The a_2'' distortion is anti-symmetric with respect to $3\hat{C}_2$, $\hat{\sigma}_h$ and $2\hat{S}_3$. As a result when the mode is launched, all these symmetry elements will be destroyed, and the symmetry reduces to the subgroup C_{3v}. In general, the result of a distortion will always be the maximal subgroup for which the distortion is totally symmetric [14].

4.12 The group of this fullerene is D_{6d}. The 24 atoms separate into two orbits: a 12-orbit containing the top and bottom hexagons, and another 12-orbit containing the crown of the 12 atoms, numbered from 7 to 18. In both cases the site group is only C_s, hence both orbits will span the same irreps:

$$a'C_s \uparrow D_{6d} = A_1 + B_2 + E_1 + E_2 + E_3 + E_4 + E_5$$

Quite remarkably the Hückel spectrum for this fullerene has a non-bonding level of E_4 symmetry.

4.13 Label the atoms in the $2N + 2$ cyclic polyene from 0 to 2N + 1, with the bridges at positions $j = 0$ and $j = N + 1$. The unnormalized Hückel orbitals for the cyclic polyene are then given in the standing wave format as

$$\Phi_k = \sum_j \cos\frac{2\pi kj}{2N+2}|\phi_j\rangle$$

$$\Psi_k = \sum_j \sin\frac{2\pi kj}{2N+2}|\phi_j\rangle$$

for integer values of k in the interval $[0, N + 1]$. The number of Φ eigenkets in this interval is $N + 2$, versus N eigenkets of Ψ, totaling $2N + 2$, which is the dimension of the basis space. To isolate solutions that apply to the linear N-chain, it is required that the bridge atoms correspond to nodes of the wavefunction. This eliminates the Φ waves. The SALCs for the linear chain thus are restricted to the Ψ waves:

$$\Psi_k = \sum_{j=1,N} \sin\frac{\pi kj}{N+1}|\phi_j\rangle$$

where k is an integer in the interval $[1, N]$. Note that these wavefunctions correspond to projections on the basis space of the wavefunctions for a particle in a box, with a length of $N + 1$ bonds. It is remarkable that cyclic symmetry can be used here for a finite chain. The nearest neighbor clause in the tight-binding model is responsible for this artificial symmetry.

5.1 Let $\mathbf{r}_i$ and $\mathbf{r}_j$ denote the position vectors of electrons i and j. The electron repulsion operator contains the distance between both electrons as $|\mathbf{r}_i - \mathbf{r}_j|$. The matrix $\mathbb{D}(R)$ expresses the transformation of the Cartesian coordinates under a

rotation. This matrix will also rotate the coordinate *differences*:

$$\hat{R}\begin{pmatrix} x_i - x_j \\ y_i - y_j \\ z_i - z_j \end{pmatrix} = \mathbb{D}(R)\begin{pmatrix} x_i - x_j \\ y_i - y_j \\ z_i - z_j \end{pmatrix}$$

Exactly as in the derivation for problem 1.2 the square of the distance between the two electrons is then found to be invariant under any orthogonal transformation of the coordinates.

5.2 For the G irrep, it is noted from Appendix C.1 that a tetrahedral splitting field will branch G into $A + T$. It thus acts as a splitting field to isolate the unique Ga component. Symmetry adaptation to $\hat{C}_2^z$ will yield two totally symmetric components, one of which will be the Ga already obtained. The remaining one is then Gz. The corresponding Gx and Gy may then be found by cyclic permutation under the $\hat{C}_3^{xyz}$ axis.

For the H irrep, one may make use of the $\hat{C}_3^{xyz}$ axis again. It resolves H into $A_1 + 2E$. This unique A_1 component will be the sum of $H\xi + H\eta + H\zeta$. We can project the $H\zeta$ component out of this sum, by using the $\hat{C}_2^z$ axis. Although the H level subduces three totally symmetric irreps in C_2 there will be no contamination with $H\theta$ and $H\epsilon$, since these were already removed in the first step by projecting out the trigonal A_1.

5.3 The total number of nuclear permutations and permutation-inversions for CH_3BF_2 is 24. This is the product of six permutations of the protons, two permutations of the fluorine nuclei, and the binary group of the spatial inversion. However, as the fluxionality of this molecule is limited to free rotations of the methyl group, the operations should be limited to those permutations or permutation-inversions that lead to structures that can be rotated back to the original frame, *or to a rotamer of this frame*. Only half of the operations will comply with this requirement. As an example, the odd permutations of the protons are not allowed, since the resulting structure cannot be turned into the original one by outer rotations, or rotations of the methyl group around the C-B bond. However odd permutations of the protons in combination with spatial inversion are allowed. The results are given in [15]. The corresponding symmetry group is isomorphic with D_{3h}.

5.4 Ferrocene is a molecule with two identical co-axial rotors. Its nuclear permutation-inversion group consists of 100 elements. It has a halving rotational subgroup of 50 proper permutations: for each of the cyclo-pentadienyl rings there are 5 cyclic permutation operations, yielding a total of $5^2 = 25$ operations, and this number must be doubled to account for the permutation of the upper and lower rings. In addition, there is a coset of improper permutation-inversions, containing another 50 elements. This cosets also contains two kinds of elements. In the table, we summarize the structure of the group. The carbon atoms are numbered 1, ..., 5 in the upper ring and 6, ..., 10 in the lower ring.

Nuclear permutation-inversion group for ferrocene (u and l refer to upper and lower rotor)		
	$\hat{R}$	#
$C_5^u \times C_5^l$	(12345)	25
(ul)	(16)(2, 10)(39)(48)(57)	25
$5\hat{\sigma}_v^u \times 5\hat{\sigma}_v^l$	(25)(34)(7, 10)(89)*	25
$(ul)^*$	(16)(27)(38)(49)(5, 10)*	25

6.1 The $(t_{1u})^2$ configuration gives rise to 15 states. The direct product decomposes as follows (see Appendix D):

$$T_{1u} \times T_{1u} = [A_{1g} + E_g + T_{2g}] + \{T_{1g}\}$$

The symmetrized part will give rise to 6 singlet functions, while there are 9 triplet substates, forming a ${}^3T_{1g}$ multiplet. Since the 3-electron Ψ state is a quartet, the singlet states cannot contribute, and we only need to couple the triplet to a ${}^2T_{1u}$ state, resulting from a $(t_{1u})^1$ configuration. The orbital part of the triplet is obtained from the $T_1 \times T_1 = T_1$ coupling table in Appendix F:

$$\begin{aligned}
|T_{1g}x\rangle &= 1/\sqrt{2}\,[-y(1)z(2) + z(1)y(2)] \\
|T_{1g}y\rangle &= 1/\sqrt{2}\,[x(1)z(2) - z(1)x(2)] \\
|T_{1g}z\rangle &= 1/\sqrt{2}\,[-x(1)y(2) + y(1)x(2)]
\end{aligned}$$

The coupling with the third electron can yield A_{1u}, E_u, T_{1u}, and T_{2u} states. Our result is based on the A_{1u} product. This yields

$$\begin{aligned}
A_{1u} &= 1/\sqrt{3}\left[|T_{1g}x\rangle|x(3)\rangle + |T_{1g}y\rangle|y(3)\rangle + |T_{1g}z\rangle|z(3)\rangle\right] \\
&= -\frac{1}{\sqrt{6}}\begin{vmatrix} x(1) & y(1) & z(1) \\ x(2) & y(2) & z(2) \\ x(3) & y(3) & z(3) \end{vmatrix}
\end{aligned}$$

This should be multiplied by the product of the three α-spins, $\alpha_1\alpha_2\alpha_3$, to obtain the ${}^4A_{1u}$ ground state of the $(t_{1u})^3$ configuration.

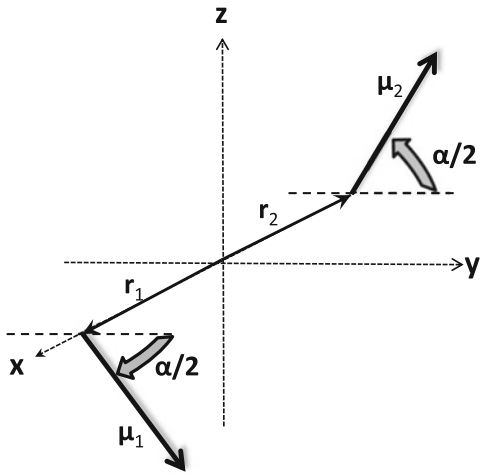
6.2 The JT problem is determined by the symmetrized direct product of T_{1u}. As we have seen in the previous problem, this product contains $A_{1g} + E_g + T_{2g}$. Since A_{1g} modes don't break the symmetry the JT problem is of type $T_1 \times (e + t_2)$. In the linear problem, only two force elements are required. The distortion matrix is thus as follows:

$$\mathcal{H}' = \frac{F_E}{2}\begin{pmatrix} -Q_\theta + \sqrt{3}Q_\epsilon & 0 & 0 \\ 0 & -Q_\theta - \sqrt{3}Q_\epsilon & 0 \\ 0 & 0 & 2Q_\theta \end{pmatrix} + \frac{F_T}{\sqrt{2}}\begin{pmatrix} 0 & -Q_\zeta & -Q_\eta \\ -Q_\zeta & 0 & -Q_\xi \\ -Q_\eta & Q_\xi & 0 \end{pmatrix}$$

6.3 The magnetic dipole operator transforms as T_{1g}, while the direct square of e_g irreps yields $A_{1g} + A_{2g} + E_g$. Since the operator irrep is not contained in the product space, the selection rules will not allow a dipole matrix element between e_g orbitals.

6.4 We first draw a simple diagram, representing the R-conformation. The point group is C_2. The twofold-axis is oriented along the y-direction, and the centers of the two chromophores are placed on the positive and negative x-axis. The dipole moments with (x, y, z) components are then oriented as

$$\boldsymbol{\mu}_1 = \mu\left(0, \cos\frac{\alpha}{2}, -\sin\frac{\alpha}{2}\right)$$
$$\boldsymbol{\mu}_2 = \mu\left(0, \cos\frac{\alpha}{2}, \sin\frac{\alpha}{2}\right)$$



The exciton states on both chromophores are interchanged by the twofold axis, and can be recombined to yield a symmetric and an anti-symmetric combination, denoted as A and B, respectively. One has

$$|\Psi_A\rangle = \frac{1}{\sqrt{2}}\left(|\Psi_1\rangle + |\Psi_2\rangle\right)$$
$$|\Psi_B\rangle = \frac{1}{\sqrt{2}}\left(|\Psi_1\rangle - |\Psi_2\rangle\right)$$

The corresponding transition dipoles are oriented along the positive y- and negative z-direction, respectively:

$$\boldsymbol{\mu}_A = \sqrt{2}\mu\left(0, \cos\frac{\alpha}{2}, 0\right)$$
$$\boldsymbol{\mu}_B = \sqrt{2}\mu\left(0, 0, -\sin\frac{\alpha}{2}\right)$$

The dipole–dipole interaction is given by

$$V_{12} = \frac{1}{4\pi\epsilon_0}\frac{\cos\alpha}{R_{12}^3}$$

For $\alpha < \pi/2$, the dipole orientation is repulsive. As a result, the in-phase coupled exciton state $|\Psi_A\rangle$ will be at higher energy than the out-of-phase $|\Psi_B\rangle$ state. Finally, we also calculate the magnetic transition dipoles, using the expressions from Sect. 6.8:

$$\mathbf{m}_A = \frac{i\pi\nu}{\sqrt{2}}\left(\mathbf{r}_1 \times \boldsymbol{\mu}_1 + \mathbf{r}_2 \times \boldsymbol{\mu}_2\right) = \frac{i\pi\nu\mu}{\sqrt{2}} R_{12}\sin\frac{\alpha}{2}\,(0, 1, 0)$$
$$\mathbf{m}_B = \frac{i\pi\nu}{\sqrt{2}}\left(\mathbf{r}_1 \times \boldsymbol{\mu}_1 - \mathbf{r}_2 \times \boldsymbol{\mu}_2\right) = \frac{i\pi\nu\mu}{\sqrt{2}} R_{12}\cos\frac{\alpha}{2}\,(0, 0, 1)$$

These results are now combined in the Rosenfeld equation to yield the rotatory strength of both exciton states:

$$\mathcal{R}_A = \frac{\pi\nu\mu^2}{2} R_{12}\sin\alpha$$
$$\mathcal{R}_B = -\frac{\pi\nu\mu^2}{2} R_{12}\sin\alpha$$

This result predicts a normal CD sign, with a lower negative branch (B-state) and an upper positive branch (A-state) [16]. This is typical of a right-handed helix, corresponding to a rotation of the dipoles in the right-handed sense when going from chromophore 1 to chromophore 2 along the inter-chromophore axis. In the S-conformation the sign of α will change, and the CD spectrum will be inverted.

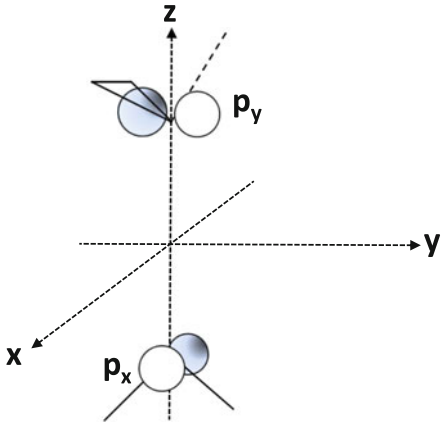
6.5 The direct square of the e-irrep in D_{2d} yields 4 coupled states:

$$e \times e = A_1 + A_2 + B_1 + B_2$$

The corresponding coupling coefficients are given in the table below. This table is almost the same as the table for D_4 in Appendix F, but note that B_1 and B_2 have been interchanged. Such details are important, and therefore we draw again a simple picture of the molecule in a Cartesian system. Both in D_4 and in D_{2d}, the B_1 and B_2 irreps are distinguished by their symmetry with respect to the $\hat{C}_2'$ axes.

$\mathbf{D_{2d}}$				
$E \times E$	$\frac{A_1}{a_1}$	$\frac{A_2}{a_2}$	$\frac{B_1}{b_1}$	$\frac{B_2}{b_2}$
$x\ x$	$\frac{1}{\sqrt{2}}$	0	0	$-\frac{1}{\sqrt{2}}$
$y\ y$	$\frac{1}{\sqrt{2}}$	0	0	$\frac{1}{\sqrt{2}}$
$x\ y$	0	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0
$y\ x$	0	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	0

In the orientation of twisted ethylene, as indicated in the figure below, the directions of these axes are along the bisectors of x and y. In contrast in the standard orientation for D_4, they are *along* the x and y axes, while the bisector directions coincide with the $\hat{C}_2''$ axes, hence the interchange between B_1 and B_2.



Note that the two-electron states are symmetrized, except the A_2 combination. The symmetrized states will combine with singlet spin states, while the A_2 state will be a triplet. One thus has

$$
\begin{aligned}
{}^1A_1 &= \frac{1}{\sqrt{2}}\left(x(1)x(2) + y(1)y(2)\right)\frac{1}{\sqrt{2}}\left(\alpha(1)\beta(2) - \beta(1)\alpha(2)\right) \\
&= \frac{1}{\sqrt{2}}\left(|(x\alpha)(x\beta)| + |(y\alpha)(y\beta)|\right) \\
{}^1B_1 &= \frac{1}{\sqrt{2}}\left(|(x\alpha)(y\beta)| + |(y\alpha)(x\beta)|\right) \\
{}^1B_2 &= \frac{1}{\sqrt{2}}\left(-|(x\alpha)(x\beta)| + |(y\alpha)(y\beta)|\right) \\
{}^3A_2 &= |(x\alpha)(y\alpha)|
\end{aligned}
$$

The ${}^1A_1$ and ${}^1B_2$ states are the *zwitterionic states*, while the ${}^1B_1$ and ${}^3A_2$ states are called the *diradical* states. It is clear from the expressions that in both cases the two radical carbon sites are neutral. The zwitterionic states are easily polarizable though.

6.6 The carbon atoms in TMM form two orbits. The p_z orbital on the central atom is in the center of the D_{3h} symmetry group and transforms as a_2''. The three methylene orbitals are in C_{2v} sites, transforming as the b_2 irrep of the site group, i.e., they are anti-symmetric with respect to $\hat{\sigma}_h$ and symmetric with respect to $\hat{\sigma}_v$. The induced representation is:

$$b_2 C_{2v} \uparrow D_{3h} = a_2'' + e''$$

The SALCs are entirely similar to the hydrogen SALCs in the case of ammonia; this implies for instance that the e-component labeled x is symmetric under the vertical symmetry plane through atom A. It will be anti-symmetric for the twofold-axis going through atom A, since the relevant orbital is of p_z type.

$$
\begin{aligned}
|\Psi_a\rangle &= \frac{1}{\sqrt{3}}\left(|p_A\rangle + |p_B\rangle + |p_C\rangle\right) \\
|\Psi_x\rangle &= \frac{1}{\sqrt{6}}\left(2|p_A\rangle - |p_B\rangle - |p_C\rangle\right) \\
|\Psi_y\rangle &= \frac{1}{\sqrt{2}}\left(|p_B\rangle - |p_C\rangle\right)
\end{aligned}
$$

The a_2'' orbitals interact to yield bonding and antibonding combinations at $E = \alpha \pm \sqrt{3}\beta$. Since the graph is bipartite the remaining e'' orbitals are necessarily non-bonding, and will be occupied by two electrons. The direct square of this irrep yields symmetrized A_1' and E' states, and an anti-symmetrized A_2' state. The expressions for these states are obtained from the coupling coefficients for D_3 in Appendix F:

$$
\begin{aligned}
{}^1A_1' &= \frac{1}{\sqrt{2}}\left(x(1)x(2) + y(1)y(2)\right)\frac{1}{\sqrt{2}}\left(\alpha(1)\beta(2) - \beta(1)\alpha(2)\right) \\
&= \frac{1}{\sqrt{2}}\left(|(x\alpha)(x\beta)| + |(y\alpha)(y\beta)|\right) \\
{}^1E_y' &= \frac{1}{\sqrt{2}}\left(|(x\alpha)(y\beta)| + |(y\alpha)(x\beta)|\right) \\
{}^1E_x' &= \frac{1}{\sqrt{2}}\left(-|(x\alpha)(x\beta)| + |(y\alpha)(y\beta)|\right) \\
{}^3A_2 &= |(x\alpha)(y\alpha)|
\end{aligned}
$$

Note that the distinction between zwitterionic and diradical states does not hold in this case. Formally TMM can be described as a valence isomer between three configurations in which one of the peripheral atoms has a double bond to the central atom, and the other two sites carry an unpaired electron.

7.1 In a cube the d-shell also splits in $e_g + t_{2g}$, but the ordering is reversed. Explicit calculation of the potential shows that the splitting is reduced by a factor 8/9:

$$\Delta_{\text{cube}} = -\frac{8}{9}\Delta_{\text{octahhedron}}$$

7.2 Perform the matrix multiplication and verify that the product matrix is of Cayley–Klein form. The multiplication is not commutative.

$$
\begin{pmatrix} a_1 & b_1 \\ -\overline{b}_1 & \overline{a}_1 \end{pmatrix} \times \begin{pmatrix} a_2 & b_2 \\ -\overline{b}_2 & \overline{a}_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 - b_1\overline{b}_2 & a_1b_2 + \overline{a}_2b_1 \\ -\overline{a}_1\overline{b}_2 - a_2\overline{b}_1 & \overline{a}_1\overline{a}_2 - \overline{b}_1b_2 \end{pmatrix}
$$

7.3 The double group D_3^* contains 12 elements. In the table, we list the six representation matrices for the elements on the positive hemisphere. The $\hat{C}_2^A$ axis is along the x-direction, $\hat{C}_2^B$ is at 120° and $\hat{C}_2^C$ is in between. The derivation of the multiplication table and the underlying class structure (see Table 7.6) is then straightforward.

Representation matrices for the spinor basis in D_3^* ($\omega = \exp(\pi i/6)$)

$$\mathbb{D}(E) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \mathbb{D}(C_2^A) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$$

$$\mathbb{D}(C_3) = \begin{pmatrix} -i\overline{\omega} & 0 \\ 0 & i\omega \end{pmatrix} \qquad \mathbb{D}(\aleph C_2^B) = \begin{pmatrix} 0 & \overline{\omega} \\ -\omega & 0 \end{pmatrix}$$

$$\mathbb{D}(C_3^2) = \begin{pmatrix} -i\omega & 0 \\ 0 & i\overline{\omega} \end{pmatrix} \qquad \mathbb{D}(\aleph C_2^C) = \begin{pmatrix} 0 & -\omega \\ \overline{\omega} & 0 \end{pmatrix}$$

Multiplication table for the double group D_3^*

D_3^*	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\aleph\hat{C}_3$	$\aleph\hat{C}_3^2$	$\aleph$	$\hat{C}_2^A$	$\hat{C}_2^B$	$\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\aleph\hat{C}_2^C$
$\hat{E}$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\aleph\hat{C}_3$	$\aleph\hat{C}_3^2$	$\aleph$	$\hat{C}_2^A$	$\hat{C}_2^B$	$\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\aleph\hat{C}_2^C$
$\hat{C}_3$	$\hat{C}_3$	$\hat{C}_3^2$	$\aleph$	$\aleph\hat{C}_3^2$	$\hat{E}$	$\aleph\hat{C}_3$	$\hat{C}_2^C$	$\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\hat{C}_2^B$
$\hat{C}_3^2$	$\hat{C}_3^2$	$\aleph$	$\aleph\hat{C}_3$	$\hat{E}$	$\hat{C}_3$	$\aleph\hat{C}_3^2$	$\aleph\hat{C}_2^B$	$\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\hat{C}_2^A$
$\aleph\hat{C}_3$	$\aleph\hat{C}_3$	$\aleph\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3^2$	$\aleph$	$\hat{C}_3$	$\aleph\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\hat{C}_2^B$	$\hat{C}_2^C$	$\hat{C}_2^A$	$\aleph\hat{C}_2^B$
$\aleph\hat{C}_3^2$	$\aleph\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\aleph$	$\aleph\hat{C}_3$	$\hat{C}_3^2$	$\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\hat{C}_2^C$	$\aleph\hat{C}_2^A$
$\aleph$	$\aleph$	$\aleph\hat{C}_3$	$\aleph\hat{C}_3^2$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\aleph\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\hat{C}_2^A$	$\hat{C}_2^B$	$\hat{C}_2^C$
$\hat{C}_2^A$	$\hat{C}_2^A$	$\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\aleph\hat{C}_2^B$	$\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\aleph$	$\aleph\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\hat{C}_3$	$\aleph\hat{C}_3^2$
$\hat{C}_2^B$	$\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\hat{C}_2^C$	$\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\hat{C}_3^2$	$\aleph$	$\hat{C}_3$	$\aleph\hat{C}_3^2$	$\hat{E}$	$\aleph\hat{C}_3$
$\hat{C}_2^C$	$\hat{C}_2^C$	$\hat{C}_2^A$	$\hat{C}_2^B$	$\aleph\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\aleph\hat{C}_3$	$\aleph\hat{C}_3^2$	$\aleph$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$
$\aleph\hat{C}_2^A$	$\aleph\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\hat{C}_2^C$	$\hat{C}_2^B$	$\aleph\hat{C}_2^C$	$\hat{C}_2^A$	$\hat{E}$	$\hat{C}_3$	$\aleph\hat{C}_3^2$	$\aleph$	$\aleph\hat{C}_3$	$\hat{C}_3^2$
$\aleph\hat{C}_2^B$	$\aleph\hat{C}_2^B$	$\hat{C}_2^C$	$\hat{C}_2^A$	$\aleph\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\hat{C}_2^B$	$\aleph\hat{C}_3^2$	$\hat{E}$	$\aleph\hat{C}_3$	$\hat{C}_3^2$	$\aleph$	$\hat{C}_3$
$\aleph\hat{C}_2^C$	$\aleph\hat{C}_2^C$	$\aleph\hat{C}_2^A$	$\aleph\hat{C}_2^B$	$\hat{C}_2^A$	$\hat{C}_2^B$	$\hat{C}_2^C$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{E}$	$\aleph\hat{C}_3$	$\aleph\hat{C}_3^2$	$\aleph$

7.4 The action of the spin operators on the components of a spin-triplet can be found by acting on the coupled states, as summarized in Table 7.2. As an example, where we have added the electron labels 1 and 2 for clarity:

$$\begin{aligned} S_x|+1\rangle &= S_x\left[|\alpha_1\rangle|\alpha_2\rangle\right] = \left[S_x|\alpha_1\rangle\right]|\alpha_2\rangle + |\alpha_1\rangle\left[S_x|\alpha_2\rangle\right] \\ &= \frac{\hbar}{2}\left[|\beta_1\rangle|\alpha_2\rangle) + |\alpha_1\rangle|\beta_2\rangle\right] = \frac{\hbar}{\sqrt{2}}|0\rangle \\ S_y|-1\rangle &= -\frac{i\hbar}{\sqrt{2}}|0\rangle \end{aligned}$$

These results can be generalized as follows:

$$\begin{aligned} S_z|M_S\rangle &= \hbar M_S|M_S\rangle \\ \left(S_x \pm iS_y\right)|M_S\rangle &= \hbar\left[(S \mp M_S)(S \pm M_s + 1)\right]^{1/2}|M_s \pm 1\rangle \end{aligned}$$

The action of the spin-Hamiltonian in the fictitious spin basis gives then rise to the following Hamiltonian matrix (in units of μ_B):

$\mathcal{H}_{Ze}$	$\|0\rangle$	$\|+1\rangle$	$\|-1\rangle$
$\langle 0\|$	0	$g_\perp \frac{1}{\sqrt{2}}(B_x + iB_y)$	$g_\perp \frac{1}{\sqrt{2}}(B_x - iB_y)$
$\langle +1\|$	$g_\perp \frac{1}{\sqrt{2}}(B_x - iB_y)$	$g_\parallel B_z$	0
$\langle -1\|$	$g_\perp \frac{1}{\sqrt{2}}(B_x + iB_y)$	0	$-g_\parallel B_z$

We can now identify these expressions with the actual matrix elements in the basis of the three D_3 components, keeping in mind the relationship between the complex and real triplet basis, as given in Eq. 7.39. One obtains:

$$\langle 0|\mathcal{H}_{Ze}|+1\rangle = -\frac{1}{\sqrt{2}}\langle A_1|\mathcal{H}|E_x + iE_y\rangle = \frac{1}{\sqrt{2}}[-a + d + i(-b - c)]$$
$$\langle 0|\mathcal{H}_{Ze}|-1\rangle = \frac{1}{\sqrt{2}}\langle A_1|\mathcal{H}|E_x - iE_y\rangle = \frac{1}{\sqrt{2}}[a + d + i(b - c)]$$
$$\langle \pm 1|\mathcal{H}_{Ze}|\pm 1\rangle = \frac{1}{2}[\langle x|\mathcal{H}|x\rangle + \langle y|\mathcal{H}|y\rangle \pm i(\langle x|\mathcal{H}|y\rangle - \langle y|\mathcal{H}|x\rangle] = \pm f$$

From these equations the parameters may be identified as follows:

$$a = 0$$
$$b = -g_\perp B_y$$
$$c = 0$$
$$d = g_\perp B_x$$
$$e = 0$$
$$f = g_\parallel B_z$$

The Zeeman Hamiltonian does not include the zero-field splitting between the A_1 and E states. This can be rendered by a second-order spin operator, which transforms as the octahedral $E_g\theta$ quadrupole component:

$$\mathcal{H}_{ZF} = \frac{D}{3\hbar^2}\left(2\widetilde{S}_z^2 - \widetilde{S}_x^2 - \widetilde{S}_y^2\right) = \frac{D}{\hbar^2}(\widetilde{S}_z^2 - {}^1\!/_3\widetilde{S}^2)$$

One then obtains:

$$D = 3\Delta$$

7.5 The action of the components of the fictitious spin operator on the Γ_8 basis can immediately be derived from the general expressions for the action of the spin operators on the $|SM_S\rangle$ basis functions. It is verified that the spin-Hamiltonian that generates the J_p part of the matrix, precisely corresponds to

$$\mathcal{H}_p = J_p \mathbf{B} \cdot \widetilde{\mathbf{S}}$$

The fictitious spin operator indeed transforms as a T_1 operator, and has the tensorial rank of a p-orbital. However, as we have shown, the full Hamiltonian also includes a J_f part, which involves an f-like operator. To mimic this part by a spin-Hamiltonian one thus will need a symmetrized triple product of the fictitious spin, which will embody an f-tensor, but still transforming in the octahedral symmetry as the T_1 irrep. These f-functions can be found in Table 7.1, and are of type $z(5z^2 - 3r^2)$. But beware! To find the corresponding spin operator, it is not sufficient simply to substitute the Cartesian variables by the corresponding spinor components, i.e z by $\widetilde{S}_z$ etc.; indeed while products of variables x, y and z are commutative, products of the corresponding operators are not. Hence when constructing the octupolar product of the spin components, products of non-commuting operators must be fully symmetrized. For the f_{z^3} function, this is the case for the functions $3zx^2$ and $3xy^2$, which are parts of $3zr^2$. As an example, the operator analogue of $3zx^2$ reads

$$3zx^2 \rightarrow \widetilde{S}_z\widetilde{S}_x\widetilde{S}_x + \widetilde{S}_x\widetilde{S}_z\widetilde{S}_x + \widetilde{S}_x\widetilde{S}_x\widetilde{S}_z$$

One then has for the operator equivalent of $3z(x^2 + y^2)$:

$$\begin{aligned}\widetilde{S}_z\widetilde{S}_x\widetilde{S}_x + \widetilde{S}_x\widetilde{S}_z\widetilde{S}_x + \widetilde{S}_x\widetilde{S}_x\widetilde{S}_z + \widetilde{S}_z\widetilde{S}_y\widetilde{S}_y + \widetilde{S}_y\widetilde{S}_z\widetilde{S}_y + \widetilde{S}_y\widetilde{S}_y\widetilde{S}_z = \\ 3\widetilde{S}_z(\widetilde{S}_x^2 + \widetilde{S}_y^2) + i\hbar\left(\widetilde{S}_x\widetilde{S}_y - \widetilde{S}_y\widetilde{S}_x\right) = 3\widetilde{S}_z(\widetilde{S}_x^2 + \widetilde{S}_y^2) - \hbar^2\widetilde{S}_z\end{aligned}$$

where we have used the commutation relation for the spin-operators:

$$S_xS_y - S_yS_x = i\hbar S_z$$

The octupolar spin operator will then be of type:

$$\begin{aligned}\mathcal{H}_f = \frac{\mu_B}{\hbar^3}g_f B_z\left(\widetilde{S}_z^3 - {}^3\!/_5\widetilde{S}_z\widetilde{S}^2 + {}^1\!/_5\hbar^2\widetilde{S}_z\right) + B_x\left(\widetilde{S}_x^3 - {}^3\!/_5\widetilde{S}_x\widetilde{S}^2 + {}^1\!/_5\hbar^2\widetilde{S}_x\right) \\ + B_z\left(\widetilde{S}_y^3 - {}^3\!/_5\widetilde{S}_y\widetilde{S}^2 + {}^1\!/_5\hbar^2\widetilde{S}_y\right)\end{aligned}$$

In order to identify the parameter correspondence, let us work out the action of this operator on the quartet functions. As an example for a magnetic field along the z-direction, the matrix is diagonal, and its elements (in units of μ_B) are given by

$$\begin{aligned}\langle \pm{}^3\!/_2|\mathcal{H}_f|\pm{}^3\!/_2\rangle &= \pm g_f B_z\frac{3}{2}\left(\frac{9}{4} - \frac{45}{20} + \frac{1}{5}\right) \pm \frac{3}{10}g_f B_z \\ \langle \pm{}^1\!/_2|\mathcal{H}_f|\pm{}^1\!/_2\rangle &= \mp\frac{9}{10}g_f B_z\end{aligned}$$

By comparing these elements to the results in Table 7.8, we can identify the parameter correspondence as

$$J_f = -\frac{3}{10} g_f$$

8.1 Parameter expressions for the cylindrical coordinates:

$$z_t = z_0 + t\frac{H}{n}$$
$$\phi_t = \phi_0 + t\frac{2\pi}{n}$$

Along the left-handed helix n turns over an angle $2\pi/m$ are needed to reach the helical translation H, hence:

$$\chi' = \arctan\frac{mH}{2n\pi R} = \arctan\frac{H}{2(n-1)\pi R}$$

The sum rule then yields

$$\frac{1}{\tan\chi} + \frac{1}{\tan\chi'} = \frac{2n\pi R}{H}$$

Normally the option with the steepest inclination determines the handedness of the helix.

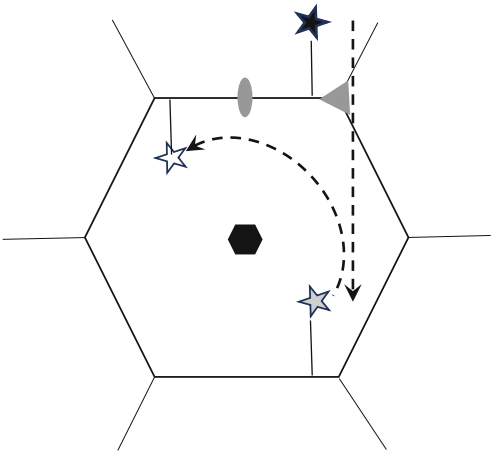
8.2 Let $(U|0)$ denote the Umklapp axis through A, and $(U'|0)$ the axis through B. Then one has

$$(U'|0) = (E|1)(U|0) = (U|0)(E|-1)$$

Both Umklapp axes thus belong to the same coset.

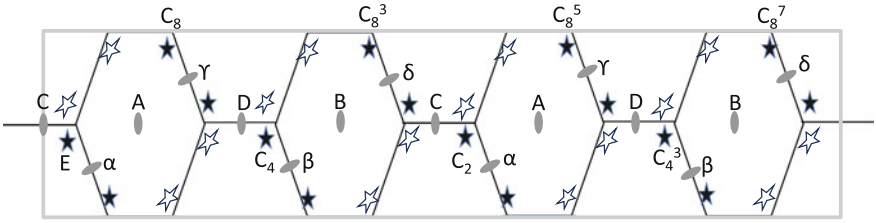
8.3 In the figure below, the twofold edge axis rotates the black star to the white star. This can be done via the intermediate gray star, which is reached by a vertical translation, followed by the third power of the sixfold axis. The twofold edge axis is thus an element of the $(C_6^3|0)$ coset:

$$(C_2^{edge}|0) = (C_6^3|0)(E|-1,+1)$$



8.4 Figure 8.18 shows that every atom with relative coefficient $+1$ is surrounded by two $+1$ atoms and one -1 atom, and vice versa every atom with relative coefficient -1 is surrounded by two -1 atoms and one $+1$ atom. Inserting these values in Eq. 8.73 immediately yields eigenvalue $\lambda = \beta$.

8.5 The figure below shows the proper rotation elements. The orbit of black stars which are reached by the eightfold axis, starting from the E unit element, are labeled with the appropriate powers of C_8, C_4, and C_2. The perpendicular twofold axes are labeled as $A - D$ for the C_2' class, and as $\alpha - \delta$ for the C_2'' class. Improper symmetry elements are found by combining the rotations with the horizontal symmetry plane.



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